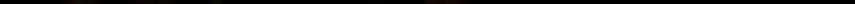


ng.
ns, sy
led they
ween text and
produced by so
st not violate why
nd the analysis of text
ars have provided essen
t to be inherent in a text, 2. I
at take content to emerge in t
e kinds of definitions leads to
n analysis. Berelson's original
d content analysis as "a resear
t content of communication
ed under the dual requirem
be governed by rules that
systematicity" in order to
ther than against ther
ing, reading, and a
testable, but objec
ts. One is his ins
ic endeavors, q
ganda, in psy
n computer
er program
nd of the
t analysis
to the
ceab
in

Content Analysis



Klausur Krippendorff



Content Analysis

An Introduction to Its Methodology

Fourth Edition

Sara Miller McCune founded SAGE Publishing in 1965 to support the dissemination of usable knowledge and educate a global community. SAGE publishes more than 1000 journals and over 800 new books each year, spanning a wide range of subject areas. Our growing selection of library products includes archives, data, case studies and video. SAGE remains majority owned by our founder and after her lifetime will become owned by a charitable trust that secures the company's continued independence.

Los Angeles | London | New Delhi | Singapore | Washington DC | Melbourne

Content Analysis

An Introduction to Its Methodology

Fourth Edition

Klaus Krippendorff
Annenberg School for Communication
University of Pennsylvania



Los Angeles | London | New Delhi
Singapore | Washington DC | Melbourne



FOR INFORMATION:

SAGE Publications, Inc.
2455 Teller Road
Thousand Oaks, California 91320
E-mail: order@sagepub.com

SAGE Publications Ltd.
1 Oliver's Yard
55 City Road
London EC1Y 1SP
United Kingdom

SAGE Publications India Pvt. Ltd.
B 1/1 Mohan Cooperative Industrial Area
Mathura Road, New Delhi 110 044
India

SAGE Publications Asia-Pacific Pte. Ltd.
3 Church Street
#10-04 Samsung Hub
Singapore 049483

Acquisitions Editor: Terri Accomazzo
Editorial Assistant: Erik Helton
Production Editor: Andrew Olson
Copy Editor: Michelle Ponce
Typesetter: C&M Digital (P) Ltd.
Proofreader: Ellen Howard
Indexer: Jeanne Busemeyer
Cover Designer: Anupama Krishnan
Marketing Manager: Liz Thornton

Copyright © 2019 by SAGE Publications, Inc.

All rights reserved. No part of this book may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying, recording, or by any information storage and retrieval system, without permission in writing from the publisher.

Printed in the United States of America

Library of Congress Cataloging-in-Publication Data

Names: Krippendorff, Klaus, author.

Title: Content analysis : an introduction to its methodology / Klaus Krippendorff, University of Pennsylvania.

Description: Fourth Edition. | Los Angeles : SAGE, [2018] | Includes bibliographical references and index.

Identifiers: LCCN 2017050739 | ISBN 9781506395661 (Paperback : acid-free paper)

Subjects: LCSH: Content analysis (Communication)

Classification: LCC P93 .K74 2018 | DDC 302.2—dc23
LC record available at <https://lcn.loc.gov/2017050739>

This book is printed on acid-free paper.

18 19 20 21 22 10 9 8 7 6 5 4 3 2 1

BRIEF CONTENTS

Preface to the Fourth Edition	xii
Introduction	1
PART I • CONCEPTUALIZING CONTENT ANALYSIS	
Chapter 1 • History	10
Chapter 2 • Conceptual Foundation	24
Chapter 3 • Uses and Inferences	51
PART II • COMPONENTS OF CONTENT ANALYSIS	
Chapter 4 • The Logic of Content Analysis Designs	86
Chapter 5 • Unitizing	102
Chapter 6 • Sampling	115
Chapter 7 • Recording/Coding	128
Chapter 8 • Data Languages	157
Chapter 9 • Analytical Constructs	178
PART III • ANALYTICAL PATHS AND EVALUATIVE TECHNIQUES	
Chapter 10 • Analytical/Representational Techniques	196
Chapter 11 • Computer Aids	215
Chapter 12 • Reliability	277
Chapter 13 • Validity	361
Chapter 14 • A Practical Guide	383
Glossary	407
References	414
Index	433
About the Author	453

DETAILED CONTENTS

Preface to the Fourth Edition	xii
Introduction	1
PART I • CONCEPTUALIZING CONTENT ANALYSIS	
Chapter 1 • History	10
1.1 Some Precursors	10
1.2 Quantitative Newspaper Analysis	11
1.3 Early Content Analysis	12
1.4 Propaganda Analysis	14
1.5 Content Analysis Generalized	17
1.6 Computer Text Analysis	18
1.7 Qualitative Approaches	21
Chapter 2 • Conceptual Foundation	24
2.1 Definition	24
2.2 Epistemological Elaborations	27
2.3 Examples	31
2.4 Framework	37
2.4.1 Texts	38
2.4.2 Research Questions	39
2.4.3 Context	40
2.4.4 Analytical Constructs	42
2.4.5 Inferences	43
2.4.6 Validating Evidence	45
2.5 Contrasts and Comparisons	46
Chapter 3 • Uses and Inferences	51
3.1 Traditional Overviews	51
3.2 Extrapolations	54
3.2.1 Trends	55
3.2.2 Patterns	56
3.2.3 Differences	58
3.3 Standards	61
3.3.1 Identifications	61
3.3.2 Evaluations	62
3.3.3 Judgments	64
3.4 Indices and Symptoms	65
3.5 Linguistic Re-Presentations	69

3.6 Conversations	73
3.7 Institutional Processes	74
3.8 Areas of Likely Success	80

PART II • COMPONENTS OF CONTENT ANALYSIS

Chapter 4 • The Logic of Content Analysis Designs 86

4.1 Content Analysis Designs	86
4.1.1 Components	87
4.1.2 Quantitative and Qualitative Content Analysis	91
4.2 Designs Preparatory to Content Analysis	94
4.2.1 Operationalizing Available Knowledge of the Context	94
4.2.2 Testing Analytical Constructs as Hypotheses	95
4.2.3 Developing Analytical Constructs by Trial and Error	96
4.3 Designs Going Beyond Content Analysis	98
4.3.1 Comparing Similar Phenomena Inferred From Different Bodies of Texts	99
4.3.2 Testing Relationships Among Phenomena Inferred From One Body of Texts	99
4.3.3 Testing Hypotheses Concerning How Content Analysis Results Relate to Other Variables	100

Chapter 5 • Unitizing 102

5.1 Units	102
5.2 Types of Units	103
5.2.1 Sampling Units	103
5.2.2 Recording/Coding Units	104
5.2.3 Context Units	105
5.3 Ways of Defining Units	107
5.3.1 Physical Distinctions	107
5.3.2 Syntactical Distinctions	108
5.3.3 Categorial Distinctions	109
5.3.4 Propositional Distinctions	110
5.3.5 Thematic Distinctions	111
5.4 Productivity, Efficiency, and Reliability	112

Chapter 6 • Sampling 115

6.1 Sampling in Theory	115
6.2 Sampling Techniques Applicable to Texts	117
6.2.1 Random Sampling	118
6.2.2 Systematic Sampling	118
6.2.3 Stratified Sampling	119
6.2.4 Varying Probability Sampling	119
6.2.5 Cluster Sampling	120
6.2.6 Snowball Sampling	121
6.2.7 Relevance Sampling	122
6.2.8 Census	123
6.2.9 Convenience Sampling	124

6.3 Sample Size	124
6.3.1 Statistical Sampling Theory	124
6.3.2 Sampling Experiments	125
6.3.3 The Split-Half Technique	127

Chapter 7 • Recording/Coding **128**

7.1 The Function of Coding and Recording	128
7.2 Coder Qualifications	130
7.2.1 Cognitive Abilities	130
7.2.2 Background	130
7.2.3 Frequency	131
7.3 Coder Training	131
7.4 Crowdcoding	134
7.4.1 Methodological Advantages	135
7.4.2 Efforts to Assure Crowdcoding Quality	136
7.4.3 Uses of Reliability Tests	136
7.4.4 Limitations and Warnings	137
7.5 Approaches to Defining the Semantics of Data	138
7.5.1 Verbal Designations	139
7.5.2 Extensional Lists	140
7.5.3 Examples	141
7.5.4 Decision Schemes	142
7.5.5 Magnitudes and Scales	143
7.5.6 Simulation of Hypothesis Testing	145
7.5.7 Simulation of Interviewing	146
7.5.8 Constructs for Closure	148
7.6 Records	149
7.6.1 Administrative Information	151
7.6.2 Information on the Organization of Records	151
7.6.3 Substantive Information About the Phenomena of Interest	153

Chapter 8 • Data Languages **157**

8.1 The Place of Data Languages in Analytical Efforts	157
8.2 Definitions	159
8.3 Variables	161
8.4 Nominal Variables	167
8.5 Ordered Variable	167
8.5.1 Chains	168
8.5.2 Recursions	169
8.5.3 Cubes	169
8.5.4 Trees	171
8.5.5 Multi-Valued Sets	171
8.6 Metrics	172
8.6.1 Ordinal Metrics	173
8.6.2 Interval Metrics	175
8.6.3 Ratio Metrics	176
8.7 Mathematical Operations	176

Chapter 9 • Analytical Constructs	178
9.1 The Role of Analytical Constructs	178
9.2 Sources of Certainty	180
9.2.1 Previous Successes and Failures	180
9.2.2 Expert Knowledge and Experience	181
9.2.3 Established Theories	183
9.2.4 Embodied Practices	185
9.3 Types of Constructs	186
9.3.1 Extrapolations	186
9.3.2 Applications of Standards	186
9.3.3 Indices and Symptoms	186
9.3.4 Re-Presentations	188
9.3.5 Conversations/Interactions	189
9.3.6 Institutional Processes	189
9.4 Sources of Uncertainty	191
9.4.1 Variance of the Target Domain	191
9.4.2 Confidence Levels	192
9.4.3 Stability of the Analytical Constructs Model	192

PART III • ANALYTICAL PATHS AND EVALUATIVE TECHNIQUES

Chapter 10 • Analytical/Representational Techniques	196
10.1 Counts	197
10.2 Cross-Tabulations, Associations, and Correlations	200
10.3 Multivariate Techniques	203
10.4 Factor Analysis and Multidimensional Scaling	205
10.5 Images, Portrayals, Semantic Nodes, and Profiles	207
10.6 Contingencies and Contingency Analysis	210
10.7 Clustering	212
Chapter 11 • Computer Aids	215
11.1 What Computers Do	215
11.2 How Computers Can Aid Content Analyses	216
11.3 Text Analyses	220
11.3.1 Accounts of Character Strings	220
11.3.2 Text Mining	227
11.3.3 Mining Textual Evidence for Theories	236
11.3.4 Analysis of Networked Texts	239
11.4 Computational Content Analyses	246
11.4.1 Coding/Dictionary Approaches	248
11.4.2 Machine Learning of Discriminant Functions	254
11.4.3 Statistical Association Approaches	256
11.4.4 Semantic Network Approaches	258
11.4.5 Memetic Approaches	264
11.5 Qualitative Data Analysis Support	268

11.6 Frontiers	272
11.6.1 Intelligent Browsers	272
11.6.2 Common Platforms	272
11.6.3 Computational Theories of Meaning	273
11.6.4 Utilization of Intertextualities	274
11.6.5 Analyzing Networked Text	274
11.6.6 Natural Interfaces	275

Chapter 12 • Reliability **277**

12.1 Why Reliability?	277
12.2 Reliability Designs	280
12.2.1 Types of Agreement; Types of Reliability	280
12.2.2 Conditions for Generating Reliability Data	283
12.2.3 Reliability Data	285
12.3 Agreement on Coding Predefined Units	289
12.3.1 The Direct Path to α -Agreement	293
12.3.2 The Path to α via Coincidence Matrices	296
12.3.3 Difference Functions for Data With Various Metrics	299
12.3.4 Some Typical Examples and Important Properties	306
12.3.4.1 <i>Binary Data, None Missing</i>	306
12.3.4.2 <i>Coincidences and Contingencies</i>	308
12.3.4.3 <i>α's Decomposition of Observed Coincidences</i>	310
12.3.4.4 <i>Two Observers, Several Nominal Values, No Missing Data</i>	310
12.3.4.5 <i>Uninformative Values in Ordered Variables</i>	312
12.3.4.6 <i>Trading Information for Reliability</i>	314
12.3.5 Some Contested Coefficients and Correspondences	315
12.4 Accuracy, Surrogacy, and the Decisiveness of Majorities	323
12.4.1 Accuracy of or Surrogacy for Crowd Members' Judgments	325
12.4.2 Accuracy of or Surrogacy for Majorities or Aggregates	326
12.4.3 The Decisiveness of Majorities or Aggregates	328
12.4.4 Numerical Examples	329
12.5 The Reliability of Text Mining and Information Retrieval	331
12.5.1 For Replications of Searches	334
12.5.2 For Replications of Judgments on the Relevance of Retrieval Results	336
12.6 Agreement on Unitizing and Coding Finite Continua	337
12.6.1 Reliability Data From Unitizing	338
12.6.2 Observed and Expected Coincidences	340
12.6.3 Agreement on the Partitions of a Continuum Into Unequal Segments	342
12.6.4 Agreement on the Distinction Between Relevant and Irrelevant Segments	343
12.6.5 Agreement on the Coding of Single-Valued Segments Independent of Unitizing	344
12.6.6 Agreement on Individual Values Among Valued Intersections	346
12.7 Agreement on Multi-Valued Coding	347
12.7.1 Two Reasons for Multi-Valued Coding	348
12.7.2 Reliability Data for Multi-Valued Coding	349
12.7.3 Multi-Valued Difference Functions	349

12.7.4 Disagreements for Multi-Valued Data	350
12.7.5 ϵ_α for Multi-Valued Coding	351
12.8 Statistical Properties of α	351
12.8.1 Insufficient Variation	351
12.8.2 Statistical Significance	353
12.8.3 Sampling Considerations	354
12.8.4 Standards for the Reliability of Data	356
Chapter 13 • Validity	361
13.1 Validity Defined	361
13.2 A Typology for Validating Evidence	365
13.2.1 Sampling Validity	368
13.2.2 Semantic Validity	370
13.2.3 Structural Validity	376
13.2.4 Functional Validity	377
13.2.5 Correlative Validity	378
13.2.6 Predictive Validity	381
Chapter 14 • A Practical Guide	383
14.1 Designing an Analysis	384
14.1.1 Text-Driven Analyses	384
14.1.2 Problem-Driven Analyses	386
14.1.2.1 <i>Formulating Research Questions</i>	387
14.1.2.2 <i>Ascertaining Stable Correlations (With the Research Questions)</i>	388
14.1.2.3 <i>Locating Relevant Texts</i>	390
14.1.2.4 <i>Defining and Identifying Sampling Units Among Relevant Texts</i>	392
14.1.2.5 <i>Sampling a Sufficiently Large Number of These Units</i>	393
14.1.2.6 <i>Developing Appropriate Coding Categories and Recording Instructions</i>	394
14.1.2.7 <i>Selecting Appropriate Analytical Procedures</i>	395
14.1.2.8 <i>Adopting Standards for the Reliability of Generated Data and Statistical Significance Levels for the Results</i>	396
14.1.2.9 <i>Allocating Resources</i>	397
14.1.3 Method-Driven Analyses	398
14.2 Writing a Research Proposal	400
14.2.1 Rhetorical Function	400
14.2.2 Contractual Function	401
14.2.3 Outline for a Research Proposal	402
14.3 Applying the Research Design	403
14.4 Narrating the Results	404
14.4.1 Outline for a Research Report	405
Glossary	407
References	414
Index	433
About the Author	453

PREFACE TO THE FOURTH EDITION

Content analysis is one of the most important research techniques in the social sciences. It acknowledges that society is enacted in talk, texts, and other modalities of communication and that understanding social phenomena cannot be achieved without understanding how language operates in the social world. Content analysts inquire into social phenomena by treating data not as physical events but as communications that are created and disseminated to be seen, read, interpreted, enacted, and reflected upon according to the meanings they have for their recipients. Interpreting communications as texts in the contexts of their social uses distinguishes content analysis from other empirical methods of inquiry.

Investigative methods in the natural sciences, by contrast, rarely are concerned with meanings, contents, intentions, references, communications, and what they do. Natural scientists may have good reasons to presume that the objects of their attention are not aware of how they are being observed and do not speak a language to articulate how they want to appear, but they also rarely reflect on how their own writings shape their conception of nature and what it allows them to do or prevents them from studying. The conception of nature as preexisting and independent of scientific attention is convenient for it relieves natural scientists from being held accountability for the realities that their theories construct, but it also prevents them from contributing to understanding social phenomena that arise in how they are described and enacted. Where social scientists adopt natural scientific methods of inquiry, it is the epistemology inscribed in these methods that prevents them from addressing what matters in the social life of people, what motivated and inspires them, and how they enact what they know. Certainly, content analysis is not the only research method that takes meanings seriously, but it is a method that has the additional advantage of being unobtrusive—not affecting the sources or receivers of what is being analyzed—able to process large volumes of data—providing insights that exceed the comprehension of ordinary readers of textual matter—and its findings can mediate between individuals and challenge the operation of institutions.

In the first edition of *Content Analysis*, published in 1980, I suggested that content analysis was at a crossroads. Content analysts at that time had a choice: They could continue their shallow counting game, motivated by a journalistic fascination with numbers and a narrow conception of science in which quantitative measurements provided the only evidence that counted (Lasswell, 1949/1965b), or they could redirect their attention to social phenomena that are both generated by and constituted in texts and images and, hence, needed to be understood through their written and pictorial manifestations. The latter motivated the first edition, and I am pleased to say that the epistemology and logic I developed therein have survived critical assessments and the challenges posed by the radical transformations that the information revolution and social media have introduced into contemporary social

practices. The widespread use of electronic, and hence computer-readable, texts concerning virtually everything that matters to us as social beings has moved content analysis, particularly computer-aided text analysis, into the center of how society examines itself.

In the 1970s, content analysis was a research method that had entered the psychological and social sciences but was used mainly by journalists and communication researchers. During that time, the amount of human effort required to collect, transcribe, and code textual data made content analysis a time-consuming, labor-intensive, and often costly effort that limited the scope of what it could address. However, it became an efficient alternative to public opinion research and a method of tracking markets, political influence, and emerging ideas. It was used to detect biases in reporting of facts, settle legal disputes, understand how institutions establish themselves in the texts they produce, and explore the mind of individuals through what they said or wrote.

The first edition of *Content Analysis* reported numerous analytical innovations but barely met the challenges of the new era of electronic communication. It taught a generation of social scientists about the methodology of this research technique. It soon became the leading text on the subject and was translated into Italian, Persian, Japanese, Spanish, Indonesian, and Hungarian.

Although the outline of the second, 2004 edition of *Content Analysis* remained essentially unchanged from that of the first, the book clarified numerous methodological issues and responded to the technique's latest challenges. All chapters were substantially rewritten, addressing developments that had taken place since 1980, especially concerning computer-aided text analysis. It added a practical guide that incorporated input from my students, colleagues, and own practical and consulting experiences in various academic and commercial research projects. In 2004, it received the International Communication Association's Fellows Book Award for its lasting contribution to the advancement of communication studies.

To broaden the range of examples and recommend solutions to practical problems, Mary Angela Bock and I coedited *The Content Analysis Reader* (2009). This reader was meant to be a companion to the methodologically oriented *Content Analysis*. I have been using the *Reader* with considerable success, heard from others about its added benefits, and recommend it to students, teachers, and practitioners of content analysis.

The third, 2013 edition of *Content Analysis* continued the careful and critical examination of how texts and images can support inferences about phenomena not accessible to independent observations and updated several chapters, notably on computer aids. It also added a convenient glossary. In 2014, it appeared in its Chinese translation. In 2017, my work on content analysis was honored by receiving the first methodological innovation award from the Mass Communication Division of the International Communication Association.

The outline of this fourth edition is kept essentially as it was before. It corrects a previously published error in the formula for computing the reliability of unitizations. Its discovery by a group of French scholars is just one example of the many fruitful collaborations that the book encouraged. It includes a section on crowdsourcing and expands Krippendorff's Alpha in several dimensions: for assessing the reliability of text retrievals, crowdcoding, and of multi-valued coding of predefined units as well as

of unitized textual continua. And it points to software among many versions known to be correct.

I thank my students at the University of Pennsylvania's Annenberg School for Communication for their interest and for giving me useful feedback and my colleagues for presenting me with the challenging methodological problems of their content analyses. I would also like to thank numerous readers of the earlier editions—both students and practicing content analysts—for sharing their comments and recommendations and Sage Publications for its continuing support of content analysis literature.

During the 38 years of publications of *Content Analysis*, its three previous editions have reached an enormous audience. It has been widely adopted as a text in social science, humanities, and business curricula, served researchers as a guide for the design and execution of large and small content analyses, and provided epistemologically sound standards for justifying as well as critically evaluating content analysis findings. When I travel to national and international conferences, I continue to be amazed and pleased to meet researchers from all over the world who tell me how studying this text has helped them in their current inquiries. I hope this fourth edition will withstand critical examinations and continue to advance scholarship based on texts, images, and communications in all fields that acknowledge to operate within language.

—Klaus Krippendorff

*Professor of Communication and
Gregory Bateson Professor Emeritus, for
Cybernetics, Language, and Culture*

*The Annenberg School for Communication,
University of Pennsylvania*

INTRODUCTION

The term *content analysis* is about 70 years old. *Webster's Dictionary of the English Language* included the term in its 1961 edition, defining it as “analysis of the manifest and latent content of a body of communicated material (as a book or film) through classification, tabulation, and evaluation of its key symbols and themes in order to ascertain its meaning and probable effect.” The intellectual roots of content analysis, however, can be traced far back in human history, to the beginning of the conscious use of symbols and voice, especially writing. This conscious use, which replaced the magical use of language, has been shaped by the ancient disciplines of philosophy, rhetoric, and cryptography. It has also spawned religious inquisitions and political censorship on the part of ruling establishments. Today, symbolic phenomena are institutionalized in art, literature, education, and the mass media, including the Internet. Theoretical and analytical concerns are found in such academic disciplines as anthropology, linguistics, social psychology, sociology of knowledge, and the comparatively younger field of communication studies. Many practical pursuits have grown from these fields: psychotherapy, advertising, politics, the arts, and so on. Virtually all disciplines within the whole spectrum of the humanities and the social sciences, including those that seek to improve the political and social conditions of life, are concerned with the functions and effects of symbols, meanings, and messages. In recent years, the emergence of the information society has moved the minutiae of communication—texts, contexts, images, interfaces, and, above all, information—into the very center of researchers’ attempts at self-understanding.

However ancient the roots of analyzing symbolic and textual matter might be, today’s content analysis is significantly different, in aim and in method, from that of the past. Contemporary content analysis has three distinguishing characteristics.

First, content analysis is an *empirically grounded method*, exploratory in process, and predictive or inferential in intent. Many of our current concepts relating to language are of Greek origin; for example, the words *sign*, *significance*, *symbol*, and *logic* all have Greek roots. However, the ancient Greeks’ interest in language was largely prescriptive and classificatory, not empirical. Aristotelian logic set the standards for clear expression, and much of rhetorical theory was directed toward a normative conception of persuasive argumentation. Science that explores rather than declares is a relatively recent accomplishment. Only a century ago, George Boole and his contemporaries believed that the brain works according to (Boolean) logic and that human conduct is entirely rational. However, computers built on this logic turned out to be rather disappointing thinking machines. Empirical research in psychology is replacing Aristotelian categories in favor of a “psycho-logic.” And

we no longer measure human communication against the ideal of transmitting information. Instead, we inquire into what happens to the relationships between people who converse with one another.

With new conceptualizations and an empirical orientation, contemporary content analysts join other researchers in seeking valid knowledge or practical support for actions and critique. However, unlike researchers who employ other empirical techniques, content analysts examine data, printed matter, images, or sounds—texts—in order to understand what they mean to people, what they enable or prevent, and what the information conveyed by them does. These are questions for which natural scientists have no answers and for which their methods are generally insensitive.

Second, contemporary content analysis *transcends traditional notions of symbols, contents, and intents*. This may be seen in the evolution of the concept of communication, in how the development of media technologies has shaped our attention to communication, and in the role of culture in assigning significance to what is being analyzed. I would argue that in recent years our awareness of communication has undergone six conceptual revolutions, as described below, and it probably is not stopping with these:

- The idea of *messages*: the early awareness not only that verbal discourse is portable when written but also that writing has predictable effects. This awareness emerged in ancient Greece when messengers were used as the carriers of significance; history became documented; laws of the land were laid down in writing; and written instructions built organizational structures, directed events, and influenced (and possibly deceived) their receivers or the public. The concept of a message was a precursor of the rhetorical exploration of language. Tropes, syllogisms, and meanings came to be thought of as inherent qualities of speeches, letters, or documents. But a message is the metaphorical container of all these, a “container of content,” a vehicle for shipping meanings from one place to another—for example, when we now leave a message for someone on voice mail or say that a message was meaningful (full of meanings) or meaningless (void of meanings).
- The idea of *channels*: the awareness of constraints that every medium imposes on human communication. This awareness came with the increased reliance on different media of communication and served to explain their limitations: The alphabet limits what one can say in writing; the telephone confines communication to sound; and a television station can air no more than what is transmittable without interference from other stations, appealing to large audiences, and deemed profitable by sponsors. The channel metaphor conjures images of canals and pipes with restricted capacities for shipping messages (with their contents) of certain forms and volumes.
- The idea of *communication*: the awareness that it establishes relations between senders and receivers, enables the negotiation of complementary roles, constitutes social structures, and communities whose members have a sense of knowing each other. This awareness has ancient roots in the conception of dialogue but has entered current awareness through the

limitations experienced in the use of mass communication. By producing and disseminating identical messages—news and entertainment—to everyone, the mass media promised to be an agent of sharing, of building publics, ideally beyond borders. Modeled on the success of mass production and ever-expanding markets for useful goods, the mass media also made us aware of where this one-way model failed: making people aware of often amplifying social differences, limiting face-to-face conversations, curtailing public deliberations, and addressing only commercially viable issues. In U.S. culture, mass-media technology has become synonymous with progress, and the lack of communication is often blamed for most interpersonal problems or national conflicts.

- The idea of *systems*: the awareness of global, dynamic, and technologically supported interdependencies. This idea emerged with the growth of communication networks—telephone nets, wire services, mass-media systems, and most recently the Internet—transforming commerce, politics, and interpersonal relationships, creating networks whose properties have largely defied theoretical accounts. Unlike the one-way mass media, systems are marked by the interactivity and simultaneity of parallel communication on a massive scale and with the potential of near universal participation.
- The idea of *computation*: the awareness of the increasing replacement of communicators by algorithms. Messages may be generated by robots and addressed to computers, whether it involves the use of automatic teller machines, online shopping, robo-calls, or computer-generated messages on the Internet. This changes the conception of messages as understandable by senders and receivers to what has computable consequences. Letter writing is being replaced by e-mailing, and interpersonal relations are shaped by the platforms that mediate them. Computation has expanded access to textual matter—consider the amazing power of search engines, the individual ability to travel to distant places virtually—but it precludes computationally unavailable experiences. Computation not only has made social life more variable but also has introduced complexities that easily escape understanding.
- The idea that reality is *discursively co-constructed*: the recognition that we live together in language, can never think alone (Sloman & Fernbach, 2017), and constitute our institutions as interdependent realities through what we say to and do with each other. Wittgenstein’s (1958) notion of language as consisting of numerous language games that coordinate actions in the world comes to mind. The empirical fact that almost all innovations are the product of collaborations, that even scientific accomplishments arise and are vetted in discourse communities renders texts as actively supporting or challenging social realities.

This rather sketchy history of communication suggests that researchers who are concerned with texts can no longer treat them as symbolic or representational, nor can they limit themselves to questions about “who says what, through which channels, to whom, and with which effects” (Lasswell, 1960). The popular and simplistic notion of “content” has outlived its explanatory value (Krippendorff, 2017): *content*, the *what*

of a communication, something that authors are conceived of as *entering* into the texts they are producing, something that is *preserved* in texts while *shipped* to remote receivers, who are expected to *remove* its *content* as intended and henceforth *share it* with others through still other containers. This bizarre notion of message content renders authors as authorities of what messages contain and to the conception of *content analysts* as experts, able to objectively account for what messages *really contain* or are *intended to convey*, distinct from ordinary and usually inaccurate readings by less qualified readers.

The absence of authors, especially historical ones, makes the use of this container metaphor nonverifiable. Mass media messages tend to replace individual authors by institutional constructs, and their effects are more important than what someone may have intended. Trust in them has become trust in the technology of transmission. The mass media coordinate the lives of their audience members. Their practices challenge old distinctions among channels of communication, obviating physical distances, and pushing capacities of the human participants to their limits. This erodes the validity of traditional communication theories, all the while enabling computer systems to thrive in this new environment. It is these computer systems that simulate and coordinate parts of the very social processes that researchers wish to understand. This is a radically changing world in which texts play distinctly new roles. Newspaper accounts, public opinion polls, corporate reports, files in government agencies, credit information, bank transactions, and, above all, textual data archives of social media—all are now linked into networks that can be analyzed from numerous positions. In effect, the social systems that we conceived of as explaining society are now holographically retreating into our computers. This development calls for a redefinition of content analysis, one that aligns content—the presumed target of the research effort—with how contemporary society operates and understands itself through its texts.

With the container metaphor rendered useless, perhaps the term *content analysis* no longer fits the realities of contemporary society. For better or for worse, I continue to use the term in this book, but I also plead with readers to unflinchingly oppose the naive and misleading entailments of this still pervasive metaphor.

Third, *contemporary content analysis has been forced to develop a methodology of its own*, one that enables researchers to plan, execute, communicate, reproduce, and critically evaluate their analyses whatever the particular results. Content analysts have had to develop such a methodology for three reasons:

- *Content analysts now face larger contexts.* The shift in interest from small collections of printed messages to systems and then to electronic texts and images circulating in the environment of content analysts is tied less to the nature of textual data than to the increasingly complex worlds that produce and are sustained by these data. This shift calls for theories and conceptions that earlier content analysts did not need. Although content analysts have frequently lamented the lack of general theories that could justify their work, progress in implementing more specific or microlevel theories is encouraging. This is especially true where content analysis has migrated through disciplines that were not previously concerned with textual data, such as the cognitive sciences and artificial intelligence.

- *Greater numbers of researchers need to collaborate in the pursuit of large-scale content analyses.* This observation is a correlate of the growing sample sizes of relevant texts, the analysis of which easily exceeds what individual analysts can handle. It implies that content analysts must work together, in parallel, and as research teams. Teamwork, however, needs to be organized reliably. Both the social problem of coordinating researchers and the methodological problem of assuring replicability tend to be solved through the adoption of a language whose vocabulary enables researchers to clarify the analytical procedures they use, negotiate the individual responsibilities of the participants, assure agreement on the analytical categories, and evaluate the performance of team members. A novel solution of the human problem of analyzing increasing volumes of textual data is to *outsource the coding of textual matter to crowds who can interpret data reliably*. The methods to cope with this practice are currently at their infancy. Developing them moves content analysis closer to other social science methods such as survey research. Economy and speed are noticeable benefits of outsourcing the coding of textual data, but reliability and depth are still issues to be improved upon.
- *The large volumes of electronically available data call for computer aids.* Such aids need to convert large bodies of electronic text into more manageable representations that preserve the ability to answer the research questions that content analysts seek to derive from available data. Currently, two competing approaches are pursued to handle overwhelming volumes of data. One is to develop text analysis software capable of computing complex statistics of the character strings of text or pixelized images. However, exactly how such software transforms raw texts into abstract representations is often uncertain and the subject of exaggerated claims, for example, being content extractors or promising to mine meanings. The other approach is to develop less ambitious computer aids that merely replace the labor-intensive and recurrent parts of more complex content analyses. For example, qualitative data analysis software tends to allow recursive coding decisions to be automatically extended to large documents. Software that could serve as aids to parts of content analytical steps requires some transparency, enabling analysts to assess their appropriateness by tracing and evaluating the transformations that texts are undergoing in the process.

To be clear, *methodology* is not a value in itself. The purpose of methodology is to enable researchers to plan and examine critically the logic, composition, and protocols of research procedures; to evaluate the performance of individual techniques; and to estimate the likelihood of particular research designs to contribute to knowledge. All researchers must become proficient in defining the terms of their analysis and justify the analytical steps taken to a skeptical friend or critical colleague. Methodology provides a language for talking about the process of research, not about subject matters. In the history of scientific pursuits, the development of methodology has always been a major accomplishment. For example, for thousands of years, humans preserved history by retelling or chanting stories, for example, of the *Iliad* before Homer put it in writing. It was only a century ago that the historian Leopold von Ranke gave the

“document” the methodological status it now has in the academic study of history. Similarly, scholars practiced “content analysis” well before Berelson and Lazarsfeld (1948) undertook the first codification of this method. Although many critics claim that each content analysis is unique, possibly by focusing mainly on its subject matter, I would argue that all content analyses share a procedural logic that can be examined and is used to justify their procedures in terms of socially acceptable criteria. What content analyses have in common form the substance of this book.

I disagree with the frequent contention that content analysis is “nothing more than what everyone does when reading a newspaper, except on a larger scale.” Content analysis may have been that way, in its early, journalistic stage, and its methodology does not rule out such readings, but this narrow definition is no longer sufficient today. As newspaper readers, we are perfectly justified in applying our individual worldviews to texts and enacting our interest in what those texts mean to us; in fact, as citizens, we can hardly do otherwise. But as content analysis researchers, we must do our best to explicate what we are doing and describe how we derive our judgments, so that others—especially our critics—can replicate our results or build on them.

This book, then, introduces readers to ways of analyzing meaningful matter, texts, images, and voices—that is, data whose physical manifestations are secondary to what they mean to particular populations of people. The chapters are grouped into three main parts. Part I, “Conceptualizing Content Analysis,” begins with a brief chapter on the history of content analysis. In Chapter 2, I develop a definition of content analysis that distinguishes this technique from other methods of inquiry, and in Chapter 3, I present a discussion of some of the ways in which content analysis has been applied. The chapters in Part II, “Components of Content Analysis,” outline the procedures used in content analyses, beginning with their procedural logic and moving naturally from “unitizing,” “sampling,” “recording/coding” in terms of formal “data languages,” and “analytical constructs.” The chapters in Part III, “Analytical Paths and Evaluative Techniques,” trace several paths through content analysis protocols. In this part of the book, I discuss analytical constructs that enable researchers to draw inferences from data, the use of computers and computational techniques, and the two principal criteria used in evaluating content analyses: reliability and validity. In the final chapter, I provide a practical guide that summarizes the foregoing discussion from a practitioner’s perspective.

Readers who have never undertaken a content analysis may want to begin by reading Chapter 1, on the history of content analysis, and Chapter 3, on the uses of this technique, to get a sense for whether it suits their research interests. If it does, they might want to familiarize themselves with the conceptual foundations of content analysis by reading Chapter 2. Beginners in content analysis are advised to start with a small pilot project, to get a feel for what is involved in undertaking a larger project. Methodology without some practice is empty. The guidelines in Chapter 14, although written as a summary, could also serve as a start. In this chapter, readers will find many helpful references to pertinent chapters in this volume, which may answer emerging questions and place these answers within the context of larger methodological issues. Beginning researchers will soon realize that content analyzing text is not a mechanical task and neither is designing of a project. Both undertakings require creativity and competence.

Readers who have had some experience with coding will acquire a larger perspective on what they had been doing. As the table of contents suggests, coding

is only a small part of content analysis—despite popular misconceptions. In fact, only Chapter 7 is devoted to issues of coding and recording content analysis data. Attention to their problems is particularly important when data or texts are unwieldy, complex, and deserving of careful attention. By coding/recording textual matter, one learns to appreciate both the conceptual problems involved in imposing analytical categories on ordinary readings of text and the ways in which competent researchers have managed to solve such problems. I like to mention *The Content Analysis Reader* (Krippendorff & Bock, 2009), which provides many more examples of solutions to practical problems than this book could discuss in sufficient length. Designing a content analysis is something different, however. I recommend that readers who have had experience with coding expand on that experience by examining the chapters offered here about all the other components of content analysis, adding these to the conceptual framework they seek to operationalize enriches an analysis. Such readers might well look into Chapter 11, on computer aids, to gain an alternative perspective on manual unitizing and coding.

Readers who have already undertaken content analyses or similar text-based research will discover in this book alternative paths for inquiries of texts and a vocabulary that can be used in deliberating about what is involved in analyzing texts—unlike observational studies of naturalistic phenomena—as data whose significance stems from the meanings that others bring to their readings. Those who think they know what content analysis is are advised to start with Chapter 2, on the conceptual foundations of content analysis. This chapter discusses the ways that researchers talk about content and exposes readers to the larger perspective they will need in order to conceive a content analysis or critically evaluate the content analyses of others. As a condition for publication, scholarly journals increasingly demand some demonstration of why a content analysis should be taken seriously. In the past, content analysts relied heavily on conceptions of content as “contained” in messages, as discussed above, or “inherent” to texts. This settled the thorny issue of multiple text interpretations by fiat and consequently disabled explicitness about the researchers’ procedures. Several research traditions—such as interpretive research, discourse analysis, literary scholarship, and rhetoric—tend to be plagued by similar conceptions. Researchers from these traditions would greatly benefit from explicating their approaches, checking their results against the work of others, and evaluating the social consequences of their findings outside their own schools of thought—as I am suggesting.

For experts in content analysis, this book raises several epistemological questions that practitioners rarely ask, transforms them into methodological ones, and provides new solutions to practical problems.

Readers who must make decisions concerning whether or not to trust the findings of content analyses and other text-based research—for instance, judges in courts of law, practitioners in the fields of public relations and advertising, and reviewers of research submitted for funding or publication in scientific journals—will find the vocabulary of this book useful as they need to weigh the quality of findings and make informed recommendations for improvements. Such readers will find the discussions in Chapters 2 (on the conceptual foundations), 12, and 13 (on reliability and validity) especially applicable to evaluative endeavors.

While this book may serve as a handbook for various practitioners, it grew out of my experiences in teaching courses, seminars, and workshops in content analysis. In the context of education, I conceive of it foremost as a textbook for advanced

undergraduate and beginning graduate students. Teachers and their students may not want to work through all the chapters in their numerical order. For instance, those intending to use computers will find Chapter 11 more important than Chapter 7, on recording/coding, and may omit Chapter 12, on reliability, which is not a problem for software applications; but they ought to consider the easily ignored validity of computer uses, discussed in Chapter 13. Students with specific projects in mind may pass over sections that may not be useful to their projects. However, readers should not rule out chapters as irrelevant before knowing what they offer.

Finally, for me, the book will have achieved its purpose if it helps to make the newly available wealth of electronic texts accessible to systematic analysis, if it improves the social significance of research in the humanities and the social sciences, and if it encourages developments beyond the methods discussed therein. Understanding the realities that social actors construct with the textual matter they produce and transmit is an ongoing challenge.

CONCEPTUALIZING CONTENT ANALYSIS

PART

I

1

HISTORY

Empirical inquiries into the meanings of communications date back to theological studies in the late 1600s, when the Church found the printing of nonreligious materials to be a threat to its authority. Such inquiries have since mushroomed, moving into numerous areas and becoming the backbone of communication research. This chapter discusses several stages in the history of content analysis: quantitative studies of the press; propaganda analysis during World War II; social scientific uses of the technique in studies of political symbols, historical documents, anthropological data, and psychotherapeutic exchanges; computer text analysis and the new media; and qualitative challenges to content analysis.

1.1 SOME PRECURSORS

Content analysis entails a systematic reading of a body of texts, images, and symbolic matter, not necessary from an author's or user's perspective. Although the term *content analysis* did not appear in English until 1941 (Waples & Berelson, 1941, p. 2; cited in Berelson & Lazarsfeld, 1948), the systematic analysis of text can be traced back to inquisitorial pursuits by the Church in the 17th century. Religions have always been captivated by the written word, so it is not surprising that the first known dissertations about newspapers were defended in 1690, 1695, and 1699 by individuals pursuing academic degrees in theology. After the advent of the printing press, the Church became worried about the spread of printed matter of a nonreligious nature, and so it dealt with newspaper content in moralizing terms (Groth, 1948, p. 26). Surprisingly, in spite of the rhetorical tradition of ancient Greece, which was normative and oral in orientation, the 17th century contributed very little to the methodology of content analysis.

Probably the first well-documented quantitative analyses of printed matter occurred in 18th-century Sweden. According to Dovring's (1954–1955; see also Krippendorff & Bock, 2009, Chapter 1.1) account, these analyses were undertaken as the result of the publication of the *Songs of Zion*, a collection of 90 hymns of unknown authorship. The collection had passed the Royal Swedish censor, but soon after its publication it was blamed for undermining the orthodox clergy of the Swedish state church. When the collection became popular, it was said to be “contagious” and was accused of aiding a dissenting group. Outstanding in this case is the fact that literary scholars of good reputation participated in the controversy, which crystallized around the question of whether

the songs harbored dangerous ideas and, if so, how. Scholars on one side made a list of the religious symbols in the songs and became alarmed. Those on the other side, however, found the very same symbols in established songbooks and so discounted the claimed difference. Then some scholars noted that the symbols in the songs occurred in different contexts and had acquired meanings that were different from those taught in the official church. A debate arose about whether the meanings should be interpreted literally or metaphorically. The interpretations came to be compared with the results of a German study of the outlawed Moravian Brethren, a religious sect whose members later emigrated to the United States. This process—of revising a method in response to criticism—continued until it became clear to both sides in the debate how the symbols in the *Songs of Zion* differed from the symbols used in the official songbooks and how this (in the end political) phenomenon could be explained. The controversy generated many ideas that are now part of content analysis and stimulated debates about methodology that continue today.

In 1903, Eugen Löbl published in German an elaborate classification scheme for analyzing the “inner structure of content” according to the social functions that newspapers perform. His book, which became well-known in journalistic circles, contributed to the idea of *Publizistik*, or newspaper science, and foreshadowed functionalism, but it did not stimulate empirical investigations.

At the first meeting of the German Sociological Society in 1910, Max Weber (1911; see also Krippendorff & Bock, 2009, Chapter 1.2) proposed a large-scale content analysis of the press, but for a variety of reasons, the research never got off the ground. During the same period, Andrei Markov (1913), who was working on a theory of chains of symbols, published a statistical analysis of a sample of Pushkin’s novel in verse, *Eugene Onegin*. These inquiries were discovered only recently or influenced the content analysis literature only indirectly. For example, Weber is celebrated as one of the great sociologists, but his advocacy of the use of content analysis as a method for understanding the mass media is relatively unknown. And Markov’s probability theories entered the content analysis literature only through Shannon’s mathematical theory of communication (see Shannon & Weaver, 1949), which influenced Osgood’s (1959) contingency analysis and cloze procedure.

1.2 QUANTITATIVE NEWSPAPER ANALYSIS

The beginning of the 20th century saw a visible increase in the mass production of newsprint. In the United States, the boom in newspapers created mass markets and interest in public opinion. Journalism schools emerged, leading to demands for ethical standards and for empirical inquiries into the phenomenon of the newspaper. These demands, plus a somewhat simplistic notion of scientific objectivity, were met by what was then called *quantitative newspaper analysis*.

Probably the first quantitative newspaper analysis, published in 1893, asked the rhetorical question, “Do newspapers now give the news?” (Speed, 1893). Its author showed how, between 1881 and 1893, New York newspapers had dropped their coverage of religious, scientific, and literary matters in favor of gossip, sports, and scandals. In a similar but far more simplistic study published in 1910, Mathews

attempted to reveal the overwhelming space that one New York daily newspaper devoted to “demoralizing,” “unwholesome,” and “trivial” matters as opposed to “worthwhile” news items. By simply measuring the column inches that newspapers devoted to particular subject matters, journalists in the early 20th century attempted to reveal “the truth about newspapers” (Street, 1909). Some believed that they had found a way of showing that the profit motive was the cause of “cheap yellow journalism” (Wilcox, 1900); others became convinced that they had established “the influence of newspaper presentations on the growth of crime and other antisocial activity” (Fenton, 1910). At least one concluded that a “quarter century survey of the press content shows demand for facts” (White, 1924).

Quantitative newspaper analysis seemingly provided the needed scientific ground for journalistic arguments. The respect for numbers has a long history, and facts that could be quantified were considered irrefutable. In a footnote, Berelson and Lazarsfeld (1948) quote from a source published more than 200 years ago:

Perhaps the spirit of the battle over ratification is best reflected in the creed ironically attributed to each of the contending parties by its opponents. The recipe for an Anti-Federalist essay which indicates in a very concise way the class-bias that actuated the opponents of the Constitution, ran in this manner: “wellborn, nine times—Aristocracy, eighteen times—Liberty of the Press, thirteen times repeated—Liberty of Conscience, once—Negro Slavery, once mentioned—Trial by Jury, seven times—Great men, six times repeated—Mr. Wilson, forty times . . .—put them together and dish them up at pleasure. (p. 9; quoted from *New Hampshire Spy*, November 30, 1787)

Quantitative newspaper analysis led to the development of many valuable ideas, however. In 1912, Tenney (see also Krippendorff & Bock, 2009, Chapter 1.4) made a far-reaching proposal for a large-scale and continuous survey of press content to establish a system of bookkeeping of the “social weather” “comparable in accuracy to the statistics of the U.S. Weather Bureau” (p. 896). He demonstrated what he had in mind with an analysis of a few New York newspapers for different ethnic groups, but his proposal exceeded the scope of what was then feasible. Quantitative newspaper analysis culminated in sociologist Malcolm M. Willey’s 1926 book *The Country Newspaper*. In this model study, Willey traced the emergence of Connecticut country weeklies, examining circulation figures, changes in subject matter, and the social role these papers acquired in competition with large city dailies.

When other mass media became prominent, researchers extended the approach first used in newspaper analysis—measuring volumes of coverage in various subject matter categories—initially to radio (Albig, 1938) and later to movies and television. Content analysis in subject matter categories continues today and is applied to a wide variety of printed matter, such as textbooks, comic strips, speeches, and print advertising.

1.3 EARLY CONTENT ANALYSIS

The second phase in the intellectual growth of content analysis, which took place in the 1930s and 1940s, involved at least four factors:

- During the period following the 1929 economic crisis, numerous social and political problems emerged in the United States. Many Americans believed that the mass media were at least partially to blame for such problems as yellow journalism, rising crime rates, and the breakdown of cultural values.
- New and increasingly powerful electronic media of communication, first radio and later television, challenged the cultural hegemony of the newspapers. Researchers could not continue to treat these new media as extensions of newspapers, because they differed from the print media in important ways. For example, users of radio and television did not have to be able to read.
- Major political challenges to democracy were linked to the new mass media. For example, the rise of fascism was seen as nourished by the as-yet little-known properties of radio.
- Perhaps most important, this period saw the emergence of the behavioral and social sciences as well as increasing public acceptance of the theoretical propositions and empirical methods of inquiry associated with them.

In the 1930s, sociologists started to make extensive use of survey research and polling. The experience they gained in analyzing public opinion gave rise to the first serious consideration of methodological problems of content analysis, published by Woodward in a 1934 article titled “Quantitative Newspaper Analysis as a Technique of Opinion Research.” From writings about public opinion, interest in social stereotypes (Lippmann, 1922) entered the analysis of communications in various forms. Questions of representations were raised, with researchers examining topics such as how Negroes were presented in the Philadelphia press (Simpson, 1934); how U.S. textbooks described wars in which the United States had taken part, compared with textbooks published in countries that were former U.S. enemies (Walworth, 1938); and how nationalism was expressed in children’s books published in the United States, Great Britain, and other European countries (Martin, 1936).

One of the most important concepts that emerged in psychology during this time was the concept of “attitude.” It added evaluative dimensions to content analysis, such as “pro-con” or “favorable-unfavorable,” that had escaped the rough subject matter categories of quantitative newspaper analysis. Attitude measures redefined journalistic standards of fairness and balance and opened the door to the systematic assessment of bias. Among the explicit standards developed, Janis and Fadner’s (1943/1965) “coefficient of imbalance” deserves mention. Psychological experiments in rumor transmission led Allport and Faden to study newspaper content from an entirely new perspective. In their 1940 article, “The Psychology of Newspapers: Five Tentative Laws,” they attempted to account for the changes that information undergoes as it travels through an institution and finally appears on the printed page.

The interest in political symbols added another feature to the analysis of public messages. McDiarmid (1937), for example, examined 30 U.S. presidential inaugural addresses for symbols of national identity, of historical significance, of government, and of fact and expectations. Most important, Lasswell (1938), viewing public communications within his psychoanalytical theory of politics, classified symbols into such categories as “self” and “others” and forms of “indulgence” and

“deprivation.” His symbol analysis led to his “World Attention Survey,” in which he compared trends in the frequencies with which prestige newspapers in several countries used national symbols (Lasswell, 1941; see also Krippendorff & Bock, 2009, Chapter 5.3).

Researchers in several disciplines examined the trends in scholarship, as reflected in the topics that representative journals published. Rainoff’s (1929) Russian study regarding physics was probably the first of this kind, but the most thorough analyses were conducted in the field of sociology (Becker, 1930, 1932; Shanas, 1945) and later in journalism (Tannenbaum & Greenberg, 1961).

Several factors influenced the transition from quantitative newspaper analysis, which was largely journalism driven, to content analysis:

- Eminent social scientists became involved in these debates and asked new kinds of questions.
- The concepts these social scientists developed were theoretically motivated, operationally defined, and fairly specific, and interest in stereotypes, styles, symbols, values, and propaganda devices began to replace interest in subject matter categories.
- Analysts began to employ new statistical tools borrowed from other disciplines, especially from survey research but also from experimental psychology.
- Content analysis data became part of larger research efforts (e.g., Lazarsfeld, Berelson, & Gaudet, 1948), and so content analysis no longer stood apart from other methods of inquiry.

The first concise presentation of these conceptual and methodological developments under the new umbrella term *content analysis* appeared in a 1948 mimeographed text titled *The Analysis of Communication Content*, authored by Berelson and Lazarsfeld, which was later published as Berelson’s *Content Analysis in Communications Research* (1952). This first systematic presentation codified the field for years to come.

1.4 PROPAGANDA ANALYSIS

Berelson described content analysis as the use of mass communications as data for testing scientific hypotheses and for evaluating journalistic practices. Yet the most important and large-scale challenge that content analysis faced came during World War II, when it was employed in efforts to extract information from propaganda. Before the war, researchers analyzed texts in order to identify “propagandists,” to point fingers at individuals who were attempting to influence others through devious means. Fears concerning such influence had several origins. Propaganda was used extensively during World War I (Lasswell, 1927), and the years between the two world wars witnessed the effective use of propaganda by antidemocratic demagogues in Europe. In addition, Americans tend to have deep-seated negative attitudes toward religious fanatics, and the lack of knowledge concerning what the extensive use of the

new mass media (radio, film, and television) could do to people raised concerns as well. According to the Institute for Propaganda Analysis (1937), propagandists reveal themselves through their use of tricks such as “name-calling,” employing “glittering generalities,” “plain folks” identifications, “card stacking,” “bandwagon” devices, and so on. Such devices could be identified easily in many religious and political speeches, even in academic lectures, and this approach to propaganda analysis led to a kind of witch hunt for propagandists in the United States. Theories concerning subliminal messages, especially in advertising, raised widespread suspicion as well.

In the 1940s, as U.S. attention became increasingly devoted to the war effort, the identification of propagandists was no longer an issue. Nor were researchers particularly interested in revealing the power of the mass media of communication to mold public opinion; rather, military and political intelligence were needed. In this climate, two centers devoted to propaganda analysis emerged. Harold D. Lasswell and his associates, having written on political symbolism, worked with the Experimental Division for the Study of Wartime Communications at the U.S. Library of Congress, and Hans Speier, who had organized a research project on totalitarian communication at the New School for Social Research in New York, assembled a research team at the Foreign Broadcast Intelligence Service of the U.S. Federal Communications Commission (FCC). The Library of Congress group focused on analyzing newspapers and wire services from abroad and addressed basic issues of sampling, measurement problems, and the reliability and validity of content categories, continuing the tradition of early quantitative analysis of mass communications (Lasswell, Leites, & Associates, 1965).

The FCC group analyzed primarily domestic enemy broadcasts and surrounding conditions to understand and predict events within Nazi Germany and the other Axis countries and to estimate the effects of Allied military actions on the war mood of enemy populations. The pressures of day-to-day reporting left the analysts little time to formalize their methods, and Berelson (1952) thus had little to say about the accomplishments of the FCC group. After the war, however, Alexander L. George worked through the volumes of reports that resulted from these wartime efforts to describe methods that had evolved in the process and to validate the inferences the researchers had made by comparing them with documentary evidence now available from Nazi archives. These efforts resulted in his book *Propaganda Analysis* (1959a; see also Krippendorff & Bock, 2009, Chapter 1.5), which made major contributions to the conceptualization of the aims and processes of content analysis.

The assumptions that propagandists are rational, in the sense that they follow their own propaganda theories in their choice of communications, and that the meanings of propagandists’ communications may differ for different people reoriented the FCC analysts from a concept of “content as shared” (Berelson would later say “manifest”) to conditions that could explain the motivations of particular communicators and the interests they might serve. The notion of “preparatory propaganda” became an especially useful key for the analysts in their effort to infer the intents of broadcasts with political content. In order to ensure popular support for planned military actions, the Axis leaders had to inform, emotionally arouse, and otherwise prepare their countrymen and -women to accept those actions; the FCC analysts discovered that they could learn a great deal about the enemy’s intended actions by recognizing such preparatory efforts in the domestic press and broadcasts.

They were able to predict several major military and political campaigns and to assess Nazi elites' perceptions of their situation, political changes within the Nazi governing group, and shifts in relations among Axis countries. Among the more outstanding predictions that British analysts were able to make was the date of deployment of German V weapons against Great Britain. The analysts monitored the speeches delivered by Nazi propagandist Joseph Goebbels and inferred from the content of those speeches what had interfered with the weapons' production and when. They then used this information to predict the launch date of the weapons, and their prediction was accurate within a few weeks.

Several lessons were learned from these applications of content analysis, including the following:

- Content is not inherent to communications. People typically differ in how they read texts. The intentions of the senders of broadcast messages may have little to do with how audience members hear those messages. Temporal orderings, individuals' needs and expectations, individuals' preferred discourses, and the social situations into which messages enter are all important in explaining what communications come to mean. Interpretations on which all communicators readily agree are rare, and such interpretations are usually relatively insignificant.
- Content analysts must predict or infer phenomena that they cannot observe at the time of their research. The inability to observe phenomena of interest tends to be the primary motivation for using content analysis. Whether the analyzed source has reasons to hide what the analyst desires to know (as in the case of an enemy during wartime or the case of someone needing to impress) or the phenomena of interest are inaccessible in principle (e.g., an individual's attitudes or state of mind, or historical events) or just plain difficult to assess otherwise (such as what certain mass-media audiences could learn from watching TV), the analyst seeks answers to questions that go outside a text. To be sure, the questions that a content analyst seeks to answer are the analyst's questions, and as such, they are potentially at odds with whether others could answer them and how. Quantitative newspaper analysts made inferences without acknowledging their own conceptual contributions to what they thought they found but actually inferred. Content is not the whole issue; rather, the issue is what can be legitimately inferred from available texts.
- In order to interpret given texts or make sense of the messages intercepted or gathered, content analysts need elaborate models of the systems in which those communications occur (or occurred). The propaganda analysts working during World War II constructed such models more or less explicitly. Whereas earlier content analysts had viewed mass-produced messages as inherently meaningful and analyzable unit by unit, the propaganda analysts succeeded only when they viewed the messages they analyzed in the context of the lives of the diverse people presumed to use those messages.
- For analysts seeking specific political information, quantitative indicators are extremely insensitive and shallow. Even where large amounts of

quantitative data are available, as required for statistical analyses, these tend not to lead to the “most obvious” conclusions that political experts would draw from qualitative interpretations of textual data. Qualitative analyses can be systematic, reliable, and valid as well.

Convinced that content analysis does not need to be inferior to unsystematic explorations of communications, numerous writers in the postwar years, such as Kracauer (1947, 1952–1953) and George (1959a), challenged content analysts’ simplistic reliance on counting qualitative data. Smythe (1954) called this reliance on counting an “immaturity of science” in which objectivity is confused with quantification. However, the proponents of the quantitative approach largely ignored the criticism. In his 1949 essay “Why Be Quantitative?” Lasswell (1949/1965b) continued to insist on the quantification of symbols as the sole basis of scientific insights. His approach to propaganda analysis produced several working papers but very few tangible results compared with the work of the FCC group of scholars. Today, quantification continues, although perhaps no longer exclusively.

1.5 CONTENT ANALYSIS GENERALIZED

After World War II, and perhaps as the result of the first integrated picture of content analysis provided by Berelson (1952), the use of content analysis spread to numerous disciplines. This is not to say that content analysis emigrated from mass communication. In fact, the very “massiveness” of available communications continued to attract scholars who looked at the mass media from new perspectives. For example, Lasswell (1941) realized his earlier idea of a “world attention survey” in a large-scale study of political symbols in French, German, British, Russian, and U.S. elite press editorials and key policy speeches. He wanted to test the hypothesis that a “world revolution” had been in steady progress for some time (Lasswell, Lerner, & Pool, 1952). Gerbner and his colleagues pursued Gerbner’s (1969) proposal to develop “cultural indicators” by analyzing, for almost two decades, one week of fictional television programming per year, mainly to establish “violence profiles” for different networks, to trace trends, and to see how various groups (such as women, children, and the aged) were portrayed on U.S. television (see, e.g., Gerbner, Gross, Signorielli, Morgan, & Jackson-Beeck, 1979).

Psychologists began to use content analysis in four primary areas. The first was the inference of motivational, mental, or personality characteristics through the analysis of verbal records. This application started with Allport’s (1942) treatise on the use of personal documents, Baldwin’s (1942) application of “personal structure analysis” to cognitive structure, and White’s (1947) value studies. These studies legitimated the use of written material, personal documents, and individual accounts of observed phenomena as an addition to the then-dominant experimental methods. A second application was the use of verbal data gathered in the form of answers to open-ended interview questions, focus group conversations, and verbal responses to various tests, including the construction of Thematic Apperception Test (TAT) stories. In the context of TAT stories, content analysis acquired the status of a supplementary technique. As such, it allowed researchers to use data that they could gather without imposing too much structure on subjects and to validate findings they had obtained

through different techniques. Psychological researchers' third application of content analysis concerned processes of communication in which content is an integral part. For example, in his "interaction process analysis" of small group behavior, Bales (1950) used verbal exchanges as data through which to examine group processes. The fourth application took the form of the generalization of measures of meaning over a wide range of situations and cultures (which derived from individualist notions of meaning or content). Osgood (1974a, 1974b) and his students found numerous applications for Osgood, Suci, and Tannenbaum's (1957) semantic differential scales and conducted worldwide comparisons of cultural commonalities and differences.

Anthropologists, who started using content analysis techniques in their studies of myths, folktales, and riddles, have made many contributions to content analysis, including the componential analysis of kinship terminology (Goodenough, 1972). Ethnography emerged in anthropology, and although ethnographers often interact with their informants in ways that content analysts cannot interact with authors or readers, after ethnographers gather their field notes they start to rely heavily on methods that are similar to those that content analysts use.

Historians are naturally inclined to look for systematic ways to analyze historical documents, and they soon embraced content analysis as a suitable technique, especially where data are numerous and statistical accounts seem helpful. Social scientists also recognized the usefulness of educational materials, which had long been the focus of research. Such materials are a rich source of data on processes of reading (Flesch, 1948, 1951) as well as on a society's larger political, attitudinal, and value trends. In addition, literary scholars began to apply the newly available techniques of content analysis to the problem of identifying the authors of unsigned documents.

On one hand, this proliferation of the use of content analysis across disciplines resulted in a loss of focus: Everything seemed to be content analyzable, and every analysis of symbolic phenomena became a content analysis. On the other hand, this trend also broadened the scope of the technique to embrace what may well be the essence of human behavior: talk, conversation, and mediated communication.

In 1955, responding to increasing interest in the subject, the Social Science Research Council's Committee on Linguistics and Psychology sponsored a conference on content analysis. The participants came from such disciplines as psychology, political science, literature, history, anthropology, and linguistics. Their contributions to the conference were published in a volume titled *Trends in Content Analysis*, edited by Ithiel de Sola Pool (1959a). Despite obvious divergence among the contributors in their interests and approaches, Pool (1959a, p. 2) observed, there was considerable and often surprising convergence among them in two areas: They exhibited (a) a shift from analyzing the "content" of communications to drawing inferences about the antecedent conditions of communications and (b) an accompanying shift from measuring volumes of subject matter to counting simple frequencies of symbols and then to relying on contingencies (co-occurrences).

1.6 COMPUTER TEXT ANALYSIS

The late 1950s witnessed considerable interest among researchers in mechanical translation, mechanical abstracting, and information retrieval systems. Computer languages suitable for literal data processing emerged, and scholarly journals started

to devote attention to computer applications in psychology, the humanities, and the social sciences. The large volumes of written documents to be processed in content analysis and the repetitiveness of the coding involved made the computer a natural but also a difficult ally of the content analyst.

The development of software for literal (as opposed to numerical) data processing stimulated new areas of exploration, such as information retrieval, information systems, computational stylistics (Sedelow & Sedelow, 1966), computational linguistics, word processing technology, and computational content analysis. New software also revolutionized tedious literary work, such as indexing and the creation of concordances. Probably the first computer-aided content analysis was reported by Sebeok and Zeps (1958), who made use of simple information retrieval routines to analyze some 4,000 Cheremis folktales. In a Rand Corporation paper titled *Automatic Content Analysis*, Hays (1960) explored the possibility of designing a computer system for analyzing political documents. Unaware of both these developments, Stone and Bales, who were engaged in a study of themes in face-to-face interacting groups, designed and programmed the initial version of the General Inquirer system. This culminated in a groundbreaking book by Stone, Dunphy, Smith, and Ogilvie (1966) in which they presented an advanced version of this system and demonstrated its application in numerous areas, ranging from political science to advertising and from psychotherapy to literary analysis.

The use of computers in content analysis was also stimulated by developments in other fields. Scholars in psychology became interested in simulating human cognition (Abelson, 1963; Schank & Abelson, 1977). Newell and Simon (1963) developed a computer approach to (human) problem solving. Linguistics researchers developed numerous approaches to syntactic analysis and semantic interpretation of linguistic expressions. Researchers in the field of artificial intelligence focused on designing machines that could understand natural language (with very little success).

In 1967, the Annenberg School of Communications (which later became the Annenberg School for Communication) sponsored a major conference on content analysis. Discussions there focused on many areas—the difficulties of recording nonverbal (visual, vocal, and musical) communications, the need for standardized categories, the problems involved in drawing inferences, the roles of theories and analytical constructs, what developments content analysts could expect in the near future—but the subject of the use of computers in content analysis permeated much of the conference. Stone et al.'s (1966) book on the General Inquirer had just been published, and it had created considerable hope among content analysts. The contributions to the 1967 conference are summarized in a 1969 volume edited by Gerbner, Holsti, Krippendorff, Paisley, and Stone, the publication of which coincided with Holsti's (1969) survey of the field.

In 1974, participants in the Workshop on Content Analysis in the Social Sciences, held in Pisa, Italy, saw the development of suitable algorithms for computer content analysis as the only obstacle to better content analyses (Stone, 1975). Since that time, computational approaches have moved in numerous directions. One has been the development of customizable content analysis packages, of which the General Inquirer was the most important precursor. Attempts to apply the General Inquirer system to German texts revealed that software's English-language biases and led to more general versions of General Inquirers, such as TextPack. The basic ingredient of the General Inquirer and TextPack is a dictionary of relevant words. In the 1980s,

Sedelow (1989) proposed the idea of using a thesaurus instead, as a thesaurus might be more accurate than a dictionary in reflecting “society’s collective associative memory” (p. 4; see also Sedelow & Sedelow, 1986). In the 1990s, George Miller initiated a major research effort to chart the meanings of words using a computer-traceable network called WordNet (see Miller et al., 1993). In the 1980s, some authors observed that the enthusiasm associated with large systems that had appeared in the 1960s was fading (see Namenwirth & Weber, 1987), but today the development of text analysis software is proliferating, fueled largely by the historically unprecedented volumes of electronic and digital texts available for content analysis. More recently, Diefenbach (2001) reviewed the history of content analysis by focusing on four specific areas: mass communication research, political science, psychology, and literature.

Naturally, many researchers have compared computer-based content analyses with human-based content analyses. For example, Schnurr, Rosenberg, and Ozman (1992, 1993) compared the Thematic Apperception Test (Murray, 1943) with a computer content analysis of open-ended free speech and found the low agreement between the two to be discouraging. Zeldow and McAdams (1993) challenged Schnurr et al.’s conclusion, however. Nacos et al. (1991) compared humans’ coding of political news coverage with data from Fan’s (1988) computer-coded approach to the same coverage and found satisfactory correlations between the two. Nacos et al. came to the conclusion that content analysts can best use computers in their research by thinking of them as aids, not as replacements for the highly developed human capabilities of reading, transcribing, and translating written matter. As one might expect, today scholars hold many different opinions regarding the future of the use of computer-based content analysis.

Another development that has influenced how content analysts employ computers in their work is the increasingly common use of word processing software, which provides users with such features as spell-checkers, word- or phrase-finding and -replacing operations, and even readability indices. Although not intended for this purpose, ordinary word processing software makes it possible for a researcher to perform basic word counts and KWIC (keyword in context) analyses, albeit laboriously.

Word processing software is inherently interactive; it is driven by the user’s reading of the textual material, not fixed. In the absence of computational theories of text interpretation, content analysts have found the symbiosis of the human ability to understand and interpret written documents and the computer’s ability to scan large volumes of text systematically and reliably increasingly attractive. In such collaborations, human coders are no longer used as text-level content analysts; rather, they serve as translators of text or sections of text into categories that emerge during reading and then into a data language (that preserves relevant meanings), which enables various computational algorithms (that cannot respond to meanings) to do housekeeping and summarizing chores. This has given rise to a new class of software designed for computer-aided qualitative text analysis, of which NVivo and ATLAS.ti are two examples. Such interactive-hermeneutic text analysis software is becoming increasingly accessible, especially to students.

The most important stimulus in the development of computational content analysis, however, has been the growing availability of text in digital form. It is very costly to enter handwritten documents, such as transcripts of audio recordings of interviews, focus group protocols, minutes of business meetings, and political

speeches, into a computer. Scanners have vastly improved in recent years, but they are still too unreliable to be used without additional manual editing. In the 1970s, data consortia emerged through which social scientists could share costly data, but the operations of these consortia were marred by a lack of standards and the usually highly specialized nature of the data. Then, in 1977, DeWeese proposed and took the remarkable step of bypassing the costly transcription process by feeding the typesetting tapes of a Detroit newspaper directly into a computer to conduct an analysis of the paper's content the day after it was published. Since that time, word processing software has come to be an integral part of the internal operations of virtually all social organizations; personnel create texts digitally before they appear on paper, use electronic mail systems, and surf the Internet to download materials relevant to their work.

Today, a fantastic amount of raw textual data is being generated daily in digital form, representing almost every topic of interest to social scientists. Electronic full-text databases, to which all major U.S. newspapers, many social science and legal journals, and many corporations contribute all of the materials they publish, are growing exponentially and have become easily available and inexpensive to use online. Add to this the volume of electronic publications, the research potential of the Internet, data available from online multiuser discussions (MUDs) and newsgroups, and online survey systems, which may well replace focus groups and interviews in certain empirical domains, and it is clear that the landscape of how society presents itself has been altered drastically. With more and more people interested in this wealth of digital data, there is a corresponding demand for increasingly powerful search engines, suitable computational tools, text base managing software, encryption systems, devices for monitoring electronic data flows, and translation software, all of which will eventually benefit the development of computer-aided content analysis. The current culture of computation is moving content analysis into a promising future.

1.7 QUALITATIVE APPROACHES

Perhaps in response to the now dated “quantitative newspaper analysis” of more than a century ago or as a form of compensation for the sometimes shallow results reported by the content analysts of 60 years ago, a variety of research approaches have begun to emerge that call themselves *qualitative*. I question the validity and usefulness of the distinction between quantitative and qualitative content analyses. Ultimately, all reading of texts is qualitative, even when certain characteristics of a text are later converted into numbers. The fact that computers process great volumes of text in a very short time and represent these volumes in ways someone can understand does not remove the qualitative nature of the texts being analyzed and the algorithms used to process them: On the most basic level, computers recognize zeros and ones and change them as instructed, proceeding one step at a time. Nevertheless, proponents of qualitative approaches to content analysis offer alternative protocols for exploring texts systematically.

Discourse analysis is one such approach. Generally, *discourse* is defined as text above the level of sentences. Discourse analysts tend to focus on how particular phenomena are represented. For example, Van Dijk (1991) studied manifestations of racism

in the press: how minorities appear, how ethnic conflicts are described, and how stereotypes permeate given accounts, for example, in advertisements during sports events (Wonsek, 1992). Other discourse analysts have examined how television news programs and other TV shows in the United States manifest a particular ideological vision of the U.S. economy (Jensen, 2006), the components of “age markers” in the humorous context of the TV series *The Golden Girls* (Harwood & Giles, 1992), and the portrayal of the peace movement in news editorials during the Gulf War (Hackett & Zhao, 1994).

Researchers who conduct *social constructivist analyses* focus on discourse as well but less to criticize (mis)representations than to understand how reality comes to be constituted in human interactions and in language, including written text (Gergen, 1985). Such analysts may address how emotions are conceptualized (Averill, 1985) or how facts are constructed (Fleck, 1935/1979; Latour & Woolgar, 1986), or they may explore changing notions of self (Gergen, 1991) or of sexuality (Katz, 1995).

Rhetorical analysis, in contrast, focuses on how messages are delivered and with what (intended or actual) effects. Researchers who take this approach rely on the identification of structural elements, tropes, styles of argumentation, speech acts, and the like; Kathleen Hall Jamieson’s book *Packaging the Presidency* (1984) is an example of such an analysis. Efforts to study negotiations (Harris, 1996), what works and what doesn’t, might be described as rhetorical analyses as well.

Ethnographic content analysis, an approach advocated by Altheide (1987), does not avoid quantification but encourages content analysis accounts to emerge from readings of texts. This approach works with categories as well as with narrative descriptions but focuses on situations, settings, styles, images, meanings, and nuances presumed to be recognizable by the human actors/speakers involved.

Conversation analysis is another approach that is considered to be qualitative. The researcher performing such an analysis tends to start with the recording of verbal interactions in natural settings and aims at analyzing the transcripts as records of conversational moves toward a collaborative construction of conversations. This tradition is indebted to the work of Harvey Sacks, who studied numerous interactive phenomena, including the collaboration among communicators in the telling of jokes (Sacks, 1974). Goodwin (1977, 1981) extended conversation analysis by incorporating video data in his groundbreaking study of turn taking.

Qualitative approaches to content analysis have their roots in literary theory, the social sciences (symbolic interactionism, ethnomethodology), and critical scholarship (Marxist approaches, British cultural studies, feminist theory). Sometimes they are given the label *interpretive*. They share the following characteristics:

- They require a close reading of relatively small amounts of textual matter.
- They involve the rearticulation (interpretation) of given texts into new (analytical, deconstructive, emancipatory, or critical) narratives accepted within particular scholarly communities that are sometimes opposed to positivist traditions of inquiry.
- The analysts acknowledge working within hermeneutic circles in which their own socially or culturally conditioned understandings constitutively

participate. (For this reason, I refer to these approaches as *interactive-hermeneutic*, a description that speaks to the process of engaging in systematic interpretations of text.)

One could summarize and say that content analysis has evolved into a repertoire of methods of research that promise to yield inferences from all kinds of verbal, pictorial, symbolic, and communication data. Beyond the technique's initially journalistic roots, the past century has witnessed the migration of content analysis into various fields and the clarification of many methodological issues. After a short period of stagnation in the 1970s, content analysis is today growing exponentially, largely due to the widespread use of computers for all kinds of text processing. As of February 2011, an Internet search for "*content analysis*" using the Google search engine found 1,650,000 documents. By comparison, "*survey research*" turned up 275,000 hits and "*psychological testing*," 894,000. Since the first casual mention of "content analysis" in 1941—that is, seventy years ago and with a frequency of one—the public interest in the body of content analysis research has clearly grown to an astonishing extent.

2

CONCEPTUAL FOUNDATION

Content analysis has its own approach to analyzing data that stems largely from how the object of analysis, content, is conceived. This chapter introduces an epistemologically grounded definition of content analysis that guides the remainder of the book. It discusses its relationship to other definitions, other methods of social inquiry, and where and how it deviates from them. A conceptual framework is developed in terms of which the purposes and processes of content analysis may be understood. It includes the researcher, the knowledge he or she needs to bring to it, and the criteria by which a content analysis can be justified, and it makes transparent the function of various methods to be discussed in subsequent chapters. This chapter concludes with the suggestion that the ongoing changes in information technology make content analysis an increasingly attractive method for understanding social phenomena.

2.1 DEFINITION

Content analysis is a research technique for making replicable and valid inferences from texts (or other meaningful matter) to the contexts of their use.

As a *technique*, content analysis involves specialized procedures. It is learnable and divorceable from the personal authority of the researcher. As a research technique, content analysis provides new insights, increases a researcher's understanding of particular phenomena, or informs practical actions. Content analysis is a scientific tool.

Techniques are expected to be *reliable*. More specifically, research techniques should result in findings that are *replicable*. That is, researchers working at different points in time and perhaps under different circumstances should get the same results when applying the same technique to the same phenomena. Replicability is the most important form of reliability.

Scientific research must also yield *valid* results, in the sense that the research effort is open to careful scrutiny and the resulting claims can be upheld in the face of independently available evidence. The methodological requirements of reliability and validity are not unique to, but make particular demands on, content analysis.

The reference to *text* in the above definition is not intended to restrict content analysis to written material. The phrase “or other meaningful matter” is included in parentheses to indicate that in content analysis works of art, images, maps, sounds, signs, symbols, and even numerical records may be included as data—that is, they may

be considered as texts—provided they speak to someone about phenomena outside of what can be sensed or observed. The crucial distinction between text and what other research methods take as their starting point is that a text means something to someone, it is produced by someone to have meanings for someone else, and these meanings therefore must not be ignored and must not violate why the text exists in the first place. Text—the reading of text, the use of text within a social context, and the analysis of text—serves as a convenient metaphor in content analysis.

In the content analysis literature, scholars have provided essentially three kinds of definitions of this research method:

1. Definitions that take content to be *contained* in a text
2. Definitions that take content to be *a property of the source* of a text
3. Definitions that take content to *emerge in the process of a researcher analyzing a text* relative to a particular context

Each of these kinds of definitions leads to a particular way of conceptualizing content analysis and, consequently, its analytical procedures.

Berelson's original definition of content analysis is an example of the first kind. Berelson (1952) defined content analysis as “a research technique for the objective, systematic and quantitative description of the manifest content of communication” (p. 18). His requirement that content analysis be “objective” and “systematic” is subsumed under the dual requirements of replicability and validity in our definition. For a process to be replicable, it must be governed by rules that are explicitly stated and applied equally to all units of analysis. Berelson argued for “systematicity” in order to combat the human tendency to read textual material selectively, in support of expectations rather than against them. Our requirement of validity goes further, demanding that the researcher's processes of sampling, reading, and analyzing messages ultimately satisfy external criteria. Replicability is measurable, and validity is testable, but objectivity is neither.

Our definition of content analysis omits three of Berelson's further requirements. One is his insistence that content analysis be “quantitative.” Although quantification is important in many scientific endeavors, qualitative methods have proven successful as well, particularly in political analyses of foreign propaganda, in psychotherapeutic assessments, in ethnographic research, in discourse analysis, and, oddly enough, in computer text analysis. The ability of computers to crunch words as well as numbers is well-known. When a computer program is used to analyze words, the algorithms that determine the program's operation must embody some kind of theory of how humans read texts, rearticulate texts, or justify actions informed by the reading of texts. Reading is fundamentally a qualitative process, even when it results in numerical accounts. By including the attribute “manifest” in his definition, Berelson intended to ensure that the coding of content analysis data be reliable; this requirement literally excludes “reading between the lines,” which is what experts do, often with remarkable intersubjective agreement (I will have more to say on this topic later in this chapter).

My chief objection to Berelson's definition, and numerous derivatives of that definition, is related to his phrase “description of the manifest content of communication.” It implies that content is obviously *manifest* in messages, waiting

to be separated from its form and described. Berelson felt no need to elaborate on the crucial concept of “content” in his definition because for him and his contemporaries, at the time of his writing, there seemed to be no doubt about the nature of content—it was believed to reside *inside* a text.

Berelson’s operationalization of the attribute “manifest” is telling. If sources, receivers, and content analysts have different interpretations of the same message, which is quite natural, Berelson’s definition restricts content to what is *common to all of these accounts*, what everyone can agree to. Gerbner (1985) starts from a similar assumption when he insists that mass-media messages carry the *imprint* of their industrial producers. For him, too, content is right there to be described for what it is. However, Gerbner goes beyond Berelson’s notion by suggesting that the messages of the mass media are revealed in statistical accounts of their contents. Mass-media audiences, he suggests, are affected by certain statistical properties of mass-produced messages of which neither mass producers nor mass audiences are conscious. This privileges content analysts’ accounts over the readings by audience members. Shapiro and Markoff’s (1997) definition equates content analysis with scientific measurement as well, specifically, with “any systematic reduction . . . of text (or other symbols) to a standard set of statistically manipulable symbols representing the presence, the intensity, or the frequency of some characteristics relevant to social science” (p. 14). Its implicit representationalism is common in several definitions of content analysis. For example, Riffe, Lacy, and Fico (1998) start their textbook with the proposition that content is central to communication research but then assert that the purpose of content analysis is to describe “it” so as to make “it” amenable to correlations with other (noncontent) variables—as if content were a variable or thing inherent to mass-media messages. These examples demonstrate that the container metaphor for meaning still abounds in much of the communication research literature (Krippendorff, 1993). The use of this metaphor entails the belief that messages are containers of meaning, usually one meaning per message, and justifies calling any analysis of any conventionally meaningful matter a content analysis, regardless of whether it counts words or offers in-depth interpretations. Clearly, this is an insufficient way to define content analysis.

Definitions of the second kind distinguished above tie the content analysis of texts to *inferences about the states or properties of the sources of the analyzed texts* (Krippendorff, 1969a, p. 70; Osgood, 1959, p. 35). These definitions tend to generalize the aims of psychological tests to reveal what is going on inside an individual: emotional states, cognitions, intentions, attitudes, and prejudices. Shapiro and Markoff (1997), among others, have criticized such definitions as too limiting. Holsti (1969, p. 25) elaborates on this idea by committing content analysis to an encoding/decoding paradigm in which message sources are causally linked to recipients through processes of encoding, transmission through channels, and decoding by a receiver. Holsti wants the content analyst to describe the characteristics of messages in terms of “what,” “how,” and “to whom” in order to infer their antecedents in terms of “who” and “why” and their consequences in terms of “with what effects.” These questions could be answered directly if sources and recipients were accessible to observation or able to inform the analyst honestly. When antecedents and consequences are not accessible to direct observation, the analyst must make inferences. I am sympathetic to Holsti’s logic, but putting sources—senders and/or receivers—in charge of the validity of the inferences may not be the best way for the content analyst to capture all of the communicators’

circumstances and intents. Moreover, describing message characteristics in terms of “what,” “how,” and “to whom” fails to acknowledge the analyst’s own conceptual contributions to what constitutes the appropriate reading of the analyzed texts and the relevance of this reading to a given research question.

The analyst’s conceptual contributions to the reading of a text are specifically recognized in a call for an *ethnographic content analysis* (Altheide, 1987); unfortunately, however, this call has not yielded specific methods of analysis. Proponents of ethnographic content analysis oppose the sequential nature of traditional content analysis, suggesting instead that analysts be flexible in taking into account new concepts that emerge during their involvement with texts. This call acknowledges the theory-driven nature of content analysis but also demands that the analytical process be closely linked to the communicators studied. Ethnographic content analysis is emic rather than etic in intent; that is, it attempts to rely on indigenous conceptions rather than on conceptions imposed by an analyst’s theory. Although the preference for communicators’ conceptions would appear to tie ethnographic content analysis to the second kind of definition noted above, by urging researchers to reflect on their involvement in the process, the approach acknowledges the possibility that researchers’ theories can play a role in how analysis proceeds. The latter ties it more closely to the third kind of definition of content analysis, which we now explore.

2.2 EPISTEMOLOGICAL ELABORATIONS

The definition of content analysis offered at the opening of this chapter is of the third kind. It focuses attention on the process of content analysis and does not ignore the contributions that analysts make to what counts as content. The key to the definition lies in the operations that define the nature of content analysis data. Most content analysts probably realize that the starting points of their analyses, *texts* (printed matter, recorded speech, visual communications, works of art, websites, and cultural artifacts), are quite unlike physical events in that they are meaningful to others, not just to the analysts. Recognizing meanings is the reason that researchers engage in content analysis rather than in some other kind of investigative method. A content analyst must acknowledge that all texts are produced and read by others and are expected to be significant to them, not just to the analyst. Inasmuch as linguistically competent communicators are able to transcend the physical manifestations of their messages and respond instead to what those messages mean to them, content analysts cannot remain stuck in analyzing the physicality of text—its medium, characters, pixels, or shapes. Rather, they must look outside these characteristics to examine how individuals use various texts in their respective worlds. It would follow that the popular measurement model for conceptualizing content analysis, borrowed from mechanical engineering and widely used in the natural sciences and behavioral research, is misleading; it implies that there is nothing inherent in text that could be measured without interpretation by competent authors, readers, users, and—we need to include—culturally competent analysts. Below, I elaborate on six features of texts that are relevant to our definition of content analysis.

1. *Texts have no objective*—that is, *no reader-independent—qualities*. Seeing something as a text entails an invitation, if not a commitment, to reading it. Regarding something as a message implies that someone is trying to make sense of

it. Accepting particular markers as data entails taking them as an unquestionable ground for subsequent conceptualizations. Thus texts, messages, and data arise in the process of *someone* engaging with them conceptually. A text does not exist without a reader, a message does not exist without an interpreter, and data do not exist without an observer. In a content analysis, it is methodologically trained researchers who, being familiar with their texts, design the analysis, instruct their coders to describe textual elements, and end up interpreting the results—always in the expectation of others' understanding. There is nothing inherent in a text; the meanings of a text are always brought to it by someone. Ordinary readers and content analysts merely read differently.

2. *Texts do not have single meanings* that could be “found,” “identified,” or “described” for what they are or correlated with states of their sources. Just as texts can be read from numerous perspectives, so signs can have several designations, and data can be subjected to various analyses. One can count the characters, words, or sentences of a text. One can categorize its expressions, analyze its metaphors, describe the logical structure of its compositions, and ascertain its associations, connotations, denotations, and commands. One can also offer psychiatric, sociological, political, or poetic interpretations of that text. All of these accounts may be valid but different. Untrained analysts may be overwhelmed by these choices. Researchers who pursue content analysis according to the first of the above definitions are led to believe that a message has but one content, all other meanings being deviant, wrong, or subjective, and hence to be excluded. This naive belief is an entailment of the unreflecting use of the container metaphor. Perhaps the term *content analysis* was ill chosen for this reason. The possibility that any text may have multiple readings renders the frequently published claims by some researchers that they have analyzed *the* content of particular bodies of text untenable by our (third kind of) definition.

3. *The meanings invoked by texts need not be shared.* Although intersubjective agreement as to what an author meant to say or what a given text means to a community would simplify a content analysis tremendously, such consensus rarely exists in fact. Demanding that analysts limit themselves to “common” or “shared ground” would restrict the empirical domain of content analysis to the most trivial or “manifest aspects of communications,” on which Berelson's definition insists, or it would restrict the use of content analysis to a small community of message producers, recipients, and analysts who happen to see the world from a common perspective. If content analysts were not allowed to read texts in ways that are different from the ways other readers do, content analysis would be pointless. In fact, psychiatrists are expected to interpret the stories they hear from their patients in ways that differ from the patients' interpretations. Anthropologists' analyses of cultural artifacts need not conform to what informants say about those artifacts, and conversation analysts have good reasons to see verbal interactions in ways conversants might not. As Gerbner and his colleagues have shown through content analyses, mass-media audiences are not aware of the statistical trends in the qualities of popular heroes, the kinds of violence depicted, and the representations of minorities in television programming. Critical scholarship would be stifled if it could not go outside of what everyone accepts as true. Content analysis is in trouble only when expert interpretations fail to acknowledge the multiplicity of uses of texts by

designated populations of readers or actors, particularly when content analysts fail to spell out the criteria for validating their results.

4. *Meanings (contents) speak to something other than the given texts*, even where convention suggests that messages “contain” them or texts “have” them. Probably the most distinctive feature of communications is that they inform their recipients, invoke feelings, or cause behavioral changes. Texts can inform their readers about events at distant locations, about objects that no longer exist, about ideas in other people’s minds, about available actions—just as symbols represent things in their absence, and stories walk their listeners through imagined worlds. Texts can also lead to responses of various kinds. All of these phenomena link the reading of present texts to something else, not necessarily present. Whether these other phenomena concern purely mental constructions, past or future experiences, or hidden causes, the analyst must be able to conceive of them and verbalize them. It follows that content analysts must look outside the physicality of texts—for example, to how people other than the analysts use these texts, what the texts tell them, the conceptions and actions the texts encourage. This requirement is a key to understanding the limitations inherent in computer text analysis. Computers can be programmed to manipulate character strings in amazingly complex ways, but their operations remain confined to the conceptions of their programmers. Without human intelligence and the human ability to read and draw inferences from texts, computer text analysis cannot point to anything outside of what it processes. Computers have no environment of their own making; they operate in the contexts of their users’ worlds without understanding those contexts.

5. *Texts have meanings relative to particular contexts, discourses, or purposes*. Although diverse readings of a text are typical, the task of content analysts is far from hopeless. Messages always occur in particular situations, texts are read with particular intents, and data are informative relative to particular problems. Statisticians, linguists, anthropologists, psychiatrists, and political analysts all have their own discipline-based reasons for interpreting given assertions differently. A therapist and a conversation analyst will view the same conversation differently. A speech on economics may be analyzed for its political implications, for how well it presents certain arguments, for what the speechwriter knows about economics, or for the emotions it arouses. We explain these differences by the *contexts* within which analysts choose to listen to that speech. Differences in interpretations do not preclude the possibility of agreements within particular contexts, however. In fact, once content analysts have chosen the context within which they intend to make sense of a given text, the diversity of interpretations may well be reduced to a manageable number, sometimes to one.

Every content analysis requires a context within which the available texts are examined. The analyst must, in effect, construct a world in which the texts make sense and can answer the analyst’s research questions. A context renders perceptual data into readable texts and serves as the conceptual justification for reasonable interpretations, including for the results of content analysis. Often, analysts presuppose particular contexts based on their own disciplinary commitments, as in the above example about a speech on economics. Analysts working within particular disciplines, such as political science, rhetoric, economics, and psychology, hold particular theories

concerning how texts are to be handled; that is, they are willing to accept only a certain context. Holsti's encoding/decoding paradigm, mentioned above, functions as a prominent analytical context in communication research, but it is by no means the only one. The contexts that psychiatrists tend to construct are very different from those that political scientists are likely to accept or within which literary scholars prefer to work. Once an analyst has chosen a context for a particular body of text and clearly understands that context, certain kinds of questions become answerable, and others make no sense.

Just as the analytical contexts that content analysts must adopt may vary from one analysis to another, these contexts may also differ from the interpretive schemes that unaided listeners, viewers, or readers employ in reading their sensory data, the characters of their texts, and the messages they receive. The same body of texts can therefore yield very different findings when examined by different analysts and with reference to different groups of readers. For a content analysis to be replicable, the analysts must explicate the context that guides their inferences. Without such explicitness, anything would go.

6. *The nature of text demands that content analysts draw specific inferences from a body of texts to their chosen context*—from print to what that printed matter means to particular users, from how analysts regard a body of texts to how selected audiences are affected by those texts, and from available data to unobserved phenomena. Texts, messages, and symbols never speak for themselves. They inform someone. Information allows a reader to select among alternatives. It narrows the range of interpretations otherwise available. For the content analyst, the systematic reading of a body of texts narrows the range of possible inferences concerning unobserved facts, intentions, mental states, effects, prejudices, planned actions, and antecedent or consequent conditions. Content analysts infer answers to particular research questions from their texts. Their inferences are merely more systematic, explicitly informed, and verifiable—ideally—than what ordinary readers do with texts. Recognizing this apparent generality, our definition of content analysis makes the drawing of inferences the centerpiece of this research technique.

The element of “making inferences” is not entirely absent from other definitions of content analysis. For example, Stone, Dunphy, Smith, and Ogilvie (1966) define content analysis as “a research technique for making inferences by systematically and objectively identifying specified characteristics within a text” (p. 5). Although their inclusion of “within a text” here would suggest a commitment to “inherentist” conceptions of meaning, Stone et al. nevertheless recognize the inferential character of the processes of coding and categorizing textual material, in their case by computer. Their dictionary of fixed linguistic classifications of word meanings leads to semantically simplified representations of a text's conventional readings. Other authors have equated inferences with statistical generalizations (e.g., Roberts, 1997), which do not, however, move into the context of textual matter. As early as 1943, Janis (1943/1965) pointed to the need for researchers to validate the results of content analyses of mass communications by relating research findings to audience perceptions and to behavioral effects. Our definition requires that content analysts be able to validate their results as well, whether those results are used to predict something,

to inform decisions, or to help conceptualize the realities of certain individuals or groups. But validation becomes an issue only where inferences are specific and thus have the potential for failing.

Regarding the drawing of inferences, Merten (1991) paraphrases the essential elements of my definition of content analysis (Krippendorff, 1980b) when he writes, “Content analysis is a method for inquiring into social reality, which consists of inferring features of a nonmanifest context from features of a manifest text” (p. 15; my translation). All theories of reading (hermeneutics) and theories of symbolic forms (semiotics), including theories of message meanings (communication/conversation theory), can be operationalized as processes of moving from texts to the contexts of the texts’ use. I would also suggest that a context is always constructed by someone, here by content analysts, no matter how hard they may try to objectify it. This is true even for ethnographers who believe that they can delegate the definition of the context to their informants’ world conceptions. It is the ethnographers who are held responsible for what they end up reporting. One cannot deny content analysts’ interest and conceptual participation in what their analysis reveals. Whether the analysts’ context coincides with the many worlds of others is a difficult question to answer. Whether the analysts’ world makes sense to their scientific peers depends on how compellingly the analysts present that world.

2.3 EXAMPLES

In this section, I offer some examples to illustrate how our definition of content analysis applies to practical situations.

Example 1. Consider the situation of wartime analysts of enemy broadcasts who want to gauge, among other phenomena, the popular support that enemy elites enjoy in their country. In peacetime, researchers could obtain such information directly, through public opinion surveys, for example, or by on-site observations. In wartime, however, information of this nature is difficult to get, if not deliberately concealed, and analysts are forced to use indirect means of obtaining it. The inability to use direct observation is an invitation to apply content analysis. Here, analysts are typically not interested in the literal meanings of enemy broadcasts, in the rhetorical devices political leaders use, or in judging whether individual citizens are being deliberately misled. In fact, wartime propaganda analysts have good reasons to overlook manifest contents and ignore their truths. To infer the extent of popular support for elite policies from enemy domestic broadcasts, the analysts must understand the role these broadcasts play in complex communication networks in which the mass-media system and political system interact with a population to make news acceptable. The propaganda analysts have to know something about the actors involved in the governing elite and in the military, about the media these actors have access to, and about other institutions that have a stake in current affairs. They must also have some knowledge of the political-economic processes that keep a country together and how the public tends to respond to mass-mediated messages. The picture they construct of what they are dealing with amounts to the context of their analysis. It connects the intercepted broadcasts to the

phenomena of interest, whether they concern popular support of the governing elite's policies, planned military actions, or evidence of war weariness.

Example 2. Historians are never mere collectors of documents. They offer reconstructions of past events that they deem consistent with current readings of all available documentary evidence. Historians are far removed from the worlds they wish to articulate. They cannot interview Julius Caesar, ask Homer about his sources for the *Iliad*, participate in the experiences of African slaves entering colonial America, or listen to conversations between Pablo Picasso and Henri Matisse. Historical figures reside in our readings of available documents, not in facts. And although some have left their writings to us, it is unlikely that they anticipated contemporary historians' readings. Past happenings become comprehensible to us only by inferences from documents that have survived to the present (Dibble, 1963). Historians who infer past events from available texts are, by our definition, involved in content analysis. It is not surprising, therefore, that historians are keenly aware of the need to place the documents they analyze within the context of other relevant documents. Without the appropriate context, a document means very little; a document placed in the wrong context acquires incorrect meanings, or at least meanings that may not make much sense. Historiographical methods organize available documents into webs of inferential relationships that may ultimately answer a historian's questions.

Example 3. Psychological researchers have a long tradition of developing theories whose generalizability is established by repeated experiments. The subjects of psychological research must be present; however, making it difficult for researchers to study developmental issues and individuals who are available only through their writings. Expanding psychological research methods, Allport (1942) added personal documents, witness accounts, and letters to the repertoire of data amenable to psychological inquiries. The research he proposed amounts to content analysis by our definition: There are texts in the form of personal documents, diaries, letters, and recorded speeches, and researchers construct the contexts for analyzing these texts with the help of available theories concerning the correlations between what people say and a variety of psychological variables (e.g., cognitive processes, attitudes, emotional arousal, personality traits, worldviews, or psychopathologies). Different schools of psychology direct their researchers to different questions, but they all are interested in inferring psychological variables of authors from the texts they left behind. In the course of analyzing personal documents, psychologically oriented content analysts have developed a variety of inferential techniques (e.g., type/token ratios of key concepts, the discomfort/relief quotient, graphological interpretations, readability yardsticks, Thematic Apperception Tests, and personal structure analysis). In individual psychology, content analysis has become an established method of inquiry since Allport's (1965) pioneering work.

Example 4. For good reasons, interview and focus group data are frequently subjected to content analysis. Structured interviews generate predefined question-answer pairs, and the researcher then analyzes their distributions. Researcher's conceptions are imposed on the interviewees, who cannot express the reasons for

their choices among predefined answers and whose individual conceptions are ignored. In open-ended interviews and focus groups, in contrast, participants are allowed to speak freely and in their own terms. To explore the conceptions that are manifest in such conversations, researchers need to perform what amounts to content analysis on the transcripts of these conversations. In a breast cancer study, for example, patients were asked about their lives after they had received treatment (Samarel et al., 1998). The answers were naturally freewheeling, as expected, enabling the researchers to adapt their theory of “coping” to the transcripts at hand. The researchers’ reformulated theory then provided the context for a subsequent content analysis. Armed with questions derived from the researchers’ theory, coders looked for and identified answers within the transcripts, and by tabulating these, the researchers provided frequencies and statistical accounts that the funders of the research required. In this study, the qualitative inferences were made during the process of coding, not based on the resulting frequencies, which merely summarized these inferences.

Example 5. Mass communication is the archetypal domain of content analysis. Communication researchers tend to be interested in communicator conceptions, media biases and effects, institutional constraints, implications of new technologies, audience perceptions, public opinion, and how certain values, prejudices, cultural distinctions, and reality constructions are distributed in society—relying on mass-media messages as their causes or expressions. Typically, mass-media material calls for more reading than any single person can handle. Its analysis thus requires a framework, a theory, a vocabulary, and an analytical focus in terms of which the researcher can construct a suitable context for analysis and collaborate with other researchers on the same project. Different contexts answer different research questions, of course.

A stereotypical aim of mass-media content analysis is to describe how a controversial issue is “depicted” in a chosen genre. Efforts to describe how something is “covered” by, “portrayed” in, or “represented” in the media invoke a picture theory of content. This approach to content analysis decontextualizes the analyzed text and thus reverts to the first kind of definition of content analysis distinguished above. It conceals the researchers’ interest in the analysis and hides their inferences behind the naive belief that they are able to describe meanings objectively while rendering the results immune to invalidating evidence. Consider common findings of political biases, racial prejudices, and the silencing of minorities on television as such issues. Although counts of evident incidences of such phenomena can give the impression of objectivity, they make sense only in the context of accepting certain social norms, such as the value of giving equal voice to both sides of a controversy, neutrality of reporting, or affirmative representations. Implying such norms hides the context that analysts need to specify. Unless analysts spell out whose norms are applied, whose attitudes are being inferred, who is exposed to which mass media, and, most important, where the supposed phenomena could be observed, their findings cannot be validated. Berelson and Lazarsfeld (1948, p. 6) noted long ago that there is no point in counting unless the frequencies lead to inferences about the conditions surrounding what is counted. For example, counting the numbers of mentions of *Microsoft* or *AIDS* or the term *road rage* over time in, say, the *New York Times* would

be totally meaningless if the observed frequencies could not be related to something else, such as political, cultural, or economic trends. That something else is the context that lends significance to quantitative findings.

Example 6. Content analysis has many commercial uses. For example, word-association databases (which collect huge numbers of pairs of words that consumers associate in their minds, as determined through word-association experiments) can serve as the context within which advertising researchers can infer chains of associations for new products, services, or brand names. In another, very different application, Michael Eeley and I studied how publicity generated by the Public Broadcasting Service about its programming ended up in newspaper articles (Krippendorff & Eeley, 1986). The purpose of the study was to enable PBS analysts to infer how the Public Broadcasting Service is perceived by newspaper editors in different regions of the United States and to assess the effectiveness of PBS's publicity efforts. Here the context was very simple. It included what we knew about newspaper editors' access to wire services and press releases, their newspapers' coverage of PBS programming, and certain theories and assumptions about the difference between the two, which led us to infer the (controllable) persuasive force of PBS publicity and the (uncontrollable) attitudes and competencies of the journalists, further differentiated by region and size of the newspaper.

Example 7. Content analysis essentially places texts in the contexts of their use. The emergence of numerous social media platforms—Academia for academics and researchers, Facebook for sharing photos; blogs and apps among so-called “friends”; Identi.ca for connecting hackers and open source advocates; LibraryThing for book-lovers; and Twitter to name only a few—pose considerable conceptual challenges. Their widespread use has changed the nature of public discourse, business practices, and individual connections, and is driving social realities in unprecedented ways. They affect the conduct of content analysts in at least four ways. One is the massive amounts of easily accessible digital images and texts. This has encouraged statistically oriented social scientists to create theories that serve equally statistically oriented political, commercial, and economic institutions. Content analysts, by contrast, will have to develop computational methods to reveal the underlying structures of these media. A second is the schism between the global nature of Internet technologies and users experiencing only the microcosms of their individual participation. The latter has the effect of creating virtual communities of like-minded individuals that may well close themselves off from equally closed but other-minded communities, consequently fostering new kinds of social divisions and conflicts. Facing such differences, content analysts can no longer apply the same analytical categories to all texts. They will have to adopt a systems perspective that can account for the emerging differences in the use and interpretation of texts. A third is the dynamics that the use of social media exhibit. While printed texts have always afforded unlike readings by different communities, their mode of production, their materialities, and lifespans are unlike digital texts. Some social media messages are inconsequential, others are reproduced in great numbers, become viral, capture the imagination of large interest groups, and encourage spontaneous social actions. While media activists tend not to keep long records of their participation, content analysts can often draw on the data that the hosts of social media platforms keep. But merely celebrating the size of such

databases as goldmines for traditional content analyses hides the dynamics of the role of social media in society.

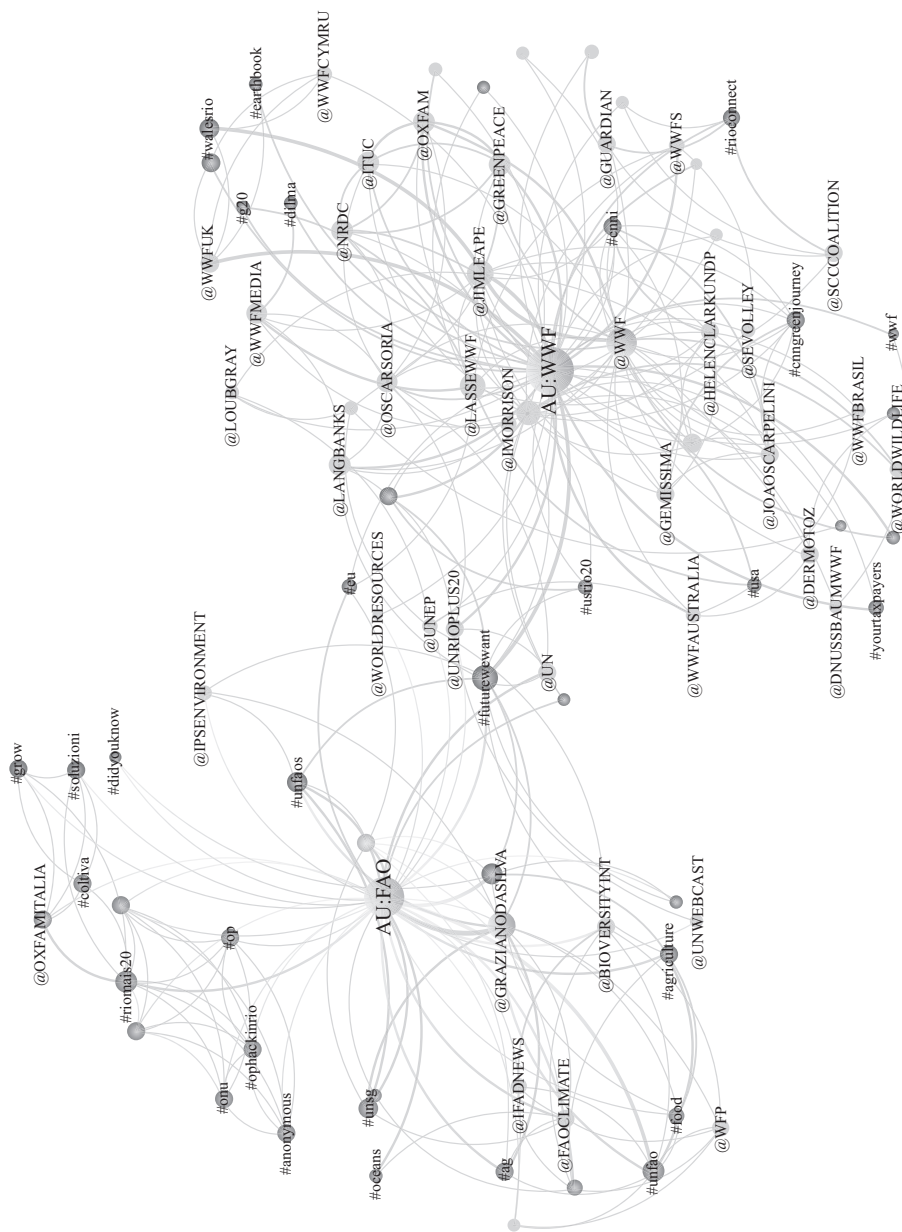
Fourth, and perhaps most important, is the new kind of texts that social media generate. Their messages are short. They make little sense in isolation, even as statistical aggregates. Their meanings cannot be found in dictionaries or thesauri. They reside in how participants respond to previous messages and what happens to them after being posted. Although the technicalities of different platforms play a significant role in how users network texts (Niederer, 2016), the connections among social media messages provide the context of their networked quality. One of these connections, already mentioned, is the emergence of community-specific vocabularies, abbreviations, emojis, and identifiable topics that develop in particular networks. Another is the identity of the participants—assumed or authentic—and references in one message to another. Such fluid connections are the dynamic analogues of what makes a written text coherent and interpretable by single readers.

Iina Hellsten and Loet Leydesdorff (2017) provide an excellent example of how a content analysis can cope with such networked text. They combined two well-known computational analytical methods into one: semantic network analysis, which maps patterns of co-occurring words to identify clusters of common meanings; and actor networks analysis, which addresses paths of influences in social organizations and distinguishes between key and peripheral players. They analyzed 72,077 English language tweets out of 100,073 generated during the United Nations Conference on Sustainable Development Rio+20, June 20–22, 2012. Twitter enables their users to direct short, then up to 140 character messages, recently doubled, to specific others, using @ before their user name, to retweet messages followed by the mark RT, and to hashtag a word within tweets marked by #. Hashtags enable users to highlight what is important to them and provide users access to other messages containing them. The absence of rules for identifying hashtags (save for their limitation to a single character string) render hashtags categories of topics, judgments, or feelings indigenous to a user group. It is these three text markers that enable users to navigate and contextualize their contributions and content analysts to link social media messages into networks.

Hellsten and Leydesdorff's (2017) content analysis exemplifies the shift of attention from analytically defined units and context-sensitively coded texts to connecting naturally occurring units of text, tweets, by the markers they carry, establishing each other's contexts. Their work shows a way to fulfill an important promise of content analysis to develop perspectives larger than what individual readers can acquire, in their case suggesting inferences of interest to activists, policy makers, and communication scientists. Figure 2.1 focuses on two organizations, the World Wildlife Foundation (WWF) and United Nations' Food and Agriculture Organization (FAO). They are linked by the hashtag #futurewewant and have essentially separate networks. WWF is more connected to particular users tagged @ while FAO is more substantially focused on topics manifest in hashtags #. Hellsten and Leydesdorff used Kamada and Kawai (1989) for the layout and VOSviewer (<http://www.vosviewer.com>) for visualizing the network.

The foregoing suggests that purely descriptive intents, manifest in claims to have analyzed "*the content* of a newspaper," to have quantified "*the media coverage* of an event," or to have "*found* how an ethnic group is *depicted*," fail to make explicit the very contexts within which researchers choose to analyze their texts. Content analysts have to know the conditions under which they obtain their texts, but, more

FIGURE 2.1 ■ An Analysis of the Network of Twitter Messages During the Rio-20 Meeting Focusing on Two Organizations, the World Wildlife Foundation (WWF) and United Nations' Food and Agriculture Organization (FAO)



important, they also have to be explicit about *whose* readings they are speaking of, *which processes or norms* they are applying to come to their conclusions, and *what the world looks like* in which their analyses, their own readings, and their readings of others' readings make sense to other content analysts. Explicitly identifying the contexts for their analytical efforts is also a way of inviting other analysts to bring validating evidence to bear on the inferences published and thus advance content analysis as a research technique. In the 7th example, the categorization of texts by hashtags is left to their authors, and the contexts in which social media messages are used is utilized to generate a larger picture of what happened during this international meeting. The framework presented in the next section is intended to help content analysts to conceptualize the content analytical process so that their results are arguably acceptable.

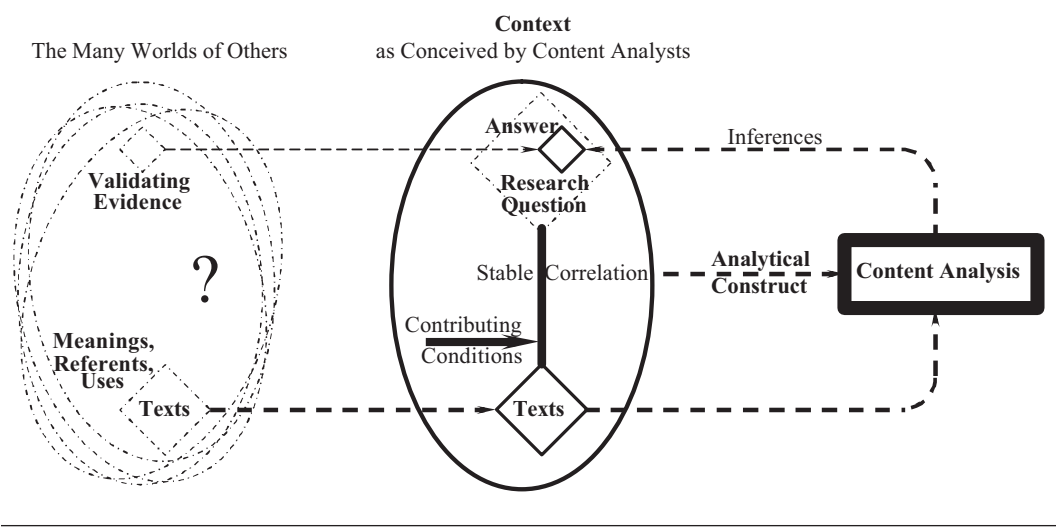
2.4 FRAMEWORK

The definition of content analysis offered at the opening of this chapter and illustrated with the above examples emphasizes the drawing of inferences of a certain kind. It also assigns content analysts a particular role vis-à-vis their objects of inquiry. Following from the above and previous work (Krippendorff, 1969b, pp. 7–13; 1980b), I offer a conceptual framework for content analysts within which their role becomes clear, and specialized methods can be located. This framework is intended to serve three purposes: Its *prescriptive* purpose is to guide the conceptualization and design of practical content analytic research; its *analytical* purpose is to facilitate the critical examination and comparison of the published content analyses; and its *methodological* purpose is to point to performance criteria and precautionary standards that researchers are invited to apply when evaluating ongoing content analyses. It is hoped that the use of the framework will lead to long-term systematic improvements of the method.

The framework, which is depicted in Figure 2.2 and will be revisited in Chapter 4's discussion of the logic of content analysis designs, is simple and general, employing only a few conceptual components:

- A body of text, the data that a content analyst has available to begin an analytical effort
- A research question that the analyst seeks to answer by examining the body of text
- A context of the analyst's choice within which to make sense of the body of text
- An analytical construct that operationalizes what the analyst knows about the context of the body of text
- Inferences that are intended to answer the research question, which constitute the basic accomplishment of the content analysis
- Validating evidence, which is the ultimate justification of the content analysis

FIGURE 2.2 ■ A Framework for Content Analysis



2.4.1 Texts

Data are the starting point of any empirical research. Data are *taken as givens*—that is, the researcher is not in doubt as to what they are. In surveys, focus groups, and psychological experiments, researchers attempt to control the generation of their data, thereby assuring that they know what the data mean, largely, if not exclusively, in the researchers' terms. Most content analyses start with data that are not intended to be analyzed to answer specific research questions. They are *texts* in the sense that they are meant to be *read, interpreted, and understood by people other than the analysts*. Readers may decompose what they read into meaningful units, recognize compelling structures, rearticulate their understandings sequentially or holistically, and act on them sensibly. When we are capable of this kind of rearticulation, we attribute textuality to what we see as writing, pictorial images, gestures, web pages, videos, musical compositions, even behavioral sequences. Text results from reading and rearticulation.

One could speak of symbolic qualities instead of text, but it is preferable not to assume such qualities to exist without reference to who regards them as such. An analyst's reading—the units, syntax, and narrative structures that constitute the texts for the analyst—naturally differs from the readings that initiate the interpretations of ordinary readers, including the texts' authors. It follows that an analyst's reading must never be taken as the only legitimate one, nor should content analysts assume the sole power to determine the form of the texts they analyze. They would then be examining only themselves. We presume that all authors write in the expectation of being understood by self and by others, and it is the implication of others that renders a text socially significant. Although content analysts are not bound to analyze their data with reference to the conceptions or intended audiences of their texts' authors, they must at least consider that texts may have been intended for someone like them. We know that interviewees answer questions differently when they know how the

research findings could affect them, and so we need to read interview results in the context of possible self-interests. We know that when politicians speak, they anticipate being scrutinized by the public, and so we cannot take their speeches at face value, as natural objects. Content analysts have to acknowledge that the textuality they rely on is not the only one that counts.

Content analysts' best guarantee against the contamination of texts by the stakes their sources have in how their texts are analyzed is to focus on textual features of which their sources are unconscious or to apply categories that the sources of their texts use habitually or are unable to control. This is most obviously possible when the sources of texts are of the past (historical), when they are unaware of how their texts are being analyzed, or when communication to the analysts is one-way, without feedback. However, given that the results of most content analyses are published, and that the categories that analysts use have the potential of becoming known to the text sources as well, content analysts are justified in applying unconventional categories, that is, in looking at textuality in ways others may not. As Figure 2.2 illustrates, texts occur in the analyst's world but acknowledge their origins in the worlds of others.

2.4.2 Research Questions

Research questions are the targets of the analyst's inferences from available texts. Generally, such questions delineate several possible and initially uncertain answers. In this respect, a research question is analogous to a set of hypotheses. However, in contrast to scientific hypotheses, which are pitted against direct observational evidence, the research questions of content analysis must be answered through inferences drawn from texts. The difference between testing scientific hypotheses and selecting an answer to a research question is crucial. Whereas observations are registered or measured for what they are, and hypotheses about observational phenomena amount to generalizations from observations, texts inform an analyst about extratextual phenomena, about meanings, consequences, or particular uses. Thus, whereas scientific hypotheses are accepted on account of a preponderance of evidence in favor of one at the expense of other hypotheses, an ideally large number of observations that support one and rule out others, inferences from texts (although large numbers may play a role here as well) pertain to phenomena that are not observed during a content analysis, phenomena that are outside the texts and thus retain their hypothetical character until confirmed by validating incidences.

There are two reasons for content analysts to start with research questions, ideally in advance of undertaking any inquiries: *efficiency* and *empirical grounding*. One can surely explore the meanings that come to mind while reading a text, following the threads of the inferences to wherever they may lead or engaging in so-called fishing expeditions. Hermeneutical, interpretive, and ethnographic approaches to reading cherish such open-endedness. However, when research is motivated by specific questions, analysts can proceed more expeditiously from sampling relevant texts to answering given questions. Content analysts who start with a research question read texts for a purpose, not for what an author may lead them to think or what they say in the abstract.

The pursuit of answers to research questions also grounds content analysis empirically. All answers to research questions entail truth claims that could be supported, if not by direct observation then at least by plausible argumentation from

related observations. Our framework suggests that content analysis compensates for analysts' inability to observe phenomena in which they are interested, whether these phenomena pertain to the characteristics of writers or readers, to happenings hidden behind intentional information barriers, or to events in a distant past or future.

Formulating research questions so that the answers can be validated in principle protects content analysts from getting lost in mere abstractions or self-serving categorizations. For example, the question of how frequently a particular word occurs in a text can be answered by counting. Counting is what analysts do. Counts cannot be validated by independent evidence; to assure that counts are correct, analysts must repeat them, perhaps employing different persons as counters. The same is true for questions concerning whether one can categorize, measure, or analyze something. Their answers lie in a researcher's ability to execute these processes reliably. These questions cannot be answered by research. Questions concerning the statistical generalizability of textual attributes or "contents" (in the sense of the first kind of definition of content analysis discussed above) from a sample to a population from which this sample was drawn are not suitable content analysis research questions either, but for a different reason. Although their answers do rely on empirical evidence, without abductive inferences to phenomena outside the texts being analyzed, generalizations are inductive and cannot answer content analysis research questions. Thus, in content analysis, research questions have the following characteristics:

- They are believed to be answerable (abductively inferable) by examinations of a body of texts. (In Figure 2.2, this is indicated by the bold dashed arrows.)
- They delineate a set of possible (hypothetical) answers among which analysts select. (In Figure 2.2, an answer is indicated by the unlabeled diamond.)
- They concern currently inaccessible phenomena.
- They allow for (in)validation—at least in principle—by acknowledging another way to observe or substantiate the occurrence of the inferred phenomena. (In Figure 2.2, this is indicated by the thin dashed arrow from the worlds of others to the answer to the research question.)

2.4.3 Context

I have argued above that texts acquire significance (meanings, contents, symbolic qualities, and interpretations) in the contexts of their use. Although data enter a content analysis from outside, they become texts to the analyst within the context that the analyst has chosen to read them—that is, from within the analysis. A context is always someone's construction, the conceptual environment of a text, the situation in which it plays a role. In a content analysis, the context explains what the analyst does with the texts; it could be considered the analyst's best hypothesis for how the texts came to be, what they mean, or what they can tell or do. In the course of a content analysis, the context embraces all the knowledge that the analyst applies to given texts, whether in the form of scientific theories, plausibly argued propositions, empirical evidence, grounded intuitions, or knowledge of reading habits.

The context specifies the world in which texts can be related to the analyst's research questions. This world is always one of many. Political analysts construct worlds that differ from those of politicians, often embracing additional perspectives,

but those worlds also differ from the worlds of psychologists, journalists, historians, psychotherapists, scholars of literature, and—naturally—communication researchers, who pursue their own research agenda and approach texts with their own questions, concepts, models, and analytical tools. Scholars in different disciplines tend to place the same texts in different contexts but rarely without acknowledging that there are other readings, other contexts, other worlds, within which given texts function as well—authors, audiences, users, and beneficiaries, for example. In Figure 2.2, these worlds are shown in the ovals embracing texts and their multiple meanings.

Knowledge of the context for content analyzing given texts can be separated into two kinds:

- The network of stable correlations, which are believed to connect available texts to the possible answers to given research questions, whether these correlations are established empirically, derived from applicable theory, or merely assumed for the purpose of an analysis
- Contributing conditions, which consist of all the factors that are known to affect that network of stable correlations in foreseeable ways

In Figure 2.2, these relationships are shown by a bold line and a bold arrow.

To use an example that is far from simple, in an ordinary conversation, what is observed and heard as being said at any one moment (the data) is understandable only in the context of what has been said before, by whom and to whom, the responses it elicited from the participants, and how it directed the conversation. This is an observer's account of a conversation, from outside of it. To participants, their version of what is going on (the contexts that include the other participants) is not necessarily shared. In fact, there would be no point in conversing if all participants saw their worlds, thought, and spoke alike. A conversation analyst contextualizes the transcript of a conversation (the text) in yet another way, by constructing a world (the analyst's context) within which the participants appear to "speak" in the analytical terms that the conversation analyst is familiar with and brings to the analyzed transcript. Whether a conversation analyst wants to infer the intentions of the participants to initiate certain moves (turn taking, for example) or how addressees will respond to a string of "he said–she said" (the evolution of a topic), the analyst draws on knowledge of the empirical relationship between these speech acts (the correlations that connect one to another) and the strengths (perlocutionary forces) of particular utterances, the network of connections that leads, hopefully, from texts to answers to the research question.

A conversation is not a mechanical system. Participants alter the rules of their engagement as it unfolds. This leaves outside observers uncertain as to what the participants mean, how they understand what is going on, and which rules govern the conversation at any one moment. Because conversation analysts tend not to participate in the conversations they analyze, and therefore have no way of asking the interlocutors how they see their situation, the analysts have to acknowledge other determining variables (the contributing conditions) and find ways to ascertain how they affect the correlations relied upon to lead to the intended inferences.

Inasmuch as a context stands in place of what is momentarily inaccessible to direct observation, there is no limit to the number of contexts that may be applicable in a given analysis. Unless told, readers of the conclusions of a content analysis may

not know the context that the analyst was using and may come to seriously misleading interpretations. In view of this possibility, *content analysts need to make their chosen contexts explicit, so that the results of their analyses will be clear to their scientific peers and to the beneficiaries of the research results.* Without explication of the context, the steps that a content analyst takes may not be comprehensible to careful readers, and the results to which they lead may not be validatable by other means.

2.4.4 Analytical Constructs

Analytical constructs operationalize what the content analyst knows about the context, specifically the network of correlations that are assumed to explain how available texts are connected to the possible answers to the analyst's questions and the conditions under which these correlations could change. Analytical constructs represent this network in computable forms. Extracted from the known or assumed context and entered into the research process, analytical constructs *ensure that an analysis of given texts models the texts' context of use*, which means that the analysis does not proceed in violation of what is known of the conditions surrounding the texts. Procedurally, analytical constructs take the form of more or less complex "if-then" statements, much like those used in computer programs. These "if-then" statements amount to *rules of inference* that guide the analyst, in steps, from the texts to the answers to the research questions. They also render knowledge of the context *portable to other content analyses* of similar contexts and make it possible for students and critics to examine the procedures that a content analyst has been using. In this respect, analytical constructs function much like testable mini-theories of a context, with the provision that they are computable on the coded features of available texts.

For example, a computer-aided content analysis might employ a dictionary of tags that mimics how competent speakers of a language categorize words into classes with similar meanings. Such a dictionary assumes linguistic stability, which may not be warranted, but it at least models a standard competence of language use. Another approach that an analyst might take is to adopt a computational theory of a context—a neuronal network model, for instance—that promises to explain how people form categories from words that occur in proximity to each other. Of course, labeling an analytical construct a "model" does not guarantee that it accurately represents the network of relationships that are relevant to readers and writers. More often, content analysts draw on empirically obtained correlations between observed and currently unobserved variables. Correlations measure the extent of a linear relationship between variables—for example, between the rate of recorded speech disturbances and anxiety—which, if sufficiently general, could in turn be applied to individual cases, here yielding a prediction of a speaker's anxiety. However, as linguistic variables are rarely describable in intervals, and linear regression equations tend to hold only under restricted conditions, the use of such constructs typically requires that the analyst have additional information about the conditions under which the construct is predictive of that behavior. Similarly, knowing that public agendas are influenced by the mass-media coverage of pertinent events may give a content analyst the idea of an analytical construct for analyzing media coverage in place of public opinion surveys. Such research, which has been done, requires a fairly detailed operationalization of the conditions under which verbal or pictorial elements influence particular public conversations.

Analytical constructs need not be perfect, of course, but unfortunately, many text analysts employ computational procedures that have no obvious relationship to any context in which given texts would arguably make sense. Counting units of text or applying sophisticated statistical techniques will always yield something, but this does not guarantee that the results will refer to anything. Content analysts must make sure that their analytical constructions model the contexts they have chosen. The purpose of all analytical constructs is *to ensure that texts are processed in reference to what is known about their use*.

2.4.5 Inferences

The inferential nature of content analysis should by now be obvious. Content analytic inferences may be hidden in the human process of coding. They may be built into analytical procedures, such as the dictionaries in computer-aided text analyses or well-established indices. Sometimes, especially after complex statistical procedures have been applied, inferences appear in the analyst's interpretations of the statistical findings. Figure 2.2 depicts the path that an inference takes with bold and broken lines, with the inference motivated or explained by an analytical construct that enters the analysis as a representation of the chosen context.

Because the word *inference* has several meanings, it is important to distinguish the meaning that is relevant to this discussion from others that are perhaps more familiar to readers. In logic, at least three types of inferences are distinguished:

- Deductive inferences are implied in their premises. For example, if all humans speak a language, then John, being human, must speak one as well. Deductive inferences are logically conclusive. They proceed from generalizations to particulars.
- Inductive inferences are generalizations to similar kinds. For example, I might infer from the fact that all of my neighbors speak English that all humans do. This inference is not logically conclusive, but it has a certain probability of being correct. Statistical generalizations from smaller samples to larger populations (typical of social research) and the idea of measuring the statistical significance of scientific hypotheses involve inferences of this kind. They proceed from particulars to generalizations.
- Abductive inferences proceed across logically distinct domains, from particulars of one kind to particulars of another kind. (These are the kinds of inferences of interest to content analysis, where they proceed from texts to the answers to the analyst's questions.) Consider linguistic competence and age. Logically, neither implies the other. However, if one has practical experience with infants' language acquisition, one might be able to infer children's ages from the sounds they make or from the vocabulary they use. Of course, one can make such inferences only with a certain probability, but the probability may be strengthened if one is able to take other variables (contributing conditions) into account.

Deductive and inductive inferences are not central to content analysis. The following examples of inferences employed in content analysis are all abductive in nature:

- One might date a document from the vocabulary used within it.
- One might infer the religious affiliations of political leaders from the metaphors used in their speeches.
- One might infer the readability of an essay from a measure of the complexity of its composition.
- One might infer whether someone is lying from his or her nonverbal (facial) behavior.
- One might infer the problems of a city from the concerns expressed in letters written to the city's mayor's office.
- One might infer the prevailing conceptualizations of writers and readers from the proximities of words in frequently used texts.
- One might infer editorial biases from a comparison of the editorial pages of different newspapers.
- One might infer a writer's psychopathology from the images used in her prose.
- One might infer the identity of the author of an unsigned document from the document's statistical similarities to texts whose authors are known.
- One might infer the political affiliations of citizens from the TV shows they choose to watch.
- One might infer an individual's propensity to engage in a hate crime from the ethnic categories he uses in ordinary speech.
- One might infer the likelihood of war from the coverage of international affairs in the elite newspapers of neighboring countries.

According to Eco (1994):

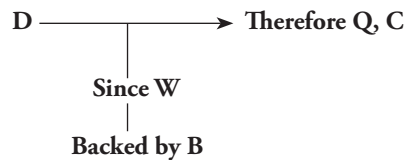
The logic of interpretation is the Peircean logic of abduction. To explain a conjecture means to figure out a law that can explain a Result. The "secret code" of a text is such a Law. . . . In the natural sciences the conjecture has to try only the law, since the Result is under the eyes of everybody, while in textual interpretation only the discovery of a "good" Law makes the Result acceptable. (p. 59)

For Josephson and Josephson (1994, p. 5), abduction starts with a body of data (facts, observations, givens)—our text. A hypothesis—our analytical construct—if true, would explain these data. No other hypothesis can explain the data as well as the chosen one does. Therefore, the hypothesis is probably true and can be used to deduce other entailments—that is, answer our research questions.

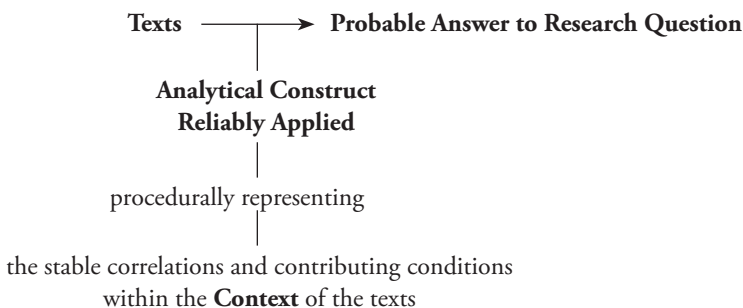
Abductive inference is Sherlock Holmes's logic of reasoning as well (Bonfantini & Proni, 1988; Truzzi, 1988). Holmes's creator, Sir Arthur Conan Doyle, always lets him find empirical connections and apply bits of common knowledge in the context of established facts that he is then able to weave ingeniously into an inferential network

containing the initially unrecognizable chain of logical steps from known facts to the perpetrator of an unobserved crime. Content analysts are in a similar position of having to draw inferences about phenomena that are not directly observable, and they are often equally resourceful in using a mixture of statistical knowledge, theory, experience, and intuition to answer their research questions from available texts.

In this respect, the whole enterprise of content analysis may well be regarded as an argument in support of an analyst's abductive claims. In Toulmin's (1958) theory of argumentation, which applies not just to abductions, the move from data (D) to conclusions or claims (C) must be justified by a suitable warrant (W). In his example, learning that "X is a Swede," the inference that "X most likely is a Protestant" is warranted by the knowledge that "most Swedes are Protestants." Because this inference is not without exceptions, it includes a qualification (Q) of the conclusion (C) (i.e., "most likely"). The warrant provides the logical bridge between the data and the conclusion. Toulmin also introduces another element: the ground on which the warrant may be justified, or the backing (B). In Figure 2.2 we may recognize the diagram that Toulmin (p. 104) uses to show the relationships among the above-mentioned parts of arguments:



In moving from texts to the answer to a research question, as illustrated in Figure 2.2, it is the assumptive analytical construct plus the assurance that the analysis has been performed reliably that warrants that inference, which in turn is backed by the analyst's knowledge of the context in which the texts occur or are interpreted:



2.4.6 Validating Evidence

Any content analysis should be validatable in principle. Because the *raison d'être* of content analysis is the absence of direct observational evidence, validation may be difficult or infeasible, if not impossible, in practice. It is infeasible when a content analysis is to be acted upon in the absence of direct observational evidence—for example, in wartime analysis of planned military activities from domestic propaganda

or in assessments of whether a politician is lying. It is impossible when research questions concern past or future happenings, such as inferences from surviving documents to historical facts, inferences from the works of deceased authors to their intentions, or inferences from psychological tests to an individual's aptitude for a particular job. The point of requiring that content analyses be "validatable in principle" is to prevent analysts from pursuing research questions that allow no empirical validation or that yield results with no backing except by the authority of the researcher. For example, a conclusion from an analysis of television fiction that hedonism is on the rise in the United States means nothing unless those who claim such findings can show that this conclusion is not merely their abstraction from fictional programming but also has some independently observable reality—that is, unless they can show that a rise in hedonism is manifest in something other than television fiction.

Ex post facto validation of content analysis is not merely a matter of curiosity. It can increase confidence in the results of future content analyses of similar texts and in similar contexts, but only if the categories of analysis and the analytical constructs are used repeatedly, so that successes and failures can be weighed against each other and used to advance the technique in the long run. Much too often, researchers design content analysis studies ad hoc and conduct them without any thought of validation; such research contributes little to the literature on content analysis.

A good example of ex post facto validation is George's (1959a) effort (mentioned in Chapter 1) to examine documents captured after World War II to see whether they matched what the Federal Communications Commission propaganda analysts had inferred during the war and to evaluate the FCC researchers' techniques for use by future analysts. In 1943, Janis (1943/1965) proposed an indirect method of validation, suggesting that the results of mass-media content analyses should at least correlate with audience verbal reports or observed behaviors (e.g., public opinion polls, voting, consumption, aggression). Thus Gerbner and his colleagues sought to correlate the amount of violence seen on television with survey data on audience members' perceptions of how violent their world "really" is (see, e.g., Gerbner, Gross, Signorielli, Morgan, & Jackson-Beeck, 1979; Gerbner, Gross, Morgan, & Signorielli, 1994).

As noted above, our framework demands merely that a content analysis be validatable *in principle*. For example, if a content analyst infers what a particular group of voters learned from TV campaign ads about a candidate for political office, then, potentially, a survey of those exposed to the ads could validate or invalidate this inference. However, if a content analyst insists that such TV campaign ads have certain contents, there is no way for others to validate this "finding." Repeating this particular content analysis would merely indicate the degree to which the original analysis was reliable. Similarly, finding that a certain word occurs with a certain frequency does not constitute an abductive inference. Recounting cannot validate what a frequency is inferred to mean.

2.5 CONTRASTS AND COMPARISONS

Every research technique has its powers and its limitations, and content analysis is no exception. A researcher can misapply a technique or use a technique that is ill suited for a particular purpose, in ignorance of better ones. In this section, I contrast content

analysis with other techniques used in social research, paying special attention to the four distinguishing features of content analysis.

Content analysis is an unobtrusive technique. As Heisenberg's uncertainty principle tells us, acts of measurement interfere with the phenomena being assessed and create contaminated observations; the deeper the observer probes, the greater the severity of the contamination. For the social sciences, Webb, Campbell, Schwartz, and Sechrest (1966) have enumerated several ways in which subjects react to being involved in scientific inquiries and how these can introduce errors into the data that are analyzed:

- Through the subjects' awareness of being observed or tested
- Through the artificiality of the task or the subjects' lack of experience with the task
- Through the expectations that subjects bring to the role of interviewee or respondent
- Through the influence of the measurement process on the subjects
- Through stereotypes held by subjects and the subjects' preferences for casting certain responses
- Through experimenter/interviewer interaction effects on the subjects

Controlled experiments, interviews, focus groups, surveys, and projective tests are especially vulnerable to such errors. By contrast, content analyses, computer simulations, research using already available statistics, and interpretive research (in cultural studies, for example) are nonreactive or unobtrusive. Researchers using ethnographic methods subscribe to the unobtrusive ideal as well, but while conducting fieldwork, even the most careful ethnographers cannot escape influencing their informers.

Social researchers may want to avoid reactive situations for two primary reasons. The first is that *undue influence on the situation that gives rise to the data may distort the data, jeopardizing the validity of the research*. For this reason, ethnomethodologists prefer to obtain data in natural settings, psychiatrists avoid asking their patients questions that might induce false memories, and economists investigate mathematical models rather than experiment with the real economy. The second reason is that *researchers need to conceal their interest in the data for fear of being manipulated by their sources*. Instrumental assertions are difficult to analyze (Mahl, 1959). Had Goebbels, the Nazi-era minister of propaganda in Germany, known how, by what methods, and for what purposes American analysts were examining his broadcasts during World War II, he would have found ways to deceive the analysts. Individuals can be taught how to achieve high scores on aptitude tests, and those who believe that success in their chosen career paths depends on their scoring well on these tests eagerly seek appropriate education. The extent to which preparatory instruction improves students' scores on a given test is also the extent of that test's invalidity. As an unobtrusive technique, content analysis can avoid such biases altogether.

Content analysis can handle unstructured matter as data. For efficiency's sake, researchers gain a considerable advantage if they can impose a structure on the

data-making process so that the results are readily analyzable. Surveys, mail questionnaires, and structured interviews typically offer respondents predefined choices that are easily tabulated, coded, or processed by computer. But they thereby also prevent the respondents' individual voices from being heard. Subjects in laboratory experiments are often taught what amounts to a highly artificial data language: pushing buttons, scaling their opinions numerically, identifying shapes or forms they may never have seen before, or administering electric shocks to fellow subjects in place of less clearly measurable expressions of violence. These techniques are successful because they allow researchers to suppress unwieldy variations, which are due largely to the fact that ordinary human subjects see, talk, and behave in many different ways.

Typically, content analysts become interested in data only after the data have been generated. They have to cope with texts in a diversity of formats associated with different purposes, do not always find what they are looking for, and cannot fully anticipate the terms and categories used by the sources of their texts. This puts content analysts in an analytical position that is less than advantageous, a condition they share with ethnomethodologists, anthropologists doing fieldwork, historiographical researchers, and researchers using hermeneutical or interpretive approaches (such as those used in studies of politics, psychotherapy, feminist scholarship, and social constructionism). The chief advantage of the unstructuredness of content analysis data is that *it preserves the conceptions of the data's sources*, which structured methods largely ignore.

Content analysis is context sensitive and therefore allows the researcher to process as data texts that are significant, meaningful, informative, and even representational to others. Context-insensitive methods, such as controlled laboratory experiments, surveys, structured interviews, and statistical analyses, generate data without reference to their original contexts, thus disembodiment observations, unitizing complex and contiguous events, and taking single words out of their contexts of use and representing them as data points in the analysts' theoretical spaces. In such methods, it no longer matters what gave rise to the data, how various elements in the data relate to each other, how others understand the data, or what the data mean to their sources. Context-sensitive methods, in contrast, acknowledge the textuality of the data—that is, they recognize that the data are read by and make sense to others, and they proceed by reference to contexts of their own. Inferences drawn through the use of such methods have a better chance of being relevant to the users of the analyzed texts.

Content analysts may not always be as qualitative as are political analysts, who live in the very process they analyze. Nor are they quite as free as cultural studies researchers and conversation analysts, who contextualize their texts in a vocabulary that may appear alien to the people they speak for or of. Finally, content analysts may not be quite as limited in scope as the users of projective tests, who confine themselves to inferences concerning individuals' psychological characteristics (much as in content analyses of the second kind of definition discussed above).

Content analysis can cope with large volumes of data. Much of ethnomethodology as well as case study approaches, historiographical methods, and interpretive research relies on small samples of text; the volume of data is limited largely by what a researcher can read reliably and without losing track of relevant details. Although content analysis can be used to analyze small samples of texts (in fact, this is quite common, especially in the academic world, where funds are few and stakes are not as high

as in politics, commerce, or medicine), such uses do not realize the technique's full potential. The ability to process large volumes of text in content analysis is paid for by the explicitness of the method's procedures, which, if clearly stated, can be applied repeatedly, by many coders or by computer software. As noted above, Berelson and Lazarsfeld (1948) stated long ago that content analysts must be systematic in their reading of texts and suggested that category schemes be devised that can be applied to every unit of text equally and without exception. Explicit vocabularies enable content analysts to employ many coders and pool their readings, which allows them to process quantities of text that far exceed what single individuals can analyze reliably. Consider the following numbers of units of analysis processed in early content analyses, largely without the aid of computers:

- 481 personal conversations (Landis & Burt, 1924)
- 427 school textbooks (Pierce, 1930)
- 4,022 advertising slogans (Shuman, 1937; cited in Berelson, 1952)
- 8,039 newspaper editorials (Foster, 1938)
- 800 news of foreign language radio programs (Arnheim & Bayne, 1941)
- 19,553 editorials (Pool, 1952a)
- 15,000 characters in 1,000 hours of television fiction (Gerbner et al., 1979)

These numbers, considered impressive in 1980 when the first edition of *Content Analysis* was published, were the result of physically collecting these documents or actually taping radio programs and TV shows. They are now dwarfed by the sizes of electronic full-text databases that have since emerged. At the time of this writing, 2017, in the United States:

- LexisNexis, originally developed to assist legal research, now provides online access to five billion searchable documents from more than 40,000 legal, news, and business sources, some accumulating since the late 1980s.
- ERIC, the Educational Information Resources Center, provides more than 1.3 million records accessing education literature and resources.
- Factiva.com, started jointly by Dow Jones and Reuters, lists 36,000 sources of premium content from 200 countries, in 28 languages. Users have access to a wide range of information from newspapers, newswires, industry publications, websites, blogs, company reports, and more.
- YouTube provides access to over 60 hours of video uploaded per minute, yes, per minute. Its video database measures around 45 terabytes.
- The core collection of the Vanderbilt News Archive consists of regularly scheduled newscasts from ABC, CBS, NBC, CNN, and Fox News. Its database currently includes 1,097,178 records, including abstracts at the story level of regular evening newscasts and catalog records for each special news report.

- Academic Search Complete includes nearly 6,600 active full-text journals and magazines.
- Proquest NewspaperARCHIVE is an online database that provides searchable access to local historical newspapers dating back to the 17th century. The full collection of its Library Edition contains more than 130 million digitized pages from over 6,000 newspapers throughout North America, the UK, and other select countries worldwide.
- As of October 15, 2015, Google Books has scanned 25 million books since 2004.
- The INTERNET ARCHIVE currently contains
 - 279 billion web pages
 - 11 million books and texts
 - 4 million audio recordings (including 160,000 live concerts)
 - 3 million videos (including 1.5 million television news programs)
 - 1 million images
 - 100,000 software programs

The electronic availability of such enormous volumes of analyzable texts is changing the ways content analysts can gather its raw data. For example, Rogers, Jansen, Stevenson, and Weltevrede (2010) collected 653,883 tweets, tagged “iranelections RT,” exchanged among Iranians during a 20-day period in June, 2009, to analyze the chronology of changing concerns right after Iran’s contested election. Dodds, Harris, Kloumann, Bliss, and Danford (2011) created a lexicon of 10,222 words by merging the 5,000 most frequently occurring words in a corpus of tweets. Leskovec, Backstrom, and Kleinberg (2017) relied on 90 million documents (blog posts and news articles) from 1.65 million different sites, or 390GB of texts, essentially all social media activity during a three-month period in 2008, and cluster analyzed 47 million phrases extracted from them. These are unimaginably large numbers. The blogosphere alone is currently estimated to double in size every six months. The expanding popularity of social networks and the instantaneous availability of electronic texts have had the effect of bringing content analysis closer to large population surveys—less expensive, yielding faster results, based on large numbers of short texts, and being only minimally obtrusive.

However, big data do not guarantee big insights. The ability to search and retrieve such volumes of text has certainly shifted the traditional bottleneck of content analysis of human coding of individually collected documents to the need for good theory, sound methodology, and software that is capable of coping with the larger contexts in which texts circulate. Here, pioneering work is needed and emerging.

3

USES AND INFERENCES

This chapter reviews the breadth of applications of content analysis, not in terms of subject matter or the disciplines that engage the technique but in terms of the kinds of inferences that enable content analysts to accomplish their tasks. Several types of logic capable of relating data to their contexts are distinguished here: systems, standards, indices, representations, conversations, and institutions. These outline frameworks for theories that content analysts may draw upon or need to develop.

3.1 TRADITIONAL OVERVIEWS

Content analysis has an important place in the wide range of investigative tools available to researchers. As noted in Chapter 2, content analysis is an unobtrusive technique that allows researchers to analyse relatively unstructured data in view of the meanings, symbolic qualities, and expressive contents they have and of the communicative roles they play in the lives of the data's sources. The combination of these features is unique among research methods. Because virtually all social processes can be seen as transacted through matter that is meaningful to the participants—symbols, messages, images, performances, and organizational phenomena, even nondiscursive practices—the widest use of content analysis is found in the social sciences and humanities, although legal, political, and commercial applications are rising in number as well. Scholars who have surveyed content analysis research have used a variety of categories to describe the growing diversity of research techniques used under the umbrella of content analysis. Janis (1943/1965) offers the following classification:

- (1) *Pragmatical content analysis*—procedures which classify signs according to their probable causes or effects (e.g., counting the number of times that something is said which is likely to have the effect of producing favorable attitudes toward Germany in a given audience).
- (2) *Semantical content analysis*—procedures which classify signs according to their meanings (e.g., counting the number of times that Germany is referred to, irrespective of the particular words that may be used to make the reference).
 - (a) *Designations analysis* provides the frequency with which certain objects (persons, things, groups, or concepts) are referred to, that is, roughly

speaking, subject-matter analysis (e.g., references to German foreign policy).

- (b) *Attribution analysis* provides the frequency with which certain characterizations are referred to (e.g., references to dishonesty).
 - (c) *Assertions analysis* provides the frequency with which certain objects are characterized in a particular way, that is, roughly speaking, thematic analysis (e.g., references to German foreign policy as dishonest).
- (3) *Sign-vehicle analysis*—procedures which classify content according to the psychophysical properties of the signs (e.g., counting the number of times the word “Germany” appears). (p. 57)

Leites and Pool (1942; cited in Berelson & Lazarsfeld, 1948) describe four functions of content analysis:

- To confirm what is already believed
- To correct the “optical illusions” of specialists
- To settle disagreements among specialists
- To formulate and test hypotheses about symbols

Berelson (1952) lists 17 uses:

- To describe trends in communication content
- To trace the development of scholarship
- To disclose international differences in communication content
- To compare media or levels of communication
- To audit communication content against objectives
- To construct and apply communication standards
- To aid in technical research operations (to code open-ended questions in survey interviews)
- To expose propaganda techniques
- To measure the readability of communication materials
- To discover stylistic features
- To identify the intentions and other characteristics of the communicators
- To determine the psychological state of persons or groups
- To detect the existence of propaganda (primarily for legal purposes)
- To secure political and military intelligence
- To reflect attitudes, interests, and values (cultural patterns) of population groups
- To reveal the focus of attention
- To describe attitudinal and behavioral responses to communications

Stone, Dunphy, Smith, and Ogilvie (1966) note that although the historical origins of content analysis lie in journalism and mass communication, they found applications of the technique in the following empirical domains:

- Psychiatry
- Psychology
- History
- Anthropology
- Education
- Philology and literary analysis
- Linguistics

I have already mentioned Holsti's (1969) commitment to an encoding/decoding paradigm, which, much like Janis's approach, places message content in the context of communication between senders and receivers. Consequently, Holsti surveys content analyses in terms of three principal purposes:

- To describe *manifest characteristics* of communication—that is, asking what, how, and to whom something is said
- To make inferences as to the *antecedents* of communication—that is, asking *why* something is said
- To make inferences as to the *consequences* of communication—that is, asking *with what effects* something is said

The way in which I categorize the content analyses discussed in this chapter deviates from the ways used by the authors cited above in that it focuses on how researchers *use* content analytic techniques and on how researchers then justify the *inferences* they draw in their analyses. The categories addressed are as follows:

- Extrapolations
- Standards
- Indices and symptoms
- Linguistic re-presentations
- Conversations
- Institutional processes

Not all of the content analysts whose work is reviewed here have explicated the logic of their inferences as we would hope they would. In some cases, this logic is embedded in the notions of meaning that the analysts have subscribed to. In others, the logic can be found in the more or less explicit assumptions that the researchers have made regarding the contexts of their data. Often, this logic remains hidden because researchers take it for granted, presuming that their own tacit knowledge should be obvious to everyone. I have already noted that Berelson

(1952) did not even feel the need to define content. Analysts need to render their assumptions, the logic they employ, examinable. The appropriateness of particular forms of reasoning is an empirical question, of course, not a logical one, and analysts need to judge the appropriateness of their inferences on a case-by-case basis. I have chosen to review content analyses in these terms because such a review will lead most naturally to an understanding of the notion of analytical constructs—but that is another chapter.

3.2 EXTRAPOLATIONS

Extrapolations are inferences of unobserved instances in the intervals between or beyond the observations (data points). Some well-known kinds of extrapolations are interpolations, predictions, extensions, derivations of theorems from other theorems, and systems. Let's take the notion of a system as a general case. A system is a conceptual device, a "complex variable," one might say. The reality that may be described in a system's terms is not part of the definition of a system, although its construction may well be so motivated. Mathematics supplies systems. Its axioms are mathematical, not empirical. Minimally, a system consists of the following:

- *A set of components* whose states are variable
- *Relations* that are manifest in constraints on the co-occurrence of the components' states
- *Transformations* according to which some relations imply other relations in time or in space

One example of a system is our solar system, in which celestial bodies move in relation to each other. The configurations of planets follow a temporal sequence. For someone who knows the system's transformation rules, data on one configuration imply all succeeding configurations. This is a classical Newtonian system. Kinship terminology also constitutes a system, although it is far from deterministic, not as dynamic as the solar system is conceived to be. It defines kin in terms of certain relations between individuals—in English, according to gender, descent, and marriage—and prescribes rights, obligations, modes of address, and so on among kinfolk toward each other. The system allows "extrapolations" in the sense of extending this terminology to individuals entering the system, whether as spouses, children, or adoptees, and it transforms the roles of these individuals relative to each other throughout their lifetimes within the system. Another example of a system is found in the treatment of language as a system of signs, as in the work of Ferdinand de Saussure. The components of language (words and sounds) are thought to be combinable into larger units (sentences and utterances), following grammatical rules. Knowledge of the system enables the knower to generate novel strings of words that are all considered well formed, such as English sentences. Grammar, it should be noted, is not a "natural" kind of system. It is constructed by academic linguists under the assumption that language is a system in its own right.

Some kinds of systems, especially social systems, can be quite complex. The inferences of interest to sociological content analysts are grounded in knowledge

of a society's transformations, which enables analysts to extrapolate features of the system beyond the time and space of available texts—but always within the domain of the system's description. As in the case of grammar, the “rules” by which a social system works are not natural. They are sociological constructions. Although systems can be elaborate, in comparison with other kinds of inferences, extrapolations are relatively simple.

In content analysis, the idea of studying systems goes back to Tenney (1912), who asked,

Why should not society study its own methods of producing its various varieties of thinking by establishing [a] . . . careful system of bookkeeping? . . . What is needed . . . is the continuous analysis of a large number of journals. . . . The records in themselves would constitute a series of observations of the “social weather,” comparable in accuracy to the statistics of the United States Weather Bureau. (p. 896)

Tenney described systematic relations between subject matter categories within the newspapers he analyzed, noted changes in their distribution over time, and explored, especially, the ethnic characteristics of these publications. He equated the dynamics of press coverage in an entire country with the thinking processes of that country's population, but he lacked methods that were adequate to process the great volume of data that the construction of such a system would require. Tenney made his proposal not only well before computers existed but also before systems theory had been developed.

Rapoport (1969) prepared the ground for a systems theory of “verbal corpuses” as he sought answers to questions such as what it means to describe a large body of verbal data as behaving, changing, and evolving, and what the suitable components, relationships, and laws of interaction within such corpuses might be. Although he was aware that our symbolic world both mirrors and constitutes human existence, and that it can be both enriched in talk and polluted by institutional policies, he suggested that researchers could most fruitfully pursue the large-scale study of verbal corpuses, at least to start out with. This study would be done without reference to speakers, symbol users, and meanings—that is, as a system with its own autonomous regularities. From this perspective, content analysis could be said to probe selectively into what Vernadsky (1945) called a “noosphere,” a sphere of human knowledge distinct from the “biosphere” in which humans live qua organisms.

3.2.1 Trends

The prototype of a systems approach in content analysis is the extrapolation of trends. In one of the earliest content analyses ever conducted, Speed (1893) compared several New York dailies published in 1881 with the same newspapers published 12 years later and observed changes in the frequencies of subject matter categories. Of course, data on only two points in time hardly lend themselves to solid predictions, but Speed's lamenting of the continuing decline of newspaper coverage of literary matters and the increase in gossip, sports, and fiction, raising the question of where this trend would lead, is a clear indication of his desire to predict how newspaper publishing

was changing. Lasswell (1941) proposed a study and presented preliminary findings on trends in the frequencies with which references to various countries occurred in different national presses. Loeventhal (1944) studied the changing definition of heroes in popular magazines and found a drift (still ongoing today) away from working professionals and businessmen as heroes and toward entertainers.

Other trend studies have concerned values in inspirational literature, advertising themes, and political slogans, as well as the frequency of the use of the word *mainstreaming* in several subcategories of educational research (Miller, Fullmer, & Walls, 1996). Researchers have also undertaken numerous analyses of trends in scholarly literature, from sociology (Shanas, 1945) to content analysis (Barcus, 1959), to ascertain the directions in which particular fields seem to be moving. Shanas (1945), for example, analyzed emerging interests in the field of sociology in the United States by examining the distribution of articles in the *American Journal of Sociology* over a 50-year period. Scholars in many academic disciplines have, from time to time, made efforts to review their literature to assess the directions in which their fields were moving and to identify new frontiers. One of the more extensive content analyses conducted to date using a time-series analysis approach is Namenwirth's (1973) analysis of value changes in U.S. political party platforms over a 120-year period. Namenwirth and Weber (1987) also applied time-series analysis to a study of all speeches made by British monarchs between 1689 and 1972 (see also Krippendorff & Bock, 2009, Chapter 3.8). Both studies revealed two independent cycles of value changes in the data, a short-term cycle and a long-term cycle (see Figure 10.5). Such findings amount to descriptions of the dynamics of autonomous systems. Thome and Rahlf (1996) analyzed these same data using a "filtering" methodology instead of time-series analysis, but both methods operate within the notion of a system that enables the analyst to interpolate between data points and extrapolate its ups and downs into the future.

Political party platforms, policy positions, and campaign materials are natural candidates for this kind of content analysis because they are recurrent and there is an interest in knowing what comes next. The European Consortium for Political Research, constituted in 1979, has undertaken numerous trend studies of how different political systems behave over time, in various dimensions, and concerning a variety of emerging issues (see, e.g., Budge, Robertson, & Hearl, 1987). For more than two decades, consortium researchers have coded nearly 2,000 party manifestos using a single coding scheme based on 56 categories and performed numerous trend analyses; they are now experimenting with computerizing this approach (Pennings & Keman, 2002).

3.2.2 Patterns

Another kind of content analysis involves the extrapolative use of patterns. In folklore, for example, researchers have conducted structural analyses of riddles, proverbs, folktales, and narratives with the aim of identifying patterns that have a high degree of commonality within genres, regardless of particular contents (Armstrong, 1959), and can therefore be regarded as generative of those genres. Such analysts begin by identifying the constituent elements within a body of literature and then seek to describe the logic that relates these elements. Thus Sebeok and Orzack (1953), analyzing Cheremis charms, found that in such charms, a

“purely factual statement” about the world is followed by a “motif of an extremely improbable eventuality.” Labov (1972) found a set of components that accounted for the narratives he had elicited and considered these the building blocks for the construction of narratives generally.

Another example of the extrapolation of patterns is the analysis of genealogies within a body of literature through citation networks. Scholarly works tend to cite previous publications on which they relied or which they criticized, which in turn cite earlier publications, and so on. Citation networks enable the analyst to trace present ideas through these networks to their origins, and what is equally interesting, to infer the importance of scholars or particular conceptions by the number of works that cite them or the networks they grow. Citation networks enable content analysts to identify separate discourse communities who generate tightly interconnected literatures without much connection to the literature of other discourse communities. Or they identify canonical texts that everyone in a particular field has read and draws upon. Garfield (1979) relied on this simple idea when he developed his “citation index” as an alternative to information retrieval by keywords. Communication researchers have charted communication channels among members of organizations as senders and receivers and have analyzed those connections in terms of typical network features that organizations tend to reproduce. On a totally different level, word co-occurrences within sentences or paragraphs have also revealed networks of association that individual contributors of a discipline mindlessly reproduce, patterns that can permeate a genre.

Combined interest in trends and patterns has led to many interesting content analyses. Bales’s (1950) “interaction process analysis” yielded patterns of communication, evaluation, control, decision making, tension reduction, and reintegration, all of which were identified within 12 basic categories of verbal exchanges in small groups. Holsti, Brody, and North (1965) studied public statements made by major decision makers in the United States and the Soviet Union during the 1962 Cuban missile crisis and distinguished perceptions and expressions within these successive statements that they described in terms of Osgood’s semantic differential dimensions: evaluative, strength, and potency. With the help of a dynamic interdependency model, Holsti et al. found that these data proved moderately predictive of the pattern of emotional responses each group of decision makers made to the other.

Possibly the most fruitful recent reliance on patterns has emerged in studying networked text in social media and on the Internet. Obviously, but not too often taken seriously, everything said is said to someone else and establishes a relationship. When Chapter 2 suggested that content analysts treat texts neither (or at least not exclusively) in terms of their authors’ intentions, as many literary scholars do, nor as symbolic representations of physical phenomena, as many semioticians do, but in the contexts of their use, this implies a conception of texts that circulate in communities, enliven interconnected realities, and do something that content analysts could possibly infer. Social media can be considered micro blogs in which users create or reproduce texts and disseminate them within networks of friends or communities that form around particular texts—of identity and of coordinating actions, like the Arab spring, which is known to have resulted from the high-speed, emotionally engaging, and politically focused communication on the Internet. (See Section 11.3.4 for the use of software to reveal networked texts enabled by hyperlinked URLs, Twitter use, and what can be inferred from them.)

3.2.3 Differences

Differences are central to all communication studies. Gregory Bateson (2000, p. 459) defined “the elementary unit of information (as) a difference which makes a difference” and demonstrated the explanatory power of this human-centered conception of information by tracing the flow of differences through social systems of actions. The differences of interest here also include differences between communicators; the communication of ethnic, gender, religious, or national differences; the polarization of politics; and their consequences. For example, content analysts have studied how unlike expectations of what an audience is composed of changes how politicians address them. In one of the earliest content analyses, Berelson and Salter (1946) compared the racial composition of the population of fictional characters in magazines with that of the U.S. population. This comparison may be too simplistic. If the mass media would be asked to accurately represent the population of the public, we would not be able to see magicians, learn about breakthroughs by outstanding scientists, and hear exceptional entertainers. However, differences in the news coverage of public issues do have consequences on differences in the public domain. They have been correlated with editorial endorsements (Klein & Maccoby, 1954) and in covering civil rights issues with newspapers’ geographic location, ownership, and political orientation (Broom & Reece, 1955). Differences in newspaper content have been correlated with whether or not newspapers face competition within their regions (Nixon & Jones, 1956).

Gerbner (1964) demonstrated how different ideological and class orientations are reproduced in the stream of French news media messages in the reporting of an apolitical crime. Researchers have also shown how the sources of political messages become aligned with the audience expectations, by comparing, for example, the political speeches that John Foster Dulles made before different kinds of groups (Cohen, 1957; Holsti, 1962). Research has linked differences in television sports reporting of men’s and women’s athletics to prevailing cultural values (Tuggle, 1997) and revealed differences in works of fiction written for upper-, middle-, and lower-class readers (Albrecht, 1956) as well as in advertisements in magazines with predominantly Black and predominantly White readerships (Berkman, 1963). Allport and Faden (1940) examined how the published newspaper content related to the number of sources of information available to newspapers. How stories changed as they were translated from one medium to another was illustrated in Asheim’s (1950) analysis of what happens when books are adapted to become movie scripts. Similar are studies that have compared scientific breakthroughs, discovering how unanticipated differences appear in popular media.

The Hoover Institution’s study titled *Revolution and the Development of International Relations (RADIR)* combined the analysis of differences between media and the analysis of trends. The RADIR researchers identified so-called key symbols such as *democracy*, *equality*, *rights*, and *freedom* in 19,553 editorials that appeared in American, British, French, German, and Russian prestige newspapers during the period 1890–1949. Analyses of these data led Pool (1951) to correlations that he sought to generalize. He observed, for example, that proletarian doctrines replaced liberal traditions, increasing threats of war are correlated with growths in militarism and nationalism, and hostility toward other nations is related to perceived insecurity. Although these symbols refer to aspects of political realities and never

left their connections to the contexts in which they were published and read, Pool and other researchers did not need these references to conduct their analyses. They tried to establish which differences were maintained over time, which differences increased or decreased whether correlated or not, and how they compensated for or amplified each other. For example, Pool (1952b) observed that symbols of democracy become less frequent where representative forms of government are accepted rather than in dispute. It should be noted that the knowledge of whether a particular form of government is generally accepted or in dispute comes from outside the system of selected symbols the RADIR researchers were studying. To the extent external variables can explain a system's behavior, in the form of the contributing conditions illustrated in Figure 2.2, such systems are not entirely autonomous. However, content analysts who study such systems have some analytical freedom to include categories of dissent, defiance, and struggle in their analysis, even if they are not recognized within such systems as long as they explain how a system copes with such differences.

In a very different approach, Gerbner and his colleagues accumulated a very large database on television violence in fictional programming that enabled them to make extrapolations (recommendations) of interest to policy makers (see, e.g., Gerbner, Gross, Morgan, & Signorielli, 1994; Gerbner, Gross, Signorielli, Morgan, & Jackson-Beeck, 1979). Gerbner's (1969) "message systems analysis" proposes to trace the movement of mass-media culture through time by means of a system consisting of four kinds of measures of any category of content (component):

- The frequencies with which a system's components occur, or "what is"
- The order of priorities assigned to those components, or "what is important"
- The affective qualities associated with the components, or "what is right"
- The proximal or logical associations between particular components, or "what is related to what"

One might question Gerbner's equation of frequencies with "what is," how stable these quantitative measures really are, and whether the system is sufficiently autonomous. The point, however, is that any system of measurements, when observed long enough, will allow analysts to make predictions in the system's own terms, whatever they mean.

Simonton (1994) has made an interesting and rather unusual use of the content analysis of systems of differences in his analysis of musical transitions in melodies. He analyzed 15,618 melodic themes in the works of 479 classical composers working in different time periods. Simonton was interested in the relationship between originality and success, and he inferred originality from the unusualness of the transitions in particular works and for particular composers relative to the pool of all melodic themes. For example, he found that Haydn's "Symphony no. 94" employs transitions found in 4% of the theme inventory, whereas Mozart's "Introduction to the Dissonant Quartet" uses transitions that occur in less than 1% of this inventory.

Up until the civil rights movement in the United States gained momentum in the 1950s, its Afro-American population was almost invisible on television. Berelson and Salter (1946) showed that much. In his dissertation, Cedric Clark (1966) observed that Afro-American presence in the public media had increased in numbers but

merely as negative contrast to White leads, as the target of demeaning comments and ridiculing jokes. Public dissemination of such differences, particularly to audiences who do not have direct experiences with minorities, creates expectations of differences that, when enacted, reproduce the very differences displayed in the media. Fictional differences tend to become real in the form of self-fulfilling prophesies. This process has been exploited by totalitarian regimes to create internal or external enemies they need to fight in order to maintain the fiction of their own superiority. It can be found also in contexts where ethnic, religious, national, and gender prejudices are created and amplified.

A well-received study of intelligence in American society provides a good example of ignoring the effects of publishing differences. Herrnstein and Murray (1994) applied the standard IQ test to a cross section of the U.S. population, partitioned by race, socioeconomic status, job performance, criminal records, family stability, and more. They found that its African-American population was less intelligent, less achieving economic success, more prone to commit crimes, and so on. Moreover, they observed little change in IQ from one generation to the next, which led them to conclude that intelligence is genetically fixed. The two authors did not reflect on where they learned to make the distinctions they were applying to their data. They made no references to how White-dominated mass media in the United States presented members of the Black community. They did not acknowledge that IQ tests embodied the values of the educated White population and ignored proposals of other conceptions of intelligence, but most importantly, that their findings could have been explained by how this minority is treated by a majority enacting their expectations in hiring practices, admissions to higher education, and suspecting Blacks to be prone to commit criminal offenses. Widely communicating such differences and enacting them amounts to a circular causality that makes it hard for the population of African-Americans to escape that self-validating confinement. Herrnstein and Murray's *Bell Curve*, claiming genetic explanations of population differences when sociolinguistic ones are at their core, merely feeds these prejudices (Naureckas, 1995). Critical content analysts ought to be aware of the contexts in which their analyzed texts reside, and this awareness ought to include the effects of their own publications.

Although studying the textual differences in the context of their causes and consequences is a fruitful application of content analysis, it is not always easy to find appropriate texts to support such inferences. For example, Silicon Valley and mathematics-intense fields like engineering and computer science have been accused of thriving in a sexist culture, without much proof. Representatives of these disciplines blame lack of mathematical competence or disinterest among women for this situation. In economics, too, gender inequality is well documented (Wolfers, 2017) but not what sustains it. What economists actually say when hiring or promoting women candidates to faculty positions takes place largely behind closed doors. Alice H. Wu (2017) took advantage of the intersection of two technologies, online messaging and computer-aided content analysis, and scraped over 1,143,416 posts across 131,913 threads between 2014 and 2016 from Economics Job Market Rumors (EJMR), an online forum where economists in the job market post news, rumors, and advice about open academic positions. Posts are anonymous, which removes the constraints of political correctness, but makes it difficult to identify the gender of their

contributors. Wu therefore focused on what posts were about. She used four levels of gender-identifiers, starting with personal pronouns “he” and “she” and ending with “guy” and “bro” among seven others. About 31% of the posts were female related, and 61% concerned males. She used a machine learning algorithm (see Section 11.4.2) to identify the 30 most prominent words that differentiated posts discussing females and males. A comparison of these two lists is shocking. Whereas the words that are unique to female posts refer to body parts and unspeakable sexist descriptions, the words that are unique to male posts refer to economic concepts, professional designations, and publications. She also grouped the 10,000 most frequent topic words into 15 categories, among them academic/professional (economics, academic, and professional) and personal/physical (marital status, attractiveness, and gender related attributes), and found additional evidence for the above gender inequality. Posts about females contained 43% fewer academic or professional terms than expected and 192% more personal terms and physical attributes. Among the most prominent authors in economics, men outnumber women by 12 to 1. Several women economists who read her work reported experiences that resonated with her findings (Wolfers, 2017).

Textual differences matter. Their analyses have to address the richly interactive textual dynamics and the social realities they bring about (Krippendorff, 2005).

3.3 STANDARDS

Humans measure observed phenomena against standards to establish (a) the kinds of phenomena they are (identifications), (b) how good or bad the phenomena are (evaluations), and (c) how close the phenomena come to expectations (judgments). I discuss each of these three uses of standards below. The facts that identities do not reveal themselves (they require someone to identify them as such), that evaluations are not objective or natural (they are the products of someone’s values), and that audits by themselves are inconsequential (unless someone can invoke institutional consequences) reveal that standards facilitate inferences of a certain kind. In content analysis, standards are often implicit. People are quick to be for or against something without any clear idea of why. As suggested in Chapter 2, content analysts should take care to make explicit why they infer what they do, and this includes defining the standards they apply in their studies.

3.3.1 Identifications

Identification concerns *what something is*, what it is to be called, or to what class it belongs. Identifications are “either/or” inferences—that is, something either is or is not of a certain kind. Most basically, all computer text analyses start with the identification of character strings, not meanings. Any two strings are either the same as or different from each other. In his above-cited typology of content analyses, Janis (1943/1965) calls one type “sign-vehicle analysis.” In this type of analysis, researchers use procedures that classify content according to the psychophysical properties of the signs (e.g., by identifying the word *Germany* and then counting how often it appears). Dibble (1963), who analyzes the kinds of inferences that historians habitually make in their work, includes “documents as direct indicators” as one kind of inference. For

example, suppose a historian wants to know whether the British ambassador to Berlin communicated with England's foreign ministry the day before World War I began; a letter from the ambassador in the file of that ministry would provide direct evidence of its having been sent and received. Because identifications are often obvious, it is easy to overlook their inferential nature. In content analysis, the simplest task requires that a decision be made concerning whether something has occurred, was said, or has been printed. For example, when officials of the Federal Communications Commission are alerted that certain four-letter words have been broadcast over the public airwaves, they need definite proof that those words have been aired before they can consider suspending the offending station's broadcasting license. Identifications are rarely so simple, however.

The legal system's use of content analysis as an evidentiary technique provides us with many examples of identifications (see "Content Analysis," 1948; Lashner, 1990). Tests aimed at establishing whether a particular publication is defamatory, whether a given political advertisement is based on facts, whether a certain signature is real, and whether a given painting is the work of a particular artist all involve either/or-type inferences about identities or class memberships, but not all of them are simple and obvious. For example, to identify a statement as defamatory in the context of a legal proceeding, an analyst must show that all components of the applicable legal definition of defamation are satisfied.

3.3.2 Evaluations

Well before the term *content analysis* appeared, at a time when media research was equated with the journalism-inspired premises of quantitative newspaper analysis, the evaluation of press performance was an important issue, as it still is. Early concerns about changes in newspaper publishing (Speed, 1893), which surfaced in public criticisms of increases in the coverage of "trivial, demoralizing, and unwholesome" subject matter at the expense of "worthwhile" information (Mathews, 1910), were certainly motivated by largely unquestioned ideals, evaluative standards, and norms couched in seemingly objective frequency measures. Some cultural critics today may share the concerns expressed by the authors of these early studies, but journalism has changed in the intervening years and has shown itself to be responsive to the evolving cultural climate and to shifting political and economic conditions.

Evaluative studies of newspaper reporting have focused largely on two kinds of bias: the bias in accuracy (truth) of reporting and the bias in favoring one side of a controversy over the other. For example, Ash (1948) attempted to determine whether the U.S. public was given a fair opportunity to learn about both sides of the controversy that accompanied the passage of the Taft-Hartley Labor Act. Accuracy in reporting and favoritism in reporting can be difficult to separate, however. During election campaigns, for instance, most politicians allege that some segments of the media display bias in their election coverage. The more popular candidates, who enjoy frequent attention from the press, tend to complain about inaccuracies in reporting, whereas the less popular candidates, struggling for publicity, are more likely to complain about inattention. Because journalists are committed to being fair to all sides in their reporting, many are defensive when the press is accused of taking sides and take the measurement of bias quite seriously.

In practice, evaluative studies of journalistic practices have not solved the now century-old problem of the lack of unquestionable criteria. Janis and Fadner

(1943/1965) sought to put this deficiency to rest with their publication of a coefficient of imbalance, in which

f = the number of favorable units,

u = the number of unfavorable units,

r = the number of relevant units = $f + u$ + the number of neutral units, and

t = the total number of units = r + the number of irrelevant units.

The “coefficient of imbalance C ,” which Janis and Fadner derived from 10 propositions intended to capture the prevailing intuitions regarding (im)balance in reporting, measures the degree to which favorable statements, f , outnumber unfavorable statements, u , relative to the two ways of assessing the volume of a text, r and t :

$$C = \begin{cases} \frac{f^2 - fu}{rt} & \text{when } f \geq u \\ \frac{fu - u^2}{rt} & \text{when } f < u \end{cases}$$

This coefficient ranges in value from -1 to $+1$. It is a good example of an evaluative standard that enables the kind of inferences we often make without much thinking: It defines an ideal (here a balance between positive and negative evaluations), and it measures deviations from that ideal in degrees (here in either the positive or the negative direction). The reality of evaluative standards is far from clear, however. For example, whether journalists can always be impartial is an unsettled issue; some would argue that there are circumstances under which they may not have to be or under which impartiality may not be possible. In the last days of Nixon’s presidency, for example, it was difficult for journalists not to take the side of the public. And so it is in situations of war, where loyalty tends to outweigh fairness to both sides. To give one’s nation’s enemies a fair hearing might be an intellectual challenge, but in practice, it is utterly unpopular. In the early 1960s, Merrill (1962) tried to differentiate dimensions of evaluative standards for journalistic practices. He proposed a battery of evaluative criteria to be applied to journalistic presentations (attribution bias, adjective bias, adverbial bias, contextual bias, photographic bias, and outright opinion), but his catalog is far from complete.

To assess accuracy in reporting, one must have standards against which to judge representations. Insofar as the reality we know is always already described, accuracy amounts to correspondence with sources that are deemed authentic. In a landmark study, Berelson and Salter (1946) compared the racial composition of the population of fictional characters in magazines with that of the U.S. population. The statistical operationalization of “representativeness” that they used has also been employed in many subsequent evaluative studies (Berkman, 1963). But whether the population of fictional characters in magazines, in plays, or in television programs should be statistically representative of the audience in characteristics such as ethnicity, age, occupation, and artistic capability remains debatable. The community television projects of the 1970s died precisely because audience members did not find it particularly entertaining to look into the lives of their ordinary neighbors. The

“reality” TV shows of today may give the impression of being representative of real life, but they actually amount to contrived games played by carefully selected people. A bit less controversial are comparisons of the contents of narratives with those of other narratives. The Council on Interracial Books for Children (1977) has proposed and demonstrated a method for evaluating history texts in the United States by comparing the information in them with known historical facts. Here too, however, matters are not as simple as they seem. In the presentation of history, some selectivity is unavoidable; such evaluative efforts should aim to discover systematic exclusions and overstatements, not variations around an ultimately arbitrary standard. In journalism, the standard of truthful reporting is almost universally subscribed to, but it often conflicts with journalists’ responsibility for the consequences of their reporting—for example, preventing fair trials, stimulating public fears, hyping people into action, and creating scandals.

3.3.3 Judgments

Like identifications and evaluations, judgments are based on standards but with the additional provision that they are *prescribed or legitimated by institutions*, and research using such standards tends to have institutional implications. For example, when the United States Federal Communications Commission (FCC) grants licenses to television stations, the stations are obligated to maintain certain proportions of news, community, and public service programming; that is, the FCC sets explicit criteria with which broadcasters must comply. Content analysts have measured the proportions of different kinds of programming aired on some stations and, in effect, have influenced FCC decisions regarding the status of the stations’ broadcasting licenses.

Social scientists have long been fascinated with social deviance, and many have theorized about crime, pornography, obscenity, and the like. In doing so, they have influenced the community of their peers and undoubtedly affected public opinion. However, for content analyses to have institutional implications, their results must be presented in the target institutions’ terms; otherwise, they do not have any effect. Content analysts may study such social problems as plagiarism, discriminatory communication practices, and the effects of fictional programming on particular kinds of crimes, but their findings are not likely to support judgments with consequences unless the researchers use the concepts, categories, and language of laws, enforceable agreements, or other institutional standards that are applicable to the institutions concerned with these problems. For example, organizational communication researchers are often asked to perform so-called communication audits of industrial or business organizations, in which they ask what is being said, how, and to whom, and what function it serves. Such an audit is usually driven not by scientific curiosity or public concerns but by expectations from within the organization that the results will be useful, solve problems, or inform effective actions. Early communication audits often failed because they were conducted by academics who measured their findings against communication theories that had little to do with how organizations have to function. If the results of organizational communication research are to lead to consequences, they must be couched in the studied organization’s terms and be measured against the standard of communication structures known to be successful.

3.4 INDICES AND SYMPTOMS

An index is a variable whose significance rests on its correlation with other phenomena. According to the semiotician C. S. Peirce, an index must be causally connected to the event it signifies, as smoke indicates fire. This presumes an underlying mechanism such that the relation between an index and what it signifies is a matter of necessity rather than convention (symbol) or similarity (icon). Indices are so conceived in medicine, where they are called *symptoms*. To diagnose, a physician looks for visible or measurable manifestations of an illness. However, even in medicine, symptoms have their histories, and medical practitioners must be educated to recognize them for what they are, which makes symptoms a property of the institution of medicine as much as of the phenomena the symptoms are supposed to indicate. In the social domain, where physical mechanisms (causalities) tend to be absent, the observer-dependent nature of indices is even more prominent. As Rapoport (1969) has noted, “An index . . . does not depend on (or should not be confused with) the physical entities or events from which it is derived” (p. 21).

In content analysis, indices of unobservable or only indirectly accessible phenomena are most common. Typically, analysts use measures of textual (verbal and paralinguistic), visual (gestural and pictorial), and communicational characteristics to address extratextual phenomena. For example, the ratio of disturbed speech to normal speech (speech-disturbance ratio) may serve as an index of a patient’s anxiety during psychiatric interviews (Mahl, 1959); the frequency of a category of assertions or images related to action, goals, and progress is understood to indicate their producer’s achievement motive (McClelland, 1958); and the frequencies of expressed concerns for an issue and the typographical positions of its expressions in a medium (e.g., in newspapers: size of headlines, front or inside pages, lead paragraphs of stories or mere mentions) are seen as indices of the amount of public attention to that issue (e.g., Budd, 1964). Gerbner et al. (1979) created a television violence index based on the numbers of violent scenes in fictional TV programs. Krendel (1970) developed an index of citizen dissatisfaction based on letters of complaint to city halls. Flesch’s (1948, 1951, 1974) “readability yardstick” is derived through a formula that, after several incarnations, responds to two factors: average sentence length (in number of words) and average number of syllables per word. Danielson, Lasorsa, and Im (1992) used Flesch’s criteria in their comparison of the readability of newspapers and novels. Government contractors are required to apply a version of Flesch’s yardstick before finalizing instructions to military personnel, and insurance companies use it to evaluate contracts. Hawk (1997) extended Flesch’s criteria for readability to evaluate the “listenability” of television news. Jamieson (1998) has constructed a campaign conduct index that takes into account Americans’ expressed concerns about how much money politicians spend on campaigns, what candidates say to get elected, candidates’ ethics and morals, and the proportion of negative ads used in political campaigns. Broder’s (1940) adjective-verb ratio has been employed as an index of schizophrenia (Mann, 1944), and above-chance co-occurrences of nouns have been interpreted as indicators of associations in speakers’ and receivers’ minds (Osgood, 1959).

In mass communication research, five indices have had a long history of use:

- *The presence or absence* of a reference or concept is taken to indicate the source’s *awareness* or *knowledge* of the object referred to or conceptualized.

- The *frequency* with which a symbol, idea, reference, or topic occurs in a stream of messages is taken to indicate the *importance of, attention to, or emphasis on* that symbol, idea, reference, or topic in the messages.
- The *numbers of favorable and unfavorable characteristics* attributed to a symbol, idea, or reference are taken to indicate the *attitudes* held by the writers, the readers, or their common culture toward the object named or indicated.
- The kinds of *qualifications*—adjectives or hedges—used in statements about a symbol, idea, or reference are taken to indicate the *intensity, strength, or uncertainty* associated with the *beliefs, convictions, and motivations* that the symbol, idea, or reference signifies.
- The *frequency of co-occurrence* of two concepts (excluding those that have grammatical or collocational explanations) is taken to indicate the *strength of associations* between those concepts in the minds of the members of a population of authors, readers, or audiences.

The use of such easily computable quantities as indices is not without its problems. Chomsky (1959) took Skinner to task for suggesting that promptness of response, repetition, and voice volume are natural indices of the intensity of motivation and that meanings can be discerned from the co-occurrence of words with the objects they refer to. He observed that most words are uttered in the absence of what they mean. Rapoport (1969) compares two hypothetical women, each of whom has just received a luxurious bouquet of flowers. The first woman, upon seeing the flowers, shouts, “Beautiful! Beautiful! Beautiful! Beautiful!” at the top of her lungs, thus giving evidence, according to Skinner’s criteria, of a strong motivation to produce the response. The second woman says nothing for 10 seconds after she first sees the flowers, then whispers, barely audibly, “Beautiful.” Frequency and voice volume would not be good indications of the importance of these flowers or, in Skinner’s terms, the motivation to respond.

In content analysis, as in many social scientific inquiries, researchers often simply *declare* indices without demonstrating their empirical validity, especially when the phenomena to be indicated are abstract and far removed from validating data. Obviously, a researcher would not declare a measure to be an index if his or her claim is unlikely to be convincing (i.e., to have face validity) to scientific peers. Simple declarations, however, do not constitute an index as defined above. A declaration is discursive in nature and should not be confused with a correlation between an index and what it claims to indicate. A correlation needs to be demonstrated or at least hypothesized, so that it is testable in principle. Take, for example, a researcher’s declaration that the frequency of violence in TV fictional programming is a measure of attention to violence (in real life). To make this claim, the researcher must first clarify whose attention this frequency is supposed to indicate. The authors’ or editors’? The audience members actually exposed to the violence so measured or the audiences that producers had in mind attracting, the public at large, or the culture in which these kinds of mass communications are circulating? Given the target of the intended inferences, the researcher must also describe how the attention to be indicated will manifest itself—directly (by observation of TV-related violence) or

indirectly (by correlation with other observable phenomena, such as [in]tolerance for otherness, domestic/disciplinary violence, or crime rate). Counting, emphasizing, paying attention to, and expressing concerns about something are four wholly different things. Their correlation is an empirical question.

Quantification is not an end in itself. Researchers must distinguish between quantifications that lead to the testing of a statistical hypothesis and quantifications that indicate something other than what is counted. These two uses are often confused in the early content analysis literature. For example, in his famous essay “Why Be Quantitative?” Lasswell (1949/1965b) celebrates quantification as the only path to scientific knowledge, by which he means the testing of statistical hypotheses; however, in most of his content analyses, Lasswell used frequency measures as declared indices of extracommunicational phenomena.

In a study of the indicative power of frequencies of mentions, a student of mine used a book on U.S. presidents that was written by a scholar who was available on our university’s campus. The student examined the book thoroughly, counting the numbers of mentions of the different presidents; the numbers of chapters, pages, and paragraphs in which each president is mentioned; and the numbers of sentences devoted to each president. He then asked the author to rank the U.S. presidents according to their importance and according to their contributions to U.S. history. He also asked the author how other scholars might rank the presidents and how the public might rank them. Finally, the student even asked the author how much attention he thought he had paid to each of the presidents in his book. Surprisingly, all correlations were very low, to the point that probably none of the measures could serve as a valid index of the author’s attention or emphasis. The tentative insight we may derive from this exploratory study is that frequencies may not be good indicators of conceptual variables, such as importance or favoring one side over the other in a complex political controversy. Frequency measures are more likely to succeed as indicators of frequency-related phenomena—for example, the number of mentions of crime and the number of people believing crime to be an issue (not to be confused with actual crime statistics, which can be very detailed and may not correlate with public concerns), the number of favorable references to a political candidate and the number of votes that the candidate is likely to attract (not to be confused with how much the candidate has done for his or her constituency), or the proportion of unfavorable letters written to city hall (Krendel, 1970) and the likelihood that the mayor will not be reelected.

The use of Dollard and Mowrer’s (1947) discomfort-relief quotient demonstrates some of the difficulties involved in establishing an index. Dollard and Mowrer applied learning theory in deriving this very simple quotient as an index of the anxiety of speakers. The quotient is computed as the proportion of the number of “discomfort” or “drive” words and the sum of this number and the number of “comfort” or “relief” words. Despite Dollard and Mowrer’s sound theoretical arguments and careful definitions of the two kinds of words, tests of the indicative power of this quotient have led to mixed results. Significant correlations with palm sweating have been reported, but correlations with other measures of anxiety seem to be demonstrable only in very restricted circumstances. Murray, Auld, and White (1954) compared the discomfort-relief quotient with several other motivational and conflict measures applied during therapy and found that the quotient was not sensitive to changes in therapeutic progress. What the quotient indicates is therefore far from clear and simple.

The empirical evidence for indices of readability is more convincing. Clearly, sentences that include foreign expressions, long and compound words, complex grammatical constructions, and many punctuation marks are more difficult to read than simpler sentences. The success of Flesch's readability formula may well lie in two of its features: (a) Overall judgments concerning the readability of a piece of writing are formed cumulatively, with each encountered difficulty reducing the readability score, and (b) the indices are validated by the judgments of a population of readers. Both of these features are frequency related (Krippendorff & Bock, 2009, Chapter 3.9). Many word processing programs now are capable of providing not only counts of the numbers of characters, words, paragraphs, and pages in a document but also a readability score. Such scores might lend themselves to interesting correlational studies.

Researchers have also used indices successfully to settle disputes about authorship. In the 1940s, Yule (1944), an insurance statistician, reconsidered whether Thomas à Kempis, Jean Gerson, or one of several others wrote *The Imitation of Christ*. He correlated frequencies of nouns in works known to have been written by each prospective author and thereby developed discriminating indices to their identities, which he then applied to the disputed work (the inference was in favor of à Kempis). Mosteller and Wallace (1964), arguing that the choices of nouns are more specific to content than to author identity, found function words to be far more distinctive in their effort to settle the disputed authorship of 12 of the *Federalist Papers*. Evidence from their analysis favored Madison as the author, a finding that historians increasingly believe to be correct.

Again, declarative definitions are not sufficient. Declaring frequencies to be a measure of attention does not make them a statistical index that correlates with attention as measured by any other means. Even where correlations are found between an index and what it is said to indicate, there remains the problem of generalizability. For example, Morton and Levinson (1966) analyzed Greek texts by known authors and extracted seven discriminators of style that, according to the researchers, tap the unique elements of any person's writing: sentence length, frequency of the definite article, frequency of third-person pronouns, the aggregate of all forms of the verb *to be*, and the frequencies of the words *and*, *but*, and *in*. Morton's (1963) analysis of the 14 Epistles attributed to Paul in the Bible led him to conclude that six different authors wrote these works and that Paul himself wrote only four of them. Ellison (1965) then applied the constructs that Morton used to texts by known authors, which led to the inference that James Joyce's novel *Ulysses* was written by five different authors, none of whom wrote *A Portrait of the Artist as a Young Man*. Ellison found in addition that Morton's own article was written in several distinct styles. This research casts serious doubt on the generalizability of Morton's stylistic indices of an author's identity.

The inability to demonstrate high correlations should not prevent analysts from using quantitative measures, however. Researchers may be able to strengthen the indicative capabilities of such measures by adding independent variables, or they may observe these measures for long periods of time and then construct regularities that can be extrapolated into yet-unobserved domains. In addition, researchers may vindicate their construction of such measures by successfully correlating them with other phenomena not initially anticipated (correlative validity). In any case, it is always advisable to use indices cautiously.

In a self-reflective moment, Berelson (1952) wondered what Martians might infer from the high frequencies of love and sex found in modern Earth's mass-media

recordings: Would they infer a promiscuous society or a repressive one? As noted above, Pool (1952b) has observed that symbols of democracy occur less frequently where democratic processes govern than where they are in question; thus they represent something other than the degree to which democracy is accepted. Although most learning theories suggest that repetition strengthens beliefs, repetition is also known to lead to semantic satiation—not only a loss of interest but also a loss of meaning. Thus it is not a simple matter to determine what it is that frequency measures indicate, and it is certainly not an issue that can be settled by proclamation.

3.5 LINGUISTIC RE-PRESENTATIONS

In language, the analogue of indicating is *naming*. Both establish one-to-one relationships—in the case of indices, relationships between two kinds of variables, and in the case of naming, relationships between words and particular persons, things, concepts, or experiences. A name recalls the named. Although narratives use names, naming is not sufficient to allow us to understand what narratives do. Narratives conjure, bring forth, and make present (re-present as they are reread, hence *re-presentation*, with a hyphen) rich worlds consisting of people in relationships with each other: objects that do things: and ideas, morals, and perspectives that guide observations. Narratives are imaginable and, under favorable circumstances, realizable through actions. Thus, texts do not merely map, speak about, or indicate features of an existing world, they can *construct worlds* for competent speakers of a language to see, enact, and live within. To analyse texts as *re-presentations*—not to be confused with picture-like representations—is to analyse the conceptual structure that a text invokes in particular readers, the worlds they can imagine, make into their own, and consider real.

Written text is not just a collection of words; rather, it is sequenced discourse, a network of narratives that can be read variously. Hays (1969) provides the following examples of some typical streams of text that social or political scientists may be interested in understanding:

- *A sequence of editorials:* The staff of a newspaper, experiencing an epoch, produces a series of essays that recapitulate some of the day's events, placing them in context with respect to historical trends, theory, and dogma. The essays express opinions about the true nature of situations that are necessarily not fully comprehended and about the responses called for.
- *International exchanges of an official character:* This kind of correspondence is comparable to a sequence of newspaper editorials as described above, except that there are two or more parties involved, each pursuing its own policy.
- *Personal documents:* These may be letters, diaries, or written materials of other kinds. Such materials differ from newspaper editorials or official governmental exchanges in the particularity of their content.
- *Interview transcripts:* Usually in an interview situation there are two parties, one naive and the other sophisticated. The purpose of the interview may be, for example, therapeutic or diagnostic.
- *Social interaction:* Two or more persons participate, discussing a fixed task or whatever other topic they deem suitable.

Such streams of texts, which could be extended to include types of literature, folktales, reports of scientific findings, and corporate reports, have several characteristics in common. For instance, they are all sequential in nature. Narratives respond to each other and are no longer individual accomplishments. The structures of interest are not manifest in vocabularies of words or in sentential constructions but in larger textual units, in intertextualities. An analysis of texts as re-presentations has to acknowledge the connectedness of these larger textual units. The container metaphor that informed early conceptions of content analysis continues to influence many content analysts, making them most comfortable with classifications of content and indices that tend to ignore linguistic or narrative structures. Because such textual data tend to stem from several narrators, not one, analysts cannot presume consistency from narrator to narrator. Nevertheless, inconsistencies make sense as motivators of interactions and as causes of evolution. Re-presentations essentially provide *conceivable worlds*, spaces in which people can conceptualize reality, themselves, and others. An analysis of these re-presentations proceeds with reference to designated readers, the imaginability of actors and actions, and how each datum contributes to the unfolding of the data stream.

A simple yet generic example of such content analysis is the development of maps. Maps are not just descriptive. The user of a map needs to understand that map in order to participate in the alleged reality that the map depicts. A road map aids a driver in seeing the possibilities for realizing self-chosen goals. Without a map, the probability of the driver's reaching his or her destination would be no better than chance. But maps not only enable, they also constrain thought and enforce coordination of their users relative to each other. Inferences drawn from maps should concern what their users do or could do with them. Lynch (1965), an architect, placed verbal statements of what informants recalled seeing when moving within a city onto a composite map of that city as seen by its residents. He wanted to infer what city planners should do to provide citizens with needed orientations but found also how and where people would go when they had particular goals in mind. In his book, *Letters From Jenny*, Allport (1965) reported the results of an analysis of personal correspondence, showing what the world of its author looked like and what kind of psychological insights one could derive from her reality constructions. Gerbner and Marvanyi (1977) developed maps of the world based on their analysis of news coverage in U.S., East European, West European, Soviet, and some Third World newspapers; they distorted the sizes of the regions in the maps to correlate with the volume of news devoted to the regions. So (1995) developed maps of the field of communication research based on the titles of papers presented at several of the International Communication Association's annual conferences and on the sources cited in the papers in order to infer the "health" of the discipline. Although all the studies I have mentioned here as examples lacked good ways of tapping into complex linguistic structures, the researchers who conduct such studies tend to compensate for this shortcoming by providing rich interpretations of their findings.

Qualitative content analysts clearly recognize the need to respond to texts as connected discourse. Such researchers have examined the social construction of emotions in everyday speech (Averill, 1985), the metaphorical notion of facts in scientific discourse (Salmond, 1982), the prejudicial path toward an institutionally acceptable understanding of the causes of AIDS in medical writing (Treichler, 1988), the role of psychotherapists as depicted in fictional literature featuring psychotherapists

(Szykiersky & Raviv, 1995), the portrayal of African Americans in children's picture books in the United States (Pescosolido, Grauerholz, & Milkie, 1996), the construction of natural disasters in U.S. print media (Ploughman, 1995), and the depiction of women in the media, to name a few topics. To be clear, many of these qualitative studies have lacked formalization, and so the findings are difficult to replicate or validate. Many of these studies have also had avowedly descriptive aims; in some cases, the researchers have stated their intent to reveal biases in representations. For example, Gerbner and Marvany (1977) created the maps mentioned above with the intention of appealing to a fairness standard of equal attention. The use of content analysis to describe how particular media depict members of certain professions, people from certain nations, or certain social problems or political figures usually amounts to the development of maps in which the concepts of interest occupy certain places.

Analysts of re-presentations seek to rearticulate relevant portions of texts to make the readers of their analyses aware of alternative readings or readings by particular others. For example, critical discourse analysts offer accounts of the roles of language, language use, and (in)coherences and of the communicative uses of texts in the (re)production of dominance and inequalities in society (see Van Dijk, 1993). Critical discourse analysis also includes an element of self-reflexivity in that it may be applied to its own text—asking what critical analysis is, what its practitioners do to a text, and so on. Such analyses have been characterized as explorations of social cognition and the public mind. However, in the absence of the reality that re-presentations bring forth, the only criteria applicable to the analyses of re-presentations are whether they *answer* informed readers' *questions*, whether they can *withstand critical examination* from the perspective of individuals who are familiar with the context of the data, and whether *the worlds they rearticulate resemble or add to the worlds of specified readers* of the analyzed texts or of other content analysts.

Examples of analyses of re-presentations that start from the other end of this spectrum of complexity are found in simulations of cognitive processes (Abelson, 1968) and in applications of such simulations to aid political campaigns (Pool, Abelson, & Popkin, 1964). In such research, analysts use a large number of grammatically simple propositions, goals, and scripts that people know how to follow—for example, how to order a meal from a menu, how to drive a car, or how a kinship system works (Wallace, 1961)—and compute entailments from the way they hang together semantically. Without the use of computers, but certainly with that in mind, Allen (1963) proposed a logical content analysis of legal documents that demonstrated, by means of a formal procedure, which options (loopholes) existed for the signatories of an arms limitation agreement. This led Allen to infer the directions in which the parties to this agreement could, and probably would, move, given appropriate incentives, and the conflicts that could be expected to emerge. Emphasizing constraints rather than options, Newell and Simon (1956) proposed a “logic theory machine” that shows how a sequence of logical implications (a proof) from available evidence (premises, axioms) may lead to decisions within an unknown problem area (the validity of a theorem). Danowski (1993) used the data obtained from a semantic network analysis to arrive at recommendations concerning how persuasive messages ought to be constructed. Semantic network analysis is the content analysts' version of expert systems that artificial intelligence researchers aim to build in various empirical domains.

Hays (1969) developed a vision for this kind of content analysis, calling it *conversationalist*. It would accept a stream of linguistic data—dialogue, diplomatic exchanges, treaty negotiations, and the like. It would recognize that an understanding of any linguistic form presumes a great deal of background knowledge, including knowledge about beliefs and assumptions, and it would allow for such knowledge to be added to the linguistic data. If several interlocutors populate the context of an analysis, which is typical, the analysis must acknowledge differences in their background knowledge as well. The analysis would also recognize that meanings change over time and would place every assertion in the context of previous assertions. A content analysis of re-presentations, Hays’s conversationalist, would *answer questions of interest to the analyst that are not literally found in the text*. The conversationalist is an engine that computes a text’s implications that answer the questions given to it.

In the terms employed in our framework, as described in Chapter 2, the context of such content analyses is the reality that available texts make present to a specified community of readers. The stable relations are manifest in the reasons that the community of readers would accept for answering specific questions from specific texts, for pursuing the logical implications of these data to a chosen target. Although many content analyses of re-presentations are not so clear about their aims and rarely care to go as far, this idea is being realized, at least in part, in fifth-generation computers, so-called expert systems. The discussion of expert systems has been overshadowed by interest in search engines for the Internet, computer networking, and collaborative systems, to name just a few, but the fact that we now have large volumes of textual data available in computer-readable form makes the content analysis of re-presentations increasingly possible and a challenge.

Sherlock Holmes’s detective work, already mentioned in section 2.4.5, provides a literary example of the use of linguistic re-presentations. Content analysts merely deal with larger volumes of textual matter, pursue different kinds of questions, and, for the process to be examinable or replicable, need to proceed transparently.

In this context, I might mention various visualizations of less directly language-related textual relationships such as webgraphs, exemplified by Figures 11.5 and 11.6. Webgraphs depict the links between one website and another, between a source URL and a destination URL. They exemplify a hypertextual conception of text that content analysts have to cope with. Ordinary text is expected to be read as written. Hypertext gives the reader far more freedom to navigate a textual multiverse, and gives content analysts considerable headaches for how to account for the many ways a hypertextual multiverse can be and is, in fact, visited. While the use of URLs in websites is a technological modification of the traditionally linear conception of text, the networking of what is said is ancient, of course. It may not always be recognized. However, I would say that everything said is addressed to someone; and when an addressee listens and responds to what is heard, what is said, the text, has established a relationship. This is what the literary theorist Mikhail Bakhtin (Todorov, 1988) taught in opposition to the linguistic conception of language—which blatantly omits relational, social, and dynamic conceptions of texts, moving from reader to reader, being transformed in that process, but establishing, maintaining, and replacing networks of text-based relationships. This is a good introduction to the next section.

3.6 CONVERSATIONS

When children in well-to-do families say they are hungry, they may well want to have something to eat, but they may just as well be trying to avoid going to bed, gain attention, prevent their parents from doing something, or so on. In the context of a lived history of interacting with their children (knowing when they last ate, for example), parents tend to know how to respond when their children claim to be hungry. In such a situation, the propositional content of an utterance is secondary to the role that utterance plays in an ongoing interaction. In an attempt to infer anxiety from speech, Mahl (1959) addressed the difficulties of analyzing this kind of instrumental use of language, but he ended up bypassing the problem in favor of developing nonverbal indicators of anxiety instead. The path he took demonstrates the limitation of content analyses that are guided by a representational concept of content. Already in the 1950s, Bateson (2000; Ruesch & Bateson, 1951) had suggested that all messages convey content *and* relational information (a concept addressed by many researchers since, from Watzlawick, Beavin, & Jackson, 1967, to Baxter & Montgomery, 1996). When we view utterances as only representations, we ignore their relational or conversational functions. The essential feature of *conversational interactions* is that they *take place in and create interpersonal relations and define their own conditions for continuing the process*. When we blame someone for lying, we invoke the standard of representational truths, which is only one of many possible conversational frames interlocutors can adopt and one that makes continuing a conversation less important than being right. In content analyses of conversations, *inferences concern the continuation of the process*. Indexical and re-presentational aspects (content in the sense of what is conveyed in processes of communication) are at best a means to that end.

Conversation analysis has emerged as one approach to the study of talk in natural settings (Atkinson & Heritage, 1984; Goodwin, 1981; Hopper, Koch, & Mandelbaum, 1986; Jefferson, 1978; Sacks, 1974; ten Have, 1999). Unlike discourse analysts, who start with written texts (Van Dijk, 1977, 1993), regard a discourse as a string of sentences, and aim to account for what the discourse (re)presents, as well as how and why it (re)presents what it does, conversation analysts tend to start with voice or video recordings of naturally occurring speech. They then proceed by transcribing conversational interactions, using highly specialized transcription conventions that enable them to capture not only words and who uttered them but also intonations, overlaps, and incompletions, as well as nonverbal behaviors such as gaze and especially silences and turns at talk. Broadly speaking, conversation analysts aim to understand the structure of naturally occurring speech, which necessarily includes two or more of its participants. Their methods of study are intended to preserve as much of the richness of human communication as possible. One typical analytical strategy is to differentiate among speech acts, or utterances that do something, such as questions, requests, promises, declarations, and expressions of feelings that are constitutive of relationships between the conversants.

Although conversation analysts are beginning to address reliability issues in their studies (e.g., Carletta et al., 1997; Patterson, Neupauer, Burant, Koehn, & Reed, 1996), efforts to establish the validity of conversation analyses have been marred by a lack of consensus concerning what constitutes supporting evidence. Most published

reports of conversation analysis research can be characterized as “show and tell.” In these reports, researchers reproduce exemplary fractions of transcribed dialogue to demonstrate their explanations of “what is ‘really’ going on.” It is generally futile to ask conversants to confirm conversation analysts’ claims, as ordinary speakers engage each other “on the fly” and without access to or understanding of the analytical tools that conversation analysts have developed to transcribe and examine verbal interactions in great detail. However, inasmuch as conversations involve several participants whose utterances are made in response to previous utterances and in anticipation of future responses (thus the process is directed from within a conversation), researchers have the opportunity to understand conversations as cooperatively emerging structures that are, at each point in the process, responsive to past interactions and anticipatory of moves to come. A content analysis of data as conversation could involve (a) *inferring from any one moment of a recorded history of interactions the range of moves that could follow*, (b) *reinterpreting that history from the moves that actually did follow*, and (c) *systematically applying this explanatory strategy to all moments of naturally occurring conversations*.

This form of analysis is applicable not just to everyday conversations but also to exchanges between actors in organizational roles or as representatives of national governments. In exchanges between managers and employees, just as between therapists and their clients or between professors and their students, power issues enter through the speech acts the interlocutors choose, accept, or deny each other. Power relationships have become a favorite topic of critical scholarship among conversation analysts. Social organizations can be seen as reproducing their members’ commitment to the preservation of the organizations’ form. Commitments need to be asserted, heard, believed, and enforced. Thus, organizations reside in certain speech acts, in how members respond to each other’s talk. This makes organizations analyzable as networks of conversations of a certain kind. Analyses of exchanges between representatives of nations are not new, but conversation analyses of the unfolding dynamics in such exchanges offer a new approach to international relations. Content analyses of negotiations have advanced an understanding of the process (Harris, 1996). Pathologies of communication gain new currency when analyses reveal restrictions or constraints on conversation. Some scholars have called for the quantification of interactions (e.g., Hopper et al., 1986). This has been demonstrated, for example, regarding doctor-patient interactions (Ford, Fallowfield, & Lewis, 1996; Katz, Gurevitch, Peled, & Danet, 1969).

3.7 INSTITUTIONAL PROCESSES

The foregoing discussion has moved beyond the traditional notion of content in three senses: in the assertion that content analysis may be applied to any kind of data—texts, images, sounds, music, artifacts, anything that humans are able to vary for effect or unconsciously reproduce, in the assertion that analysts may draw inferences from textual data to features of any specifiable context, and in the assertion that texts are part of relational dynamics, of networks of relationships. In this section, I discuss a further expansion of the scope of content analysis to include inferences about institutional phenomena of which the institutions’ constituents may be only dimly aware. Much communication that takes place within institutions is routine,

relational, and coordinative, and valued as such, even enforced, without apparent reason. Moreover, institutions reside in particular qualities of communication. Because communication in institutions tends to go beyond unaided readers' scope of comprehension, content analyses that probe into institutional properties call for *analytical instruments and theories* that, like microscopes, telescopes, and computer intelligence, *provide inferential access to social realities that are too complex to be accessible otherwise*.

Berger and Luckmann (1966) outline the context of this kind of content analysis. To start, they suggest that *habitualization* is an important prerequisite of institutionalization:

Any action that is repeated frequently becomes cast into a pattern, which can then be reproduced with an economy of effort and which, *ipso facto*, is apprehended by its performer *as* that pattern. Habitualization further implies that the action in question may be performed again . . . with the same (or reduced) economical effort. . . . Habitualization carries with it the important psychological gain that choices are narrowed. (p. 53)

An example is the pattern of grammar, which directs our thoughts and actions in ways we rarely notice. For instance, the English language recognizes just two genders. Consequently, native English speakers tend to find the distinction between males and females natural and obvious. In turn, many institutions in English-speaking societies are built on this distinction. Gender is not a given phenomenon. It is an artifact of enacting grammar and vocabulary, which change only slowly. Because of the linguistic distinctions we confuse with facts, numerous problems arise, for example, by providing little if any space between the two gender identities. Such problems manifest themselves in gay bashing, the shunning of transvestites, and the difficulties that androgynous people face in their lives. That members of other cultures draw different distinctions demonstrates the institutional nature of such grammatical categories.

The ways in which we greet each other every day, the repetitive and utterly predictable categories of television programming and news coverage that we have come to take for granted, the ceremonial nature of the political process, the pervasiveness of climates of power in everyday life—all such patterns, weathered in the process of apparent successes, are the backbone of the institutionalization of human behavior. However, the comfortable certainties that this kind of habitualization offers also suppress our ability to see untested alternatives. Content analysts who study institutions can infer habitualization from repetition and the narrowing of choices from the absence of mention of alternative ways of being or doing things. Surprisingly, Shannon's information theoretical notions of redundancy (a quantification of the nonuse of otherwise available alternatives) and information (a measure of surprise in the context of available messages) can be seen to have institutional interpretations (see Shannon & Weaver, 1949).

Regarding habitual *patterns*, to the extent that people are concerned, they talk of them in a language that *categorizes* (typifies) not only the *actions* that constitute these patterns but also the *actors/participants* involved. Teaching is what teachers do in front of students. Entertaining is what entertainers do for their audiences. Such obvious and semantically tautological propositions involve categories that furnish people with spaces they may occupy or grow into, not only to fit the categories but also *to see*

each other in terms of those categories. Berger and Luckmann (1966, p. 54) identify institutions in terms of such *reciprocal categorizations*. For instance, knowing what a bank is enables the individuals in the context of a bank to interact with each other as certain categories of people—customers, tellers, guards, bank presidents, even bank robbers—regardless of who they are as individuals and regardless of whether the bank is in Philadelphia or Hong Kong. The same applies to understanding what is being said as proper. Reciprocal categorizations provide a key to how institutions are constituted, and such categorizations are easily identified in the texts that institutions generate—employee handbooks, memos on hiring practices, minutes of meetings, reports to shareholders—especially those generated by mass-media entertainment, which is highly institutionalized.

Berger and Luckmann (1966) note that *we grow into a world already constructed by others*, without knowing why things have become the way they now are. This lack of knowledge of the histories of institutions leads to the belief that “things are good because this is how they have ‘always’ been.” Such a belief discourages, if not punishes, deviations from established patterns. Thus, our lack of knowledge of history fuels institutional controls. Moreover, playing down human participation, without which institutions cannot exist, we tend to transfer agency to institutions, as when we say, “Science says . . .,” “The media show . . .,” or “The military discriminates against . . .” We consider institutions to be capable of preserving themselves, as when we speak of the “interests of government.” Institutions do not really control deviance from institutional patterns, nor do they assign powers to the roles people play in them. Individual participants police each other. Talk and texts that prevents people from realizing the roles they play in maintaining institutional practices is one target of content analyses of institutions.

Institutions tend to remain hidden behind habitual practices until flaws emerge and certainties break down. Families, bureaucracies, and nations are unthinkable without routine forms of communication. Family therapy is justified by the experience of routines whose mindless enactment within a family gets their members into trouble. Most institutions are taken for granted. Whereas the culturally dominant individualist approach to therapy encourages families to identify one family member as the one who needs help, working with that person as the identified patient rarely changes the dynamics of interactions within the family, perpetuating its dysfunctionality.

The emergence of international conflicts, which are rarely ever intended and in which nobody really likes to participate, especially when they may die as a result, is evidence of the fact that people can get involved in such events without knowing how. It thus appears that the way individuals create and preserve the institutional structures they live with are barely recognized and are given free range to play out. Institutions are not transparent unless we witness moments of breakdown, when something gets its members into unknown trouble in need of explanation, fixing, or trying to escape from its hold on them and are prevented from doing so. Content analyses of institutions often focus on communications at moments of such breakdowns. For instance, Berelson’s (1949) study of “what ‘missing the newspaper’ means,” conducted during a newspaper publishing strike in New York, revealed numerous previously unrecognized uses of newspapers and how their absence made people aware of these institutionalizations.

It is at moments of emerging difficulty or fear of the consequences of deviation from normalcy that the human constituents of institutions invent, appeal to, and apply institutional rules of conduct. Often such rules grow into systems of institutionalized explanations that become accessible through the very communications that invoke them. Mass communications researchers who have examined institutional processes have focused on legal, economic, political, technological, that is, structural explanations for those processes.

Legal explanations emphasize that communicators operate under certain legal conditions; for example, they may have to be licensed or must comply with or avoid violating contractual arrangements. Communicators may speak in an official capacity for particular social organizations, or they may question the legitimacy of certain practices. Texts obtained in legally regulated contexts reflect the legal constraints under which the institutional communicators who are being studied operate.

Economic explanations emphasize that when communication (production, transmission, and consumption) incurs costs, someone must pay them in some form, which creates networks of stakeholders with economic interests in what is being communicated. For example, in the United States, the traditional mass media are paid for largely by advertising; thus what is aired must be profitable in some way and cannot offend its sponsor. The effects of media ownership—in particular, the effects of monopolies and certain commercial interests—on communications have been a frequent target of content analyses.

Political explanations emerge when communications are disseminated widely and become of concern to competing public interests. The institution of ethical standards may result from debates about apparent problems, such as the violation of individuals' privacy by photographers (paparazzi) or reporters in the competitive pursuit of sensationalist material. Intentional misrepresentations by journalists and campaign advertisements perceived as "dirty," "slandorous," or "unfair" may lead to legal actions. Newspaper publishers, television stations, and other kinds of organizations cannot afford to displease vocal publics if they expect to have some longevity, whether the publics that concern them are made up of governing elites or masses of audience members. Thus, communications reflect as well as enact the prevailing configurations of rhetorical power. In conducting content analyses aimed at examining power relationships, researchers have to be careful not to fall into the trap of believing that everyone perceives power as they do. Instead, they may want to look at how power is enacted, received, or undone (Krippendorff, 1995b).

Technological explanations of institutional processes emphasize that all communications must be producible, recordable, distributable, and accessible through various technological means, and that actual communications are not only shaped in that process but also shape the institutions in whose terms they are processed. The film and television industries employ techniques of mass production that are vastly different from those employed by newspapers. This is a matter not of intention but of the nexus between technologies and the institutions that thrive on them. An even greater difference exists between newspapers and what computer-mediated communication—the Internet, for example—can make available. Content analyses have shed light on the systematic changes in content that take place when a book is made into a film (Asheim, 1950), on the role of gatekeepers in news flow (e.g., what happens to controversial content; White, 1964), on how news is made

as opposed to reported (Gieber, 1964), on the social role of the magazine cover girl as a function of channels of distribution (Gerbner, 1958), and on how expectations about institutions shape the forms of petitions directed to those institutions (Katz, Gurevitch, Danet, & Peled, 1969). In addition, based on an examination of mass communication from a technological perspective, Adorno (1960) has contended that the institutionalized repetitiveness of the mass production of messages preserves and strengthens social stereotypes, prejudices, and ideologies rather than corrects them (see Section 3.2.3).

There are a few fundamental generalizations from which content analyses of institutionalized texts may start. One is that everything said, written, listened to, or read—every communication—not only says something to its receiver but also institutes the very pattern of which it is a part. For example, a person who cashes a check at a bank is not merely taking part in a mutually beneficial transaction; his or her action also manifests trust in money and supports banking as an institution. If people did not bank regularly, the banking industry could not exist. Banks are instituted in our trust in money, in our belief in the safety of banks, in the belief that one should earn interest on savings, and so on. When people turn on their television sets to see certain shows, they are not only gaining entertainment but also supporting the shows they watch by increasing the programs' ratings. Their actions also legitimate the mass media as an institution to provide such entertainment. If nobody were to watch television for a while, the mass media could not exist as usual. The use of racial categories—whether on television, in everyday talk, or in survey questions—demonstrates that they are important, of public or interpersonal concern, and the very use of these categories invariably strengthens ethnic prejudices and makes them real. People's participation in a pattern of reciprocal categorization is an essential requirement for institutions to persist, and this applies also to issues of race. Therefore, analyses of such communication phenomena cannot stop at what is being said or heard. What matters is that the very act of communication strengthens that act, allows for repetition, and keeps people in attendance. Researchers conducting content analyses of institutionalized texts—which most mass communications are—have to observe whether communications constitute new patterns, strengthen what has been said before through repetition, or weaken a pattern by omission or attention to alternatives.

The properties of the medium of communication in which an institution is constituted have profound effects on the development of that institution. Innis (1951) compared oral communications with written communications and concluded that writing has the effect of freezing traditions, rendering institutions more permanent and reliable; thus written communications can support empires that extend control over larger geographic areas. Radio and television, with their virtually instantaneous transmission over vast distances, tend to support the development of geographically widely dispersed organizational forms, but because such media do not leave many records behind, these forms are far less stable than those supported by written media. Oral and audiovisual media are also less controllable than written ones. The revolution against the shah of Iran succeeded largely because of the support generated among the people of Iran through the distribution of audiotapes that escaped the control of the state, unlike all other media in that country. The short-lived pro-democracy movement in China was organized largely through fax communication, which recognized no national boundaries. The Internet now provides vast numbers

of geographically dispersed users with almost instantaneous access to computer-mediated communications; moreover, it is capable not just of disseminating data but of facilitating discussion among parties to negotiations and commitments as well as the rise of commercial enterprises and virtual communities that can undermine more traditional institutions. These diverse spatial, memory-related, and coordinative properties of communications media have profound effects on institutional dynamics, and research that focuses on single messages or on readings of individual users cannot possibly reveal those effects. Content analyses in institutional contexts can lead to inferences regarding the weakening or strengthening of certain institutions, and frequencies play important roles in such inferences.

Finally, *communications tend to reinforce the very institutional explanations and rules by which they are created and disseminated*. For one example, the traditional mass media, which operate in conformity with the one-way theory of communication, produce communications that demonstrate the workings of this theory and are likely to encourage the widespread use of this theory at the expense of alternative ways of thinking about human communication. It is therefore not surprising that from its inception, the field of communication research has been fundamentally committed to a model of communication that consists of a sender, messages, and receivers—as if no others, bystanders and judges, were relevant and worthy of attention. One-way communication technology has given rise to totalitarian regimes but also to disciplines such as advertising and causal theories such as of persuasion, influence, and effects. The emergence of interactive media has challenged this paradigm somewhat, but even today many researchers who seek to understand computer-mediated communication, which is well instituted, start with mass communication models. To infer institutional controls, content analysts have to observe what is not said, what happens when institutions are challenged or break down, and what is done to those who dare to deviate from institutionalized practices. Content analyses of textual data in view of such phenomena can add to our understanding of the workings of institutionalized patterns of thinking and acting. All of these phenomena become evident through analyses of the use of language.

Lasswell (1960) sought to clarify the institutional roles of communication by distinguishing among three functions:

- Surveillance of the environment
- Correlation (coordination) of a society's parts in response to the environment
- Transmission of social heritage from one generation to the next (culture)

To these, Wright (1964) added one more function:

- Entertainment

Echoing Parsons's (1951) sociological systems theory, both Lasswell and Wright argue that any society has to develop institutions that specialize in performing these functions. In the United States, journalism could be seen as serving the surveillance function in that journalists report publicly on events that take place, and politics could be regarded as serving the correlation function by mobilizing individuals to behave in ways that serve society as a whole and coordinating the distribution of resources (Lasswell, 1963). Functionalist accounts of institutions, to which this classification

is indebted, are not the only accounts, however. Nor can they be accepted without question, for they preserve sociological theorists' vision of what society is. Instead of imposing theoretical categories from outside or from a position of authority, content analysts attempt to understand institutions through how the participants in them talk with each other and about their own institutional involvements, how they participate in maintaining these institutions through talk and writing, and how they judge the legitimacy and appropriateness of institutions' actions. Institutions are constituted, constructed, and reconstructed in language use and in the distribution of narratives of that use through particular media of communication; hence the need for content analyses of this kind. Lasswell's classification at least acknowledges that institutions are manifest in all communicative practices and serve as the glue that holds a society together.

3.8 AREAS OF LIKELY SUCCESS

Having completed the above survey of areas in which content analysis may be applied, I conclude this chapter by offering a brief answer to the question of where content analysis might be used most fruitfully. To this end, I draw on the conceptual issues discussed in Chapter 2, including my argument against the use of the container metaphor, which entails the conception of content as a tangible entity, contained in messages and shipped from one place to another, that researchers presume to be able to analyze through the use of objective (i.e., observer-independent) techniques. I have argued instead for the metaphor of reading, which shifts attention from what content is to what readers do with texts, how they relate texts to the contexts of their use—individually, politically, socially, and culturally—and what this means for various social phenomena. In this shift, readers and their communities become central, whether the readers are authors, users, bystanders, or content analysts. Content analysts cannot exclude themselves from the population of readers, albeit their reading is aided by systematic methods of careful inquiry.

To determine where content analysis is most fruitfully employed, we should consider texts as the by-products of ongoing conversations. We have to acknowledge that people learn to read and write a language only after they have learned to speak it, that texts build on the experience of speech. They can substitute for speech, as when people write letters. They can extend the range of speech, as in the use of communication technology—mass and computer-mediated communication—and they usually revert back to speech, by being read, interpreted, talked about, accepted, dismissed, or selectively enacted. Texts are more durable than speech. Texts may be reread in support of gaining new insights, and they may be analyzed repeatedly and replicated by several analysts. This is why conversation analysts, for example, record and transcribe speech before they analyze it or talk of it. Whether texts are written for fear of forgetting something; to assure their writers desirable certainties, as when signing a contract; or to grant certainties to others, as when writers delegate their agency to a lawyer or government to act on their behalf, most writing shifts some kind of authority. Literacy, the ability to read and act on a text like everyone else, is expected by all members of a speech community. This stability is questionable, of course, as are the institutions whose members are dedicated to preserve them.

From my perspective, texts are always rooted in the dialogical context of conversations, subject to adaptation to changing situations, open to reinterpretations in dialogue among community members, even if such changes remain unnoticed. Content analysts may be able to chart these changes, even if they are not recognizable by those who reproduce the interpretations of texts.

Scholars have studied the effects of reading and writing, and the uses of the mass media, for some time, although this research has always lagged behind technological developments (information technology, for example). As noted above, typical topics of research have included deception, attitude change, message effects, uses and gratifications, technological biases, rational decision making, institutionalization, and causal connections between textual and nontextual phenomena.

Regarding assessments of crime, unemployment, and the economy, for example, studies have repeatedly demonstrated that correlations between what the mass media present—text in our generalized sense—and what public opinion polls find or individuals express as concerns are higher than the correlations between either of these and actual statistics on crime, unemployment, and the economy. This suggests that content analyses are generally more successful when they stay close to the uses of language—after all, public and individual opinions involve talk, not physical measurement. Another example is the well-studied phenomenon of agenda setting by the mass media (McCombs & Shaw, 1972; McCombs, Shaw, & Weaver, 1997)—that is, the fact that themes and issues distributed by the mass media in the form of print, speech, and images have a good chance of becoming topics of public conversations and, in turn, affect civic actions, inform political decisions, and stimulate artistic rearticulations. The simple reason for this phenomenon is that widely distributed texts enter, are adopted into, and come alive in conversations, not only in conversations between interviewers and interviewees but also conversations in public places ranging from side-street cafés to political demonstrations. If new words and expressions resonate with readers' or listeners' previously acquired language habits, they may take hold in the public imagination and become part of many people's vocabularies.

Content analyses are most successful when they focus on facts that are constituted in language, in the uses of the very texts that the content analysts are analyzing. Such linguistically constituted facts can be broken down into four classes:

- *Attributions*: Concepts, attitudes, beliefs, intentions, emotions, mental states, and cognitive processes ultimately manifest themselves in the verbal attributes of behavior. They are not observable as such. The words that make them real are acquired, largely in conversations but also through reading and attending to various media of communication. The attribution of competence, character, morality, success, and belongingness to particular categories of people enables or discourages actions, makes or breaks politicians, creates heroes and demonizes villains, identifies leaders and marginalizes minorities. These facts cannot exist without language, and to the extent that texts are instrumental in disseminating and creating such attributions, they are natural targets of successful content analyses.
- *Social relationships*: Noting that statements or questions can be uttered either subserviently or authoritatively, Bateson introduced the distinction between the content and the relationship aspect of all human communication

(Ruesch & Bateson, 1951, pp. 179–181). Relationships may be established implicitly or taken for granted in how communication takes place. For example, by offering third-person plural accounts of observed “Others,” scientific observers set themselves apart from their subjects and assume a position of superiority. This is manifest in the grammar of talk or writing. But relationships may also be negotiated, unilaterally imposed, and explicitly accepted or rejected. Authority, power (Hillman, 1995; Krippendorff, 1995b), contractual agreements, and inequalities are all constituted primarily in *how* language is used and only secondarily in *what* is said. Content analyses tend to be more successful when they focus on how language is used, relying on social grammars of recorded speech or written communication of which speakers or writers may not be fully aware.

- *Public behaviors:* Individuals’ values, dispositions, conceptions of the world, and commitments to their way of being surface in conversations that involve repeated confirmations. Without such repetition, individuals drift apart, their behaviors become no longer coordinated, and they experience difficulties in understanding each other. To the extent behavior is public, and hence observed and judged by others, it is brought into the domain of language. Narratives too, are essentially public. They may inspire individuals to act, but they are always told by someone and listened to by others, rendering inspiration a social experience. Reading a newspaper may be an individual act, but not only do newspapers print what editors consider to be of public interest, newspaper readers also talk to others about what they read, and so make newspaper reading a public activity. The vocabularies we use are all acquired from others who have used the words before us. Inasmuch as a vocabulary suggests the range of what a person can talk about and conceive, the conceivable is transmitted from parents to children, from speakers to listeners, and from writers to readers. All uses of language ultimately are public—not shared but in the open. Content analyses are more likely to succeed when they address phenomena that are of a public, social, or political nature or concern phenomena of individuals’ participation in public, social, or political affairs. Cognition, for example, the supposed crown of individualism, is never an exclusively individual phenomenon. It always reflects the contexts of others, much as texts do.
- *Institutional realities:* We often overlook the institutional nature of social realities—of marriage, money, government, history, illness, and even scientific pursuits. Public opinion, for example, is a construction that relies heavily on the language of social science, on statistics in particular, but it also depends crucially on being taken as a political reality and acted upon. Without the institution of free speech, the authority of journalism, and constitutional democracy, public opinion research would not make much sense. Mental illness has an institutional reality as well. It is projected onto identified patients in terms of categories that mental health professionals and insurance companies have developed for their convenience. The factuality of these phenomena derives from certain institutionalized texts, such as the *Diagnostic and Statistical Manual of Mental Disorders (DSM-IV-R)*,

which is published by the professional authority on mental illness, the American Psychiatric Association (2013). These texts legitimate numerous therapeutic interventions. For still another example, consider how a social organization such as a family or a corporation constitutes itself. Members of an organization coordinate their activities through communication and continually affirm their membership, often shielding vital stories about inside practices from outsiders. When exchanges within organizations take place in written form, they stabilize organizational memories, identities, and practices. Disrupting an organization's network of communication can cause the organization to collapse. Organizational communication research has successfully inquired into how organizations arise in the communications among members and develop nourishing organizational cultures. Content analysis of what is said and written within an organization provides the key to understanding that organization's reality, but it is most likely to succeed if it considers the more stable categories in which the organization constitutes itself.

In sum, content analyses are most likely to succeed when analysts relate textual matter—understood quite broadly to include images and traffic on the Internet—to social realities that enact them in their daily life, in conversations that yield consensus on how they are to be read or should no longer be read, that is, in conversations that keep texts alive. Repetitive, routine, public, and institutionalized phenomena are easier to infer than are rare and unconventional ones. Moreover, because content analysis presupposes familiarity with, if not literacy in, the language of the analyzed texts, the more cognizant content analysts are of vocabulary and subtle discursive conventions, including their own, the more they are able to reflect on what they do with texts that had or still have a life for others and the more likely they are to succeed and leave their contributions for everyone to read and interpret.

COMPONENTS OF CONTENT ANALYSIS

PART



4

THE LOGIC OF CONTENT ANALYSIS DESIGNS

As a technique, content analysis relies on several specialized procedures for handling text. These can be thought of as tools for designing suitable analyses. This chapter outlines the key components of content analysis and distinguishes among several research designs, especially designs used in the preparation of content analyses and designs for content analyses that collaborate with other research methods to contribute to larger research efforts.

4.1 CONTENT ANALYSIS DESIGNS

The very idea of *research*—a repeated *search* within data for generalizations, patterns that appear to permeate the data—presupposes explicitness about methodology. Unless researchers explain clearly what they have done, how can they expect to be able to replicate their analyses or to process more texts than an individual can read? Beyond that, how can they convince others that their research was sound and thus their results should be accepted?

A *datum* is a unit of information that is recorded in a durable medium, distinguishable from and comparable with other data, analyzable through the use of clearly delineated techniques, and relevant to a particular problem. Data are commonly thought of as representing observations or readings, but they are always the products of chosen procedures and geared toward particular ends—in content analysis, data are justified in terms of the procedures the researcher has chosen to answer specific questions concerning phenomena in the context of given texts. Hence data are made, not found, and researchers are obligated to say how they generated their data.

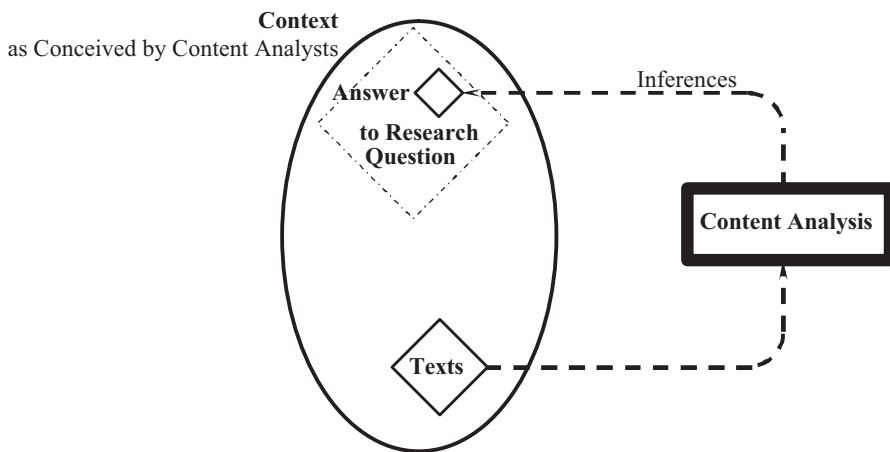
The network of steps a researcher takes to conduct a research project is called the *research design*, and what knits the procedural steps into the fabric of a coherent research design is the design's *logic*. Generally, this logic concerns two qualities: the efficiency of the procedural steps (avoiding structural redundancies while preventing “noise” from entering an analysis) and the evenhandedness of data processing (preventing prejudices from biasing the data or favoring of one outcome over another). This logic enables analysts to account to their scientific community for how the research was conducted. For a research design to be replicable, not merely understandable, the researcher's descriptive account of the analysis must be complete enough to serve as a set of *instructions* to coders, fellow researchers, and critics—much as a computer program determines what a machine is to do. Although the thoroughness of a computer program may serve as a scientific ideal, in social

research the best one can hope for is an approximation of that ideal. Content analysts in particular must cope with a good deal of implicitness in their instructions. (I will return to this topic in subsequent chapters.)

Traditional guides to research methods tend to insist that all scientific research tests hypotheses concerning whether or not patterns are evident in the data. Content analysis, however, has to address prior questions concerning why available texts came into being, what they mean and to whom, how they mediate between antecedent and consequent conditions, and, ultimately, whether they enable the analysts to select valid answers to questions concerning their contexts. Hence the logic of content analysis designs is justifiable not only according to accepted standards of scientific data processing (efficiency and evenhandedness) but also by reference to the context in relation to which texts must be analyzed.

Figure 2.2 contextualizes content analysis and content analysts most comprehensively. It may be seen to contain Figure 4.1, which represents the simplest content analysis design. Here, the analyst relies solely on available texts to answer a research question. Although this figure locates texts and results—inputs and outputs of the analysis—in a chosen context, it suggests nothing about the nature of the context that justifies the analysis (discussed in Chapter 3) or about the network of needed analytical steps, which I address below.

FIGURE 4.1 ■ Content Analysis: Answering Questions Concerning a Context of Texts



4.1.1 Components

Here we open the “content analysis” box in Figure 4.1 and examine the components the analyst needs to proceed from texts to results. Listing these components is merely a convenient ways to partition, conceptualize, talk of, and evaluate content analysis designs step by step. As accounts of what the components do must also serve as instructions for replicating them elsewhere, each component has a descriptive and an operational state:

- *Unitizing*: relying on definitions of relevant units
- *Sampling*: relying on sampling plans
- *Recording/coding*: relying on coding instructions
- *Reducing* data to manageable representations: relying on established statistical techniques or other methods for summarizing or simplifying data
- Abductively *inferring* contextual phenomena: relying on established analytical constructs or presumed models of the chosen context as warrants
- *Narrating* the answer to the research question: relying on narrative traditions or discursive conventions established within the discipline of the content analyst

Together, the first four components constitute what may be summarily called *data making*—creating computable data from raw or unedited texts. In the natural sciences, these four are embodied in physical measuring instruments. In the social sciences, the use of mechanical devices is less common—often impossible—and data making tends to start with observations. The fifth component, abductively inferring contextual phenomena, is unique to content analysis and goes beyond the representational qualities of data. I describe each of the components in turn below.

Unitizing draws systematic distinctions within a continuum of otherwise undifferentiated text—documents, images, voices, videos, websites, and other observables—that are of interest to an analysis, omitting irrelevant matter but keeping together what cannot be divided without loss of meaning. In Chapter 5, I discuss different kinds of units—sampling units, recording units, context units, units of measurement, units of enumeration—and the different analytical purposes they serve. With these diverse uses, unitizing may occur at various places in a content analysis design. Content analysts must justify their methods of unitizing, and, to do so, they must show that the information they need for their analyses is represented in the collection of units, not in what unitizing omits and not in the relationships between the units, which an analysis discards by treating them independent of each other.

Sampling allows the analyst to economize on research efforts by limiting observations to a manageable subset of units that is statistically or conceptually representative of the set of all conceivable relevant units, the population or universe of interest. Ideally, an analysis of a whole population and an analysis of a representative sample of that population should come to the same conclusion. This is possible only if the population manifests redundant properties that do not need to be repeated in the sample drawn for analysis. But samples of text do not relate to the issues that interest content analysts in the same way that samples of individuals relate to populations of individuals of interest in surveys of public opinion, for example. Texts can be read on several levels—at the level of words, sentences, paragraphs, chapters, or whole publications; as literary works or discourses; or as concepts, frames, issues, plots, genres—and may have to be sampled accordingly. Hence creating representative samples for content analyses is far more complex than creating samples for, say, psychological experiments or consumer research, in which the focus tends to be on one level of units, typically individual respondents with certain attributes (I discuss the issues involved in sampling for content analysis in depth in Chapter 6). In

qualitative research, samples may not be drawn according to statistical guidelines, but the selective use of quotes and examples that qualitative researchers present to their readers are intended to serve the same function as do samples. Quoting typical examples in support of a general point implies the claim that they are fair representations of the phenomena of concern.

Recording/coding bridges the gap between texts and someone's reading them, between distinct images and what people see in them, or between separate observations and their situational interpretations. One reason for this analytical component is researchers' need to create durable and analyzable records of otherwise transient phenomena, such as spoken words or passing visual events. Once such phenomena are suitably recorded, analysts can compare them across time, apply different methods to them, and replicate the analyses of other researchers. Written text is always already recorded in this sense, and, as such, it is rereadable. It has a material base—much like an audiotape, which can be replayed repeatedly—without being in an analyzable form, however. The second reason for recording/coding is, therefore, content analysts' need to transform unedited texts, original images, and/or unstructured sounds into analyzable representations. The coding of text is mostly accomplished through human intelligence. I discuss the processes involved in recording and coding in Chapter 7, and then, in Chapter 8, I discuss the data languages used to represent the outcomes of these processes. In content analysis, the scientific preference for mechanical measurements over human intelligence is evident in the increasing use of computer-aided text analysis (discussed in Chapter 11); the key hurdle of such text analysis software, not surprisingly, is the difficulty of programming computers to respond to the meanings of texts. It often confuses character strings with what they mean to particular readers or users of documents.

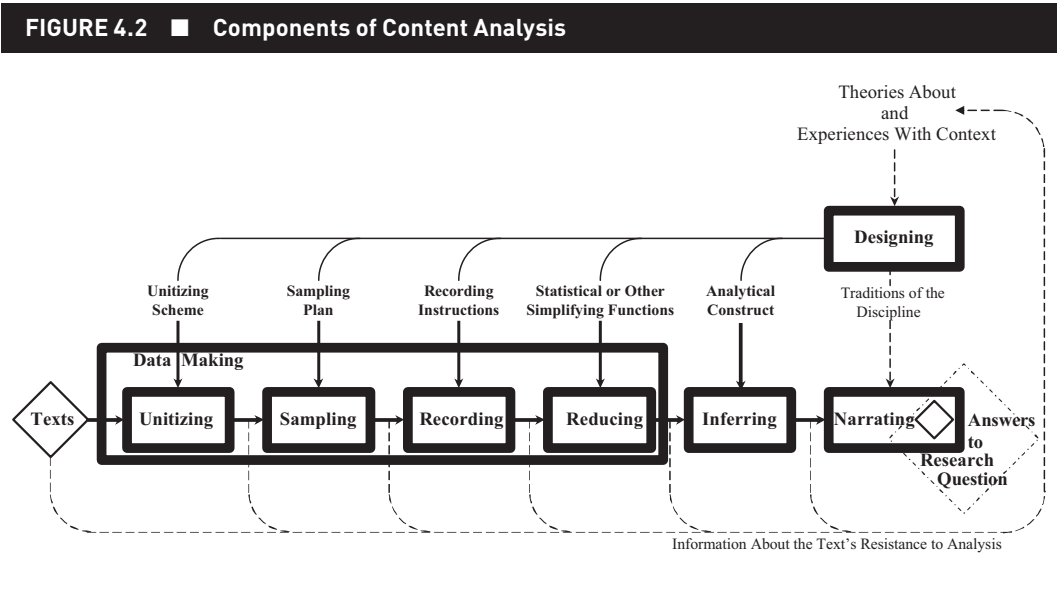
Reducing data serves analysts' need for efficient representations, especially of large volumes of data. A type/token statistic (a list of types and the frequencies of tokens associated with each), for example, is a more efficient representation than a tabulation of all occurrences. It merely replaces duplications by a frequency. Because one representation can be created from the other, nothing is lost. However, in many statistical techniques for aggregating units of analysis—correlation coefficients, parameters of distributions, indices, and tested hypotheses—information is lost. In qualitative pursuits, rearticulations and summaries have similar effects: They reduce the diversity of text ideally to what matters.

Abductively *inferring* contextual phenomena from texts moves an analysis outside their data. It bridges the gap between descriptive accounts of texts and what they mean, refer to, entail, provoke, or cause. It points to unobserved phenomena in the context of interest to an analyst. As I have noted in Chapter 2, abductive inferences—unlike deductive or inductive ones—require warrants, which in turn may be backed by evidence. In content analysis, such warrants are provided by analytical constructs (discussed in Chapter 9) that are backed by everything known about the context. Abductive inferences distinguish content analysis from other, largely inductive modes of inquiry.

Narrating the answers to content analysts' questions amounts to researchers' making their results comprehensible to others. Sometimes, this means explaining the practical significance of the findings or the contributions they make to available literature. At other times, it means arguing the appropriateness of the use of content analysis rather than direct observational techniques. It could also entail making

recommendations for actions—legal, practical, or for further research. Narrating the results of a content analysis is a process informed by traditions that analysts believe they share with their audiences or the beneficiaries of their research (clients, for example). Naturally, most of these traditions are implicit in how social scientists conduct themselves. Academic journals may publish formal guidelines for researchers to follow in narrating their results and let peer reviewers decide whether a given content analysis is sound, interesting, and worthwhile.

The six components of content analysis do not need to be organized as linearly as suggested by Figure 4.2. A content analysis design may include iterative loops—the repetition of particular processes until a certain quality is achieved. Or components may recur in various guises. For example, unitizing may precede the sampling of whole documents, but it may also be needed to describe the details of their contents. Thus, coding instructions may well include unitizing schemes. Moreover, a content analysis could use components that are not specifically highlighted in Figure 4.2. Decisions, to mention just one analytical action, typically direct the content analysts along an inferential path with many forks and turns toward one or another answer to the research question. Here, decisions are part of the inference component. Finally, it is important to note that there is no single “objective” way of flowcharting research designs.



The analyst’s written instructions (represented in boldface type in Figure 4.2), which specify the components in as much detail as feasible, include all the information the analyst can communicate to other analysts so that they can replicate the design or evaluate it critically. The traditions of the analyst’s discipline (in medium type in Figure 4.2) are an exception to the demand for explicitness. Most scientific research takes such traditions for granted.

Any set of instructions, it must be noted, imposes a structure on the available texts. Ideally, this structure feels natural, but it may feel inappropriate or forced, if not alien, relative to the analyst’s familiarity with the texts’ context. Take unitizing, for example. Texts may be cut into any kind of units, from single alphabetical characters to whole publications. Unitizing is arbitrary but not for a particular content analysis.

For example, if an analyst wants to infer public opinion from newspaper accounts, stories may be more natural for an examination of what readers think and talk about than, say, value-laden words that occur in these accounts. The use of inappropriate units leads analysts to experience conceptual troubles. Or an analyst may apply a particular sampling plan and then discover, perhaps too late, not only that the sampled documents are unevenly relevant but also that the sampling plan has excluded the most significant ones. Finally, in reading given texts, an analyst may encounter important concepts for which the coding instructions fail to provide suitable categories. Such a discovery would render the recording/coding task irrelevant or of uncertain validity. During the development phase of content analysis design, a sensible analyst should “resist the violence” that poor instructions can inflict on the texts and attempt to reformulate instructions as needed so that they are appropriate to the texts at hand. The path that a sensible approach might take is illustrated in Figure 4.2 by the dashed lines, which show another flow of information that is motivated by the analyst’s resistance to inappropriate analytical steps. The instructions in good content analysis designs always take such information into account.

A final point regarding Figure 4.2: As noted in Chapter 2, texts are always the observable parts of a chosen context. The context directs the analysis of a text, and the results of the analysis contribute to a (re)conceptualization of the context, redirecting the analysis, and so forth. This reveals the essentially recursive nature of the process of designing content analyses. This recursion contrasts sharply with the application of a content analysis design, which is essentially a one-way transformation of available texts into the answers to the analyst’s research questions. We must therefore distinguish between the *development* of a content analysis, during which a design with context-sensitive specificity should emerge, and the *execution* of a content analysis, during which the design is relatively fixed and ideally replicable, regardless of what the texts could teach the analyst. Interestingly, the context-sensitive path that the content analyst takes while developing the design may no longer be recognizable when the finished design is applied to large volumes of text and/or replicated elsewhere.

4.1.2 Quantitative and Qualitative Content Analysis

In Chapter 2, I noted that quantification cannot be a defining criterion for content analysis. Text is always qualitative to begin with, categorizing textual units is considered the most elementary form of measurement (Stevens, 1946), and a content analysis may well result in verbal answers to a research question. Using numbers instead of verbal categories or counting instead of listing quotes is merely convenient; it is not a requirement for obtaining valid answers to a research question. In Chapter 1, I suggested that the quantitative/qualitative distinction is a mistaken dichotomy between the two kinds of justifications of content analysis designs: the explicitness and objectivity of scientific data processing on one side and the appropriateness of the procedures used relative to a chosen context on the other. For the analysis of texts, both are indispensable. My view is the subject of a continuing debate. Proponents of quantification, starting with Berelson and Lazarsfeld (1948), but most explicitly Lasswell (1949/1965b), equate quantification with science and dismiss qualitative analyses as mere literature. This view has been rightly criticized for restricting content analysis to numerical counting exercises (George, 1959b) and for uncritically buying into the measurement theories of the natural sciences or commercial or political

interests in large markets. Proponents of qualitative approaches, who have come largely from the traditions of political analysis, literary scholarship, ethnography, and cultural studies (Bernard & Ryan, 1998), have been criticized for being unsystematic in their uses of texts and impressionistic in their interpretations. Although qualitative researchers compellingly argue that each body of text is unique, affords multiple interpretations, and needs to be treated accordingly, there is no doubt that the proponents of both approaches sample text, in the sense of selecting what is relevant; unitize text, in the sense of distinguishing words, propositions, or larger narrative units and using quotes or examples; contextualize what they are reading in light of what they know about the circumstances surrounding the texts; and have specific research questions in mind. Thus, the components of content analysis in Figure 4.2 are undoubtedly present in qualitative research as well, albeit less explicitly so. I think it is fair to state the following:

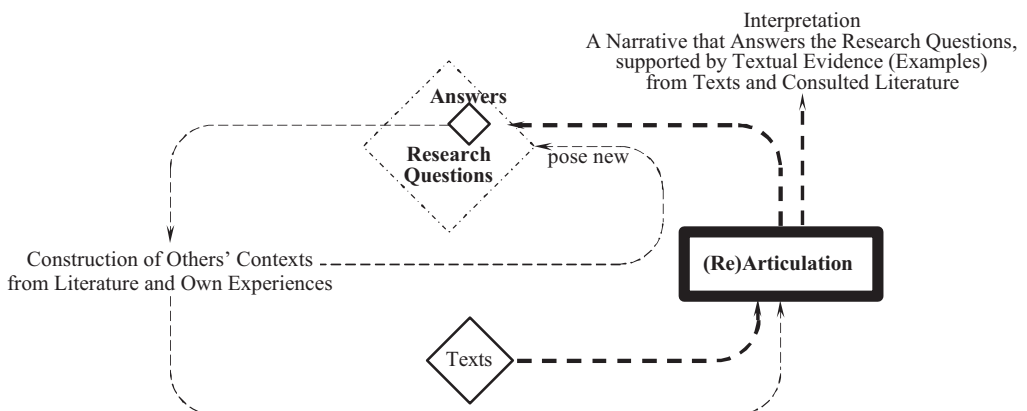
- Avowedly qualitative scholars tend to find themselves in a hermeneutic circle, using known literature to contextualize their readings of given texts, rearticulating the meanings of those texts in view of the assumed contexts, and allowing research questions and answers to arise together in the course of their involvement with the given texts. The process of recontextualizing, reinterpreting, and redefining the research question continues until some kind of satisfactory interpretation is reached (see Figure 4.3). Scholars in this interpretive research tradition acknowledge the open-ended and always tentative nature of text interpretation. Taking a less extreme position, content analysts are more inclined to limit such hermeneutic explorations to the development phase of research design.
- Qualitative scholars resist being forced into a particular sequence of analytical steps, such as those illustrated in Figure 4.2. Acknowledging the holistic qualities of texts, these scholars feel justified in going back and revising earlier interpretations in light of later readings; they settle for nothing less than interpretations that do justice to a whole body of texts. As such readings cannot easily be standardized, this process severely limits the volume of texts that a single researcher can analyze consistently and by uniform standards. Because this process is difficult to describe and to communicate, qualitative studies tend to be carried out by analysts working alone, and replicability is generally of little concern. By contrast, faced with larger volumes of text and working in research teams, content analysts have to divide a body of texts into convenient units, distribute analytical tasks among team members, and work to ensure the consistent application of analytical procedures and standards. For these reasons, content analysts have to be more explicit about the steps they follow than qualitative scholars may need to be.
- Qualitative researchers tend to acknowledge the possibility of multiple interpretations of textual units by considering diverse voices (readers), alternative perspectives (from different ideological positions), oppositional readings (critiques), or varied uses of the texts examined (by different groups). In Figure 2.2 these are referred to as the many worlds of others. This conflicts with the measurement model of the natural sciences—the assignment of unique measures, single values, typically numbers, to distinct

objects—but not with content analysts’ ability to use more than one context for justifying multiple inferences from texts.

- Qualitative content analysts support their interpretations by weaving quotes from the analyzed texts and literature about the contexts of these texts into their conclusions, by constructing parallelisms, by engaging in triangulations, and by elaborating on any metaphors they can identify. Such research results tend to be compelling for readers who are interested in the contexts of the analyzed texts. Quantitative content analysts, too, argue for the context sensitivity of their designs (or take this as understood), but they compel readers to accept their conclusions by assuring them of the careful application of their design.
- Committed qualitative researchers tend to apply criteria other than reliability and validity to their results. It is not clear, however, whether they take this position because intersubjective verification of their interpretations is extraordinarily difficult to accomplish or whether the criteria they propose are truly incompatible with making abductive inferences from texts. Among the many alternative criteria qualitative scholars have advanced are, according to Denzin and Lincoln (2000, p. 13), trustworthiness, credibility, transferability, embodiment, accountability, reflexivity, and emancipatory capabilities.

In other words, qualitative approaches to text interpretation are not incompatible with content analysis. The recursion (hermeneutic circle) shown in Figure 4.2 is visible in Figure 4.3 as well, although the former figure provides more details and is limited to the design phase of a content analysis. Multiple interpretations are not limited to qualitative scholarship either. Content analysts can adopt multiple contexts, pursue multiple research questions, and apply multi-valued codes to the phenomena of interests. Content analysis researchers’ reflexive involvement—systematically ignored in naturalist inquiries, often acknowledged in qualitative scholarship—manifests itself in the awareness that they are the ones who construct contexts for their analysis, acknowledge the worlds of others in the pursuit of their own research questions, and adopt analytical

FIGURE 4.3 ■ Qualitative Content Analysis



constructs based on available literature or prior knowledge about the contexts of given texts. Whether a close but uncertain reading of small volumes of text is superior to a systematic content analysis of large bodies of text is undecidable in the abstract.

The reason this book appears to rely most heavily on examples involving quantitative content analyses is that researchers working within this tradition have tended to encourage greater explicitness and transparency than have qualitative scholars.

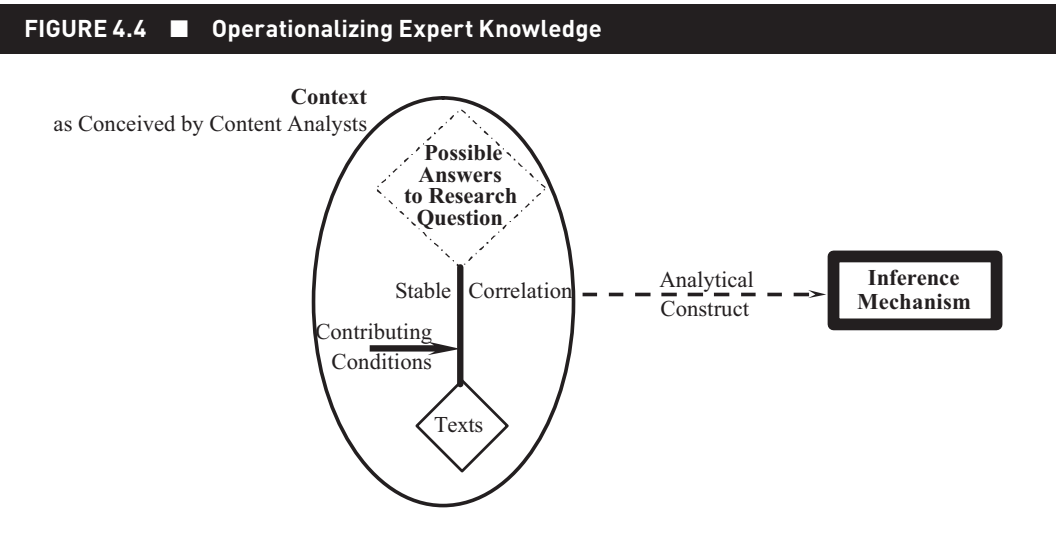
4.2 DESIGNS PREPARATORY TO CONTENT ANALYSIS

Generating data—describing what was seen, heard, or read—is relatively easy. Content analyses succeed or fail, however, with the validity of the analytical constructs that guide their coding instruction and inform their inferences. Once established, analytical constructs may become applicable to a variety of texts and may be passed on from one analyst to another, much like a computational theory concerning the stable features of a context. Below, I discuss three ways of establishing analytical constructs.

4.2.1 Operationalizing Available Knowledge of the Context

Content analysts, by their very ability to read and have an interest in given texts, acknowledge at least cursory knowledge of their sources: who else reads, appreciates, or uses the texts (note that authors always are the first readers of what they write); what the texts in question typically mean and to whom; which institutionalized processes are invoked in publishing and disseminating the texts; and what makes the texts hang together. Knowledge of this kind, intuitive as it may seem in the beginning, concerns the stable features surrounding given texts. Figure 4.4 suggests that such knowledge needs to be rearticulated into an inference mechanism. Without clarifying one’s conceptions of the context of given texts, the analytic procedure employed may not qualify as a “design.” I provide more specific discussions of this process in Chapter 9, but because the three preparatory designs all yield the same result, an analytical construct, I present them here for comparison.

Operationalizing available knowledge may be as simple as equating the frequency of co-occurrence of two categories of text with the strength of the association between



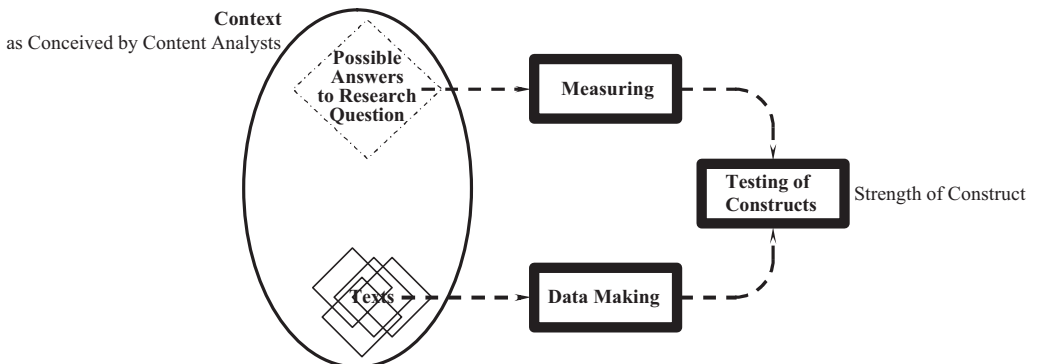
the two conceptual categories in an author's mind. Other operationalizations include the construction of the tagging dictionary for a computer program, which requires extensive knowledge of language use; formulating an algorithm that formalizes the propositions found in the message effects literature; and a computer program written to trace the linguistic entailments of selected political slogans through a body of texts. Operationalizations must be justified, of course, and available theory, literature, or acknowledged experts may be consulted for this purpose. Ultimately, the application of any operationalization of analysts' knowledge to given texts must yield valid inferences.

4.2.2 Testing Analytical Constructs as Hypotheses

The most traditional way to establish a valid analytical construct is to test it empirically. To develop alternative hypotheses of relations between textual and contextual variables and let empirical evidence select the most predictive one. This is how researchers establish psychological tests, validate behavioral indices, and develop predictive models of message effects. Once the correlations between textual and extratextual features are known, content analysts can use these correlations to infer contextual correlates from given texts—provided the correlations are sufficiently decisive and generalizable to the current context. This is why we speak of stable or relatively enduring relations operating in the chosen context. Osgood (1959), for example, conducted word-association experiments with subjects before building the correlation he found between word co-occurrences in text and patterns of recall into his contingency analysis (see also Krippendorff & Bock, 2009, Chapter 3.1). In a carefully executed study, Phillips (1978) established a correlation between reports of suicides of important celebrities and the fatality rate due to private airplane crashes. He found that the circulation of such suicide reports did predict an increase in airplane crashes (see also Krippendorff & Bock, 2009, Chapter 2.4). Whether such an index has practical consequences is another matter.

To test such statistical hypotheses empirically, data must not only be large enough to support statistically sound analytical constructs, but they must also be generalizable to the textual matter currently content analyzed. It follows that this design applies mainly to repeatedly asked research questions or situations in which the relations between texts and the answers to these questions concerning their context are stable, not unique (see Figure 4.5).

FIGURE 4.5 ■ Testing Analytical Constructs as Hypotheses



4.2.3 Developing Analytical Constructs by Trial and Error

Most traditional content analysis designs develop their analytical constructs from sound background knowledges or theories of the context of texts and apply them to obtain potentially valid inferences of interests. However, there are many occasions where individual content analysts or larger scholarly communities stay with an analytical effort for longer periods of time and systematically work on improving the quality of their inferences. Such efforts may start by making tentative inferences from given texts, comparing them to validating evidence and using observed discrepancies to alter their analytical construct for subsequently improved inferences from similar texts. In the absence of validating evidence, the corrective feedback may also come from misleading decisions to which the inferences lead or from respected critics. The iterative process of this content analysis design converges toward a “best fit” of the analytical construct and the context within which the analysts work. This is how intelligent content analysts learn from their failures and improve the quality of their research over time, as did the Federal Communications Commission propaganda analysts during World War II, who simply became better analysts as they repeatedly corrected the assumptions of their predictions (George, 1959a).

A particularly transparent example of developing of analytical constructs is the iterative computation of discriminant functions, sometimes called “machine learning” (Alpaydin, 2004). A discriminant function is an analytical construct, whether in the form of a procedure, decision rule, or algorithm which, when applied to a set of texts, distinguishes among them according to well-defined criteria. In content analysis, these criteria are tied to the inferences that a researcher wishes to make from texts.

Perhaps Earl B. Hunt was the first who developed the methodology of (1) starting with a sample of documents categorized in ways to be generalized to other documents; (2) using an algorithm that iteratively searches for the strongest relationships between the presence or absence of textual qualities in these documents to their categorization; (3) applying the thereby acquired discriminant function to distinguish among documents whose categories are known to the analyst but withheld from the function; and, if successful, (4) inferring the categories of other documents. Hunt (Stone & Hunt, 1963) had two sets of suicide notes. One collected by Shneidman (Shneidman & Farberow, 1957) at the San Francisco Suicide Prevention Center, hence real. The other generated by a panel of writers from Osgood’s research group, hence simulated. Hunt relied on Stone’s (Stone & Hunt, 1963) dictionary-based general inquirer, which gave him the frequencies of semantically related groups of words for each note. His discriminant function emerged in the process of examining 15 pairs of notes, testing numerous hypotheses regarding how the real and simulated notes differed. After obtaining a good fit, the difference function was applied to another set of notes and inferred the psychological states of the writers in 17 out of 18 cases. This was a remarkable success, achieved without suicide-related preconceptions.

It is important to note that this discriminant function emerged during a “learning” period prior to its application, and was limited to a very small universe of existing notes with known authenticity. I do not know whether it found applications outside

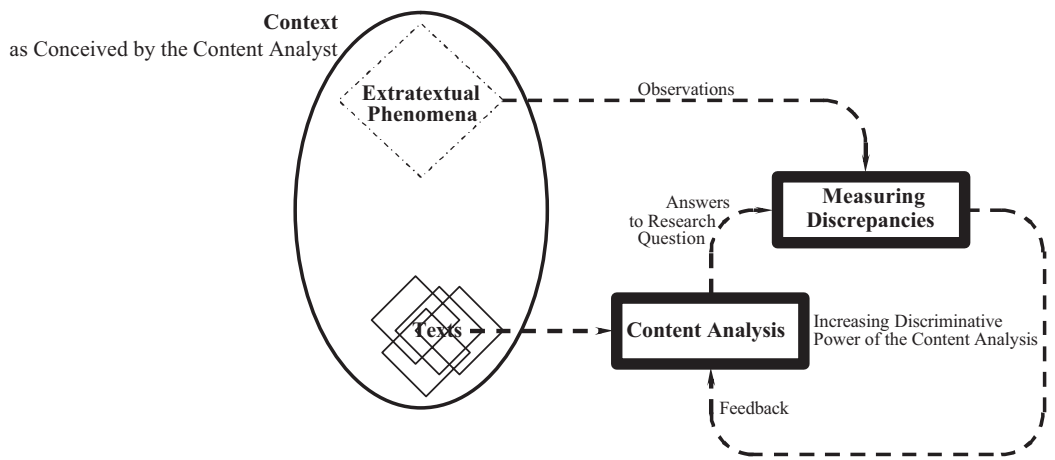
the scholarly effort. However, the idea of developing an analytical construct by trial and error before applying it to generate inferences contrasts with the most common practice of developing analytical constructs from sound preconceptions or theories of the context under consideration. For example, Osgood, who had posed the problem of distinguishing real from fake suicide notes and had generated the simulated notes, before Stone and Hunt got hold of his data, applied an analytical construct derived from a psychological theory to distinguish between these notes. Stone and Hunt reported that Osgood's content analysis was far less successful than theirs.

While Hunt's discriminant function contributed little to understanding the motivation of the authors of these notes (see Section 9.2.1), the process of developing it became the prototype for formulating algorithms capable of distinguishing among textual characteristics from which inferences could be justified, now called supervised machine learning algorithms (see Section 11.4.2).

Hunt's design was limited to very small sets of texts, one research team, and little interest in theories underlying their inferences, but the process of developing analytical constructs by trials and learning from errors is quite common, especially in larger contexts and when time for learning is available. Besides the example of slowly improving predictions from domestic enemy propaganda during World War II, already mentioned, there are numerous occasions where analytical constructs evolve within research communities facing and coping with emerging challenges. Efforts to distinguish between good and bad journalistic practices have been around for a century. Media critics and governmental regulators of the public use of radio frequencies had a long history of content analyzing media presentations using objectivity, public service, and balanced reporting as the criteria for such evaluations. The measurability of these criteria became challenged by new forms of communication media and institutions with commercial or political interests in bypassing criticism and regulations. Today, content analyses that measure the quality of journalism in these three dimensions are undermined by the very journalistic practices to be judged. The meaningfulness of the concept of objectivity is challenged when journalists correctly quote lies by politicians. Balanced reporting faces limits when applied to giving criminals and their victims equal attention. Whether news outlets report their sources is easy to establish. However, the distinction between trustworthy and unreliable sources is not always easy, and when it is, reporting their identity may violate their privacy. When reporting on controversies, trusted sources may well become the target of harassment by opponents when identified by name. Violating their privacy may conflict with the value of transparency. In the United States, free speech is protected by the Constitution, but public responsibility calls for excluding hate speech, the promotion of ethnic prejudices, and use of slander, a not always easy distinction. Journalists are not mere reporters of facts; good journalistic research needs to uncover barely noticeable and often deliberately hidden connections between them. Unless evidence of these connections is unquestionably compelling, opponents can free themselves of apparent wrongdoing by dismissing them as conspiracy theories or witch hunts. In other words, the analytical constructs that could enable the community of media scholars to evaluate journalistic practices have become exceedingly complex, multidimensional, and require openness to continuous improvements as new media and rhetorical strategies are invented that make journalistic accountability difficult.

In all of these examples, content analyses, designed to improve their analytical constructs in the face of failures, need to be iteratively coupled with validating information from the contexts they are addressing, as shown in Figure 4.6. Whether this feedback results in computable algorithms (discriminant functions, further discussed in Section 11.4.3), analytical techniques, or improved abilities by scholarly communities to make valid or usable inferences from available texts to their contexts of use, it always involves time and clear criteria of successes and failures, even when their analytical objectives and contexts are evolving.

FIGURE 4.6 ■ Developing Discriminant Functions



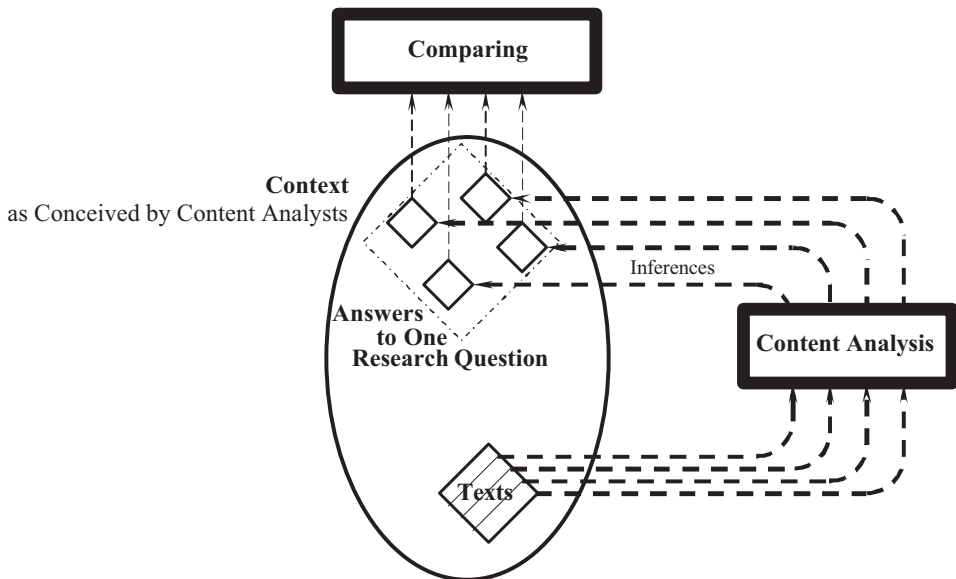
4.3 DESIGNS GOING BEYOND CONTENT ANALYSIS

Unfortunately, starting with Berelson’s (1952) account, the content analysis literature is full of insinuations that content analyses are aimed at testing scientific hypotheses, which brings us back to the notion of content as something inherent in or indistinguishable from text, a conception we abandoned in Chapter 2. According to the definition of content analysis employed in this book, content analysts rely on hypothetical generalizations in the form of analytical constructs. But the proof of these generalizations lies in the validity or usefulness of the inferences they support. It becomes evident mainly after content analysts have answered their research questions, made their abductive inferences, or interpreted their texts systematically. For example, to test a hypothesis concerning the behavioral correlates of anxiety, one must know the level of anxiety and separately observe the behavioral correlates of interest. By inferring the level of anxiety from an individual’s talk—from accounts of feelings, distress vocabulary, or speech disturbances (Mahl, 1959)—the content analysis becomes a necessary part of a larger research effort. Despite what Figure 4.1 might suggest, content analyses do not need to stand alone, and they rarely do. Below, I briefly discuss three research designs in which content analysis is instrumental.

4.3.1 Comparing Similar Phenomena Inferred From Different Bodies of Texts

In this design, researchers have reasons to draw distinctions within a body of text and apply the same content analysis to each part (see Figure 4.7). For example, to study speeches made before, during, and after a given event—or trends—analysts must distinguish texts according to time periods. To compare the treatment of one event in different media, analysts would have to distinguish texts by their sources. To examine how candidates for a political office tailor their promises to different audiences, analysts would want to distinguish texts according to audience demographics. And to test hypotheses regarding the impacts of competition between newspapers on the papers' journalistic qualities, analysts would want to distinguish texts by how their sources are situated. What content analysts compare—the hypotheses they test—in this design do not concern differences among textual properties but differences among the inferences drawn from texts, which are a function of the assumed context, not directly observed.

FIGURE 4.7 ■ Comparing Similar Phenomena Inferred From Different Texts

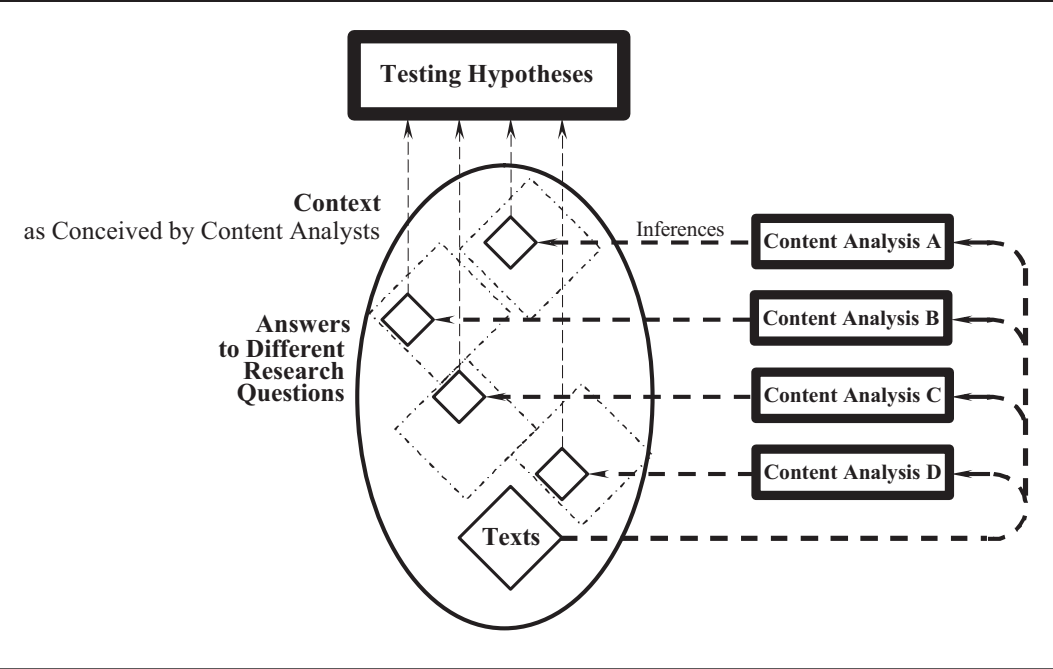


4.3.2 Testing Relationships Among Phenomena Inferred From One Body of Texts

In this design, the researcher analyzes one body of texts from different perspectives, with reference to different contexts, through different analytical constructs, or addressing different dimensions of meaning, and then correlates the results (see Figure 4.8). In behavioral research, such separately inferred phenomena tend to appear as different variables, which can be compared, correlated, or subjected to hypothesis testing. On a micro level, examples of such designs are found in analyses of attributions (multiple adjectives that qualify nouns), co-occurrences of concepts (inferred from word co-occurrences), KWIC lists (keywords in their textual contexts),

contingencies (Osgood, 1959), and conversational moves (adjacency pairs or triplets). On a macro level, examples include efforts to understand how public concerns—crime, environment, health, unemployment, and politics—compete with or stimulate each other in the mass media. Such designs also enable an analyst to compare different readings of the same texts by readers of unlike gender or from divergent socioeconomic, educational, ethnic, or ideological backgrounds. Here, the content analyst would define diverse contexts in reference to which texts are being read and analyzed.

FIGURE 4.8 ■ Testing Hypotheses Concerning Relations Among Various Inferences From One Body of Texts



4.3.3 Testing Hypotheses Concerning How Content Analysis Results Relate to Other Variables

Typically, this kind of design brings communicational or symbolic and behavioral variables together. For example, the cultivation hypothesis, which asserts that there are correlations between media coverage and audience perceptions, calls for comparing the results of a content analysis of mass-media presentations with interview data on audience members’ perceptions of everyday reality. Gerbner and his colleagues have explored the relationship between the “world of TV violence” and how TV audiences perceive the world outside television (see, e.g., Gerbner, Gross, Morgan, & Signorielli, 1995; and the debate reproduced in Krippendorff & Bock, 2009, Chapter 6.6). In comparing newspaper coverage of crime with crime statistics and public opinion, Zucker (1978) found that the frequency of crime reports in the media correlated more highly with public opinion than with official crime statistics. Conversation analysts usually are satisfied with their own accounts of what they see in the transcripts of naturally occurring conversations; thus their approach conforms to the design illustrated in Figure 4.8. However, if they were to relate their

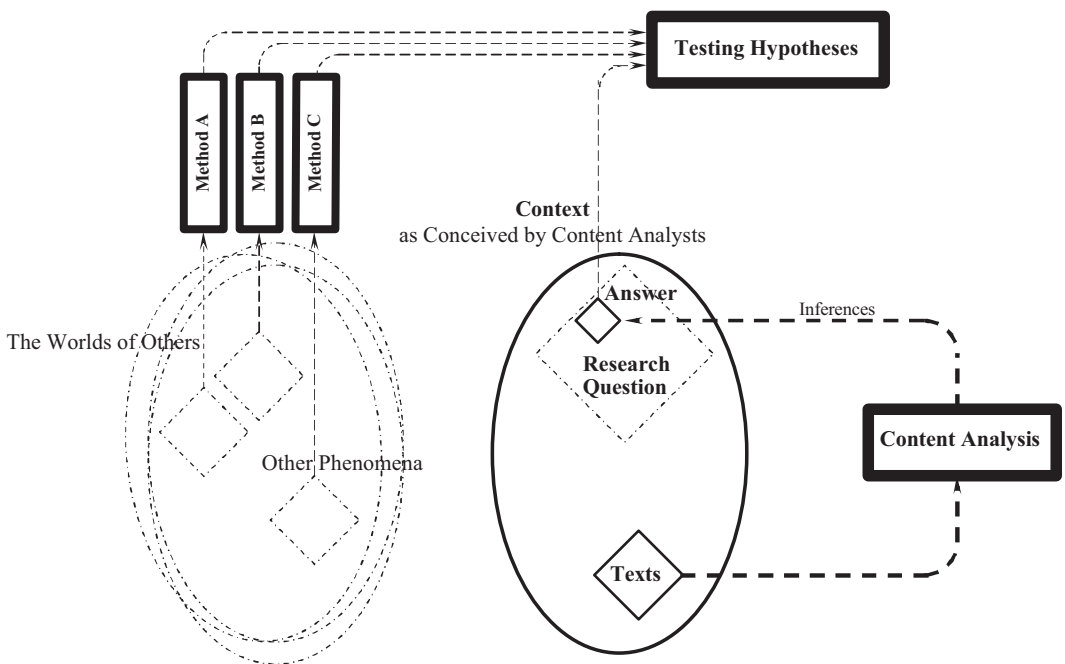
interpretations to participants' awareness of the phenomena being inferred, then they would compare inferences from texts with other accounts.

Such designs have three primary aims:

- To provide variables about the nature of communications that enable the testing of hypotheses concerning the causes, correlates, and effects of such communications
- To enrich indicators of observed behavioral phenomena by adding measures that concern the meanings of these phenomena (multiple operationalism), especially concerning individuals' perceptions or interpretations of social phenomena, which cannot be observed as such
- To substitute more economical measures for measures that are cumbersome (for example, using content analysis of TV news instead of surveys of what the public knows)

This design is represented in Figure 4.9.

FIGURE 4.9 ■ Testing Hypotheses Concerning Relations Between Observations and Inferences From Texts



I should emphasize that content analysts are not limited to the research designs distinguished above. Researchers can combine designs to obtain more complex forms that embrace many variables, and they can use any design in tandem with other techniques. There is no methodological limit to the use of content analysis in large social research projects.

5

UNITIZING

This chapter discusses the units of analysis used in content analysis: sampling units, recording units, and context units. It also addresses the purposes of unitizing and discusses five ways of defining units so as to increase the productivity, efficiency, and reliability of content analysis research.

5.1 UNITS

The first task in any empirical study is to decide what is to be observed as well as how observations are to be recorded and thereafter considered data. Empirical research needs to rely on a multitude of observational instances that collectively support an often statistical hypothesis or conclusion or that exhibit patterns that single cases cannot reveal. In unitizing, the researcher draws relevant distinctions within an observational field. This creates a multiplicity of observations, information-bearing instances, or units for short, and readies that multiplicity for subsequent analysis.

Generally, units are wholes that analysts distinguish and treat as independent elements. For example, in the operation of counting, the objects that are counted must be distinct—conceptually or logically, if not physically—otherwise the numerical outcome would not make sense. Thus we can count pennies but not water; we can count words or sentences but not text. The counting of meanings is problematic unless it is possible to distinguish among meanings and ensure that one does not depend on another. The wholeness of a unit of analysis suggests that it is not further divided in the course of an analysis or at a particular stage of an analysis. Physical definitions of units demand that their boundaries not overlap. Statistical definitions of units stress that “there is very little freedom for variation within [a unit] but much freedom at its boundaries” (Pool, 1959b, p. 203). However, in content analysis, units need not be motivated physically or statistically—these are mere options.

A political speech may serve as an example. Whereas ordinary listeners may well respond to the speech as a whole experience, being inspired or bored by it, or find the speaker impressive or unconvincing, political analysts may see the speech as addressing several distinct public issues. They may thus divide the speech into different parts and, ignoring the connections between the parts, probe, one at a time, how these issues are defined, what solutions the politician offers, and whether the discussions of the issues in the speech are used to attack, acclaim, or defend (Benoit, Blaney, & Pier, 1998). Linguists, in contrast, are likely to break the speech into sentences. Because there are no grammatical rules that would make one sentence construction dependent on another, there would seem to be no need to consider units

larger than that. Computer text analysis software that identifies character strings as words, on the other hand, would produce units of a vastly different size and kind. Linguists would have little interest in a list of words, just as political analysts would find a collection of sentences insufficient to help them understand what is important about a speech. Content analysts might collect not one speech but many speeches delivered during a particular political campaign and identify different kinds of units in them so as to compare them with one another or relate them to other data.

Although text, being ultimately composed of characters, would seem to be unitizable naturally, the unitizing of text poses many epistemological questions that I cannot attempt to address here. For purposes of this discussion, suffice it to say that units should not be considered givens. They emerge in processes of reading and thus implicate the experiences of the analyst as a competent reader. Units are often regarded as a function of the empirical tenacity of what is observed, but it is the act of unitizing that creates them and recognizes them as such. This act crucially depends on the analyst's ability to see meaningful conceptual breaks in the continuity of his or her reading experiences, on the purposes of the chosen research project, and on the demands made by the analytical techniques available to date.

5.2 TYPES OF UNITS

In content analysis, three kinds of units deserve distinction: sampling units, recording/coding units, and context units. I elaborate the analytical purposes and uses of each kind below, after which I discuss five different ways of defining such units.

5.2.1 Sampling Units

Sampling units are *units that are distinguished for selective inclusion in an analysis*. Someone who claims to have analyzed a certain number of issues of a newspaper talks of sampling units. These units may have been drawn from a larger population of issues of that newspaper, or they may include every issue ever published. I present a full discussion of sampling in Chapter 6; here, I want merely to address the purpose of sampling units relative to the other units. In survey research, where units tend to be individuals who are capable of answering questions, sampling from a population is straightforward, the difference between sampling units and recording units disappears, and context units are irrelevant. In content analysis, these three kinds of units have different functions.

The use of inferential statistics (*inductive* inferences from data, not abductive ones) to test statistical hypotheses, for example, is predicated on the mutual independence of sampling units. Frequencies, probabilities, and the computed likelihood that a sample represents a population would be meaningless without that mutual independence. Thus, survey researchers take great pains to ensure that their interviewees are not aware of each other's answers, experimenters make sure that the stimuli they manipulate are unrelated, and sociologists conducting research into causes make certain that their dependent and independent variables are clearly distinct. Without such precautions, the statistical correlations that result could be spurious and difficult to interpret.

The kinds of data of interest to content analysts are not so controllable. It is quite natural for people to create meaningful connections among almost any variety of

things, including among the units that a content analyst has sampled. When analysts sample issues of newspapers, for example, it could be argued that such issues are not truly independent because most news events unfold in time and over several issues, building on what was published previously; and thus issues of newspapers are not strictly independent of each other. Indeed, reading just one newspaper issue out of context—say, an issue from 50 years ago—makes one realize how little one can understand without knowing what happened before. Similarly, during an election campaign, the speeches delivered by candidates for political office tend to refer or respond to other speeches, and some may even result from strategic alliances among speakers. If such connections are relevant to an analysis, sampling separate speeches would not only prevent the researcher from recognizing the connections in the data but also confound the findings. Content analysts must define sampling units so that (a) connections across sampling units, if they exist, do not bias the analysis; and (b) all relevant information is contained in individual sampling units, or, if it is not, the omissions do not impoverish the analysis. It is not easy to break a highly interconnected stream of messages into separate sampling units.

5.2.2 Recording/Coding Units

Recording/coding units are *units that are distinguished for separate description, transcription, recording, or coding*. Whereas sampling units are distinguished for inclusion in or exclusion from an analysis, ideally in a way that acknowledges natural boundaries, recording units are distinguished to be separately described or categorized. Thus, recording units are typically contained in sampling units, at most coinciding with them, but never exceeding them. Holsti (1969), relying on the image of categorization, defines a recording unit as “the specific segment of content that is characterized by placing it in a given category” (p. 116).

The text included in any one recording unit need not be contiguous. Suppose an analyst samples fictional narratives with the aim of studying the populations of characters occurring in them. Narratives have definite beginnings and definite endings; thus, they constitute natural sampling units among which the analyst can easily make decisions regarding inclusion or exclusion in a sample, even without much reading. In a typical narrative, however, the characters are rarely dealt with one at a time, or one per paragraph, for example. They tend to interact and evolve over the course of the narrative, and information about them emerges in bits and pieces, often becoming clear only toward the end. To be fair to the nature of narratives, the analyst cannot possibly identify one unit of text with each character. Thus, information about a recording unit may be distributed throughout a text. Once the analyst has described the recording units, it is these descriptions, the categories to which they are assigned, that are later compared, analyzed, summarized, and used as the basis for intended inferences.

A good reason for choosing recording units that are significantly smaller than the sampling units is that sampling units are often too rich or too complex to be described reliably. For example, whole movies are conventionally labeled “documentary,” “fiction,” “comedy,” “tragedy,” “popular,” “class A (or B or C),” “R-rated,” and so on. Such categories are very superficial and say little about what movies of each type mean in the lives of viewers and other stakeholders. To capture meaningful variations among movies, analysts would surely need to have a rich vocabulary of descriptive

terms at their disposal. But so do film critics, who claim to be able to describe movies without ever expecting to agree among themselves on all details. To obtain reliable accounts of larger units of text, content analysts have found it convenient to describe smaller units on which they can more easily agree and then use analytic procedures to obtain descriptions of larger units. To ensure agreement among different analysts in describing the coding/recording units of a content analysis, it is desirable to define these units of description as the smallest units that bear all the information needed in the analysis, words being perhaps the smallest meaningful units of text.

Recording units may also be distinguished and described on several *levels of inclusion*. In recording newspaper data, for example, an analyst may have one set of categories for describing the newspapers included in the sample, such as cosmopolitan versus local, or having a certain circulation; a second set of categories for addressing the actual newspaper issue being included in the sample, weekday or Sunday edition, or consisting of a certain number of pages; a third set of categories concerned with a particular article printed in that issue, its writer or source, its placement in the issue (front, middle, or last page), and its length; and a fourth that is concerned with the individual propositions in that article. These multilevel recording units form inclusion hierarchies. Newspapers publish many issues. Each issue contains many articles. Each article is made up of a number of sentences or propositions. Ingenious definitions of recording units can open the door to many interesting content analyses.

5.2.3 Context Units

Context units are *units of textual matter that set limits on the information to be considered in the description of recording units*. In the above example of describing the characters in a narrative, a natural choice for the context unit might be the whole narrative in which the characters play the roles they do. However, when analysts are trying to ascertain particular characters' development—where the characters emerge, what they do, and what happens to them—maybe chapters would be a better choice as context units, as this choice would allow characters to vary in the course of the narrative. More generally, the meaning of a word typically depends on its syntactical role within a sentence. To identify which meaning applies to a word from a list of dictionary entries, one must examine the sentence in which the word occurs. How else would one know, for example, whether the word *go* is meant to denote a game, an action, or a command? Here, the sentence is the context unit, and the word is the recording unit. Sentences are the minimal context units for individual words, but sentences may not be enough. To identify the referent of a personal pronoun, for instance, an analyst may need to examine a few sentences preceding that pronoun. To judge whether a political commentary is positive or negative for a candidate, an analyst might need to examine even larger context units, such as a paragraph or a whole speech.

Unlike sampling units and recording units, context units are not counted, need not be independent of each other, can overlap, and may be consulted in the description of several recording units. Although context units generally surround the recording units they help to identify, they may precede the occurrence of a recording unit (as in the example concerning personal pronouns) or be located elsewhere, such as in footnotes, indices, glossaries, headlines, or introductions. There is no logical limit to the size of context units. Generally, larger context units yield more specific

and semantically more adequate accounts of recording units than do smaller context units, but they also require more effort on the part of analysts.

Geller, Kaplan, and Lasswell (1942) have demonstrated how the characterization of a recording unit (and ultimately the research results) depends on the size of the context unit. They had their subjects judge how positively or negatively “democracy” was presented in an article, using a sentence, a paragraph, three sentences, and the entire article as context units. Although the results from the four methods were generally in agreement as to the direction of the bias (favorable, neutral, unfavorable), they differed in extent. As the size of the context increased, the number of neutral evaluations decreased significantly. Evidently, the context of a symbol contains a great deal of information, especially concerning affect.

The size of context units also affects the reliability and efficiency of the descriptive effort. To describe the treatment of fictional characters in the context of a whole novel, an analyst would need to read the entire book first and then assign each character to the appropriate categories. Not only would this process be time consuming, but it would also invite unreliability, because different individuals approach a novel differently, and the analyst would have to keep the whole in mind when making judgments. Going through a document sentence by sentence, or watching one scene of fictional programming at a time (perhaps even in slow motion), or recording dramatic encounters within their immediate setting, or looking for the characterization of a concept within a context no larger than a paragraph might be more reliable and more efficient, provided what is lost is not too significant. The best content analyses define their context units as large as is meaningful (adding to their validity) and as small as is feasible (adding to their reliability).

Conceivably, context units could be defined so that the original text can be reconstructed without loss. This would require that all relevant information about the organization of the text be retained in the description of each recording unit, much as in a hologram. A complete concordance that retains the positions of all the words it lists, including function words and punctuation marks (which most concordances omit), would exemplify this possibility. Some computer text analysis software stores texts in this manner. However, as soon as fewer and more conceptual recording units are applied to a text and an analyst’s conception of the context of the analysis enters their description, this goal becomes less likely to be achievable. Reconstructibility, even when not perfect, preserves the possibility of analyzing data in several ways and on several levels.

Some authors who write on content analysis (e.g., Holsti, 1969) also mention *units of enumeration*. The importance given to these units derives largely from the early definitional requirement of content analysis that it be quantitative (Berelson, 1952; Lasswell, 1949/1965b), which simply meant that textual units had to end up being categorized or measured in numerical terms. In content analysis, quantities may be of three different kinds and uses:

- Quantities that are extraneous to but associated with sampling units, such as a newspaper’s circulation figures, a television show’s Nielsen ratings, or a movie’s box-office figures. These numbers are *descriptive*, here of *recording* units that happen to coincide with sampling units. The difference of describing units in numbers or in categories is not important methodologically.

- Quantities that measure a recording unit, such as the size of a photograph, the column inches of an article, or the length of a speech, or enumerate something contained in a recording unit, such as number of speech disturbances encountered within a period of time. These measures or counts are also *descriptive*, here explicitly of *recording units*, and do not need any special attention either.
- Quantities that result from the counting of recording units, especially within the categories to which they are assigned. These express the *sizes or magnitudes of classes of units*, whether they are expressed relative to a sample, within a hierarchy of recording units, or within cross-tabulations of coincidences. These numbers refer to classes that are formed in the process of analysis. They do not describe units of text.

In early content analyses, these rather different kinds of quantities were often confused. Researchers tried to satisfy Berelson's (1952) and Lasswell's (1949/1965b) quantification requirements without considering the functions the rather different quantities served. Quantities of the first kind tend to come into play, for example, when analysts are deciding on which newspapers to sample or how to weigh the importance of TV shows. I discuss these in Chapter 6. Quantities of the second kind are measures that are no different from coding recording units in numerical categories, using a yardstick instead of qualitative judgments, or recording a count rather than a category. Coders need to be instructed, and these measures are parts of a data language. I discuss these issues in Chapters 7 and 8. In contrast, quantities of the third kind are of no concern to coders. They emerge after the analyst has put recording units into categories or measured them, as a matter of convenience for summarizing data or applying suitable statistical techniques: cross-tabulations of frequencies, correlations, variance analyses, factor analyses, and so forth. These numbers are useful computational artifacts, and I discuss some of the statistical techniques for which they are useful in Chapter 10.

The three principal kinds of units distinguished above all serve different analytical functions. Sampling units are units of selection and may provide an analyst with a basis for judging the statistical representativeness of data. Recording units are units of description that collectively bear the information that content analysts process and provide the basis for statistical accounts. Context units are units that delineate the scope of information that coders need to consult in characterizing the recording units.

5.3 WAYS OF DEFINING UNITS

Content analysts identify units according to one or more of five kinds of distinctions: physical, syntactical, categorical, propositional, and thematic. I discuss each of these in turn below.

5.3.1 Physical Distinctions

Physical distinctions arise in *the use of mechanisms to sever a physical medium*. Being incapable of understanding and insensitive to meanings, yet repetitive and

systematic, mechanisms unitize by imposing their own structure onto the material being unitized. The units resulting from such a process coincide only accidentally with the discontinuities that humans typically recognize in a medium. An everyday example is the digitization of photographic images. If the resolution of a digitized photo is very fine, a viewer can distinguish between background and foreground and make out objects, people, or shapes. Newspaper images are of a quality barely above where a grid is noticeable but not too distracting. However, if a photo's resolution is very low, showing a grid of uniformly colored squares, the viewer is aware of the fact that the grid has nothing to do with the nature of the original image. Diagonal lines appear jagged, colors are averaged, and details that do not coincide with the grid disappear. Digitization is the result of a mechanism that recognizes neither wholes nor shapes but imposes its own distinctions.

In content analysis, physical distinctions partition a medium by time, length, size, or volume but not by the information it would provide analysts. Osgood (1959) sampled pages of Goebbels's diary. Ekman and Friesen (1968) used frames of film as their smallest recording unit. Dale (1937) analyzed newsreel film foot by foot, and Albig (1938) provided his observers with a clock and requested that they summarize each minute of broadcasting. Cappella, Turow, and Jamieson (1996) coded public radio broadcasts in 30-second intervals. Time units are also common in studies of interpersonal behavior (Weick, 1968). Similarly, in my own work, I have applied a grid to photographs of crowds to count how many people appeared in each cell.

5.3.2 Syntactical Distinctions

Syntactical distinctions are "*natural*" relative to the grammar of the medium of the data. They do not require judgments on meaning. Being "natural," they seem hardly questionable: books, issues of a newspaper, letters, poems, posters, theatrical performances, television shows. Their naturalness stems from the content analyst's familiarity with the data source's grammar, often because both share the same culture and find multiple ways to distinguish the same. For example, TV shows are listed in the *TV Guide*, separately produced, selectively watched, individually named, and so on. Such distinctions are reproduced in so many contexts that the analyst can hardly ignore them.

Words are the smallest and, as far as the reliability of their identification is concerned, the safest recording unit for written documents. Lasswell's (1941; Lasswell, Lerner, & Pool, 1952) World Attention Survey, many literary detection efforts (e.g., Mosteller & Wallace, 1964; Yule, 1944), the analysis of style (Herdan, 1960; Miles, 1951), the analysis of psychodiagnostic inferences (Dollard & Mowrer, 1947), and research on readability (Flesch, 1948, 1951; Taylor, 1953) all rely on words or symbols. For computers, words are easily recognizable, so they often serve as a first step in computer-aided analyses (see Chapter 11). However, there are other easily recognizable syntactically defined units in text: sentences, quotations, paragraphs, chapters, journal articles, monographs or books, series, collections, and so on. In the nonverbal media, we have such units as the acts in theatrical performances, news items in television broadcasts, and editing shots in films. In conversation analysis, we have utterances between turns of talk.

The so-called proximity operators of queries in text searches (see Chapter 11) define "natural" units as well, relying on typographical conventions. Thus, words may

be identified as character strings that are bracketed by blank spaces or punctuation marks, sentences may be identified as strings of words that begin with a capital letter and end with a period, paragraphs may be identified as text fragments bracketed by carriage controls (§), and chapters may be identified as text bracketed by headings or the end of a written work. These typographical definitions may not always coincide with the distinctions that knowledgeable readers make in text, but they clearly rely on the conventions of the medium under consideration.

A good example of unitizing visual media is found in Kracauer's (1947) analysis of propaganda films. Kracauer presented good reasons to reject the familiar dramaturgical distinctions of a film into sequences, scenes, and shots in favor of three classes of units of analysis:

- *The commentary*: separable verbal statements and occasional captions
- *The visuals*: picture units consisting of one or more shots that represented a symbolic unity of subject, place, time, or action
- *The sound track*: sound units whether sound effects, musical motifs, or songs

These three kinds of units are media-specific; Kracauer preserved their co-occurrence by quoting and describing them in spreadsheet-like scripts, and his analysis involved correlating them in time (see Krippendorff & Bock, 2009, Chapter 2.6).

5.3.3 Categorical Distinctions

Categorical distinctions define units by their *membership in a class or category*—by *their having something in common*. A common reference is typical: any character string that refers to a particular object, event, person, act, country, or idea. For example, “the 37th president of the United States” may also be referred to by personal pronouns “he” or “him,” as “the first U.S. president to visit China,” as “Richard M. Nixon,” as “Tricky Dick,” or as “the occupant of the White House between 1969 and 1974.” All of these character strings designate the same person. If U.S. presidents are the target of a content analysis, units may be defined in terms of all of their manifestations. Whether it is direct or indirect, which grammatical form it takes, or the political perspective it entails then becomes secondary to the definition of the unit. Aside from synonyms, categorical distinctions tend to rely on taxonomies. Coders ignore any subdivisions when they are asked to identify a politician, a nation, a hero, or an animal.

Categorical distinctions can also result from a theory that has been adopted for an analysis. For the psychiatric profession, mental illnesses are defined in the fourth revised edition of the *Diagnostic and Statistical Manual of Mental Disorders* (American Psychiatric Association, 2000), which provides the official recording units for psychiatrists to use in making insurance claims and justifying appropriate treatment. Sociologists may define a family as a group in which members are related through marriage or descendance. This definition may be at variance with how the members of a family define their family, but it may serve analysts well when they are recording how families appear in texts. Early content analysts defined symbols (usually single words) by their denotations but categorized them according to the values, attributes, and the qualifications associated with them (e.g., Pool, 1959a). For them, categorical distinctions required not only a theory-driven reference but also the presence

of adjectives (or else one could not describe them in the proper terms). A content analysis of how doctors are portrayed in nonmedical literature (Posen, 1997; Turow, 1989) has to rely on recording units that contain information about doctors and may not apply institutional definitions but popular versions of “doctor.” Bales (1950), studying interactions in various small groups, defined twelve interaction categories and instructed coders to identify units as the smallest contiguous segment that could be described in his terms. When an Internet search engine retrieves articles according to a query, it effectively identifies units that meet one or more search criteria, which presumably embody the relevant criteria for a category of meaning. McMillan (2000) describes the challenges of unitizing the World Wide Web.

5.3.4 Propositional Distinctions

Propositional distinctions delineate units according to *particular constructions*, such as those that have a particular *propositional form* or exhibit certain *semantic relations* between conceptual components. For example, in his proposal for a linguistic content analysis, Roberts (1989) suggests using clauses as units. He defines clauses as sentences that include an inflected verb and, optionally, a subject, object, and related modifiers. The coder would have four types of clauses from which to select: perception, recognition, justification, and evaluation. A perception clause describes an activity (e.g., “Businesspeople vote predominantly Republican”). A recognition clause classifies a phenomenon as belonging (or not belonging) in a category (e.g., “He is a politician,” or “This is not a scientific statement”). A justification clause claims that an action is reasonable or unreasonable, and an evaluation clause asserts how well a phenomenon fits a particular category.

Another example comes from Osgood, Saporta, and Nunnally’s (1956) evaluative assertion analysis, which I describe further in Chapter 9. All verbal material that enters this form of analysis must be reduced to a set of propositions of two kinds:

Attitude Object / Verbal Connector / Common (Evaluative) Meaning Term
 Attitude Object₁ / Verbal Connector / Attitude Object₂

Because natural discourse rarely comes this way, Osgood et al. felt the need to develop and publish explicit rules by which analysts could decompose compound sentences into these basic kernels. According to these rules, the sentence “He had AIDS and lived in fear of not being able to finish his only novel” would become four units:

he / has / AIDS
 he / is writing / (his only) novel
 (the) novel / is / desirable
 (the) novel / may not be / completed

One may note that the word *fear* is omitted from this kernelization, but the concept of fear is represented in the last two forms. Conventions vary regarding how many liberties analysts can take in this process, including whether or not they may add implicit propositions. What the personal pronoun *he* represents in the sentence

above would have to be obtained from the linguistic environment of the sentence. *He*, *AIDS*, and *novel* are attitude objects. *Has*, *is writing*, *is*, and *may not be* are verbal connectors, the first three associative, the last one dissociative. *Desirable* and *completed* are common meaning terms of a favorable nature.

Similarly, in analyses of human interactions, including conversation analysis, it is common for researchers to decompose long sequences of verbal exchanges into collections of so-called interaction triplets. In an interaction between A and B, for instance, an interaction triplet would consist of the following:

A's utterance

B's response to A's utterance

A's acceptance or rejection of B's response to A's utterance

It is presumed that any interaction sequence between two people, A and B, can be analyzed into a series of such triplets, from which the original sequence can be reconstructed without loss. Although reconstructibility is neither a goal nor a requirement of content analysis, it is worth noting that a mere collection of the statements made by A or B would lose information on the interaction and prevent a reconstruction of the interaction sequence.

Propositions are elementary statements—basic sentences, complete claims, whole assertions, not yet analyzed—that can be strung together with the logical connectors *and* or *or*, much as a text can be constructed through the sequencing of separate sentences. Conversely, a compound statement can be thought of as decomposable or analyzable into basic propositions or kernels.

5.3.5 Thematic Distinctions

According to Smith (1992a), “The term *thematic* . . . connotes the analysis of storylike verbal material, and the use of relatively comprehensive units of analysis such as *themas* (Murray, 1943), *themes* (Holsti, 1969), . . . combinations of categories (Aron, 1950)” (p. 4), motifs (Thompson, 1932), imagery, and thoughts. Through their work on Thematic Apperception Tests, Smith and his colleagues realized the virtue of unitizing freely generated narratives thematically and analyzing them toward ends that are comparable to the ends of such tests.

For example, in their attempts to infer the achievement motives of subjects, McClelland, Atkinson, Clark, and Lowell (1992) start by searching stories for statements of goals, assertions of individuals' being or failing to be successful in competing with other individuals according to their own standards of excellence. These researchers then characterize the individuals in the stories as having *needs* or *motives*, as anticipating the accomplishment of *goals* or the frustration of failing to reach their objectives, and as engaging in *instrumental activities* in attempts to reach their goals, which may be either *blocked* by obstacles or *helped* by circumstances or other people, resulting in negative or positive feelings. The rearticulation of any part of a story in these terms constitutes a thematic unit. McClelland et al. then score the thematic units to infer the achievement motive of the subject—but recording/coding them is another matter.

Another example of the use of thematic units is Katz, Gurevitch, Danet, and Peled's (1969) analysis of letters of appeal to shed light on how emigrants use Israel's

administrative services. These researchers defined their thematic units as requests to authorities for favors or exemptions that included as constituent elements descriptions of the writers' personal qualifications and the reasons their requests should be granted.

Thematic unitizing of folkloristic material goes back to Thompson (1932), who listed and described motifs that fill six large volumes and aimed at an exhaustive coding scheme. Armstrong (1959) reviewed some of the problems with using thematic units in folkloristics. Almost 20 years later, the Council on Interracial Books for Children (1977), treating historical writing as a kind of folklore, published a list of sexist and racist themes—themes in which recurring stereotypes, distortions, and omissions are manifest—that the council's researchers applied to U.S. history texts.

Holsti (in North, Holsti, Zaninovich, & Zinnes, 1963, p. 137) instructed coders to edit and rephrase political documents in terms of an action framework containing the following components:

- The perceiver and incorporated modifiers
- The perceiver other than author of the document and incorporated modifiers
- The perceived and incorporated modifiers
- The action and incorporated modifiers
- The object acted upon (other than an actor-target) and incorporated modifiers
- The auxiliary verb modifier
- The target and incorporated modifiers

Accordingly, his actor-action-target theme has up to seven components, and the textual material that specified these units could be distributed over several sentences. Readers of political accounts seem to have no difficulty thinking in these terms. (For other operationalizations of actor-action-target units, see Heise, 1995; Kleinnijenhuis, 1990; Kleinnijenhuis, De Ridder, & Rietberg, 1997. See also Chapter 11, section 11.4.3.)

Although the choice of units is always dictated by the purpose of an analysis, because of the descriptive richness of thematic units and their link to readers' understanding, many content analysts with representational aims find thematic definitions of units attractive. However, because thematic units may have to rely on textual features that are distributed throughout a text, even carefully trained coders can easily be led in different directions, making reliability difficult to achieve. Themes, even when they are relatively formalized or limited in scope, are not as easily analyzed as simpler units.

5.4 PRODUCTIVITY, EFFICIENCY, AND RELIABILITY

The five ways of defining units just discussed differ mainly in the kinds of cognitive operations coders must go through to identify units within a text. Generally, the simpler and more "natural" these operations are, the more efficient and reliable unitizing is, and the easier it is for the analyst to formulate reliable instructions and program computers to aid in the task. But simple units may not be the analytically

most productive ones. Content analysis researchers have to optimize productivity without losing too much in efficiency and reliability.

Physical distinctions emerge from strictly mechanical operations, as noted above. Mechanical devices are reliable by nature, so when humans identify units in physical terms, errors arise mainly from the careless application of formal rules. However, physically distinguished units are totally oblivious to meanings. When the material to be unitized requires a complex reading and relevant meanings span several units, physical distinctions can discard relevant information and invite unreliability into subsequent coding. For example, coders of 30-second intervals of talk on talk radio shows may end up coding incomplete utterances or finding the answer to a question in different 30-second intervals that, because they are separate units, may no longer be seen as connected. Seconds of talk are unnatural units, much as are lines of printed text or the individual frames of a movie. Physically distinguished units may be better suited to definitions of sampling units: time periods in years, articles containing keywords, or sampling every fifth issue of a daily newspaper, for example. Such uses may not interfere much with how meanings are read.

Syntactical distinctions tend to be efficient and reliable, but they are not always productive in subsequent analyses. This is especially so when the source employs units on various levels of inclusion, whereas the analyst operates on only one level. For example, if an analysis proceeds sentence by sentence, as is customary in linguistics, hierarchies of syntactical units are ignored: paragraphs making a point, chapters elaborating a topic, and a book addressing a theme. It may be unproductive to define sentences as units of text, at least by standards of the source's or other readers' conceptions. The use of context units that are larger than the recording units eases this problem by allowing at least two levels of information to enter the units' subsequent descriptions: information from the recording unit and information from its surroundings, both encoded in the recording unit. But the distinction between recording and context units may not capture how a source organizes its contents and how ordinary readers understand the text. Whether reference to a text's source or to specific readers is relevant to an analysis depends on how the analytical context of the content analysis is defined.

To make *categorical distinctions*, one must be familiar with the meanings of character strings, references of names, connotations of symbols, contents of short expressions, and the like. Because the definitions of such units depend on interpretations, the identification of units becomes unreliable when multiple interpretations are possible. A stretch of text may not be simultaneously a unit and not a unit. Context units can improve the reliability of unitization. (I should note that multiple descriptions of recording units are not excluded; however, my concern here is with distinguishing units, with deciding what is to be sampled, the size of the context to be consulted, and the recording units to be described.) In content analysis, categorical distinctions among units are most commonly used for defining sampling—but especially for defining recording units.

Propositional distinctions require considerable clarity about the formal requirements of an analysis, as illustrated by the examples given above concerning the extraction of rather specific propositional forms from complex or compound sentences. They call for familiarity with the logical syntax and semantics of source language

expressions and require mastery of the rules for kernelizing and rearticulating these expressions. Unless the latter are commensurate with the verbal production or thought processes of a source, propositional units often seem artificial and contrived although clearly focused on the analysts' purpose. Although propositional distinctions lead to very rich and interesting content analyses, the process of using such distinctions can be quite inefficient, largely because the identification of such units can be tedious, even when coders have well-formulated rules to guide them.

Thematic distinctions are rich in information and potentially very productive, and they would therefore be preferable to all other kinds of distinctions if their use did not make it so difficult to achieve reasonable levels of reliability. Research communities whose members have worked together and refined their methodology for a long time often report remarkable reliability in using thematic distinctions. This is the case with one group of scholars that has analyzed achievement motives for years, starting with McClelland's work in the 1940s (see Smith, 1992b). Beginners are often attracted to thematic content analysis because it seems to preserve the richness of textual interpretations, but they often fail to satisfy the reliability requirements and give up.

In unitizing, analysts aim to select the empirically most meaningful and informative units that are not only efficiently and reliably identifiable but also well suited to the requirements of available analytical techniques. To achieve these often-conflicting objectives, analysts must be prepared to make compromises. Most often, this means letting unreliable information go, unitizing by propositional distinctions instead of by thematic ones, unitizing by categorial distinctions instead of by propositional ones, or redefining the target of inferences so as to be able to make sense of the data.

Calculating the reliability of unitizing may not be simple (see Chapter 12, section 12.6), but this difficulty is no excuse for ignoring it. Bypassing the need for it, for example, by adopting physical definitions of units, may be superficially convenient but only at the expense of obtaining meaningful units of analysis, which ultimately reduces validity.

The distinctions introduced in this chapter are intended to optimize productivity, efficiency, and reliability, always with an eye to the ultimate validity of a content analysis.

6

SAMPLING

The universe of available texts is too large to be examined as a whole, so content analysts need to limit their research to a manageable body of texts. Although attempts to answer research questions from a limited set of data introduce the specter of sampling bias, it is possible to collect data by means of sampling plans that minimize such bias. This chapter extends the theory of sampling from populations of individuals to the sampling of texts. It discusses available sampling techniques and makes suggestions concerning how analysts can determine adequate sample sizes.

6.1 SAMPLING IN THEORY

Printing, sound and video recording, photocopying, word and image processing, digital storage media—from floppy disks to whole libraries—and the worldwide dissemination of texts in electronic form have exploded the availability of content-analyzable matter. Thus, when communication researchers ask questions that available texts could answer, they can easily become overwhelmed by volumes of relevant data. This situation creates a challenging problem for researchers: how to answer their research questions from a small body of texts. Even when researchers generate their own data—say, by videotaping speeches or verbal interactions—transcribing and analyzing such data in sufficient detail can consume 10 to 100 times the hours spent taping these situations. Researchers who are unfamiliar with the theory and techniques of sampling might realize at some point during data analysis that their task exceeds available resources and may be forced to terminate data analysis prematurely, leaving their results incomplete or biased by the researchers' own limitations.

Statistical sampling theory gained prominence when researchers began to address the problems associated with measuring public opinion through survey research. Survey researchers attempt to estimate the properties of an entire population by observing or asking questions of only a select subset of individuals drawn from that population. At one extreme, if all individuals in a given population were identical, a sample of one would be sufficient. This assumption guides much of engineering and consumer research, where the qualities of one product from an assembly line are tested and the results are assumed to be true of all the products coming from the same assembly line. At the other extreme, if each individual in a population were unique, no sample would be able to represent the whole population. A researcher

would have to study every member of the population. The challenges of sampling arise between these extremes. Usually, there are similarities *and* differences within any population, and research findings need to be perfect only within certain limits. A sample is said to be representative of a population if studying it leads to conclusions that are approximately the same as those that one would reach by studying the entire population. Thus, fundamentally, sampling theory is concerned with the ability to generalize the properties found in a sample to the population from which the sample is drawn. It relies on the law of large numbers to estimate the bias introduced by generalizations from inadequate sample sizes, and it provides justifications for several sampling techniques aimed at minimizing such biases.

Sampling theory, as outlined above, does not, however, fully map out the sampling problems that content analysts face. Four of its assumptions prevent its wholesale application to the *sampling of texts*:

- In the above-outlined sampling theory, *sampling units are individuals*—actual or, when applied elsewhere, metaphorical—that is, they are indivisible unities, independent of each other, and hence individually countable by their properties, opinions, or behaviors. Texts, in contrast, may be variously conceptualized and unitized. For example, textual units could be conceived of in terms of hierarchies in which one level includes the next (film genre, movie, scene, episode, encounter, shot, assertion/action, frame, and so on; see Chapter 5). They could be read as sequentially ordered events, jointly constituting narratives whose integrity would be lost if the components were permuted or as networks of intertextualities (co-occurring, making reference to, building on, or erasing each other). There is no single “natural” way of counting texts.
- In the above sampling theory, *the units sampled are the units counted*. In content analysis, this is rarely the case. Content analysts may sample letters, issues of newspapers, or time periods of movie production, but they find answers to their research questions by enumerating sentences, categorizing references, or interpreting the details of visual images. In Chapter 5, I distinguished between sampling units and coding/recording units. The sampling of sampling units constrains the sampling of recording units, which tend to be the ones counted.
- *Survey researchers control the questions* asked of their interviewees and *decide on the legitimacy of the interviewees’ answers*. By virtue of their membership in a population of interest, all sampled individuals are considered *equally informative* about the survey researchers’ questions. In contrast, the texts that content analysts use typically are generated for purposes other than being analyzed, and it is rare for different textual units to have equal relevance for a content analyst’s research question.
- Traditional sampling theory is a theory of representation, in the sense that the sample drawn from a population has the same distributional properties as the population. It offers all members of *that (single) population an equal chance of being included in the sample*. Content analysts, in contrast, have to consider at least two populations at once: the population of answers to a

research question and the population of texts that contains or leads to the answers to that question. Therefore, content analysts are rarely interested in accurate representations of the textual universe; rather, their concern is that the texts of interest are relevant to the research question and help to answer it fairly. Texts must be sampled in view of what they mean, the interpretations they enable, and the information they contain. Thus, content analysts *have to sample their texts to give their research questions a fair chance of being answered correctly*. Sampling from one population in view of another is a problem that differs radically from the problems addressed by statistical sampling theory.

6.2 SAMPLING TECHNIQUES APPLICABLE TO TEXTS

All content analyses are (or should be) guided by research questions. Sampling problems do not arise when analysts can answer their research questions by examining all texts of a particular population of texts, such as all of a given writer's works, all issues of a newspaper within a chosen period, all documents generated by a legal proceeding, the complete medical record of a patient, or all e-mails received and answered by a certain office, on a certain issue, and during a certain period in time. When researchers analyze a sample of texts in place of a larger population of texts, however, they need a *sampling plan* to ensure that the textual units sampled do not bias the answers to the research question.

Sampling from the Internet presents special problems. If one samples with criteria from offline sources (lists of URLs, directories, governmental agencies), sampling is relatively unproblematic, and traditional sampling techniques suffice. However, offline lists are typically out of date. Online sources are continuously updated and thus current, but they rarely appear in the form of lists from which to sample. One has to use search engines instead. Using criteria derived from a research question, one now faces the problem of finding sites with vastly different texts and graphics, hypertext-linked to each other and to other sites, and usually in the process of continuous reconstruction. A content analysis of online text usually represents what has already moved on. This property may be shared by many social phenomena, speeches, and political events, which come and go and can be sampled mainly from records. However, it is difficult, often impossible, to sample from websites with links to other websites, from a whole network of texts. The changing nature of texts on the Internet calls for sampling plans with narrow time windows. McMillan (2000) found that content analysts of Internet texts use between a couple of days and a couple of months to sample and code online texts. Rogers's (2002, 2004, 2010) webgraph analysis (see Chapter 11, section 11.3.4) reduces the time to sample linked websites to a few hours—without coding, however. Taking time to sample such texts causes problems of interpreting unreliabilities, not merely from coder disagreement but now also from changes in websites while their texts are coded by different coders, from unknown delays in how soon a new text is discoverable by a search engine, and from changing links, as in social networks—all of which increase the inability to determine which population a sample of online texts does represent. In addition, websites, blogs, and e-mail exchanges are vastly uneven in volume, format, and relevance.

Only when all sampling units are *equally informative* concerning a research question is sampling in content analysis the same as sampling in survey research. For such situations, statistical sampling theory offers three sampling techniques,

summarily called *probability sampling* because they are designed to ensure that all sampling units have the same chance to be included in the sample. In the following subsections, I describe these sampling techniques first. When sampling units are *unequally informative*, which is far more typical in content analysis than in survey research, the sampling of texts becomes a function of what is known about the distribution of information (content) within a textual universe. I describe below four sampling techniques that respond to this condition.

In addition to the distinction between equal and unequal informativeness of sampling units, there are situations in which researchers *know their populations of texts* well enough to *enumerate* (assign numbers to) or comprehensively *list* the members of those populations. Regular publications, for example, have sequential dates of publication that are known before sampling. Many institutions keep accounts of texts in various forms, including library catalogs; books in print; professional guides to scholarly journals; records of legal transactions; variously kept logs, diaries, chronicles, histories, and almanacs; reference works such as dictionaries and encyclopedias; and alphabetical directories. Among the many existing and widely used systems for enumerating texts are the International Standard Book Numbers (ISBNs) of books, Uniform Resource Locators (URLs) of Web pages on the Internet, telephone numbers, and product numbers in catalogs—all the way down to the page numbers of books, paragraphs, and propositions. The first four sampling techniques reviewed below rely on systems of this kind. (Enumeration systems may not be of uniform quality; for example, most URLs do not name active Web pages, and for some daily newspapers there may be publication gaps.)

A more challenging situation is one in which a population of text has a conceptual boundary but no enumerable members, for example, when a researcher is interested in information on a certain issue that could appear in a rather diverse population of texts. Cluster sampling, the fifth technique described below, is useful in situations where sampling units can be listed in larger chunks, or clusters. Cluster sampling also may be used in situations in which sampling units and recording units differ in kind and/or in number. Following the discussion of cluster sampling, I address three sampling techniques that deviate even further from the idea of selecting a representative subsample from a population. And the final technique discussed, convenience sampling, contradicts the most important features of statistical sampling theory.

6.2.1 Random Sampling

To draw a simple random sample, a researcher must enumerate (or list) all sampling units for potential inclusion in or exclusion from the analysis (issues of journals, authors, Web pages, speeches, turns at talk, sentences). The researcher then applies a randomization device—a device that grants each unit the same probability of being included in the sample—to the enumerated units to determine which will be analyzed. Throwing dice is one way of selecting units at random, but a random number table is more versatile. Often the population of all possible sampling units is not known, however, or listing them is impractical.

6.2.2 Systematic Sampling

In systematic sampling, the researcher selects every k th unit from a list after determining the starting point of the procedure at random. In content analysis,

systematic samples are favored when texts stem from regularly appearing publications, newspapers, television series, interpersonal interaction sequences, or other repetitive or continuous events. The interval k is a constant, so it will create a biased sample when it correlates with a natural “rhythm” in a list of units, such as seasonal variations or other cyclic regularities. For example, if a researcher examining issues of newspapers were to select every seventh day of the week, the *New York Times* science section, which is published every Tuesday, would be overrepresented if sampling commenced on a Tuesday and never included otherwise. For this reason, researchers should take care not to select every seventh issue of a daily publication or every even (as opposed to odd) turn at talk in two-person conversations. Hatch and Hatch’s (1947) study of marriage announcements in Sunday editions of the *New York Times* unwittingly demonstrated this kind of bias. The researchers systematically sampled all June issues between 1932 and 1942 and found an absence of announcements concerning marriages performed in Jewish synagogues; however, they failed to realize that all the issues sampled coincided with a period during which Jewish tradition prohibits marriages (Cahnman, 1948).

Systematic sampling can be applied to any kind of list; the units need not necessarily be consecutive events.

6.2.3 Stratified Sampling

Stratified sampling recognizes distinct subpopulations (strata) within a population. Each sampling unit belongs to only one stratum, and the researcher carries out random or systematic sampling for each stratum separately. Thus, stratified samples represent all strata either in equal numbers (i.e., in proportion to their actual size) or according to any other a priori definition, whereas the properties within individual strata are sampled without a priori knowledge. Newspapers, for example, may be stratified by geographic area of distribution, by frequency of publication, by size of readership, or by audience composition as obtained from readership surveys.

For many years, Gerbner and his colleagues analyzed a “typical week of U.S. television programming” each year, constructing that typical week through stratified sampling from the entire year’s programming by the three major TV networks (see, e.g., Gerbner, Gross, Morgan, & Signorielli, 1995; or Krippendorff & Bock, 2009, Chapter 6.6). The strata were the networks’ programming slots, much as they are listed in *TV Guide*. For each year, the researchers obtained a “typical week” by randomly selecting one out of the 52 programs aired over the year for each programming slot of each weekday. This “week” had no empty periods or duplications, and the sampling method granted each program aired on the networks the same probability of inclusion.

6.2.4 Varying Probability Sampling

Varying probability sampling recognizes that textual units are unequally informative about the answers to analysts’ research questions and so assigns to each sampling unit an individual probability of contributing to any one answer. In pursuit of answers to research questions about public opinion, for example, analysts may sample newspapers according to their circulation figures. In such a sample, large-circulation newspapers, which presumably affect more people, would have to be overrepresented relative to low-circulation newspapers in order for their contents

to relate to public opinion variables. Thus, when Maccoby, Sabghir, and Cushing (1950) were interested in the information that newspaper readers were exposed to, they listed all dailies within each of nine census districts (strata) in descending order of their circulation figures and assigned a probability to each newspaper according to its share in total circulation. Here, readership determined the likelihood that any given newspaper would be included in the sample.

Analysts may not find it easy to assign probabilities to sources of text in terms of their importance, influence, or informativeness. One strategy that has been used in such cases is to have experts rank the sources. In surveying psychological literature, for instance, Bruner and Allport (1940) enlisted professional psychologists to rank publications in order of their importance to the field. In studying newspaper coverage, Stempel (1961) relied on journalists. Some other kinds of evaluative sources that analysts might consult when sampling with unequal probabilities include best-seller lists, reviews (of books, plays, or films) in prestige journals, book awards, and lists showing frequencies of citations.

Researchers may also use varying probability sampling to reverse certain known statistical biases in representations of reality. For example, the mass media are likely to air the voices of celebrities and to suppress unaffiliated voices that may not fit the media's own conceptions of the stories being reported. To infer what might exist outside such selective reporting, an analyst might need to give the rare occasion of normally silenced views more weight than uninformed reiterations of mainstream ideas. For discussion of correcting biases in available sampling units, see Chapter 13, section 13.2.1, on sampling validity.

6.2.5 Cluster Sampling

Cluster sampling is the technique of choice when analysts cannot enumerate all units of analysis but find lists of larger groups of such units, or clusters. Analysts start by listing available clusters, then select among them randomly, systematically, or stratificationally and bring all units of analysis contained in those chosen into the analysis. In fact, wherever sampling units and recording units (see Chapter 5) differ, cluster sampling is taking place. Because the units that are contained in the sampled clusters are unknown, not only in kind but also in number, the probability that a particular unit will be included in an analysis depends on the size of the chosen cluster. In content analysis, cluster sampling is used far more often than many realize.

Since the early days of quantitative newspaper analysis, communication researchers have sampled among issues of newspapers but then measured, coded, and analyzed every article, paragraph, or proposition contained in the chosen issues. If such sampling is done correctly, every issue will have the same chance of being included in the sample. And if the sample is large enough, it should also accurately represent the population of newspapers from which the sample was drawn, but it will not represent the population of units contained in the newspapers, because the probability of particular units' inclusion in the analysis depends on such factors as where the newspaper is printed, which newspapers publish which kinds of articles, and which tend to reflect which kinds of perspectives, discourses, or attitudes. In content analysis, cluster sampling is convenient because text tends to be organized in relatively large units—journals containing articles, television shows featuring casts of characters, news broadcasts presenting issues, conversations occurring among

participants—that address different topics. Analysts handle these large units (each of which consists of material that was printed, recorded, or aired in one piece) as wholes; they give the units names or label them by dates, keywords, headlines, author names, or genres and catalog them for easy retrieval. The text’s constitutive elements, usually the primary focus of an analysis, thereby become secondary or implied by the way the large textual units, the clusters, are handled.

From the perspective of statistical sampling theory, the variance within cluster samples is likely to be exaggerated, and sampling error remains uncontrolled. In content analysis, where researchers choose texts according to the texts’ likely ability to contribute to decisions on rather specific research questions, sampling by clusters is more economical than sampling from a list of all available recording units. If the recording units are very unevenly distributed across the sampled clusters, the researcher will find it difficult to justify statistical generalizations about these units. However, because generalization is not a very important issue in content analysis, it is usually sufficient for a researcher to take precautions to prevent the uneven distribution of recording units.

6.2.6 Snowball Sampling

Snowball sampling is a multistage technique. Analysts start with an initial sample of units to which they repeatedly apply a given set of sampling criteria. This recursion produces a sequence of additions of sampling units that cause the sample to grow in size until a termination criterion is reached. A good example is the sampling of the literature on a particular subject. Researchers may start with a recent text, note its references, examine the cited works for their references, and so on. If the field examined is a close-knit one, the researchers will find themselves in a dense network of duplicate citations. Snowball sampling naturally terminates when the process generates no new references. In the case of a study of the content analysis literature, the trail stops with an obscure 1690 dissertation referred to by a historian of *Publizistic* (German for *newspaper science*) named Otto Groth (1948). One could complement this snowball sampling criterion by adding the requirement that the term *content analysis* be used and thus get up to 1941 (Waples & Berelson, 1941) as probably the earliest use of the term. This example illustrates snowball sampling that relies on citations of one work in another. The *Science Citation Index* (Garfield, 1979) has expanded snowball sampling of scholarly literature into the other direction, by iteratively generating lists of published articles in which particular works are cited.

Another example of snowball sampling can be taken from Web searches on the presence of political issues. One may start with the URLs of a few well-known, politically opposed websites—Democratic and Republican, pro-life and pro-choice, or pro-war and anti-war—list all URLs occurring in them, add to this list the URLs occurring in the latter, and so forth, perhaps limiting the sample to those websites that link to each other reciprocally, until the sample is large enough, even closed, and the issues the sites promote or reject can be analyzed comparatively.

Underlying all snowball sampling is the idea of intertextuality, the notion that units of text are connected, that they form actual or virtual networks within natural boundaries. The network of scientific references is just one example. The unfolding in time of a story in the news, which makes one news item dependent on a preceding one; the reproduction of information from news wire services to public conversations;

networks of literary relationships within which ideas but also plagiarisms travel; hypertext links connecting one text to another and one Internet site to another—all of these may be used as bases for snowball sampling. Sociologists have studied the effects of social networks, such as how the buddy system in an organization influences promotions, how a subject is able to get a message to a famous person via a chain of acquaintances, and how rumors spread. Analysts could use all such intertextualities to sample relevant texts naturally.

Snowball sampling starts with an initial set of sampling units, as I have noted—and it is important that researchers choose these units wisely. Snowball sampling ends when it reaches natural boundaries, such as the complete literature on a subject. When it reaches its boundaries, the importance of the starting sample diminishes in favor of the sampling criteria that recursively create the boundaries. (All rumors, for example, have origins, but their transmission quickly renders those origins unimportant. The limits that rumors reach have much to do with the networks through which they travel and the needs they serve in a population.) But snowball sampling can also explode, growing sample sizes exponentially, like an avalanche, in which case the researchers need to accept some constraints (e.g., requiring that chosen samples conform to more stringent inclusion criteria—that citations be multiple, for instance, not casual—or that the sample not exceed a manageable size).

6.2.7 Relevance Sampling

In the sampling techniques reviewed above, texts are sampled according to their sources, situations, time periods, genres, and intertextualities—all of these can be used without significant reading or analysis of the sampled texts. Relevance sampling, in contrast, aims at selecting all textual units that contribute to answering given research questions. Because the resulting sample is defined by the analytical problem at hand, relevance sampling is also called *purposive sampling* (see, e.g., Riffe, Lacy, & Fico, 1998, p. 86).

It is important to remember that the use of random samples always entails the admission that one does not have a clue regarding what the population of interest looks like or where to find the needed information. In content analysis, this is rarely the case. Cluster sampling already acknowledges that the universe of texts is partitioned onto large clusters and makes use of this knowledge. Snowball sampling presumes knowledge of the network-like organization of this universe of texts. When using relevance sampling, analysts proceed by actually examining the texts to be analyzed, even if only superficially, often in a multistage process. Suppose researchers are interested in alcoholism in the United States; more specifically, they want to find out what conceptions drive the use of alcohol on college campuses, what makes this a problem, and for whom. A random sample drawn from all that people read, write, and talk about would certainly contain answers to these research questions, but the task of sorting through the mostly irrelevant records in the sample would be a hopeless undertaking. Perhaps the researchers' first step in reducing the task would be to think about where they might find relevant documents and what those documents are likely to contain. When searching the Internet for *alcohol*, using the Google search engine, the researchers may find, say, 7,230,000 mentions of the word. They then narrow the search to find documents relevant to alcohol consumption, say, on campuses: "*alcohol* +

students” yields 1,140,000 hits; *alcoholism*, 658,000; “*alcoholism + students*,” 131,000; “*alcoholism + students + academic*,” 40,000; “*alcoholism + students + academic + rehabilitation*,” 10,500; and so on. Thus, the size of a universe of possible texts is reduced to a sample containing, ideally, a manageable number of relevant texts. Of course, relevance sampling is not limited to Internet searches, nor does it require electronic texts and their containing keywords as criteria for relevance. In the case of research into alcoholism on college campuses, possibly the most relevant data are recorded interviews of students by students, reports by student counselors, accounts of fraternity parties, and medical and police reports.

Relevance sampling is not probabilistic. In using this form of sampling, an analyst proceeds by following a conceptual hierarchy, systematically lowering the number of units that need to be considered for an analysis. The resulting units of text are not meant to be representative of a population of texts; rather, they are the population of relevant texts, excluding the textual units that do not possess relevant information. Only when the exclusion criteria have exhausted their ability to shrink the population of relevant texts to a manageable size may the analyst apply other sampling techniques. Issues of accurate representation may arise at that point but only relative to the relevant units from which the sample was drawn, not relative to the whole population of possible texts.

Relevance sampling is so natural that it is rarely discussed as a category of its own. It has motivated political scientists since Lasswell’s (1941) World Attention Survey, which compared the political climates of several countries; Lasswell restricted his analysis to the “prestige” newspapers in these countries (ignoring the “less influential” local papers). In a study of the coverage of foreign affairs during the 1990 U.S. congressional campaign, Wells and King (1994) used the same logic to limit their content analysis to the *New York Times*, the *Washington Post*, the *Los Angeles Times*, and the *Chicago Tribune*. They reasoned that these newspapers include extensive international coverage, have their own news-gathering abilities, and serve as the main channels of knowledge about other countries for U.S. political elites as well as other newspapers. Most researchers adopt some kind of relevance criteria for defining the populations from which they sample.

The problems associated with relevance sampling have gained in importance with the increasing use of very large electronic text databases and the Internet, where irrelevant texts are vast in number. Relevance sampling selects relevant data in ways that statistical sampling theory has not yet addressed.

6.2.8 Census

A body of texts that includes all of its kind is called a *census*. Studying the collected works of a particular author requires no sampling. The analysts may have to exert some effort to get hold of these works, but that is a clerical task; the analysts do not make any choices concerning what to include or exclude. For another example, if content analysts want to know something about the press coverage of a given event and collect all newspaper articles pertaining to the event, that complete set of texts constitutes a census. Because it is complete, the analysts have no need to expand the number of texts by snowballing, and if the set of texts is manageable in size, they have no need to reduce it by using relevance or random sampling.

6.2.9 Convenience Sampling

A convenience sample is motivated by analytical interest in an available body of texts that is known not to include all texts of the population that the analysts are concerned with. Such a sample is convenient in the sense that the analysts do not care to make an effort or find it too difficult to sample from that population. By proceeding from available texts without any sampling effort, analysts leave the matter of how and why the data—and which data—get into the sample to circumstances out of their control, to the interests of the texts' channels or sources, whether or not the latter are aware of how their texts will be analyzed.

The idea of sampling entails choosing to include or exclude data, with the intent of being fair to all possible data. Convenience samples do not involve such choices and leave uncertain whether the texts that are being analyzed are representative of the phenomena that the analysts intend to infer. Convenience samples may contain biases, or, worse, the analysts may be deceived or used by others in ways they may not understand. For example, personal diaries are written for many reasons, such as to preserve the writer's ideas for posterity, to impress a particular community, or to revise history in the writer's favor. Without the benefit of other corroborating texts and without knowledge of why the diaries were written, analysts of diaries may be unwittingly drawn into the project of the writer—which may be, for example, to assure the writer's place in history.

Examples of convenience samples are many: enemy broadcasts, which are produced for propaganda purposes; psychotherapeutic conversations, which contain only features that therapists and patients consider relevant to the therapy; and election campaign speeches, which are unlikely to mention issues, intentions, and knowledge that the candidates believe will cost them the election. Historical accounts rarely are fair representations of what happened (Dibble, 1963). All documents from which we might infer past events are those that have survived for particular physical, personal, political, and institutional reasons. Consider how few witness accounts are available from the victims of the Holocaust, the Napoleonic campaign in Russia, or the enslavement of Africans in America.

Convenience samples present content analysts with the potential problem of having to undo or compensate for the biases in such data, taking into account the intentions that brought these texts into being and into the analysts' hands.

6.3 SAMPLE SIZE

After an analyst decides on a sampling plan, the question that naturally follows concerns how large the sample must be to answer the research question with sufficient confidence. There is no set answer to this question, but the analyst can arrive at an appropriate sample size through one of three approaches: by reducing the research question so that it can be answered, given statistical sampling theory; by experimenting with the accuracy of different sampling techniques and sample sizes; or by applying the split-half technique.

6.3.1 Statistical Sampling Theory

As noted above, the sampling of texts may not conform to the assumptions of statistical sampling theory. Sampling units and recording units tend to differ. Texts

have their own connectivity, and recording units may not be as independent as the theory requires. Textual units tend to be unequally informative, and the researcher must sample them so as to give the research question a fair chance of being answered correctly. Nevertheless, there is one solid generalization that can be carried from statistical sampling theory into content analysis concerns: *When the units of text that would make a difference in answering the research question are rare, the sample size must be larger than is the case when such units are common.*

This proposition is illustrated in Table 6.1, which lists the sizes of samples required to “catch” rare units on different levels of significance. For example, assuming the probability of the rarest relevant instances to be 1 in 1,000, or .001, and the desired significance level of the answers to research questions to be 0.05, a sample of 2,995 would give the analyst 95% certainty that it includes at least one of these instances. This logic is applicable not only to the sampling of rare incidences but also to critical decisions. When an election is close and its outcome depends on very few voters, political pollsters need larger sample sizes in order to predict the results accurately than they do when candidates’ levels of popularity are wide apart. Although this generalization is sound, researchers who rely on the actual numbers in this table should understand that they derive from statistical sampling theory, from the binominal distribution in particular. Thus, an analyst should use these figures only if the assumptions on which they are based do not violate the research situation in major ways.

TABLE 6.1 ■ Sample Size: Least Likely Units and Significance Level (all sampling units informative)

		Probability of Least Likely Units in the Population				
		.1	.01	.001	.0001	.00001
Desired Level of Significance	.5	7	69	693	6,931	69,307
	.2	16	161	1,609	16,094	160,942
	.1	22	230	2,302	23,025	230,256
	.05	29	299	2,995	29,955	299,563
	.02	37	390	3,911	39,118	391,198
	.01	44	459	4,603	46,049	460,512
	.005	51	528	5,296	52,980	529,823
	.002	59	619	6,212	62,143	612,453
	.001	66	689	6,905	60,074	690,767

6.3.2 Sampling Experiments

Analysts may elect to experiment with various sample sizes and sampling techniques in order to find the combination best suited to answering their research questions. Stempel (1952), for example, compared samples of 6, 12, 18, 24, and 48 issues of a newspaper with issues from an entire year and found, when he measured the average proportion of subject matter in each sample, that increasing the sample size beyond 12 did not produce significantly more accurate results. Riffe et al. (1998, pp. 97–103) have reported replications of these early studies as well as the results

of experiments designed to determine how the use of different sampling techniques affects how well a sample represents a population. In one study, Riffe et al. used local stories printed in a 39,000-circulation daily over a 6-month period as the closest practical approximation to the population. They then drew 20 samples for each of three methods, selecting issues at random (random sampling), in fixed intervals (systematic sampling), and by constructing artificial weeks (stratified sampling) “with 7-, 14-, 21-, and 28-day samples.” The researchers defined sufficiency of a technique as follows:

A sampling technique was sufficient when the percentage of accurate sample means fell within the percentage for one and two standard errors found in a normal curve. In other words, if 68% of the 20 sample means fell within plus or minus one standard error of the population mean and 95% of the sample means fell within plus or minus two standard errors of the mean, a sampling technique was adequate. (p. 98)

Riffe et al. found remarkable differences among the methods:

It took 28 days of editions for simple random sampling to be adequate, and consecutive-day sampling never adequately represented the population mean. One constructed week adequately predicted the population mean, and two constructed weeks worked even better. . . . one constructed week was as efficient as four, and its estimates exceeded what would be expected based on probability theory. (p. 98)

It follows that different sampling techniques yield samples of different degrees of efficiency. It is wise, however, to be wary of unchecked generalizations. Different media may have different properties, and results like Stempel’s (1952) and Riffe et al.’s (1998) actually reflect measuring frequencies of content categories and may be generalizable only within a genre. If newspapers were to change their reporting style and feature, say, more pictures, many more sections, and shorter stories (as is typical among today’s tabloid papers), or if content analyses were to use measures other than proportions of subject matter or frequencies, the findings noted above may no longer be generalizable.

What is common to experimental generalizations regarding adequate sample sizes is the researchers’ approach, which involves these steps:

- Establish a benchmark against which the accuracy of samples can be assessed, usually by analyzing a very large sample of textual units, thereafter taken as the population of texts. Obtain the standard error of this large sample for the adopted benchmark.
- Draw samples of increasing sizes and, if appropriate, by different sampling techniques, and test their accuracy by comparing the measures obtained for them with the confidence interval of the benchmark.
- Stop with the combination of sample size and sampling technique that consistently falls within the standard interval of the method (see Riffe et al.’s [1998] criteria above).

Such experiments require a benchmark—that is, the results from an analysis of a reasonably large sample of data against which smaller sample sizes can be measured. Researchers can conduct experiments like these only when they have a reasonable idea of the population proportions and they intend to generalize statements about the minimal sample sizes needed. The former is rarely available, hence the following recommendation.

6.3.3 The Split-Half Technique

The split-half technique is similar to the experimental method described above, except that it does not require a population measure against which the adequacy of samples is assessed and does not allow generalizations to other samples drawn within the same genre. It does not even require knowledge of the size of the population from which samples are drawn. The split-half technique calls for analysts to divide a sample randomly into two parts of equal size. If both parts independently lead to the same conclusions within a desired confidence level, then the whole sample can be accepted as being of adequate size. Analysts should repeat this test for several equal splits of the sample, as it is expected to yield the same results for as many splits as are demanded by the confidence limit. If such tests fail, the content analysts must continue sampling until the condition for an adequate sample size is met.

7

RECORDING/CODING

In making data—from recording or describing observations to transcribing or coding texts—human intelligence is required. This chapter addresses the cultural competencies that observers, interpreters, judges, or coders need to have; how training and instruction can help to channel these to satisfy the reliability requirements of an analysis; and ways in which the syntax and semantics of data languages can be implemented cognitively. It also suggests designs for creating records of texts in a medium suitable for subsequent data processing.

7.1 THE FUNCTION OF CODING AND RECORDING

Research—that is, *re-search*—requires the ability to search repeatedly within available data, not only to assure that the claims advanced about them are warranted but also to enable other scholars to evaluate and even replicate the process. Clearly, re-search is possible only if its ground does not shift while being examined, which means that the data are recorded in a medium that is durable enough to withstand recurrent examinations. Human speech vanishes unless it is audio-recorded (taped) or written down (transcribed). Social situations are lost unless witness accounts of them are preserved. And even written texts and photographic images will defy content analytic techniques that cannot recognize at least some of their features. Transcribing speech, describing observations, creating field notes, interpreting messages, judging performances, categorizing television presentations—all of these are ways of recording or coding transient, unstructured, or fuzzy but otherwise perfectly meaningful phenomena into the terms of a data language, ways to make transient phenomena available to appropriate analytical techniques.

As Figure 4.2 indicates, recording/coding is one among several procedural components of content analysis. In practice, however, it represents a major problem for analysts, who must formulate recording instructions that they and other researchers can reliably execute. The recognition of the rather unique role that coding plays in content analysis explains why older definitions of the technique virtually equate content analysis with coding, as if this were all there is to it. For example, Janis (1943/1965) provides this definition:

“Content Analysis” may be defined as referring to any technique (a) for the classification of *the sign-vehicles* (b) which relies solely upon the *judgments* (which theoretically may range from perceptual discrimination to sheer

guesses) of an analyst or group of analysts as to which sign-vehicles fall into which categories, (c) provided that the analyst's judgments are regarded as the report of a scientific observer. (p. 55)

Another early characterization of content analysis comes from Miller (1951):

In order to handle larger blocks of verbal material in a statistical way, it seems necessary to reduce the variety of alternatives that must be tabulated. This can be accomplished by putting a wide variety of different word patterns in a single category. (p. 95)

Although Janis's conception of recording—categorizing sign-vehicles—is severely limited by the semiotic terminology of his time, he nevertheless acknowledges the role of specially trained analysts (as noted in Chapter 3) and different levels of what I refer to in this volume as *data languages* (see Chapter 8). Miller's assertion invokes measurement theory, the simplest form of which is categorization (Stevens, 1946).

Recording takes place when observers, readers, or analysts interpret what they see, read, or find and then state their experiences in the formal terms of an analysis; *coding* is the term content analysts use when this process is carried out according to observer-independent rules. The preference in the natural sciences for data making by mechanical instruments privileges the latter; thus, researchers attempt to formulate recording instructions that contain explicit and detailed rules that coders can apply reliably, just as mechanical devices would.

However, where texts and images are involved, or, more generally, where the phenomena of interest to analysts are social in nature, mechanical measurements exhibit serious shortcomings that only culturally competent humans can overcome. Notwithstanding the many advances that have been made in computer-aided text analysis in recent years (see Chapter 11), in most content analyses, researchers come to a point at which they need to rely on human interpretive abilities (Shapiro, 1997; see also Krippendorff & Bock, 2009, Chapter 4.3). This said, I use the term *coder* in this volume merely as a convenient designation for a person employed in the process of recording, transcribing, or coding observations, perceptions, and readings of texts. Coders are particular kinds of observers. They follow the recording or coding instructions that content analysts create to minimize idiosyncratic judgments in the recording process. Unrecognizable judgments create uncertainties and make it difficult to know what the records are about. Coding instructions provide rules for coders to follow and for analysts to know what the data they examine stand for. These rules do not deny the participation of widely shared and largely taken-for-granted human abilities. After all, even very strict instructions need to be read, understood, and followed by humans, and coders are humans even when they are asked to act like computers.

The recording instructions for a content analysis should spell out everything needed to replicate the analysis elsewhere and with equally qualified coders. In the following pages, I recommend that such instructions include specific information in four major areas:

- The qualifications that coders need to have
- The training that coders must undergo in preparation for the task of recording

- The syntax and semantics of the data language, preferably including the cognitive procedures that coders must apply in order to record texts and images efficiently and reliably
- The nature and administration of the records to be produced

Specifying the recording process is only one function of the instructions that content analysts need to develop. Another is assuring that the meanings of the resulting records are available to others, to their analysts, to the beneficiaries reported research, and to its critics who ought to be able to interpret what the content analysis inferred. The check marks on a data sheet, the numbers entered into boxes, the annotations written in the margins of a text, the transcription symbols used by conversation analysts, the scales used to indicate extents—all provide information as long as their connections to the recorded phenomena are clear. If a study's data-making instructions, the codebook, or scale definitions are lost—which sometimes happens—the data are left without their semantic relations to the phenomena to be presented in them, and the records that a study has generated are reduced to nothing more than a collection of meaningless marks or numbers—computable but no longer interpretable.

Probably, the most important motivation for generating data by relying on several human coders and making sure they follow the same rules is that content analyses have to process texts whose volume easily exceeds the ability of single scholars to read and interpret them. If everyone would be left to their own devices, the data they generate would be inconsistent and uninterpretable and could not possibly have social consequences.

7.2 CODER QUALIFICATIONS

The coders involved in a content analysis must have the necessary cognitive abilities, but what is perhaps more important is that they have appropriate backgrounds. In addition, the qualifications they bring to a content analysis must be shared by a sufficiently large population of potential coders and those who intend to make use of the data they generate.

7.2.1 Cognitive Abilities

Even where recording is reduced to coding—that is, to the seemingly mechanical application of stated rules for mapping textual units into the terms of a data language—coders must be capable of *understanding these rules and applying them consistently* throughout an analysis. Recording is a highly repetitive analytical task that requires strenuous attention to details. Not everyone is capable of maintaining consistency under these conditions.

7.2.2 Background

In selecting coders, content analysts should not underestimate the importance of coders' familiarity with the phenomena under consideration. In order to read and interpret texts, or even observe visual images, coders need a level of familiarity with what they are looking at that usually cannot be made explicit. Literacy, for example, is a social ability. It is acquired through a lifetime of using texts within

a particular community. It would be impossible to convey all that is involved in reading, observing, and understanding in a document or instruction. When it comes to interpreting what local folks are saying to each other, coders who lack familiarity with the local vernacular may feel that they are able to understand what is being said, yet their understanding may not correspond to how those who speak that vernacular conceptualize their world accordingly. We rarely know what we do not know, and we are generally unable to articulate how our understanding differs from that of others. The joint *familiarity* with a subject matter is one qualification that coders need to bring to a content analysis. Ultimately, the sharing of similar *backgrounds*—similar histories of involvement with texts, similar education, and similar social sensitivities—is what facilitates generating reliable data.

Even the most detailed recording/coding instructions take for granted that coders and content analysts are able to interpret the written instructions alike. To ensure high reliability of coding, it makes sense, therefore, for content analysts to employ coders from the same cultural/educational/professional background (Peter & Lauf, 2002). In analyses of therapeutic discourse, licensed therapists are an obvious choice. In literary applications, English majors are likely to do well, whereas in analyses of intricate visual images, graphic artists or connoisseurs of film might do better. The challenge for content analysts is to find *clear and communicable descriptions of coders' backgrounds* so that other analysts can select coders with backgrounds similar to those in the original research.

7.2.3 Frequency

Scientific research demands an intersubjective understanding of the process as well as of its results. Thus, content analysts must allow other scholars to replicate their analyses. Adequate instructions are easy enough to communicate, but coders are not. Analysts attempting to replicate previous research need to choose coders from the same population that provided the coders for the original research. To ensure the availability of potential coders, content analysts must make sure that the above-noted coder qualifications (suitable cognitive abilities and appropriate backgrounds) are common—that is, that they *occur with sufficient frequency* within the population of potential coders. If they do not, the recording process may not be replicable, and the research results become questionable. Any researcher who claims to be the only one who is capable of reading a text correctly in fact denies the possibility of replicating the research elsewhere. The requirement of frequency might be disheartening to those who consider themselves outstanding experts, but even experts need to be able to communicate, and the requirement that particular coder qualifications occur with the necessary frequency ensures that they will.

7.3 CODER TRAINING

Recording and coding are not natural and everyday activities. They may be motivated by abstract theory about the context of available texts or by the necessities of a complex research design. Coders may be asked to interpret texts in terms that are unfamiliar or difficult, even seemingly contrived or meaningless to persons without knowledge of the research questions. Although instructions ideally should be understood as

written, it is typical for content analysts to provide coders with additional training in using the recording instructions.

Content analysts have reported spending months in training sessions with coders, during which time they refined categories, altered instructions, and revised data sheets until the coders felt comfortable with what was expected of them and the analysts were convinced they were getting the data they needed. It is typical for analysts to perform reliability tests during the development of the coding instructions until the reliability requirement is met as well. Singer's (1964) report on his study of Soviet-American attitudes provides a good example of how definitions of a data language emerge during the training of coders:

The purpose of the study was to generate an accurate picture of Soviet and American foreign policy goals and strategies as far as they might be reflected in elite articulations regarding (A) the international environment, (B) the distribution of power, (C) the other's operational code, and (D) their own operational code.

The procedure followed two main phases: designing and refining our coding procedure and applying it. The first phase followed six more or less distinct steps:

1. The questions that seemed most germane to the study at hand were compiled. These were, of course, based on a multiplicity of sources: The author's general knowledge of the subject, the parameters of his own social science conceptual schemes, and those dimensions of foreign policy suggested by the writings and research of others in the field.
2. Once a tentative set of essentially a priori dimensions was set up and arranged, these dimensions were discussed, criticized, and modified by the author, his assistants, some consultants, and several professional colleagues.
3. This set of dimensions was then applied by the coders to a sample of the material to be coded, resulting in the deletion of some dimensions, the rephrasing of others, and the addition of a few new dimensions.
4. The author then re-appraised the dimensions and further tightened up the three categories under each dimension, in order to maximize mutual exclusiveness as well as exhaustiveness of the categories under each dimension.
5. The dimensions and their categories were then pre-tested by the coders themselves to ensure that:
 - a. The literature to be coded made frequent enough reference to the dimensions to be worth coding,
 - b. The dimensions themselves did not overlap one another (except in a few cases where some subtle shadings of attitude were being sought),
 - c. The dimensions themselves were clear and unambiguous enough to assure that independent coders would have a high agreement that a specific article should or should not be coded along that dimension,

- d. The three category alternatives under each dimension were as mutually exclusive as possible, yet exhaustive of the possible ranges of relevant response.
6. When the pre-tests had demonstrated (by agreement between two or more independent coders) that the dimensions and categories were adequately refined and clarified, they were settled upon as final. (pp. 432–433)

In this example, the analyst achieved closure. However, a methodological problem lies in the implicitness of the process. During the negotiations that Singer (1964) describes, the boundaries of categories shifted until their meanings could accommodate what Singer wanted and what the coders were able to code with agreement and ease. When coders participate in such conceptual development, it becomes difficult to know whether they have merely become more careful in applying the coding instructions as written or developed coder-specific and nonreplicable consensus on how to interpret them in order to achieve the needed level of agreement among them. Summarizing the use of content analysis in psychotherapy, Lorr and McNair (1966) observe the effects of such implicit adjustments on replicability:

Although most investigators publish respectable indices of inter-rater agreement in categorizing the responses, these are open to serious questions. Usually the published inter-rater agreement is based on two people who have worked together intimately in the development of a coding scheme, and who have engaged in much discussion of definitions and disagreements. Inter-rater agreement for a new set of judges given a reasonable but practical period of training with a system would represent a more realistic index of reliability. Trials with some existing systems for content analysis suggested that reliabilities obtained by a new set of judges, using only the formal coding rules, definitions, and examples, are much lower than usually reported. Often they do not meet minimum standards for scientific work. (p. 583)

Ideally, the individuals who take part in the development of recording instructions should not be the ones who apply them, for they will carry the history of their development into the process which other coders do not have access to and that future scholars who may wish to use the instructions cannot imagine much less replicate. The best recording instructions incorporate everything that has been agreed upon, everything that coders have been exposed to during training, every test applied to them to establish whether they are sufficiently qualified to be employed as code, and the final form of the coding instructions should be tested for their replicability with a fresh set of coders.

Coders need to learn to work independently of each other with the recording instructions as their sole guide. Relying on extraneous sources of information (e.g., about the development of the instructions or the intentions of the researchers), or consulting with each other about problems they experienced during the coding process and how they solved them, undermines the exclusive governance of the recording instructions and corrupts the independence of individual coders. Working independently is a precondition for the ability to interpret observed inter-coder agreement as indicating that the data making process is replicable. Extraneous

information and any help coders might receive from each other is not communicable to future coders. If analysts see the need to correct, amend, or modify the recording instructions for their project, they must assure that these changes are consistently employed and communicable, that is, put in writing for future coders to be able to replicate them. Also, if analysts see the need to provide coders with additional training material, they should report on what they provided them so that the calibration of coders can be replicated elsewhere.

To assure the replicability of given coding instructions, my colleagues and I once devised a detailed self-teaching program for the coders we hoped to employ in a study of television violence: Initially, the trainees were briefed about the clerical aspects of the task, the equipment they had to operate, and the test they faced to qualify. Thereafter, potential coders worked on their own, watching a preselected television show, identifying incidences of violence, the units of our content analysis, and applying the written coding instructions to generate records of what they saw. After they evaluated one preselected TV program, noted incidences of violence and recorded their categories on one data sheet, they had access to the ostensibly correct scores (established by a panel of experts). Comparing their own performance with what we had determined to be correct provided immediate feedback to the trainees and enabled them to reevaluate their conceptions before examining the second television show, and so on, until we could see that they could work on the television shows we intended to analyze by means of the data they provided us. This training program allowed coders to perfect their understanding of what we wanted them to do. It allowed us to monitor their progress and decide when trainees would be ready to code the television programs we intended to analyze. Some students progressed quickly, some required more time, and some were just incapable of becoming more reliable. A self-teaching program like this is easily communicable and replicable elsewhere and provided us the capable coders we needed to complete our project.

I should mention more humbling experiences with this program. There were a couple of incidences when coders told us convincingly that the standard we supplied was wrong. Even panels of experts can be mistaken. When this happened, we had to change the coding instructions and the training program and recode all television shows with the corrected instructions.

Early content analysts often devised and applied their own categories to the texts they were analyzing. Regarding the reliability of the resulting data, this practice is probably worse than allowing coders to consult each other on difficulties they experience. Publishing the categories used in their analysis is no substitute for demonstrating their replicability. Without comparing the use of these categories with independently working coders, reliability remains uncertain. Claiming to be an authority on the phenomena studied does not produce reliable data. In fact, self-applied recording instructions are notoriously unreliable. Content analysts should be able to find other coders who are able to understand and reliably apply the recording instructions before claiming that the data they generate do account for the phenomena under considerations.

7.4 CROWDCODING

Jeff Howe (2006, 2008) is credited with coining the term *crowdsourcing* for processes of outsourcing tasks, usually done by the employees of a company, to an anonymous

and typically large group of contractors who can perform them as well if not better, cheaper, and faster. Crowdsourcing may not be a substitute for the capabilities of employees. Its results may well exceed their abilities. Crowdsourcing has succeeded in accessing public sentiments, finding acceptable solutions to common problems, creating grassroots marketers for campaigns, and it has opened up even Fortune 100 companies to directions not imagined from their insiders.

Crowdsourcing presupposes access to crowds, usually via the Internet, and its success assumes that crowd members possess unequally distributed competencies that can be mobilized and combined to yield answers to questions not easily imagined by any one member of the crowd, least of all by outsiders. This “wisdom of crowds” is evident in numerous collaborations ranging from conversations in design offices, focus groups, political deliberations, peer reviews of academic publications, and markets regulating prices. But the systematic use of Internet users is new.

Content analysts’ initial excitement about the availability of big electronic data has become muted by the realization that tabulating and analyzing character strings of texts resulting from powerful search engines provides data stripped of their context and meanings. Interpretations of the results of their statistical analyses tend to be unclear, subject to debates and uncertain claims. The costs of training literary competent coders and the time it would take for them to code large volumes of textual data, easily retrieved from electronically available text, is generally unaffordable. The idea of outsourcing the task of coding texts to large numbers of capable readers—crowdcoding (Guo, Mays, Sameki, & Betke 2017)—has become an attractive possibility.

The empirical justification for attributing wisdom to crowds stems from the recognition that “we never think (and read) alone” (Sloman & Fernbach, 2017). Tapping into this wisdom would call for qualified readers to overcome their individual shortcomings by collaborations that could combine each other’s strengths, fill in evident gaps of knowledge, and correct apparent misconceptions, all of which could lead to collectively satisfactory interpretations of texts that exceed individual crowd members’ competencies.

To get a sense the potential of crowdcoding, consider how many people play intelligent games on the Internet that could be directed to gain collective insights. If an appropriately designed platform can motivate a crowd of 5,000 to label the about 400 million images on Google in just 31 days, as Ahn and Dabbish (2004) observed, imagine what collaborative crowds could contribute to answer content analysis questions on far smaller data. Wang (2017) reviewed the literature of collaborative crowdsourcing and concluded that while social media do have the potential to network people into teams, current crowdsourcing platforms rarely encourage collaborations among crowd members by limiting communication to coders and researchers. This does not take full advantage of the wisdom of crowds that resides in their diversity.

7.4.1 Methodological Advantages

Among the half a dozen crowdsourcing platforms that have been used for crowdcoding of selected texts, Amazon’s Mechanical Turk, *mTurk* for short, is favored by most content analysts. It recruits what are called “crowdworkers” on the Internet market place—anyone willing to examine given texts according to instructions

provided by what *mTurk* calls “requestors” and return them to the researcher for a small financial reward.

There is a major methodological advantage of outsourcing written coding instructions. Whereas the usually intensive training of a few research assistants almost inevitably leads to noncommunicable consensus among present coders on how the written coding instructions are to be interpreted and applied, rendering the observed intercoder agreement no longer interpretable as measuring the replicability of the coding process elsewhere, the agreement observed among crowdworkers is not so polluted. It measures the replicability of completely transparent coding instructions as members of a crowd read, interpret, and enact them. However, the very anonymity that provides this methodological benefit poses other problems to content analysts.

7.4.2 Efforts to Assure Crowdcoding Quality

An obvious but in due time diminishing problem is that *mTurk*-recruited crowdworkers have to be Internet users. This sampling bias disappears as more and more people connect with each other through the Internet.

Another problem is that the answers content analysts receive to judge the qualifications of volunteers for crowdcoding may well be self-serving if not dishonest. What works in favor of self-selecting quality coders is their awareness that the quality of their performance will be tested. The questions posed by the analysts may serve as clues to what is required of the volunteers, giving them the option of dropping out if they are not confident that they qualify. Qualifying questions typically aim at eliminating crowdworkers who are not native speakers of the language of the texts they are asked to code, are too young to understand what they are asked to do, or do not have the experiences needed to evaluate the texts in question. Avid readers of literature tend to be more reliable, especially when they are in command of relevant background knowledge. The latter can be tested by knowledge questions, for example, about nutrition, politics, education, legal issues, affirmative action, or whatever crowdcoding is intended to reveal. Weber et al. (2017), seeking to obtain reliable data on the moral foundation of political narratives, pretested their coders' political knowledge, moral intuitions, and political views by administering established knowledge indices in all three areas.

7.4.3 Uses of Reliability Tests

Besides the uncertain validity of self-reports, what qualifying questions cannot reveal are individual coders' stabilities of judgments, intellectual honesty, and commitment to doing a good job. People tend not to be aware of their own flaws, and when they are, are unwilling to report them. Moreover, there are some who do not really care about what they are paid to do and who respond to questions without much attention to the instructions or the nature of the texts to be interpreted. Whether their coding is the product of intellectual limitations, individual idiosyncrasies, or unwillingness to thoroughly attend to the assignments, this results in arbitrary coding and becomes evident in data with near chance observed disagreements.

In order to identify individuals whose participation is likely to introduce too much uncertainty or noise into the data, it is standard practice to put all candidates for crowdcoding to an initial test that compares coders' performance against known

results and hire only those whose coding does not deviate, or only minimally deviates from what it should be. Inasmuch as the researcher can estimate how much time a coder spends on a given task, it is also common to dismiss candidates who respond before they could have examined the texts, suggesting their responses do not have anything to do with the text. Lehmann and Malisazobel (2017) built a test question into each coding task, not recognizable by hired crowdcoders; and they disregarded all of their data when the crowdcoders failed to correctly answer that question.

In my experiences, content analysts are rarely infallible, as the above accuracy tests presume. Belief in the wisdom of crowds would suggest another way to separate reliable from unreliable coders without imposing a priori coding standards: Start with the coder pair with the highest α -agreement (see section 12.3). If this α -value is below a chosen standard, then it is the content analysts' task to improve the coding instructions from which crowdcoders work. If their α -value is well within what one is willing to accept as reliable, because α can cope with any number of coders, including their uneven participation, add the coder who does not affect, or only minimally affects the α -agreement of the now-expanded group. By iteratively expanding the group of coders in this manner until their joint agreement threatens to fall below what one is willing to accept, the result operationally separates the grain from the chaff.

Reliability measures can give rise also to predictive criteria for selecting qualified coders. Weber et al. (2017) performed a linear regression analysis of 9,869 pairwise reliability measures and the characteristics that each pair of coders shared, and they derived predictors that were specific to their project. Obviously, pairwise coder similarities, for example, older and gender-homogeneous coder pairs, are more reliable compared with those who had little in common. The possibilities of correlating reliabilities with shared coder qualities provides empirical evidence for what traditional content analysts, working with only a few coders, could merely intuit.

7.4.4 Limitations and Warnings

mTurk and similar platforms offer inexpensive and efficient ways to obtain content analysis data from typically large volumes of textual and pictorial matter. Julien Cornebise (personal communication, 2017) faced 7,572,703 binary judgments on 1,384,130 one square kilometer satellite images of war ravaged Darfur in Sudan, judged by between 1 and 56 coders. Hsueh, Melville, and Sindhvani (2009) extracted 631,224 snippets from over 500,000 blog posts spanning the activity of 16,741 political bloggers from August 15, 2008, to election day. These are big numbers indeed.

The reliability of such data depends on the shared abilities of crowd members to understand what they are asked to do, agree on their reading of the given texts, and record their interpretations. The key term in the previous sentence is *shared abilities*. Such abilities define the size of the crowd attended to. While most examples of crowdcoding draw on the vast community of Internet users, crowdcoders could also be invited from communities of relevant experts, scientists, designers, medical professionals, political activists, or particular consumer groups. However, if all Internet users are invited to collaborate in generating the needed data, their interpretations have to be common, limited to simple concepts, manifest contents as Berelson (1952) would have said—for example, whether the national flag appears on a picture, whether the dangers of smoking are mentioned, whether a text depicts a public issue as positive or negative, or whether the strengths and weaknesses

attributed to candidates for political office are explicitly mentioned. These are not only relatively small bits of information, usually far removed from the theories that social scientists seek to test; but by relying on individual crowdcoders, they do not really justify claims to have tapped into a crowd's wisdom.

Significant progress would come from the development of platforms for generating data that enable crowdcoders to work in teams or small networks; to able to communicate, correct or expand on each other's interpretations; and to address larger issues than any one crowd member could imagine.

Many researchers confuse the wisdom of crowds to come to judgments superior to those of its individual members, with mathematically created aggregates of diverse individual responses, taking majorities or computed averages as true values, as the gold standards, from which individuals' deviations are considered inferior. While there is undeniable evidence that individual estimates of lengths, weights, heights of buildings, and population sizes of cities tend to vary around their true values, justifying taking averages of crowd judgments, such evidence is limited to quantitative phenomena that have simple objective truths. When it comes to more complex social, political, or planning decisions, majorities and computed averages tend to be less meaningful. Mathematical aggregates ignore, and reliance on them dismisses, the importance of social diversity and the possibilities of communication to alter realities. While 51% of a population is an undisputable majority, taking it as the true characterization of that population would ignore or silence the remaining 49%. We know of the political implications of the indecisiveness of such proportions. The wisdom of crowds is manifest in developing conceptions, decisions, and actions that are superior to majorities or averages. Many content analyses do have political implications, and ignoring what crowds can do may not be wise.

In response to such frequent misinterpretations, section 12.4 develops agreement measures to assess three reliabilities for crowds: 12.4.1 the accuracy of or surrogacy for crowd members; 12.4.2, the accuracy of aggregates (majorities or averages); and 12.4.3, the decisiveness of majorities or averages in crowds.

7.5 APPROACHES TO DEFINING THE SEMANTICS OF DATA

The reliability of recording is greatly enhanced if the task that an instruction delineates is natural, relies on familiar conceptual models, and remains close to how the texts to be recorded would be read ordinarily.

The two requirements that categories be mutually exclusive and exhaustive (see Chapter 8) are important not only because of the syntactical requirements of subsequent computation; they are of semantic concern as well. Coders must be able to conceptualize clearly what they read. *Exhaustive* refers to the ability of a data language to represent all recording units, without exception. No relevant unit of analysis must be excluded because descriptive terms are not available. *Mutually exclusive* refers to the ability of a data language to make clear distinctions among the phenomena to be recorded. No recording unit may fall between two categories or be represented by two distinct data points. These two requirements assure that the resulting records represent texts completely and unambiguously.

A set of categories that lacks exhaustiveness may be rendered exhaustive through the addition of a new category that represents all units not describable by the existing ones. Such fail-safe categories typically are labeled “not applicable,” “none of the above,” or simply “other.” Because such categories are defined by their absence of categories that are informative to the research questions under consideration, they tend to contribute little, if anything, to an analysis.

Situations in which two or more categories turn out to be unreliable for being not mutually exclusive are difficult to resolve. The well-intended practice of adding such categories as “undecidable,” “ambiguous,” or “applicable to two or more categories” to a variable does not alter its categories’ fundamental indistinctiveness; it invites indecision on the part of coders and rarely renders a variable sufficiently reliable. When content analysts use such categories, they reveal more about their own unclear conceptions than about the properties of texts, and they bias their research results in the direction of easily describable phenomena. There is no real remedy for ambiguous conceptions.

Little has been written about how coders actually read texts or perceive visual phenomena in order to record them. Instead of outlining a theory of purposeful reading, or coding, here, I shall approach this problem from the other end, by distinguishing a few cognitive devices that analysts have used to delineate meanings within texts: verbal designations, extensional lists, decision schemes, magnitudes and scales, simulation of hypothesis testing, simulation of interviewing, and constructs for closure and inferences. Analysts may use these devices in their written instructions for coders with some degree of confidence that adherence to the instructions will yield reliable records.

7.5.1 Verbal Designations

It is most common for content analysts simply to name their categories, using verbal designations that are typical and widely understandable—ideally dictionary definitions, perhaps with some additional technical terms. For example, in English, gender is either *male* or *female*. Although we may therefore easily put living beings in either category, actual texts may not do us the favor of revealing what we need to know to make this distinction. Missing information on gender may call for the addition of a third gender category, such as “gender neutral” or “unspecified.” In recording the actions of TV characters, my colleagues and I have found many instances of roles unmarked by gender: babies, mummies, robots, cartoon characters, and abstract persons referred to by their dramatic functions, such as murderers, mayors, managers, and doctors. Ordinary language tends to provide no places for uncommon categories. Note the enduring discrimination against homosexuals and lesbians, due mainly to the rigid adherence to binary gender conceptions. Not only for the purposes of coding texts, analysts of social phenomena are well advised to expand conventional distinctions to include all conceivable alternatives. Even fiction creates unanticipated imaginations that may become real.

Single-word designations for categories—proper nouns—are easy to understand but often inadequate for recording more complex meanings. By using longer definitions of concepts, content analysts gain more freedom in asking coders to make theoretically motivated distinctions rather than common ones. Mahl (1959)

developed the following set of eight categories to identify indicators of psychiatric patients' anxiety levels. Note the absence of abstractions in these definitions:

1. *"Ah"*: A definite "ah" sound occurs in speech.
2. *Sentence correction* (SC): Any correction in the form or content of an expression within the word-word progression. Such a correction must be sensed by the listener as an interruption in the word-to-word sequence.
3. *Sentence incompleteness* (Inc): An expression is interrupted, clearly left incomplete, and the communication proceeds without correction.
4. *Repetition* (R): The serial superfluous repetition of one or more words, usually of one or two words.
5. *Stutter* (St).
6. *Intruding incoherent sound* (IS): A sound that is absolutely incoherent as a word to the listener. It merely intrudes without itself altering the form of the expression and cannot be clearly conceived of as a stutter, omission, or tongue-slip (although some may be such in reality).
7. *Tongue-slip* (T-S): Includes neologisms, transpositions of words from their correct serial position, and substitutions of unintended words for intended words.
8. *Omission* (O): Parts of words or, rarely, entire words are omitted (contractions are exempted). Most omissions are terminal syllables of words.

7.5.2 Extensional Lists

Extensional lists become important when the analyst's conceptions are difficult to communicate to coders. In such lists, the analyst enumerates all the instances that define each category. Extensional lists are essential to computer-aided text analysis, in the construction of computer dictionaries (see Chapter 11, section 11.4.1), in particular. Coders tend to find extensional lists awkward to use, and content analysts often find it difficult to anticipate in advance of an analysis all occurrences of the desired kinds. For conceptually difficult tasks, however, the use of extensional lists may be a technique of last resort.

An interesting example is provided by O'Sullivan (1961), who attempted to quantify the strength of relationships reported to hold between variables in theoretical writings on international relations. Prior commitments to factor analysis required him to conceptualize "strength of relation" as a "correlation coefficient" between two conceptual variables. The conceptual variables were easily identifiable, but early on, during coder training (all coders were well-informed graduate students), O'Sullivan realized that the idea of the strength of a relationship, expressed in words, was incompatible with the formal requirements of the statistical definition of a correlation. Mapping the former onto the latter was totally unreliable. After much experimentation, O'Sullivan came up with the following extensional lists for each of six correlation coefficients:

- 0.2 is less likely to; in certain situations induces; may lend some; may be due to; may be, to the extent that; can be used without; possible consequences seem to follow
- 0.3 has introduced additional; not merely a function of, but of the; is a factor of; will depend not only on but upon; depends in part on; possibility of
- 0.4 leads; is likely to be; tends to produce; would tend to; will tend to induce; tends to; tends toward; tends to introduce
- 0.5 makes it improbable that; strongly affects; is most likely to result from; is most likely to occur; creates essentially; depends primarily on; depend primarily on; is a major source of; creates a problem of
- 0.6 will heighten; requires at least; will enhance; necessitates; will determine; produces; depends on; is inevitable; produces; depends; is the result of; will reflect; will impose; prevents; will override; weakens; strengthens; offers maximum; will be less; will add to
- 0.7 will; any must first; are least when; as will be; puts a; has; is a; is less when there has been; if it is this is; there is; there has been, and is; is directly related to; will be enhanced in direct relation to; is inversely related to; will influence in direct proportion to; is directly related to; there is a direct relationship between; stand in marked contrast to; to the extent that; the longer the more; the greater; the greater the greater the more; the greater the less the greater; the greater the greater the greater; the greater the more; the wider the less; the more the less; the more the more; the more the larger the more; the more the greater; the more the less likely; more than; the wider the greater; the wider the more; the higher the greater; the longer the less; the shorter the greater must be; the fewer the greater; becomes more as the; is more likely to be the more; the less the fewer; the less the less; will be more the larger; the larger the more

7.5.3 Examples

It is rarely possible to anticipate all possible occurrences of concepts of analytical interest. Some have extremely complex structures. Anger, a simple and common five letter word, may be expressed in a variety of ways. Lakoff (1987, p. 360–415) notes that *Roger's Thesaurus* lists over 300 synonyms for anger, including a range of abstract metaphors conveying anger and a class of proverbial scenarios in which anger is manifest. To list these scenarios in a coding instruction is nearly impossible. One may ask potential coders to read conceptual treatises on the categories to be identified in text, such as Lakoff's on anger, but many coding tasks do not have such literature, and defining categories that help to answer a particular research question may well conflict with intuitions that coders bring to their assignments. Consider hard-core pornography. Everyone thinks he or she knows what it is when facing an example of it, but even the U. S. Supreme Court could not define it. In its famous 1964 decision,

Justice Potter Stuart admitted that he was unable to define it intelligently but did not doubt that “I know it when I see it.” Pornographic images may well be an extreme case, but there are other examples where the boundaries of a concept are not easily definable, yet demonstrable by examples—whether while training coders or in formulating coding instructions.

It is always helpful to accompany definitions of categories with examples for what they should apply to and what they ought to exclude. Defining concepts by example is complementary to defining concepts by listing the qualities that determine their membership in a category. We often do not know where one category ends and another starts. We know of ambiguous colors, talk of cross-over vehicles, and calling something bordering insanity. Rosch (1983) observed that we do not ordinarily conceptualize our world in terms of logically defined categories, in terms of sets of elements with shared attributes in well-defined borders. She convincingly distinguished concepts that are abstract from those we can visualize and the latter by qualifying our visualizations. For example, even children can imagine and draw a chair, a house, a person, or a spoon. They are basic prototypes. Some chairs are more typical than others. We never know all possible chairs but use a prototype of it to judge how closely something we see resembles it. The category of chair is defined by its prototype. There are subordinates to the prototype of chair, like baby chairs, dining room chairs, office chairs, or three-legged chairs. The chair prototype can become part of various superordinate and no longer visualizable categories, like furniture. Similarly, when we hear or read “bird,” we more likely visualize a robin, not a penguin, scientific definitions notwithstanding. The robin is closer to our prototypical bird. The best examples of a category should be close to their proto- or ideal type.

Offering visual, situational, or narrative examples in coding instructions facilitates the reliable coding of phenomena that are more complex than implied in lists of mutually exclusive names of categories. It calls on coders to do what actually is most natural in human cognition, to compare what they read to what it most obviously resembles, what it could mean, and how one could describe it to others. Relying on examples calls on content analysts to choose those that are close to basic prototypes and omit distracting details as possible (or leave them for distinguishing among subordinate categories).

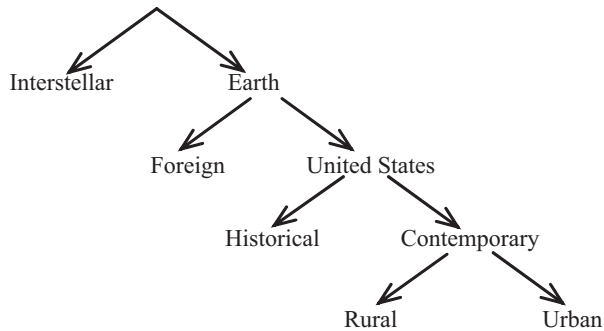
7.5.4 Decision Schemes

In a decision scheme, each recorded datum is regarded as the outcome of a predefined sequence of decisions. Decision schemes are uniformly reliable for four reasons.

First, it comes naturally to most people to organize complex judgments in terms of what has to be decided first, second, third, and so on. When coders take each of these steps with separate criteria in mind, criteria confusion is minimized.

Second, it is always difficult for coders to consider fairly large numbers of categories. As a rule of thumb, humans cannot keep the meanings of more than seven (plus or minus two) alternatives in mind simultaneously. Larger numbers encourage coding habits to form and allow preferences to develop. Following decision schemes can drastically reduce consideration of large numbers of alternatives to smaller numbers that coders can conceptualize and tackle sequentially.

Third, decision schemes can prevent unreliabilities due to categories that are defined on different levels of generality or that overlap in meaning. Schutz (1958) has demonstrated how the drawing of a decision tree can clarify the meanings of seemingly confusing categories used in a content analysis of comic strips: “United States,” “Foreign,” “Rural,” “Urban,” “Historical,” and “Interstellar” (Spiegelman, Terwilliger, & Fearing, 1953a). He organized the categories in terms of dichotomous decisions (and added two logically helpful verbal designations, “Contemporary” and “Earth,” to preserve the logic of these distinctions), which eliminated the confusion of logical levels:



Fourth, and finally, when recording involves several dimensions of judgments, decision schemes offer coders the opportunity to decide each separately.

Figure 7.1, which comes from Carletta et al.’s (1997) analysis of conversational moves, illustrates several of these advantages. As one can see, this decision tree has 12 terminal categories. If the analysts had attempted to write a definition for each—similar to, say, the above example from Mahl (1959)—the results would have been lengthy, confusing, and probably unreliable.

7.5.5 Magnitudes and Scales

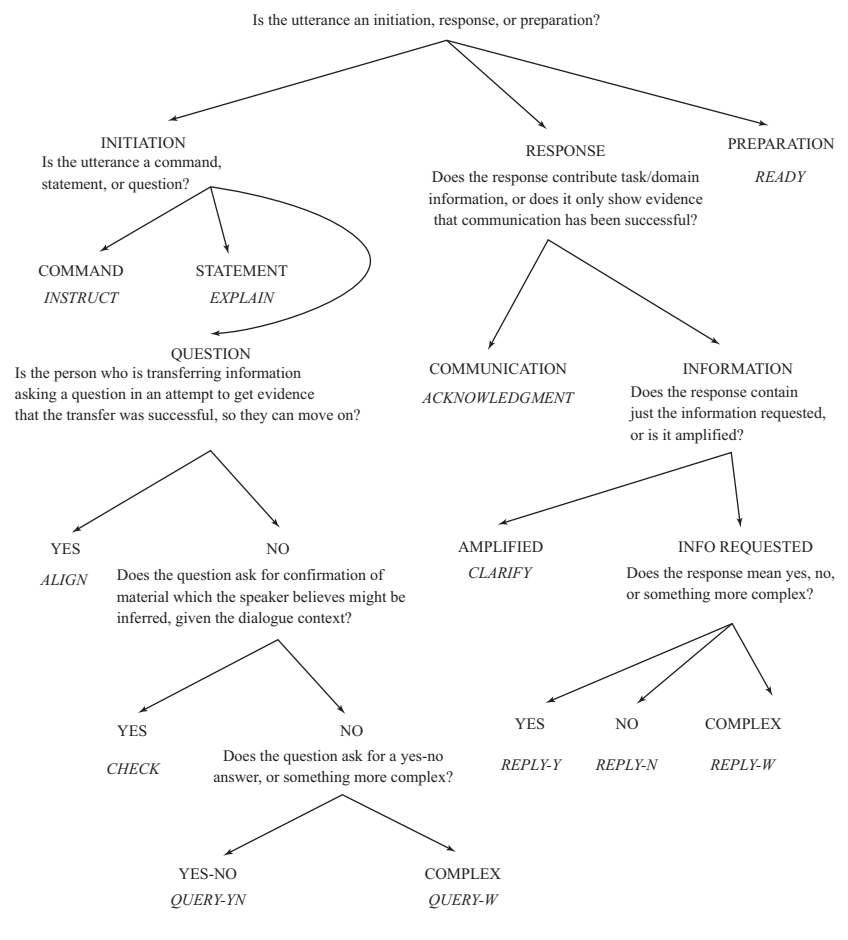
When magnitudes and scales are used as recording media, coders are expected to conceptualize the meanings of texts as continua, as having more or less of something, as possessing a metric. Osgood, Suci, and Tannenbaum’s (1957) widely used semantic differential scales serve as an example:

Good : ____ : ____ : ____ : ____ : ____ : ____ : ____ : **Bad**
Active : ____ : ____ : ____ : ____ : ____ : ____ : ____ : **Passive**
Strong : ____ : ____ : ____ : ____ : ____ : ____ : ____ : **Weak**

Semantically, each scale is anchored by the common meanings of two opposing adjectives. The intermediate scale points remain undefined except for the suggestion of equal intervals between the named extremes. Coders are asked to conceptualize a recording unit according to the semantic dimension that these opposites share and to judge its proper place along this presupposed continuum.

Seven-point semantic differential scales are widely used in psychological research, where experimenters can control their subjects’ responses. In this research, the above three scales—potency, activity, and evaluative—explain much of the variation in

FIGURE 7.1 ■ Categories for Coding Conversational Moves



Source: Carletta, Jean; Isard, Amy; Isard, Stephen; Kowtko, Jacqueline; Doherty-Sneddon, Gwyneth; & Anderson, Anne. The reliability of a dialogue structure coding scheme. *Computational Linguistics*, 23: 1 (March, 1997), pp. 13–31. © 1997 by the Association for Computational Linguistics.

what scholars in psychology refer to as human affective cognition (Osgood, 1974a, 1974b; Osgood et al., 1957). In content analysis, however, such scales are somewhat problematic. Text is not always scalable. Legal procedures distinguish between *legal* and *illegal*, and between these there are no intermediate points. The polarity of *news* and *fiction* may not be unidimensional, as a scale with these two words as endpoints would imply. And if coders are instructed to mark the midpoint of a scale whenever they encounter something that is not codable along such semantic dimensions, uncodability comes to be confused with perfect balance.

Demanding the application of a scale that does not resonate with the phenomena to be recorded causes meaningless data, which are manifest in unreliabilities. In content analyses, semantic differential scales turn out to be unreliable mainly when information about the attributes to be recorded is absent or unclear. For example, in fictional narratives, characters become known only in the dimensions that are relevant to their roles in that narrative. Fictional characters may not have all the attributes of real people. Naturally, the less that is known about a character, and the more coders need to guess, the greater the

unreliability of a scale that requires coders to make choices among polar attributes. In 1964, Zillmann introduced a scale that avoids this difficulty, his “semantic aspect scale.” It is a 7-point unipolar scale that ranges from zero, the absence of an attribute, to 6, the pervasive presence of that attribute. The use of such a scale is appropriate when attributes, qualities, or phenomena can be more or less, including absent—more or less significant to a character, more or less present in an assertion, or more or less frequent. The following is an example:

	Absent							Very much present							
Honesty:	:	0	:	1	:	2	:	3	:	4	:	5	:	6	:

7.5.6 Simulation of Hypothesis Testing

The recording devices discussed above rely on cognitive models of labeling, categorizing, deciding, and interpreting what is read within a framework of established natural language definitions. Simulation of hypothesis testing addresses a text’s presuppositions, implications, and omissions over and above its explicit meanings. For example, if someone is presented as a Swede, we might assume several things about that person: There is a good chance that he or she is Protestant, understands other Scandinavian languages, has not fought in or experienced war, is blond, enjoys nature, and so on. These characteristics may not be mentioned, can be presupposed, and would not be surprising if found true. A subscriber to the *New York Times* most likely knows English, is old enough to be interested in what happens in society, has some political sophistication and cosmopolitan interests (the newspaper has no comics page!), and so on. These are presuppositions, things that “go without saying.”

Television commentators on the speeches of political leaders, for example, are adept at revealing the speeches’ implications, which might escape ordinary audience members’ attention otherwise. The implications of a political speech are not only more interesting than what was said, they usually are the point of the speech. Of particular interest are omissions, what a politician should have said but did not, what was conveniently left out—for example, about contested issues such as abortion, women’s rights, gay marriage, religious commitment, or the candidate’s own less desirable history—that could cost the candidate reelection. Such omissions say a lot about the climate of political expectations and about how the candidate perceives the might of his or her community. One cannot count what is not there, but one can ask content analysts to address such implications of language use.

In pursuit of such implications, it would be impossible for content analysts to list all conceivable interpretations or omissions reliably, but it is quite feasible for them to ask coders the more limited question of whether a textual unit can be read as supporting or opposing a stated set of alternative propositions. These propositions function similarly to a set of hypotheses about what a text tells the coder, who records his or her judgment of *each* textual unit in categories such as these:

- a. Affirmed
- b. Implicitly affirmed by not denying the proposition when it would have been easy to do so (e.g., by not arguing against it or opposing alternative propositions)

- c. Neither affirmed nor denied—irrelevant
- d. Implicitly denied by not affirming the proposition when it would have been easy to do so (e.g., by not arguing in favor of it or talking about alternative propositions)
- e. Denied

In fact, answering such questions is the qualitative analogue of testing statistical hypotheses. It proceeds by verbal logic—the truth of each proposition is rejected by counterexample, by disproof, or by evidence in favor of the opposite—not by the frequency of confirming cases. Such a testing of mutually exclusive propositions (hypotheses) is nothing but a disciplined way of recording what ordinary readers do when reading, say, a detective story, weighing the evidence against each of a set of suspects. Back to the point, this cognitive device calls on coders to look for any evidence, within a specified context unit, for whether a recording unit speaks in favor of or against either of the stated hypotheses—for example, when scanning the literature for statements about the connection between smoking and lung cancer, the connection between oil consumption and global warming, the connection between hate talk and ethnic violence, the connection between the curbing of civil liberties and homeland security, or the attribution of guilt and innocence following a human-made disaster.

A classic example of the simulation of hypothesis testing is found as early as in Lasswell's (1965a) effort to detect foreign propaganda in domestic German broadcasts during World War II. Lasswell presumed that the Nazi elites pursued four basic propaganda aims, stated them in his terms, and asked his coders to judge whether radio news items, public pronouncements, and commentaries about events implicitly supported or undermined any one or more of these aims. Coders could thus record what was not explicit, insinuated, or implied, as long as it was relevant to the propositions. A hypothetical example of a beneficial use of this recording device would be an analysis aimed at inferring ethnic prejudices from writings by authors on entirely unrelated topics. For the past 50 years, authors have rarely expressed ethnic prejudices explicitly—in fact, hate talk is a crime in many U.S. states, so those who hold such prejudices are forced to express them indirectly or to hide them deliberately in their constructions.

Given the implicitness of much of ordinary writing, this underused recording strategy should appeal to psychotherapists, who must attempt to diagnose their patients' mental illnesses; to political analysts, who look for the public implications of campaign speeches; to public opinion researchers, who seek to understand the public perceptions of particular events; and to medical discourse analysts, who attempt to ascertain the cognitive models that underlie patients' accounts of their illnesses.

7.5.7 Simulation of Interviewing

Interviewing is a way to come to know other persons—their beliefs, attitudes, and expectations—and to understand the cognitive models that shape their worldviews. It is a common device used by journalists and public opinion researchers for information gathering. In practice, interviewing is limited to people who actually are available to answer questions, which excludes historical figures, people who do not have time to answer detailed questions, and people who prefer to hide behind their writing.

The simulation of interviews from available texts offers content analysts a means of obtaining answers to questions that they could conceivably have asked the authors of these texts, had the authors been accessible. When using this device, content analysts have coders start by familiarizing themselves with a particular author's writing, a book or article, which, having been written by one individual, would be the recording unit. Then coders go through the author's text a second time, this time looking for evidence from anywhere within it that would indicate how the author might feel about certain issues and how that author would be likely to answer the content analysts' questions.

A good example of the use of such a simulation is found in Klausner's (1968) content analysis of a stratified sample of 199 out of 666 child-rearing manuals published in the United States over a period of two centuries. The attitudes toward child rearing and conceptions of child-rearing practices of each of the manuals' authors were recorded in terms of sets of predefined answers to 80 questions. One of these questions and its possible answers was as follows:

Question 32: **How does the book legitimate the authority of the parent in the parent's eyes? (The basis on which the author appeals to parent to attend the child)**

- Answers:
- 1 Not discussed
 - 2 Legitimation assumed, but no specific basis given
 - 3 The parent has knowledge superior to the child
 - 4 The parent is morally superior to the child (appeal to sense of personal responsibility)
 - 5 The parent is a moral representative of the community
 - 6 The parent influences the child morally, intellectually whether or not he wills it and so has the responsibility for the consequences of his own acts
 - 7 Parent influences the child psychologically whether or not he wills it
 - 8 Other
 - 0 NA (question not applicable and does not deal with question)

Note that in this case, the recording unit is a whole manual. Each question amounts to one variable of multi-variable data language, and Question 32 has nine numerical values with the above-stated meanings. These meanings define unordered categories.

Like the simulation of hypothesis testing, the simulation of interviews relies on the coder's logical and interpretive abilities, but the simulation of interviews relies additionally on the coder's ability to assume an author's role and answer as the author would, given what that author wrote. Assuming the author's position is a cognitive device that literary scholars commonly use in their efforts to infer authors' intentions—what they had in mind—as well as what authors stood for, valued, justified, and hoped to accomplish.

Typically, the simulation of interviewing becomes unreliable when the writing is voluminous and the informative passages are scarce and therefore easily overlooked.

In such a situation, content analysts would be wise to use context units of sizes smaller than the whole work.

Interview simulations can be used in analyses of texts that were written long ago or by authors currently unavailable, ranging from foreign dignitaries to indicted criminals. Unlike real-life interviews, they can be repeated as often as needed. For the latter reason, content analysts may use such simulations because simulated “interviewees” are unaware of how they are being questioned and thus unable to speak to the analysts’ intentions. Because content analysts can define the context of the analysis, they can place simulated interviewees in situations where they can answer embarrassing questions (Barton, 1968) with ease and without moral qualms.

7.5.8 Constructs for Closure

Experienced therapists often advise their students to discount what their patients say and to listen instead to what they omit. We have many metaphors for this epistemologically questionable but common practice: “reading between the lines,” “detecting latent meanings,” “hearing silences,” and “discovering hidden motivations.” Conspiracy theorists thrive on such metaphors, and the above-noted advice legitimates therapists’ denial of their patients’ stories. The well-studied phenomenon of “induced memories”—“filling gaps” in recollections—and the public demonization of slightly deviant groups both result from undisciplined interpretations of what is not said. Such practices may result in public consensus, but that consensus is usually highly unreliable. Under certain conditions, however, it is quite possible to “complete the picture” from what is said (to use still another metaphor) and specify in advance, and without devious intentions, the abstract organization of the whole that would enable content analysts to infer the missing parts, to obtain closure and accomplish this quite reliably.

One, perhaps procedurally less specific, example is found in George’s (1959a) account of the U.S. Federal Communication Commission’s inferences from domestic enemy broadcasts during World War II. In the course of their work, in fact during several war years, the analysts developed elaborate constructs that they believed explained why these broadcasts came into being and what perceptions and antecedent conditions had caused them. I discuss their approach in more detail in Chapter 10; here, it suffices to say that the analysts developed and used highly specific constructs of the network of the political and military players in Germany and generalizations regarding the Nazi elite’s political and propaganda behavior that allowed the analysts to obtain military intelligence and to predict political changes in the Axis countries. George provides a good description of the analytical constructs that were developed in this situation and the cognitive processes the analysts employed to make rather specific recommendations that were not obvious in the domestic broadcasts. He suggests:

The analyst’s reasoning takes the form of filling in, or assigning a value to, each of the major unstable variables, which are not already known, and supporting this reconstruction both by generalizations and by logic-of-the-situation assessments. This type of inferential reasoning may be likened to an effort to reconstruct the missing pieces in a mosaic. Certain parts of the mosaic are given or readily assumed. Other pieces in the mosaic, however

(including the conditions which the analyst particularly wants to clarify), are missing. In effect, therefore, the analyst rehearses in his mind the different possible versions of each particular missing variable which he wants to infer, trying to decide which version is the most plausible, given the known value of the content variable and the known or postulated values of other antecedent conditions. (p. 61)

Another example of recording absences comes from Shneidman's (1966, 1969) effort to analyze suicide notes collected by a suicide prevention center in San Francisco. Shneidman started with the fair assumption that each individual is logically coherent relative to his or her own world constructions. Readers as well as the analysts know only the asserted premises and the conclusions that the writer draws from them. Shneidman calls the particular logic by which a person thinks and argues that person's *idio-logic*. Accordingly, writers are assumed to accept their own idio-logical conclusions, even when they are fallacious relative to standard textbook logic. Adopting this textbook logic as a construct, Shneidman asked his coders to focus on a suicide letter's manifest fallacies of reasoning and then identify all the unwritten propositions that the writer must have taken for granted in order for his or her conclusions to be coherent in terms of that textbook logic. In a second step, Shneidman inferred from the coders' list of hidden assumptions how someone would be able to communicate (reason) with the writer and enter his or her world—a *pedago-logic*, in Shneidman's terms.

Incidentally, the examples of logical fallacies, the idio-logical propositions that would seem to make these fallacies acceptable to the writer, as well as the pedago-logical recommendations in Shneidman's (1966) codebook are mostly drawn from political speeches. This is due to Shneidman's (1963) parallel interest in studying political communication, especially the worldviews inferable from the logical fallacies that national leaders find acceptable, to explain why leaders are prone to misunderstanding each other and what one could recommend to either side. Here, too, content analysts record and infer what omissions entail under the assumption of a very detailed framework that assures some reliability.

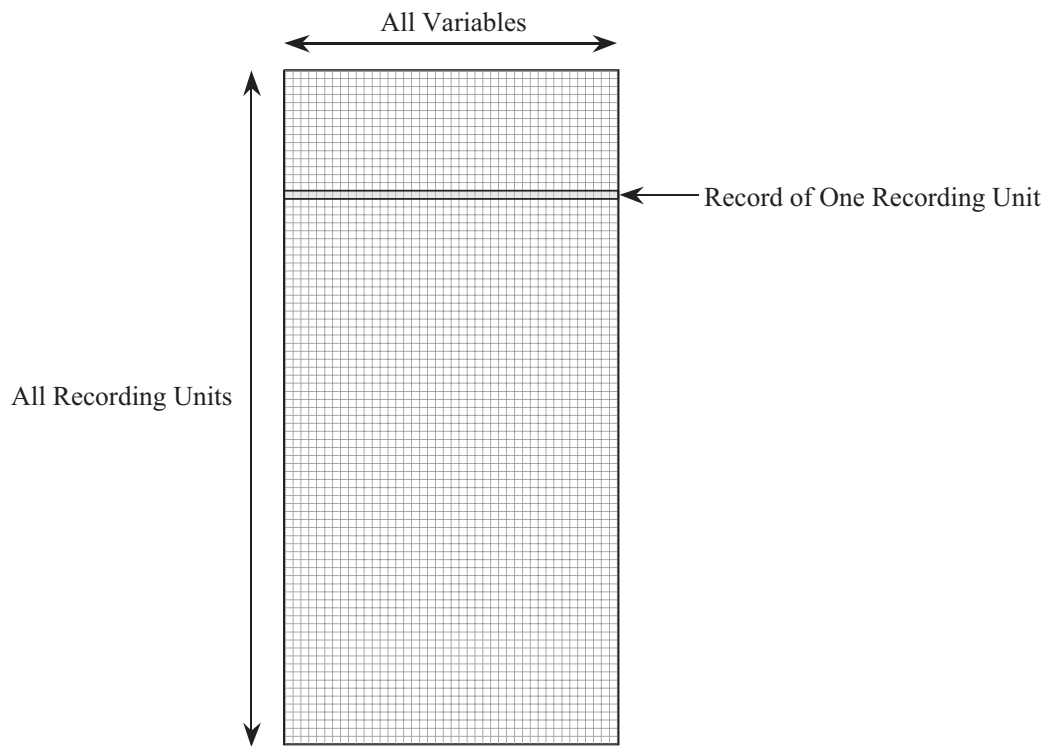
The foregoing discussion of strategies for operationalizing the semantics of a data language is by no means complete—I have presented only the major approaches here. Nor is it my intention to create the impression that the tools discussed are mutually exclusive alternatives. Content analysts can draw on any of these as well as others—as long as they rely on cognitive models that coders are familiar with and can learn to use. Familiarity and specificity have a chance to ensure the efficiency and reliability of recording.

7.6 RECORDS

The computational part of content analysis begins after recording is completed, with the records it has produced. Records are the most basic and explicit representations of the phenomena being analyzed. Records may come in many forms, from notations in the margins of written documents to tags entered into electronic text (Stone, Dunphy, Smith, & Ogilvie, 1966), binary data stored on optically readable microfilm (Janda, 1969), codes added to searchable images (Ekman & Friesen, 1968), and coding sheets modelled after interview schedules. Figure 7.2 shows the

most general form of a collection of records imaginable (but only imaginable); it depicts a huge spreadsheet of all recording units of the analyzed body of text by all the variables of the applicable data language, containing transcriptions, categories, or numbers in its cells.

FIGURE 7.2 ■ Most General Structure of Data



Designing records for a suitable storage medium—data sheets, coding forms, questionnaires, or excel files—requires much ingenuity on the part of the content analyst. Because the demands made on recording texts are so varied, it is impossible for anyone to suggest a standard or optimal form to be followed. A few recommendations may be made, however. The most general derives from the common practice of tabulating and enumerating recording units. To be countable, units must be described separately and in comparable terms, the categories of the same set of variables. It makes sense, therefore, for analysts to create records, one for each recording unit, that have the same organization of variables, into which coders enter the appropriate values. This is analogous to answering all applicable questions on an interview form. When records are organized in this way, a content analysis needs as many forms as there are units to be recorded, which may be many—hence the need for simplification.

The records of a content analysis—an array of descriptions in the form of alphabetical characters, check marks, or numbers—should contain three kinds of information: administrative information, information on the organization of the

records, and information on the phenomena represented in the records. The last of these is obvious; the first two are often overlooked.

7.6.1 Administrative Information

Administrative information guides the handling of data. It is impossible to overstate its importance in most research efforts. For example, data sheets can get out of order, and unless analysts can find out how those sheets were generated, they have no way to know what the data mean. Coders make clerical mistakes, do not record all required variables, and unless the records include information about who coded them and where to locate the recording unit that was coded, there may be no simple way to correct even simple oversights. Much too often, analysts lose valuable time when they find sets of completed forms in their possession and are unable to determine whether they came from a pretest, whether the data have been verified, entered into a computer, when, by whom, and so on. Electronic forms of data may become useless when they do not carry information on what they represent, who created them, when, and according to which coding instructions.

Typically, administrative information includes the following:

- The name of the content analysis *project* to which the data belong
- The *version of the recording instructions* (e.g., the first or second version) used to generate the record
- The *kind of texts* the record represents (e.g., a test sample or the main body of text)
- The *state of the record* (e.g., whether it has been completed, verified, entered into a computer, or otherwise processed), including information on what is still to be done with it
- The *identities of the coder* who created the record and *others* who checked or processed it
- The serial number of the recording unit or any way to get to it
- If multiple units are used, the names or numbers of the *recording units* the coded one *contains*

The first three pieces of information on this list are common to a larger set of recording units, and these may be preprinted on all data-entry forms used in a particular recording effort or head a data file. I discuss the final item on the list below. Complete administrative information is essential to efficient data management. When using electronic records, analysts may have fewer opportunities to mess them up, but they must ensure that their programmers have made provisions for recording the information listed above and conform to the requirements of the software to be used.

7.6.2 Information on the Organization of Records

One of the features that content analysts face, rarely found in other kinds of research, is the use of several levels of recording units that reflect the organization of the texts being analyzed. Cluster sampling, discussed in Chapter 6, section 6.2.5,

may not be a mere convenience; it can generate nested recording units that need to be coped with. Consider the following example:

The newspapers coded

The issues of a newspaper coded

The articles in an issue of a newspaper coded

The paragraphs in an article in an issue of a newspaper coded

The propositions constituting a paragraph in an article in an issue of a newspaper coded

Similar hierarchies can be found in levels of discourse (e.g., language, meta-language, meta-meta-language), of scientific findings (e.g., hypotheses, theories, paradigms, and epistemologies), of geographical locations (e.g., cities, districts, states, countries continents), and of decisions made on different levels in social organizations. On each level, a different set of categories applies, and following their nesting, categories applied to higher level units implicitly apply the units they contain. In the above newspaper example, newspapers may be characterized by circulation figures, prestige, access to news services, and ownership. Issues may be coded by publication date, day of the week, and size. Articles may be distinguished in terms of kind, placement, length, and so forth, until one comes to, words, even characters.

Not all multiple-unit content analyses produce such neat hierarchies of inclusion. Web page links and hypertext documents operationalize relationships of inclusion as well but allow recursions—that is, the possibility that one unit of text contains references to another, and another, and so forth, but also back to the text where one started. Such organizations are not hierarchical but heterarchical, or network like. When working with such multilevel data, analysts must keep track of whether and how the categories that are applied on one level of units relate to the categories applied to subordinate units or to units that can be reached from the former, directly or indirectly. There are essentially three ways to accomplish this:

- The analysts might keep a *master file*, a separate file that preserves *all connections among separately coded recording units*, whether these connections form hierarchies or heterarchies. The above newspaper example involves a hierarchy of inclusions, and the appropriate master file would tell the analysts where each recording unit belongs in relation to all others.
- The analysts might code *each recording unit* separately but include *references to all units in which each occurs* (where it can be reached from) and *references to all the units each contains* (the units that can be reached from it).
- The analysts might keep a *complete but redundant record* of all variables by all smallest recording units identified in the body of texts, similar to the data structure in Figure 7.2. Here, the category assignments of any one unit are duplicated for each unit it contains and hence are redundant in parts.

To cross-tabulate or correlate categories on different levels of description, such as the circulation figures of newspapers and favorable or unfavorable mentions of a

public personality, content analysts working with multilevel recording units need to connect the categories of (large in total number of recorded details and numerically few) newspapers included in the study and the (small and numerous) recording units asserting the evaluations of interest. Analysts using the first method above will have to consult the master file, those using the second may need to trace the connections from one recording unit to the next, and those using the third may want to ensure that frequencies of characteristics on one level are not confused or exaggerate the frequencies on another.

7.6.3 Substantive Information About the Phenomena of Interest

Generating analyzable data is, of course, the *raison d'être* of the recording process. Whatever the device, coders must be able to record information with ease, verify instantaneously what they have entered, and correct their mistakes. Each medium for recording data has its own properties and makes special demands on human coders. Optical scanners call for the use of pencils of a particular kind, otherwise some uncertainty may arise about the intended meanings of the marks. A long time ago, the accuracy of punch cards was difficult to verify without another mechanical reader. Spreadsheets offer convenient overviews of whole data arrays, but they often make it difficult for analysts to connect cell contents to recording units and available categories. Although software is available that allow coders to generate electronic data files on the fly—during telephone interviews or while watching television, for example—such tools must be carefully designed so that they interface easily with coders, minimize mistakes, and provide ample feedback for verification, much as traditional paper data sheets do. Qualitative data analysis software, *Atlas.ti*, *NVivo*, and *MAXQDA*, for example, tends to preserve the relationship of codes and segments of text coded but makes it not easy to move to formal analyses of the records it provides.

Most Americans are familiar with the conventions of filling out questionnaires and comfortable with using a mouse to point and click on a computer screen. Analysts have to rely on coders' competencies in the chosen recording medium. The more natural a recording medium is to coders, the fewer clerical errors occur.

The above outlined several proven approaches to defining the semantics of a data language. I want to conclude this chapter by pointing out some easily avoidable errors that content analysts make when designing the instruments that coders use to record what they are observing, reading, categorizing, judging, or scaling.

One frequent source of errors is the overuse of numbers. Numbers are short and concise; but when they are used for everything, they can become confusing. Content analysts tend to number their categories, their variables, their coders, the units to be recorded, the pages of instructions, where the numbered values of numbered variables are defined, and so on. In most cases, the designers of content analysis instructions could specify the required organization of data by using descriptive words instead of numbers whose meanings must be learned, by using typographical or spatial arrangements instead of paragraphs of prose, or even by using icons, which may cause less confusion than numbers.

A second source of errors is the inconsistent use of category names or numbers across different variables. For example, when the default category of “not applicable”

or “other” is coded “0” for one variable and “9” or “99” for another, confusion is bound to arise. The same is true when analysts use the same words but with different intended meanings in different variables. Explicitly defined differences across variables are easily forgotten.

A third source of errors is the adding of comments to records. This is one reason various qualitative software packages allow users to highlight text, assign codes, and cut and paste text rather than rewriting pertinent text. These features significantly reduce the chance of using synonyms and even spelling errors, which cause spurious differences to emerge during subsequent analyses.

A fourth source of errors is poor design of the presentation of options on the recording medium. In widely used graphic computer interfaces, users are asked either to check “boxes” on or off, click on alternative “radio buttons,” to select options, or copy and paste segments to defined areas. These are logically different operations. And whereas a computer interface can be designed to force users to comply with a designer’s intentions—for example, by disabling unavailable options—paper instruments are not so intelligent. Nevertheless, the designer of a recording medium can do much to discourage coders from recording data incorrectly and thus avoid unreliability and polluting the data with illegitimate values. Consider the following three ways of recording the outcome of an interpersonal interaction:

Enter the appropriate number	Encircle one only	Check ☒ as many as applicable
☐ 0 - favorable to neither	<i>favorable to</i> neither	
1 - favorable to recipient	recipient only	☐ favorable to recipient
2 - favorable to initiator	initiator only	☐ favorable to initiator
3 - favorable to both	both	

Although these three alternatives record the same information, in effect, they differ in the kinds of errors they invite. In the version on the left, the coder is asked to enter one of four numbers in a box. There is nothing to prevent a coder from writing a number larger than 3 in that box, which would be undefined, regardless of what the coder had in mind, and cause subsequent analytical confusion. Leaving the box blank is not a legitimate option either, although it might make sense to a coder who found nothing favorable to record. This version is also sensitive to bad handwriting. Finally, it is not uncommon for coders to confuse category numbers with, for example, coder ID numbers, unit numbers, variable numbers, or scale points, as mentioned above. The middle version is designed to prevent the latter confusion, but it does nothing to discourage the coder from circling more than one option, circling two equally imperfect alternatives, or failing to circle the category “neither” when none is evident. The version on the right resists illegitimate entries altogether, but this solution is limited to binary distinctions—to being checked or not checked, present or absent, yet favorability could be a scale with more or less, a property not afforded by either alternative. Checking or not checking a box is a simple, unambiguous alternative. Analysts can reduce recording errors by phrasing the recording options so that they require a minimum of writing; the best way to do this is to provide a list of appropriate alternatives and instruct coders to “check all that apply”—not burdening

the coders with the information that each then becomes a binary variable on its own (see Chapter 8).

The most common source of coding errors is evident when the recording instructions are not well spelled out, especially when the values of a variable are not clearly distinguished, and coders need to consult, even puzzle over the meanings of, the instructions and possibly come to conclusions not shared with other coders. One extreme solution for this kind of problem is to merge the recording instructions with the recording medium, so that the recording medium is similar to a questionnaire in survey research. The high level of consistency this could ensure, however, is counterbalanced by the fact that using such a medium can be tedious. Having to reread a lengthy recording instruction for each recording unit can be time consuming and becomes excessive when recording units are small and numerous (e.g., words, frames of videotape, seconds of verbal interaction)—even when recording instructions are reproduced electronically and choices are indicated on a computer screen. The other extreme of overloading coders with duplicate instructions is when coders are given a spreadsheet—a large grid of recording units by variables, as in Figure 7.2—to be completed according to separately provided instructions. This invites a host of confusions; for instance, while consulting the instruction manual to resolve indecision, coders may lose track of which row they are coding or may enter the categories (numbers) for one variable into the cells of another. In addition, most of these kinds of errors are difficult to detect. The following recommendations chart a middle course between these two extremes:

- At each data entry point, the designer of coding instructions should present the coders with adequate verbal descriptions of each variable and the list of its mutually exclusive options. In an instructive report excerpted in *The Content Analysis Reader* (Krippendorff & Bock, 2009, Chapter 4.1), MacQueen, McLellan, Kay, and Milstein (1998) suggest that each coding category (or scale value) of a codebook should specify the following:
 - The marker (code) that actually is recorded
 - A brief definition
 - A full definition
 - When to apply that marker
 - When not to apply that marker
 - Examples
- In situations in which the choices coders are asked to make are deemed obvious, such elaborate instructions should remain hidden but be immediately available when uncertainties need to be resolved. This assures coders work efficiently and make use of the competencies they inevitably acquire in the process.
- The designer of coding instructions should supply coders with a well-defined set of alternatives to choose from—in computer applications, pull-down menus, for example, or a row of bull's-eyes. Ambiguities—that is, the possibility that coders may mark something between two alternatives—should be prevented. Ideally, coders should not need to write anything.

Asking coders to enter numerical or alphabetical characters into openings is common but problematic, especially when these characters have no intrinsic relation to the phenomena to be recorded. Having coders check well-labeled boxes may avoid the need to teach them the meanings of written markers or characters.

- Where possible, the designer of coding instructions should create visual analogues (mappings) to the phenomena to be recorded, showing the relationship between the way the analyzed text is organized and the way the recording medium is laid out. For example, when recording the spatial locations of articles on the front pages of newspapers, it should not be difficult to visualize the locations of interest (above or below the center fold, in the center, or on the side). When recording interactions in small groups, it is relatively easy to graph the networks among members and allow coders to pick who says what to whom. Or when sentences are parsed into actor-action-target-effect statements (see Chapter 8, section 8.3), the semantic conventions for such statements should be preserved in the visualization of a coding structure for these statements.
- The general rule is to remove as many unrelated or clerical complications as possible from the work coders are asked to do without making it difficult to reexamine the data they generate.

The availability of computer software that presents coders with choices and instructions of optional depth enables content analysis designers to take advantage of creating digital records directly. Coding software is being developed that will enable analysts to check reliability on the fly and make content analysis data immediately available.

8

DATA LANGUAGES

Categories and measurements are the entry points to empirical research. We treat their particular organization as a data language, which is conceived to have a syntax and a semantics. The semantics of a data language ties data to the phenomena of the observed world, to coders' readings of texts, and the syntax of a data language links the data to the computational processes of an analysis. This chapter is concerned with the forms that satisfy the syntactical requirements of data languages for content analyses. It provides definitions of terms related to the construction of such languages, illustrates the essential features of data languages, and distinguishes variables—categories, measurement scales, and the like—according to the orderings and metrics they exhibit. This chapter extends Coombs's (1964) A Theory of Data into the domain of content analysis.

8.1 THE PLACE OF DATA LANGUAGES IN ANALYTICAL EFFORTS

A data language is the descriptive device in which terms analysts cast their data. For natural scientists, a data language is a system of physical measurements and records of basic observations. For content analysts, who start with textual matter, images, verbal exchanges, transmissions, and records of observed phenomena, a data language describes how all the categories, variables, notations, formal transcripts, and computer-readable accounts hang together to form one system. For both kinds of researchers, data languages mediate between otherwise unstructured phenomena and the scientific discourse about them, and in the case of content analysis, they mediate between the experiences of reading text, interpreting images, and observing transient social phenomena of interest on one hand and the formal demands made by available analytical or computational procedures on the other.

Treating a system of categories and measurements as a data language allows analysts to separate syntactical considerations from semantic ones. The semantics of a data language delineates the meanings of its terms, operationalized in the process of coding or recording, whereas the syntax of a data language must satisfy the formal operations required in the scientific handling of data. When the semantics of a data language is ill defined, one does not know how to interpret the marks or numbers that observers or coders left behind, and when the syntax of a data language is incompatible with the demands made by the analytical technique employed, computational results are unintelligible.

I have discussed the problem of operationalizing the semantics of a data language in Chapter 7, will address at least some of the demands that analytical techniques make in the data they accept in Chapter 10, and will consider the consequences of ambiguities in the semantics of a data language in Chapter 12. This chapter concerns mainly the *syntax* of the data languages that are of interest to content analysts. Regarding their syntax, data languages must meet three criteria:

- They must be *free of syntactical ambiguities* and inconsistencies.
- They must *satisfy the requirements of the analytical techniques* to be used.
- They must *transmit enough information* about the phenomena of interest.

In order to meet the first of these three requirements, data languages should be formal or formalized. Formal languages are computable in principle. Humans, being naturally sensitive to contexts and bringing their own experiences to any readings of texts, are well equipped to cope with syntactical ambiguities. Explicit analytical techniques are not. For example, an ordinary reader with access to the context of the sentence “They are flying planes” rarely encounters difficulty in deciding whether *they* refers to a group of pilots or to several objects seen in the sky. In fact, when reading the sentence in context, an ordinary reader would rarely notice its syntactical ambiguity. In a content analysis, such syntactical ambiguities have to be removed through human editorial interventions—for example, for the above sentence, analysts need to specify whether *flying* is a verb or an adjective. Similarly, “Jim or Joe and Mary are coming” can be read either as “(Jim or Joe) and Mary are coming” or as “Jim or (Joe and Mary) are coming.” Naturally occurring texts are full of such ambiguities, which are rarely problematic for ordinary readers. Content analysts are well-advised to design coding sheets, checklists, and rules for transcribing text or kernelizing sentences in ways that will prevent syntactical ambiguities and inconsistencies from entering the analysis.

The second demand on data languages stems from the formal requirements imposed by the analytical techniques that analysts intend to use. Although this may seem obvious, it is amazing how often researchers generate data on a very interesting phenomenon only to discover, usually too late, that the way data are recorded makes it impossible to process them. A few examples should suffice: Factor analysis requires multiple correlations, which in turn presupposes interval data on several dimensions; multidimensional scaling techniques start with distances between pairs of data points; causal connections can be shown only in time-series data that allow the analyst to check for spurious correlations. Although most analytical techniques accept numbers as inputs, the mere fact that data are in numerical form is no guarantee that the analysis will make sense. Applying a variance analysis on ranks, which do not live up to the requirements of such an analysis, produces results that are difficult to interpret (for an opposing view, see Tukey, 1980). Researchers make more drastic mistakes when they analyze nominal data as if they were ordered: Analyzing people according to their Social Security numbers or using the numbers that football players wear as interval data is bound to produce garbage.

Researchers have attempted to design computer software for text analysis (see Chapter 11) in order to circumvent the problems addressed by the first two criteria above. Computational efforts assume a data language that recognizes a text as a finite string of characters, recording words, for example, or pairs of words occurring within

a window of a finite number of characters that slides over a text. This is an easily computable data language that bypasses, however, more complex structures in which sentential meanings of text reside for a reader.

The third requirement that data languages must meet derives from the target of content analysis, selecting among appropriate inferences from text. Lasswell (1960), paralleling similar questions for political science (Lasswell, 1963), once characterized communication research as asking, “Who says what, in which channel, to whom, and with what effect?” He then suggested that content analysis answers the “says what” part of the question, audience research answers the “to whom” part, and effects research answers the “with what effects” part. In so distinguishing among analytical approaches, Lasswell failed to see that separate answers to the questions of “who,” “what,” “to whom,” and “what effects” cannot provide the information that analysts require in order to say anything meaningful about processes of communication, the influence asserted, the relationships established, the coordination accomplished, and so on (Krippendorff, 1970d). A data language may fail to provide enough information by assuming a perspective from which the whole cannot be comprehended—as in Lasswell’s separation of content analysis from analyses of other facets of communication—by leaving out important variables, ignoring the correlations between them, or making too few distinctions. The information flow through an analysis can be traced and measured (see Krippendorff, 1991). The amount of information that analysts need to ultimately select a defensible answer to a given research question can often be spelled out in advance. An appropriate data language must provide at least as much to answer a research question. Redundant information (distinctions, correlations, and variables) is better than insufficient amounts.

Given the requirements discussed above, we can define a data language in terms general enough to cover most content analysts’ concerns.

8.2 DEFINITIONS

The data language used in an analysis prescribes the form in which the data are recorded. The syntax of a data language consists of the following:

- *Variables* whose *values* represent the variability within one conceptual dimension
- The *values* within variables, which may be *ordered* and/or exhibit a *metric*
- *Constants* whose operational meanings are fixed within the data language and specify how the values of different variables are related to each other
- A *grammar* whose rules govern the construction of well-formed expressions (data records or descriptions)
- A *logic* that determines how the expressions of the data language imply each other or are equivalent, specifying logical (a priori) dependencies among these expressions

For example, in the algebraic formula

$$A \cdot X + B = C,$$

A , B , C , and X are *variables*, each of which is a placeholder for a numerical *value*. The symbols $+$ and $\cdot$ have the operational meanings of addition and multiplication, respectively, and are invariant to the values in the variables. In the process of recording data, and in order to apply analytical procedures to a data language, analysts have to enter values into the places that the variables provide.

The *grammar* of a data language makes certain combinations of values within variables illegitimate or ill formed. According to the rules of algebra, for example, both sides of the above equation are well formed, whereas the string “ $ABXC = +$ ” would not be and must therefore not occur.

The symbol $=$ is a logical sign that defines the two sides of the formula as numerically equivalent and as mutually substitutable. The *logic* of a data language defines the relationships between combinations of values from different expressions: equality, entailment, or orderings.

In many content analyses, the syntax and logic of data languages are so simple that they may not be recognizable as such. The most basic form of a data language consists of the product of a finite set of, say, n variables, such as:

$$A \cdot B \cdot C \cdot D \cdot E \dots$$

The product sign between these variables allows the values of different variables to co-occur freely. In effect, for n logically independent variables, this defines an n -dimensional space. Raw data then take the form of a collection of “ n -tuples” of values $a, b, c, d, e, \dots$, one value for each variable, $a \in A, b \in B$, and so on. A collection of r such n -tuples could be listed as an r -by- n matrix:

$$\begin{array}{c} \langle a_1, b_1, c_1, d_1, e_1, \dots \rangle \\ \langle a_2, b_2, c_2, d_2, e_2, \dots \rangle \\ \langle a_3, b_3, c_3, d_3, e_3, \dots \rangle \\ \dots \\ \langle a_r, b_r, c_r, d_r, e_r, \dots \rangle \end{array}$$

They could also be seen as listed in a spreadsheet similar to Figure 7.2. This basic data language can be thought of as defining an n -dimensional space in which each unit, described as an n -tuple, finds a unique cell to occupy and the data as a whole define a particular distribution in this n -dimensional space. In this basic data language, the values from different variables can co-occur without constraint. There is no particular logic by the above definition.

I mention the grammar and logic of a data language here mainly because exciting developments are taking place in fields related to communication research, notably in linguistics and cultural anthropology, which content analysts need to consider. For example, transformational grammars, whose syntax includes rewrite rules that are aimed at characterizing natural language expressions, cannot be represented spatially and without logic. But even relatively unambitious content analyses may include recursions that violate the idea of multidimensional representations of data. For instance, in a content analysis of native–foreigner attitudes in Africa, Piau (1965) recorded answers to open-ended questions in terms of the following:

- (a) An ordered set of variables concerning social characteristics of the individuals X and Y
- (b) The origin of a judgment, using statements of the following form:
 - X judges Y to be []
 - X talks about Y judging X to be []
 - X talks about Y talking about X . . .
- (c) Relations between X and Y , relative to the origin of the judgment
- (d) Three kinds of *themes* (i.e., arguments) associated with each judgment
- (e) A lexicon consisting of 675 terms, variables (which note presences or absences), and constants (in the form of Boolean operators AND and OR and qualifiers) in terms of which arguments are entered in the places provided by []

Here, the first item associates a social variable with the individuals who are speaking or being talked about. The second item describes who judges whom, allowing for recursive judgments, . . . , and the fifth item preserves the original attributes used by the respondents as an open-ended variable that can be searched with Boolean operators. This data language met the syntactical demands of the information retrieval routines that Piault used in the course of her analysis.

Research may be viewed as a series of systematic transformations of one data language into another. For example, counting the recorded n -tuples eliminates redundant listings and adds an additional ($n + 1$ st) variable to the array, their observed frequency. Developing an index maps a set of variables into one, the index, whose variability is a function of the various original data it represents. Applying a replacement dictionary to text reduces the great diversity of expressions to fewer and more relevant kinds. Data languages do not need to be confined to traditional statistical distributions in multidimensional geometric spaces. With linguistically sensitive analytical techniques increasingly available, sophisticated grammatical rules and forms of logic have become increasingly important, especially in analyses of themes. I will not develop this topic here, however; instead, I focus on what is common to all data languages: variables.

8.3 VARIABLES

A variable is a concept that allows for variations of its instances. In the above, we took a variable as a placeholder for any one of several mutually exclusive values. In *Merriam-Webster's Collegiate Dictionary* (11th edition), the adjective *variable* is defined as “able or apt to vary,” and the noun *variable* is defined as “something that is variable” and as “a quantity that may assume any one of a set of values.” Variation is what enables data to be “informative.” Indeed, the variable *sex* has no descriptive significance unless one can distinguish between males and females, and the notion of bias in journalism is meaningless unless journalists have the option of leaning toward one or the other side of a controversy. In other words, if the units of a content analysis do not exhibit variation in their description, analysis of the units cannot inform anything.

The individual values of a variable must be *mutually exclusive* relative to each other. This satisfies the requirement that a data language be unambiguous and in effect *partitions the set of recording units* (the sample) *into mutually exclusive subsets*. Jointly, the values of a variable must provide an *exhaustive* account of all units, which means that the partition of the sample should leave nothing unaccounted for. In content analysis, the requirement of exhaustiveness is sometimes relaxed when it comes to irrelevant matter. In this respect, the social sciences deviate from physics, for example, which assumes that all physical objects have the same dimensions.

The idea of a variable of mutually exclusive and descriptively exhaustive values is so general that it occurs in numerous intellectual endeavors, albeit with different names. Some correspondences are presented below. The set theoretical expression “ $a \in A$ ” probably is the most general one; it simply states that the element a is a member of the set A of elements.

Values	of a	Variable
a	$\in$	A
Elements	$\in$	Set
Categories	$\in$	Set of categories
Points	$\in$	Scale
Members	$\in$	Family or class
Position	$\in$	Dimension
Locations	$\in$	Space
Measures	$\in$	Gauge
States	$\in$	System
Tokens	$\in$	Type
Possible Sets	$\in$	Super set, set of all sets

The concept of a variable with mutually exclusive values does not mean that content analysts are limited to single-valued descriptions, to assigning one and only one value to each recording unit. Text typically affords multiple interpretations, whether because readers with different backgrounds and interests come up with unique but, in the aggregate, divergent interpretations or because ambiguity leads a single reader to alternative and equally valid interpretations. A variable that records possible sets (the last on the above list), possible patterns, or possible connections affords multivalued descriptions. Multiple interpretations of text may present problems for coding—for the semantics of a data language and for reliability—and for the analytical techniques available for handling such data, but they are not incompatible with the notion of variables.

Variables may be *open* or *closed*. At least in principle, numerical variables are open—there is no largest or smallest number. Open variables require conceptual clarity on the part of coders or, in the natural sciences, knowledge of the construction of the measuring instrument. For instance, when coding instructions call for the rephrasing of a text into the form

[] says [] to [],

such as “[Jim] says [hi] to [Mary],” coders are guided by concepts of what would fit in the open places. In this context, one could easily rephrase the openings by “who” says

“what” to “whom.” For Piault (1965), they were personal qualities that X attributed to Y. But she had no idea which would show up. The values in open variables are taken from the observed phenomena or units of text without preconceptions by the research designer.

When variables are closed, analysts may define them *implicitly*, by *specifying their range*, or *explicitly*, by *listing all alternative values*. Many social variables are defined by concepts that imply definite ranges. For example, the concepts of gender, marital status, and kin offer closed vocabularies to describe all possible kinds. Sometimes institutions limit the ranges of variables (e.g., kinds of criminal offenses in a legal code or kinds of mental illnesses in the *DSM-IV-R*) and sometimes particular theories do (e.g., dramaturgical roles in fiction or types of personalities). Somewhat more tailored for quantification are verbally anchored scales of measurement, such as 7-point semantic differential scales. Semantic differential scales show a pair of polar opposite attributes of recoding units. For example,

Good	+3	—	+2	—	+1	—	0	—	-1	—	-2	—	-3	Bad
Strong	+3	—	+2	—	+1	—	0	—	-1	—	-2	—	-3	Weak
Fast	+3	—	+2	—	+1	—	0	—	-1	—	-2	—	-3	Slow

Or a pain scale:

No Pain 0 — 1 — 2 — 3 — 4 — 5 — 6 — 7 — 8 — 9 — 10 **Excruciating Pain**

When using such scales, coders are encouraged to create a continuum of intermediate and equally spaced meanings between the designated extremes and then judge where an observed phenomenon or verbal expression would belong. The endpoints of the scale are defined by the conceptions that readers have of these adjectives; the remainder is semantically implicit in the use of the continuum, which pertains to the syntax of the data language using this variable.

Finally, analysts may define variables *explicitly*, in terms of exhaustive *lists of values*. For example, Searle (1969) claimed to have identified a mutually exclusive and exhaustive set of five values for the variable “speech acts”:

Representatives

Directives

Commissives

Expressives

Declaratives

The three ways of defining variables noted above may also be recognized in the design of coding sheets, recording devices, and computer interfaces for coding. In the past, coders typically checked boxes or typed the values of open-ended variables into prepared openings; indicate variables defined implicitly by turning knobs, arranging objects to indicate how different they are, marking or clicking on appropriate scale points, or selecting listed alternatives on computer interfaces.

Given the demands of a research question, analysts may choose among different kinds of data languages for recording the same kind of information. For example, one could ask coders to identify which of 20 logically possible communication networks are operating within a five-person group or which of 10 possible communication channels between pairs of members of that group are being used. These two ways provide the same information. The choice of one data language over another may be informed by differences in the coding effort required, perhaps by issues of reliability, but certainly by differences in the amount of information the data languages provide. In general, content analysts should construct data languages that are as detailed and basic as they can possibly be and leave as much to computation as possible for subsequent processing. For example, Budd's (1964) attention measure, Gerbner, Gross, Signorielli, Morgan, and Jackson-Beeck's (1979) violence index, and Hawk's (1997) listenability scores for TV programming, modeled after Flesch's (1974) readability yardstick (see also Krippendorff & Bock, 2009, Chapter 3.9), are all computed on the results of numerous simple coding judgments. An analyst could, of course, define a far simpler higher-level data language—a ratio scale for recording attention, violence, and listenability as an overall coder judgment, for example—but it has been shown that abstract measures rarely achieve the reliability obtainable for data close to the text, and they cannot provide the fine distinctions that aggregate measures produce.

Unfortunately, content analysts often publish their results without making their “conceptual schemes” or “systems of categories” explicit, perhaps because they are not clear themselves as to the nature of the data languages they employed. Consider the following scheme, reported in the literature by Herma, Kriss, and Shor (1943):

Standards for rejecting Freud's dream theory:

- A. Depreciation through value judgment
 - 1. Ridicule and mockery
 - 2. Rejection on moral grounds
 - 3. Denial of validity
- B. Denial of scientific character of theory
 - 1. Questioning analyst's sincerity
 - 2. Questioning verification of theory
 - 3. Questioning methodology
- C. Exposure of social status of theory
 - 1. Disagreement among experts
 - 2. Fashionableness
 - 3. Lack of originality

This scheme looks more like the outline for a paper than a coding instruction. Its publication does not make clear how the analysts used it. One interpretation could be that it defines but one variable consisting of nine values, *A1*, *A2*, . . . , through *C3*, with a merely convenient grouping of these values into *A*, *B*, and *C* kinds that could aid conceptualizing these nine values, having conceptual but no descriptive significance. A second interpretation is that there are three variables (*A*, *B*, and *C*) with three values (*1*, *2*, and *3*) each, with different meanings for each variable. This

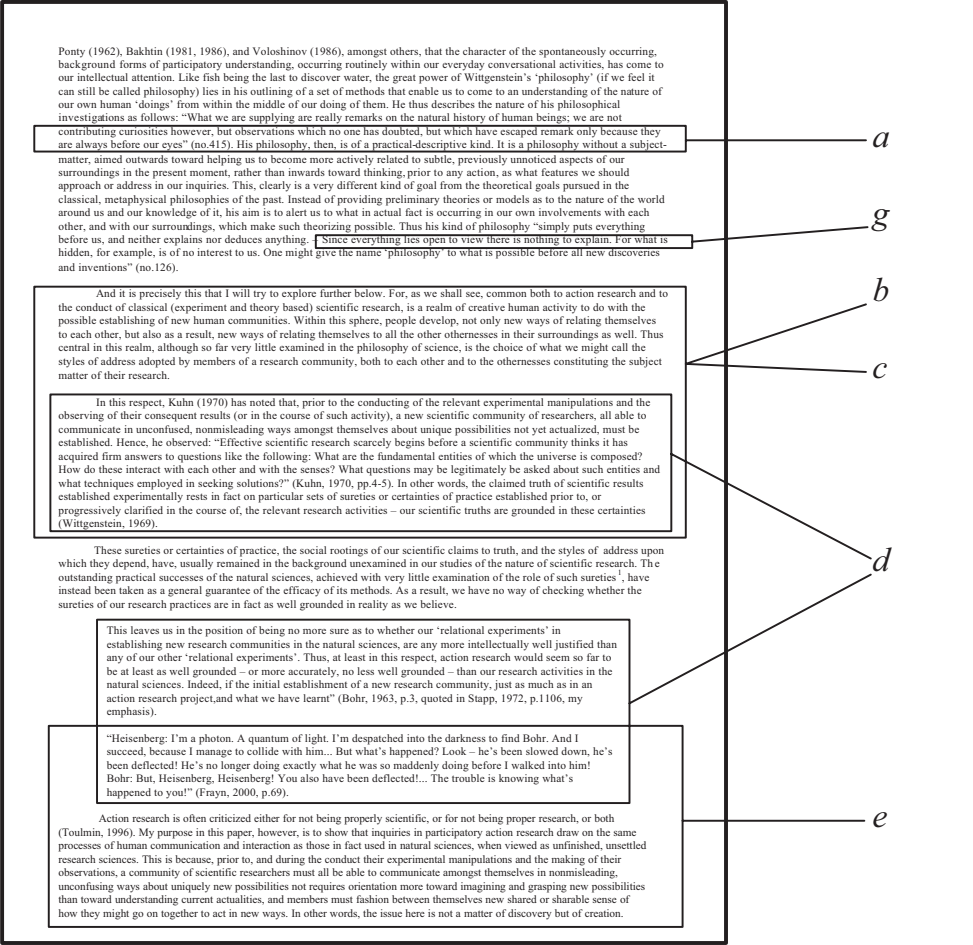
interpretation would suggest that the researchers regarded the arguments against Freud's dream theory as having valuational, scientific, and social dimensions. A third interpretation is that there are nine variables whose values are *present* and *absent*, with the breakdown into *A*, *B*, and *C* providing three convenient conceptual contexts for their individual definitions.

Readers of content analysis research may find important clues to the data language in use by examining how data are treated. An invariant organization of the data suggests constants. Variables, by contrast, vary, allowing coders to record different kinds of observations or readings. Separate judgments suggest separate variables. Inasmuch as the mutually exclusive values of a coding instrument partition a sample into mutually exclusive sets of units, the summing of frequencies to a total always points to the assumption of mutual exclusivity of values or combinations of values. Several ways of summing frequencies suggest independent variables—in cross-tabulations, for example. In the example of coding for Freud's dream theory, a careful reading of Herma et al.'s (1943) report of their study reveals the first of the three interpretations offered above to be correct, only because the frequencies reported for the nine values sum up to 100%. The grouping into *A*, *B*, and *C*, defined within the same variable, allowed the researchers to lump findings into simpler categories later. This, however, is not evident from their published conceptual scheme.

There have been unfortunate misunderstandings among content analysts and sometimes even resistance to being clear about the data language employed. For example, in interactive-hermeneutic explorations of texts, specifically when the researchers are using computer-aided text analysis software (see Chapter 11, section 11.6), coders are given considerable freedom to highlight any relevant portions of a document and to assign any number of codes to them (see Figure 8.1). Subsequent to the coders' unitization and categorization, the qualitative researchers can retrieve, reorganize, and tabulate the coded sections for further examination. Note that in the example in Figure 8.1, the first and second textual units are assigned one code each, *a* and *g*, respectively. The third unit is assigned two codes, *b* and *c*. That third unit also contains a fourth unit that is assigned the code *d*. If all the highlighted text segments were as separate as the first two units, their codes could be treated as the values of a variable. In this case, *a*, *b*, *c*, and *d* are all binary variables, not categories in the technical sense.

Because overlapping units cannot be enumerated, researchers with quantitative research questions shy away from double coding. In contrast, qualitative content analysts find uniform unitizations irresponsible to the nature of the texts they study, and so they tend to avoid more formal analyses. Both attitudes limit the analysis of textual matter, which often is complex in structure. However, when an analyst treats each code as a binary variable (i.e., either present or absent) and keeps references to the beginnings and ends of the highlighted text (on the reliability of unitizing, see Chapter 12, section 12.4), this constitutes a data language that would enable the analyst to correlate these codes and apply more complex analyses. In Figure 8.1, the third highlighted text segment would contribute to a correlation between *b* and *c*, and the overlapping text segments would count toward correlations between *b* and *d*, between *c* and *d*, and between *d* and *e*. As the highlighting and coding of text is implemented in a computer, there are no ambiguities or inconsistencies. By being clear about the data language by which data are created from raw text,

FIGURE 8.1 ■ Highlighted and Coded Text Sections



content analysts make their analysis more transparent and reexaminable, enabling others to learn from them and encouraging the development of suitable analytical techniques.

The values of variables may be unordered or ordered, and, in the latter case, they may exhibit one of several metrics. *Ordering* refers to a system of relationships between the values of a variable, determining which pairs of values are neighbors and how far they are apart. For example, in a hierarchy, one value may have several neighbors that are not neighbors of each other, and each of the latter may have other neighbors that are not neighbors of each other either. In a chain, each value has two neighbors, except for the values at the beginning and end of the chain, which have only one neighbor each.

Metrics define quantitative differences between all pairs of values in a variable. We distinguish several kinds of metrics according to the *mathematical operations applicable to these differences*. Thus, two dollar amounts may be added or subtracted, representing the experience of earning or spending, but addition and subtraction would not make sense when the values to be compared are qualitative attributes

such as individuals' emotional states, citizenship, or occupation. When qualitative attributes are expressed numerically—telephone numbers, Social Security numbers, the numbers on the jerseys of basketball players—addition and subtraction are mathematically possible but do not make sense semantically. Being concerned here only with the syntax of data languages, I distinguish among the possible metrics of variables by the operations that are applicable to their values. The metric of money differs from the metric of telephone numbers, for example. I will start with the simplest of all variables whose values are unordered—nominal variables—and then introduce several orderings and several metrics.

8.4 NOMINAL VARIABLES

Nominal variables, the most basic kinds of variables, are defined by the absence of both ordering and metric. Their values are merely distinct from each other, and hence, unordered. The mathematics for nominal variables is set theory, a calculus concerned with unordered entities. The adjective *nominal* suggests the “by name only” nature of these variables. Calling nominal variables “nominal *scales*” is a misnomer, because a “scale” conjures images of a linear ordering of values, which is precisely what the values of nominal variables do not possess. They may be arranged in any way conceivable without making a difference. Data recorded in nominal categories are also called *qualitative* because the difference between any two values of a nominal variable is the same for all possible pairs of values.

The nine standards for rejecting Freud's dream theory listed above constitute one nominal variable. Other examples are alphabetical characters, speech acts, forms of government, ethnic identities, and Social Security numbers. Analysts must take care not to be misled by the use of numbers as names for nominal categories. Numerical listings of bank customers' PINs or of the numbers on the jerseys of athletes have no operational significance. A more technical way of stating this property is to say that *the distinctions within a nominal variable are preserved under all permutations of its values*.

All variables reduce to nominal variables when their orderings and their metrics are removed from them. In the following sections, I discuss what distinguishes other variables from nominal variables. Table 8.1 shows, orderings by metrics, the types of variables to be discussed and useful in content analysis.

8.5 ORDERED VARIABLES

For a variable to make sense, any ordering of its values must somehow be appropriate to the phenomena the variable is to record. Something that varies along one dimension, such as length, audience size, or positive or negative evaluation, is very different from daily time, which repeats over and over again and is circular, or individual names, which are either this or that but nothing in between. Networks of concepts extracted from a writer's work (Baldwin, 1942), the semantic connections within a text as stored in a computer (Klir & Valach, 1965), and the hierarchy of organizing a piece of writing (from the work as a whole to its chapters, down to individual sentences)—these exhibit other orderings. Below, I discuss four common orderings of values: chains, recursions, cubes, and trees. These are not intended to constitute an exhaustive classification; I have chosen them merely to expand the

conventional limitation to linear scales of measurement, so-called measuring scales, which are favored by statisticians but rarely capture the meanings of text.

TABLE 8.1 ■ Types of Variables by Orderings and Metrics

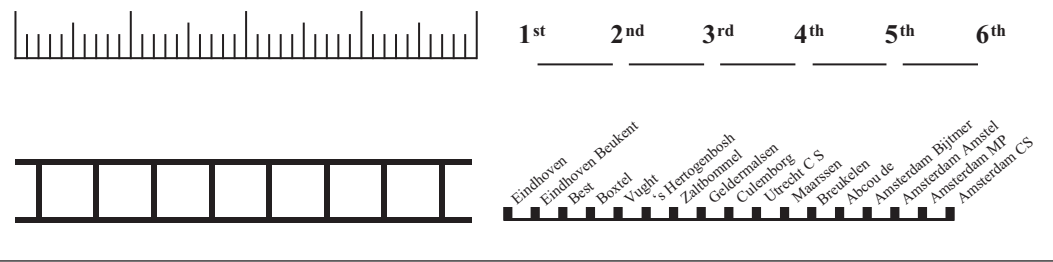
	Metrics				
Order	None	Chains	Recursions	Cubes	Trees
None	Nominal variable				
Ordinal	Grouping	Ordinal scale	Loop	Cross-tab of ord. variables	Typology
Interval	Network of distances	Interval scale	Circle π	Geometric space	Interval tree
Ratio		Ratio scale		Vector space	Ratio tree

8.5.1 Chains

Chains are linearly ordered sets of values, as in scales of measurement. The values of a chain are transitive in the sense that $a \rightarrow b$ and $b \rightarrow c$ implies $a \rightarrow c$ for any three values of a chain. In speaking of body temperature, for instance, we have a conception of what is normal, and we conceive of temperature as going up or down in degrees. Temperature is a one-dimensional variable. It can move through all of its values between extremely high and extremely low, never moving sidewise, never bypassing or jumping over any one temperature. The actual unit of measurement (degrees Fahrenheit, degrees Celsius, or kelvins) is secondary to the conception that it moves to one or the other of two neighbors. When we talk of more or less, before or after, or changes, we tend to imply chains, even when we use relative terms such as *wealthy*, *intelligent*, *successful*, or *progressive*. Chains may be open or bounded. Polar adjective scales, introduced in Chapter 7 and mentioned above, have defined beginnings and ends. Chains may also be conceived of as emanating from one outstanding value in one direction, as in the size, readership, or frequency of a newspaper or Zillmann’s (1964) semantic aspect scale, or in two directions, as in the positive or negative bias of reporting. Figure 8.2 adds a train schedule and a ladder conception to the examples of chains.

The familiar ordinal scales, interval scales, and ratio scales are all chains to begin with; the difference between these scales and chains is one of metrics. As

FIGURE 8.2 ■ Chains

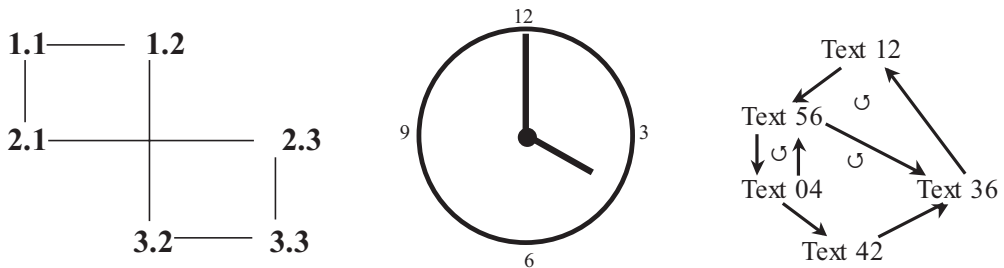


noted above, the term *nominal scale* is a misnomer, as nominal variables exhibit no ordering at all.

8.5.2 Recursions

Recursions are circular connections between values. Recursions can be conceived of as chains whose ends are seamlessly joined. Each value has exactly two immediate neighbors; there is no end and no outstanding value. Moving from any given value in one direction eventually brings one back to where one began—which was arbitrary to begin with. Transitivity applies locally but not to all values. Figure 8.3 shows this graphically. Examples of recursively ordered phenomena include seasonal fluctuations, ecological cycles, and human-computer interactions. Namenwirth and Weber (1987) demonstrated cyclicity in the use of political values and adopted a recursive notion of time (see Figure 10.5 and Krippendorff & Bock, 2009, Chapter 3.8). Biologists describe biological phenomena in terms of life cycles, and cyberneticians have identified the stabilizing efforts of complex systems in terms of circular causal feedback loops. In accounts of how social prejudices take hold in a population, how political candidates get elected, and how the “spiral of silence” affects public opinion, recursive variables are indispensable.

FIGURE 8.3 ■ Loops and Circles



The practice of cutting recursions into more easily analyzable linear continua usually destroys their circular essence. For example, some social psychology researchers have cut speech acts out of the ongoing circularity of human interactions that realize their meanings, and this may account for the rather artificial causal conceptions that dominate social psychological explanations of language use. In the same way, when one describes a computer interface in terms of the graphics involved, one hides the dynamic nature of the interface. Many so-called inconsistent preferences, such as $a \rightarrow b$, $b \rightarrow c$, and $c \rightarrow a$, in fact define recursions. These are far from irrational or abnormal; rather, they belong to a nonlinear ordering.

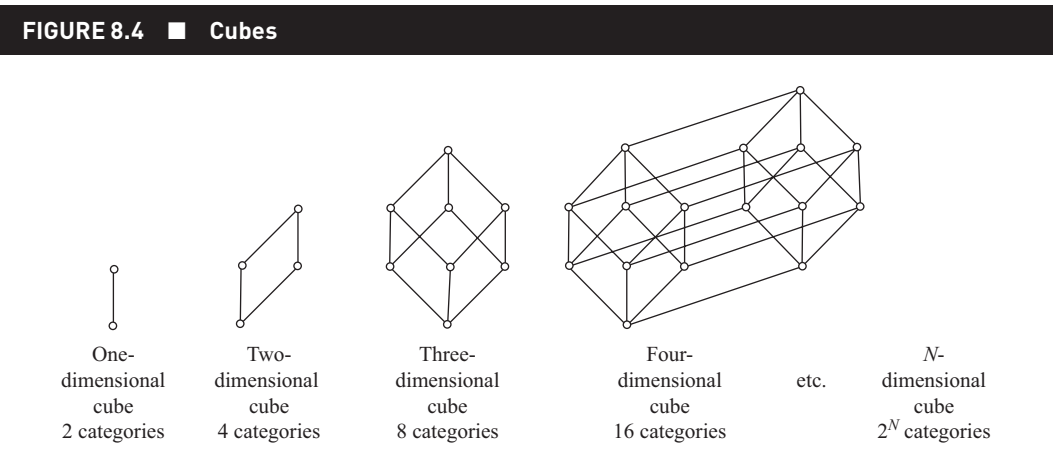
8.5.3 Cubes

Cubes depict multidimensional variations. The values in cubes are ordered with immediate neighboring values differing in one of a cube's dimensions. Cubes often arise by conceptual default. Consider Lasswell and Kaplan's (1950) so-called eight

values. When listed as follows, they give the impression of constituting one variable with eight categories:

- Power
- Rectitude
- Respect
- Affection
- Wealth
- Well-being
- Enlightenment
- Skill

Within any one variable, the choice of any one of its categories precludes all others. However, conceptually, one can value *Wealth*, *Affection*, and education (or *Enlightenment*), which would speak against treating these eight terms as categories of a variable called “values.” Rather, they constitute an eight-dimensional space in which all of these qualities can be cherished independently of each other. Indeed, Lasswell and Kaplan (1950) allowed their recording units—persons, symbols, and statements—to score high on more than one of these values. If the eight values were to be taken as categories of a nominal variable, the permission to record combinations of them would violate the mutual exclusivity requirement of variables and render the data so recorded no longer analyzable as a nominal variable. In fact, any instruction to coders to “check as many as applicable” signals a data structure other than a scale. When any of the eight values could be present or absent independent of each other, the values define an eight-dimensional cube consisting of eight binary variables giving the coder not eight but $2^8 = 256$ options. Figure 8.4 shows cubes of increasing dimensionality created by the presence or absence of independent qualities.

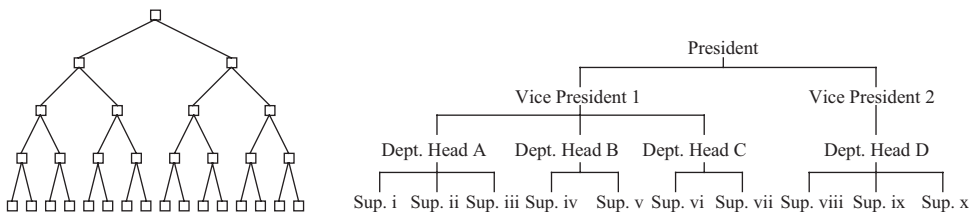


8.5.4 Trees

Trees have one origin and two kinds of values, branching and terminal. All values are available for coding. Trees show no recursions, as Figure 8.5 illustrates. Each value in a tree can be reached from its one origin by a separate path that passes through a number of branching values. Trees are basic to the recording of linguistic expressions and conform to one of the earliest theories of meaning. Aristotle's notion of a *definition*, for example, requires naming the *genus* (the general class) to which the *definiens* (the word to be defined) belongs and distinguishing the latter from all other species of that genus. Moving from genus to genus describes moving through the branching points of a tree. The system of categories in the Linnean classification in biology—not the organisms it classifies—constitutes a tree. Closer to content analysis, a reference to Europe implies references to France, Italy, Germany, and so on. A reference to France implies references to the regions of that country. The relation connecting “Europe,” “France,” and “Provence” is one of inclusion and defines a path or chain through a tree. France and England are on different paths, as neither includes the other.

Most content analyses fix the level of abstraction on which countries, populations, products, or mass-media programs are coded. By allowing variable levels of abstraction, trees offer a richer alternative. The family of trees has many members: decision trees, telephone trees, the trees that the rules of a transformational grammar generate, and social hierarchies in business organizations, in the military, and in government. (I discuss the possible confusion of trees with groupings in section 8.6.1. Note here only that each branching point of a tree is a value that is legitimate for coding data.)

FIGURE 8.5 ■ Trees



8.5.5 Multi-Valued Sets

While most statistical methods require units to be characterized by mutually exclusive categories or values of any number of variables, there are good reasons to allow a unit to be characterized by more than one code, by sets of them. Literary theory assumes that all texts have multiple interpretations, not one. Word dictionaries list semantically similar meanings among which readers may choose depending on their syntactical function, context of use, or the discourse they are pursuing. Efforts in computational linguistics to disambiguate sentence structures acknowledge ambiguities. Evidently, a single phenomenon may be described in terms of multiple terms. One can distinguish several conditions under which a variable needs to be

defined not in terms of mutually exclusive categories but in terms of distinct sets of them with the possibility of sharing them across units. There are many occasions to generate such multi-valued data:

- Ambiguity:
A single statement may be variously interpretable.
- Indecision:
It may be obvious what someone accomplished but not whether it was due to *luck* or *calculation*.
- Simultaneous qualities:
Someone may be a *father*, a *baseball player*, and a *cook*.
Someone may have several hobbies: *playing cards*, *reading fiction*, *eating in expensive restaurants*.
- Multiple indicators:
A narcissistic personality disorder is defined by *an exaggerated feeling of self-importance*, *need for admiration*, and *lack of understanding others' feelings*. (APA, 1954).
- Guessed ranges:
Her dress has *blue* and *white* stripes. The confidence interval of α is between *0.860* and *0.935*.
- Statistical distributions:
A document maybe represented by a word cloud of the most frequent words.
- Clusters:
A grouping of objects with some shared qualities, such as the set of people in a photo or the books on a reading list

When the values of multivalued descriptions of units can be assigned to different variables, it is conceivable to treat them as single-valued descriptions of units in multi-dimensional vector spaces. However, when they are all part of a common concept or type, they will have to be taken as multi-valued sets with the possibility of intersections among them (see sections 12.6 and especially 12.7), able to form many structures.

8.6 METRICS

Any two values may differ quantitatively—whether they are neighbors or not. In a semantic network, singular concepts (its nodes) are linked to each other by relational concepts (for example, [Joe] <runs for> [Governor]; [Joe] <was> [an accountant]). A metric defines how closely any two concepts are associated, how similar they are, or how much they have to do with each other, not whether they are neighbors or where their ordering locates them. A metric recognizes various kinds of differences and specifies what an analyst can do with them. The literature on metrics developed largely on chainlike variables and in the context of a measurement theory that distinguishes among nominal, ordinal, interval, and ratio scales (Stevens, 1946). These four metrics (listed here in the order of their increasing power) differ in the

information they can represent and are therefore often called levels of measurement. In the following, I describe the three principal metrics and then compare their mathematical properties in Table 8.2. In Chapter 12, section 12.3.4, I discuss how they variously affect reliability.

8.6.1 Ordinal Metrics

Ordinal metrics describe recording units in such relational terms as “larger than,” “more than,” “precedes,” “causes,” “is a condition of,” “is a refinement of,” “is contained in,” “supervises”—in short, in terms of *ranks*. Ordinal scales (chains with ordinal metrics among values) are probably most common in the social sciences, largely because relationships between people and objects tend to occur in language, spoken or written, and are then also more easily recorded in words. When the stock market is said to “gain,” an ordinal metric is implied. When it is said to “gain 5 points,” an interval metric is invoked. Ordinal scales using 3, 5, and 7 ranks are popular in content analysis because language rarely has more words available to distinguish among them. Polar opposites lend themselves to 3-point scales (e.g., a scale from *good* to *bad*, with *neutral* as its midpoint); the addition of simple adjectives, such as *more* or *less*, results in 5-point scales, and the addition of superlatives (e.g., *most* and *least*) leads to 7-point scales.

In content analysis, ranks may be variously operationalized. Newspaper editors, for example, employ several typographical devices to express the importance they assign to the news items they publish. Suppose that interviews of a sample of newspaper editors revealed that the importance of the news items of the day is expressed in the following placement of articles:

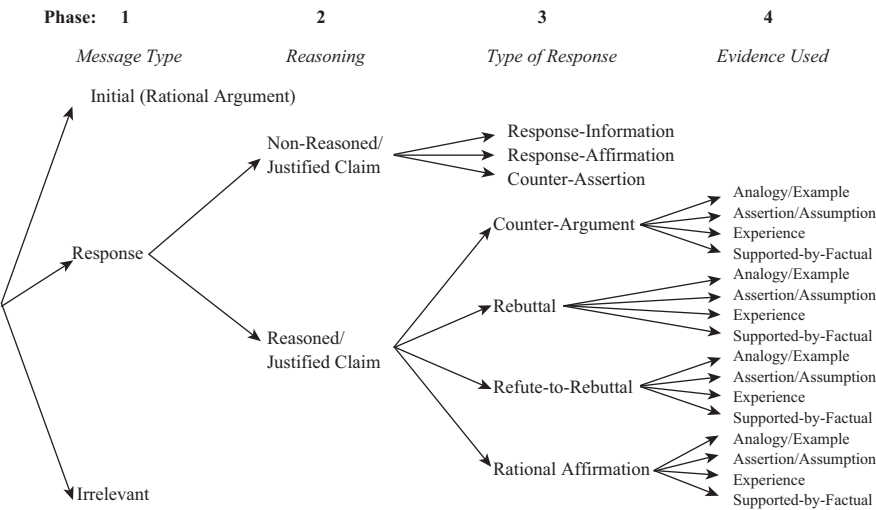
- 1st: Largest multicolumn headline above the center fold of the front page
- 2nd: Any other headline above the center fold of the front page
- 3rd: Any headline below the center fold of the front page
- 4th: Any multicolumn headline on the second, third, or last page
- 5th: Any other headline for single-column articles above the center fold of any other inside page
- 6th: Any other headline below the center fold of any other inside page
- 7th: Any other news item

Assuming that the editors’ indications of the importance of news items are those of a somewhat stable journalistic culture, content analysts may use this construct to infer the importance of news items by ranking them with this 7-point rank order. Note the frequent reference to “any other” in the above descriptions. It refers to the previous and higher ranked qualities, which is typical of rank-orders, whether of pain scales from 0 to 10 or the cultural importance of cities. Using numbers to indicate ranks does not mean that one can add, subtract, or multiply them.

As Table 8.2 suggests, ordinal metrics are not limited to chains. Grouping an unordered set of values into conceptual categories introduces inequalities between the otherwise pairwise equal differences. Groupings suppose that the values

within one group have more in common with each other than with the values in different groups. The above-mentioned standards for analyzing Freud’s dream theory represent a grouping: The difference between A.1 and A.2 is smaller than the difference between A.1 and B.1, but nothing indicates by how much. Graham and Witschge (2003) introduced such differences in ranks by categorizing messages, the posts to online discussion groups, in four convenient phases, effectively grouping 21 categories on four levels. Figure 8.6 shows the researchers’ process. In Phase 1, messages were distinguished into three groups, two of which were coding categories. In Phase 2, messages that responded to previous messages were grouped into two kinds, depending on whether they manifested reasons. In Phase 3, nonreasoned claims led to three coding categories while reasoned claims were divided into four types of responses, and, in Phase 4, each led to four groups indicating the kind of evidence used in the arguments. Frequencies were obtained for each of these 21 coding categories, which could be summed in the reverse order of the distinctions that led to them. Although the data are qualitative, showing no particular ordering, the grouping imposed a metric that rendered messages in the same group more similar to each other than messages in different groups. Figure 8.6 suggests that messages that manifest reasons and those that do not are more different than messages that differ in whether they contain counterarguments, rebuttals, refusals to rebut, or rational affirmations.

FIGURE 8.6 ■ A Grouping of Messages From Online Deliberations



Source: Adapted from Graham and Witschge (2003, p. 181, fig. 1).

Groupings reflect conceptual hierarchies that are defined on top of a set of originally nominal values. When used repeatedly, any decision tree—for example, that depicted in Figure 8.6 but also the one in Figure 7.1—creates groupings.

Decision trees proceed from rougher to finer distinctions and from larger and less differentiated sets of units of analysis to smaller and more specialized sets. One analytical implication of grouping is that it suggests the order in which the frequencies of values may be summed, undoing decisions one by one. Hierarchical clustering procedures, for instance, proceed that way as well. They capitalize on unequal differences between elementary qualities to develop a hierarchy, represented by a dendrogram, which could explain the collection of these qualities as groupings (for instance, see Figure 10.10).

Groupings and trees are easily confused, and as both are important in content analysis, I want to highlight their distinction. As already noted, groupings provide convenient conceptualizations of a given set of values, the terminal points of a decision tree, like the outline of a book in chapters and sections. Groups are not values, however. An outline is not the text it organizes. In contrast, the values of a tree are not limited to the terminal values of the tree; they include its branches as well. Thus, their values are not merely different across the boundary of groups; they may include each other, enabling the coding of different levels of inclusion, abstraction, or entailments. Take the above-mentioned Linnean classification system as an example. It groups organisms into classes and subclasses and provides concepts that label these groups on different levels. For instance, *mammal* is not an organism but the name of a group that includes humans, whales, and mice. The Linnean system *groups organisms*, but it defines a *tree for the names of groups* of organisms, which describe organisms on different levels of commonalities.

8.6.2 Interval Metrics

Interval metrics represent quantitative differences between recording units. Measures of time, distance, and volume as well as changes in quantities and movement in space all assume that intervals are meaningful. When applied to chains, an interval metric creates interval scales and enables the addition or subtraction of differences between scale points. In psychological tests, subjects are often asked to use rating scales with equal intervals to answer questions. In content analysis, the semantic differential scales that are used to record judgments of biases, personality traits of characters, and so on are often conceptualized as equal-interval scales. Intervals do not need to be equal, however.

Interval data are the preferred kind in empirical social research, largely because of the wealth of statistical techniques that are available and accessible to process them, especially techniques that require the calculation of differences, as in variance calculations, correlation methods, factor analyses, multidimensional scaling, and clustering. Interval metrics might well be an artifact of these techniques. In the natural sciences most measures, except for time, have ratio metric properties, and in content analysis, interval scales tend not to be as reliable as data with a less powerful metric. For example, in research on the personality characteristics of fictional characters on television, semantic differential scales, which are treated as interval scales, have been notoriously unreliable. This has been so not only because language is rarely as precise as would be necessary for differences to be calculable but also mainly because personality characteristics that are irrelevant to a plot may not be present at all, causing coders to guess when they are forced to choose among interval values. Nevertheless, many secondary measures that content analysts provide—quantitative

indices of phenomena, geometric depictions of findings—have valid interval qualities.

8.6.3 Ratio Metrics

Ratio metrics are defined from absolute zero points relative to which all differences between values are expressed. Lengths, weights, speeds, masses, and absolute temperatures in kelvins (but not in degrees Fahrenheit or Celsius) exemplify ratio scales in the physical sciences, none of which can go below their absolute zero point. There are also many examples of ratio-level measurements of text, such as column inches of newsprint, sizes of photographs, frequencies of publication, audience sizes, and Nielsen ratings, as well as amounts of information and costs. These have no negative values either. In content analysis, these measures may have less to do with what a text says or the role it plays in a particular context than with how prominent recording units are or how much they say to the analyst.

The list of metrics is far from settled, and far more orderings are available than are relevant for content analysis. Regarding data languages—of which variables, orderings, metrics, and relationships or connections are the most prominent features—it is important to keep their Janus-faced character in mind. Any data language must be appropriate to the phenomenon being recorded—and from this perspective, the best data language is the raw text itself. Any data language must also render the data amenable to formal analyses. Given the currently available analytical techniques, the gap between how texts are read and acted upon and the forms these techniques require often seems large. For content analysts, the challenge is to develop computational techniques whose requirements come closer to naturally occurring phenomena. Data processing has advanced toward making more and more natural phenomena electronically available—fingerprint, face, and voice recognition, for instance. However, the ability to store, retrieve, and compare electronic records tends to fall far short of how intelligent humans identify, interpret, conceptualize, communicate, and act—hence the importance of creating sufficiently rich but also computable data languages.

8.7 MATHEMATICAL OPERATIONS

As noted above, a metric is defined by the mathematical operations under which the relations between the recorded units remain invariant. Adding 4 to the values of a semantic differential scale transforms a -3-to-+3 scale into a 1-to-7 scale, yet the algebraic differences between the scale's values remain exactly the same. However, when these values are multiplied by 4, the numerical differences between neighboring values become very uneven; only their ordering remains unchanged (i.e., it remains a chain). This property is important in determining the applicability of analytical techniques to the recorded data. Thus, I distinguish two kinds of operations on the values of a variable, one preserving the original numerical relationships between the values of a variable and the other preserving their orderings. Table 8.2 lists these functions. The table amounts to permission for analysts to apply analytical techniques that employ these transformations and suggests the kinds of relationships in texts that various operations preserve or omit.

TABLE 8.2 ■ Operational Properties of Metrics

Metric	Relations R_{xy}	Relation-Preserving $f()$ s $R_{xy} = R_{f(x)f(y)}$	Order-Preserving $f()$ s $f(R_{xy}) > f(R_{wz}) \Leftrightarrow R_{f(x)f(y)} > R_{f(w)f(z)}$
None	Distinctions $x \neq y$	1:1 Permutations	1:1 Permutations
Ordinal	Ranks $x \geq y$	Monotonically increasing f s	Monotonically increasing f s
Interval	Differences $x - y$	$x' = x + b$	Linear functions $f()$: $x' = ax + b$
Ratio	Proportions x/y	$x' = ax$	Exponential functions $f()$: $x' = bx^a$

Table 8.2 lists metrics in the increasing order of their *power*. Ratio metric data, which are most powerful and potentially most informative, may be computed as interval data at the expense of all information about the location of values relative to their absolute zero point. Using the more readily available variance-type statistics on ratio-level data, for example, discards this information. Ratio and interval data may be computed with ordinal techniques, which treat them as ordinal data, but only at the additional expense of all information about the numerical differences between values. Finally, ratio, interval, and ordinal data may be computed as nominal data, at which point all information about their orderings and metric qualities is lost. Losses of relational information are irreversible.

Going in reverse through these metrics, applying an ordinal technique on nominal data produces uninterpretable results. Applying an interval technique on ordinal data yields spurious findings. The lesson to be learned from the above is that

*the power of a data language must match or exceed
the power of the analytical procedures to be employed.*

9

ANALYTICAL CONSTRUCTS

Following the discussions in previous chapters of different uses of content analysis and the kinds of inferences they make, this chapter illustrates several ways of operationalizing analytical constructs from various ways of knowing the contexts of given texts. Analytical constructs are adopted to justify the abductive inferences content analysts make. This chapter presents examples of the forms that such constructs might take.

9.1 THE ROLE OF ANALYTICAL CONSTRUCTS

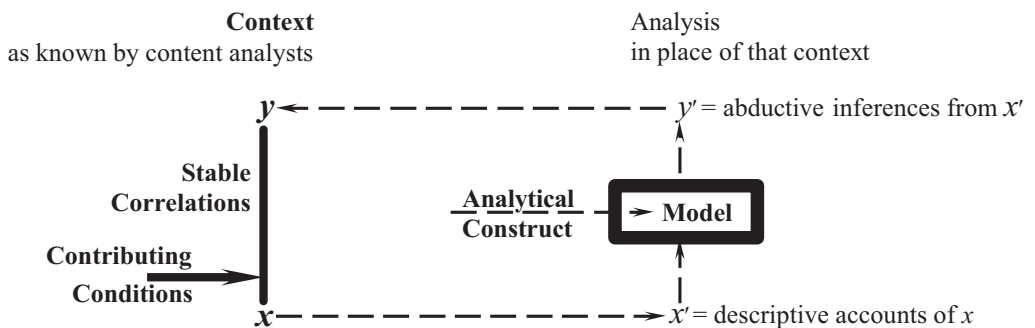
An analytical construct accounts for what the content analyst knows, suspects, or assumes about the context of the text, and it operationalizes that presumption procedurally in order to produce inferences from that text. Figure 4.2 shows the role of the analytical construct among other analytical components of content analysis. In its simplest form, an analytical construct is a function, a collection of “if-then” statements, or a computer program that defines at least one path from available text to the answers sought.

In Chapter 2, I identified the inferential step from text to the answer to a research question. This inference is *abductive* because the two domains—texts (along with their descriptions or transcriptions) and what these texts imply—are logically independent of each other, and bridging this logical gap requires justification. So conceived, an analytical construct functions as a hypothesis, the best hypothesis or explanation that the analyst can imagine and defend, of how a body of text is read, what it does, or to what use it may be put in a context of the analyst’s choice. Appropriating Toulmin’s (1958) terms, Chapter 2 suggested that analytical constructs, if reliably executed, *warrant* the intended inferences (guide the analyst along a logical path), but such inferences must in turn be *backed* by knowledge of the context of the analyzed texts (assure the analyst that the path leads to valid conclusions). I will discuss the justifications for analytical constructs that underlie content analyses in Chapter 13; here, my focus is on how content analysts can derive these constructs.

In the processes of operationalizing an analysis, ideas, which may well emerge during the reading of samples of the texts to be analyzed; hypotheses, derived from the literature of the context chosen for the text; or established theories are formalized, “tamed,” or computationally structured so that they meet the researcher’s need to answer research questions from given text—much as a computer could be programmed to accept data of a certain kind and yield usable results. For a content

analysis to proceed relative to a context, its analytical construct must also be a *model of the relationships between the texts and the target of intended inferences* about unobserved variables of that context. What makes drawing these inferences practically possible is the explicit nature of that model—allowing these inferences to be drawn repeatedly and reliably, as is expected of all scientific procedures. What backs these inferences is a demonstration, or at least an arguable assumption, that the analytical construct is empirically rooted in the context of the given body of text, that it represents the stable correlations within a context (as in Figure 2.2), leaving two main uncertainties or variables: (a) the contributing conditions under which these correlations are presumed stable (and the analytical construct applicable) and (b) the available texts by means of which the research questions are to be answered. Either of these two uncertainties may require qualifications (in Toulmin’s sense) of the research results.

FIGURE 9.1 ■ Analytical Construct as Specifying the Model of a Context



To be specific, Figure 9.1 extracts the analytical construct from Figure 2.2 and presents it as a model of the relevant variables of the context. In Figure 9.1, x is the sampled texts. The arrow from x to x' defines a mapping that summarizes the processes of unitizing, sampling, recording, and reducing, described as data making in Chapter 4. The analytical construct computes y' , which remains indeterminable or open until x' is known (much as in algebra, where a function, such as squaring, has no answer unless the number to which it is applied is known). If there is a mapping from y to y' on which grounds y' can be said to point to, represent, or at least correlate with specific variables of the context to which y belongs as well, then y' is valid. The abduction is justified by the assumption that the analytical construct is a true or heuristic model of the context. In adopting an analytical construct for the purpose of an analysis, a researcher is led to distinguish, according to George (1959a), between the stable or unchanging conditions, which are the ones modeled, and the unstable or variable conditions, which may become fixed once the analyst obtains a body of relevant text, x , analyzes it, and infers y' from it. The motivation for engaging a content analysis, y , always remains more or less uncertain, y' being its best approximation.

Below, I address some ways in which content analysts can develop analytical constructs, operationalizing $x' y'$ relationships, that, when implemented, have a good chance of selecting valid answers, y' , to the content analysts’ questions concerning y .

9.2 SOURCES OF CERTAINTY

Analysts need to justify their analytical procedures, not only in their final form but also at each step taken during their development. In constructing an analysis, the analyst needs to preserve the modelling relationship to the chosen context. The structure of the texts themselves, significant as it is, shows up in the analyst's appropriate choices of a data language and suitable recording instructions, which I have addressed in Chapters 7 and 8; here, the focus is on analytical constructs only.

Content analysts rely on one or more sources of certainty in developing analytical constructs:

- *Previous successes and failures* of content analyses to argue for functional correspondences between the construct and a chosen context
- *Expert knowledge and experience* with a context to argue for structural correspondences between the construct and the context
- *Established theories*, including about a context, which requires arguing that the structural correspondences between the construct and that context is governed by them
- *Embodied practices*, sampled from a context, to argue for the representative nature of the inferences obtained from these practices

I discuss each of these sources in turn below.

9.2.1 Previous Successes and Failures

Previous successes and failures provide an analyst with very practical reasons for developing and adopting a particular analytical construct. The contention is obvious: What succeeded in the past must have had something to do with the context in which it worked, and, unless that context has changed, one might as well continue to use what worked—until it runs into failures. History is no theory for the success of an analysis. It provides no justification for the structure of its construct except for its having worked before.

Recall that in Chapter 4, I mentioned one design (development of a discriminant function) that increases the success of a content analysis incrementally. Stone and Hunt's (1963) computer analysis of real versus simulated suicide notes proved extraordinarily successful despite its a-theoretical way of proceeding. The real notes in their study came from court records in Los Angeles. The simulated notes were written by a panel made up of individuals who matched the authors of the real notes in population characteristics (sex, age, race, occupation, and so on). The researchers' decision to analyze these notes was motivated by a suicide prevention center's interest in knowing whether the center's personnel should take particular suicide notes seriously. Stone and Hunt analyzed 15 pairs of notes with known identity (real or simulated) for what differentiated them and came to three conclusions:

1. The frequency of references to concrete things, persons, and places is higher in real notes.
2. The frequency of the actual word *love* is higher in real notes.
3. The total number of references to thought and decision processes is higher for simulated notes.

From these findings, the researchers developed an index by simply subtracting the total number of references to thought and decision processes (conclusion 3) from the sum of the frequency of references to concrete things, persons, and places (conclusion 1) and the frequency of the word *love* (conclusion 2). This discriminant function enabled them to infer the alleged authenticity of the notes. In fact, it correctly differentiated 13 of the 15 pairs of notes used in this preparatory analysis.

Having succeeded thus far, albeit with some uncertainty remaining in the results, Stone and Hunt (1963) then applied the discriminant function to 18 more pairs of notes whose identity, real or simulated, was not revealed to the two researchers, and they were able to determine correctly the authenticity of 17 out of 18 notes. In addition to being statistically significant, the performance of the discriminant function developed solely on past successes and failures also turned out to be significantly better than human judgments obtained in separate experiments.

Obviously, Stone and Hunt (1963) reached the three conclusions listed above without having any particular theory in mind. It would indeed be a challenge for researchers to find a rationale for how these three variables of suicide notes are connected to feeling suicidal. The adding and subtracting of scores, for example, would make sense if there were hidden quantities that cooperated (measured by variables 1 and 2) or competed for each other (measured by variable 3 versus variables 1 and 2). However, such quantities are difficult to imagine as the cause for a writer's suicidal versus nonsuicidal inclinations. Nevertheless, the analytical construct in the form of this discriminant function succeeded statistically, and this demonstration would be a sufficient argument for its use.

Ultimately, all content analyses must demonstrate success to a degree better than chance. Unfortunately, opportunities for the repeated availability of validating evidence are rare—which brings us to the next path to analytical constructs.

9.2.2 Expert Knowledge and Experience

Familiarity concerning the chosen context is a valuable asset to all content analysts, of course. It can provide analysts with important face validity checks on the decisions they are making, but familiarity may not be sufficient. Even hard-nosed researchers occasionally get carried away by conceptions that make sense to them but, in reality, have little to do with the world of others in the contexts they are analyzing. Yet, when content analysts are faced with a novel situation, with having to design a content analysis procedure without precedents to rely on, or without history of research and theorizing about that context, personal knowledge, perhaps of known experts, may be all the analysts have to start with.

Leites, Bernaut, and Garthoff (1951; see also Krippendorff & Bock, 2009, Chapter 3.3) provide a fascinating example of the development of an analytical construct out of researchers' experiences with the context of given texts. These analysts, all experts on the politics of the Soviet Union, were interested in the distribution of power within the Kremlin and particularly in predicting the succession in Soviet leadership. In governments in which succession is not regulated politically, such as in the Soviet Union of the 1950s, succession remains largely hidden to outside observers. Predicting the succession of leadership in such governments is both a favorite game of political analysts and important to foreign policy decision makers, for example, in the United States. Leites et al. obtained the public speeches made by politburo members on the occasion of Stalin's 70th

birthday in 1949, all of which expressed adulation of Stalin, as would be expected. Initially, political scientists attributed differences among the speeches to individual speakers' styles and therefore found them of no help.

The key to finding politically relevant differences among these speeches, Leites et al. (1951) surmised, might lie in linguistic modes of expressing nearness, for which Soviet political discourse seemed to offer two distinct approaches. The analysts noted that one set of "symbols of nearness and intimacy (father, solicitude, and so on) appears most frequently in the popular image of Stalin and [is] stressed for that audience which is far removed from him." The other set of symbols derived from the prevailing "depreciation of such nearness in political relationships. The ideal party member does not stress any gratification he may derive from intimacy for political ends. . . . Those closer to Stalin politically are permitted to speak of him in terms of lesser personal intimacy ('leader of the party,' etc.)" and are thus privileged to refrain from the crudest form of adulation. Leites et al. conclude their argument for the inferential strategy they adopted by suggesting that the relative emphasis on the "Bolshevik image" as opposed to the "popular image" of Stalin "not only reflects the Bolshevik evaluation of the party as distinguished from and superior to the masses at large, but also indicates the relative distance of the speaker from Stalin" (pp. 338–339).

Leites et al. (1951) tabulated the numbers of both kinds of references and ranked the speakers according to their relative emphasis on one kind versus the other. They found the speeches of Molotov, Malenkov, and Beria (in that order) to have the highest numbers of references to Stalin's Bolshevik image, and from this they inferred that these three politburo members were probably closest to Stalin. The power struggle that ensued immediately after Stalin's death clearly confirmed the validity of their inferences. Just out of curiosity, I translated the logic of Leites et al.'s arguments into a simple construct in the form of a distance function (Krippendorff, 1967, pp. 118–119):

$$D_{\text{to Stalin}} = \frac{N_{\text{popular}} + \frac{N_{\text{ambiguous}}}{2}}{N_{\text{popular}} + N_{\text{ambiguous}} + N_{\text{Bolshevik}}},$$

where N s are the frequencies of a politburo member mentioning popular, ambiguous, and Bolshevik images in their speeches. The ranking obtained by this construct replicated the inferences on the top of Leites et al.'s (1951) list but ended differently. It suggested that Khrushchev was the furthest removed from Stalin, who was immediately succeeded by Bulganin and Kosygin. As we now know, those furthest removed from Stalin ended up playing major roles in the fight to overcome Stalinism in the Soviet Union. The content analysis did not foretell the actual political events, but it led to the correct inference of an otherwise unobservable relationship among the politically significant players.

The preceding example is intended to suggest that the experts of a context for content analysis can often provide a wealth of perhaps disconnected propositions that content analysts may be able to sort out and assemble into constructs for analyzing available texts. Unless researchers put such expert knowledge into the formal terms of an analytical construct, texts may remain silent about the questions that analysts are asking.

Without adding another example, I will merely note here that fifth-generation computers, so-called expert systems, contain huge collections of propositions that represent what experts know about particular subjects. They are designed to answer knowledge questions by finding inferential links (not only abductive ones) between factual givens and the possible answers to users' questions. Such computers could be used for content analysis as well.

In moving from incomplete, perhaps even contradictory, expert knowledge to a construct that is suitable for a content analysis, analysts need to preserve the structural correspondence between what they do and that context at each analytical step they take. Analysts using expert knowledge proceed from the bottom up, so to speak, whereas those using established theories proceed from the top down.

9.2.3 Established Theories

If a context is well researched and theorized, especially including the role that available texts play in a context and the relationship between texts and the research questions the analyst seeks to answer, then the analyst may derive analytical constructs from available generalizations about that context. In Chapter 4, section 4.2, I outlined how a researcher might test analytical constructs as theories about a context. Here I address the converse of this activity, the derivation of analytical constructs from available theories and from the results of related research.

Theories come in diverse forms, of course. Sometimes theories are fairly specific propositions that have been tested in a variety of situations—for example, concerning correlations between speech disturbances and the anxiety levels of speakers, between the frequency of reported crimes and public concerns with law-and-order issues, or between the number of arguments people advance for or against a public issue and the level of their political competence and knowledge. Berelson and Steiner (1964) have published an inventory of 1,025 scientific findings in the social and behavioral sciences that content analysts might consult. Inventories specializing in content analysis are not available, but propositions of a comparable nature can be found in handbooks on psychiatry, social psychology, sociology, and sociolinguistics. Sometimes such propositions derive from more general theories, such as theories regarding the expression of emotions, linguistic manifestations of psychopathologies, how and according to which criteria the mass media select and their audiences receive or make use of news, or the frequency of exposure to particular propositions and the resulting belief in them. The “uses and gratifications” approach to media research exemplifies the latter. Perhaps for good reasons, a general theory of communication that includes interpretations of texts and perceptions of images has not been formulated. Content analysts have lamented this deficit for years.

An example of how a particular theory became an analytical construct is found in Osgood, Saporta, and Nunnally's (1956) “evaluative assertion analysis.” (See also Holsti's account in Krippendorff & Bock, 2009, Chapter 3.5.) I present only the result of its development here. The technique is derived from a version of dissonance theory, which assumes the following:

1. Concepts (attitude objects) are valued, “liked” or “disliked,” in degrees, ranging from positive through neutral to negative.
2. All linguistic assertions can be decomposed into one or more pairs of concepts (attitude objects) whose connections are accounted for by relations of association or dissociation, expressed in degrees. *Is, has, likes, supports, belongs,* and *cooperates* are strongly associative, whereas *is not, dislikes, opposes, fights,* and *denies* are strongly dissociative, with several shades of strengths between these extremes.
3. Some concepts, called *common meaning terms*, are unalterable in value. These consist mostly of adjectives such as *good, bad, dishonest, ugly,* and *mean*. Others are valuationally variable, for example, *United States, wine, psychotherapy, teacher,* and *Richard Nixon*. Their valuations depend on how references to them are used, which is the point of the analysis.
4. Individuals accept concept pairs that are balanced—that is, assertions containing associations between two similarly valued concepts (“I like my friend”) and dissociations between two dissimilarly valued concepts (“I hate my enemy”). Individuals reject concept pairs that are imbalanced or modify them so as to achieve balance. These are assertions containing dissociations between two similarly valued concepts (“I hate my friend”) and associations between two dissimilarly valued concepts (“I love my enemy”). Graphically, let the evaluation of attitude objects be represented in parentheses, (+) and (–), and let associations be represented as positive links, — + — and dissociations as negative links, — — —.

(+) — + — (+),
 (+) — — — (–), and
 (–) — — — (+) are balanced,
 Whereas (+) — — — (+),
 (+) — + — (–), and
 (–) — + — (+) are imbalanced.

From the psycho-logic sketched above, and under the assumption that individuals seek to avoid imbalance and move to achieve balance, it would follow that concepts with open valuation are valued implicitly, and analysts can therefore infer their value from suitably recorded texts:

From (+) — + — (?) one can infer (?) = (+),
 from (+) — — — (?) one can infer (?) = (–), and
 from (–) — — — (?) one can infer (?) = (+).

Evidently, this is made possible by the assumption that balance is the usual way of speaking and thinking, effectively ruling out the three imbalance triplets.

Osgood et al. (1956) moreover stipulate a quantitative relationship between the degrees of positive/negative valuation of the concepts involved and the

degree of association/dissociation between them. This motivates a calculus of valuation, which enabled Osgood et al. to infer not only the direction of a concept's (implicit) valuations but also its extent. It also leads to a statistic of these valuations, conceived of as describing the attitude structures of individual writers or readers. This may be a long story, but content analysts who work in well-researched contexts may have to develop similar constructs in support of their intended inferences. Osgood et al.'s calculus is derived from a psycholinguistic theory. Theories surely must exist in other empirical domains that could be so adopted in content analysis.

Deriving analytical constructs from established theories does not guarantee that the constructs will be flawless. Some theories have not been formulated with large bodies of text in mind. An analyst may close gaps in a theory with assumptions that send the construct astray. A theory might not be as general as the analyst assumes. In the case of evaluative assertion analysis, irony has no place, asserted changes in evaluation (e.g., I am starting to like my opponent) are difficult to deal with, reflections on a connection (e.g., Why should I love him?) cannot be handled, and other forms of resolving imbalances, including tolerating them (e.g., agree to disagree), mess up the inferences. Some scholars have raised doubts as to whether it makes sense to decompose a whole text into pairs of concepts and whether the notion of an attitude structure predicts anything. Be that as it may, analytical constructs that are derived from a valid theory are valid to the extent the theory accounts for the structural correspondence with that context. Although evaluative assertion analysis is far from perfect, it has passed several tests.

9.2.4 Embodied Practices

Without history of other content analyses to rely on and no theory or expert knowledge about the selected analytical context, content analysts may be tempted to sample individuals who are known to embody the needed analytical constructs, who are competent readers, interpreters, and users of the texts in the chosen context. Researchers use this strategy far more often than is noticed. For example, when researchers employ coders because of their familiarity with the language or subject matter of the texts to be analyzed, this amounts to importing their knowledge without having to theorize or translate it into coding instructions or analytical constructs. It is assumed that other readers, not involved in coding, share the background and familiarity that the sampled coders bring to the research. In fact, the idea that we can understand each other's communications is based on this notion.

The key to the methodological problem with this strategy is found in the difference between content analysis and direct observational techniques. If the questions that content analysts are asking could be answered through direct observation or interviewing subjects, content analysis would be superfluous. The point of content analysis is not to study observable behavior or common interpretations but to answer questions concerning events that are not accessible at the time, actions that have not yet been taken, large-scale social phenomena that escape individuals' unaided perceptions, or evidence in court for something otherwise difficult to ascertain. Although most content analyses do draw on the embodied experiences and qualifications of coders, and their employment is justified in these terms, the virtue of

content analysis lies in going beyond individuals' interpretive competencies. Coding is not the same as interpreting texts by a multitude of competent readers. Content analysis is fundamentally abductive. It must not be confused with psychological experiments that attempt to generalize responses to stimuli, including to texts, from a sample to a population of individual readers or interpreters.

9.3 TYPES OF CONSTRUCTS

In Chapter 4, I discussed the following kinds of inferences:

- Extrapolations
- Applications of standards
- Indices and symptoms
- Re-presentations
- Conversations/interactions
- Institutional processes

Analytical constructs can be grouped into these types as well. Below, I offer a few comments on each, with the bulk of the discussion reserved for indices and symptoms on which content analyses most commonly depend.

9.3.1 Extrapolations

Extrapolations of trends, patterns, and differences call for analytical constructs that take the form of *recursive* or *autocorrelative functions*. Underlying extrapolations is the idea of *systems* that determine their own behavior within self-maintained boundaries. The behavior of a system—or, if time is not a factor, the entailments of a system—can be inferred from recurrent interactions or stable relations *within* the system, hence *autocorrelatively*.

9.3.2 Applications of Standards

The application of standards for identifying, evaluating, or auditing involves two steps: (a) comparing a textually derived variable with a given or assumed standard for that variable, and (b) judging what this entails. As examples I have mentioned the diagnosis of psychopathology according to manuals of professional standards; the evaluation of press performance, journalistic bias, and codes of conduct in the mass-media industry against established ideals or in reference to tolerable limits; and the audit of communication or accounting practices within an organization. Generally, the analytical constructs for the use of standards are embodied in institutional practices. They are invariant to the extent that institutions manage to enforce them.

9.3.3 Indices and Symptoms

Indices and symptoms are variables that are claimed to correlate with other variables of interest to analysts. The most basic form of an analytical construct for

what George (1959b) called *direct* indicators is a one-to-one relationship, a mapping or mathematical function from a variable (often numerical), the supposed index, to the phenomenon it is supposed to indicate. Such simple relationships are found when anthropologists date artifacts by measuring the potency of radiocarbon in them; they are also present when physicians link medical symptoms to diagnoses and treatments. In content analysis, indicative functions are often tied to frequencies: the proportion of discomfort words as an index of anxiety, the relative frequency and space devoted to a topic as an index of an author's knowledge or interest, or the importance that the mass media attached to that topic, or the change in frequency of value words in party platforms as an indicator of changes in a country's political climate (see Namenwirth, 1973; also in Krippendorff & Bock, 2009, Chapter 3.8). Analysts can establish these simple analytical constructs, for example, through the use of regression equations or even agreement coefficients.

In addition to needing to correlate with the phenomena they claim to represent, indices must satisfy two additional conditions. First, indices should *not* correlate with phenomena that are considered to be independent of the indicated phenomena. In other words, indices must not only point to but also distinguish among phenomena: A chosen answer to a research question must exclude other answers. For example, a therapist cannot identify a patient's mental illness from the way the patient talks unless there is enough variation in the ways different people talk to allow the therapist to draw distinctions and exclude some illnesses from the list. If all textbooks had identical readability scores, the readability measure for a particular text would not mean anything. Even drawing inferences regarding who may be the author of an anonymously written book means making informed choices. Paisley (1964; also in Krippendorff & Bock, 2009, Chapter 3.7), who reviewed the literature on research efforts to infer the authors of unsigned documents, found the following conditions necessary for indices to distinguish among alternative authors:

- Indices should exhibit low variance within a communicator's known work.
- Indices should exhibit high variance among the works of all communicators being compared.
- The frequencies contributing to the value of an index should be high relative to the sampling error.

I shall continue this thread in Chapter 13.

Second, indices should not be affected by variables that are accidental or irrelevant to the phenomenon indicated. For example, in their attempt to infer the authorship of the *Federalist Papers*, Mosteller and Wallace (1963) argued against using Yule's (1944) method of counting nouns, because individuals may write on different subjects, and the choice of content words may hence contaminate what could otherwise reveal an author's identity: "The words we want to use are non-contextual ones, words whose rate of use is nearly invariant under change of topic. For this reason, the little filler words, called function words, are especially attractive" (Mosteller & Wallace, 1963, p. 280).

The shift from counting frequencies of words, symbols, references, or subject matter to counting frequencies of pairs of words, co-occurrences of symbols, patterns of references, or relationships within texts does not affect how indices are developed,

but it took 50 years for content analysts to develop an interest in co-occurrences and patterns. Baldwin (1942) had explored such ideas in his analysis of personality structures from autobiographies. Pool (1952b) observed that symbols tend to occur together or to come in clusters, but he was unable to realize this observation analytically. Osgood (1959) built the co-occurrences of words into his contingency analysis, demonstrating its power in his study of Goebbels's diary, and conducted experiments to determine what co-occurrences indicate. Now, computing indices from co-occurrences has become a standard option in several computer aids to content analysis. What these indices indicate, however, often remains an open question.

It is important that I warn against a conceptual confusion here. In content analysis, frequencies are used in two ways: as indices for magnitudes and as bases for testing the significance of hypotheses. For example, Berelson (1952) justifies his insistence on quantification largely in terms of the need to test statistical hypotheses, but all of his examples concern frequency indicators of other phenomena—attention, emphasis, and bias, to name but three.

In opposing the use of frequencies as *direct* indicators, George (1959a, 1959b) considered analytical constructs for *indirect* forms of inferences. These recognize variable institutional conditions under which the use of correlations might be warranted; I take up this thread in section 9.3.6.

9.3.4 Re-Presentations

It is popular to construe texts as containing meanings that represent something in the real world. Semiotics is a theory that explains the relationship between signs and what they represent. Chapter 2 suggested that texts have no content, that their meanings are the result of someone with literary competencies to read them. It also insisted on denying texts the agency to point to something else, which is what representational theories of language allege. In contrast to both epistemologically dubious conceptions, I argue that texts can enable readers to recall something seen before, to make presents again or for the first time, or engaged in communication with others, realize the intentions of their partners. If readers use texts to re-present something, the analytical constructs capable of inferring what readers take away from reading texts have to do with readers' literary capabilities to image the use of texts. Inferences about re-presentations demand analytical constructs that are discourse specific.

Analyses of texts as re-presentations involve knowledge of how readers or users understand the language of a text and, most important, how they conceptualize its subject matter, which the preceding constructs hardly need. Such analyses usually employ several components: (a) an operationalization of the linguistic structure of the texts, yielding syntactical accounts of the units of text (usually sequences of sentences or propositions) being analyzed; (b) a listing of the possible meanings of words (which one might obtain from a discourse-specific dictionary), sentences (as analyzed by semantic parsers), and their semantic entailments on interpretations; and (c) a mapping of these semantic interpretations onto a world model or territory of the larger discourse, whose logic allows the analyst (d) to obtain answers to questions that could be inferred from the information entered into this model (Hays, 1969; Krippendorff, 1969b).

The expert systems mentioned previously aim at constructing these very components for computing linguistic re-presentations. So far, such systems have been

successful only in relatively small and well-structured worlds (medical diagnosis, mathematics, chemistry, and event analysis) and in cases where the vocabulary of the analyzed texts is not too rich or fuzzy. An interesting commercial application is the automatic generation of answers to frequently asked questions posed by clients of software companies, who describe their problems in their own everyday language. Although the syntactical analysis component may be minimal here, there surely is a dictionary-like component that looks for words and phrases with relevant meanings, which are then entered into a map of the software from which help can be made available. The world model of a particular software usually is far smaller and more structured than the world model of, say, a particular international situation with treaties in force, deception possible, and threats responded to.

9.3.5 Conversations/Interactions

Analytical constructs for verbal exchanges recognize meanings to be interactively established. According to Wittgenstein (1958), the meanings of assertions reside in the responses they elicit. Accordingly, the analytical constructs that account for social interactions partition texts—transcripts or other records of interactions—into turns at talk or similar units of interactions, such as e-mail messages, reactions to public performances, or responses to political events. These are sequenced so that each unit of text is seen as someone's response to another's unit of text. In conversations, inferences from texts stay within a conversation and concern the possible responses that the units of text entail. Most important, each such unit is seen as expanding or constraining the space of possible continuations of the participants of that conversation. Holsti, Brody, and North (1965) developed a crude interaction model into which they mapped measures of affect and actions obtained from messages exchanged internationally during the Cuban missile crisis. Conversation analysis (see Chapter 3) has made major strides in suggesting concepts that might well be operationalized for content analyses of recorded interactions, whether between individuals, between institutional communicators, such as during political campaigns or during negotiations of international treaties. Niederer (2013) studied the unfolding interactions on the Internet among major players in the climate change debate. She could construct a network of how their writing resonated with each other, which sources they relied on, and which they avoided. This kind of analysis is becoming known as networked text analysis.

9.3.6 Institutional Processes

Institutional processes do not follow easily generalizable formats. Analytical constructs for such processes are highly specific to the institutions involved and tend to have the following characteristics:

- They are qualitative, using institutionalized modes of reasoning as well as statistical ones.
- They rely on multiple methods for interpreting texts, as texts are generated and responded to by diverse participants, not one or a few.
- They consider known laws, operating rules, and regulations, each entailing constraints on vocabulary use, strategic moves, and instrumental actions,

thus accounting for spaces of possibilities before locating any one move (text) in them.

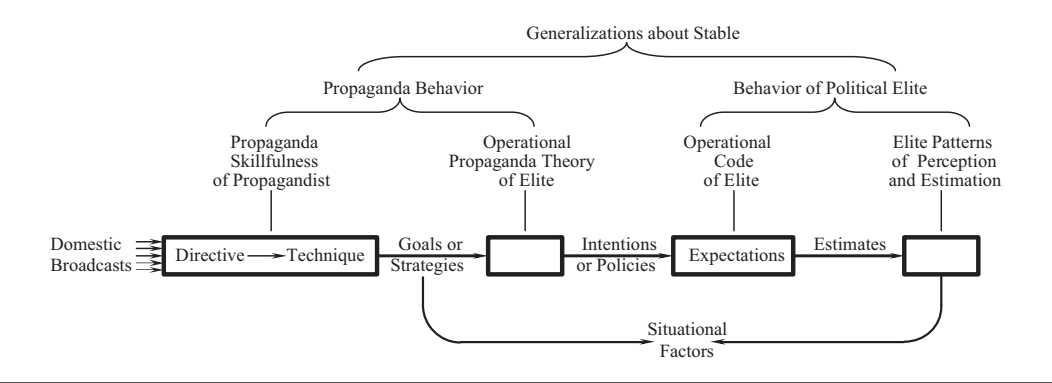
- They recognize that any text at one point in time can have the effect of changing the context for analyzing future texts.

Although the seeming lack of formal rigor might discourage some content analysts from facing the uncertainties apparent in such constructs, researchers have reported considerable success with them, especially when analyses are made over time, allowing partial validations to take place as institutional interactions are unfolding.

A good example of such an analytical construct is found in George’s (1959a; see also Krippendorff & Bock, 2009, Chapter 3.4) account of the inferences made by the U.S. Federal Communications Commission (FCC) regarding domestic enemy broadcasts during World War II. The analytical construct is sketched in Figure 9.2. In the course of their regular inferences, the FCC analysts developed elaborate constructs that were intended to explain why certain enemy broadcasts came into being and what, therefore, the antecedent conditions, perceptions of their producers, and possible military plans were. In this particular situation, the constructs were built on generalizations about the following:

- The propaganda skillfulness of the major propagandist
- The operational propaganda theory of the elites—that is, the role they assigned to the mass media in pursuing their policies
- The elites’ operational code—that is, how the elites translated their estimates of the situation and their capabilities into policies
- The elites’ pattern of perceiving and estimating themselves and their environment

FIGURE 9.2 ■ Analytical Construct for Political Elites’ Stable Behavior Patterns



Source: Adapted from George, A. L. (1959a, p. 53).

Figure 9.2 illustrates how the analysts assembled these generalizations (about the stable features of the source) into a model, linking the major (unstable) variables of the context, especially the broadcasts, to their research questions. In this diagram,

horizontal arrows indicate the order in which inferences were made, which is the inverse direction of the causalities that were presumed to work in this context. The model that these propaganda analysts used was evident in the justifications of the inferences they were asked to make in regular intervals, which George (1959a) analyzed after the war (an institutional content analysis in its own right). He points out that these justifications were far more complex and not quite as linear as the model suggests. Moreover, as the analysts monitored these broadcasts and correlated them with known events, the analytical construct evolved as the war unfolded. Institutions are not carved in stone, and any analysis of government intentions and actions needs to keep up with how they develop.

9.4 SOURCES OF UNCERTAINTY

Published content analyses often offer a false sense of conclusiveness of the reported inferences. In fact, the contexts of analyzed texts rarely cooperate with the scholarly objectives of content analyst. This sense of conclusiveness tends to be conveyed by conveniently ignoring three common sources of uncertainties. It is advisable to take them seriously already during the development phase of a content analysis, as not attending to them may weaken eventual findings. They concern the variance of the target of these inferences, the confidence levels (validity) of the analytical constructs, and the lack of attention to what could influence the correlations that these constructs claim to model. Analysts might not be able to cope with all of these issues as they involve circumstances beyond their control.

9.4.1 Variance of the Target Domain

Content analysts may underestimate the variability of the targets of their research questions, for several reasons:

- They may not be able to imagine all relevant implications of the analyzed texts. Coders may have difficulties detecting unusual interpretations of texts, and the coding instructions given to them often discourage noticing rare deviations from what is expected. Consequently, content analysts may be surprised to find that they failed to capture all voices that turn out to be consequential.
- Theories are always simplifications. Analytical constructs that are derived from theories tend to be skeletal as well, accounting for far smaller amounts of variation than may be evident empirically. For example, in the propaganda analysis effort sketched in Figure 9.2, subsequent verification of predictions of Germany's war efforts revealed that the analysts assumed far more rigidity and determinism than were present.
- Computations are deterministic and everywhere defined. Ideally, they are mathematical functions with multiple inputs but one output. To the extent that analytical constructs specify computations, analysts have a tendency to make inferences that are too precise and more single valued than warranted.

- Statistically oriented content analysts are inclined to predict the means of distributions at the expense of their variances. Researchers with qualitative orientations tend to talk of “common readers,” “typical phenomena,” or “dominant forces.” All of these simplifications discount existing diversities. Both kinds of analysts thus ignore actual variations in their targets, in the first case because of methodological commitments and in the second because of scholarly preferences for linguistically guided abstractions.

While underestimating the variance in the target of the research question is common in much of academic research, content analysts may be able to cope with this variance by matching the variance in the target with the variance that the coding instructions can record. It is well advised to consider the analytical procedure as a communication channel that has enough capacity to allow data to ultimately select among the possible answers to the research question.

9.4.2 Confidence Levels

Here, *confidence* refers to the inductive probability that an analytical construct affords the content analyst. It is the probability not to be an accidental product of the circumstances of its construction, that it is the demonstrably best one available.

- For a statistical interpretation of this uncertainty, the analyst may apply the procedure (described in Chapter 4) of testing analytical constructs as hypotheses. Generally, the smaller the sample of data collected in support of a chosen analytical construct, the lower is the claimable confidence in the construct selected.
- For qualitative interpretations of this uncertainty, the analyst may establish the confidence in the chosen analytical constructs by compellingly arguing that the structural or functional correspondence of one construct is better than others, that the number of theories that lead to the same construct is larger than the number leading to another, or that research has brought forth more evidence in favor of the construct than against it.

The classic solution to low confidence levels is larger sample size, a luxury that most content analysts may not be able to afford, or working well established theories into its construction.

9.4.3 Stability of the Analytical Constructs Model

The correlations that an analytical construct is to describe may not be as stable as the analyst assumes. In Figures 2.2 and 9.1, the possibility that contributing conditions could alter the text/target relationships is acknowledged. This may happen in several ways:

- Analytical constructs developed under laboratory conditions or with samples made up of undergraduate subjects, for example, may not be generalizable to the circumstances in which the actual content analysis is conducted.

- A context in which a content analysis proved successful at one time may have changed. Analytical constructs of institutional processes typically embrace such changes, whereas analytical constructs for computing indices and symptoms are likely to be at a loss when such changes occur.
- Content analysts whose work concerns historical documents often make the mistake of ignoring the fact that linguistic conventions are unstable over long periods of time, that past readings could dramatically differ from contemporary ones, and that the institutions that accounted for the documents' creation are not comparable to the ones with which the analyst is familiar.
- Content analysts may simply be mistaken in their choice of analytical constructs. For example, the first three types of constructs described above assume that the context is void of human intelligence. Suppose a source has a sense of how it is being analyzed and tailors the texts it makes available so as to achieve favorable results. Content analysts who employ constructs that cannot cope with such intentions are blinded by their own methodological commitments and become unwitting instruments of their text sources' goals.
- The publication of especially large-scale content analyses of social significance may affect the analyses' own contexts and weaken subsequent findings. For example, once the television violence indicators that Gerbner and his colleagues developed were published (see, e.g., Gerbner, Gross, Signorielli, Morgan, & Jackson-Beeck, 1979), the three major television networks at that time had the opportunity to respond to the way the researchers measured violence in TV programming. If violence indicators had then declined over time, it would not have been clear whether there was actually less violence on television or whether the networks had found ways to mislead Gerbner et al.'s measures.

Without invalidating evidence, content analysts may never know whether any of their analytical choices succeeded in coping with the underlying uncertainties. However, when invalidating evidence emerges, even only in small hints, analysts may be able to solve some of the problems noted above by altering their analytical constructions, usually in the direction of expanding the variance they can cope with.

ANALYTICAL PATHS AND EVALUATIVE TECHNIQUES

PART



10

ANALYTICAL/
REPRESENTATIONAL
TECHNIQUES

Methods in content analysis largely address the making and processing of data and the application of analytical constructs that preserve some of the data's meanings, leading to valid inferences. This chapter discusses ways in which researchers can represent the results of content analyses such that they may recognize patterns and discover new ways of exploring their findings. Such representations are informative relative to often-implicit standards, several of which are reviewed in this chapter.

After texts have been recorded and analytical constructs have been applied, the content analyst needs to do the following:

- Summarize the inferences from text so that they are easily understood, interpreted, or related to intended decisions
- Discover patterns and relationships within findings that an unaided observer would otherwise easily overlook, to test hypotheses concerning various relationships
- Compare the findings with data obtained by other means or from other situations to support conclusions drawn from other research (multiple operationalism), to gain confidence in the validity of the content analysis at hand, to add another dimension to the intended inferences, or to provide missing information

In practice, these three tasks are not entirely distinct. They are not entirely unique to content analysis either. Much scholarly work, especially in statistics, is concerned with summarizing large bodies of data, making various comparisons, and testing statistical hypotheses. I cannot possibly review all techniques that content analysts might use, so I focus in this chapter on a few that benefit content analysts especially. Moreover, I will not attempt to discuss these techniques in such detail that readers can replicate them—some require expertise found in common textbooks on research methods, and others are built into readily available statistical packages. Rather, my aim in this chapter is to suggest ways of analyzing and representing results tied to texts.

10.1 COUNTS

Owing to the large amount of text that content analysts typically consider, counting parts of it is by far the most common technique used to reduce its volume to something manageable and presumably still comprehensible. The qualification “presumably” is intended to warn novices that counting always eliminates all relationships between the units counted and the simplicity of numerical accounts is always purchased at the cost of omitting the contextually complex meanings between the units in text. It is important not to conflate counting with the use of scientific methods. The early content analyses described by Berelson (1952) and justified by Lasswell (1965b) did just this. In this section, counting is taken as a practical way of coping with large volumes of text, as a method of data reduction, described in Chapter 4, section 4.1.1. With this use in mind, counting is justifiable only when the resulting frequencies mean something, or have something to do with the context of texts, including selecting quantitative answers to research questions.

Referring to the distinction between open and closed variables (introduced in Chapter 8, section 8.3), it is important to distinguish counts of mentions—that is, of textual units, words and phrases, that occur as such in a body of text, unanticipatedly—from counts of coded, categorized, scaled, or tabulated textual elements, whose terms are exhaustively defined by the analyst.

For content analysts, counting is justified only when the resulting frequency accounts can somehow be related to what a body of text means in the chosen context, whether it leads to answering a research question. Counting mentions is most elementary, and, when it comes to the use of computers, it is the easiest account obtainable. Not only because synonyms, grammatical variations of words, and typos are not distinguished in counts of mentions, such counts have difficulties keeping intact stereotypical phrases such as “content analysis,” “computer literacy,” or “museum of modern art” and can also include words with purely grammatical meanings. So-called word clouds are popular visualizations of frequencies of mentions. In 2005, Amazon.com added such accounts to its “inside the book” feature. They can be obtained from any text and for all languages with clearly distinguished words.

The following example, a word cloud of Chapter 1 of this book, contains the 50 most frequent words in the chapter (omitting function words) and depicts their frequency by the size of the type. As Mary Bock observes, “These counts make no scientific claims or inferences; instead, they offer their readers the possibility of unconstrained interpretations, based solely on the assumption that word frequencies mean something” (Krippendorff & Bock, 2009, p. 38). In the absence of inferences from text, Bock calls such accounts “impressionistic content analyses.” Those who have read Chapter 1 may find this visualization plausible, but from its word cloud, it would be impossible to reconstruct the content of this Chapter 1. Accounts of mentions amount to the lowest-level abstraction from text.

Word Cloud of Chapter 1

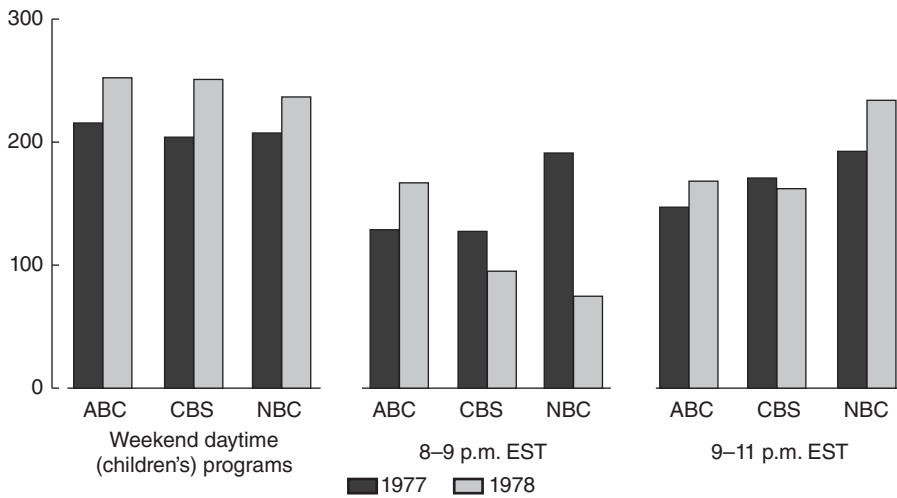
analyses **analysis** analysts analyzing approach became berelson
 communications computer **content** data development different
 documents emerged example group interact interest MASS **matter** meanings media messages **newspaper**
 numerous **political** press processing propaganda psychology public published qualitative
 quantitative **research** results **social** software study subject survey symbols **text** theory
 used volumes WAR work world

Source: Generated at <http://tagcrowd.com>; accessed June 20, 2011.

Counts of textual units in terms of coding categories of closed variables are more interesting. In fact, coding, categorizing, or scaling relevant textual recording units resembles acts of abstracting textual matter on which qualitative content analysts rely heavily. Although frequencies are often celebrated for their precision and simplicity, they should not be granted any special scientific significance. In comparing the results of counts in terms of coding categories, analysts typically apply several interpretive standards, often without being explicit about them. Content analysts should recognize and note their standards explicitly. I discuss three standards in the current section and one in the next.

Content analysts apply the standard of a *uniform distribution* when reporting that the frequency in one category is larger or smaller than the average frequency for all categories. The idea of bias in reporting, such as attending to one candidate for political office more than to another, exemplifies the implicit use of this standard. If favorable and unfavorable accounts were the same for both candidates, analysts would not call this bias and probably would not bother writing about it—except perhaps in surprise, because equality in coverage rarely happens. Figure 10.1, which is taken from Gerbner, Gross, Signorielli, Morgan, and Jackson-Beeck's (1979) work on television violence, invites questions about such issues as why weekend children's programs are so much more violent than other programs and which networks increased or decreased the violence in their programming from 1977 to 1978. When content analysts find observed frequencies noteworthy, then the frequencies deviate from what would not be noteworthy, and that usually means deviations from uniform distribution of categories. The bar graph in Figure 10.3, which comes from Freeman's (2001) study of letters to auto industry shareholders, does not even show frequencies, displaying only deviations from their average, just what is significant.

When analysts observe changes in frequencies over time, they are likely to ask why some changes are irregular and deviate from what would be expected if changes were regular and predictable. They then refer to a *stable temporal pattern* as an interpretive standard; deviations from that pattern are noticed and considered important. Figure 10.2, which comes from an analysis conducted by Strodthoff, Hawkins, and Schoenfeld (1985), shows trend lines for environmental content, environmentalism, and substantive content of special-interest and general-audience channels, largely magazines. The researchers also list four kinds of events in hopes that these might explain the deviations from the otherwise smooth increase over time.

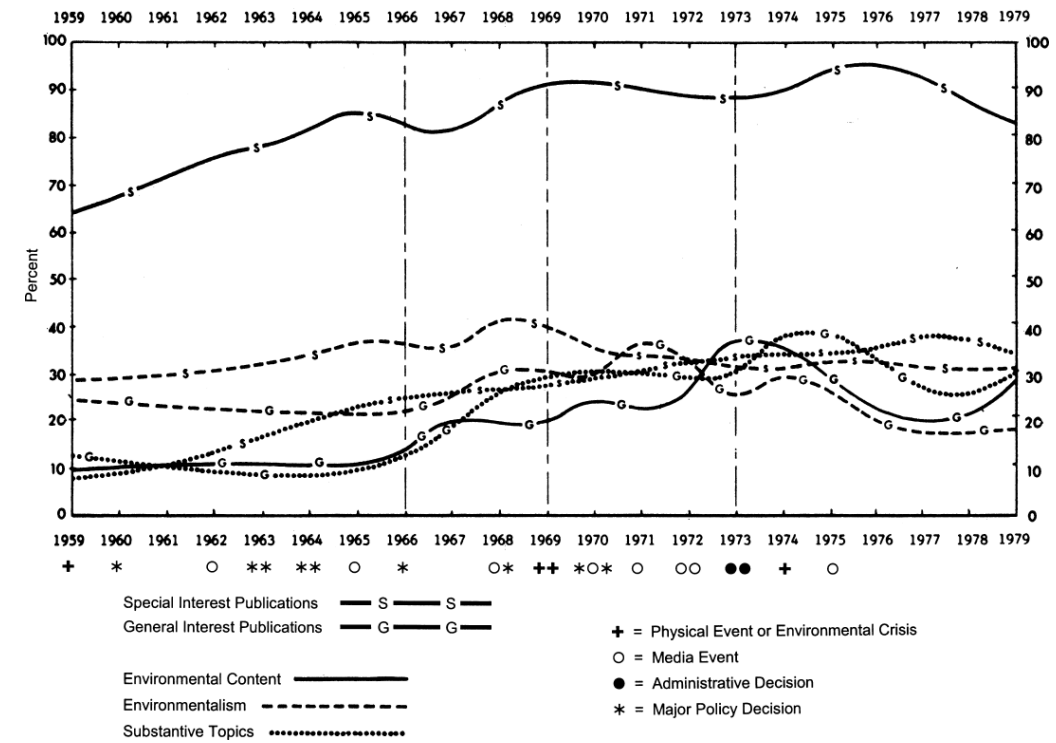
FIGURE 10.1 ■ Bar Graph Representation of Frequencies of Television Violence

Source: Gerbner et al. (1979).

Equally important and perhaps more typical in the content analysis literature is the standard of *accurate representation*, which is implied when an analyst notes that the relative frequencies differ from what would be expected if the data were a statistically correct representation of a population. This standard was introduced by Berelson and Salter (1946), who compared the population of characters featured in magazine fiction with the known demographics of the U.S. population. They found that minorities and poor people were all but absent in magazine fiction, and that popular heroes were overrepresented. Many critics of the mass media have noted that the population of television characters is not representative of the U.S. population or of the members of the mass-media audience. In early television research, analysts demonstrated this lack of representativeness especially for ethnic groups but also for people in particular occupations, people with low socioeconomic status, women in positions of leadership, and elderly people, and these studies were used to infer social prejudices, economic interests, and technological biases.

Whether a population of audience members is the appropriate standard against which the population of television characters should be judged is debatable. Many popular figures, from film stars to television commentators, exist only in the media, not in any unmediated population, and there are good reasons popular talents are more likely to be shown on the screen and attended to than are ordinary people doing ordinary things. Yet, the application of the standard of accurate representation can have political consequences. For example, content analysis research in the late 1950s demonstrated the systematic underrepresentation of African Americans on U.S. television, and these research findings contributed to the eventual achievement of at least some racial balance on U.S. television.

FIGURE 10.2 ■ Trend Lines for Environmental Content, Environmentalism, and Substantive Content of Special-Interest and General-Audience Channels Juxtaposed With “Real World” Events



Source: Strodthoff et al. (1985, fig. 4).

10.2 CROSS-TABULATIONS, ASSOCIATIONS, AND CORRELATIONS

The standard of *chance* is probably most common in statistical accounts of content analysis findings. It arises from analysts’ efforts to cross-tabulate the frequencies of several variables and to observe the frequencies of co-occurrences of values or categories rather than of simple categories. For example, a content analysis of 2,430 acts performed by television characters yielded the observed/expected frequencies of co-occurrences shown in Table 10.1 (Brouwer, Clark, Gerbner, & Krippendorff, 1969). Simple frequencies say nothing about relationships between content variables. Table 10.1, for example, shows that good characters are the origin of most acts, a total of 1,125, followed by 935 acts by bad characters and 370 by neutral ones. Out of the 1,125 acts by good characters, most (751) are unrelated to the law. Although these are large frequencies, and they are far from being uniformly distributed, the actual numbers say little about the relationship between the favorable-unfavorable evaluation of television characters and their association with the law. If one is interested in the statistical relationship between two variables, one must compare the observed frequencies of co-occurrences with those obtained by chance. In

cross-tabulations, frequencies are at the level of chance when all columns and all rows are proportional to their respective margins, which means that the marginal frequencies explain the distribution of frequencies within the table. In Table 10.1, the frequencies obtainable by chance are depicted in italics, directly below the corresponding observed frequencies in bold typeface.

TABLE 10.1 ■ Cross-Tabulation of Frequencies of Acts Engaged in by Characters in Fictional Television Programming

Numbers of Acts Initiated by Fictional Characters who are:

	Good	Neutral	Bad	Totals:
Associated with Law Enforcement	369	27	23	419
	<i>194</i>	<i>64</i>	<i>161</i>	
Unrelated to Law	751	328	454	1,533
	<i>710</i>	<i>233</i>	<i>590</i>	
Criminals	5	15	458	478
	<i>221</i>	<i>73</i>	<i>184</i>	
Totals:	1,125	370	935	2,430 Acts

Source: Brouwer et al. (1969).

What is noteworthy in such a table are co-occurrences of categories whose observed frequencies deviate significantly from what would be expected when variables were independent and co-occurrences were chance events. In Table 10.1, the largest observed frequency of 751 is also nearly as expected, 710, and thus does not contribute to the significance of the relationship between the two variables. In fact, when one uses a χ^2 test of this significance, the cells that make the largest contribution to this relationship are the four corner cells, which indicate the extremes of good and bad and of upholding and breaking the law. The differences between the observed and the expected frequencies in these cells tested statistically significant and thus can be interpreted as supporting the statistical hypothesis that the *good guys* are more likely acting on the side of the law, whereas the bad guys are acting in opposition to it. I say “statistical hypothesis” here because the table shows that there are exceptions, although significantly fewer than chance.

Cross-tabulations are not limited to two or three variables, but they are more easily visualized and interpreted when the number of variables is small. Multivariate techniques are available for testing complex structures within multidimensional data (Esbensen, 2010; Reynolds, 1977).

When variables are nominal (an unordered set of categories), we speak of associations, as shown above, but when they consist of numerically ordered values, we speak of correlations. The standard of chance is common to both, but the use of correlation coefficients adds another standard to that of chance, the standard of *linearity*. Correlation coefficients are zero if data are as expected by chance, and they are unity when all data fall on a straight line, a regression line (see also Chapter 12, section 12.7). Otherwise, correlation measures the degree to which data resemble a

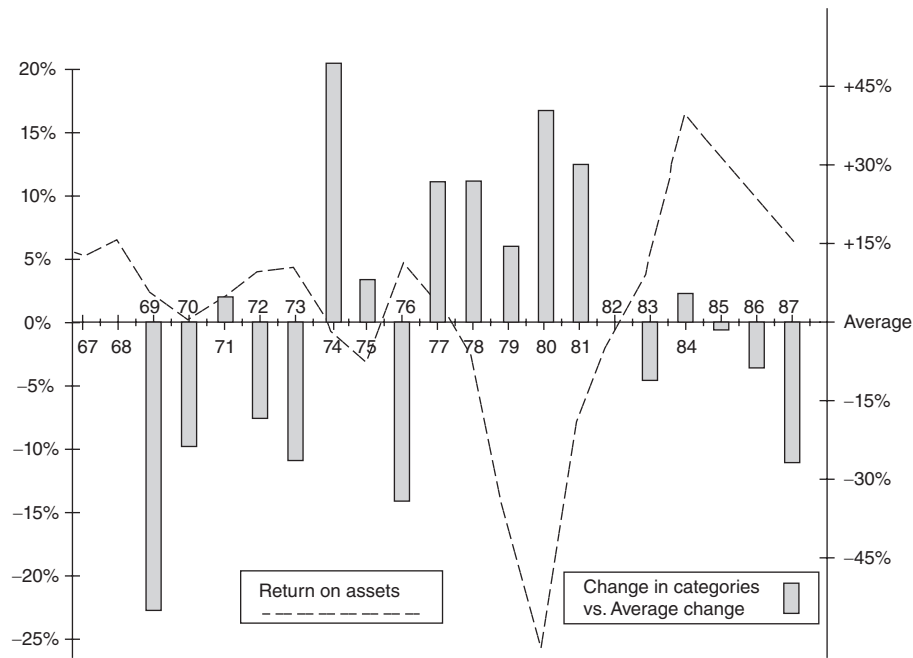
regression line as opposed to chance. Above-chance statistical relations—associations and correlations—may be of two kinds:

- Within the results of a content analysis, as in Table 10.1
- Between the results of a content analysis and data obtained independently, as in Figure 10.3

Because content analysts control the definitions of their variables, there is always the danger that the statistical relations within content analysis results are artifacts of the recording instrument. In Table 10.1, the positive association (good cops, bad criminals) is notable because the underlying relation could have gone in the other direction (bad cops, good criminals). But a positive association between, say, feminine-masculine personality traits (gender) and sex (its biological manifestation) is expected in our culture precisely because these two variables are semantically related. Association and correlation coefficients do not respond to semantic relationships between variables, and if such relationships do exist, these correlation measures are partly spurious and uninformative by themselves.

Correlations between the results of a content analysis and data obtained by other means are less likely so affected because the two kinds of variables differ in how the data are generated. Figure 10.3 comes from Freeman’s (2001) study of U.S. auto industry letters to shareholders. Freeman compared the attention paid to a set of

FIGURE 10.3 ■ Correlation Between Chrysler’s Return on Assets and Year-to-Year Attention to Functional Categories in Chrysler’s Letters to Shareholders



Source: Freeman S. F. (2001, fig. 5).

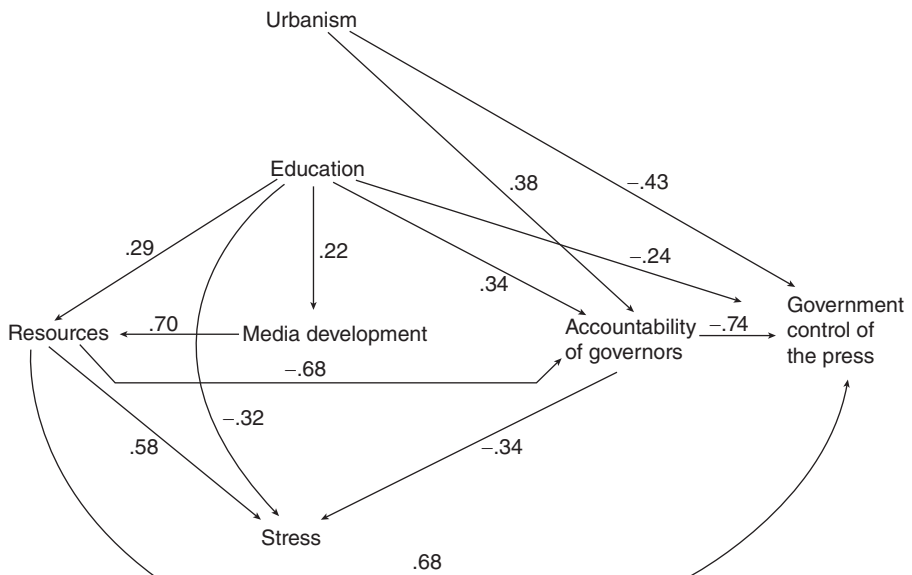
categories functional to a corporation in Chrysler's letters to shareholders with the company's return on assets and found a strong negative correlation between these variables.

10.3 MULTIVARIATE TECHNIQUES

The standard of chance underlies most multivariate techniques of data analysis and representation. Correlations are worth reporting only when the data deviate significantly from chance, ideally approximating linearity. One prominent technique is multiple regression analysis. It presupposes that the variables being analyzed are of two kinds: independent and dependent. The variation in the dependent variables is to be explained, and the variation in the independent variables serves as the explanation. Indeed, many questions that content analysts pursue are reducible to problems of regression. For example, which characteristics of novels predict their popularity? A clear answer to that question would please authors and publishers alike. Or which factors explain media content—government actions, interest groups, economics (advertising), technology, or artistic talent? Or which features of messages are effective in encouraging members of a target population to change their health care habits? The most common kind of regression analysis orders a number of independent variables according to how much they contribute to predicting the values of one chosen dependent variable.

Another multivariate technique entails the use of structural equations. Each variable is considered a dependent variable of all other variables. Only under certain conditions can such a network of multivariate correlations be interpreted in causal

FIGURE 10.4 ■ Paths for Predicting Governmental Control of the Press for Third World Countries, 1950–1979

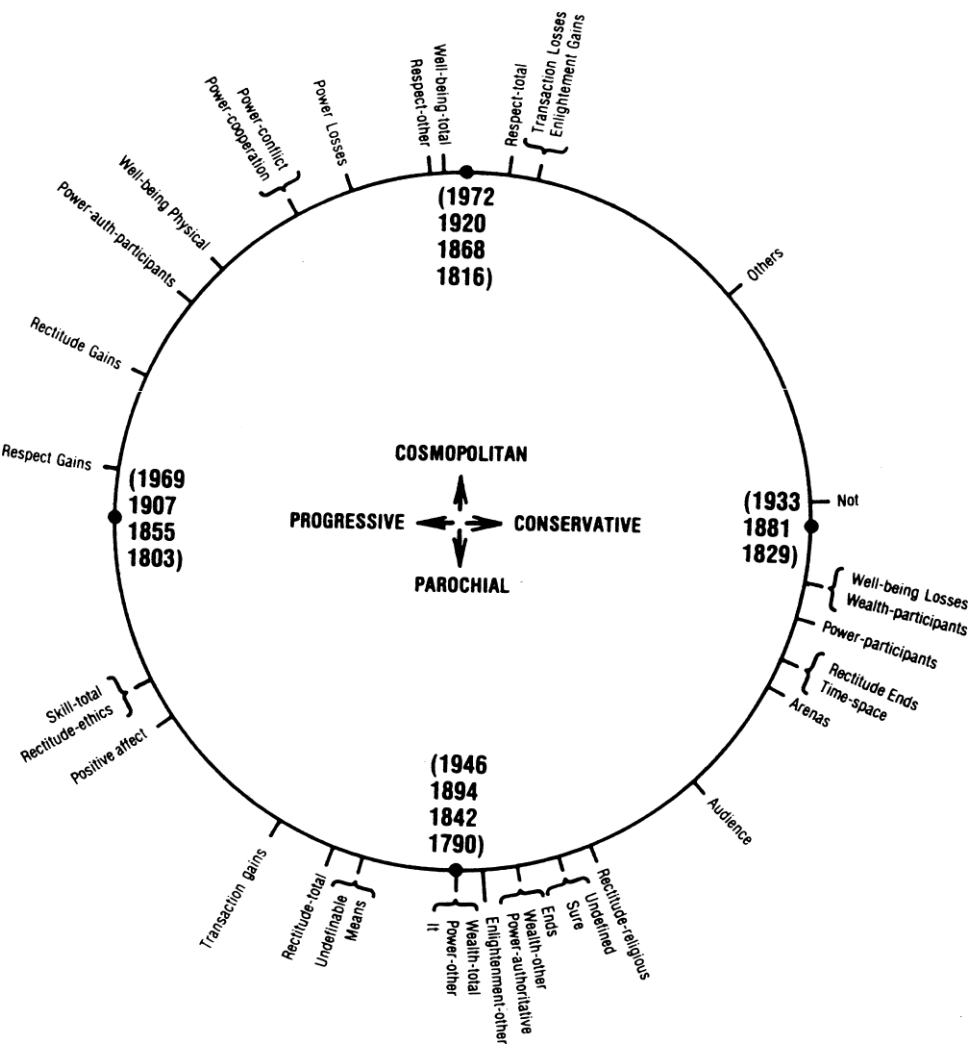


Source: Weaver et al. (1985, fig. 2).

terms. Constraints of space prevent me from discussing these conditions here, but I must note that it is extremely difficult to establish causality from exclusively textual data. One important ingredient of the use of causal explanations is time. Figure 10.4 shows the results of a path analysis conducted by Weaver, Buddenbaum, and Fair (1985) that features regression correlation (beta) coefficients above 0.20 between variables whose relationship to the development of the media in Third World countries was suspected. Weaver et al. compared this path analysis with one that used the same variables but also included data concerning all countries and concluded that in most Third World countries, the media tend to be used to facilitate the functioning of the economy and to perpetuate the power of the rulers.

Correlational techniques are not limited to linear relationships, however. A good illustration of this is found in the work of Namenwirth (1973; Namenwirth &

FIGURE 10.5 ■ A 52-Year Cycle of Values Found in Speeches From the British Throne, 1689–1772



Source: Namenwirth and Weber (1987, p. 139, fig. 5.5; also in Namenwirth, 1973).

Weber, 1987), who analyzed the Lasswellian values (see Chapter 8, section 8.5.3) in speeches from the British throne between 1689 and 1972, covering the British mercantilist and capitalist periods. Over such a long period, fluctuations in the frequencies of values are to be expected, but instead of correlating these with external events that the British Empire had to face, Namenwirth considered values as expressing the workings of an autonomous cultural/political system in which the frequencies of one kind decline as others rise, in endless cycles. To test this hypothesis, he applied a kind of Fourier analysis to these fluctuations. A Fourier analysis decomposes the complex fluctuations of a measure—whether of waves of light or of economic activity—over time into a series of additive sinus curves. Namenwirth identified at least three concurring cycles that turned out to explain much of the variance in the data: a 146-year cycle, a 52-year cycle, and a 32-year cycle.

Figure 10.5 depicts the internal structure of the 52-year cycle. Categories of values that peak are listed at the rim of the circle. Accordingly, the “6 o’clock” position, corresponding to the years 1790, 1842, 1894, and 1946, witnesses concern about the poor performance of the economy, prevailing (un)certainities, and search for knowledge (enlightenment). In the “9 o’clock” position, rectitude and respect gain, and concerns for social welfare and conflict grow. In the “12 o’clock” position, welfare and respect reach their peaks, and enlightenment co-occurs with an international orientation. At the “3 o’clock” position, wealth, trade, and conflict become issues, and well-being is feared to degenerate. In the center of this figure, Namenwirth summarizes this dynamic as a thematic progression from parochial to progressive, to cosmopolitan, to conservative, and back to the beginning. This interpretation loosely follows Parsons and Bales’s (1953) theory suggesting that every society, facing four functional requirements, cycles from an expressive phase to an adaptive phase, then to an instrumental phase, then to an integrative phase, and then back to an expressive phase, and so on. One could describe this technique as one of curve fitting. Here the usual linearity assumptions of correlation coefficients are replaced by sinus curves.

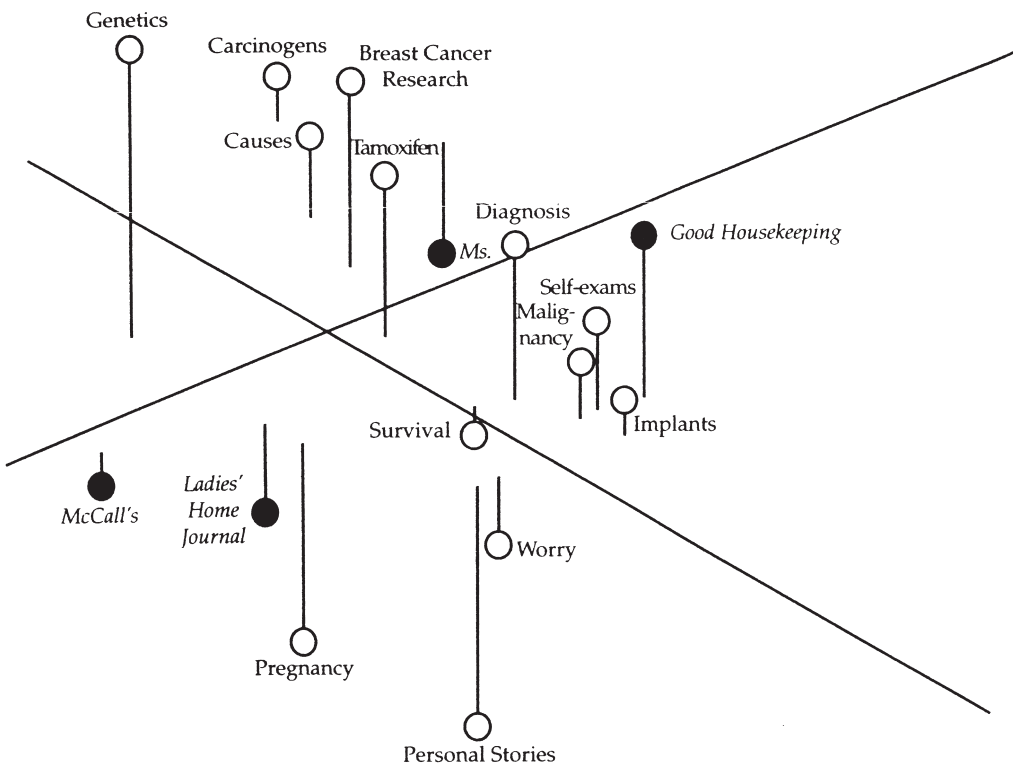
10.4 FACTOR ANALYSIS AND MULTIDIMENSIONAL SCALING

Factor analysis, a favorite method of behavioral scientists in the 1960s and 1970s, is a way to summarize the correlations among many variables by constructing a space with fewer dimensions in which these data might be represented with a minimum of loss in explanatory power. It computes a set of hypothetical and ideally orthogonal dimensions or variables and offers measures of how closely the original variables are correlated with these. These correlations (of the original variables with the virtual dimensions) provide clues that help analysts to make sense of the virtual dimensions. This is the path that Osgood (1974a, 1974b) took to obtain what he called the “basic dimensions” of affective meaning. He used data in the form of numerous semantic differential scales and found three basic dimensions that explain between 50% and 75% of the variance. After examining which of the original scales correlated highly with these, he called them the “evaluative” (good-bad), “activity” (active-passive), and “potency” (strong-weak) dimensions of affective meaning (see also Chapter 7, section 7.4.4).

Whereas factor analysis reduces the dimensionality of the original data while trying to preserve their variance, multidimensional scaling (MDS) reduces the dimensionality of the original (geometric) distances between data points, trying to preserve their positions relative to each other. It requires data on how far apart pairs of elements, concepts, and even variables are. The analyst can fulfill this condition in various ways, such as by measuring differences, dissimilarities, disagreements, dissociations, or lack of co-occurrences between all pairs, whether using objective measurements or subjective judgments. Even correlation coefficients can be and have been converted into distances and subjected to MDS techniques.

MDS starts out with a space of as many dimensions as there are data points, which usually escapes human comprehension. It then attempts to remove one dimension at a time, so as to represent the data in fewer dimensions with a minimum of adjustments to the distances between the data points—much as when one attempts to represent a three-dimensional distribution of points in two dimensions. Figure 10.6 displays the MDS results of a content analysis conducted by Andsager and Powers (1999), a three-dimensional representation of a set of frames used by four women’s magazines in discussing breast cancer. The point of this presentation is to suggest which concepts, ideas, and media sources—here called “frames”—are similar, which cluster in small

FIGURE 10.6 ■ Three-Dimensional Representation of the Frames Used by Four Women’s Magazines to Discuss Breast Cancer



Source: Andsager and Powers (1999, p. 541, fig. 1).

areas, and which are far apart. If all data points were equidistant from one another to begin with, there would be no point in scaling down the dimensionality of these data. Apparently, the standard against which MDS results become noteworthy is that of equal differences.

10.5 IMAGES, PORTRAYALS, SEMANTIC NODES, AND PROFILES

Content analysts often focus on one or a few concepts, persons, or events and seek to ascertain how they are depicted or portrayed in text and what symbolic qualities readers might find associated with them. In the content analysis literature, titles like “Medicine in Medieval Literature,” “The Role of Scientists in Popular Media,” “The Human Body in Women’s and Men’s Magazines,” “How the Portrayal of the United States Has Shifted in Arab Dailies,” and “The Public Image of AT&T” abound. Researchers seek to answer questions of this kind by analyzing the linguistic or textual contexts in which references to the selected ideas occur.

In attribution analysis, the researcher tabulates the adjectives used to describe a chosen concept. A single list of attributes is quite uninformative, however, unless it is compared with some other list that provides a standard against which deviations may be noted. In a comparative attribution analysis, at least two lists are contrasted—for example, the image of one candidate for political office may be compared with the image of his or her opponent; or the portrayals of a country in textbooks before a war may be compared with those after that war; or the way one medium depicts a political scandal may be compared with how another medium covers that scandal. The analyst compares the lists to find out what attributes they share and what attributes distinguish among them. If all the attributes are shared among all the lists, there is little for the analyst to say. This reveals the standard that is common to this kind of analysis, the *sharing of attributions* against which differences in portrayals become noteworthy. Some researchers who conduct attribution analyses use expectations as a basis for comparison, reporting on how and how much a given image deviates from the typical or usual. Unless a researcher has data on such expectations, formal tests may not be applicable. However, verbal highlights of what is unexpected or abnormal are common to many interpretations of images, portrayals, and the like (see the discussion of interactive-hermeneutic explorations in Chapter 11, section 11.6).

Another standard, common largely in linguistics, appears in the comparison of the linguistic context of one word or expression with the set of all grammatically and semantically acceptable contexts in which that word or expression can occur. The subset of actually used linguistic contexts is then equated with the meaning of the word or expression. This idea can easily be expanded to the meanings of politicians, professionals, academic disciplines, and countries.

Thus, the notion of “attribute” should not be construed too narrowly. The image of a U.S. president that spin doctors and advertisers are so worried about can hardly be reduced to a list of adjectives. This would be a convenient but limited operationalization. It may have to include a president’s speeches, editorials discussing what the president does, opinion polls, even cartoons presenting that president’s

public or private life. What is particular about the image of U.S. presidents is how what is said about them differs from what is said about comparable other personalities. Similarly, the image of, say, human genetics in science fiction makes sense only in comparison with how other scientific theories enter this genre.

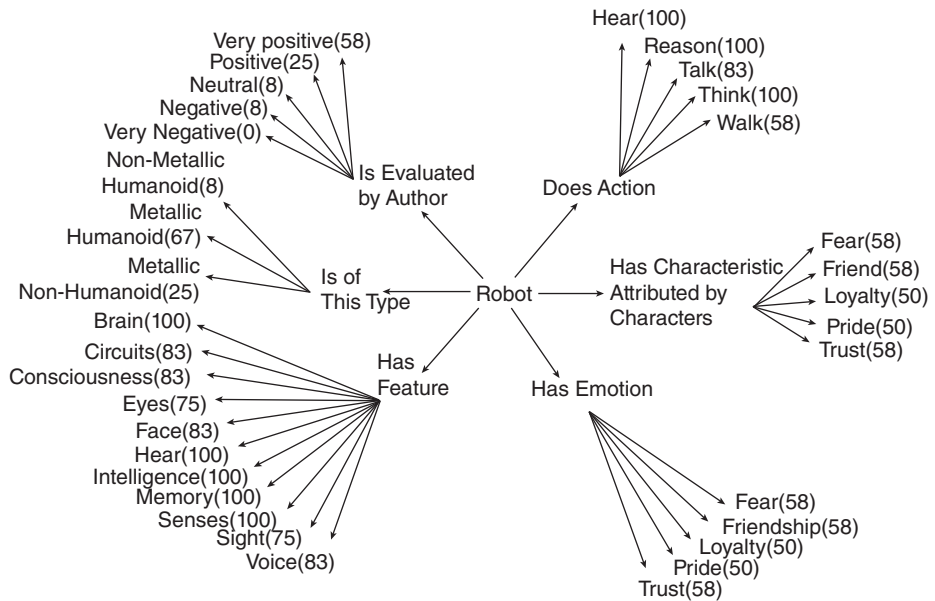
Computer-aided text analysis (CATA), which I discuss in depth in Chapter 11, has provided us with several useful devices for the analysis of images and portrayals. One is the KWIC (keyword in context) list, a tabulation of sentences or text fractions that contain a particular word or phrase. Figure 11.1 shows such a tabulation for the word *play*. Weber (1984, p. 131) compared the KWIC lists for the word *rights* as used by Republicans and Democrats and found significant differences in how the two groups employed the word; these differences are what make Weber's findings interesting. (See Chapter 11 for a fuller discussion of Weber's study.) Researchers can examine the contexts of keywords or key phrases by using the "find" function of ordinary word processing programs, although this is a cumbersome method. Qualitative text analysis software moves from listing the contexts of single words to listing the contexts of categories of textual units (see Chapter 11, section 11.6).

Analyzing the nodes of semantic networks in terms of how one node is connected to others follows the same logic. For example, Figure 11.7 depicts the concept "hacker" as it appears in the narratives of students describing each other. In such networks, nodes are typically characterized in one of two ways:

- They may be characterized in terms of measures that describe their position within a network—for example, with how many other nodes they are connected, their centrality or peripheralness, or how often they occur. Carley (1997) has measured the positional properties of nodes in terms of density, conductivity, and intensity.
- They may be characterized in terms of the semantic connections between them and any other nodes. Figure 10.7, for example, depicts the semantic connections among nodes found by researchers who examined the characteristics attributed to robots in post-1960s texts (Palmquist, Carley, & Dale, 1997). This figure also displays the percentages of texts in which the connections occur.

Comparison of the linguistic environments in which a concept occurs gives rise to a variety of analytical possibilities. Two concepts that occur (or can occur) in the same linguistic environment are interchangeable, have the same meanings, and are considered synonymous. Concepts that mediate between many other concepts are the central concepts of a belief system, a story, or a discourse, which a network represents. An analysis of the environments that two concepts do not share elucidates differences in their meanings. "True" opposites share the environments of their genus but differ in everything else.

Figure 10.7 is one of several "maps" that Palmquist et al. (1997) compared in their examination of texts involving robots written before, during, and after 1960. What they found supported their hypotheses about changes in emotions associated with the robot image—over time, the texts showed emerging trust, loyalty, and friendship that increasingly counterbalanced persistent fears.

FIGURE 10.7 ■ The Robot Image: Robot Types, Features, and Actions From Post-1960s Texts

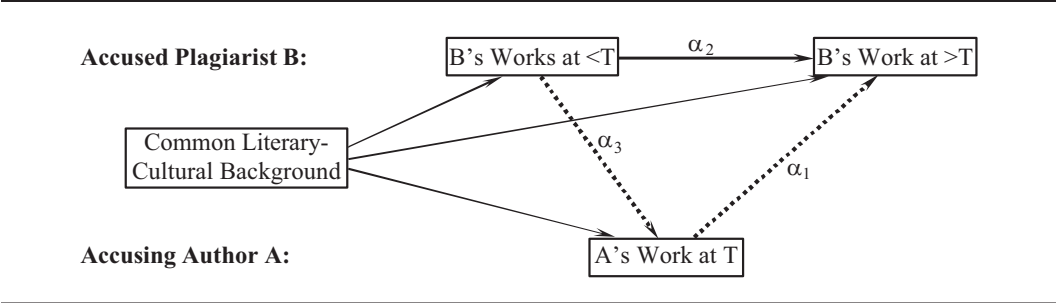
Source: Palmquist et al. (1997, p. 178, fig. 10.4).

When analysts use profiles—whether of potential authors of unsigned documents, applicants for a certain job, or persons with mental illnesses—they apply the same interpretive standard but with the addition that the attributes, correlates, or linguistic environments must be predictive of something. That is, they must answer a question, such as, Who wrote the unsigned document? Who is most likely to succeed in a given job? How should a therapist treat a patient who has a particular manner of talking?

Take the analysis of plagiarism as an example. Suppose that there are two literary works by different authors, A and B, and B is alleged to have plagiarized A's work. Suspicions that one author has plagiarized the work of another are usually grounded in the recognition of surprising similarities between the two works in word choices, grammatical constructions, plot structure, outline, and so on. Even if the similarities are compelling, they do not constitute sufficient evidence of plagiarism. Before content analysts can enter a plagiarism dispute, it must be established that B had access to A's work before or while writing the disputed work. This is the easy part, to be addressed by a court of law. Once the accessibility of A's work to B has been established, analysts can focus their attention on how the similarities between the two works can be explained. Figure 10.8 diagrams the relationships that content analysts may have to consider. The similarity or agreement a_1 could be due to B's shameless copying of A's work, B's creativity (chance), or A's and B's common literary and/or cultural background. If the similarity a_1 can be shown to exceed a_2 substantially, this would add weight in favor of plagiarism on B's part. If a_3 exceeds a_1 to a degree better than chance, then A may actually be the plagiarist of B's previous work, rather than B having plagiarized A.

Authors, by definition, create new literature, and A and B could have come to these similarities on separate paths, especially if they are acquainted with each other's

FIGURE 10.8 ■ Comparisons of Works Needed to Establish or Dispel Accusations of Plagiarism



previously published work. Previous works may not be available for comparison or may not be considered relevant when the similarities being examined concern content, subject matter, or unusual personal experiences. But even the most imaginative writers rely on a background of literature, education, cultural practices, media exposure, and common sense that they share widely with others—otherwise their works would be unintelligible. This common background provides authors with a vocabulary of metaphors, sayings, myths, and themes that they weave into their writing. Most similarities between different authors' works are due to the background they share without realizing it. In a famous plagiarism case concerning a book about teaching in New York, the similarity turned out to be explained by the fact that, unknown to plaintiff and defendant, they had both taught in the same classroom in different years. If one subtracts the vocabulary and background that A and B share from the profiles of the two works, one is left with two profiles whose similarity or difference can be explained by creativity (or chance) or plagiarism. If the remaining similarities are well above chance, this finding might support a charge of plagiarism. The analysis of images, portrayals, semantic nodes, or profiles can lead in numerous directions.

10.6 CONTINGENCIES AND CONTINGENCY ANALYSIS

Contingency analysis is a technique that enables researchers to infer networks of associations from patterns of co-occurrences in texts, whether they are generated by a source or attended to by readers. Contingency analysis started with the observation that symbols often occur in pairs of opposites, that concepts or ideas form clusters. Contingency analysis is based on the assumption (analytical construct) that concepts that are closely associated cognitively will also be closely related proximally. Content analysts have successfully applied this assumption to individual authors, to social groups with common prejudices or ideological commitments, and to whole cultures permeated by cultural stereotypes or conventions. Experiments have shown not only that statistical contingencies in messages are a reflection of the associations in the mind of their authors but that exposure to them can cause corresponding associations in their receivers as well, followed by the reproduction of these contingencies in speech, so that contingency analysis can be used to infer associations not only in the sources of texts but also in the audiences that are exposed to such statistical contingencies (Osgood,

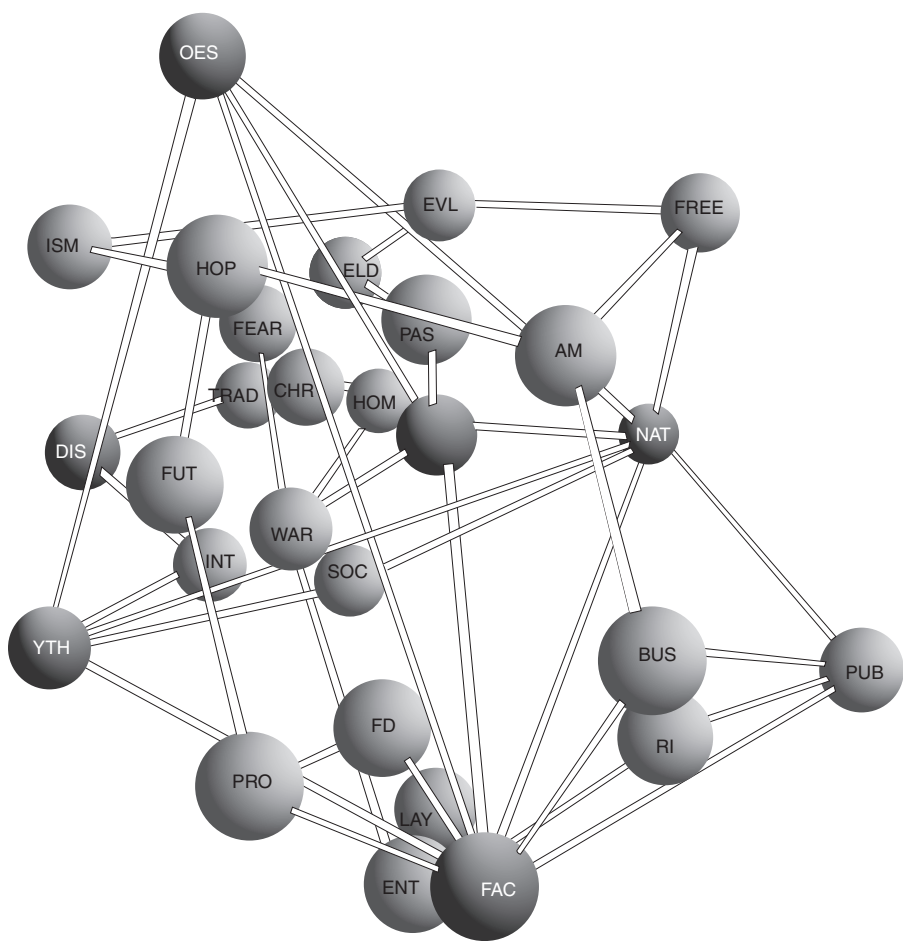
1959, pp. 55–61). Regardless of these correlational validations (see Chapter 13, section 13.2.5), contingency analysis is an analytical technique in its own right.

Contingency analysis starts with a set of recording units, each of which is characterized by a set of attributes, concepts, or features that are either present or absent. The choice of recording units is important insofar as such units must contain sufficient numbers of co-occurrences. A word is too small a unit. A sentence usually contains several concepts, but units larger than sentences tend to be more productive. Osgood (1959), who first outlined this analysis, illustrated the steps involved in his analysis of 38 talks given by W. J. Cameron on the *Ford Sunday Evening Hour* radio program. First, Osgood regarded each talk as one recording unit and recorded the presence or absence of 27 conceptual categories in each talk. In the second step, he counted the co-occurrences of these categories and entered them in a square matrix of all pairs of categories (attributes, concepts, or features). In the third step, he tested the statistical significance of these co-occurrences. Co-occurrences that are significantly above chance suggest the presence of associations, whereas co-occurrences that are significantly below chance suggest the presence of dissociations.

The interpretive standard built into this technique is that of co-occurrences by chance, of course, indicating neither association nor dissociation and therefore not supporting inferences about the cognition of authors or readers. Osgood plausibly argues that both directions of deviation from chance are of psychological importance. The association pattern that Osgood inferred from this rather small data set is depicted in Figure 10.9. Here, mentions of factories, industry, machines, production, and so on (FAC) tended to be associated with mentions of progress (PRO); Ford and Ford automobiles (FD); free enterprise and initiative (ENT); laymen, farmers, shopkeepers, and the like (LAY); and business, selling, and the like (BUS). But when Cameron used these concepts, he *avoided* talking about (to dissociate them from) such categories as youth (YTH), intellectuals, lily-livered bookmen, and so on (INT), and disease (DIS), which form another cluster of associations, dissociated from the former cluster. The figure shows also associations among violence and destruction (DES); assorted “isms,” such as communism, fascism, and totalitarianism (ISM); fear and bewilderment (FEAR); and sundry evils (EVL) (Osgood, 1959, pp. 67–68). Even without having heard these speeches, one can get a sense of the mentality of the speaker and of the times in which these speeches were broadcast.

The fundamental assumption underlying the analysis of contingencies is that co-occurrences in texts indicate associations in someone’s mind or in existing cultural practices. This assumption, along with the idea of neuronal networks, has motivated Woelfel (1993, 1997) to develop software that allows a researcher to tabulate all co-occurrences of words within a sliding window of a specified length (e.g., 100 characters) and then compute clusters of contingencies. (I discuss this software, CatPac, in more detail in Chapter 11, section 11.4.2.) Incidentally, this idea underlies computational procedures that have been given the fancy name of “data mining.” Text searches can identify contingencies as well within other kinds of windows, sentences, paragraphs, and documents. Thus, several computer-aided approaches to text attend to contingencies within linguistic contexts. What distinguishes the latter from what Osgood had proposed is the absence of human readers, coders, or transcribers of the categories that are subjected to contingency analysis. Woelfel’s aim is to bypass human readers or coders altogether, but the results that such software

FIGURE 10.9 ■ Spatial Representation of an Association Structure



Source: Osgood (1959, p. 68, fig. 4).

produces are bound to be shallow compared with analyses in which intelligent human readers are involved.

When tables of possible co-occurrences become very large, analysts may find it difficult to conceptualize the results. Examining a matrix of something like 200×200 associations between concepts, which is not unusual in content analysis, is a formidable task, and analysts trying to discover patterns in such a flood of numerical data are likely to overlook important relationships. Then clustering becomes important.

10.7 CLUSTERING

Clustering operationalizes something humans do most naturally: forming perceptual wholes from things that are connected, belong together, or have common meanings, while separating them from things whose relationships seem accidental or meaningless. Clustering is closely allied with the conception of content as a representation, inviting

abstraction, producing a hierarchy of representations that, on any one level, preserve what matters and omit only insignificant details from the original data. Procedurally, clustering either works from the bottom up, by lumping together objects, attributes, concepts, or people according to what they share, or proceeds from the top down, by dividing sets of such entities into classes whose boundaries reflect the more important differences between them. The direction that clustering takes results from the analyst's choices of the similarity measure and the clustering criterion. Clustering techniques differ widely regarding these. Contingency is but one similarity measure; others are agreement, correlation, proximity, the number of shared attributes, and common meanings, either by semantic definition or by relations within a thesaurus.

The choice of a clustering criterion is decisive for the kind of clusters a particular analysis provides. Some clustering criteria create long and snakelike clusters, whereas others produce compact and circular clusters. Some are sensitive to how much diversity accumulates within a cluster, others are not, assuring only the largest dissimilarities between the forming clusters (Krippendorff, 1980a). Under ideal circumstances, a clustering criterion reflects the way clusters are formed in the reality of the data source and relies on semantic similarities rather than purely syntactical ones. Content analysts must bear in mind that different clustering procedures may yield vastly different results; thus, to avoid ending up relying on arbitrary findings, they must always justify their use of particular clustering techniques in relation to the contexts of their analyses.

The most common of the available clustering procedures consists of the following iterative steps:

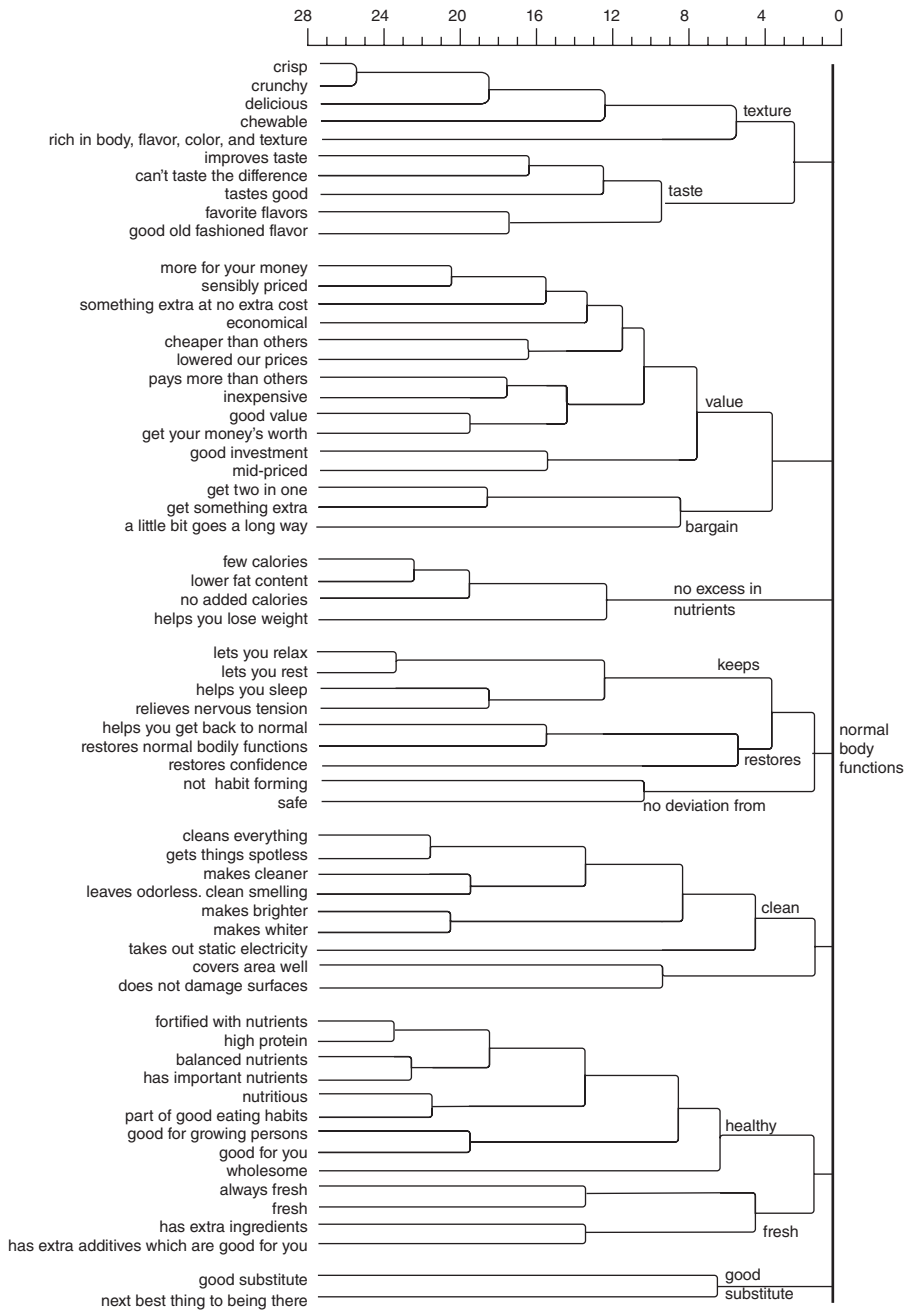
1. Within a matrix of similarity measures, search for two clusters (initially of two unclustered objects) that are, by the chosen criterion, most similar and the merger of which will least affect the overall measure of the differences in the data.
2. Lump these, taking into account the losses incurred within the newly formed cluster.
3. Recompute all measures of similarity with the newly formed cluster, thereby creating a new matrix of similarity measures within which the next two candidates for lumping are to be found.
4. Record the clustering step taken and the losses incurred for the user to retrace.
5. Repeat steps 1 through 4 until there is nothing left to merge (see Krippendorff, 1980a).

For a small amount of data and simple criteria, an analyst may even do clustering by hand.

Clustering steps are typically recorded in the form of so-called dendrograms, which are treelike diagrams that indicate when and which objects are merged and the losses the clustering incurred. Figure 10.10 shows a fraction of Dziurzynski's (1977) analysis of some 300 television advertising appeals. The resulting classification of appeals into those referring to texture, taste, value, and bargain appears to have considerable face validity.

As suggested above, clustering is popular in content analysis because, unlike factor analysis and multidimensional scaling, it is based on intuitively meaningful similarities among units of analysis, and its resulting hierarchies resemble the conceptualization

FIGURE 10.10 ■ Part of a Large Dendrogram Resulting From Clustering Advertising Appeals



Source: Adapted from Dziurzynski (1977, pp. 25, 39, 40, 41, 50, figs. 3, 6, 11, 12, 13, 14, 53).

of text on various levels of abstraction. This is why so many clustering algorithms are available. Often, however, the creators of these algorithms do not reveal how the algorithms work, and that puts the burden of proving their structural validity on the content analysts who use them (see Chapter 13, section 13.2.3).

11

COMPUTER AIDS

This chapter describes how computers can aid content analysis research. Computers have been hailed as reliable, fast, and increasingly inexpensive tools for processing large volumes of textual data, and a great deal of progress is being made in the development of software that can assist content analysts in finding statistical patterns in so-called “big data.” This chapter reviews the key procedures and approaches that underlie most currently available software and are likely to be the basis of future software. Although the use of computers allows content analysts to circumvent the tedium involved in manual data handling and virtually eliminates the problem of unreliable coding, computer applications are approaching other kinds of limits, and these are discussed here as well.

11.1 WHAT COMPUTERS DO

Content analysis, done entirely “by hand,” is often time-consuming, and unreliability is a persistent problem. Before the advent of digital computers, texts were exclusively written, typed, or printed, and they had to be read and coded. The widespread use of computers has revolutionized the manipulation of texts in ways that are attractive to content analysts. The following characteristics of computers deserve special attention here:

- Computers are sequential machines.
- Computers can process large volumes of numerical and textual data with astonishing speed.
- Computers apply logical or algebraic operations to the internal representations of data, entered as inputs, and produce new representations, outputs, some of which are available for human inspection—for example, on a computer screen or in print.
- Computers must be programmed. Software packages, mostly developed and sold separately from computer hardware, tend to provide bundles of computer programs whose details escape the comprehension of ordinary computer users. However, such software allows users to make choices concerning what a computer is to do with the data in hand, using a mouse to point and click on menu options or keying in character strings, for example.

At any one point in time, the computer program in use is a theory of what that computer does.

- Computers' operations are always deterministic and hence perfectly reliable. Within a computer, ambiguities and uncertainties do not exist. This is not to say that a computer reads data the way humans do. For a computer, meanings mean nothing—unless the computer is specifically programmed to recognize what humans recognize.

Several compelling analogies can be made between the way computers work and what content analysts do. The sequential input of discrete characters into a computer resembles the lines of text that readers follow with their eyes. Computers perform logical operations much as rational human beings are thought to do in order to comprehend, draw conclusions, make decisions, and argue. The programs within a computer resemble the instructions (unitization schemes, sampling plans, recording instructions, analytical procedures—see Figure 4.2) that content analysts give to their assistants or hired coders with the expectation that they will be followed completely and reliably—that is, “mindlessly.” In descriptions of how data enter a computer, the metaphor of “reading” is common, and terms like *coding*, *information processing*, *symbol manipulation*, and *computation* have migrated back to the domain of human intelligence. It is widely believed that if an adequate theory of how readers interpret text existed and could be formulated in a suitable computer language, computers could be made to read text just as intelligent readers would, only faster and more reliably. Although there are good reasons why adequate computational theories of literacy are unavailable to date, many developments of computer content analyses are fueled by the hope that such theories are possible and in sight. But there are less ambitious aims.

11.2 HOW COMPUTERS CAN AID CONTENT ANALYSES

The most important reason for using computers in content analysis is their ability to process large volumes of data at high speed. A pioneering example of research demonstrating this capacity is DeWeese's (1977) online analysis of newsprint. DeWeese developed a device that converted typesetting instructions for newsprint (which, at the time of her research, came in the form of a paper tape that created the templates for rotary printing presses) into a computer-readable form, which enabled her to analyze the text of a newspaper virtually while it was being printed. DeWeese's remarkable innovation happened 40 years ago. Today, journalists and printing presses communicate with one another electronically, as do individuals in virtually all spheres of text production and communication. Newspapers do not merely publish papers; they make additional texts available online and hence available to computer-aided content analysis.

Another example of the ability of computers to process large amounts of data quickly is found in the creation of concordances for literary works. Before the advent of computers, the task of creating a list of all significant words in the works of a productive writer and keeping track of their locations in a body of writing could easily occupy the lifetime of a literary scholar plus numerous assistants. Now, for any computer-readable text, the creation of an alphabetical list of key words is a matter

of hours at most. Every day, more and more books, journals, and research reports are available in electronic form, and more and more people are able to access ever-growing electronic full-text databases. In addition, extraordinary progress has been made in the reliability of optical scanning for converting written and printed materials into electronic form. In virtually all domains of public interest—events, issues, topics, controversies, records, and schedules, including scientific findings—large volumes of text are available not only on individual computer screens but also for processing by fast and powerful computers.

A second reason computers are useful in content analysis is their ability to process textual material reliably. This is a blessing with a hitch, however. Mechanical computation virtually eliminates coding errors and nourishes the dreams of some content analysts to bypass the reading of text by human coders altogether. However, it is easy to overlook the fact that computers recognize only character strings. Literate humans are astonishingly proficient at intuiting the etymology of words or the syntactical rules that explain complex expressions, and they have little difficulty verbalizing the point that the author of a text seems to be making or knowing how other members of their community would read given texts. All texts are created to be read by someone. Humans cannot help but render texts meaningful while reading, and they may not always do so reliably. The reliability that computer analysis offers, in contrast, lies in the processes of character string manipulation, which is often far removed from what humans do when interpreting texts.

Unlike humans, computers are deterministic machines. They cannot *not* process text reliably. Computers have no sense of what they do, who their users are, or what the character strings they are processing may mean to human readers, nor are they sensitive to the shifting cultural contexts relative to which humans read and understand text. Computers do not even know the difference between randomly generated character strings and words or symbols that have meaning to humans unless someone tells them how they differ—for example, by comparing them with a dictionary of legitimate words. Therefore, it is seriously misleading to characterize computers as being able to “read” texts or data the way we do. When we say that computers “read,” we are simply using a metaphor drawn from what we humans think we do with texts. Ordinarily, the use of this metaphor may not matter much, but content analysts who are contemplating the use of computer aids in their research would be wise to consider the above and not let themselves be misled into believing that computers could read text as humans do. Programming a machine to mimic how humans so effortlessly understand, interpret, and rearticulate text turns out to be an extraordinarily difficult, often impossible, undertaking.

What, then, are the specific demands of computer-aided content analysis? Recall that we have defined content analysis in terms of the drawing of inferences from available text—abductive inferences, to be specific. A computer-aided content analysis should do the same. Content analysis results, it is important to recognize, are texts as well (see Chapter 14), texts that answer analysts’ research questions about particular unobserved phenomena. Thus, in addition to being understandable to the analysts, their scientific peers, and the beneficiaries of the research results, the path that a content analysis takes must be related to how the analyzed body of text is, was, or will be used in the context relative to which that text is analyzed. Therefore, in the process of any content analysis, conducted with or without the aid of computers, relevant readings or uses of the original texts should be preserved.

To satisfy this criterion, a computer analysis of text should in effect represent, model, or embody at least some of the processes of a text's contextualization (see Figure 9.1). In the extreme, this condition would be satisfied when a computer responds to a body of text in the same way a selected community of readers would. But, as noted above, it is extremely difficult to program a computer to do what ordinary readers of text do naturally.

It follows that the use of computers in content analysis invokes a shift in methodological emphasis, from solving the human problem of achieving reliable coding (see Chapter 12) for large volumes of text at a reasonable rate to solving the computational problem of preserving relevant readings of the texts. In traditional content analysis, semantically valid reading is intuitively satisfied. Coders resist violating the tacit understandings of their speech community. The use of computers in content analysis is limited by the difficulty of achieving semantic validity (see Chapter 13, section 13.2.2).

Semantic validity amounts to preserving the ability to make sense of the text throughout the analysis. It is violated if nothing of the original meaning of the analyzed text is recognizable in the research results. Computer applications have difficulty satisfying this criterion. Content analysts who seek the aid of computers for processing texts tend to accept three compromises on the preservation of the texts' readability:

- Analysts may be satisfied with word meanings, compromising on *what words mean in the context of their use*. I have mentioned counts as an example. Frequency accounts of words or designated phrases eliminate all meanings that stem from the contexts in which they occur. Only their dictionary meanings remain recognizable.
- Analysts may use computer applications that preserve textual meanings in *specialized contexts*. Even traditional content analysts typically ask highly specific research questions—for example, in the context of a particular psychological theory, of election campaigns, or of crime investigations. There is a place for the development of computer applications for content analyses in well-structured contexts and for simple variables. DICTION (Hart, 1985), for example, is a text analysis software that is designed to infer something like the rhetorical tone of political speeches. For political rhetoricians who know the original texts, the results that DICTION produces make quantitative sense, hence DICTION retains some meanings associated with political speeches but only in this very specific context. This software would be utterly useless to investigations of plagiarism, attempts to infer biases in reporting, or attempts to infer the psychological states of a mental patient. Although DICTION is billed as content analysis software, its results make sense only to political rhetoricians who are asking the very questions the software can answer.
- In the course of their work, content analysts tend to engage numerous separable steps (see Figure 4.2); some are clerical and relatively unproblematic, and others call for human intelligence, which is difficult to specify in advance. The use of computers is most appropriate for recurrent and repetitive tasks that can be conceptualized without uncertainty. Searching,

coding, sorting, listing, and counting are obvious candidates. Their steps are small, and their operation is transparent. Content analysts have no difficulty comprehending what these procedural components do and can check whether and what of the original meanings are preserved by them—that is, *for each procedural step*—hoping that this assures semantic validity of the final results. A search engine that can make a body of potentially relevant texts available for human analysts to do the rest is one such tool. When an analytical task is divided into what humans do best and what computers do best, computers do not perform the analysis but aid it; thus, I refer to such approaches as *computer-aided text analysis*, or CATA for short.

To justify the use of CATA software, content analysts must assure themselves, as well as the community of their peers, that the way a software package processes the data is compatible with what is known about the context of the texts, how texts are read, what they mean in actual use, and the role they are expected to play in a context of the analyst's choice. For example, if the meanings of interest are tied to sentence constructions, a computer program that cuts a large body of text into a bag of words will account for that body of text in terms of word frequencies that will not retain the sentential meanings (Krippendorff, 1969a). If the research question concerns political categories, an analysis of texts in psychological terms is irrelevant. If textual meanings are changing over time, by context, or for different communities of readers, an analysis that treats all textual units alike will be misleading. If a diagnosis is wanted, a clustering program is irrelevant. In Chapter 13, I address these issues in terms of validity, semantic validity being most important to this discussion. There can be no universal computer content analyzer.

Potential CATA users must be careful not to fall prey to abstract concepts and fancy claims about what sophisticated software is supposedly able to do. Vendors of CATA software tend to market their products in terms of metaphors that suggest much but mean less in practice. For example, the claim that a software package can “extract content,” with no explication of how it does what, leaves naive users thinking that it can do something they cannot. Similarly, without a specified context, the promise to perform a “concept analysis” is as empty as the assurance that the software does a content analysis (see Chapter 2, especially on the use of the content metaphor). Concepts are always someone's concepts and may have interesting implications but only if embodied somewhere. Claims of “theme identification,” “auto-categorization,” “information mining,” “knowledge discovery,” and “relational text analysis” may leave the novice user of CATA software in awe but say little about what features of text the software responds to or preserves. Beginners may find themselves wondering what their computational results could possibly mean and, being human, undoubtedly find explanations. Software developers rarely provide explicit information about the algorithms of their software, which would enable researchers to trace samples of text through the analytical process to see for themselves how the samples are transformed or analyzed.

Because the CATA software market is evolving rapidly, I cannot possibly provide a comprehensive survey of the currently available computational aids for content analyses—such a survey would quickly be out of date—but I cannot avoid mentioning some of the most widely used packages. For reviews of CATA software, see Tesch (1990), Weitzman and Miles (1995), Popping (1997, 2000), Alexa (1997),

Alexa and Züll (1999), and Lowe (2003). The Text Analysis Info website (<http://textanalysis.info>), maintained by one software developer, is another resource.

Up-to-date information about available CATA software can be found online through the use of search engines, but researchers must bear in mind that software developers rarely include fair comparisons, provide critical evaluations, and commit themselves to methodological criteria for selection. In the remainder of this chapter, my focus is on helping potential CATA users to ask appropriate questions about various kinds of software, so that they do not invest their time in mastering one software package only to find out that it cannot provide them with the answers they seek. I distinguish and discuss three broad categories of computer aids:

- *Text analysis software*: Such software essentially works with the raw text, words, and character strings of which documents consist. It does not operationalize a particular theory of meaning. The abductive inferences supported by text analysis software rely on the analyst's ability to read and interpret the units of text processed for convenience. In this category of computer aids, I shall discuss *accounts of character strings, text mining, mining textual evidence for theories, and webgraph analyses*.
- *Computational content analyses*: Software of this type embodies some kind of theory of meaning or model of how texts are used in a chosen context and transforms a body of text into representations that bring it closer to answering a researcher's question. Here, I will distinguish among *coding/dictionary approaches, statistical association approaches, semantic network approaches, and memetic approaches*.
- *Support for qualitative data analyses*: This refers to software that enables single analysts to manage their interpretations (or readings) of analytically relevant text segments. The interpretation is hermeneutic, and the software enables analysts to interact with its otherwise systematic accounts of these text interpretations.

I conclude this chapter with a discussion of the frontiers of CATA software in which I offer some suggestions to users concerning what they should look for and to developers concerning what would be desirable to work toward.

11.3 TEXT ANALYSES

11.3.1 Accounts of Character Strings

As noted above, this kind of software partitions a typically large body of text into mutually exclusive parts and gives the user various accounts of these partitions, usually in the form of lists of types and/or counts of tokens. These accounts are straightforward and simple, largely because they use syntactical criteria, not meanings.

Most word processing programs make available to their user's information about the numbers of characters, words, paragraphs, and pages in any given file. In fact, they enumerate certain designated characters: alphabetical, numerical, and typographical ones, including wingdings, spaces, punctuation marks, paragraph signs, line divisions, page divisions, and so on. Because these characters are purely

typographical or syntactical, the frequencies computed from them cannot reveal anything about the role of a file, how it could be read, or what it is about. Most people would not call such an accounting a content analysis. Mary Bock calls it “impressionistic” because it relies solely on the ability of researchers to identify and read the meanings of character strings whose frequencies are provided (Krippendorff & Bock, 2009, Chapter 1.7).

The operation of counting otherwise meaningful character strings, such as lists of words, phrases, or sentences, tends not to be based on any theory of meaning with reference to the source, the reader, or the analyst. It is a mechanical task, and a list of counts typically violates the reasons a text was written. For example, if one chops a large body of text into mutually exclusive text segments—say, into words—the segments’ positions within the text (at the beginning, in the middle, or toward the end) are lost, their grammatical functions are no longer recognizable (is *bear* a verb or a noun?), the relations between segments are gone (is *health* affirmed or denied, as in *not healthy*?), personal pronouns become empty (who is *she*?), and dialogical distinctions, such as between questions and answers, are irrecoverably obliterated. For computers, character strings are either the same or different, nothing else. Computers treat grammatical variations of a word (e.g., *bear*, *bore*, *borne*, and *born*) or stylistic variations that express the same idea (e.g., *empty*, *unfilled*, *vacant*, and *void*) as different character strings.

How can such accounts aid content analysis? The answer, already given in the definition of text analysis, is surprisingly simple: They can do so only when they preserve the analyst’s ability to recognize relevant meanings in the accounts of the character strings as tabulated and counted. However, reading a list of words of which a text is composed is obviously not the same as reading a coherent narrative. The difference between the two unequivocally demonstrates what an account of character strings omits. A primary motivation for accounting for the parts instead of the whole is expediency. Frequencies simplify a text. Whether such accounts are justified depends on the questions the analyst needs to answer. Listing and counting have no virtue in the abstract.

In the preparation of accounts of character strings, even the establishment of the syntactical categories of what is to be distinguished and counted is not entirely unproblematic. Take the simple idea of a word, for example. In computer accounts, a word tends to be defined as a character string bracketed by blanks, punctuation marks, or paragraph signs. This definition relies on known typographical conventions, not meanings. Anyone who has seen computer-generated frequency lists of words so defined can attest that such lists tend to include rather odd entries that intelligent readers would never make. Numbers, typographical characters, and misspellings are on equal footing with legitimate words. Plural and singular forms and other kinds of grammatical variations of one word appear as distinct words. Hyphenated words such as *co-occurrence*, compound expressions such as *high school*, *North Pole*, or *weapons of mass destruction*, abbreviations such as *WMDs*, *St. Paul*, and *PhD*, and numerical expressions such as *\$2,578.30* would all be cut into nearly meaningless pieces, as would colloquial expressions such as *run of the mill*—which has little to do with running or with mills. Nevertheless, such word lists can offer content analysts a sense of the vocabulary they are facing.

Most CATA software packages provide accounts of words. Among the better-known systems capable of handling very large data sets are VBPro, a simple freeware

program; WordStat; ZyINDEX, now part of ZyLAB; dtSearch, used largely in the legal profession; and Concordance, widely used in literary research. For the above-mentioned reasons, the lists of word frequencies produced by any one such software are rarely exactly the same. For example, the General Inquirer (Stone, Dunphy, Smith, & Ogilvie, 1966), the pioneer content analysis software (which is now freely available as well), counts idioms and labels such as *United States* as single words and counts hyphenated words that are not in its dictionary, such as *response-seeking*, as two words. CATA software users need to understand how the software they are using distinguishes words or other character strings before they can make use of their results.

Among the most useful kinds of word lists an analyst can expect CATA software to provide are the following:

- Standard (left-to-right) alphabetical ordering
- Reverse (right-to-left) alphabetical ordering
- Ordering in ascending or descending frequencies

Standard alphabetical word lists have the advantage of being easily comparable with other alphabetical lists, including dictionary entries. Lists in reverse alphabetical order (that is, in which the first word ends with an *a* and the last word ends with a *z*) enable analysts to examine redundant endings, plurals, tenses, and suffixes, which aids in the construction of dictionaries and search terms with wildcards, both to be discussed below. Lists of words in ascending or descending order of frequencies typically conform to Zipf's (1935) law: Word frequencies decline exponentially with increasing word length. That is, the most frequent words are short words, typically function words, such as articles, prepositions, logical connectors, and pronouns. The least frequent words, unique in the extreme, are long and statistically unrepresentative words. Misspellings, being infrequent, do not conform to this law. Words that distinguish most clearly between different texts tend to be located somewhere between these extremes (Rijsbergen, 1979, fig. 2.1). Word lists ordered by frequencies can help identify these. Some CATA software allows users to set limits, so that the only words listed are those within frequencies that are likely to matter most.

Whereas human readers can easily correct typos and overlook misspellings, CATA software, operating on character strings, not meanings, has difficulties processing character strings not on a list of proper words, whether due to typographical errors, belonging to another language, or because they are rare, for example, from discipline-specific vocabularies. Computational spell-checkers tend to rely on the most rudimentary correction mechanisms. They relate the character strings of "nonword" to the character strings of legitimate words and select the one with the closest Humming distance (the number of characters with respect to which they differ). There is the possibility that a nonword is close to several legitimate words, posing a computationally unresolved dilemma. Some CATA software packages employ word lists not to identify and correct errors but as "filters" on relevant character strings. Such filters may take one of the following forms:

- A list of "go-words," or keywords to be included in the account (All words not on this list are ignored.)

- A list of “stop-words,” or keywords to be excluded from the account (All words not on this list are counted.)
- A list of character strings (of alphabetical and other characters) to be regarded as single words or phrases, such as compound words, abbreviations, and colloquial expressions (A special case of this is the treatment of negations. A list can be provided of negative forms of adjectives or expressions—for example, *not good*, *no problem*, or *rarely accurate*—that do not separate the negator from the negated [Péladeau, personal communication, 2003].)
- A list of prefixes, suffixes, and grammatical markers to be removed from character strings, appearing in the account without the omitted parts

All of these devices are intended to eliminate less informative words and phrases or irrelevant variations of keywords from an analysis. The first three devices are self-explanatory. The fourth facilitates a process called *stemming*—the removal of grammatical endings to get to the “stem” or core of a word. For example, stemming reduces *talking*, *talked*, *talker*, *talks*, and *talkative* to *talk* (something that could also be accomplished through queries using wildcards; see section 11.4). Stemming can create many oddities that analysts need to examine carefully. For example, when the grammatical endings *ed* and *ing* are removed from all words, the words *red* and *ring* both become *r*, which makes them indistinguishable. And removing *er* from *porter* makes it impossible for one to know if the remaining word refers to the job of guarding a door, a harbor for shipping, or a kind of wine. (A process similar to stemming but more powerful is lemmatization, which I discuss in section 11.4.1 as a dictionary application.)

Comparing the vocabularies of different documents or bodies of documents has had a long history in literary scholarship and content analysis. This interest goes back to Yule (1944), who presumed that the use of nouns would be for writers as a color palette is for painters and compared the vocabularies of several writers to whom the authorship of a religious text titled *Imitatio Christi* was attributed. He came to a convincing conclusion. However, 20 years later, in an effort to establish the authorship of the *Federalist Papers*, Mosteller and Wallace (1963) showed that it is the habitual use of semantically insignificant words, called function words, that is more clearly linkable to author identity (Paisley, 1964; see also Krippendorff & Bock, 2009, Chapter 3.7). In another comparison of vocabularies, Koppel, Argamon, and Shimoni (2002) found statistically significant differences in certain classes of words used by male versus female authors. Men emphasize noun phrases, numbers, determiners, and modifiers, whereas women rely more heavily on pronouns, particular prepositions, and negations. Using Pennebaker’s dictionary approach, Cheng, Chandramouli, and Subbalakshmi (2011) were able to differentiate male and female authors from the vocabularies used in e-mail messages. In addition to authorship identification, comparisons of vocabularies can reveal political affiliations, membership in particular discourse communities, straight and deceptive speech, and more.

Aside from comparing the word frequencies in vocabularies of different bodies of text, content analysts have found it useful to compare distributional indices computed on such frequencies. Clement So (1995) applied entropy measures, also called measures of text temperature (Krippendorff, 1986, pp. 15–20), to assess the diversity

of vocabularies and to compare the coverage in different media regarding different issues and in different geographic regions (So in Lee, Chan, Pan, & So, 2002). Several CATA software packages and even some word processors feature versions of Flesch's (1974) measure of readability. These programs assign readability scores to texts based on the frequencies of certain categories of words, punctuation, sentence lengths, and so on—all of which are distributional characteristics of character strings (see Krippendorff & Bock, 2009, Chapter 3.9). The words of a vocabulary may be ranked also by the information they contribute. Articles such as *the* and *a* are frequent but of little interpretive value. The logarithm of the relative frequency of a word brings rare and presumably more significant words on top of a list.

Whereas lists of words and frequencies take words out of their original linguistic environments and thereby prevent content analysts from recognizing differences in use, so-called KWIC (keyword in context) lists provide inventories of the linguistic environments of selected words. A KWIC list enables content analysts to examine different uses of the same word. Users may want to restrict such lists to keywords, the go-words on a list, because a complete KWIC listing of all words explodes a text in size. KWIC lists can facilitate analysts' development of categories, whether for subsequent manual coding or for the construction of computer dictionaries. They can also aid analysts in formulating rules for distinguishing among unanticipated kinds of meanings. In Figure 11.1, which reproduces a fraction of a KWIC list for the keyword *play*, several different meanings of the keyword are apparent:

- To play something (an instrument)
- To play with something (a ball)
- To play with someone (with a friend) or by oneself
- To play versus to be serious
- A play (as in a theatrical performance)

Before he analyzed campaign speeches made during the 1980 U.S. election, Weber (1984, p. 131) examined a KWIC list of these speeches using the keyword *rights*. This enabled him to observe that when Republicans used the word, it was in the context of discussing law-abiding citizens, state and local authorities, parents, and would-be Soviet immigrants. In the context of women, most occurrences of the word *rights* in Republican speeches came within claims that the Equal Rights Amendment was not needed. In contrast, when Democrats used the word, it tended to be in contexts concerning the rights of working women, minority women, striking workers, pregnant women, industrial laborers, farm laborers, the disabled, the victims of civil rights violations, and Black South Africans. A KWIC list can bring the diversity of word senses, especially of homonyms, to the analyst's attention and suggest further analytical steps.

As Figure 11.1 shows, KWIC lists also provide references to the locations of words within a body of text. Such references are essential for another well-known type of textual account, the *concordance*. As noted above, concordances link important words in a body of texts (for example, the complete works of a given author) to their locations in the works. Not all CATA systems provide KWIC lists that are as easily readable as the fragment shown in Figure 11.1. The software packages Concordance and WordStat are notable exceptions. In these programs, the user can select a word

from an alphabetical list of words together with their frequencies and see displayed all of the contexts and references to the word's occurrence. Others show one linguistic context at a time, like the “find” command in word processing systems.

FIGURE 11.1 ■ Partial KWIC Listing for the *Keyword Play*

```

A VIOLINIST OF THE HIGHEST CALIBRE PLAY A CONCERTO. JIMMY HAS BEEN TAKING
IN HIS ROOM, HE WOULD HALF-HEARTEDLY PLAY A FEW SCALES, AND THEN WOULD PUT
LEADERSHIP, THAT SCOTT WILL PROBABLY PLAY A LARGER ROLE IN THIS AREA, AND
BEST BUDDY. , , MCM ABOUT COMING OVER TO PLAY A LITTLE BALL, , , HIS FRIEND
WHILE HE IS YOUNG HE SHOULD LEARN TO PLAY A MUSICAL INSTRUMENT, THE VIOLIN.
A GREAT DEAL TO US) COULD CONCEIVABLY PLAY A SIGNIFICANT PART. IT SURPRISES ME
AN AIR OF DISDAIN. MCM CAN HE POSSIBLY PLAY A SOLO IN THE CONCERT TOMORROW
COME TO PLAY A TUNE I HAVE COME TO PLAY A SONG. WILL YOU GIVE ME YOUR
MIGUELITO MCM ARE YOU. I HAVE COME TO PLAY A TUNE I HAVE COME TO PLAY A SONG.
A VIOLIN, HE WANTED TO LEARN HOW TO PLAY A VIOLIN. GO TO THE NEXT ONE! HE
SEEP, YOU KNOW, ENCOURAGED ENOUGH TO PLAY AN INSTRUMENT AND I DO NOT THINK HE
PARENTS HAVE DECIDED THAT HE SHOULD PLAY AN INSTRUMENT, AND HAVE BOUGHT HIM
WELL GROWN THEY USED TO GO OUTSIDE TO PLAY AND HAVE FRESH AIR. THEY WENT ON IN
THINGS, THE CHILDREN WOULD ASK HER TO PLAY AND THAT IF SHE SEES THAT IT IS
BCY, WHEN I WAS DOWN SOUTH, I USED TO PLAY AND THEN I WOULD GET MAD AND GO IN
HE WANTED SOMETHING, AND SO HE WOULD PLAY AND THROW BOXES ALL OVER THE ROOF.
I-VE DECIDED TO TAKE YOUR OFFER AND PLAY AT THE , , POPS, , TONIGHT. HOW MUCH
YOU SEE I WILL BE BACK I AM GOING TO PLAY AT THE HOUSE OF OUR FRIEND MANUEL
AND SCOLD HIM AND TELL HIM HE CANNOT PLAY BALL AT ALL TODAY SINCE HE DID NOT
IS A NICE SPRING DAY AND HE WANTED TO PLAY BALL INSTEAD. HE IS NOW TRYING TO
CASUALTY HIMSELF BY GOING OUT TO PLAY BALL, AND RESIGNEDLY RETURN TO
HE IS IN THE ATTIC, WATCHING SOME BOYS PLAY BASEBALL ON THE OPPOSITE HILL. HE
I GO AROUND THERE WITH MY FRIENDS AND PLAY BASKETBALL ALL THE TIME. THAT IS
AFRICANS IN BASKETBALL, AND I OFTEN PLAY BASKETBALL WITH THE LEBANESE.
WORK AND SO THEN I GOES DOWN THERE AND PLAY BASKETBALL, BECAUSE THAT HELP FIND
BIT COCKY + LUCYAL TO MY FRIENDS + A PLAY BOY + LENTIENT TOWARD SOCIAL 20191
HE GREW A LITTLE BIGGER HE LEARNED TO PLAY BY HIMSELF AND WHEN HE GOT BIGGER
HAD TO SIT THERE AND LET THE PIANO PLAY BY ITSELF. HE WAS IN SCHOOL AND THE
IN THE EVENINGS HE TALK, WRITE, PLAY CARDS OR SCRABBLE. WE HAVE A NEW
(THAT HE/SC HAS ROBBED/5 HIS FATHER). PLAY EENY MEENY MINEY MOO WE/1 HAVE/3
BE THAT IN SIX OR SEVEN YEARS HE WILL PLAY FAIRLY WELL AND BE EXTREMELY
OURS THOU MCM ARE IN PANTAL. DO NOT PLAY FALSE BUT TURN THOU SOON ASIDE.
JUNIOR YEAR + ON THE PULSE COMMITTEE + PLAY FOOTBALL + PLAY TENNIS + ENJOY
I SELDOM PLAY I DON'T KNOW AND I PLAY I PLAY FOR MY OWN NO I CANT NOW I LEARNED
I WAS ON THE PITCHING MOUND. I PLAY FOR THE DETROIT TIGERS. WELL, I
TWO TO TWELVE. WE COME THEIR HAIR, PLAY GAMES WITH THEM, SEW CLOTHES FOR
TO TALK WITH THEM, SHOW AND PLAY GAMES, SING AND WATCH THEIR FACES
OBSERVE THE RITUAL PROHIBITIONS. THEY PLAY GONGS AND DANCE. ON THE EIGHTH DAY
WE WILL GET SUGAR CAME TODAY. THEY PLAY GONGS FOR THE SUGAR CAME. THEY CUT
WENT STRAIGHT TO MHEMENDEGAS HOUSE TO PLAY HAVOC. A LONG BLAST FROM A HORN WAS
PIGUEL HE SAID. NOW BASTARD LET US PLAY HE SAID. TAKE OFF YOUR CLOTHES HE
TO HIS MOMAS HOUSE HE WOULD COME TO PLAY HIS GUITAR. WELL NO THE OLD MAN
EVEN LEARN THE PIANO AND I I I SELDOM PLAY I DON'T KNOW AND I PLAY I PLAY FOR
I I I SELDOM PLAY I CANT KNOW AND I PLAY I PLAY FOR MY OWN NO I CANT NOW I
AND SADDLES. THAT WAS OUR GREATEST PLAY IN OUR VILLAGE IN BASUTOLAND. THE
501 WILLIAMSON UNDERGRAD
501 WILLIAMSON UNDERGRAD
2M4 SHAPIRO SHALL GROUP
501 WILLIAMSON UNDERGRAD
501 WILLIAMSON UNDERGRAD
2M5 SHAPIRO SHALL GROUP
501 WILLIAMSON UNDERGRAD
0A3 ZINACANTAN DREAMS
0A3 ZINACANTAN DREAMS
5A1 HARTMAN DETROIT GANGS
5A1 HARTMAN DETROIT GANGS
501 WILLIAMSON UNDERGRAD
0J1 KIKUYU
1A1 FAM INTERACTION FRISBIE
5A2 HARTMAN DETROIT GANGS
5A2 HARTMAN DETROIT GANGS
501 WILLIAMSON UNDERGRAD
0A3 ZINACANTAN DREAMS
501 WILLIAMSON UNDERGRAD
501 WILLIAMSON UNDERGRAD
5F1 N ACH TAT
501 WILLIAMSON UNDERGRAD
5A2 HARTMAN DETROIT GANGS
2C2 PEACE CORPS
5A2 HARTMAN DETROIT GANGS
9C2 WHO AM I
5G2 NAYAH TATS
0B2 CHILDREN S DREAMS
2K1 RAMALLO
1G1 MILLS
501 WILLIAMSON UNDERGRAD
0E1 IFUGAO
9C4 WHO AM I
1F1 JAFFE CASE
5A1 HARTMAN DETROIT GANGS
2C5 PEACE CORPS
2K2 RAMALLO
0E1 IFUGAO
0E1 IFUGAO
0J1 KIKUYU
0A1 ZINACANTAN DREAMS
0A1 ZINACANTAN FOLK
1F1 JAFFE CASE
1F1 JAFFE CASE
5B1 SUZMAN SOUTH AFRICAN

```

Source: Dunphy (1966, p. 159).

A somewhat forgotten but still useful account of character strings in text is the construction of the *map of a text* or body of texts. In such a map, relevant textual units—keywords, sentences, or paragraphs—are represented by outstanding characters and irrelevant units by visually less prominent ones. Such a map conveys a sense of where relevant textual matter is located in the texts being examined, how that matter is distributed, and which categories of words cluster and thus directs analysts' attention to sections worthy of further examination. Sedelow (1967) pioneered the use of such maps. Figure 11.2 shows the spatial distribution of eight categories (three binary components) of informers for a huge lexical research project; this map was created using software called CodeMap (Montgomery, 1989, 1993; Pederson, McDaniel, Adams, & Liao, 1989). Although this map was created for a study that was not strictly a content analysis, it gives an idea of how a large textual database can be visualized in relevant categories. On his TextArc website (<http://textarc.org>), W. Bradley Paley provides an interesting demonstration of how to map a text visually; he describes his software as “a funny combination of an index, concordance, and summary; it uses the viewer's eye to help uncover meaning.” Unfortunately, this experiment hides the sophistication of Paley's mapping technique behind an unfolding artistically motivated image. Page 41 of *The Content Analysis Reader*

(Krippendorff & Bock, 2009) shows an interactive display of selected words in a map of all State of the Union addresses by President George W. Bush. By visually mapping texts into simple categories, content analysts can get a good first view of what they are facing and then decide on the course of their analysis.

FIGURE 11.2 ■ An Eight-Category Map of a Lexical Item

	Code: Race/Sex/Age							
	1 = W/F/13-65		92/152		A = B/F/13-65		16/51	
	2 = W/M/13-65		90/163		B = B/M/13-65		14/40	
	3 = W/F/66-99		80/166		C = B/F/66-99		10/53	
	4 = W/M/66-99		95/236		D = B/M/66-99		10/53	
	1	2	3	4	5	6	7	
	1234567890123456789012345678901234567890123456789012345678901234567890							
A					123143223141144.41332413			A
B					4243241.4. 2C4141133222.A3D23431341.A.244.4			B
C					31.314.2. C4.11.2.3.2.3241331..4.22A1.4334			C
D					122322..143.A1.442.3B241334.32.2.1.44414			D
E					2342341C3A.4.22.B14.324344124C32C.3			E
F					.3.A4...C2433.1.33..D.4D1A21.1B.4.3.43			F
G					411AA144..4..BD1AB1D1.C21.21..13.3.3.11			G
H		4	2B		44.B21.....A.434.1...B..4.D..1B21.4.14			H
I	4				..43131D4....2.3.44.1..2332B.3.D4.B.3..3			I
J	3	.	2		2...34.22...4..2...4..2.A.43.33131D.....			J
K	3 .A3				.. 1...3.....3.4D..4...43.2..1....4.....			K
L	12.	.2			.2 21.1..2.....2.....2.B..241...132.1....			L
M	3	1			...432...C.31.3....C..4..24121.....3..A			M
N	4				...1.....23.....1.....2..3.....			N
O	4	.			.44...2..4...3..1..C43.....2.....			O
P		2.1			. 44.....4...3.1.....			P
Q	.3	.1	.4		.2.....3.2.1... ..1.....			Q
R	.	..			.3.....2.3..B.....2 1. .			R
S					.1..2.....22...B1 3.			S
T	4	1.2			2..A2.2 . .. 2..			T
U	.. 1	.. 2						U
V	..	.2.				4.4 . .		V
W	12.	. 1				. . .		W
X	4 1A1	...						X
Y		4.24				.. .		Y
Z	4	..1				1..2 1 1		Z
AA		..						AA
AB	12	1.				2. ... 4		AB
AC	..	2				. . .		AC
AD	.4 1					1 . .		AD
AE	1.					.22.		AE
AF		.42				... AF		AF
AG						.		AG
AH						.		AH
	1	2	3	4	5	6	7	
	1234567890123456789012345678901234567890123456789012345678901234567890							

Source: Montgomery, M. (1989).

Note: On LAGS, see Pederson et al. (1989).

A useful kind of account that comes to content analysis from literary scholarship is the *table of co-occurrences* (usually of keywords). Such a table enables the analyst

to compute associations but also serves as an entry to contingency analysis (see Krippendorff & Bock, 2009, Chapter 3.1). Word co-occurrences are always counted within a larger stretch of text, which the user needs to define. Some applications register only immediate neighbors, whereas others allow the user to define a window of a certain width within which the co-occurrence of two or more words is recorded. Early implementations of such accounts defined a window of a fixed number of characters (Iker, 1975). As tables of word co-occurrences can become unmanageably large, it is essential that accounts be restricted to keywords on a go-word list. Current CATA software enables other definitions of windows, such as numbers of words, sentences, or paragraphs (see Woelfel, 1993, 1997). Proximity connectors, discussed below, further liberalize how the windows for identifying word co-occurrences are defined.

Finally, *Boolean accounts* sort the textual units of a given body of text into categories that are defined by user-supplied Boolean expressions. A Boolean expression specifies one category of textual units by the character strings that it must or must not contain. The textual units categorized thereby should be large enough to contain the character strings that identify them to be of a certain kind—whole documents, for example, but also individual paragraphs, even sentences. All textual units should be in the analyst's possession, or at least electronically accessible to be tabulated and/or counted.

One of the earliest content analyses to use Boolean accounts is attributed to Sebeok and Zeps (1958), who searched for patterns in a collection of Cheremis folktales. By contemporary standards, this collection was small; today, the availability of CATA software has expanded Boolean accounts to much larger samples of text. One way in which analysts can account for larger collections of text is by applying Boolean operators not to the texts themselves but to indices of the texts. Scholars have been devising indices for large textual databases since the late 1960s, when Janda (1969) gained quick access to relevant political documents by sorting through indices of these documents stored on microfilm. Janda's system also allowed researchers to add indices as they examined the documents, thus expanding the scope of the accounts in the process of the analysis. Up until recently, computational accounts of visual representations were virtually unthinkable without manual indexing. In their research on facial expressions, Ekman, Friesen, and Taussig (1969) coded frames of film manually and tested their hypotheses on these codes. Today, advances in image processing and voice recognition have changed such research somewhat. Although security companies have developed face recognition software, its use in searching audiovisual records, including television shows, tends to be limited to predefined logos, clearly defined images, or sounds supplied by the user. It can point to locations of where they occur within a visual record. (See section 11.3.3 for more on Boolean searches.)

11.3.2 Text Mining

Text mining involves searching typically large textual databases, indexed for electronic access by search mechanisms, for whatever could help to answer the analyst's research questions. Textual databases—electronic books, documents, Web pages, blogs, tweets, or e-mails—are not in the analyst's possession, and their volumes make it unlikely that anyone could ever read all that they contain, yet analysts have reasonable expectations that they hold relevant information. *Mining* is a metaphor. It entails that

the evidence needed to answer a research question is scarce and hidden beneath a huge “mountain” of irrelevant matter, but it also entails that this evidence is there to begin with and obtainable, albeit with effort. I prefer the reference to *text mining* to the term *information mining* for its specificity, as it is texts that mining software extracts from textual databases, not meanings, not concepts, and not information, as suggested by the concept of *information retrieval*. Character strings, texts, and data become information only when someone is capable of relating them to the questions he or she is pursuing. The validity of text mining results cannot be separated from analysts’ literary competencies and the interests they bring to text mining results.

Because interest in text mining goes beyond academic content analysis, several software packages are on the market, some at considerable cost. Market-driven commercial enterprises use text mining to obtain information on the reception of their products, on changing demands, on the composition of market niches, and on new developments they feel the need to respond to. Political organizations scan the public sphere for concerns that their candidates for office should address or avoid. Governmental agencies are known to survey the messages of suspected terrorists, and so on. Some governments use text mining algorithms to identify opposition figures. With multiple uses of and large organizational interest in text mining comes the unfortunate fact that the most powerful software remains proprietary, and the software on the market is often expensive (Leetaru, 2011). This section mainly reviews the problems and solutions that most text mining efforts encounter and that interested researchers need to be aware of when either deciding about acquiring such software or using less sophisticated computer aids to accomplish some of the same tasks.

Generally, text mining increases in importance with the size of the searchable universe of documents and the scarcity of relevant textual matter contained therein. Unlike accounts of the character strings in a body of available text (discussed in the previous section), text mining operates in a largely unknown textual universe, most of which is of no interest to the analysts and may remain unknown to them.

Text mining is accomplished through the use of mechanisms called *search engines*. Users translate their research questions into computable questions called *queries*, which must comply with a search engine’s requirements. These queries instruct the search engine to scan a chosen textual universe for particular character strings, or *textual attributes*. A search is expected to return something the analysts can make sense of. Search engines incorporate three kinds of software: one for indexing character strings in the textual universes they search, one for identifying the indices that match user-specified queries, and one for retrieving and displaying the units of text that contain these matches. Text searches can aid content analysts by accomplishing four successive text mining tasks:

- They may aid in *defining the population of potentially relevant documents*—collections of books, news releases, posts to newsgroups, judicial decisions, Web pages, blogs, e-mails, research reports from any specialized domain of science—within a searchable textual universe. (In Chapter 6, I referred to this use of a text mining as relevance sampling; see section 6.2.7.) Sometimes this population of documents is obviously implied by the analysts’ research questions and requires no search. So, when an analyst is seeking to know

how a particular candidate for political office is depicted, political speeches, newspaper accounts, and survey results during the time of the candidacy would quite obviously constitute that population. In research on how a given drug is regarded, medical journals and interviews with medical professionals would provide this population of interest. Text mining may reach beyond analysts' preconceptions and discover sources not yet imagined.

- They may provide counts of *how often textual attributes of interest occur in the scanned textual universe*, providing some kind of rank ordering of their attention. For example, in August 2003, an Internet search for “*text analysis software*” using the Google search engine yielded 3,239,000 hits—far too many to allow an individual to read through all of them. In June 2011, there were 315,000 hits. This does not mean a decline in public interest, however; rather, it may well reflect refinements of Google's search engine. If this is so, text mining results are not entirely unproblematic.
- They may provide the *uniform resource locators (URLs) of individual websites containing the documents with desired textual attributes* and of libraries and specialized collections from which relevant readable, downloadable, or printable documents could be obtained for further analysis.
- They may provide *abstracts, excerpts, or statistical accounts of the queried textual attributes* for analysts to work with. The concept of text mining often entails such promises.

Internet users are familiar with search engines and rely on them to access online textual matter of interest—but mostly to find documents that can answer a single question. The experience that the same query produces very different results when inserted in different search engines is common. This is the case partly because search engines differ in the sophistication of the queries they can handle, partly because search engines' indexing of their textual universe may not have kept up with the rate at which it changes and moves, and partly because different search engines may search different universes (about which the user usually has no clue). Currently, search engines are said to search (and thus retrieve from) only about 30% of what exists on the Internet. Statistically oriented social scientists may find that figure discouraging, but far more problematic is the uneven quality of the retrieved documents. Anyone can put anything on the Web, and it is often difficult to distinguish assertions made by experts from those made by lunatics. Moreover, the rank ordering of mined results may not mean much when URL owners can pay to have their websites appear on users' screens more readily, when the same documents are listed multiple times, when the same documents are accessible through different links, and so on. In spite of these drawbacks, the Internet represents an important inventory of documents, Web pages, blogs, e-mail exchanges in groups, official reports, and scientific findings, and content analyses of websites are becoming increasingly common (e.g., Bauer & Scharl, 2000; Evans, 2002). Most content analysts pursue research questions that require access to large bodies of documents.

In addition to the Internet, a growing number of textual databases are available for content analysts to explore, including the following:

- For communication researchers, LexisNexis is by far the most popular online searchable full-text archive of important newspapers and magazines, financial records, interview questions and results of opinion polls, legislative materials, and court decisions. I discuss this resource more fully below.
- The U.S. National Archives and Records Administration makes billions of documents (including motion pictures, sound and video recordings, and photographs) available from the three branches of the federal government, some online. The administration's website also provides links to presidential libraries.
- The Educational Resources Information Center (ERIC), established in 1966, is a national information system for education-related literature. It is supported by the U.S. Department of Education.
- The Oxford Text Archive is dedicated to the preservation of high-quality educational texts for research and teaching, largely in the humanities. It provides 2,500 resources in more than 25 languages.
- Corpus Linguistics, an online source with links to other depositories, collects literary corpora, transcripts, and samples of ordinary spoken language, including various dialects and from various eras.
- Project Gutenberg makes its large collection of books in electronic form freely available. To date, the collection includes more than 36,000 important works, ranging from light literature (Sherlock Holmes) to serious works (the Bible and classics), including reference works. The organization invites authors to share their books.
- The American Reference Library, a CD-ROM published by Western Standard Publishing Company, contains more than 55,000 documents. Kenneth Janda calls this resource "the most important single source of political texts across U.S. history" (see Janda's personal website at <http://www.janda.org>).
- PoliTxts is a digital archive of political texts. It includes the inaugural addresses of all U.S. presidents, all of the presidents' State of the Union addresses, transcripts of television debates between presidential candidates, U.S. political party platforms since 1840, and key historical documents.

As of this writing, Google searches for "*electronic text archives*" and "*digital libraries*" yield several thousand collections of texts, national as well as university-based archives, with varying levels of accessibility. Skalski (2002) describes 21 "message archives," not necessarily electronic or online, including the Annenberg Television Script Archive and the UCLA Film and Television Archive.

For communication researchers, LexisNexis is probably the most important source of textual data. Most universities, professional libraries, and institutions of law pay subscription fees for access to the archive, which includes the full texts of documents, not just headlines, abstracts, keywords, and indices. Users interact with LexisNexis via Internet connections, entering their queries and receiving information on the numbers of matches and lists of sources. They then have the option of viewing, printing, and/or saving full documents or relevant parts of

documents. The LexisNexis database includes hundreds of sources of U.S. and international news and business information, plus state, federal, and international legal materials, statutes, codes, and patents. The news sources archived include newspapers, magazines, transcripts, and the full output of news wire services such as Reuters. The database also includes bibliographic data compiled from directories as well as election-related files. Information on public opinion polls is available

- *Phrases* are searched as character strings, just as are words.
- *Proximity connectors* enable defining the size of recording units surrounding a word, searching for co-occurrences of words or phrases and noncontiguous expressions for a single concept, and can limit the meanings of homonyms:
 - pre/#** Searches for words that occur within # words of each other in the specified order
English pre/5 second language
 - w/#** Searches for words that occur within # words of each other in any order
Colorado river w/10 water rights
 - w/p** Searches for words within the same paragraph
 - w/s** Searches for words within the same sentence
 - w/seg** Searches for words within the same segment of the article
- Boolean operators, now largely embraced by above proximity connectors:
 - AND** As in *A AND B*, retrieves articles containing both words
 - OR** As in *A OR B*, retrieves articles containing either or both words
 - NOT** As in *NOT A*, retrieves articles that do not contain that word
- *Nesting of multiple operators*, altering the sequence of applying operators from the normal left-to-right to one in which operations within parentheses are applied before those outside it
(city w/3 center) OR (south w/5 Broad Street)
- *Truncation*
 - !** Searches all variations of a root. *Transport!* finds *transportation*, *transporting*, . . .
 - *** Searches without consideration of a character. *wom*n* finds *woman* and *women*.
- *Plurals* and *possessives* that are formed by adding *s* or *es* or by changing *y* to *ies* are automatically included in the search.
- *Date qualifications*:
 - and date>1/93**
 - and date<1995**
 - and date=3/3/96**
 - and date>8-21-93 and date<3-3-96**
- *Segments*. Documents are divided into segments and searches can be limited to the sections needed, for example, *headline*, *lead* (the first 50 words of a document), *publication*, *person*, *subject*, *type* (of material contained in an article, e.g., *analysis*, *letter*), *geographic* (location), *language*, *length* (of the article).

that goes beyond the poll results, including the full text of questions and responses and the names of the polling agencies that conducted the research. Among the document sources included are U.S. congressional publications, U.S. Supreme Court reports, and the *Federal Register*, as well as some European Community publications. The full texts of numerous scholarly journals—scientific, medical, and educational—are available. Notwithstanding some limitations (Neuendorf, 2002, pp. 220–221), it is easy for a user to search LexisNexis for something that the *New York Times* printed in a particular period, to track the transmissions of the news wire services concerning a particular event (say, for a week from the day it occurred), to compare what various U.S. senators have said concerning a particular piece of legislation as printed in the *Congressional Record*, to locate legal precedents for a particular case, or to find out in what publications a particular cultural event has been reviewed or discussed.

Full-text online archives are especially attractive when they—as LexisNexis does—collect trustworthy (peer-reviewed, authoritative, and authenticated) publications, systematically and over long periods in time, and when they maintain high indexing standards. The validity of the search results of an archive depends on the quality of the archive’s collections and systems of access.

Many electronic text archives have their own search engines, which can vary in sophistication. The following summarizes the rules of the search language that LexisNexis users need to master to formulate their queries. It features proximity operators, which include Boolean operators as their special cases (to be discussed below) and numerous publication-related expressions that are absent from Internet search engines.

Text mining yields matches of queries with textual attributes, not meanings. Whether text mining results do have meanings for a content analysis depends on their relevance to the present research question. The criterion of relevance goes beyond a query/text match, however, which suggests that text mining, as is true for all text analyses, uses text processing criteria that are somewhat handicapped. We measure semantic validity (see Chapter 13, section 13.2.2) of text mining by the degree to which the results identify, count, locate, and/or retrieve *all* and *only* relevant textual units or documents. Text mining results may deviate from this ideal by making two kinds of errors: errors of omission (failing to retrieve relevant textual attributes or documents) and errors of commission (including irrelevant textual attributes or documents).

When relevant units of text consist of single and context-independent words or short expressions, text mining is relatively unproblematic. For example, in a study of negativity in political campaigns in national election years, Kenski (1999) searched for the phrases “*negative campaign*” and “*negative ads*” in newspapers and in questions asked in national opinion polls during the past five U.S. presidential election years. She found frequencies that started low in 1980, peaked in 1988, and then dipped in 1996 to about half that peak. These frequencies characterized the population of texts, and Kenski obtained them without much effort. But, as she discovered, it was far from clear what these phrases actually meant. She therefore delved into polling results to gain greater clarity.

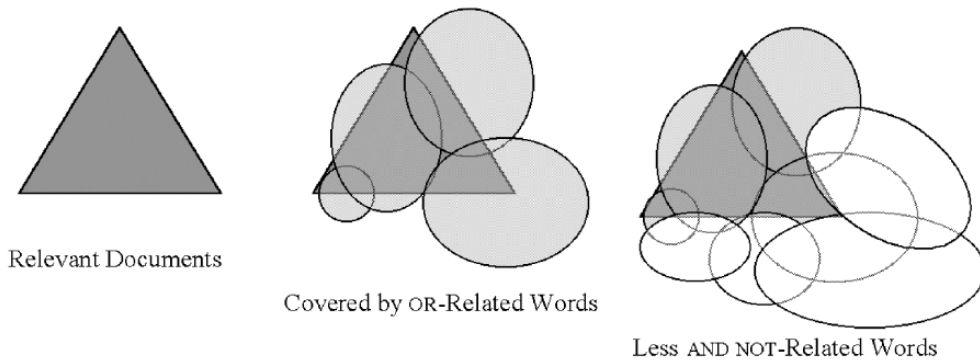
Formulating a semantically valid query is not always a simple matter. Take, for example, an Internet search for texts about the Gettysburg Address. If the user’s

query consists of the words *Gettysburg* and *Address*, entered separately, without quotation marks, an ordinary search engine will look for textual units that contain the two words, whether together or separately. In a *New York Times* article, critical of the common treatment of history on the Internet, Robert Darnton (1999) reported getting 103,467,238 matches for the two words *Gettysburg* and *Address*, including a Gettysburg realtor's mailing address, personal Web pages that mentioned Gettysburg as well as addresses, and the policy of a fast-food restaurant that stated: "At Getty's Burgers, we promise . . . When a member of our staff addresses a customer . . ." A search with a query that used quotation marks around the full string of characters, "*Gettysburg Address*," would have led to a different result. Most Internet search engines accept queries consisting of strings of words rather than individual words. This example illustrates errors of commission and how to reduce such errors. Apparently, errors of this kind are unanticipated. Reading does not work the way search engines do, and the results of text searches tend to surprise even experienced analysts. The relationship between a query and what makes a text relevant is rarely obvious.

When a concept can be represented by a single word or quotable phrase, the formulation of queries is straightforward. Bishop (1998), for instance, wanted to know when and where the concept of "road rage" emerged and how the meaning of the expression evolved. In this case, the compound expression and the concept of interest define each other. Because this was a term of recent coinage (1988, as she found out), Bishop was able to obtain all relevant documents—probably without appreciable errors of omission—by searching the LexisNexis textual database. On the other extreme, Bermejo (1997) studied self-reference in the press—that is, occasions of a newspaper's reporting on its own practices, ethical issues, the politics and economics of newspaper publishing, and any other form of self-reflection. As this was an abstract theoretical concept without an established set of words denoting self-reflexive practices, the formulation of a suitable query proved extraordinarily difficult. Bermejo ended up using a Boolean expression consisting of more than 40 nested search words that identified not only relevant articles but also many irrelevant texts such as obituaries of journalists, corrections of printing errors, reports on ownership changes of other newspapers, and references to sources that happened to be other newspapers. Because he used the text search only to identify a sample of documents, he had no problems removing these errors of commission when coding the texts.

Formulating a query is like trying to "cover" the set of relevant texts, in Figure 11.3 represented by a triangle, with a Boolean expression of search words, represented by ovals in the figure. One of the generalizations immediately apparent in this and Figure 11.4 is that OR operators widen a search, whereas AND NOT operators limit it. In attempting to formulate suitable queries, analysts are often led into a delicate dance, adding and subtracting textual attributes in an effort to approximate desired texts when they do not really know the relationships between textual attributes in the texts and the number of texts containing each. It helps if analysts have a good deal of intuition about language use, but query formulation rarely succeeds without practical explorations.

Researchers may want to develop their search queries by trying them out on small samples of texts they know well enough to learn from emerging errors of omission

FIGURE 11.3 ■ Approximation of a Query to a Set of Relevant Documents

and commission. Here are a few practical suggestions for analysts to consider when formulating queries:

- Words that are widely used (in a Venn diagram depicted as large circles) tend not to distinguish texts as well as do rarely used words (depicted as small circles in that diagram).
- Words with correlated meanings, which are likely to co-occur in the same texts (for example, logical opposites such as *liberals* and *conservatives*), are redundant search terms: They lengthen queries without making much difference in the results. (In a Venn diagram, the areas of correlated words would mostly overlap.)
- Synonyms (i.e., different words with the same meaning) have to be entered into queries one by one, as search engines cannot know their common meaning. The same applies to irregular verb forms and other mere grammatical variations, all of which need to be included as if grammar does not matter.
- Words for a general idea will not identify texts containing specific incidences of that idea. For example, the word *pets* in a query will not retrieve texts mentioning dogs or cats. Researchers interested in a general concept must include in their query all words and phrases signifying its specific instances.
- Homonyms (i.e., words that are spelled alike but have multiple meanings) in queries cause large errors of commission. The meaning of homonyms are largely context-dependent. For example, a query using the word *play* would yield many irrelevant texts, because *play* has many different meanings (as illustrated in Figure 11.1). Researchers aware of homonyms in their text searches should include words in the vicinity of the homonyms that distinguish among their meanings.
- AND operators narrow a search and tend to reduce errors of commission while increasing errors of omission. OR operators expand a search and tend to

increase errors of commission while reducing errors of omission. (One may verify this by examining Figure 11.4.)

Furthermore, researchers may find it helpful to consult one or more of the following kinds of sources as they formulate text search queries: lists of the vocabularies of the concepts in which they are interested, dictionaries of synonyms, thesauri, books of phrases and expressions, concept dictionaries such as the one developed by Laffal (1993), or, perhaps most important, lexical databases such as WordNet (Fellbaum, 1998; Miller et al., 1993). WordNet is an online lexical database maintained by Princeton University's Cognitive Science Laboratory (<http://wordnet.princeton.edu/glosstag.shtml>); it contains more than 50,000 words and 40,000 phrases collected into more than 70,000 sense meanings. WordNet stores nouns, verbs, adjectives, adverbs, compound expressions, collocations, colloquialisms, and forms of speech in the form of synonym sets (about five words or phrases per set) connected by semantic relationships, such as super/subordination, lexical entailments, part of, made of, and contexts of use. (For an excellent online visualization of the relationships between words in WordNet, see the Plumb Design Visual Thesaurus at <http://www.visualthesaurus.com/online/index.html>)

In his excellent review of text mining approaches, Leetaru (2011) mentions several problems of text analysis and some text mining solutions to these problems. One is the problem of detecting conventional word co-occurrences. By contemporary conceptions, “weapons of mass destruction” is a singular concept, coined in the United States during the George W. Bush administration. After 1941, “Pearl Harbor” acquired military and political meanings far beyond the name of a geographical location in Hawaii. “The White House,” “1600 Pennsylvania Avenue,” and “the residence of the U.S. president” denote the same building but in textually unlike forms. Single-word searches cannot recognize meaningful word combinations. Some text mining software tabulates all word coincidences, identifies those that occur above chance, and takes them as whole textual units. Such an algorithm works best in time-bound textual databases. Before 2003, the words *weapons*, *mass*, and *destruction* would not have been co-occurring above chance.

A constant source of errors of omission are misspellings, typographical errors, alternative spellings, abbreviations, and slang. Analyses that rely on well-formulated queries will not find what does not match. Search engine designers have begun to address this problem by offering fuzzy search options that can take a variety of similarities into account, such as the following:

- *Typographical similarities*: words in which characters have been omitted, duplicated, transposed, or result from the accidental hitting of a neighboring key on a keyboard.
- *Phonetic similarities*: words that are pronounced similarly but written differently (e.g., *Smith* and *Smyth*). Soundex is an algorithm that translates written words into a phonetic notation and searches on these rather than on exact spelling (Mokotoff, 1997). Tweets and e-mail messages among close acquaintances are full of phonetic abbreviations, such as “thk y” for “thank you.”

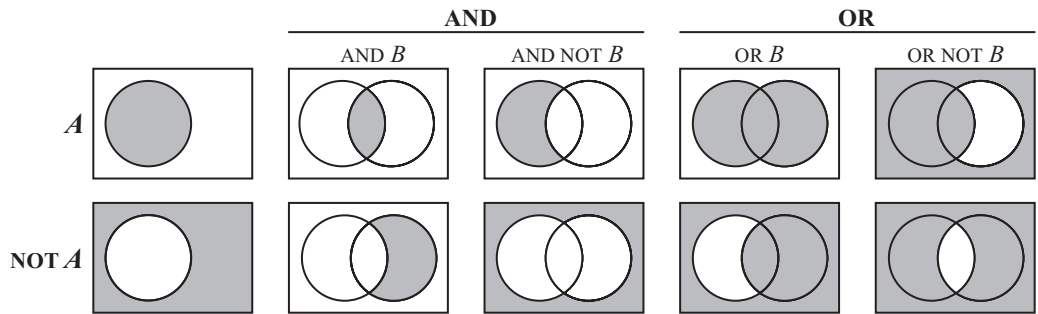
- *Stemming similarities*: words that have the same root meanings but differ in grammatical endings, suffixes, and prefixes, as for plurals, past and present tenses, and conjugations.
- *Thesaurus similarities*: words that are related in meanings, synonyms, associates, and members of word families (e.g., *go* used as a search word retrieves *going*, *goes*, *gone*, and *went*; or the search word *incendiary* brings up *arsonist*, *inflammatory*, *combustible*, and *bomb*).

The above-mentioned problems of the strict literality of search engines and ways to bypass them is carried further by algorithms that do not search for matching character strings but proceed on the basis of larger conceptual similarities between documents. The traditional approach to establishing such similarities is based on correlations of key word frequencies. Epi-search.com, a service the Institute for the Study of Coherence and Emergence (ISCE), has developed free text retrieval software that it claims outperform traditional search algorithms by allowing users to enter a block of text of up to 10,000 words, which serves as a query to retrieve so-called “more like this”-documents. As a first step, the epi-search algorithm parses the query into the most informative concepts (words and phrases), accounting for synonyms and antonyms, and excluding high frequency words, which are mostly grammatical or not sufficiently distinctive in what a document might be about. It relies on a vector space representation of the query to search for documents, decomposed into 500-word segments, that resemble the query both qualitatively and quantitatively. While its website provides free access to the software and ISCE’s extensive library, an attractive feature of the Epi-search algorithm is that its users can obtain the conceptual word cloud, derived from the users’ query, and submit it to other library resource sites.

Leetaru (2011) also mentions text mining algorithms that learn what to look for. I discussed this ability in Chapter 4, section 4.2.3, in terms of the development of a discriminant function. Such algorithms are given texts that were variously categorized by an analyst, search for textual attributes that distinguish among them, and, after having “learned” to differentiate the categorized texts, apply the discriminant function to texts whose categories are unknown. As discussed in Chapter 9, section 9.2.1, owing to the low levels of abstraction of textual matter, such discriminant functions consist of lists of words or word co-occurrences that explain very little.

11.3.3 Mining Textual Evidence for Theories

The simplest example of a Boolean account is a sorting of available texts according to whether they do or do not contain a chosen word or phrase. For example, a search for the words *alcohol* and *drugs* (which may appear singly, jointly, or not at all) in a collection of paragraphs defines two variables—mentioning alcohol or not and mentioning drugs or not—and in effect creates a cross-tabulation of these paragraphs. A Boolean account, once obtained, might be made more detailed; in this example, finer distinctions might be introduced among paragraphs mentioning the word *drugs* (e.g., illegal drugs, prescription drugs, over-the-counter drugs, and poisons), thus expanding the variable *drugs*. One might also enlarge the cross-tabulation by adding other Boolean variables; in this example, such variables might include references to users, places, or treatments, which are logically independent from drugs and alcohol but most likely empirically related.

FIGURE 11.4 ■ Effects of the Application of Boolean Operators

The Boolean operators are AND, the set theoretical intersection; OR, the set theoretical union; and NOT, the set theoretical complement. Boolean expressions can refer to many kinds of character strings, for example, those containing the word *drugs* AND *alcohol*. In so-called Venn diagrams, sets (of textual units with specific attributes) are depicted by their boundaries—by circles, for example. The effects of applying these operators to two textual attributes *A* and *B* are displayed in Figure 11.4. Not depicted in this figure is the use of parentheses, which enables the analyst to specify the order in which these operators are to be applied and thus gives rise to complex expressions. For example, one may verify in Figure 11.4 that $(\text{NOT } A) \text{ AND } B \neq \text{NOT } (A \text{ AND } B)$, which is the complement of *A* AND *B*. In the latter expression, NOT is applied to both *A* AND *B*; in the former, it is applied only to *A*, not to *B*. One may also verify that $\text{NOT } (A \text{ AND } B) = (\text{NOT } A) \text{ OR } (\text{NOT } B)$.

To mine a theory of what a textual database contains with Boolean counts, an analyst may want to take the following steps:

- Define the units of text to be individually examined, tabulated, and counted (as noted, these might be documents, paragraphs, even sentences).
- Specify relevant textual attributes, the character strings that a unit of text must contain or should not contain in order to be of the desired kind.
- Create a Boolean expression with as many textual attributes as needed, using the above Boolean operators and parentheses, and apply it to all units of text.
- Examine whether the units that the Boolean expression identifies as being of the specified kind include all and only the units of the desired kind. The result may not be satisfactory at first, violating intuition (see the discussion of semantic validity in the previous section and in Chapter 13, section 13.2.2), or may be too voluminous to handle. This calls for modifications of the Boolean expression until the identified textual units are of the desired kind, or at least approximately so.

As discussed in Chapter 8, the definition of a variable requires that its values be mutually exclusive relative to each other and that the whole set of values be exhaustive of all units of analysis. The distinction between *A* and NOT *A* is by definition a variable, albeit with just two categories. In Figure 11.4, this is depicted

in the two Venn diagrams on the left-hand side. The number of mutually exclusive categories that Boolean expressions can distinguish grows exponentially with the number of textual attributes available. Boolean expressions defined on N textual attributes can distinguish up to 2^N mutually exclusive categories. For two attributes A and B , Figure 11.4 depicts these in the four Venn diagrams under the Boolean AND operator. The four Venn diagrams shown under the OR operator, by comparison, show categories that are not mutually exclusive. Following are four Boolean variables defined on one through four attributes (defined without the use of parentheses and the OR operator for transparency):

	N = 1	N = 2	N = 3	N = 4
1	A	$A \text{ AND } B$	$A \text{ AND } B \text{ AND } C$	$A \text{ AND } B \text{ AND } C \text{ AND } D$
2	$\text{NOT } A$	$A \text{ AND NOT } B$	$A \text{ AND } B \text{ AND NOT } C$	$A \text{ AND } B \text{ AND } C \text{ AND NOT } D$
3		$\text{NOT } A \text{ AND } B$	$A \text{ AND NOT } B \text{ AND } C$	$A \text{ AND } B \text{ AND NOT } C \text{ AND } D$
4		$\text{NOT } A \text{ AND NOT } B$	$\text{NOT } A \text{ AND } B \text{ AND } C$	$A \text{ AND NOT } B \text{ AND } C \text{ AND } D$
5			$A \text{ AND NOT } B \text{ AND NOT } C$	$\text{NOT } A \text{ AND } B \text{ AND } C \text{ AND } D$
6			$\text{NOT } A \text{ AND } B \text{ AND NOT } C$	$A \text{ AND } B \text{ AND NOT } C \text{ AND NOT } D$
7			$\text{NOT } A \text{ AND NOT } B \text{ AND } C$	$A \text{ AND NOT } B \text{ AND } C \text{ AND NOT } D$
8			$\text{NOT } A \text{ AND NOT } B \text{ AND NOT } C$	$\text{NOT } A \text{ AND } B \text{ AND } C \text{ AND NOT } D$
9				$A \text{ AND NOT } B \text{ AND NOT } C \text{ AND } D$
...				...
16				$\text{NOT } A \text{ AND NOT } B \text{ AND NOT } C \text{ AND NOT } D$

In these Boolean variables, each expression describes one category, the finest distinctions possible with the given number of textual attributes. (For a graphical depiction of these binary variables, see Chapter 8, Figure 8.4.) Joining any of these expressions by OR, in effect lumping together the contents of the corresponding categories, creates other variables with fewer categories. One interesting variable can be created as follows: Take the presence of one attribute as the variable's first category, then, much as in stepwise regression analyses, consider what a second attribute adds to it, what a third attribute adds to both, and so on, until the leftover category, the absence of all attributes, is reached:

A
 $B \text{ AND NOT } A$
 $C \text{ AND NOT } (A \text{ OR } B)$
 $D \text{ AND NOT } (A \text{ OR } B \text{ OR } C)$
 $E \text{ AND NOT } (A \text{ OR } B \text{ OR } C \text{ OR } D)$
...
 $\text{NOT } (A \text{ OR } B \text{ OR } C \text{ OR } \dots \text{ OR } Z)$

Many analytically appealing variables can be created through the algebraic combination of the Boolean expressions for mutually exclusive categories. As long as they describe overlapping sets of textual units—not necessarily the exact same set—two or more Boolean variables yield Boolean cross-tabulations whose entries are the intersections of all categories in these variables. A two-way tabulation of one variable, say Z , with categories $\{z_a, z_b, z_c\}$, and a second variable, say V , with values $\{v_1, v_2, v_3, v_4, v_5\}$, would yield a table with the following cell contents:

	1	2	3	4	5	
a	$z_a \text{ AND } v_1$	$z_a \text{ AND } v_2$	$z_a \text{ AND } v_3$	$z_a \text{ AND } v_4$	$z_a \text{ AND } v_5$	$z_a \text{ AND } V$
b	$z_b \text{ AND } v_1$	$z_b \text{ AND } v_2$	$z_b \text{ AND } v_3$	$z_b \text{ AND } v_4$	$z_b \text{ AND } v_5$	$z_b \text{ AND } V$
c	$z_c \text{ AND } v_1$	$z_c \text{ AND } v_2$	$z_c \text{ AND } v_3$	$z_c \text{ AND } v_4$	$z_c \text{ AND } v_5$	$z_c \text{ AND } V$
	$Z \text{ AND } v_1$	$Z \text{ AND } v_2$	$Z \text{ AND } v_3$	$Z \text{ AND } v_4$	$Z \text{ AND } v_5$	$Z \text{ AND } V$

Like Boolean variables, Boolean cross-tabulations may list actual words or phrases—the units of text identified by the Boolean expressions—in a spreadsheet, for example. As in the steps suggested above, one may examine these in order to introduce finer distinctions, evaluate the semantic validity of the cells of the table, or interpret the relationships among the Boolean variables that emerge in the reading of the tabulated text. When textual databases are large and the cross-tabulated textual matter is voluminous, one may want to verify whether the cell contents make sense, even if the relationship between the chosen variables is statistically significant. In effect, a cross-tabulation such as this or with more variables exemplifies the testing of hypotheses within a body of text as shown in Figure 4.7 and discussed in Chapter 4 (section 4.3.2). For other Boolean analyses, less related to content analysis, see Romme (1995).

Because these accounts of textual matter are not concerned with meanings, computer aids, where available, can provide them with considerable efficiency. However, just as word frequencies mean little unless one is able to read the enumerated words, the Boolean analyses can easily escape comprehension unless one has a clear grasp of the Boolean expressions that go into them. The benefit of mining textual matter for evidence in support of stated theories is that one can check the quality of the cell contents and variables, even measure their semantic validity, but can also obtain statistical accounts of the theory underlying the qualitative distinctions.

Mining the evidence for theories is not limited to traditional Boolean queries. In the LexisNexis search engine, proximity operators have virtually replaced Boolean accounts. As long as one can define the variables of a theory of interest in logically coherent search terms, and as long as the evidence that is needed to support that theory is manifest in textual attributes, mining evidence for theories is an attractive possibility. Search engines that mine the Internet with a battery of queries, as in Rogers's IssueCrawler, are discussed in the following section.

11.3.4 Analysis of Networked Texts

The complexity of the Internet is widely acknowledged. For individual users it is deceptively easy to find answers to simple questions, search for documents of interest, visit places not seen in person, post opinions on various issues, and engage in conversations via e-mail or on a variety of platforms. Indeed, the Internet is a rich source of textual information, not only for content analysts. Traditional content analyses have focused on what can be inferred from what people say or write, what they put on their websites, write in their blogs, and post on Facebook or Twitter. Monitoring political sentiments, medical epidemics, or commercial opportunities to serve large populations is made easy by search engines that go through everything written, posted, passed on, or elaborated on particular platforms. With 80% of the

U.S. population being connected to the Internet, big data content analyses promise generalizations of large-scale concerns with texts found in tweets, posts on social media, websites, including electronic documents available in public databases of libraries and referred to in Wikipedia.

However, this is only the most conventional conception of Internet use. After all, posts on Facebook are meant to be read; and when they are, they establish links between senders and receivers. While tweets were limited to 140 characters, now 280, they allow the inclusion of hashtags by which they connect to larger conversations. Academic writing consists of interconnected texts that makes it part of the literature of particular disciplines and establishes intellectual connections among authors. The latter have long histories, but the networks and communities on the Internet are fluid and fast changing. Yet, the multitudes of connections that emerge on the Internet can make a difference beyond the social lives of their users. Studying how texts are networked in the Internet and inferring the consequences of the collective actions they support is a form of content analysis that involves the use of analytical constructs, which must account for the technologies by which connections are made possible and are established in fact.

To be sure, networks have been of interest to content analysts for quite some time. For example, recording the coincidences or co-occurrences of words or phrases, such as in contingency analyses, have supported psychological inferences about speakers or writers. There also exists a long history of studying citation networks, largely of scientific discourses (Garfield, 1988, 1955) and in literature. It enabled tracing the history of ideas to their origins, identifying influential (canonical) texts, and distinguishing discourse communities whose members are closely interconnected but distinct from other communities whose works are rarely cited. Densely networked communities tend to share distinct vocabularies, and individual authors can be recognized as being central to their communities or peripheral to them. Another root of the content analysis of networked texts can be found in the conception of social networks. Heclo (1978) introduced the term *issue networks* to describe connections among people who regard each other as knowledgeable and interested in particular public policy issues and who work these issues out essentially among themselves. Heclo saw issue networks as undermining democracy because they lacked public transparency. Others (e.g., Klein, 1999) have observed just the opposite, a revitalization of public discourse. Differences of opinion notwithstanding, they acknowledge that issue networks make a difference in society. The inability to govern public discourse is a property of social networks as well, which we are learning to recognize and to analyze (Monge & Contractor, 2003). However, by comparison with citation, issue, expert, and organizational networks, the Internet is far more dynamic, owing to the ease with which social media platforms support, maintain, or abandon intended and unintended connections.

But the Internet has other features rarely attended to in analyses of published matter: hypertextuality. Although the origin of hypertext has been traced to Ludwig Wittgenstein's (1922) hierarchical system of numbering comments on propositions and comments on such comments in his *Tractatus* (Sluga & Stern, 1996), it was not until the 1980s that digital texts enabled readers to click on links within text to explore related matter, effectively overcoming the need to read texts in the order in which they were written, and enabling them to navigate a textual universe in their

own terms. Content analysts of hypertexts began to chart the possible networks of hypertext connections but found that task daunting without computational aids. Neither psychological nor sociological explanations could fully account for their emergence, which are crucially linked to what Niederer (2013, 2017) calls the technicities of digital media, the algorithmic possibilities and constraints that afforded such networks to come about.

One of the obvious units of analysis of Internet texts is websites that house words and images and are accessible by their URL. As mentioned in section 11.3.2, ordinary search engines have no problem accessing texts on websites through their URLs. However, one of the defining features of the World Wide Web (WWW) is that its websites are hyperlinked to each other. This is what enables Internet users to travel from one website to another and from there to still more, going back and forth until they have found what they seek, bringing their hermeneutic circle to a point of closure that is crucially linked to the paths they followed. Networks based on hyperlinks from one URL to another stay entirely within the Internet, providing individual users with paths they can but do not need to take. For researchers, the multitude of such links constitute the network of potential traffic within the WWW.

Webgraphs are intended to represent the links between sites on the Web. Internet traffic can be explained in terms of at least three coordinated sources of meanings. First are web designers affording what could aid their clients and be attractive to potential visitors. Second are the paths pursued by individual users pursuing diverse motivations. And third are the virtual Internet communities whose members share similar interests, agree or oppose issues they discover on websites along the paths of their explorations, direct each other's attention to what is collectively important to that community, and encourage the coordination of individual actions, including through outside channels of communication. These coordinations plus the technicities of the WWW account for much of the effects of the Internet on political participation, educational processes, and commercial activity. Tapping into these phenomena has been a challenge for content analysts.

Probably the first algorithm designed to study the Web is attributed to Larry Page, a cofounder of Google (Brin & Page, 1998). It assigns an "importance" value to each hyperlinked document of a set of documents in the Web, depending on reciprocal quotations and references. It is said to be part of the proprietary Google search engine. Building upon Google's early search engine algorithm, Richard Rogers (2002, 2004, 2010) and the Govcom.org Foundation (Amsterdam, The Netherlands) have developed and made available a webgraph analysis tool called the IssueCrawler, actually a software suite that produces graphs that Rogers calls issue networks, (available at <http://issuecrawler.net> with instructions found at http://www.govcom.org/Issuecrawler_instructions.html) (see also Bruns, 2007).

To understand what the IssueCrawler does, it is important to be clear about the analytical steps it follows. An "issue" is any analyst's conception. It needs to be made manifest in a body of text addressing it and has the potential of circulating on the Web. A collection of such texts is entered into the IssueCrawler in the form of two or more URLs that index these issues textually. Rogers calls them "seed URLs." One way to create such seeds is to Google search for a particular issue or concern—for example, searching for two ideologically opposed positions on abortion, "pro-choice" and "pro-life." When a search produces too many hits, the analyst may have to

restrict the number of seeds to the top N hits or eliminate those deemed irrelevant by hand. The IssueCrawler harvests URLs from a collection of such seed documents by crawling each of the seed URLs and extracting their outlinks. There are three options for a crawl to proceed: “co-link” retains pages that receive at least two links from seeds; “snowball” crawls pages that receive at least one link within seeds; and “inter-actor” crawls pages that link reciprocally to any of the seeds. The results of a crawl are then fed into an analysis engine that visualizes them in the form of a webgraph for the analyst’s inspection. Sites that are more heavily interconnected appear close to one another, and those not connected at all appear in separate clusters.

Selecting an issue as a starting point is crucial for the emergence of analytically informative issue networks. It serves as the key to answer three kinds of research questions: how site owners’ concerns about the issue are distributed in the Web, or who talks to whom; to which sites the community of Internet users are most likely or least likely guided, or what opinions are likely to be found and adopted; and how the life of an issue develops, or how the owners of websites attach themselves to or detach themselves from each other over time. The self-organizing dynamics of the last question can become manifest in the construction of webgraphs crawled at different points in time.

The two webgraphs in Figures 11.5 and 11.6 provide examples of the latter, as well as of using seeds selected differently. The IssueCrawler crawled the web with the URLs of the eight Democratic Party candidates for the 2008 U.S. presidential election every second week from May 2007 to April 2008. The analysis was limited to sites that received at least two links from the seeds. The crawl involved two iterations. For URLs to be included, they had to have at least two inlinks from the linkee of the first iteration. Figure 11.5 represents the issue network on September 12, 2007, and Figure 11.6 that from March 11, 2008. In September 2007, *barackobama.com* was already the top node of the network. But by March 2008, the interlinking of the Obama-related sites crowded out the other candidates’ websites, showing a tight network of Obama constituency sites and Obama social media pages. It is interesting to note that Hillary Clinton’s campaign website did not make the cut as evident in not receiving enough direct or indirect links. This left room for only two non-Obama websites: *mydd.com* and *dailynos.com*, two leading Democratic blogs.

Webgraphs so obtained have topological qualities that can be measured and compared in terms of their density, the centralities of particular websites, the clustering of particular issues, and which sites serve as bridges between unlike clusters. Farrall explored the webgraphs obtained from a variety of sites that, distrusting U.S. government explanations of the 9/11 tragedy for serving President Bush’s agenda, pursued various alternative explanations (Farrall & Delli Carpini, 2005). He found essentially three clusters, one densely linked cluster centering in the website of the “September 11th Truth Movement,” in which various conspiracy theories were proposed; one that contained progressive left-leaning sites, addressing the political consequences of the tragedy; and a small one centering on a website that promoted American values. The last of these did not exactly bridge the two primary clusters, as it did not have links to them, but it was linked to by actors in both camps. The webgraphs pretty much reproduced the public debates at the time of Farrall’s research.

Webgraph analysis is not free of methodological uncertainties. For one, the analyst’s choices among the IssueCrawler’s analytical options affect the resulting graphs. Another is the volatility of the Internet. Connections are constantly changing,

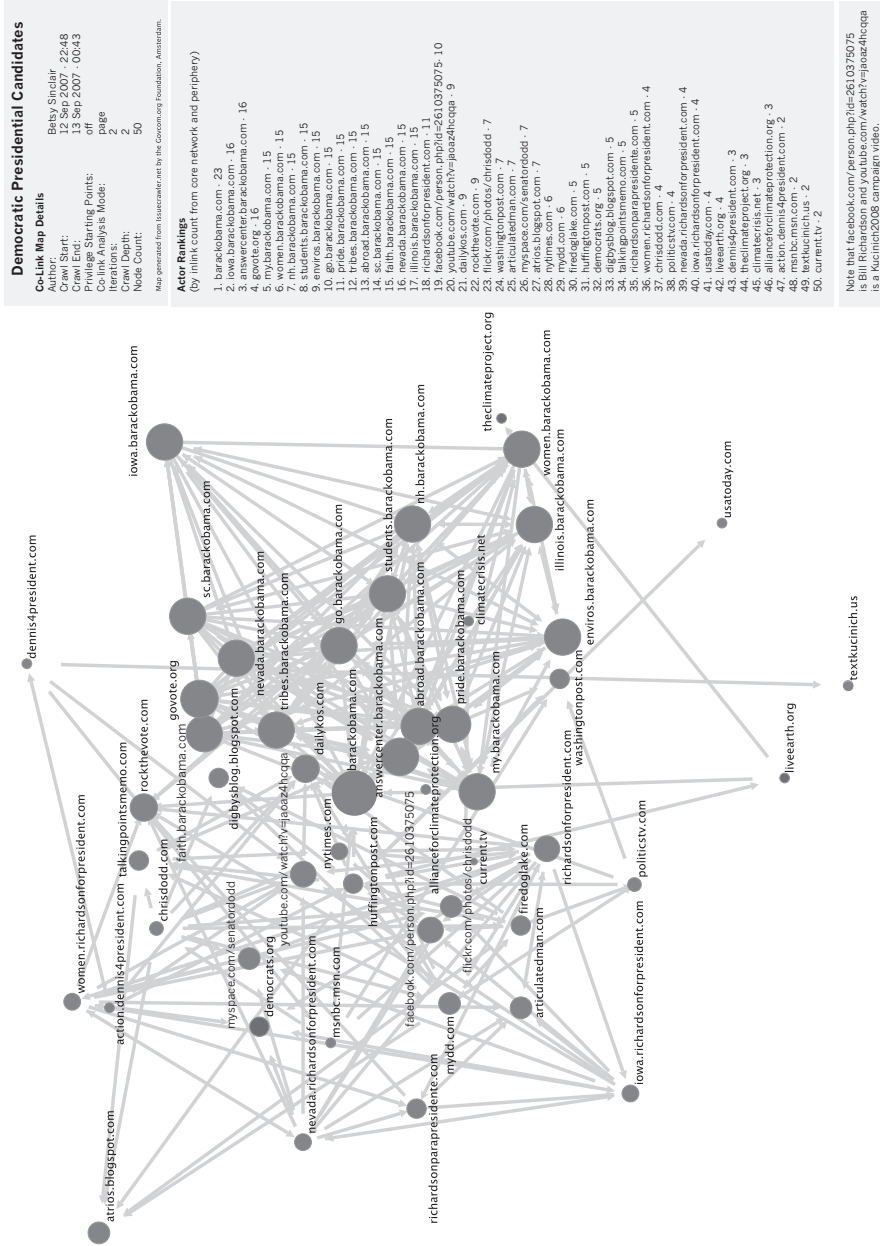
and all webgraphs remain snapshots fixed in time. A third is related to the results it provides. It produces comprehensive visualizations of complex phenomena that offer their analysts new insights or confirm their intuitions. However, correlations between recognizable properties of issue networks and social variables still need to be demonstrated. Farrall and Delli Carpini (2005) discuss some of the validity problems that need to be addressed. Nevertheless, webgraphs are impressive, not only because the enormous number of sites they are able to visualize—regarding the two webgraphs reproduced in Figures 11.5 and 11.6, in September 2007, the hosts identified by the IssueCrawler received 35,032 links, those identified in March 2008 received 35,932—but also because they have significant face validity to those familiar with the issues. It is unquestionable that much of political, commercial, and interpersonal networking makes use of the Web, blogs, and social media, and what happens electronically calls for reconceptualizing social reality in Web terms. Rogers's analysis of Obama's campaign strategy allows us to get a sense of the political capital that comes with participating in issue networks as manifest not just during elections but also in mobilizing political opposition, coping with the spread of epidemics, and campaigning for or against public concerns.

Niederer (2013, 2017) exemplified content analyses of networked texts by focusing on the climate change debate. The IssueCrawler made visible the hyperlinks among websites that address its issues. Unequal distributions of them captured affiliations and alignments between actors. Niederer performed what she calls a resonance analysis based on which websites gave voice to climate change skeptics, identified in advance by external criteria. She could also address the question of how central climate change skeptics were within the climate sciences community and found that climate change skeptics did not distinguish themselves from the topics that this community discussed but where they were published. Journals differed whether they published articles by climate change skeptics.

The latter encouraged her to employ another software, called "Lippmannian device," a tool designed to map controversies, inspired by Walter Lippmann who in 1922, wrote about partisanship and public opinion. This software is seeded by a list of URLs and a list of queries representing the key terms of a chosen controversy and returns clear and course networks of actors that resonate with either side. On one day alone, in 2008, it retrieved approximately 27,000 articles on climate change, which received at least three citations from the inserted seeds. It also provides clouds of key actors on each side of a controversy and issue clouds for selected actors' writing. Based on the latter, Niederer could obtain a network of how actors' writing resonated with each other, which sources they relied on, and which they avoided. She could observe how each actor's writings resonated with those selected for attention (Niederer, 2013). The Lippmannian device is available at <https://wiki.digitalmethods.net/Dmi/ToolLippmannianDevice> together with instructions for its use. For an overview of digital methods available to search the Internet, see Rogers (2017).

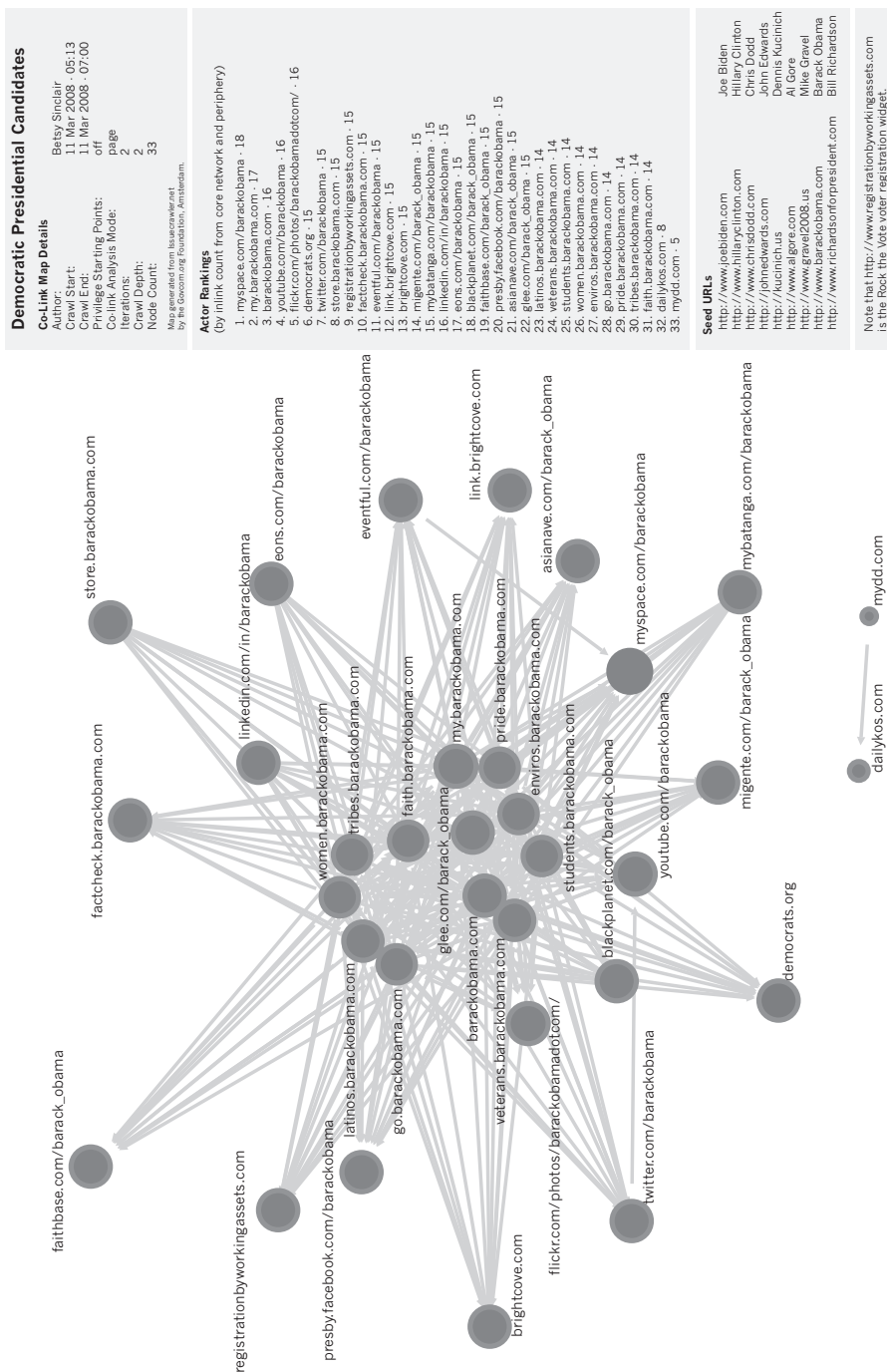
In her research, Niederer did not merely demonstrate the ability of software for harvesting hypertext networks of texts from the Internet; her methodology extended the unobtrusiveness, reader-centered conception of text, and inferential aims of content analysis, as presented here, by proposing several forward-looking generalizations. With a transcomputational number of possible networks on the Internet, Niederer suggests focusing content analytical attention to networks of key actors regarding issues chosen for analytical attention. Given the fluidity of

FIGURE 11.5 ■ U.S. Democratic Party Candidates' Issue Network, as Seen Through Hyperlink Analysis of the Eight U.S. Democratic Party Candidates' Websites, September 2007



Source: Graph by Issuecrawler.net, from the Govcom.org Foundation, Amsterdam.

FIGURE 11.6 ■ U.S. Democratic Party Candidates' Issue Network, as Seen Through Hyperlink Analysis of the Eight U.S. Democratic Party Candidates' Websites, March 2008



Source: Graph by Issuecrawler.net, from the Govcom.org Foundation, Amsterdam.

the Internet, she suggests against aiming at what I would call state descriptions and follow instead where and how actionable texts move, with whom they resonate, and what kinds of social networks they create and nourish. She also makes clear that besides starting with chosen issues, the results can no longer be captured with the predefined categories of single theoretical frameworks. Remaining open to what network texts lead to is becoming an important virtue. However, probably the most important concept Niederer developed throughout her work is that of the particular technicities of diverse media platforms: URL connection between websites, the character limitation of Tweets, how one can connect with so-called friends on Facebook—in fact, all platforms have algorithmic features that selectively attract and circulate texts of certain qualities. Traditional content analyses that hope to infer public concerns from tweets, Facebook messages, and the like ignore the fact that networked texts are filtered through the technicities of the platforms through which they circulate. Although content analyses of traditional mass media have always taken the institutional structures of the media into considerations, the technicities of Internet platforms are less obvious. Some remain hidden like Google’s proprietary search engines. They have to be studied as black boxes, by their effects. Others, like Wikipedia, a largely collaboratively created encyclopedia, is essentially transparent. However, it also employs software robots (bots), meant to correct references, detect vandalism, and ban unqualified users but also create new entries. We use Facebook to connect with “friends,” but pay for unwanted advertisements, and have only recently learned that some “friends” are bots programmed to reproduce misinformation and disrupt political processes. Niederer points out that networked texts do not simply represent the concerns of their authors and readers, these concerns are filtered by what particular technicities attract and highlight or hide. While her work has been confined to the Internet, she also recognizes that the traditional mass media become networked as well, as they maintain their own websites, publish online, and provide links to social media (Niederer, 2017).

11.4 COMPUTATIONAL CONTENT ANALYSES

Unlike accounts of character strings and text mining approaches, which employ entirely syntactical operations, computational content analyses process text according to theories of meaning that are presumed to operate within the context of the analyzed text. Theories of this kind are usually rather rudimentary, as the discussion below will show, but their roots in the chosen context of available text carry their justification.

Before I present a review of some of these computational aids, I must point out that there are vast differences among them in the demands they make on the forms of data they can handle. In fact, many computational content analysis programs attain their high level of sophistication by demanding that users perform often elaborate preediting of raw text. The following are some of the most common tasks content analysts may have to accomplish to make ordinary texts available for computer processing:

Clerical text cleaning

- Correcting misspellings (When reading, we routinely overlook orthographical errors, especially our own. A computer may not know

how to handle these. As noted in section 11.3.2, a fuzzy search, when available, is one computational solution to this problem.)

- Replacing foreign characters with characters that a given CATA system recognizes, for example, with ASCII characters (Relatively recent systems may not be so limited.)
- Introducing special markers to indicate syntactical distinctions (for instance, by adding double periods after sentences, carriage controls after paragraphs, or certain markers to distinguish headlines, abstracts, or leads from the remainder of a document)

Adding information the computer does not have

- Replacing pronouns with proper names and indirect references with direct ones; adding speakers' names in transcripts at turns of talk
- Marking the syntactical functions of words (for example, marking *bear* as a verb or as a noun)
- Introducing analytical distinctions (for example, distinguishing among speakers, actors, actions, and targets or among explanations, justifications, and excuses for actions)
- Distinguishing the meanings of homonyms (for instance, Buffalo the city, buffalo the wild oxen, buffalo the sucker, *buffalo* the verb, *buffalo* the word)

Adapting the text to the theory underlying the computational content analysis

- Dropping vocabulary that does not matter for answering the research questions (for example, creating lists of stop-words and go-words)
- Decomposing (kernelizing) longer sentences into smaller units of text, such as propositions or clauses, which often means rewriting a whole text into a computable format to circumvent the need to compute complex grammatical constructions
- “Unwrapping” the needed text through manual editing or programming (Naturally occurring text is usually packaged and so is preceded by information on sources, channels, and dates as well as by headlines, abstracts, keywords, and so on, and succeeded by appendices, references, signatures, and more. Somewhere in between is the body of text that matters.)

Packaging the text

- Many computational content analysis programs require segmenting a text into computationally convenient records: lines of a certain number of characters in length, paragraphs of a certain length, documents not exceeding a limit in size (Diction, for example, limits document length to 500 words), or batches that users must assemble to be handled one by one. The different ways in which such programs define records have important implications for the kinds of questions they can answer.

Clearly, heavy preediting requirements can easily cancel out the advantages that CATA promises. To help alleviate this problem, several computational content analysis programs include computer-aided text entry features. For example, a software called CETA (Computer-aided Evaluative Text Analysis; Cuilenburg, Kleinnijenhuis, &

De Ridder, 1988; Kleinnijenhuis, De Ridder, & Rietberg, 1997), which performs a version of contingency analysis (see Chapter 10, section 10.6), provides online support for human coders. This support takes the form of coder prompts during three preparatory stages of the analysis: while they are parsing texts, defining scoring options, and assigning numerical values to textual units. Cuilenburg (1991) has reported that in one month, 50 coders were able to code more than 400,000 textual units culled from 5,400 newspaper articles for use in CETA. In some software packages, such as Heise's (1995) Event Structure Analysis (ESA) and Schrodtt, Davis, and Weddle's (1994) Kansas Events Data System (KEDS), by far the most elaborate features are dedicated to the preparation of text for analysis.

Having established these preliminaries, I review below four kinds of computational aids, which I categorize according to the theories of meaning they embody: *coding/dictionary approaches*, machine learning of discriminate functions, *statistical association approaches*, *semantic network approaches*, and *memetic approaches*.

11.4.1 Coding/Dictionary Approaches

The simplest theory of meaning, and the one that dominates coding/dictionary approaches, derives from taxonomy, the idea that texts can be represented on different levels of abstraction, or that there are core meanings and insignificant variations of these cores.

Early on, Sedelow (1967) provided a convincing demonstration that analysts need to compare texts not in terms of the character strings they contain but in terms of their categories of meanings. She analyzed two separate English translations of a Russian book, Sokolovsky's *Military Strategy*, and found that they differed in nearly 3,000 words: 1,599 words appeared only in the Rand translations, and 1,331 words appeared only in the translation published by Praeger. Given that these were both respectable translations, differences in their vocabularies would not say little about them. Sedelow's finding demonstrates that text comparisons based on character strings can be rather shallow. Obviously, language provides for considerable stylistic variation and for many alternative forms of expressions. This observation led Sedelow to propose applying ordinary dictionary and thesaurus entries to the given texts and obtaining frequencies, not of actual character strings but of word families, that would account for texts on the basis of common meanings. This is the basic idea behind almost all coding/dictionary software.

One application of this idea involves the replacement of the words contained in texts with the entries in the dictionaries of a language, so-called lemmas. *Lemmatization* is "generally defined as the transformation of all inflected word forms contained in a text to their dictionary look-up form" (Boot, 1980, p. 175). It replaces, among others, the plurals and alternative spellings of nouns with their singular forms and the grammatical variations of verbs with their infinitive forms (for example, *is*, *was*, *will be*, *am*, *are*, *were*, *being*, and *been* are replaced by *be*). Evidently, lemmatization is more powerful than stemming. It embodies a theory of the very meaning found in the construction of ordinary dictionaries and reflects readers' use of such dictionaries. Although, operationally, lemmatization resembles a computer dictionary, where available, it tends to be used before other CATA dictionaries are applied.

Lemmatization and coding do not always require sophisticated CATA dictionaries. For example, Kim and Gamson (1999) used a word processing program

(Microsoft Word) to scan a body of text for particular character sequences. They read each sequence they found in the linguistic context in which it occurred, replaced words with similar meanings with a word that stood for their class, and counted the replacements. Through this admittedly tedious procedure, Kim and Gamson eliminated stylistic and semantic variations that they judged to be unimportant for their research, using themselves as the embodiments of a theory of how given words and phrases are read and categorized. Underlying their procedure is the idea that texts are representative of other things and can themselves be represented in simpler and more general terms, terms that eliminate the idiosyncrasies of individual readers and favor instead what the majority of readers would take away from reading them—filtered, of course, through the analysts' research questions.

Manual coding or recording of textual units into abstract categories that embrace a diversity of more specific instances of text (see Chapter 7) is not that different from what a computer thesaurus and dictionary are designed to do, representing large bodies of text in fewer and simpler terms—except that the latter applies to character strings without anyone reading them. If done well, the construction of computer thesauri and dictionaries serves as a theory of how readers rearticulate given texts in simpler terms and categorize them according to the needs of their research.

CATA dictionaries list character strings as words in categories of what they have in common, much as thesauri group words with shared meanings on different levels of abstraction. Applied to a text, a CATA dictionary either tags original words with the names of categories or replaces words that have similar meanings with more general words that stand for what they share (doing automatically what Kim and Gamson did manually in the study mentioned above). The General Inquirer (Stone et al., 1966), which was the first fully operational computer content analysis program to use a dictionary, assigns tags to words according to a theory of meaning that reflects a research question or the vocabulary of an academic discipline. For example, Stone et al.'s (1966) psychosociological dictionary tags *I*, *me*, *my*, *mine*, and *myself* as *self*, and *we*, *us*, *our*, *ours*, and *ourselves* as *selves*. From this dictionary, analysts can obtain frequencies of tags on two levels of generalization—for example, *anxiety*, *guilt*, and *depression* are counted as *distress*, and *distress*, *anger*, *pleasure*, and *affection* are counted as *emotion*. Analysts can also examine the words in a text that are assigned to a given tag to get a sense of how well the tag represents the class of character strings that are tagged alike. In addition, analysts can inspect a list of “leftover” words that occur in the text but not in the dictionary, modify and extend the dictionary by creating new entries, and apply various statistics on the tags as well as on the tagged words. WordStat, a very versatile content analysis software package developed by Péladeau (1996), conceptualizes go-word lists and stop-word lists as dictionaries, with the former replacing listed words by their categories. This creates a text with fewer types but the same number of tokens, which an analyst can then examine using a variety of statistical tools provided by WordStat's parent program, SimStat. In the General Inquirer, one word may be tagged in various ways. In WordStat, analysts have the option of using mutually exclusive or overlapping categories. The General Inquirer uses stemming and incorporates a variety of disambiguation rules for homonyms. WordStat provides a spell-checker and an English thesaurus. Many CATA programs use dictionaries in various capacities.

One important difference among CATA software packages that follow a coding/dictionary approach is whether or how easily users can tailor the dictionaries according to their own theories and context-specific research questions. The General

Inquirer dictionary is customizable, as are the dictionaries in TextPack, TextQuest, VBPro, WordStat, and many others. In contrast, Minnesota Contextual Content Analysis (MCCA; McTavish, Litkowski, & Schrader, 1997; McTavish & Pirro, 1990; <https://www.clres.com/online-papers/qq.doc>), Diction (Hart, 1985; <http://www.dictionsoftware.com>), the Regressive Imagery Dictionary (RID; Martindale, 1990), and Linguistic Inquiry and Word Count (LIWC; Pennebaker, Francis, & Booth, 2001; <http://www.liwc.net>) are all somewhat limited in this regard; each of these packages embodies one theory of meaning or a medley of several.

MCCA assigns about 11,000 words to 116 predefined categories that reflect the developers' conception of a narrative. MCCA acknowledges that word meanings differ with their linguistic environments and defines four such environments (traditional, practical, emotional, and analytic), suggesting that these are sufficient to disambiguate most homonyms concerning social behavior. MCCA produces the frequencies with which these categories are used in a text, enables the comparison of two texts based on these frequencies, and assesses the over- or underrepresentation of these categories relative to a statistical norm that has emerged over years of research with this system, much as the Diction software does.

Diction attempts to infer what its developer, Hart (1985), calls "the tone of a verbal message." It is designed to search texts of 500 or fewer words in length—a barely adequate document size for political speeches (although analysts can process several 500-word batches)—for words that its dictionary maps into five rhetorical qualities:

- *Certainty*: Language indicating resoluteness, inflexibility, completeness, and a tendency to speak ex cathedra
- *Activity*: Language featuring movement, change, the implementation of ideas, and the avoidance of inertia
- *Optimism*: Language endorsing or highlighting the positive entailments of some person, group, concept, or event
- *Realism*: Language describing tangible, immediate, recognizable matters that affect people's everyday lives
- *Commonality*: Language highlighting the agreed-upon values of a group and rejecting idiosyncratic modes of engagement

The frequencies in these categories lend themselves to comparisons of different texts and to the comparison of one text with a statistical norm obtained from a sample of some 20,000 public documents. Diction enables users to specify up to 10 additional categories, but Hart's conception of "tone" is the key to what it does.

RID is composed of about 3,200 words and word roots assigned to 29 categories of primary cognitive processes, 7 categories of secondary cognitive processes, and 7 categories of emotions. The dictionary focuses, as the name Regressive Imagery Dictionary implies, on such mental processes as the following:

- Drive (oral, anal, sex)
- Icarian imagery (ascend, descend, fire, water)
- Regressive cognition (consciousness alter, timelessness)

- Emotion (anxiety, sadness, anger, positive emotion)
- Sensation words (touch, vision, cold, hard)

Martindale (1990) developed RID to test his psychoevolutionary theory of art history, and researchers interested in psychotherapy and in tracing cognitive development have since found it useful. RID can be inserted into WordStat and TextQuest.

LIWC's dictionary can be used in WordStat as well. LIWC is designed to provide researchers with an efficient and effective method for inferring the various emotional, cognitive, structural, and process components present in individuals' verbal and written speech samples. It analyzes written text on a word-by-word basis, calculates the percentages of words in a text that match each of up to 82 language dimensions, and generates output that can be analyzed by various statistical programs. The program's basic dictionary is composed of 2,290 words and word stems, and each is included in one or more word categories. For example, *cried* is part of four word categories: sadness, negative emotion, overall affect, and past-tense verb. With each occurrence of *cried* in text, each of these registers is increased.

There are many more CATA dictionaries in existence, although some are not widely accessible. For example, dictionaries exist for coding social (or functional) versus antisocial (or dysfunctional) behaviors (Potter & Vaughan, 1997), speakers' psychological states (anxiety, hostility, cognitive impairment, and so on; Gottschalk, 1995; Gottschalk & Bechtel, 1982), Lasswell's value categories (Namenwirth & Weber, 1987), and Osgood, Suci, and Tannenbaum's (1957) semantic differential scales (Holsti, Brody, & North, 1965). For older special-purpose dictionaries, see Stone et al. (1966, pp. 169–206) and Gerbner, Holsti, Krippendorff, Paisley, and Stone (1969).

The General Inquirer was conceived to accept any kind of dictionary, but the evolution of its Harvard Psychosociological Dictionary provides an important lesson for users of coding/dictionary software. The General Inquirer dictionary took off from Bales's (1950; see also Krippendorff & Bock, 2009, Chapter 2.5) interaction process analysis but then incorporated more and more systems of categories—Lasswell's values and Osgood's semantic differential, for example. This is possible because it allows one word to have more than one tag. (WordStat, as already mentioned, provides this facility as an option.) In 1965, it had 3,500 entries and 83 tags (Kelly & Stone, 1975, p. 47). Züll, Weber, and Mohler (1989) expanded version IV of this dictionary, and its Dartmouth adaptation now contains 8,500 entries (Rosenberg, Schnurr, & Oxman, 1990). These developments demonstrate a general trend for dictionaries to become bigger and more diversified but also less changeable. Most content analyses would benefit from the construction of special-purpose dictionaries, but developing a dictionary from scratch can be a formidable task. It is not surprising, therefore, that content analysts usually try to build on available dictionaries before they attempt to develop their own. Researchers using the General Inquirer tend to add to its dictionary—if at all—rather than write their own.

How do analysts go about constructing new dictionaries? One early source of information on this subject is Stone et al.'s (1966, pp. 134–168) description of their work on the General Inquirer dictionary. Probably the best and a relatively recent account is offered by Bengston and Xu (1995; see also Krippendorff & Bock, 2009,

Chapter 5.2), who undertook a content analysis to examine changing values in the United States concerning national forests. To develop their dictionary, Bengston and Xu took the following steps:

1. They prepared a list of objects of value (both positive and negative) related to forests. Their choices of objects were of course informed by the kinds of inferences that interested them in this study.
2. They examined different size units of text regarding which would contain the needed descriptions of the objects on their list and decided on the basic unit of text that would contain the objects' valuation. They decided that a unit of text would have to contain the phrase *national forest* plus 50 words on either side.
3. They developed a list of words and expressions that could serve as indicators for text sources—writers, interviewees, or readers—to hold four basic kinds of values of interest to the researchers (these values had emerged from the first step). This amounted to four sets of words and expressions, one for each value.
4. They asked subject matter specialists to review these sets of value words and expressions (which formed the beginning of a value dictionary) and offer suggestions concerning refinement of the sets and additional words that might be included.
5. They examined the semantic validity (see Chapter 13, section 13.2.2) of the emerging dictionary entries with the help of KWIC lists applied to samples of text. In the course of this examination, they developed disambiguation rules and queries and then refined and tested these rules to see whether they correctly identified and distinguished among the four values. They then deleted any words and expressions that could not be disambiguated. (A disambiguation rule distinguishes among multiple readings. For instance, they found that economic words, by themselves, have no obvious value, but in the vicinity of words such as *devastating*, *ravaged*, and *misuse* they are clearly negative. A disambiguation rule combines two kinds of character strings to indicate a category, here of value.)
6. They sampled texts from three sources—the media, forestry professionals, and environmentalists, each over time—and applied the dictionary to them comparatively. The different sources had different relationships to forests and different ways of expressing values. The researchers continued this testing until at least 80% were correctly identified (on the semantic validity of text searches, see Chapter 13, section 13.2.2).
7. They conducted a final validity check on a representative random sample of data from these three sources.

These steps are easily replicable, and many content analysts create their own specialized dictionaries in similar ways. Others, however, rely on word databases such as the above-mentioned WordNet (Fellbaum, 1998; Miller et al., 1993), on conceptual dictionaries such as Laffal's (1993), on already available dictionaries or thesauri, on clustering techniques that reveal word co-occurrences, or on theories of the context in question.

To validate the semantics of a dictionary, analysts need to compare the dictionary's performance with categorizations obtained by other means—minimally, this means that the categorizations make sense to the researchers (face validity; see Chapter 13, section 13.1), but it would be better still if they can stand up against the judgments of human coders or the actual users of the texts, much as in the testing of a theory. The semantic validation of a dictionary ought to be a prerequisite of its use. Such an effort requires that the dictionary be open for detailed examination and that it can be applied to small texts that humans can read, tag, or categorize for comparison. The General Inquirer, for instance, aids in such examinations by offering lists of character strings that were stemmed, words with common tags, words not tagged, and more (Stone et al., 1966). WordStat produces lists of words not in the dictionary and lists of words to which the dictionary was applied. Software packages with fixed and hidden dictionaries do not offer the possibility of semantic validation.

A researcher may also determine a dictionary's functional validity (see Chapter 13, section 13.2.4) by measuring the success of the system that relies on it. Pennebaker et al.'s (2001) LIWC dictionary is a good example: It provides sufficient numbers of psychologically informed categories for analysts to define their own analytical constructs on top of the categories' frequencies of use. These analytical constructs resemble regression equations, which are selectively applied to the up to 85 dimensions that the LIWC dictionary produces and yield evidence for the presence of particular concepts. For example, Pennebaker et al. report that LIWC can identify an author's gender based on her or his word choices—a finding that is easy enough to validate against external evidence (predictive validity; see Chapter 13, section 13.2.6)—and various emotional states (Pennebaker, 1997). These researchers have developed equations for cognitive complexity, aging, depression or suicidality, lying, gender, and “talking presidential.” In a demonstration of their system, Pennebaker and Stone (2001) applied a battery of similarly derived measures (cognitive complexity, talks like a woman, depression or suicidality, talks like an old person, talks like a liar, and talks presidential) to early contenders in the 2000 U.S. presidential election campaign; they found considerable face validity, although admittedly in retrospect. Krippendorff and Bock (2009, Chapter 7.7) features a study by Pennebaker and Chung of Al-Qaeda transcripts showing the timelines for different message qualities.

Of recent, increasingly commercial interest is the use of dictionaries to evaluate emotions associated with particular attitude objects—products, people, or issues—called *sentiment analysis*. The idea of it goes back to Osgood et al.'s (1957) semantic differential of affective meanings, which suggested the existence of three dimensions of affect underlying most subjective appraisals: *Evaluation* (good–bad), *Potency* (strong–weak), and *Activity* (active–passive), the polarities implying different *intensities* (Osgood, 1974a, 1974b). Various researchers developed lists of emotion words—Heise (1965) and most famously Gottschalk and Gleser (1969)—subsequently forming the basis of computer dictionaries (Holsti et al., 1965). The most recent and extensively tested Dictionary of Affect in Language (DAL) started with research by Whissell (1980). It lists some 8,700 words that have been rated—pleasant–unpleasant; active–passive; and easy to imagine–hard to imagine—by readers of numerous texts, measuring their reliability and validity (Whissell, Fournier, Pelland, Weir, & Makarek, 1986) and applying it to large bodies of text. A well-documented user's guide is available (Whissell, n.d.). Sentiment analysis focuses

attention on selected attitude objects and can be described as a method of developing their emotional profiles (see Chapter 10, section 10.5). It is widely believed to substitute for public opinion polling and has been applied to presidential speeches, advertisements, news reporting, supreme court opinions, even lyrics of popular songs (see also Leetaru, 2011).

11.4.2 Machine Learning of Discriminate Functions

Section 4.2.3 introduced the notion of developing analytical constructs iteratively, by trial and error, without the presumption of advanced knowledge of the nature of the context of the texts to be analyzed. It discussed one example of a community of content analysts learning to improve their inferences from observable mistakes, sound criticism, and loss of relevance and another of a research project during which a discriminant function was developed and successfully applied on an albeit very small universe of textual data. The latter made use of what one might call preprocessed data in the form frequency counts of semantically related words, using Stone's (Stone & Hunt, 1963) dictionary-based general inquirer and applied traditional statistical techniques to distinguish real from simulated suicide notes.

Since that time, computational algorithms have emerged that can handle far larger volumes of text including for the development of discriminant functions capable of distinguishing among two kinds of text or assigning them to categories of analytical interests. The problem of distinguishing between real and simulated suicide notes is challenging but not too common. More obvious examples of employing discriminant functions are distinctions among news articles for whether they reveal left-leaning or right-leaning ideological biases (Gentzkow & Shapiro, 2010) or cover an issue positively or negatively (Eshbaugh-Soha, 2010). Software has been developed that can help teachers grade large numbers of essay exams after learning how that teacher graded similar ones previously or that can be taught to identify in students' written answers to exam questions, particular words and phrases that correlate with grades assigned by the instructor on previous exams.

Generally:

- (0) The data for developing discriminant functions consist of an ideally large number of independent pairs:

$$\langle \text{Text}_i, C_i \rangle$$

All Text_i have to possess sufficiently diverse characteristics in terms of their vocabularies, grammatical structures, and lengths; and the criteria C_i assigned to Text_i has to have at least two values.

Processes of developing and using discriminant functions tend to follow Hunt's (Stone & Hunt, 1963) three step protocol:

- (1) During the *machine learning phase*, a discriminant function f is developed by trial and error, as shown in Figure 4.6. In that process, all conceivable characteristics of all Text_i are candidates for discriminating among the assigned criteria C_i , and the learning algorithm iteratively selects the best ones among them. The resulting discriminant function f embraces the ones that optimally distinguish among alternative criteria C_i :

$$\langle \text{Text}_i, C_i \rangle \Rightarrow f$$

- (2) In the *reliability testing phase*, the discriminant function is applied to a set of $\langle \text{Text}_i, C_i \rangle$ pairs, not used in (1), to predict the criterion C_i' . The reliability of the discriminant function is established by paring the predicted criteria C_i' with the assigned criteria C_i for each Text_i :

$$C_i \approx C_i' = f(\text{Text}_i)$$

- (3) In the *application phase*, when results in (2) are sufficiently reliable, the discriminant function is applied to texts whose criterion is unknown but can now be predicted:

$$C_i' = f(\text{Text}_i)$$

The original conception of dichotomous discriminant analysis is due to Fisher (Cohen, Cohen, West, & Aiken, 2003) but has since become realized by different computational methods. Houle (2002) describes artificial intelligence experiments with so-called support vector machines (SVMs), which can be trained within a few seconds on 30,000 documents to develop easily comprehensible rules that distinguish whether similar documents have or do not have a given property. He reports accuracy rates as high as 90% in the SVMs' distinguishing Associated Press news wire stories in about 30 categories and as low as 60% in their distinguishing medical papers in more than 1,000 categories. In current content analyses, paths to discriminant functions have been provided by neuronal networks that "learn" the most successful connections between texts and selected contextual variables (see Chapter 11, section 11.4.2). Traditional discriminant analyses that improve the accuracy of answers to questions by combining features of text best suited to distinguish among them have also been used. Even regression analyses that attempt to predict extratextual (and dependent) variables by identifying their textual (and independent) predictors may be mentioned here as a one-step process (see the discussion of LIWC in Chapter 11, section 11.4.1; and Krippendorff & Bock, 2009, Chapter 7.7).

It is important to recognize the conceptual difference between the literality of analyzed texts and the abstract nature of the chosen criteria. Whether the author of a text is suicidal, news items are left- or right-leaning, positive or negative, or an essay deserves a passing grade, these criteria are inferences that intelligent readers would have to make to generate the data for machine learning. A discriminant function does not follow the literary competencies of readers. It merely models the relationship between textual characteristics and the differences among the assigned criteria, not what they mean to an analyst. There are diverse computational approaches available to model this relationship. They have unequal capabilities and biases. Users of available software should make an effort to understand what it aims for and how it gets there. The more abstract the assigned criteria are, the more unreliable machine learning software tends to be. Yet, there are surprising successes, as in the case of Hunt's (Stone & Hunt, 1963) ability to identify the authenticity of suicide notes. This strongly suggests not to skip phase (2) of testing the reliability of discriminant functions with texts independently read and interpreted by qualified readers but not used in phase (1).

11.4.3 Statistical Association Approaches

An unrealized dream of artificial intelligence researchers in the 1960s was the ability to generate abstracts of written documents automatically. Statistical abstracting assumed that the important words in a text were identifiable by their relative frequency; that their meanings were a function of their proximity to each other, which calls for the observance of co-occurrences; and that sentences that contained statistically prominent co-occurrences can serve as representations of a text as a whole. These efforts did not quite succeed, owing to the aforementioned difficulties involved in formulating a computational theory of how the members of a literate community summarize the community's texts. The dream did not die, however. It stimulated the development of computational content analysis programs that do not require prior categorization of textual units or predefined dictionaries and that minimize human coding efforts. These statistical association approaches, as they are called, have become popular with the use of the mining metaphor, promising to “extract” concepts or “discover” relevant information from voluminous textual databases.

The theory of meaning underlying these ideas has two roots. One goes back to association psychology. In content analysis, it entered with Baldwin's (1942) personal structure analysis, which Osgood (1959) generalized into his contingency analysis (see Chapter 10). These early analyses started with the tabulation of co-occurrences of manually coded concepts into square matrices of word pairs to compare them against the baseline of statistical expectations. The theory assumes that associations of concepts in someone's mind manifest themselves in co-occurring words. Osgood (1959, pp. 56–78) generated validating evidence for this contention in the form of experiments with human subjects and demonstrated the use of the theory by analyzing a section of Goebbels's diary.

The other root suggests that meanings do not reside in words but rather in how words relate to their linguistic environments—that is, how words relate to other words. Accordingly, two words have the same meaning if they can occur in the same linguistic environment, if one can replace the other without violating the sense of the whole. For example, the words *apple* and *orange* are interchangeable in many sentences having to do with shapes, eating, and growing on trees. Their difference in meaning becomes evident in sentences in which they are no longer interchangeable—for example, in sentences concerning where they are respectively grown. If such a theory could account for how we read, it would make references to human coders and extra linguistic phenomena such as the social/institutional/political roles of texts unnecessary.

Implementing such notions, Iker (1974, 1975) and his associates (Iker & Harway, 1969) pioneered a program called WORDS, initially to identify patterns in psychotherapeutic interview transcripts. WORDS included many now-familiar features: the stemming of words to their roots, a dictionary of the most obvious synonyms, and a stop-word list that discarded function words and ruled out very frequent and very rare words for not contributing to the statistical significance of associations. Within analyst-specified textual units of various sizes—certain numbers of words, paragraphs, pages, uninterrupted stretches of talk, answers to interview questions—WORDS computed a contingency matrix of up to 215 word types. It then factor analyzed the intercorrelations in this matrix and rotated them against a varimax criterion to obtain simple structures. These structures were considered to be the themes underlying the analyzed text. Current software packages such as CatPac,

TextAnalyst, TextSmart, and Semio create similar contingency matrices to draw statistical conclusions.

Woelfel's (1993, 1997; Woelfel & Fink, 1980) CatPac follows the initial steps of WORDS but deviates from pursuing a co-occurrence statistic by adopting a neuronal network learning model. This model begins with the assumption that all the words that have been selected as keywords are linked with equal strength. As CatPac scans the text and encounters co-occurrences within specified units of text—a window of a certain width, the number of words, for example—it strengthens the links corresponding to the observed co-occurrences at the expense of the links corresponding to words that do not co-occur. If more than two keywords co-occur, CatPac strengthens all the binary links that join them. (This approach is based on the theory that when people repeatedly think certain thoughts, read particular texts, or enact specific sequences of behavior, the neuronal connections in their brains that represent those thoughts or actions are strengthened, and thereafter they can recall, conceptualize, and enact those thoughts and actions more readily than they can recall and enact thoughts and actions experienced rarely or a long time previously.) The analysis generates clusters of words that Woelfel describes as manifestations of more or less distinct ideas. An important attribute of these clusters is that they are nonhierarchical. Unlike in hierarchical clustering, nonhierarchical clustering allows a word to belong to more than one class. So, in a content analysis of communication research literature, *communication* might well end up with *mass* in one cluster and with *theory* and *research* in a different cluster—as *mass communication* is a concept different from *communication theory* and *communication research* (Barnett & Doerfel, 1997).

Tijssen and Van Raan (1994) provide a good review of the CatPac procedure and work done by researchers using this program. For example, in a study of student essays, CatPac yielded 55 sets of frequently used words that could be grouped into five categories (McKnight & Walberg, 1994). The results identified not only such obvious concepts as “street-crime,” “gang-violence,” and “school-problem” but also stereotypical associations such as “young-bad-Black” and conceptual correlates such as “money-city.” Such results may be interpreted as some kind of common cognition, a generalization of a source characteristic, abducted by a theory of meaning—in CatPac's case, a theory of neuronal networks.

TextAnalyst, another software package that takes a neuronal network approach to text analysis, has been described as a business tool that can automatically abstract, index, store, and retrieve the volumes of documents that many businesses have to deal with. Sullivan (2001) describes this approach by using the aforementioned mining metaphors (see also the Megaputer Intelligence website, at <http://www.megaputer.com>).

The idea of neuronal learning nets is part of a more general effort on the part of researchers in artificial intelligence to develop algorithms that can simulate naturally occurring behavior, so that computers can learn to perform certain operations without being told how to do them—in the case of content analysis, the goal would be for computers to read text. One algorithm, called a support vector machine, can be trained on, say, 30 categories in 100,000 documents and then produce easily comprehensible rules for identifying these categories in other documents. Related algorithms go by the name of latent semantic analysis or latent semantic indexing (LSI). Another model is hyperdimensional analogue to language (HAL). For the latest elaborations of these ideas, interested readers should seek out sources on the Internet.

Danowski's (1982, 1993) program Wordlink uses word co-occurrences in still another way—it extracts all pairs of words within a window of specified width, except for those on a stop-word list, and records their distances from each other. With the help of NEGOPY (Richards & Rice, 1981), a software package designed for analyzing communication nets (among up to 6,000 nodes for the mainframe version and up to 3,000 for the PC version), Danowski used Wordlink to construct a very large word-distance matrix. He illustrates the information that this matrix can provide with data consisting of the answers car dealership sales and service personnel gave when asked, “When you think of ‘customer satisfaction,’ what comes to mind?” Danowski applied a “traveling salesman” algorithm to this huge distance matrix. This algorithm is designed to find the shortest path between any two nodes, passing through a number of intermediate nodes. For the most frequent words, the algorithm identified the following string: “Consumer satisfaction: good service on the new car done right the time first.” Reversing the order of just the last two words in this string yields a grammatically correct sentence that probably expresses the gist of the beliefs shared by Danowski's respondents (see Krippendorff & Bock, 2009, Chapter 7.4). In another study, he assigned values to different words and computed advertising slogans that might be worth disseminating.

Statistical association approaches differ in whether and how far analysts can supervise them to answer specific research questions. The issue of whether a neuronal network approach is superior to traditional clustering algorithms—which WordStat and several other CATA software packages enable—is undecided, and perhaps it is not too important for users, given that both rely on the same theories of meaning. The aim to substitute a general algorithm for human coding/reading has the effect of making statistical association approaches independent of particular contexts of analysis, and this raises conflicts for content analysts who are attempting to answer context-specific research questions.

11.4.4 Semantic Network Approaches

A network consists of several nodes linked by binary relations. It can be graphed from a collection of $\langle \text{node}_i\text{-connection}_j\text{-node}_k \rangle$ triplets. A network is called *semantic* when its nodes represent concepts or clauses and when these are linked to each other by more than one kind of binary relation. In the theory of meaning that underlies a semantic network approach, the meanings of nodes are a function of how they are connected with each other. The aim of semantic network approaches to content analysis is to find answers to questions that are not literally contained in a body of text but are implied by it. Hays's (1960, 1969) early content analyses of political documents are examples of such approaches, as are the current efforts of artificial intelligence researchers to design expert systems.

To be clear, the webgraphs discussed above are not semantic nets by our definition because the relationships between URLs are mentions, character strings, whose meanings require the analysts' ability to read them. They do not enter the computations. Also, the networks discussed in section 11.4.2 are not semantic, as all of their links are of the same kind: co-occurrences or proximities within a textual unit. Networks are found everywhere: causal networks in physics, road networks on drivers' maps, citation networks in literary scholarship, networks of interactions in organisms. Communication networks, for example, can be both associative and semantic. When researchers count how often people talk with each

other, communication networks are formally indistinguishable from transportation networks, cash-flow networks, or association networks. But when such networks include information about what is being said, or the kinds of social relationships within which individuals communicate with each other, their links mean different things to the communicators and therefore become semantic networks.

The idea of semantic networks probably originated in cultural-anthropological efforts to represent cognitive structures or mental models as relational graphs (D'Andrade, 1991, 1995; Quillian, 1968; Wallace, 1961). It has been facilitated by the psycho-logical theorizing of Abelson and Rosenberg (1958), by Schank and Abelson's (1977) inquiries into actionable knowledge structures, by causal inference modeling (Blalock, 1964; Simon, 1957), by graph theory (Harary, Norman, & Cartwright, 1965; Maruyama, 1963), by evaluative assertion analysis (Osgood, Saporta, & Nunnally, 1956), by studies of communication networks (Rice & Richards, 1985; Richards & Rice, 1981) and social networks (Wellman & Berkowitz, 1988), and by theories of communication networks (Rogers & Kincaid, 1981). Possibly the first mention of a computer analysis of texts using what we now call a semantic network is found in the work of Allen (1963), a lawyer and logician who proposed a system for studying the networks of propositions that constitute legal agreements, such as an arms limitation agreement, for loopholes that their signatories could use.

In artificial intelligence research, semantic networks are considered one approach to knowledge representation and natural language processing. Lindsay's (1963) demonstration of a system that "understands" natural language is an outstanding early example. His system features a relational memory, which creates what we now refer to as a semantic network. It keeps track of Basic English sentences (using 850 English words and a simplified grammar; Ogden, 1937) that mention kinship relations of the form "*a* is the *R* of *b*." As such sentences enter the system, it constructs a kinship network that the analyst can eventually question about kinship relationships not mentioned in the texts but implied by them. This is what makes semantic network approaches attractive to content analysts: the promise of finding answers that are not literally contained in a body of texts—in its readable character strings, in the abstractions that computer dictionaries can provide, or in the co-occurrences of words—by inferring what is semantically entailed by the given texts. Network conceptions of cognition and textual meanings are not entirely unproblematic, however (for a critique, see Johnson-Laird, Herrmann, & Chaffin, 1984).

As already stated, semantic networks are constructed from collections of birelational statements, two-valued predicates, or propositions containing two nodes with a particular link specified between them. The proposition "The candidate failed to clarify his position" links "the candidate" to his "position" by his "failure to clarify" it. The proposition "The U.S. women's soccer team won the 1999 World Cup" links "the U.S. women's soccer team" to the "1999 World Cup" by the team's having "won" it. And the proposition "Mary helps John" links "Mary" and "John" by her act of "helping." The first three components of Lasswell's (1960) famous formula "who says what to whom with which effect" define a connection in a semantic network of people or institutions sending and receiving particular messages, sharing "what" is being said. The <actor-action-target> triplets that some researchers employ to record interactions among individual actors and nations (Heise, 1995) constitute semantic networks, as people interact with each other in innumerable but conceptually distinguished ways.

In computing the implications of texts that constitute semantic networks, analysts need to acknowledge the relational properties of the connections between concepts. For example, “being married” is bidirectional and confined to two nodes. Being “the mother of” is unidirectional, and although it is confined to two nodes as well, it is recursively extendable (there are mothers of mothers, and so on). Subordination is unidirectional, transitive, and not confined to a pair of nodes: If *C* is subordinate to *B*, and *B* is subordinate to *A*, then *C* is subordinate to *A*. Reflexive connections turn back to the very nodes from which they originate. Moreover, there are key concepts that, if removed, would separate a network into smaller subnetworks, whereas others could drop out without making any appreciable difference. In content analysis, semantic networks are of particular interest because they preserve relationships between textual units, even when they occur in different parts of a text and thus form an interconnected web rather than separate categories of textual units.

An excellent but rather specialized example of a semantic network approach with online support for preediting is Heise’s (1995) Event Structure Analysis, mentioned above. It facilitates the analysis of texts that describe a web of events involving actors, targets, outcomes, benefits, and the like. ESA is an online computational system that joins sociological conceptions of actions with narrative analyses and mathematical graph theory (Abell, 1987, 1993; Corsaro & Heise, 1990; Doreian, 1993; Durig, 1995). The process of using ESA essentially consists of two steps. In the first, the user identifies and orders the events being described within the text. To this end, the program guides the user with pertinent instructions. It produces a diagram that depicts how some events are necessary for other events to happen and how abstract events are represented in concrete happenings, and it facilitates decisions concerning whether the network is sufficient for analysis. In the second step, the user codes information on events into an “event frame.” This frame has eight components: the agent, the action, the object (or target of the action), the instrument, the alignment (of the instrument), the setting (of other people, objects, and instruments in which the event takes place), the product, and the beneficiary. During this process, ESA makes the coding instructions for this dimensionalization of the events available to the user. After thus building the network of social events, the analyst can see how people, things, and actions are linked to what happened and can ask several analytical questions, even in quantitative terms, that would have been difficult to answer from the original text.

CETA, also mentioned above as a computer program that provides computational support for preediting of text, feeds into Kleinnijenhuis et al.’s (1997) extension of Osgood et al.’s (1956) evaluative assertion analysis. It distinguishes not two but five two-place predicate types:

- Predicates that evaluate something, implying an ideal or value (Osgood’s common meaning term)
- Predicates that claim something to be real or true in the author’s world
- Predicates that describe an actor’s actions and targets
- Predicates that claim a causal relationship between two variables, including “if-then” forms
- Predicates that aver an affective relationship between two actors or between one actor and an object

TABLE 11.1 ■ Predicate Types for Kleinnijenhuis et al.'s Semantic Network Analysis

Predicate Type	Author	Quoted Source	(IF) i	Predicate	(THEN) j
Evaluative	A:	S:	object	(dis)associated(+/-)	ideal(+/-)
Reality	A:	S:	A's or S's reality	predicate(+/-)	object
Action	A:	S:	actor	acts on(+/-)	object
Causal	A:	S:	object/variable	causes(+)/prevents(-)	object/variable
Affective	A:	S:	actor	(dis)affected(+/-)	actor/object

Source: Kleinnijenhuis et al. (1997).

These predicate types are displayed in Table 11.1. To distinguish several worlds (i.e., networks), Kleinnijenhuis et al. (1997) also attribute nuclear sentences to their authors, to speakers, to publications, and to sources quoted by authors. The positive and negative values of the ideal and the associative or dissociative properties of the link between any two concepts (e.g., “stimulates” is +, “inhibits” is –; “has” is +, “has not” is –) are expressed numerically between +1 and –1. Unlike other authors concerned with preediting efforts, Kleinnijenhuis et al. do attend to the reliability problems that such parsing necessarily poses.

In Kleinnijenhuis et al.'s (1997) analysis of these triplets, the qualitative differences among predicates are dropped in favor of the magnitude of their expressed association or dissociation, and a mathematical calculus is employed that aggregates these to form a compound network of concepts (Cuilenburg, Kleinnijenhuis, & De Ridder, 1986, 1988; Kleinnijenhuis et al., 1997). The rules of inference provided by this calculus are based on a theory of affective meaning according to which the values of ideals flow through a network along associative links and reverse their valuation as they travel along dissociative links.

In a 1990 paper, Kleinnijenhuis analyzes the world of Israeli prime minister Shamir, compares Arafat and Goebbels, and discusses Israel's relations with Western governments as presented in the Dutch press. In a 1997 paper, Kleinnijenhuis et al. demonstrate semantic network analysis with data from public and parliamentary debates on economic issues in two leading Dutch newspapers during two periods of time, 1968–1976 and 1978–1984. They consider eight political actors (e.g., government, parties, and unions) and 29 economic variables (e.g., unemployment and interest rate) as well as “ideal” and “reality.” Naturally, the directions of influence among several variables and the evaluation of several concepts changed during these periods, but Kleinnijenhuis et al. found that all changes moved away from the Keynesian policies of the first period to a neo-orthodoxy view of the role of government in the Dutch economy during the second period. The way in which these authors derived inferences, by tracing the flow of influence and affect through a semantic network, amounts to a shift from the traditionally explicit coding of concepts to latent phenomena, to a “systematic reading between the lines.” A semantic network analysis by these authors is reproduced in Krippendorff and Bock (2009, Chapter 7.5).

A third example of a semantic network approach is found in Carley's (1997) attempt to formalize social experts' knowledge. This research led Carley to the construction of semantic networks, which she calls “socio-conceptual” networks

(mentioned in Chapter 10). The software she developed, MECA, aids coders in identifying any pair of concepts that occur in the same sentence, clause, or paragraph as subject or object, respectively, and the relationship pertaining between the two concepts. Carley distinguishes three levels of strength in the relationships between nodes, including the degree of agreement or consensus encountered among the contributors of the texts (hence the reference to a society in her name for these kinds of networks). Her definitions of the three levels are as follows:

- *Definitives* are statements where one concept defines the other, such that in the society in question, if the first concept is used, the second is always implied.
- *Logicals* are statements where the concepts are logically connected, such that in the society in question, if the first and second concept are used the speaker intends a specific relation among them.
- *Simple connectives* are statements, such that in the society in question, if the two concepts are used and the speaker has not specified an alternative relation, then the socially accepted relation between them is assumed. (p. 87)

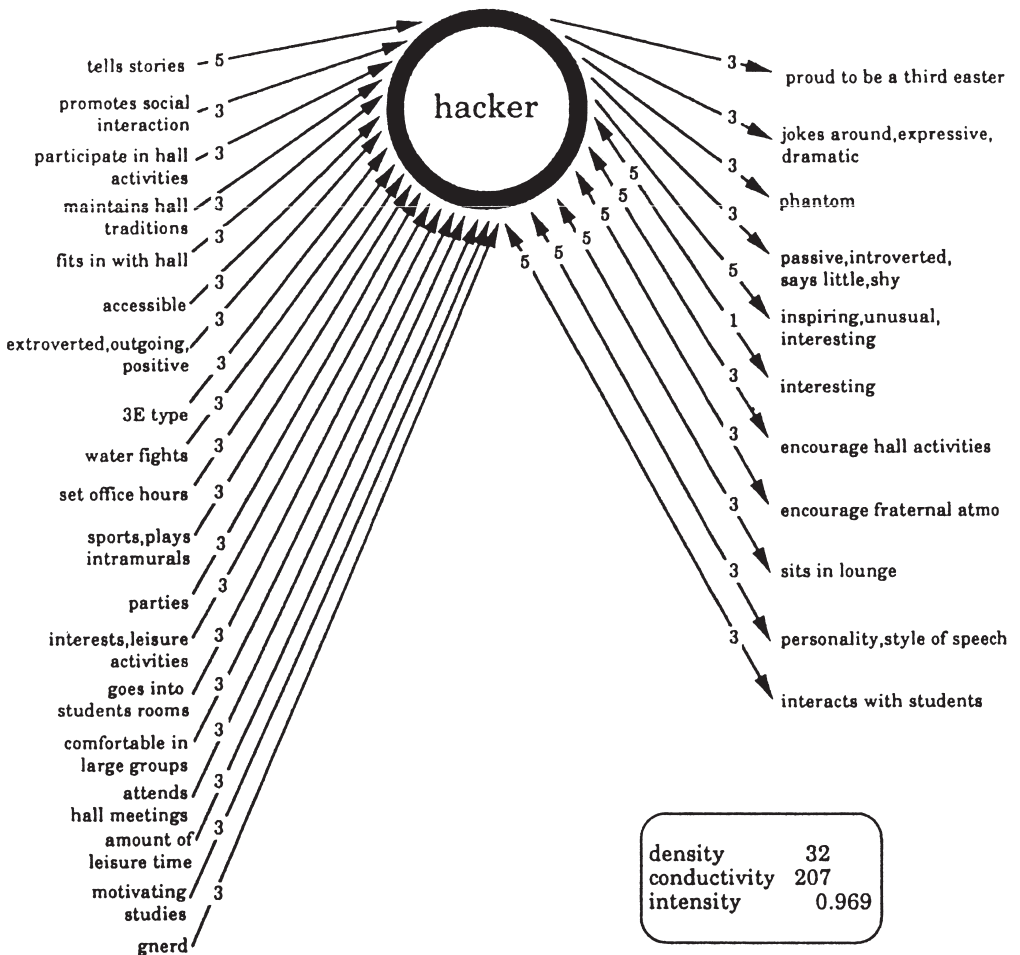
Unlike the researchers whose work provides the previous examples of semantic network approaches, Carley (1997) does not trace implications through her networks; rather, she attempts to characterize the meanings of each node (see Figure 10.7) by using five numerical indices of the role the focal concept plays within a network:

- *Imageability* is the total number of nodes connected to the focal concept by arrows emanating from it.
- *Evokability* is the total number of nodes connected to the focal concept by arrows reaching it.
- *Density* is the sum of imageability and evokability.
- *Conductivity* is the number of two-step paths through the focal concept, which amounts to the product of imageability and evokability.
- *Intensity* is the fraction of all nuclear statements that contain the focal concept.

The three dimensions of density, conductivity, and intensity give rise to a taxonomy of concepts that reflects the concepts' positions within the network. For example, ordinary concepts are low on all dimensions. Most concepts can be expected to be ordinary, and analysts use such concepts to define outstanding types. Symbols, for example, are high on all three dimensions. They represent highly embedded and multiply defined concepts on which there is often high consensus. Stereotypes are defined as concepts that are high in intensity and density but low in conductivity. Stereotypes are highly structured but change slowly due to their low level of conductivity. In addition to ordinary concepts, symbols, and stereotypes, Carley (1997, pp. 87–89) differentiates the network properties of prototypes, buzzwords, factoids, placeholders, and emblems.

In a study of MIT students' talk about tutors, Carley (1997) found that most concepts turned out to be factoids, which is not surprising, given that such talk often turns to decisions. The next-largest numbers of concepts were ordinary (e.g., "finishing thesis," "humorous comment") and stereotypes (e.g., "nerd," "hacker"). As these two concepts seem mutually distinct and opposite in evaluation, these labels would matter a great deal to tutors. Figure 11.7 illustrates the network position of the concept "hacker" in Carley's semantic network of tutor talk. I present only one such node here as an example of the analytical possibilities that semantic networks can provide for improving researchers' understanding of the relations among concepts.

FIGURE 11.7 ■ The Concept "Hacker" Within a Semantic Network of Tutor Talk



Source: Carley (1997, p. 96, fig. 4.6).

Computationally, the less diverse the properties of the relationships among nodes, the easier semantic network analysis is to handle. This is why Kleinnijenhuis et al. (1997) restrict their list of predicates to five kinds, which is only slightly more

than the two kinds of assertions of varying associative/dissociative strengths used in Osgood et al.'s (1956) evaluative assertion analysis. This is also why Carley (1997), who does not look for such simplifications, asks simpler questions to begin with, questions that can be answered through the comparison of only a few nodes at a time.

In an attempt to avoid the need for excessive preediting, Barnett and Doerfel (1997) have pursued still another interesting computational simplification. To infer scholarly groupings within the International Communication Association (ICA) from the titles of papers presented at the ICA's annual conferences from 1992 through 1996, they combined two research traditions: (a) bibliometric analyses of citations (Rice, 1990; Rice, Borman, & Reeves, 1988; So, 1988), which rely on the acknowledged use of one publication for ideas discussed in another; and (b) work in organizations conducted by Monge and Eisenberg (1987), who recommend measuring organizational networks through content analyses of members' key vocabulary, slogans, and stories. Assuming that intellectual groups sustain themselves through talk among peers, develop consensus on key concepts, and submit their papers to divisions they perceive to be sympathetic to their approaches, Barnett and Doerfel defined the connection between any two papers by the vocabulary the titles shared. Their triplets, $\langle \text{author}_i\text{-vocabulary in paper}_i \text{ and } \text{paper}_k\text{-author}_k \rangle$, made coding unnecessary and led to a semantic network of these different kinds of connections. Their preferred type of analysis is clustering, and the groupings that emerged showed interesting correlations with the ICA divisions.

11.4.5 Memetic Approaches

The CATA approaches discussed so far in this chapter all start from synchronic (i.e., atemporal) descriptions of texts and tend to provide inferences from single bodies of texts about common readings, shared categories, prevailing association structures, and semantic implications. They generally do not respond to interactions among texts over time. Yet we know that all texts are rooted in previous texts. Reading takes place in the present but always against the backdrop of previous readings, including nonverbal responses of other readers, and with an eye toward future texts. Words not only have etymologies, but their meanings change as they move through different texts, enter and leave conversations, and become reproduced, rearticulated, and used in novel texts.

Social realities, one can argue, manifest themselves within vast networks of rearticulations of texts that moreover organize themselves in the process of continuing rearticulations. The theory of meaning underlying memetic approaches attempts to relate currently available texts to the histories of their interpretations. This theory is enacted or embodied by other readers, other writers, and the media of communication disseminating the texts.

The word *meme* was coined by Dawkins (1976, 1982), a geneticist. Seeking to extend the theory of biological evolution to cultural phenomena, Dawkins conceived of the meme as a textual analogue to the biological gene. His starting point was that messages enter a brain as ideas and interact therein with other ideas, and the result of that interaction determines what the brain does. Because ideas are observable only in their consequences, if memes are to persist, they must be able to program a brain for their own replication. Memes that thus succeed can be passed on and enter other brains. Just as biological organisms can be thought of as a means for DNA replication,

so the brains of a population of people can be viewed as temporary hosts where memes replicate themselves. Memes that do not reproduce eventually die out. A meme's replicability is enhanced when the meme manages either to prevent competing memes from entering its host, a brain, or to attract cooperative memes that increase the joint probability of one another's being replicated. For memes, brains are a scarce resource. The social networks that emerge as a consequence of memes' entering and programming the brains of members of a given population constitute additional ways for memes to create support for their own replication.

Dawkins's memetics surely is a mechanistic theory, but it has fueled the development of CATA methodology. Indeed, content analysts might do well to acknowledge meanings in the interactions between different texts. The idea of plagiarism (as discussed in Chapter 10, section 10.5) is based on comparisons of two authors' texts, written at different points in time, in the context of a shared literature. History consists of a network of texts that are read in the present but focus on an unobservable past, with historians mediating the interactions among these texts. Even such a common idea as public opinion is constructed on multiple text comparisons over time, as seen in the practices of those who measure it. Although pollsters may conceptualize the public as an object to be measured, this is far from reality. Public opinion is an institutional construction based on at least three kinds of texts: pollsters' questions, multiple interviewees' answers, and the published results of this construct as accepted by readers. The first two texts grow out of conversations between interviewers and respondents who happen to be asked their opinions about issues of interest to pollsters, not what they are concerned with in their lives. Conclusions derived from the comparison of these two texts say as much about the pollsters' interests as about the respondents' answers, and it is far from clear what either has to do with what people talk about in public. Significantly, the questions asked and the responses received are derived to a large extent from agendas set in the mass media, which also publish the results of opinion polls, all with the aim of establishing what is publicly acceptable, politically feasible, or true. Thus, public opinion is embedded in texts that are both cause and consequence of an institutionalized network of textual rearticulations.

The naive view of public opinion held by many pollsters is not the only example of researchers' failure to recognize their discipline's memetic nature. Content analysts who believe they are able to analyze contents without contexts have fallen into the same trap. This is why I have argued against the conception of analyzing the content of texts and have advised analysts against reporting findings concerning "it" without making reference to readers, users, or the institutions that reproduce the texts being analyzed and without acknowledging the textual origins of the analytical constructs being used, albeit implicitly (see Chapter 2).

The study of the dynamics of very large bodies of text was envisioned by Tenney (1912), piloted by Lasswell (1941), and theorized by Rapoport (1969) and has been exemplified by Gerbner (1969, 1985) and his colleagues (Gerbner, Gross, Morgan, & Signorielli, 1994). However, all of these scholars lacked the requisite computational capabilities to process the interactions between large volumes of text and ended up making simple counts of frequencies or volumes. Confined to manual coding, Tenney could study only a few New York daily newspapers, Lasswell could study the elite newspapers of only a few countries, and Gerbner based his longitudinal studies on one week of television per year. These researchers also lacked adequate models or

analytical constructs. Tenney relied on the metaphor of weather dynamics, Lasswell and Gerbner used symbol systems, and Rapoport applied systems theory. Below, I discuss two computational content analyses of a memetic nature, one informed by ecology and the other by epidemiology, much as memetics is indebted to genetics.

The first example is Best's (1997) study of newsgroups in the popular Usenet (or NetNews) discussion system on the Internet. Within Usenet, discussion groups evolve and continue along diverse lines of subject matter, ranging from science to politics, to literature, and to various hobbies. Participants in a newsgroup read each other's contributions (called *posts*), comment on them selectively, and bring new ideas into the discussion. Naturally, contributors have limited amounts of time available to spend on the Internet, and this has the effect of limiting the participation of most to one or a few newsgroups. Posts consist of the texts that individual participants contribute in response to other posts, thus defining a forward-moving network of linked texts, an ecology of evolving textual species. In this textual ecology, newsgroups function like islands on which certain memes reproduce themselves and evolve more readily than on other islands. Usenet has grown considerably since the late 1970s and early 1980s. Currently, newsgroups create about 100,000 new posts each day, making Usenet an excellent data source for the study of cultural microevolution.

Best's pilot study, which relied on 1,793 posts made over a 10-day period, is part of a larger effort (Best, 1998). The key to Best's approach is the marriage of two kinds of software, one designed to simulate artificial life (Alife) and one designed for information retrieval (LSI; Deerwester, Dumais, Furnas, Landauer, & Harshman, 1990; Dumais, 1992, 1993; Foltz, 1990; Furnas et al., 1988). LSI represents units of text (in Best's study, the posts) as vectors, which are a function of the frequencies of word co-occurrences within them. With a principal components analysis of these vectors, Best identifies frequently recurring vector subspaces, which represent word clusters that replicate together. For example, in a collection of posts from a military discussion group, *harbor*, *Japan*, and *pearl* turn out to replicate themselves in unison and can therefore be considered a meme. Best operationally defines a meme as the largest reliably replicating unit within a text corpus and considers it as a suitable cultural replicator (Pocklington & Best, 1997). Accordingly, a post is a collection of more or less successfully replicating memes. Best also addresses the issue of mutation, or the possibility that memes, especially larger ones, do not reproduce identically. One source of mutation is memes' traveling outside the electronic system as well, through their users' brains and real-life conversations, after which they may reenter the system in rearticulated forms. What actually happens to these memes outside the system is of course not accessible to this analysis. Best considers mutations of this kind by observing memotypes, which are paraphrasings, rephrasings, or different versions of the same story.

Alife calls for the identification of "quasi-species" (Eigen, McCaskill, & Schuster, 1988), which in Best's study are defined as collections of posts with threads to common ancestors that evolve while sharing a variety of memotypes. For example, *Japan* and *Pearl Harbor* are always embedded in stories that are related in some ways and can be said to be of a certain kind. Best then examines the pairwise interactions between these quasispecies as cross-correlations over time, much as such interactions are studied in ecology.

In ecology, one population can have either a positive effect (+) on another by increasing the other in numbers (i.e., increasing members' chances of replication and

survival) or a negative effect (–) by decreasing the other’s numbers. Best has tested four kinds of interactions—mutual (+, +), competitive (–, –), neutral (0, 0), and predator/prey (+, –) (see also Boulding, 1978)—and has found strong negative cross-time correlations that account for the interactions among quasispecies in Usenet. At first glance, Best thought these interactions to be of the predator/prey variety—that is, increases in one are obtained at the expense of the other, subject to the constraint that when the prey disappears, the predator will as well. However, the negative correlations could also be explained by competition for a limited resource—in this case, the attention of human participants. Competition was especially evident for quasispecies residing in relatively narrow ecological niches. The memetic hypothesis that memes compete for the scarce resource of participants’ minds in order to replicate themselves seems to explain the speciation dynamics in Internet newsgroups.

The second example of the use of a memetic approach is the system that Fan (1988; Fan & Cook, 1997) developed for predicting public opinions and behaviors from mass-media data. Fan, a biologist by training, was not the first to see a parallel between the spread of diseases and the adoption of ideas and practices—the information diffusion literature had long thrived on this analogy (for one early example, see Rogers & Shoemaker, 1971; for one more recent, see Valente, 1993). To capture the dynamics of public opinion, Fan developed an analytical construct in the form of a complex recursive function and applied it to the day-by-day mass-media production of messages (containing memes).

Fan’s assumptions include the following: At any one point in time, public opinion equals the distribution of ideas in a population—the memes that a survey could elicit. Much as a contagious disease is conceptualized in epidemiology, a meme enters someone’s brain and either takes hold of it or does not. Memes that have entered a brain as ideas can be replicated and passed on. Fan deviates slightly from the epidemiological model by considering people not as mere passive receivers of “infectious” memes but as holding one of two opposing ideas and as being able to change their ideas in either direction. The rate at which ideas change from one to another depends on memes’ communicability.

To compute the rate of the adoption of ideas within a population of people, Fan relies on a fairly general mass-action law. If everyone reproduces memes at an equal rate, then the rate of their adoption is the product of two proportions: the proportion of people who hold the ideas in question and the proportion of people who do not. Mathematically, when these proportions are zero or one—in other words, when everyone is of the same mind—the adoption rate is zero. According to this law, the largest adoption rate can be expected when a population is evenly divided on an issue. However, when the two competing memes are equal in number, their persuasive forces cancel each other. It is only when they are unequal that opinions will shift.

Ordinarily, the distribution of memes circulating within a population reflects the distribution of public opinions. This correlation is significantly altered, however, when mass communication enters the picture, when the rate at which memes are reproduced or published is no longer equal for all but biased by the mass media, the great multiplier. This describes a memetic model of a dynamically shifting system of public opinion as it is recursively perturbed by the selective reproduction of memes by the mass media.

Mathematically, measuring public opinion at any one time and then adding the effects of mass-media coverage turns out to be equivalent to measuring a longer

history of the daily mass-media coverage and acknowledging that media coverage loses its impact on public opinion over time. Fan assumes that the persuasiveness of media coverage has a half-life of one day. This mathematical equivalence enables Fan to bypass public opinion measurement (of ideas) altogether and take the history of mass-media coverage (meme dissemination) as data for predicting public opinion. One of Fan's motivations for making this choice is that conducting public opinion polls is costly and time-consuming, whereas the history of media coverage is easily available, and such an analysis can offer instant predictions.

This analysis starts with the identification of the pros and cons of a selected issue in the media coverage of the day. For simple issues, Fan generates such data with the aid of a computer; for more intricate ones, he employs human coders. With an initial estimate of the public opinion that the media coverage is likely to meet, which may be quite arbitrary, plus the textual data of the day, the system computes a first round of predicted opinions. These plus the following day's mass-media coverage of the issue yield a second round of predictions, and so forth. As the history of mass-media coverage unfolds, the predictions that emerge become increasingly independent from the initial public opinion estimate and increasingly correlated with the actual public opinion, measured independently. The recursive function of public opinion's dependence on mass communication, the analytical construct, develops over time, continuously adjusting itself to the emerging history of the coverage of an issue. It can learn from "mistakes" when the results of available opinion surveys occasionally enter the computations.

The recursive analytical construct, which is part of what Fan calls an "ideodynamic model," has been vindicated by numerous rather successful predictions in several areas of public opinion (see Fan, 1997).

11.5 QUALITATIVE DATA ANALYSIS SUPPORT

The computer aids reviewed above support content analyses of typically large volumes of textual matter and at various stages of the analytical process—identifying relevant texts, for example, or coding. Many require users to do preparatory work: preediting texts, formulating search terms, constructing or validating dictionaries and category systems, selecting among analysis options, and defining various statistical parameters. But once a procedure is set, usually after the analyst has developed it on a small sample, it is uniformly applied to all available texts. In this section, I review computer aids to another kind of content analysis, one whose tradition originated in ethnographic, cultural-anthropological, and interpretive scholarship, often labeled *qualitative research*. I have mentioned this research tradition briefly in Chapter 1 (section 1.7) and Chapter 4 (section 4.1.2).

Computer aids in the qualitative data analysis (QDA) research tradition could be called interactive-hermeneutic—*interactive* because the categories of analysis and the choices of analytical constructs are not fixed, and content analysis categories become apparent to the analysts in the process of reading if not actively interrogating their texts; and *hermeneutic* because the process of analysis is directed by the analysts' growing understanding of the body of texts. This process enables content analysts to correct initial misconceptions or interpretive errors that might become apparent at a later point and allows for course corrections in midstream whenever interesting analytical avenues

appear. In a hermeneutic circle, text is interpreted relative to an imagined context, and these interpretations in turn reconstruct that context for further examination of the same or subsequently available text. This iteration continues (see Figure 4.3) until some satisfactory understanding is achieved. Understanding is the point at which the reading of texts resonates with the analyst's background. Understanding is always a momentary state, and the analytical results of this approach to content analysis are never final, always thought to be improvable or revisable in time.

Computer aids for QDA expand the interactive-hermeneutic process in at least four ways:

- They offer their users text manipulation routines for handling, organizing, filing, and keeping track of texts, which is especially important when texts are more voluminous than unaided analysts can handle.
- They allow analytically acceptable text alterations, spell-checking, and, most important, associating highlighted text segments with codes, summaries, or comments that making explicit and searchable what would otherwise remain implicit and hidden from an analysis.
- They provide algorithms for generalizing users' reading, for example, by enabling analysts to go through available texts and highlight sections that consist of the same or similar character sequences just identified, making analysts aware of the coding choices made previously.
- They make accessible for inspection the analytical distinctions that users have introduced, allowing them to read the texts in novel and analytically meaningful ways, ideally revealing unanticipated relationships that invite further interpretations.

Below, I review the most typical functions that QDA software tends to support selectively. No two systems enable the same functions, and no one system is best. The developers of qualitative text analysis software often compete with one another by inventing new features or making the features offered by other developers more user-friendly or efficient. As the software packages available change rather rapidly, this list is intended mainly to give potential users some idea about what features they may want to look for in a system as well as what features might render a system cumbersome for their purposes.

Text entering. Computer aids vary in the kinds of text formats they can accept, with most accepting text files with ASCII/ANSI characters. Some systems require their users to preedit text—for example, into lines (as in WinMAX Pro and AQUAD) or into paragraphs not exceeding a certain length (NUD*IST and NVivo recognize paragraphs by carriage returns). Others restrict the lengths of texts to a certain number of words, which may constrain an analysis. In addition, some systems store text files externally (e.g., HyperRESEARCH), whereas others process them after conversion into internal representations (e.g., AQUAD, ATLAS.ti, and NVivo). The ability to index raw text, a feature of most information retrieval software—ZyINDEX and dtSearch, for example—speeds up text searches but prevents text alterations; programs with this feature do not allow users to edit, correct typos, spell out abbreviations, eliminate headers, or create memos as they go on, all of which may be useful to analysts who are reading

texts in depth. ATLAS.ti offers users the option of editing original text while the program is operating in read-only mode. Systems that operate on internal representations may have to reimport edited texts, which means users lose any work already done on those texts.

Display. All QDA software allows users to scroll through original texts on their computer screens, which enables selective and nonsequential reading of any portion of the texts. Some allow users to add permanent highlighting to relevant sections, whereas others facilitate the cutting and pasting of textual units into special files. Analysts often find frequency tabulations, KWIC listings, and concordances (see section 11.3) useful to have, and some systems provide these. Software packages vary in text window sizes and in the ways they allow users to manipulate texts. There are also considerable differences among programs in how they display the connections between original textual matter and the terms in which it is analyzed and in how they keep track of the structure of assigned categories or codes.

Manual coding. Virtually all QDA software enables users to manually assign sections of text to categories or codes. Categories or codes are not predefined, so users can invent them on the fly. Some systems impose restrictions on the kinds of textual units that can be categorized or coded. WinMAX Pro and AQUAD, for example, ask users to code lines of text one by one. In more recently developed systems, such as ATLAS.ti and NVivo, users can highlight any section of text and assign it to one or more categories. The coding of noncontiguous units of text poses problems for most systems.

Automatic coding. Older QDA software requires users to read through whole texts to identify all relevant units on the screen and attach appropriate codes. Some newer systems, such as ATLAS.ti and NVivo, support automatic coding options. After users have created categories for particular character strings, they may either use the coded character strings as queries for coding all textual units with identical textual attributes or define coding categories in terms of queries that will find relevant units without the users' doing any further reading. NUD*IST and NVivo feature automatic categorization, which works similarly to a dictionary. As automatic coding can fail the semantic validity criterion, analysts should look for automatic coding options that are supplemented by features that allow users to examine the actual text segments that end up in one category and to undo inappropriate assignments.

Hierarchical/nonhierarchical categorizing. When content analysts assign codes to any relevant section of text without restrictions, nonhierarchical representations of the original text may result. This aids the retrieval and examination of textual units with the same attributes but makes classical Cartesian analyses—cross-tabulations of textual attributes, for example—difficult if not meaningless. Analysts can prevent this difficulty by using units that are mutually exclusive and by assigning each unit to one code for each variable of classification. Most QDA software packages do not require this restriction and thus limit the analytical possibilities of the emerging categorizations. NVivo, among others, allows users to categorize codes, assigning second-level codes to them (rather than to the original text), which gives the impression of a hierarchical category scheme. This hierarchy is not of mutually exclusive categories unless users restrict themselves to applying mutually exclusive categories. Nonhierarchical categories give rise to interesting measures of the strength of the categories'

relationships—for example, in terms of the amount of text they jointly represent. ATLAS.ti provides users with easily intelligible diagrams of these relationships. Systems also differ greatly in the ways they allow users to keep track of the codes used in analyses.

Memoing, commenting, and linking. An attractive feature of many QDA software packages is that they enable users to associate notes, memos, or other information with any segment of original text. Users can thus augment texts with accounts that are not explicit in the texts but are known to the analysts; these might include recollections that add to anthropological field notes, definitions of terms, names of speakers, information about nonverbal features available from video recordings or images, or the names of persons referred to by pronouns. ATLAS.ti, NUD*IST, NVivo, and WordStat include such options. Some systems allow users to add hyperlinks between files, which can accomplish similar text enrichments. It is important that any comments that users enter to become part of the text being analyzed can be easily distinguished from comments the users intend to remain private reminders of their thoughts while they were reading the text, and several text entry systems make this possible, including ESA, KEDS, CETA, and Carley's MECA.

Interoperability. Portability of CATA output across systems from different developers is rare. TextSmart and Verbastat are both SPSS products and so, naturally, they are compatible. LIWC output feeds into SPSS as well, and the LIWC dictionary can be used in WordStat and TextQuest. Files created using CATA software for qualitative data analysis are rarely shareable, even among coworkers on the same project. The coding that emerges in interaction between one analyst and given texts stays essentially within the system, reflecting the conceptual repertoire of that one analyst. Alexa and Züll (1999) conclude their survey of CATA software—including QDA software—by saying, “There is no sufficient support for either exploiting already existing data, validating the analysis performed by other researchers or performing secondary analysis using the same text material but with a different text analysis package or with other software” (p. 144).

Despite the extremely valuable procedural and visual features the software packages described in this section offer for conceptualizing and analyzing medium-size texts, these systems also have two drawbacks that software makers tend to downplay or ignore. The first is that researchers using such software are unable to share among themselves the coding/conceptual schemes that emerge during their interactive explorations or to apply those schemes to other texts or different situations. This inability effectively prevents researchers from testing the reliability of categorizations, which is an important precursor to assuring the validity of research findings (see Chapter 12, section 12.1). This is not a limitation of computing; rather, it reflects the premise of a research tradition that takes an individual analyst's understanding as the criterion for accepting analytical results, an understanding that is generalized to that analyst's scholarly community without evidence. Software packages that do not support the sharing of the systems of categories developed within them, coding schemes, and other intermediate results for reproduction elsewhere can easily become uncontested tools for privileged experts.

To qualify the above worry, I must note that reliability checks for QDA software are not inconceivable. A research project could well instruct several analysts to apply a given set of codes to the same text. Reliability calculations for unitization are described in Chapter 12, sections 12.4 and 12.5. For multiple interpretations of identified units, see Krippendorff (1992). Hermetically closed QDA software precludes this possibility, however.

The second drawback of these computer aids is a product of the theories of meaning that are built into them. Coding textual units into categories, even if intended as a convenient tool for handling large texts and for talking about them in fewer categories, amounts to abstracting or redescribing a text in simpler, more relevant, and perhaps more general terms. This advantage is gained at the cost of buying into the classical representational theory of meaning. There are obviously other ways of reading texts and other ways of supporting the analysis of texts, some of which are reviewed above, such as computing association structures, developing semantic networks, and answering research questions by following the entailments of texts. One might add to these the rather different aims of critical scholarship, such as suggesting alternative readings or conceiving of less oppressive ways of interpretation. All computer aids carry the danger of blinding their users to alternative theories of meaning and epistemologies. To encourage analysts to retain their awareness of these alternatives, we conceptualize content analysis relative to a chosen context—any social reality, conceptual framework, or period in time worthy of analysts' explication. Such a conceptualization should apply to CATA software as well.

11.6 FRONTIERS

Technological breakthroughs are hard to predict, but it is not too difficult to outline some exciting possibilities for the extension of computer-aided content analysis. Below, I briefly discuss five particular areas that are in need of further attention.

11.6.1 Intelligent Browsers

Although there have been remarkable advances in the scope of text search engines, users of existing browsers overwhelmingly experience the shallowness of their results. This is true whether users are surfing the Web, consulting full-text databases online, or using search routines within current CATA software. The Internet is full of expectations that more powerful browsers will soon be available, and CATA software developers tend to claim that they offer all kinds of features: “concept searches,” “content extractors,” “meaning finders,” “theory developers,” and many more. Unfortunately, the operationalizations of these concepts, connoting essentially human abilities, often fall far short of what software descriptions promise.

If browsers aim to find scarce information in large textual universes and content analysts hope to answer intelligent questions from available texts, the goals of both are quite compatible, if not identical. In fact, content analysis could dissolve itself into intelligent browsers if browsers could learn to model their searches by what content analysts actually do with texts: search for or sample “truly” relevant texts, interpret these by means of one or more theories of the chosen contexts of these texts, and provide textual evidence for answers to the research questions that motivate the search. Such intelligent browsers may never substitute experts' readings of textual matter or replace well-trained content analysis designers, but they certainly could be modeled after the research strategies of content analysis and could apply these to large textual universes whose volumes generally exceed human comprehension.

11.6.2 Common Platforms

On the fertile ground of naturally occurring texts, the species of CATA software, including weeds, grow wild. Most systems work essentially on their own, competing

for potential users' time and willingness to learn them. This competition is fueled by software companies' often-exaggerated claims that their products will relieve users of difficult cognitive tasks. Although all CATA software packages do process texts—a few can also handle images and sound—no widely accepted text file standards exist, and there are few ways to feed the results produced by one program into another. The software-specific preparation of raw text that most systems require is costly and discourages the use of alternative approaches. There are no interface standards that would allow software from different developers to work together. There is not even any common vocabulary for comparing various content analysis tools.

There have been a few notable, albeit small, steps toward the development of common platforms. TextSmart is distributed by SPSS and so is integrated in this statistical package. LIWC feeds into SPSS, and its dictionary also works in WordStat and TextQuest, as already mentioned. But these are very limited “collaborations.” A common platform to run various CATA procedures in parallel (comparatively) and in sequence (cooperatively) would enable developers to concentrate on refining a few good tools that can compete for users' attention along the dimensions of mutual compatibility, computational efficiency, analytical power, and user-friendliness. It would give users a box of freely combinable analytical tools with which to design all kinds of text analyses, and it would accelerate analysts' ability to use available texts in ways currently unimagined.

11.6.3 Computational Theories of Meaning

Far too often, content analyses are conceptualized—naively, one might add—in terms of traditional behavioral science methods for analyzing nontextual data: as a measuring tool for generating data amenable to inferential statistics, hypothesis testing, and modeling of causes and effects. Although the content analysis literature recognizes that analyzing meaningful text differs quite radically from analyzing observational data, computational aids, which are so much simpler when formulated as context-insensitive procedures, tend to follow in the footsteps of behaviorist assumptions. This limits CATA to very superficial meanings. CATA is in dire need of computational theories of meaning that extend theorizing about the individual cognitive ability of a typical reader to whole communities' diverse uses of texts, embracing the public processes that make particular text attributes significant as well as the social processes in which institutions reside. Computational linguistics, with its current concern for parsing sentences and disambiguating words and phrases, has made only marginal contributions to computational theories of meaning, largely because it theorizes what is general to language, not what is specific to particular nonlinguistic contexts. Content analysis, in contrast, needs to answer specific questions that have not previously been answered about extratextual phenomena of analysts' concerns. Abstractions from what a text means generally may well be a start, but they are not a useful end. Computer dictionaries often end up claiming unwarranted generalizations tied to single words, one word at a time.

Content analysis advances with the availability of rather detailed theories that, to fuel CATA development, can be converted into analytical constructs and tailored to fit specific analytical contexts. These theories may learn from ideas developed in computer science or from efforts to shed light on psychological, sociological, or political problems. They do not need to be—and in fact are better when they are not—grand theories, as they have to cope with language use “on the ground,” with processes of microevolution, as Best has suggested (see section 11.4.4). These theories

may have to be conceived as learning, adapting themselves to the context they are to model. Theories of meaning that are currently inscribed in CATA—categorizing words, mining patterns of textual co-occurrences, tracing the entailments of semantic networks, and extracting memes from textual interactions—barely describe the contours of what needs to be developed.

11.6.4 Utilization of Intertextualities

Many content analysts are content to summarize finely unitized text and measure the textual attributes that permeate these texts (Krippendorff, 1969a). Reducing a text to a large set of independently analyzed textual units robs it of its narrative qualities, of what makes it coherent, informative, compelling, and predictive. *Coherence* refers to how well parts of text—not just words or phrases—hang together or support each other in telling a larger story. For instance, most narratives start by laying a foundation, then develop a point, and end with a conclusion or coda. Their beginnings are necessarily different from their ends. Histories find coherence in the chronological order of their narrative structure, which in turn constructs the histories we come to know. Political texts might have to be analyzed as parallel constructions that acknowledge each other where their media meet. Conversations are constituted in participants' responses to previous responses. Scholarly discourse hangs together as a network of acknowledgments and citations of prior scholarly work. The abundantly obvious point of these examples is that texts are more than collections of words, phrases, paragraphs, and documents. An important CATA frontier is the development of ways to account for relationships between larger textual units—parallelisms, beginnings and ends—so that they can be treated as dependent on each other, as progressing, not as unordered entities permeating a text.

Small steps in this direction can be observed in existing systems for analyzing question-answer pairs from interview data. TextSmart and Verbatat specialize in this probably simplest coherence. They analyze questions and answers as two distinct but structurally related bodies of text, which they link largely by cross-tabulations. Code-A-Text is purportedly designed to analyze dialogue, acknowledging the interactive qualities of texts. A few years back, a system called the Coordinator came into vogue (Winograd & Flores, 1986, pp. 157–162). It did not analyze but used speech act theory to connect electronic communications to the networks of commitments that underlie social organizations. Garfield (1979) pioneered citation indexing, a system that traces chains of bibliographic citations, thus facilitating the exploration of connected bodies of scholarly discourse. Forensic agencies need to connect crime reports, media coverage of crime, telephone conversations, and other elements to the specific actions of individuals. Good scholarly work entails connecting literature from diverse sources, experiments, interviews, statistics, and theory. All of these are texts whose value lies in their connections, in their responding to each other, in their parallelisms. CATA software would advance greatly if it could explicate some of these intertextual coherences and use them to draw powerful inferences from multiple kinds of texts.

11.6.5 Analyzing Networked Text

Traditional text is linear. Narratives have a beginning and an end. Political speeches have a duration. Newspapers are printed regularly. With interest in greater volume of textual streams, content analysis managed to unitize such continua and offered quantitative accounts for comparisons across different textual continua,

generated on different occasions, communicated by different media, and over time. The first media innovation that challenged the meaningfulness of the unitizations of such textual linearities was the use of hypertext. Hypertext, the ability to refer inside one text to another, gave readers considerable control over the process of making sense in what they were exploring. Readers were able to explore texts according to their own paths, and analysts could not be sure what was read, skipped, or come back to.

Social media present an even greater challenge to content analysts. For example, Twitter messages are short. Their maximum length, 140 characters, now doubled, does not allow to say much. However, the context in which social media messages make sense resides in the interactions they initiate, what they respond to, how they multiply by being shared among many, and the social realities they set in motion. Here, relatively short character sequences are not merely intertextually meaningful, they are repeatedly reauthorized, acted upon, added to, and passed on to people known mainly as participants in creating textual networks with considerable social, political, and individual consequences.

As discussed above, Niederer (2016) analyzed how websites on climate change were linked. Their texts did not stand alone. She used a webcrawler to find their connections. Iina Hellsten and Loet Leydesdorff (2017) combined computational methods for analyzing semantic and actor-topic networks to reveal how Twitter messages built on and referred to each other. Both works begin to recognize texts as networked, not interpretable individually. On a conceptual level, this is not entirely new. Mikhail Bakhtin (Todorov, 1984) conceived of literature as articulations of the voices of their preceding contributors. François Cooren (Cooren, 2010) went so far as to describe individuals as being ventriloquized and ventriloquizing, providing the links between texts. Citation network analysis is an early effort to tap into the network qualities of academic publications. Theories aside, users of social media provide the contexts for large and volatile networks of texts. They challenge content analysts to develop ways to explore them as such with the aim of inferring the consequences of their use.

11.6.6 Natural Interfaces

Users can expect advances in the user-friendliness of CATA software in three areas:

- Ease of operating the software, in navigating an analysis through its analytical options
- Ease of handling text in various forms, including preediting
- Ease of tracing the information of texts through their analytical transformations to the answers to given research questions

CATA software packages are increasingly employing Windows-like formats. Using a mouse to point to, click on, and move text has become as conventional as making selections on menu-driven interfaces. While there are vast differences in user-friendliness among different text analysis programs, I will not discuss them here except to say that learning any system always amounts to a big investment of time, and users should be sure that the systems they choose are worth their efforts to learn how to use them.

Windows-like software is built on a metaphor of texts as movable objects, not as something that needs to be read, rearticulated, compared, questioned, triangulated, and

interpreted. QDA software, which supports interactive-hermeneutic explorations of texts, is currently leading the development of new interface metaphors. Unfortunately, QDA has not grown much beyond providing text retrieval, highlighting, coding, and summarizing aids to categorize linearly interpreted texts. Finding and implementing natural interface metaphors for users to understand and handle large bodies of text, especially networked texts, constitutes a major challenge. This is especially so when one realizes, as software developers undoubtedly do, that most texts arise out of and reenter conversations, that literacy is a social phenomenon, evident in interactions, and not reducible to individual understanding. Even content analyses tend to be developed collaboratively, relying on ethnographic researchers, linguists, coders, and academic stakeholders who may need to be listened to as texts and the computational accounts they support are interpreted. The absence of natural interface metaphors for what content analysts need to do is a major impediment to CATA use.

Users' ability to conceptualize and act on what CATA software does is the most important yet least attended aspect of user-friendliness. I have already noted the tendency of CATA software developers to use fancy labels for describing the capabilities of their software. Instead of using marketable descriptions that shortcut conceptualization, developers should provide, and users should demand, transparency of the algorithms that available software follows. As I have already suggested, CATA software can be conceptualized as a series of transformations that maintain the relevant information in reading of the original text relative to a chosen context. It would therefore be reasonable to expect that users be enabled to critically examine what happens to the initial reading and recording of text, whether the theories accounting for their analytical transformations make sense, need to be modified until they do, and satisfy at least face validity. Word frequencies are obvious examples. Analysts can interpret them only if they are able to read the tabulated words without the contexts from which they were taken. However, as soon as computational procedures are employed and decontextualization, stemming, and lemmatization take their toll on perfectly readable text, it is important that analysts are provided access to trace what happened to the words that end up informing the research questions. For example, KWIC lists of words can provide the textual context and a sense of what coded categories originally meant. Many developers of CATA software packages promise results without revealing what their algorithms actually do. To claim that a software extracts meanings from texts is most likely false. Without algorithmic transparency on the part of software developers and algorithmic literacy on the part of software users, the latter are at the mercy of often deceptive promises by software distributors. They are asked to operate by faith. Deception starts with misconceptions. For example, if a search engine retrieves texts containing character sequences that match the words of a user's query, they do not retrieve information. Whether a retrieved text informs a researcher's questions is secondary to what it does. It should not be impossible for CATA software developers to provide user-friendly ways for examining what their software does by enabling users to trace the computational steps that transform a sample of the original text to its contribution to the results—much as computer programmers can trace what happens to data when trying to find a coding error to be debugged. Transparency of CATA software not only supports user conceptualizations of what a program does with a given text, it is also a requirement for establishing semantic validity (see section 13.2.2).

12

RELIABILITY

This chapter begins by discussing the purpose of reliability in scientific research. It distinguishes among three designs for generating data to measure reliability, which leads to three kinds of reliability: stability, replicability, and accuracy or surrogacy. All of them are functions of the agreement achieved among observers, coders, judges, or measuring instruments. The agreement coefficient alpha, known in the literature as Krippendorff's α , is presented as a tool to assess such agreements, and its computation is demonstrated, starting with the simplest kind of data and moving to embrace the most common reliability analyses of nominal data, to several metrics, any number of observers, and incomplete data. Four measures of the often-ignored reliability of unitizing are introduced here, as is the reliability of text mining; multi-valued coding of data; and measures for the accuracy, surrogacy, and decisiveness of majorities. The chapter concludes with a discussion of statistical issues, adequate sample sizes, alpha's distribution, and reliability standards. It also includes a discussion of contested alternatives to α and others it subsumes.

12.1 WHY RELIABILITY?

Data, by definition, are the trusted ground for reasoning, discussion, or calculation. To stand on indisputable ground, content analysts must be confident that their data (a) have been generated with all conceivable precautions in place against known pollutants, distortions, and biases, intentional or accidental, and (b) have the same meaning for everyone who uses them.

Reliability offers an empirical ground of the confidence that researchers need to use data. There are two ways of operationalizing this confidence. In Kaplan and Goldsen's (1965) words: "The importance of reliability rests on the assurance it provides that data are obtained independent of the measuring event, instrument or person. Reliable data, by definition, are data that remain constant throughout variations in the measuring process" (pp. 83–84). Accordingly, a research procedure is reliable when it responds to the same phenomena in the same way regardless of the circumstances of its implementation. This is the measurement theory conception of reliability.

The other operationalization acknowledges that many phenomena of analytical interest are transitory. Human voices, historical events, radio transmissions, and even physical experiments disappear right after they have been observed and need

to be recorded to be subsequently analyzable. The analyst's ability to examine these phenomena in their absence, compare them with other concerns, and, particularly, discuss them with members of a community of stakeholders relies heavily on a consensual reading and use of the data that represents, points to, or invokes experiences with the phenomena of interest. Empirical inquiries into bygone phenomena have no choice other than to presume that the data they left behind can be trusted to mean the same to all of their users. In content analysis, this means that the reading of textual data as well as of the research results must be replicable elsewhere, which would demonstrate that researchers agree on what they are talking about. Here, then, reliability is the degree to which members of a designated community concur on the readings, interpretations, responses to, or uses of given texts or data. This is an interpretivist conception of reliability.

In either case, researchers need to demonstrate the trustworthiness of their data by measuring their reliability. If the results of reliability testing are compelling, researchers may proceed with the analysis of their data. If not, doubts prevail as to what these data mean, and their analysis is hard to justify.

To perform *reliability tests*, analysts require data in addition to the data whose reliability is in question. These are called *reliability data*, and analysts obtain them by duplicating their research efforts under various conditions—for example, by using several researchers with diverse personalities, by working in differing environments, or by relying on different but functionally equal measuring devices. Reliability is indicated by substantial agreement of results among these duplications.

In contrast to reliability, *validity* concerns truths. Researchers cannot ascertain validity through duplications. *Validity tests* support the claims resulting from a research effort by evidence obtained independent of that effort. Thus, whereas reliability provides assurances that particular research results can be duplicated, that no (or only a negligible) amount of extraneous “noise” has entered the process and polluted the data or perturbed the research results, validity provides assurances that the claims emerging from the research are borne out in fact. Reliability is not concerned with the world outside of the research process. All it can do is assure researchers that their procedures can be trusted to have responded to real phenomena, without claiming knowledge of what these phenomena “really” are.

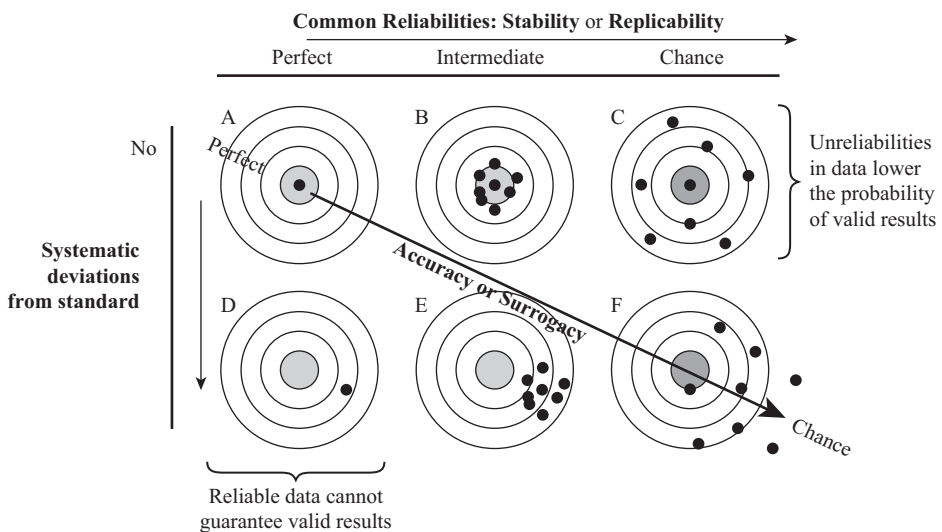
In content analysis, reliability and validity can be related by two propositions and a conjecture:

- *Unreliability limits the chance of validity.* In everyday life, disagreements among eyewitness accounts make it difficult for third parties to know what actually happened or whether the witnesses are reporting on the same event. For such accounts to be considered reliable, witnesses must concur well above chance. If the coding of textual matter is the product of chance, it may well include a valid account of what was observed or read, but researchers would not be able to identify that account to a degree better than chance. Thus, the more unreliable a procedure, the less likely it is to result in data that lead to valid conclusions.
- *Reliability does not guarantee validity.* Two observers of the same event who hold the same conceptual system, prejudice, or interest may well agree on what they see but still be objectively wrong. Content analysts are not

exempt from such concurrences. Because they have acquired a language and concepts that make them see the world from the unique perspective of their academic discipline, their observations and readings are grounded in a consensus that is not likely shared by many people outside of their scholarly community. Content analysts' shared worldview may deviate radically from the worldviews of those whose intentions, perceptions, and actions are at issue and could validate the intended inferences. A highly reliable research process may well be artificial and thus have little chance of being substantiated by evidence on the intentions, perceptions, actions, or events that were inferred. Even perfectly dependable mechanical instruments, such as computers, can be wrong—reliably. Thus, a reliable process may or may not lead to valid outcomes.

This relationship is illustrated in Figure 12.1, which depicts reliability as repeating the same score and validity as being on-target. The top part of the figure suggests that with diminishing reliability, validity increasingly becomes a chance event. The bottom part suggests that reliability does not guarantee being on-target.

FIGURE 12.1 ■ The Relationship Between Reliability and Validity



Thus, reliability is a necessary, but not a sufficient, condition for validity. The following conjecture does not have the logical force of the two preceding propositions, but it is born out of the experiences by numerous content analysts:

- *In the pursuit of high reliability, validity tends to get lost.* This statement describes the analyst's common dilemma of having to choose between interesting but difficult to replicate interpretations that knowledgeable scholars offer each other in conversations and oversimplified or superficial

but reliable text analyses generated, for example, by computers or highly trained and carefully drilled human coders. Merritt’s (1966) study of the rising national consciousness among the 13 original American colonies on the basis of newspaper accounts provides an example of a case in which complexity of interpretation was sacrificed for reliability. Because national sentiments are difficult to define and identify, Merritt elected to enumerate the mentions of American place-names instead. A shift in the use of the names of places in colonial England to the names of places in America may well be an attractive index, and counting words instead of themes causes fewer reliability problems; however, the use of place-names surely is only one manifestation of “national consciousness,” and a richer account of this phenomenon could well have led to more interesting inferences. Merritt’s index is attractive, as I have suggested, but its validity remains thin.

The use of computers in content analysis, praised for being perfectly reliable, has highlighted this dilemma even more clearly. Computers process character strings, not meanings. They sort volumes of words without making sense of them. Although it is possible to program computers to perform amazing functions, when analysts rely on them rather than on literary competent readers they run the risk of trivializing the meanings of texts (see Chapter 11, section 11.1). In content analysis, researchers should approach highly reliable procedures with as much caution as they approach fascinating interpretations that nobody can replicate.

12.2 RELIABILITY DESIGNS

12.2.1 Types of Agreement; Types of Reliability

I distinguish three types of reliability: stability, replicability, and accuracy or surrogacy (see Table 12.1). They are distinguished only in part by how agreement is measured but also by the way the reliability data are obtained. Without information about the circumstances under which the data for reliability assessments have been generated, agreement measures remain uninterpretable.

TABLE 12.1 ■ Types of Reliability			
	Agreement	Responds negatively to	Strength
Stability	(test–retest) Repetitions	Intra-observer inconsistencies	Weakest
Replicability	(test–test) Replications	Inter-observer disagreements + intra-observer inconsistencies	Intermediate but easily assessed
Accuracy or Surrogacy	(test–standard) Correspondences	Deviations from chosen standard + inter-observer disagreements + intra-observer inconsistencies	Strongest but standards are not always available

Stability is the degree to which a process is unchanging over time. It is measured as the extent to which a measuring or coding procedure yields the same results on

repeated trials. The data for such assessments are created under *test-retest* conditions; that is, one observer rereads, recodes, or reanalyzes the same text, usually after some time has elapsed, or the same measuring device is repeatedly applied to one set of objects. Under test-retest conditions, unreliability is manifest in variations in the performance of an observer or measuring device over time. With reference to humans, such variations, also called *intra-observer disagreement* or *individual inconsistencies*, may be due to insecurity, carelessness, openness to distractions, difficulties in comprehending written instructions, or the tendency to relax performance standards when tired. Even such admirable human characteristics of learning through practice, of improving one's efficiencies over time, developing expectations, and creating shortcuts can show up in increased instabilities over time.

Stability is the weakest reliability measure in content analysis. It rewards individually stable idiosyncrasies, strong prejudices, solid ideological commitments, closed-mindedness, or stubborn misinterpretations of given coding instructions and texts, high agreement measures that have little to do with the reliability of data generated by such individuals. Obviously, measures of stability are insufficient as the sole criterion for accepting data as reliable. But because test-retest data are the cheapest reliability data to generate, and internal inconsistencies reduce the stronger measures of reliability as well, repeating one's coding may be an analyst's first step in evaluating whether to give coding instructions to others.

Replicability is a measure of the degree to which a process can be reproduced by different analysts, working under varying conditions, at different locations, or using different but functionally equivalent measuring instruments. Demonstrating replicability requires reliability data that are obtained under *test-test* conditions; for example, two or more individuals, working independently of each other, apply the same recording instructions to the same units of analysis. Disagreements between these observers' performances are due to both intra-observer inconsistencies and *inter-observer differences* in the interpretation and application of given recording instructions. Compared with stability, replicability, which is also called *inter-coder reliability*, *intersubjective agreement*, and *parallel-forms reliability*, is a far stronger measure of reliability than stability.

Accuracy is the degree to which the generated data conform to given specifications. The process yields what it is designed to yield. To establish accuracy, analysts must be in the possession of an accepted standard. We say that one or more data-making procedures produce correct results when they match the given standard. Under these *test-standard* conditions, observed disagreements between a group of observers and the standard are due to intra-observer inconsistencies, inter-observer differences, and *deviations from that given standard*. Because accuracy responds to all three sources of extraneous variation, accuracy is the strongest reliability test available. It is surpassed in strength only by validity measures that conform to the test-standard design but takes truths as the standard, or at least what is known to be true—a requirement that lies outside content analysis as a method of inferring what is unknown at the time of the analysis (discussed in Chapter 2 and taken up again in Chapter 13).

When data making is merely clerical or computational, the meaning of accuracy is clear. Typos, for instance, are errors by comparison to existing spelling standards. In linguistics, conversation analysis, and therapeutic contexts, analysts perform accuracy checks by comparing novices' uses of transcription conventions,

for example, with those of acknowledged experts. In content analysis, accuracy measurements often include testing the work of coders in training against standards that have been established by panels of experienced content analysts. In the more problematic parts of content analysis, such as the interpretation and transcription of complex textual matter, suitable accuracy standards are not easy to find. Because interpretations can be compared only with other interpretations, attempts to measure accuracy presuppose the privileging of some interpretations over others, and this puts any claims regarding precision or accuracy on epistemologically shaky grounds. Thus, the use of accuracy is limited to coder training and other areas where objective standards are readily available.

Surrogacy and accuracy is measured in the same manner, however, the direction of the interpretation of surrogacy is complementary to that of accuracy. Whereas accuracy establishes the extent to which observers' categorizations or valuation conform to a chosen standard, surrogacy establishes whether an instrument for evaluating units of analysis can serve as the surrogate for the valuation by observers. This would be a question to be answered when searching for an algorithm to identify relevant elements in texts in the manner humans would be able to. Before such an algorithm can be applied, the extent to which it can substitute for human judgments needs to be shown. In that case, the multiplicity of observers serves as the standard for deciding whether a mechanical substitute can substitute for the abilities of human observers to read and interpret texts. Surrogacy is not limited to evaluating mechanical devices against human judgments. When it comes to decide which one of a set of human observers should be trusted to generate data that a group of observers rated individually or in aggregate, surrogacy is the issue. This is of particular interest when crowdsourcing and crowdcoding on the Internet. Reliability issues for this situation are discussed in sections 7.4 and 12.4.

Some scholars have mentioned the split-half technique as leading to an assessment of the reliability of data. This technique would call on content analysts to divide a sample of recording units into two approximately equal parts and have the two parts coded by different observers, one unit at a time. The analysts would then compare the frequency distributions obtained for the two parts. If the difference between the two distributions is statistically insignificant, the data would be considered reliable; otherwise, they would not be considered trustworthy. However, this measure merely assesses the degree to which two subsamples resemble each other or whether the joint data set can be considered homogeneous. (The test can also be used to determine whether a sample is large enough to represent a population; see Chapter 6, section 6.3.3.). However, as there may be good reasons why two subsamples are different, or no reasons why they should be the same, homogeneity, an aggregate measure, says nothing about whether data can be trusted in their details. In content analysis—or, more generally, when the reliability of categorizing, coding, or describing units is at issue—the split-half technique is not a suitable reliability test, and its use must be discouraged as uninformative and misleading.

I should mention a prevalent confusion of the reliability of data with the difficulties that observers experience when categorizing or valuing phenomena of analytical interest. I will not cite several misleading efforts to measure the difficulties that coders experience when facing complex coding tasks and calling it reliability. It is an empirical fact that complex coding tasks more likely lead to unreliable data than ordinary ones. The experience of coding difficulties challenges the designers of coding instructions to be less ambiguous, give more details, use better examples, and

improve ways of recoding the perceptions and readings of observers so that a greater number of them can be hired to do exactly the same thing. However, reliability measures need to assure researchers whether they can trust that the data, generated by whatever means, are about the phenomena they intend to explore. The nature of the instrument for generating reliable data—human beings, mechanical instruments, or independent research efforts—is instrumental in generating data and the form in which the phenomena of interest are manifest plays a role, but they have to be considered accidental to the reliability of the data being generated.

12.2.2 Conditions for Generating Reliability Data

By its definition (section 2.1), content analysis must be *replicable*, at least in principle. To check on this possibility, analysts must generate reliability data at least under test-test conditions and account not only for individual instabilities but also for disagreements among observers, coders, or analysts. Any effort to assess the reliability of data based on measuring the agreement among observers or instruments must meet the following requirements:

- It must employ communicable instructions—that is, an exhaustively formulated, clear, and workable data language plus step-by-step instructions on how to use it. This widely accepted requirement may need to be extended to include detailed accounts of the training programs that coders typically undergo before qualifying for the task. Without knowledge of what coders learned, the nature of the consensus achieved, one may not know what the data they generate mean and how to replicate them.
- It must employ communicable criteria for the selection of individual observers, coders, judges, or analysts from a population of equally capable individuals who are potentially available for training, instruction, and replicating the data-making process elsewhere.
- It must ensure that the observers who generate the reliability data work independently of each other. Only if such independence is assured can covert consensus be ruled out and the observed agreement be explained in terms of the given recording instructions and the phenomena observed or the texts interpreted.

Inasmuch as reliability serves as a condition for research to proceed with the data in hand, the content analysis literature is full of evidence of researchers' well-intended but often misguided attempts to manipulate the process of data generation so as to increase the appearance of high levels of agreement. Most of these involve violations of one or more of the above conditions, as the following examples may illustrate.

In the belief that consensus is better than individual judgments, some researchers have asked observers to discuss what they read or see and reach their decisions by compromise or majority vote. This practice may indeed reduce the detrimental effects of individual idiosyncrasies and take advantage of the possibility that two observers may notice more than one, but data generated in this way neither ensure replicability nor reveal its extent. In groups like these, observers are known to negotiate and to yield to each other in tit-for-tat exchanges, with prestigious group members dominating the outcome. Here, observing, identifying units of analysis, and coding come to reflect

the social structure of the group, which is nearly impossible to communicate to other researchers and replicate elsewhere. Moreover, subjective feelings of accomplishment notwithstanding, the data that are generated by consensus afford no reliability test. They are akin to data generated by a single observer. Replicability requires at least two independent observers or methods of generating data from the same set of phenomena. To substantiate the contention that consensus is superior to individual data making, a researcher would have to compare the data generated by at least two groups of at least two individuals, each group working independently of the other.

It is not uncommon for researchers to ask observers to work separately but to consult each other whenever unanticipated problems arise. Such consultation is a response to a common problem: The writers of the coding instructions may not have been able to anticipate all possible ways relevant matter is expressed, leaving coders at a loss. Ideally, coding instructions should include rules for coding or measuring every possible instance. However, the very act of discussing emerging problems creates interpretations of, if not additions to, the existing instructions that are typical of the group and not communicable to those not present in these discussions. In addition, as the written instructions become reinterpreted, the process loses some of its stability over time: Data generated early in the process use instructions that differ from those that evolve. The higher agreement achieved in the end is partly illusory of the actual reliability needed.

Because content analysts may not be able to anticipate all of the possible complications in their texts, it is a common practice to expand the written unitizing and coding instructions by adopting new and written rules as the process unfolds. The idea is that instructions evolve in practical use, eventually requiring no new rules. To avoid being misled by unwritten consensus among coders engaged in the dual task of interpreting text and expanding the common instructions to do so, content analysts should put the final instructions to a reliability test, using different coders, and reexamine the data generated before these final instructions have been reached.

Content analysts are sometimes tempted to assume, and act on the assumption, that data making is served best by experts, exceptionally acute observers, or individuals who have long histories of involvement with the subject of the research. They should be reminded, however, that replicability means the potential of reproducing the data-making process elsewhere with individuals of specified qualifications; data-making instructions that unambiguously define the units of analysis, categories, scales, and descriptive devices used in the process; and the set of phenomena to be recorded in a reliability test. If there are no other experts against whom the performance of the available expert can be checked, that expert's interpretations may be insightful and fascinating, but the analyst cannot claim that they are reliable. This is the basis of arguments against a content analyst doing his or her own coding (as a principal investigator may sometimes do, for instance) unless the analyst's performance is compared with that of at least one other person. To satisfy this requirement, a content analyst may be tempted to find coders among friends, colleagues, or close associates whom the analyst has reasons to trust to do a good job and think like the analyst, and, indeed, they may well agree—not, however, because they carefully follow the written instructions but because they know each other and the purpose of the research. The background of their agreement is not likely communicable and cannot be found elsewhere. Analysts are required to choose observers from a population of potential observers with specified backgrounds and competencies from which other researchers, intent to replicate an analysis, can select as well.

Sometimes content analysts accept as data only those units of analysis on which observers achieve perfect agreement and discard those on which agreement could not be observed. This is a particularly problematic practice, because it gives researchers the illusion of perfect reliability without affording them the possibility of separating agreement due to chance from agreement based on the sameness of reading or observation. For binary or dichotomous data, agreement by chance is at least 50%. Omitting units on which coders happen to disagree cannot change the chance nature of those on which they do agree. There is no escape from true chance events. This is true also when agreement is well above chance but not perfect. Units that are coded by chance populate both agreement and disagreement cells of a coincidence table, and in the agreement cells there is no way of separating units according to whether observers agreed by chance or by following the instructions. Most important, when analysts rely on data that are easily coded, the resulting data increasingly become an artifact of the analytical procedure, no longer representative of the phenomena the researchers hope to record and analyze.

Content analysts who employ the following two-step procedure can achieve both data whose reliability is measurable and an improvement in their confidence in the data beyond the measured reliability. First, they have to employ three or more observers working independently of one another. This yields reliability data whose reliability can be measured. Second, they reconcile discrepancies in these data either by relying on a formal decision rule—majority judgments or average scores—or by reaching *ex post facto* consensus by deliberation. The data *before* such reconciliation efforts result in proper reliability data and do yield publishable reliability coefficients. Although it is reasonable to assume that *ex post facto* reconciliation improves the reliability of the data beyond the computed reliability, this remains an assumption without measurable evidence. The only publishable reliability is the one measured before the reconciliation of disagreements. The reliability of the data after this reconciliation effort is merely arguable.

12.2.3 Reliability Data

As noted above, the data that enable researchers to assess reliability, called *reliability data*, duplicate the very data-making process whose reliability is in question. Nobody knows for sure what the phenomena of analytical interest are until they are distinguished, categorized, valued, or measured. Reliability data cannot presume truths or knowledge of what is. They can only contain evidence on whether observers concur regarding the distinctions among a diversity of phenomena, be it in identifying suitable units of analysis or describing them in the terms of a chosen data language.

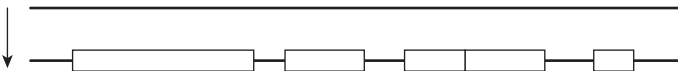
Generating content analysis data involves two processes, requiring different forms of reliability data and agreement calculations, all embraced by the coefficient **alpha** (α).

Unitizing is the process of introducing distinctions among phenomena of interests. Units of analysis are a prerequisite of counting, comparisons, and managing large volumes of textual matter. We distinguish two kinds, by fixed definition or by observers' judgments.

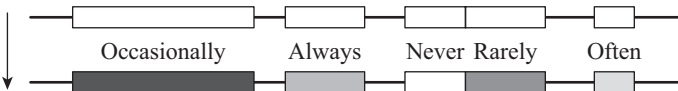
Predefined units are considered natural, created by definition, or by the uniform use of mechanical devices intended to avoid unreliabilities in the process of

unitization. When research is concerned with interviewing individuals, interviewees are predefined, mostly without question. Books, websites, and photographs may be regarded as natural or obvious. Mechanical devices are unquestionably reliable and often used to avoid the difficulty of agreeing on units of analysis. Timed 30-second segments of conversations are predefined by a clock but are far from natural.

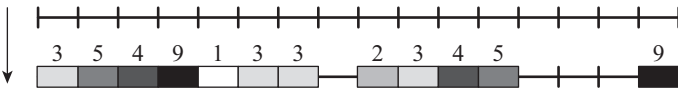
Emergent units result from processes of examining phenomena of interest and distinguishing among them in an effort to preserve phenomena-specific and analysis-relevant unities. Violent episodes on television, turn taking in conversations, and themes of a novel have conceptual boundaries that may not be operationalizable by devices or *a priori* definition. Most social phenomena have temporal extensions and proceed along a continuum, much like a text. Unitization means identifying sections that are relevant to a research question and leaving unattended those that are not. Everyday examples of unitizing include clipping relevant articles from a newspaper, earmarking pertinent sections in a video recording for subsequent analysis, isolating conversational moves within a conversation, and identifying historically significant events in time. Qualitative text analysis software operationalizes the idea of identifying relevant segments of text for subsequent categorization or evaluations. Unitizing renders a continuum analyzable. Unitizing a continuum into unequal segments can be depicted as the following:



Coding is the process of distinguishing identified units by kinds, categories, or weights or assigning values to them. It is the process of mapping a given set of descriptively unknown but distinct and enumerable phenomena into mutually exclusive categories, values on a scale, or quantitative measures of a variable pertaining to research questions presently pursued. Coding adds the information that individual observers recognize as being present in these phenomena, thereby transforming them into categorized or valued units that thereafter can be compared and analyzed. For example,

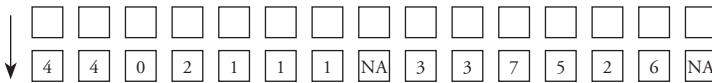


When all units are of uniform lengths, we speak of coding predefined units:



This is the target of the most common reliability measures. However, in any continuum, even predefined units occur in the context of their neighbors, and observers are often asked to take this context into consideration when categorizing or valuing predefined units. Comic strips consist of a meaningful sequence of frames. Words mean little when taken out of context. And much of research in the humanities refers to the contexts of analyzed units.

Not all content analyses are rooted in continua. For example, whole books, journal articles, references, or separately delivered speeches of politicians may be regarded as independent of each other. Independence is also assumed in many social research projects. In elections, voters are considered individual decision makers. In public opinion research, it makes sense to count answers to separate questions. Statistical software that counts observations predefines them as separate units:



Unless individual words are categorized or valued in the contexts in which they occur, content analysts tend to have considerable difficulties to make sense of computed word distributions. This applies also to computing reliability measures. When defined in terms of frequencies, all contexts of the counted units are irretrievably lost, and the calculated reliabilities necessarily omit what might have improved the reliability of content analysis results if larger contexts would have been considered.

Whether coding allows units to emerge or is applied to predefined units, it is important to distinguish between single-valued and multi-valued accounts of phenomena. This choice needs to reflect the nature of the phenomena to be preserved in the data and involves special provisions when reliability is assessed.

Single-valued data are data in which each unit of analysis receives a unique description, one category or value from each variable of a coding instrument.

Multi-valued data allow multiple descriptions or interpretations to be associated with a unit of analysis. Qualitative content analysts have good reasons to prefer multiple interpretations of tests, whether to recognize different perspectives or polysemy. As not too many statistics can handle the latter, much of this chapter addresses the reliability of single-valued data. Section 12.7 shows single-valuedness to be a special and actually rather limited case of multi-valuedness. Another distinction concerns what reliability actually addresses.

The above distinctions give rise to five agreement coefficients, which will be discussed in separate parts of this chapter:

- One α -coefficient applicable to the coding of predefined sets of units by a several trained observers. This most commonly used coefficient, will be developed in detail in section 12.3 as α_c for coding only.
- Three α -coefficients, also applicable for the coding of predefined sets of units, but by many untrained observers and of unequal numbers, such as hired by crowdsourcing. They measure not only the replicability of coding predefined units but also their accuracy or surrogacy and decisiveness of majorities and aggregates, for example by considering whether crowds have the often claimed wisdom—introduced in section 12.4 and named α^s —referring to a standard— α^m —the additional m referring to majorities— α^m — m referring to majorities in the position of a standard.
- Two measures for assessing the reliabilities of text mining and information retrievals α^{IR} . They presume predefined items and are related to measures

of accuracy, defined as retrieving all and only relevant items—discussed in section 12.5.

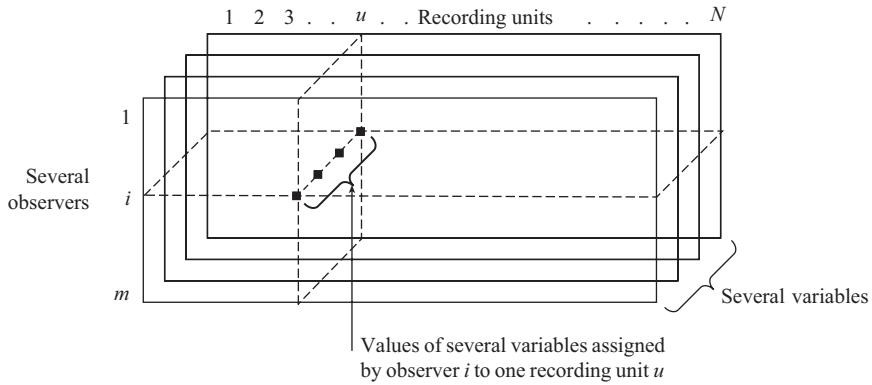
- A family of four α -coefficients for unitizing continua will be introduced in section 12.5. ${}_u\alpha$ — u for unitizing—is applicable to partitions of continua into mutually exclusive segments of different kinds. ${}_l\alpha$ reveals the reliability of distinguishing relevant segments from irrelevant ones. ${}_{cu}\alpha$ is designed to establish the reliability of coding segments of unequal lengths but ignoring the potential unreliabilities of unitizing, and ${}_{(k)u}\alpha$ to evaluate the reliabilities of individual categories or values.
- All reliability measures up to section 12.6 are defined for single-valued coding. section 12.7 introduces the provisions for the multi-valued coding applicable to the coding of predefined units and segments of a continuum of unequal lengths.

A brief comment on terminology: In discussing the generation of reliability data, I use the word *observers* in a general sense. In the literature, they are often called coders but also raters, scorers, transcribers, interpreters, unitizers, annotators, analysts, or judges. Outside of content analysis, they might be referred to as interviewers, annotators, or are acknowledged as experts. But reliability data are not limited to those generated by individual human beings. Text mining is largely considered a computational effort. Creating business accounts, medical records, and court ledgers, for example, can be interpreted as the work of institutionalized observers. And mechanical measuring instruments—which convert physical phenomena into numbers or counts—can be the subject of reliability concerns as well.

In the context of coding, I use *values* as the generic term for names, categories, ranks, scores, scale points, measurements, answers to questions, and the like—even quotes from written text without categorization may be entered in the cells of a spreadsheet and located in the coordinates of rows and columns of reliability data. Values vary along dimensions called *variables*. As discussed in Chapter 8, a variable has mutually exclusive values.

Usually, a set of mutually exclusive categories are constitutive of variables. The reliability measures developed here apply largely to one variable at a time although in section 12.8.4 I address the reliabilities of systems of variables. While the mutual exclusiveness of values is presumed throughout all reliability considerations, the strict logical independence of variables is not presumed, for example by allowing multi-valued coding. For example, a person may be honest, wealthy, good looking, a professional, and so on, all within one multi-valued description, while another person may be a stock broker, tennis player, French, and married. All of these qualities could be said to belong to different variables bundled into what the two persons are.

The general form of reliability data faced by most social science researchers consists of N recoding units u or segments of a continuum of analytical interest, each identified, coded, recorded, or measured by m observers i who assign one or more values of each variable to each unit. This is depicted in Figure 12.2.

FIGURE 12.2 ■ Reliability Data From Coding Several Variables


We measure the reliabilities for these multitudes separately for each variable. Section 12.8.4 addresses standards for accepting the reliability of multi-variable data.

12.3 AGREEMENT ON CODING PREDEFINED UNITS

For single-valued coding of predefined units of analysis, reliability calculations start with reliability data in their canonical form of an Observers-by-Units reliability data matrix for one variable containing values, as in Figure 12.3.

FIGURE 12.3 ■ The Canonical Form of Reliability Data

Units:		1	2	.	.	.	u	.	.	.	.	.	N
Observers:	1	c_{11}	c_{12}	.	.	.	c_{1u}	.	.	.	.	.	c_{1N}
	:	:	:	.	.	.	:	.	.	.	.	.	:
	i	c_{i1}	c_{i2}	.	.	.	c_{iu}	.	.	.	.	.	c_{iN}
	j	c_{j1}	c_{j2}	.	.	.	c_{ju}	.	.	.	.	.	c_{jN}
	:	:	:	.	.	.	:	.	.	.	.	.	:
	m	c_{m1}	c_{m2}	.	.	.	c_{mu}	.	.	.	.	.	c_{mN}

For N predefined units u , coded by up to m observers i or j , in terms of values c the canonical form of reliability data is defined by an N -by- m matrix containing the values c_{iu} assigned by the i th observer to unit u .

Reliability data may or may not have been generated by all m observers. When no data are missing, this matrix contains mN values. When values are missing, the

canonical form contains fewer than mN values. For values to be pairable, units must be valued by at least two observers, that is, columns must be occupied by two or more values.

Perfect reliability would be evident when all observers agree on their valuation of a diversity of units. In Figure 12.3, this would mean that the values c_{iu} within columns u are identical but differ across them. When reliability is perfect, there would be no doubt what a particular value c represents, what all m observers have seen, read, and recorded. In the total absence of reliability, an analyst would not be able to guess what prompted any one of the m observers to record a unit of analysis is c . Under these conditions, the analyst could apply all kinds of computational methods to such data but could not say anything of what he or she has been analyzing. In the total absence of reliability, the values c_{iu} in this canonical form of reliability data would be randomly distributed, correlated neither with the m observers, nor with the N units of analysis.

Researchers always aim for the highest reliability achievable, of course. As perfect reliability may be difficult to achieve, especially when coding tasks are complex and/or require elaborate cognitive processes, researchers need to assure themselves by how much the data deviate from the ideal of perfect reliability and whether this deviation is above or below accepted reliability standards. These are the two main questions that any agreement measure should answer.

The following is an example of the canonical form of reliability data generated by four Observers, categorizing 12 predefined Units of analysis with missing data:

Recording units u :	1	2	3	4	5	6	7	8	9	10	11	12
Observer A:												
Observer B:												
Observer C:												
Observer D:												

Note that

- All values in this example are categories, possess nominal qualities, that is, they exhibit no ordering, and any two values may either be same or different.
- Out of the $mN = 4 \times 12 = 48$ cells in this matrix, 7 are unoccupied or missing.
- One value, , assigned by observer *C* to the 12th unit cannot be paired with other values in that unit, cannot contribute to observed agreements or disagreements, has no place in proper reliability data, and has to be ignored in subsequent calculations.

Thus, these reliability data consist of a total of $n_{..} = 40 \leq mN$ pairable values. Unequal Participation of observers, evident in missing values are typical of many reliability data. Most if not all other reliability coefficients I know require complete information in the form the availability of all mN values.

Several coefficients for measuring agreement are available, specialized for particular kinds of data. Popping (1988) identifies 39 for nominal data alone, and

his list is blatantly incomplete. Section 12.3.5 reviews a few popular ones regarding how they live up to what their proponents claim they measure and explores their suitability for reliability assessments. Here, I am relying on α , not because I developed it over some time (see Krippendorff, 1970a, 1970b, 1978, 1980b, 1987, 1992, 2004b, 2008, 2011a) but because it is arguably the most general and flexible agreement measure with appropriate reliability interpretations in content analysis and similar efforts to generate data, as shall become clear below. To eliminate a source of possible name confusion, α -agreement should not be equated with Cronbach's (1951) alpha, which is widely used in biometric and educational research for an entirely different purpose and is unsuitable for evaluating reliability of generating data. The family of ϵ - α -coefficients allows uniform reliability standards (see section 12.8.4) to be applied to a great diversity of data:

- It corrects observed agreements by chance, defined as the absence of any statistical relationship between the generated data and a set of diverse phenomena.
- It is independent of the number of values available for coding a variable.
- It is applicable to any number of observers, not just the traditional two.
- It adjusts to unequal sample sizes of reliability data.
- It accommodates numerous metrics (scales of measurements)—nominal, ordinal, interval, ratio, and more.
- It is applicable despite missing data caused by unequal numbers of observers attending to unequal sets of recording units. When unitizing a continuum, observers always face a whole continuum, missing data appear in the form of ignoring what they consider irrelevant matter.

In its most general form:

$$\alpha = 1 - \frac{D_o}{D_e} \quad (1)$$

where D_o is a measure of the observed disagreement and D_e is a measure of the disagreement that can be expected when chance prevails. As a first of five interpretations proposed for conceptualizing the α -coefficient:

*α assesses the deviation from perfect reliability
by the proportion of observed to expected disagreement.*

This interpretation reveals the two defining reference points of reliability measures, already mentioned: When agreement is observed to be perfect and disagreement is, therefore, absent, $D_o = 0$ and $\alpha = 1$, indicating perfect reliability. When the observed disagreement is chance, $D_e = D_o$ and $\alpha = 0$, indicating the absence of reliability. α could become negative when the observed disagreement exceeds what can be expected by chance. However, inasmuch as reliability tests are undertaken to assure

that data can be trusted, that the reliability of data does not deviate too much from its ideal of perfect agreement, negative values, even zero values, are too far removed from where the trustworthiness of data is an issue. Negative values tend to have two causes: very small sample sizes and systematic disagreements. The issue of insufficient variation is discussed in section 12.8.1. For reliability considerations, α 's limits are

$$1 \geq \alpha \geq 0 \begin{cases} \pm \text{imprecision owing to small sample sizes} \\ -\text{systematic disagreements} \end{cases}$$

- *Imprecision owing to small sample sizes* happens because with small numbers of values, the calculated value of $\alpha = 0$ can only be approximated. Because observed frequencies are integers whereas expected frequencies are averages, when the latter are fractions, the observed disagreement is rarely equal to the expected disagreements, causing α values to dance above and below zero.
- *Systematic disagreements* are more serious. They occur when observers agree to disagree, unwittingly apply different coding or unitizing instructions, or consistently apply an agenda that is not shared with others. All observed disagreements distract from perfect reliability, but systematic disagreements can cause α values to drop below what would be expected by chance.

The development of the ${}_c\alpha$ -coefficient (c for coding given sets of units as opposed to u in ${}_u\alpha$ for unitizing continua—see section 12.5), stepwise elaborates (1):

$${}_c\alpha_{\text{metric}} = 1 - \frac{D_o}{D_e} = 1 - \frac{\text{Average}_{\text{metric}} \delta_{ck}^2 \text{ within all units}}{\text{Average}_{\text{metric}} \delta_{ck}^2 \text{ within all data}} \quad (2)$$

where $\text{metric } \delta_{ck}^2$ is a measure of the *difference* between two values c and k . It responds to the *metric* of these values, their levels of measurement (discussed in some detail in section 12.3.3). For simplicity, this section 12.3 begins by focusing on unordered categories or nominal data. Nominal differences have the advantage of being binary. Two values either match, $c = k$, or they are different, $c \neq k$. D_o is the *observed disagreement*, the average difference between all values c and k within units. It measures the *disagreement within units*. When that average is zero, reliability is perfect, hence any deviation from this ideal signals unreliability, unaccounted noise in the data; D_e is the *expected disagreement*, the average difference between *all values in the reliability data*, that is *within and between units*. The differences within units, which contribute to D_o , signal the existence of unreliabilities in the data, while the differences between units provide the information needed about the phenomena that the data are to represent, which the researcher needs to decide among research questions. Without the latter, data would not be useful. This brings us to the second interpretations of α that I offer here:

*$\epsilon\alpha$ is the extent to which data are dominated by the variance across units
(information about the phenomena of interest)
as opposed to the disagreements within units.*

Before I elaborate on the algebraic and computational details of α , I should mention that I assume that content analysts will use computer software to calculate the reliabilities they need to ensure that their data are worth studying. Useful SAS and SPSS macros for computing $\epsilon\alpha_{\text{metric}}$ for predefined units called KALPHA can be found in Hayes and Krippendorff (2007; and <http://www.afhayes.com> $\Rightarrow$ SPSS and SAS Macros $\Rightarrow$ KALPHA). The $\mu\alpha$ -family of coefficients for unitizing can be obtained from Krippendorff, Mathet, Bouvry, and Widlöcher (2016). There are many platforms that feature α -coefficients, some have shown to include errors. However, to understand how α works, what α is designed to do, what it responds to and what it omits, what it is that drives its values up or down, it is important for users of reliability software to check out whether calculations are correct, for example, by being able to recalculate at least elementary α -measures by hand in order to convince themselves whether seemingly odd results are correct. Much too often have researchers blindly relied on flawed software (e.g., Lombard, Snyder-Duch, & Bracken 2002; see Krippendorff, 2004b) or failed to recognize mistakes in setting up their data and report results, unknowingly mistaken. The ability to check out such agreement measures is essential, hence the computational details offered below.

Moreover, users of reliability software often find that some coefficients yield similar values, which leads them to believe they measure the same data qualities. To judge whether what is billed as reliability coefficients are sensitive to the same properties of reliability data, it is important to compare how they behave in extreme situations—many observers, different levels of measurement, missing values, large and small reliability data, units of equal or unequal sizes, and so on—but perhaps more revealing, their mathematical structures. I have seen many claims of developers of reliability statistics justifying them in terms of favorable values without understanding how they come about. This is why I prefer to develop α in small steps, starting with its application to nominal data.

I will proceed along two paths to get to $\epsilon\alpha$, one direct and general and the other using the more informative coincidence matrix representations of reliability data. Then I will discuss several typical cases that content analysts may face, give examples that demonstrate computations, suggest computationally simpler formulas where possible, and offer three more interpretations of α to reveal its multiple meanings. Once a basic understanding of how $\epsilon\alpha$ works has been acquired, readers may be more selective in exploring the following sections of this chapter.

12.3.1 The Direct Path to $\epsilon\alpha$ -Agreement

There is some virtue in representing reliability data in terms of a values-by-Units matrix, shown in Figure 12.4. Its terms are obtainable from the canonical form of Figure 12.3 by summing and thereby eliminating references to observers:

FIGURE 12.4 ■ The Values-by-Units Matrix of Reliability Data

Units:	1	2	.	.	<i>u</i>	.	.	.	.	.	.	<i>N</i>	
Values:	1	<i>n</i> ₁₁	<i>n</i> ₂₁	.	.	<i>n</i> _{<i>u</i>1}	.	.	.	.	.	<i>n</i> _{<i>N</i>1}	<i>n</i> _{.1}
:	:	:	.	.	:	.	.	.	.	.	.	:	
<i>c</i>		<i>n</i> _{1<i>c</i>}	<i>n</i> _{2<i>c</i>}	.	.	<i>n</i> _{<i>u</i><i>c</i>}	.	.	.	.	.	<i>n</i> _{<i>N</i><i>c</i>}	<i>n</i> _{.<i>c</i>} = $\sum_{u=1}^N n_{uc} \mid m_u \geq 2$
<i>k</i>		<i>n</i> _{1<i>k</i>}	<i>n</i> _{2<i>k</i>}	.	.	<i>n</i> _{<i>u</i><i>k</i>}	.	.	.	.	.	<i>n</i> _{<i>N</i><i>k</i>}	<i>n</i> _{.<i>k</i>}
:	:	:	.	.	:	.	.	.	.	.	.	:	
<i>v</i>		<i>n</i> _{1<i>v</i>}	<i>n</i> _{2<i>v</i>}	.	.	<i>n</i> _{<i>u</i><i>v</i>}	.	.	.	.	.	<i>n</i> _{<i>N</i><i>v</i>}	<i>n</i> _{.<i>v</i>}
Numbers ≥ 2 of pairable values:		<i>m</i> ₁	<i>m</i> ₂	.	.	$m_u = \sum_{c=1}^v n_{uc} \mid m_u \geq 2$	.	.	.	.	.	<i>m</i> _{<i>N</i>}	<i>n</i> _{..} = $\sum_{u=1}^N m_u = \sum_{c=1}^v n_{.c} \leq mN$

Wherein: 1, ... *c*, *k*, ... *v* are the assigned values

$m_u \geq 2$ = The number of pairable values assigned to unit *u* by $m_u \leq m$ observers

N = The number of units with pairable values, excluding units with $m_u < 2$

n_{uc} = The number of pairable values *c* assigned to unit *u*. n_{uk} by analogy

$n_{.c}$ = The number of pairable values *c* occurring in the reliability data. $n_{.k}$ by analogy

$n_{..}$ = The total number of all pairable values in the reliability data

Omitting references to observers is justified in all reliability considerations as the individuality of observers must not enter any reliability index. Observers, coders, raters, and other measuring devices have to be considered interchangeable. What matters is their collective valuation of the given units. Figure 12.4 also shows in its column sums the number m_u of observers coding unit *u*. As already mentioned, to contain pairable values, reliability data require that column sums be $m_u \geq 2$. The row sums $n_{.c}$ of the number of pairable values *c*, collectively used, is important as their distribution serves as the best estimate of the probabilities $p_c = \frac{n_{.c}}{n_{..}}$ in the data whose reliability is in question. For the agreement coefficient α , it plays an important role in defining expectations.

Only *c-k* pairs of values can lead to differences $\text{metric } \delta_{ck}^2$. With m_u values in unit *u*, one can form $m_u(m_u - 1)$ pairs of values—excluding values paired with themselves. If the m_u values in *u*, are composed of n_{uc} values *c* and n_{uk} values *k*, there will be $n_{uc}(n_{uc} - 1)$ pairs of matching values *c*, $n_{uk}(n_{uk} - 1)$ matching values *k*, and $2n_{uc}n_{uk}$ mismatching pairs of values. To allow each value in m_u to make the same contribution, we multiply the average metric difference $\text{metric } \delta_{ck}^2$ within all pairs of values, the numerator in (2), by $n_{..}$ and come to the observed disagreement D_o :

$${}_c D_o = \sum_{u=1}^N \frac{1}{m_u - 1} \sum_{c=1}^v \sum_{k=1}^v \left\{ \begin{array}{ll} n_{uc}(n_{uk} - 1) & \text{iff } c = k \\ n_{uc}n_{uk} & \text{iff } c \neq k \end{array} \right\} \text{metric } \delta_{ck}^2 \quad (3)$$

The analogue expression for the expected disagreement D_e , the denominator in (2), follows the same logic, except that D_e is not concerned with $m_u(m_u-1)$ differences within N units u rather with $n_{..}(n_{..}-1)$ differences among all pairable values in the reliability data, also excluding the pairing of values with themselves:

$${}_c D_e = \frac{1}{n_{..}-1} \sum_{c=1}^v \sum_{k=1}^v \left\{ \begin{array}{ll} n_{.c}(n_{.k}-1) & \text{iff } c = k \\ n_{.c}n_{.k} & \text{iff } c \neq k \end{array} \right\} \text{metric } \delta_{ck}^2 \quad (4)$$

To obtain ${}_c \alpha$, (3) and (4) can be inserted in (1) or (2). Fortunately, the resulting expression for ${}_c \alpha$ affords two algebraic simplifications:

- Because of differences between matching values $\text{metric } \delta_{cc}^2 = \text{metric } \delta_{kk}^2 = 0$, the numbers $n_{uc}(n_{uk}-1)$ of matching pairs $c = k$ of values in units u , and $n_{.c}(n_{.k}-1)$ of matching pairs $c = k$ of values do not need to be enumerated.
- The symmetry of differences $\text{metric } \delta_{ck}^2 = \text{metric } \delta_{kc}^2$ and of the product $n_{uc}n_{uk} = n_{uk}n_{uc}$ enables calculating only one half of the total number of mismatching values. The double sum $\sum_c \sum_k$ can be reduced to $\sum_c \sum_{k>c}$.

With these provisions, the most efficient, simple, and entirely general expression for computing ${}_c \alpha$ becomes:

$${}_c \alpha_{\text{metric}} = 1 - \frac{{}_c D_o}{{}_c D_e} = 1 - (n_{..}-1) \frac{\sum_{u=1}^N \frac{1}{m_u-1} \sum_{c=1}^v n_{uc} \sum_{k>c}^v n_{uk} \text{metric } \delta_{ck}^2}{\sum_{c=1}^v n_{.c} \sum_{k>c}^v n_{.k} \text{metric } \delta_{ck}^2} \quad (5)$$

For our example from the reliability data in canonical form, the values-by-units matrix is as follows: (Zero frequencies are omitted for simplicity):

Note that the 12th row contains only one value, $n_{12\text{☎}} = 1$. The lone ☎ cannot be paired within unit 12 and has to be omitted from the reliability data. It therefore does not enter the row and column sum. This omission is indicated by crossing out $n_{12\text{☎}}$ and m_{12} . Hence, the number of pairable values is $n_{..} = 40$, and the number of units with pairable values is $N = 11$.

Units $u =$	1	2	3	4	5	6	7	8	9	10	11	12	
Values $c :$	3	.	.	.	.	1	.	3	.	.	2	.	$9 = n_{. \text{book}}$
	.	3	.	.	4	1	.	1	4	.	.	.	$13 = n_{. \text{envelope}}$
	.	1	4	4	.	1	.	.	.	.	.	+	$10 = n_{. \text{phone}}$
	.	.	.	.	.	1	4	.	.	.	.	.	$5 = n_{. \text{laptop}}$
	.	.	.	.	.	.	.	.	.	3	.	.	$3 = n_{. \text{folder}}$
Pairable $m_u =$	3	4	4	4	4	4	4	4	4	3	2	+	$40 = n_{..}$

As already mentioned, this example consists of nominal data. A characteristic of these data is that any two values are either the same or different. They either match or do not match. The expression (5) already enumerates only values that do not match for which $\text{nominal} \delta_{c \neq k}^2 = 1$. So, in calculating ${}_c \alpha_{\text{nominal}}$ for this example, we simply enumerate the observed and expected disagreements as (5) states them:

$${}_c \alpha_{\text{nominal}} = 1 - (40 - 1) \frac{0 + \frac{3.1}{4-1} + 0 + 0 + 0 + \frac{1.1}{4-1} + \frac{1.1}{4-1} + \frac{1.1}{4-1} + \frac{1.1}{4-1} + \frac{1.1}{4-1} + \frac{1.1}{4-1} + 0 + \frac{3.1}{4-1} + 0 + 0 + 0}{9(13+10+5+3)+13(10+5+3)+10(5+3)+5.3} = 0.743$$

Expression (5) is all that is needed to obtain the most basic ${}_c \alpha$ -agreements for coding sets of predefined units directly and for all metric differences, not just for the nominal data in our example. The difference functions $\text{metric} \delta_{ck}^2$ for various metrics are discussed in section 12.3.3. Readers who do not care about the complications introduced by ordered data may want to skip to the last section 12.8, which discusses statistical consideration and standards for accepting and rejecting data as reliable. Those who are curious about how ${}_c \alpha$ relates to other agreement coefficients may consult section 12.3.5, which discusses the most popular ones. Section 12.4 defines three variations of ${}_c \alpha$ for measuring the accuracy or surrogacy, as distinguished in Table 12.1. Moreover, in response to popular hailing of majority judgments of crowds as a gold standard, it presents a measure of the decisiveness of majorities. Section 2.5 presents a way to measure the reliability of searching large databases: information retrieval and text mining. Section 12.6 concerns the reliability of unitizing textual and other continua. All reliability measures presented up to and including in section 12.6 deal with the single-valued coding of predefined units and unitized segments of continua. Section 12.7 extends all of them to capture the reliability of multi-valued coding. And section 12.8 discusses statistical properties of ${}_c \alpha$, including reliability standards.

12.3.2 The Path to ${}_c \alpha$ via Coincidence Matrices

The direct path to α is entirely general but conceptually not too transparent. It is possible and has proven to have considerable analytical as well as pedagogical advantages to break the required additions in the somewhat redundant expressions (3) into two parts, one by summing the c - k pairs of values, obtained from the values-by-units matrix over the units u , $\sum_u$, thereby creating the entries of an intermediate representation of the reliability data, a matrix of observed coincidences o_{ck} , and the other by summing $\sum_c \sum_k$ the differences between the values represented by these coincidences to obtain the observed disagreements ${}_c D_o$:

$$o_{ck} = \sum_{u=1}^N \begin{cases} n_{uc}(n_{uk}-1) & \text{iff } c=k \\ n_{uc}n_{uk} & \text{iff } c \neq k \end{cases} \quad (6) \quad \text{and} \quad {}_c D_o = \sum_{c=1}^v \sum_{k=1}^v o_{ck} \text{ metric } \delta_{ck}^2 \quad (7)$$

One should not have difficulties recognizing the expressions (6) and (7) in (3).

A matrix of observed coincidences o_{ck} defined by (6) is a v -by- v , values-by-values matrix. It is square and symmetrical around its diagonal. It tabulates the

numbers of pairs of values in units u from the values-by-units matrix, weighted by $(m_u - 1)$ so that each pairable value in unit u is represented and counted in the total $n \dots$

FIGURE 12.5 ■ The Matrix of Observed Coincidences o_{ck}

Values:	1	.	.	k	.	v	
1	O_{11}	.	.	O_{1k}	.	O_{1v}	$n_{1.}$
.	.	.	.	.	.	.	.
c	O_{c1}	.	.	O_{ck}	.	O_{cv}	$n_{c.} = \sum_{k=1}^v o_{ck}$
.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.
v	O_{v1}	.	.	O_{vk}	.	O_{vv}	$n_{v.}$
	$n_{.1}$	.	.	$n_{.k}$	.	$n_{.v}$	$n_{..} = \sum_{c=1}^v \sum_{k=1}^v o_{ck}$

The row sums $n_{.c} = \sum_{u=1}^N c_{uc}$ in Figure 12.4 equal the row or column sums $n_{.c} = n_{c.} = \sum_{k=1}^v o_{ck}$ in Figure 12.5, and because coincidence matrices are symmetrical, $o_{ck} = o_{kc}$, the marginal sums of coincidence matrices are equal, $n_{.c} = n_{c.}$ Finally, the totals $n_{..} \leq mN$ in Figure 12.4 also are the totals $n_{..}$ in Figure 12.5. The diagonals of coincidence matrices with entries o_{cc} and o_{kk} enumerate perfectly matching pairs of values. Their off-diagonal entries, $o_{c \neq k}$ represent the mismatching pairs of values.

While Figure 12.4 results from summing the references to observers from Figure 12.3, Figure 12.5 results from summing the references to units from Figure 12.4. Both are justified as neither the individuality of observers nor the particular units they code should enter measures of data reliability. The advantage of coincidence matrices lies in visualizing where deviations from the diagonal of perfect matches occurs.

To obtain the matrix of the expected coincidences from Figure 12.4, we proceed just as the observed coincidences were defined by dividing the definition of the expected disagreements (4) into the two comparable components:

$$e_{ck} = \frac{1}{n_{..} - 1} \begin{cases} n_{.c}(n_{.k} - 1) & \text{iff } c = k \\ n_{.c}n_{.k} & \text{iff } c \neq k \end{cases} \quad (8) \quad \text{and} \quad {}_c D_e = \sum_{c=1}^v \sum_{k=1}^v e_{ck} \text{ metric } \delta_{ck}^2 \quad (9)$$

(8) leads to the matrix of expected coincidences in Figure 12.6, the companion to the matrix of observed coincidences:

FIGURE 12.6 ■ The Matrix of Expected Coincidences e_{ck}

Values:	1	.	.	k	.	v	
1	e_{11}	.	.	e_{1k}	.	e_{1v}	$n_{1.}$
.	.			.		.	.
c	e_{c1}	.	.	e_{ck}	.	e_{cv}	$n_{c.} = \sum_{k=1}^v e_{ck}$
.	.			.		.	.
.	.			.		.	.
v	e_{v1}	.	.	e_{vk}	.	e_{vv}	$n_{v.}$
	$n_{.1}$	.	.	$n_{.k}$	.	$n_{.v}$	$n_{..} = \sum_{c=1}^v \sum_{k=1}^v e_{ck}$

In terms of both coincidences in Figures 12.5 and 12.6, inserting (7) and (9) into (1) or (2), again taking advantage of the symmetry of coincidences $o_{ck} = o_{kc}$, $e_{ck} = e_{kc}$, $n_{c.} = n_{.c}$, and of the differences $\text{metric } \delta_{ck}^2 = \text{metric } \delta_{kc}^2$, just as in (5), yields two expressions for ${}_c\alpha$, one with and the other without references to the expected coincidences e_{ck} :

$$\begin{aligned} {}_c\alpha_{\text{metric}} &= 1 - \frac{{}_cD_o}{{}_cD_e} = 1 - \frac{\sum_{c=1}^v \sum_{k>c}^v o_{ck} \text{ metric } \delta_{ck}^2}{\sum_{c=1}^v \sum_{k>c}^v e_{ck} \text{ metric } \delta_{ck}^2} \\ &= 1 - (n_{..} - 1) \frac{\sum_{c=1}^v \sum_{k>c}^v o_{ck} \text{ metric } \delta_{ck}^2}{\sum_{c=1}^v n_{c.} \sum_{k>c}^v n_{.k} \text{ metric } \delta_{ck}^2} \end{aligned} \tag{10}$$

From the fact that (10) can express ${}_c\alpha_{\text{metric}}$ without references to expected coincidences in (8) and Figure 12.6, one can conclude that the expected coincidences e_{ck} are computationally unnecessary, and they are. However, inasmuch as the proportion of ${}_cD_o/{}_cD_e$ is part of ${}_c\alpha$'s definition, comparing the two matrices can provide analytical insights about the reliability data, what contributes and what distracts from the measured reliability, something that the direct path to ${}_c\alpha$ hides in its definition (5). Many of the following examples will demonstrate the conceptual virtues of representing the reliability data in terms of the observed coincidences in Figure 12.5 rather than in

Observed coincidences							Expected coincidences						
Values:													
	7	$\frac{4}{(4-1)}$	$\frac{1}{(4-1)}$	$\frac{1}{(4-1)}$	0	9		$\frac{9 \cdot 8}{39}$	$\frac{9 \cdot 13}{35}$	$\frac{9 \cdot 10}{39}$	$\frac{9 \cdot 5}{39}$	$\frac{9 \cdot 3}{39}$	9
	1.33	10	$\frac{4}{(4-1)}$	$\frac{1}{(4-1)}$	0	13		3.00	$\frac{13 \cdot 12}{39}$	$\frac{13 \cdot 10}{39}$	$\frac{13 \cdot 5}{39}$	$\frac{13 \cdot 3}{39}$	13
	0.33	1.33	8	$\frac{1}{(4-1)}$	0	10		2.31	1.33	$\frac{10 \cdot 9}{39}$	$\frac{10 \cdot 5}{39}$	$\frac{10 \cdot 3}{39}$	10
	0.33	0.33	0.33	4	0	5		0.38	0.33	0.33	$\frac{5 \cdot 4}{39}$	$\frac{5 \cdot 3}{39}$	5
	0	0	0	0	3	3		0.69	1.00	0.77	0.38	$\frac{3 \cdot 2}{39}$	3
	9	13	10	5	3	40		9	13	10	5	3	40

terms of Figures 12.3 and 12.4, starting with the reliability data introduced in section 12.3.1. For these, the observed and expected coincidence matrices are as follows:

In the matrix of observed coincidences, one may notice that files, , are perfectly reliable. There is no confusion with any other values. By comparison, , , , and  exhibit some confusion among themselves. To appreciate the frequencies that go into the computation of o_{ck} , the above matrices tabulate the products and fractions that produce these coincidences in one off-diagonal triangle and their decimal equivalents in the other. When comparing the observed and the expected coincidences, one may notice that the off-diagonal cells of the observed coincidences are all lower than those expected. Because off-diagonal entries represent mismatching pairs of values that count toward the observed disagreements, and are smaller than what can be expected by chance, one can anticipate that ${}_c\alpha$ would have to have a positive value.

Indeed, according to the second expression in (10):

$${}_c\alpha_{\text{nominal}} = 1 - \frac{1.333 + 0.333 + 0.333 + 0 + 1.333 + 0.333 + 0 + 0.333 + 0 + 0}{3.000 + 2.308 + 1.154 + 0.692 + 3.333 + 1.667 + 1.000 + 1.282 + 0.769 + 0.385} = 0.743$$

The result obtained by means of the two coincidence matrices are of course identical to the one obtained directly, but the coincidence matrices facilitate comparisons and detailed examinations of what accounts for the reliability as measured.

12.3.3 Difference Functions for Data With Various Metrics

The previous two sections suggested two paths to ${}_c\alpha$. Both (5) and (10) refer to difference functions ${}_{\text{metric}}\delta_{ck}^2$, and the subscript *metric* in ${}_c\alpha_{\text{metric}}$ promises ${}_c\alpha$'s applicability to data other than the nominal data to which the computational example was limited so far.

Nominal data are simple for they exhibit no discernible ordering. Any two categories are either the same or different, not more or less. The diagonals of coincidence matrices contain perfectly matching values, and all of their differences are 0 regardless of whether we are concerned with nominal, ordinal, interval, or ratio data. But for nominal data, all mismatching pairs of values are simply unlike, not the same, and their differences are something other than 0. The simplest way is to take the differences between unequal values to be 1. This fact allows algebraic simplifications of ${}_c\alpha_{\text{nominal}}$, which we have not yet explored. The algebraic simplicity of dealing with agreement for nominal data has stimulated a large literature of agreement coefficients, some of which will be reviewed in section 12.3.5. It has blinded the developers of agreement measures to easy generalizations of reliability measures to more complex data, which is what the ${}_c\alpha$ -statistics offers by associating a variety of difference functions ${}_{\text{metric}}\delta_{ck}^2$ with the observed and expected coincidences as in (7) and (9). They are reviewed in this section.

In section 8.4, a metric was defined by the operations that are applicable to the values of a variable. Although in calculating ${}_c\alpha_{\text{metric}}$ researchers have the choice of several difference functions with which to analyze their reliability data, their choice needs to be appropriate to the nature of their data, whether the operations that define a metric make sense. For example, one cannot sum names or multiply ranks. The values of a ratio metric cannot be negative. An algebraic difference between the numbers assigned to two football players has no meaningful interpretation. Therefore, the choice of an appropriate metric for ${}_c\alpha$ must take into account the ordering that underlies the

phenomena data are to represent, which mathematical operations they afford. Equally important is the demand made by the method used to process the data generated for this purpose. An analysis of variance requires interval data, and contingency coefficients are computed from nominal data. It would not be appropriate to assess the reliability of data with an agreement coefficient for nominal data and analyze them with correlations. The agreement measured under the assumption of a ratio metric would not match what a factor analysis requires, which are interval data. The metric of ${}_c\alpha_{\text{metric}}$ needs to be the same as the metric that an analytical procedure requires its data to possess.

The ability of ${}_c\alpha$ to accommodate the most important metrics, providing reliability measures tailored to intended analyses, and allowing diverse kinds of data to be judged by the same standard, is one of the outstanding features of ${}_c\alpha$.

As already mentioned, differences have these properties:

- They have an absolute reference value: $\delta_{cc}^2 = \delta_{kk}^2 = 0$
- They are symmetrical: $\delta_{ck}^2 = \delta_{kc}^2$
- They may be standardized: $0 \leq \delta_{ck}^2 \leq 1$ for convenient comparisons.

All difference functions discussed in this section are defined for single-valued data. They measure the differences between two values c and k . According to the expressions (2) and (5), the direct path to ${}_c\alpha$ averages the differences between values found within units for the observed disagreements ${}_cD_o$, and between values found in all data for the expected disagreement ${}_cD_e$. The construction of coincidence matrices has no effect on the nature of the difference functions employed.

However, (7) and (9) offer another decoupling of two analytical conceptions: the two coincidences and the applicable differences in (10), both in the form of ν -by- ν matrices:

$${}_c\alpha_{\text{metric}} = 1 - \frac{{}_cD_o}{{}_cD_e} = 1 - \frac{\sum_{c=1}^{\nu} \sum_{k>c}^{\nu} o_{ck} \text{ metric } \delta_{ck}^2}{\sum_{c=1}^{\nu} \sum_{k>c}^{\nu} e_{ck} \text{ metric } \delta_{ck}^2} = 1 - \frac{\sum \left[\begin{array}{|c|} \hline o_{ck} \\ \hline \end{array} \right] \times \left[\begin{array}{|c|} \hline \text{metric } \delta_{ck}^2 \\ \hline \end{array} \right]}{\sum \left[\begin{array}{|c|} \hline e_{ck} \\ \hline \end{array} \right] \times \left[\begin{array}{|c|} \hline \text{metric } \delta_{ck}^2 \\ \hline \end{array} \right]} \quad (11)$$

Accordingly, coincidences and difference can be depicted as matrices of the same size whose entries are multiplied and summed to yield the observed and expected disagreements.

The following presents the four basic **metric-specific difference** functions for nominal, ordinal, interval and ratio data, both algebraically and in their matrix representation (within arbitrary limits). A difference function for multi-valued data is presented in section 12.7. However, it, too, is defined in terms of the very same single-valued difference functions defined here.

Nominal metric. For the nominal data used so far, reference to a metric was less important as they entail no particular ordering. Values are merely distinct and freely permutable. When nominal values are represented numerically—area codes, banking PINs, the numbers on the jerseys of football players—adding or subtracting them

from one another makes no sense. Any pair of values is either same or different. They match, or they do not match:

$$\text{nominal } \delta_{ck}^2 = \begin{cases} 0 & \text{iff } c = k \\ 1 & \text{iff } c \neq k \end{cases} \quad (12)$$

						
	0	1	1	1	1	1
	1	0	1	1	1	1
	1	1	0	1	1	1
	1	1	1	0	1	1
	1	1	1	1	0	1
	1	1	1	1	1	0

Evidently, in the diagonal cells of all difference matrices, $\delta_{c=k}^2 = 0$. For data with nominal metrics, all mismatches occur in the off-diagonal cells whose cells are $\delta_{c \neq k}^2 = 1$.

Binary or dichotomous data are the most reduced case of nominal data. They represent only one choice between two alternative values. So limited, binary difference functions δ_{ck}^2 behave exactly as nominal ones do. However, binary difference functions possess an important property. All difference functions become binary when their choices are limited to two alternatives, regardless what metric underlies them.

Ordinal metric. In ordinal data, values are ranked. Their differences are merely asymmetrical, smaller than, following, better, higher, as exemplified by the use of unequal signs, for example:

$$1^{\text{st}} < 2^{\text{nd}} < 3^{\text{rd}} < 4^{\text{th}} < 5^{\text{th}} < \dots$$

Ordinal data do not need to be labeled numerically as long there is a linear ordering, for example, alphabetically:

$$a < b < c < d < e < \dots$$

While principally open ended, in practice, they may have a beginning or smallest rank and an end or largest ranks as in predefined answers to polling questions, for example:

$$\text{best} > \text{just OK} > \text{no knowledge or opinion} > \text{not really OK} > \text{worst}$$

Since Spearman (1904) introduced his famous rank correlation coefficient ρ (rho), ordinal differences have been quantified by how many ranks are found between the means of any two ranks c and k . The following gives two definition, which do not influence computed α -coefficients: one raw, ranging from 0 to the largest number of ranks in given ordinal data, and one standardized, ranging from 0 to 1.

An important quality of the ordinal metric is that numerical characterizations of ranks have no arithmetic value (see Section 8.6.1). Differences between ranks depend entirely on the distribution of ranks in data. It follows that unused ranks do not contribute to differences among ranked units occurring in data. The frequencies of ten ranked units exemplify these properties:

$$n_{\text{smallest}} = n_1^{\text{st}} = 2, \quad n_2^{\text{nd}} = 3, \quad n_3^{\text{rd}} = 1, \quad n_4^{\text{th}} = 2, \quad n_5^{\text{th}} = 0, \quad n_6^{\text{th}} = 2 = n_{\text{largest}}$$

Raw definition:

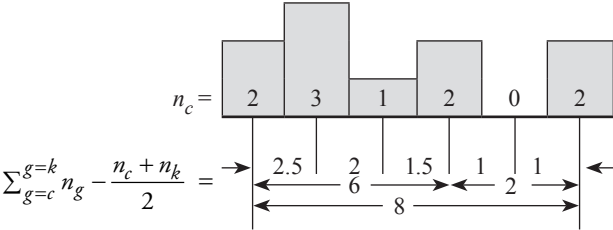
$$\text{ordinal } \delta_{ck}^2 = \left(\sum_{g=c}^{g=k} n_g - \frac{n_c + n_k}{2} \right)^2$$

Standardized definition: (exemplified): (13)

$$\text{ordinal } \delta_{ck}^2 = \left(\frac{\sum_{g=c}^{g=k} n_g - \frac{n_c + n_k}{2}}{n - \frac{n_{\text{smallest}} + n_{\text{largest}}}{2}} \right)^2$$

where: $n = n_{\text{smallest}} + \dots + n_c + \dots + n_k + \dots + n_{\text{largest}}$

Ranks:	1 st	2 nd	3 rd	4 th	5 th	6 th
1 st	0	$\left(\frac{2.5}{8}\right)^2$	$\left(\frac{4.5}{8}\right)^2$	$\left(\frac{6}{8}\right)^2$	$\left(\frac{7}{8}\right)^2$	$\left(\frac{8}{8}\right)^2$
2 nd	0.10	0	$\left(\frac{2}{8}\right)^2$	$\left(\frac{3.5}{8}\right)^2$	$\left(\frac{4.5}{8}\right)^2$	$\left(\frac{5.5}{8}\right)^2$
3 rd	0.32	0.06	0	$\left(\frac{1.5}{8}\right)^2$	$\left(\frac{2.5}{8}\right)^2$	$\left(\frac{3.5}{8}\right)^2$
4 th	0.56	0.19	0.04	0	$\left(\frac{1}{8}\right)^2$	$\left(\frac{2}{8}\right)^2$
5 th	0.77	0.32	0.10	0.02	0	$\left(\frac{1}{8}\right)^2$
6 th	1	0.47	0.19	0.06	0.02	0



The ordinal different matrix tabulates the standardized differences of the collection of eight ranks in two ways, as squared fractions. The lower left off-diagonal cells show the numerical values of this difference function. The upper tight triangle of off-diagonal cells shows how they are obtained, as the fractions of the number of ranks between any two ranks, divided by the maximum, squared. In addition, because ordinal differences are the only ones that are not associated with their differences in assigned rank, the bar graphs of the frequencies of ranks are shown below the difference matrix with unsquared differences of the raw definition of the ordinal difference function. Observe that the 5th and unused rank does not make a difference in how far the 4th and 6th ranks are apart. Only actually used ranks matter.

Interval metric. One cannot add and subtract ranks, but the more familiar interval scales do afford these mathematical operations. In interval data, it is the algebraic difference that indicates how far apart any two values are:

Raw definition (exemplified):

$$\text{interval } \delta_{ck}^2 = (c - k)^2$$

Standardized definition

$$\text{interval } \delta_{ck}^2 = \left(\frac{c - k}{c_{\text{max}} - c_{\text{min}}} \right)^2$$

where c_{max} is the largest and c_{min} the smallest value.

(14)

	-1	0	1	2	3	4
-1	0	1 ²	2 ²	3 ²	4 ²	5 ²
0	1	0	1 ²	2 ²	3 ²	4 ²
1	4	1	0	1 ²	2 ²	3 ²
2	9	4	1	0	1 ²	2 ²
3	16	9	4	1	0	1 ²
4	25	16	9	4	1	0

Again, for the convenience of reading the interval differences in this matrix, one triangle contains squared differences of the raw definition, the other their integer equivalences. Imagine drawing lines of equal differences in this matrix. For interval

data, these lines would parallel the diagonal, and their intervals would rapidly narrow with increasing distance from that diagonal.

Note that when all ranks are of uniform frequency, interval and ordinal differences are proportional, and their agreement coefficients become identical:

$$c_{\text{interval}} = c_{\text{ordinal}}$$

Ratio metric. In ratio scales of measurement, algebraic differences between two values matter only in relation to how remote they are from zero, which is their common reference point. Guessing the age of an older person $\pm$ a year may be remarkable, whereas guessing the age of a baby $\pm$ a year is not. Losing a dollar may mean nothing for a millionaire but a lot for somebody who has only one. Age and wealth are ratio scales and so are frequencies, proportions, and distances. Algebraic differences between small values weigh more than the same differences between large values. The following difference function reflects these intuitions:

$$\text{ratio } \delta_{ck}^2 = \left(\frac{c - k}{c + k} \right)^2 \quad (15)$$

	0	1	2	3	4	5
0	0	$(\frac{1}{1})^2$	$(\frac{2}{2})^2$	$(\frac{3}{3})^2$	$(\frac{4}{4})^2$	$(\frac{5}{5})^2$
1	1	0	$(\frac{1}{3})^2$	$(\frac{2}{4})^2$	$(\frac{3}{5})^2$	$(\frac{4}{6})^2$
2	1	.11	0	$(\frac{1}{5})^2$	$(\frac{2}{6})^2$	$(\frac{3}{7})^2$
3	1	.25	.04	0	$(\frac{1}{7})^2$	$(\frac{2}{8})^2$
4	1	.36	.11	.02	0	$(\frac{1}{9})^2$
5	1	.44	.18	.06	.01	0

Whereas in interval metrics the lines of equal differences are parallel to the diagonal, in ratio metrics they all join in the zero-point and extend, fanlike, into infinity, ratio differences being the tangent of the angular deviation from the 45-degree diagonal line.

Recoding the nominal values, our example by  $\Rightarrow$ 1,  $\Rightarrow$ 2,  $\Rightarrow$ 3,  $\Rightarrow$ 4, and  $\Rightarrow$ 5, yields a matrix of observed coincidences, already familiar in content from section 12.3.2:

		Observed coincidences					
Values:		0	1	2	3	4	
0	7	$\frac{4}{(4-1)}$	$\frac{1}{(4-1)}$	$\frac{1}{(4-1)}$	0	9	
1	1.33	10	$\frac{4}{(4-1)}$	$\frac{1}{(4-1)}$	0	13	
2	0.33	1.33	8	$\frac{1}{(4-1)}$	0	10	
3	0.33	0.33	0.33	4	0	5	
4	0	0	0	0	3	3	
Totals:		9	13	10	5	3	40

Applying the second form of (10) (without references to expected coincidences) to the observed coincidences in this matrix, the four difference functions defined above call for the following computations:

For *nominal data*, that is, where numerical values are treated as unordered names, and mismatches show no degrees:

$${}_c\alpha_{\text{nominal}} = 1 - (40 - 1) \frac{\frac{4}{4-1} + \frac{4}{4-1} + \frac{4}{4-1} + 0 + \frac{4}{4-1} + \frac{4}{4-1} + 0 + \frac{4}{4-1} + 0 + 0}{9(13 + 10 + 5 + 3)13(10 + 5 + 3) + 10(5 + 3) + 5.3} = 0.743$$

For *ordinal data*, that is, where numerical values are considered ranks, and differences depend on the distribution of ranks:

$${}_c\alpha_{\text{ordinal}} = 1 - (40 - 1) \frac{\frac{4}{4-1}11^2 + \frac{1}{4-1}22.5^2 + \frac{1}{4-1}30^2 + 0 + \frac{4}{4-1}11.5^2 + \frac{1}{4-1}19^2 + 0 + \frac{1}{4-1}7.5^2 + 0 + 0}{9(1311^2 + 1022.5^2 + 530^2 + 334^2) + 13(1011.5^2 + 519^2 + 323^2) + 10(57.5^2 + 311.5^2) + 534^2} = 0.815$$

For *interval data*, that is, where numerical values are understood as the points of an equal interval scale:

$${}_c\alpha_{\text{interval}} = 1 - (40 - 1) \frac{\frac{4}{4-1}1^2 + \frac{1}{4-1}2^2 + \frac{1}{4-1}3^2 + 0 + \frac{4}{4-1}2^2 + 0 + \frac{1}{4-1}1^2 + 0 + 0}{9(131^2 + 102^2 + 53^2 + 34^2) + 13(101^2 + 52^2 + 33^2) + 10(51^2 + 32^2) + 531^2} = 0.849$$

For *ratio data*, that is, when difference between numerical values of a variable have absolute zero reference points:

$${}_c\alpha_{\text{ratio}} = 1 - (40 - 1) \frac{\frac{4}{4-1}\left(\frac{1}{3}\right)^2 + \frac{1}{4-1}\left(\frac{2}{4}\right)^2 + \frac{1}{4-1}\left(\frac{3}{5}\right)^2 + 0 + \frac{4}{4-1}\left(\frac{1}{5}\right)^2 + \frac{4}{4-1}\left(\frac{2}{6}\right)^2 + 0 + \frac{4}{4-1}\left(\frac{1}{7}\right)^2 + 0 + 0}{9\left[13\left(\frac{1}{3}\right)^2 + 10\left(\frac{2}{4}\right)^2 + 5\left(\frac{3}{5}\right)^2 + 3\left(\frac{4}{6}\right)^2\right] + 13\left[10\left(\frac{1}{5}\right)^2 + 5\left(\frac{2}{6}\right)^2 + 3\left(\frac{3}{7}\right)^2\right] + 10\left[5\left(\frac{1}{7}\right)^2 + 3\left(\frac{2}{8}\right)^2\right] + 53\left(\frac{1}{9}\right)^2} = 0.797$$

One may be curious why ${}_c\alpha_{\text{interval}} = 0.849$ is larger than ${}_c\alpha_{\text{nominal}} = 0.743$. This difference is small but informative. The answer can be found in the distribution of the observed coincidences underlying both calculations. In the above matrix of observed coincidences, one may notice that observed disagreements are slightly closer to the diagonal, between neighboring values, and the extreme difference between the values 1 and 5 is absent. This suggests that there is some ordering in the distribution of disagreements that the interval ${}_c\alpha_{\text{interval}}$ is sensitive to, but the nominal ${}_c\alpha_{\text{nominal}}$ cannot recognize. This difference is small, but for other data, it may be significant (see the next section, 12.3.4, for a more pronounced example). That the interval ${}_c\alpha_{\text{interval}} = 0.849$ is very close to the ordinal ${}_c\alpha_{\text{ordinal}} = 0.815$ is not surprising as both are sensitive to an albeit slightly different linear ordering of values—interval differences by assuming equal intervals between values and ordinal differences by counting the number of ranks between any pair of ranks. In this example, the number of ranks between any two ranks is not that different especially where most of the disagreements reside. The ratio metric weights numerical difference heavier when nearer the zero point, as discussed above. In the observed coincidences, the 1-2 disagreement is larger than the 3-4 disagreement with the 4-5 disagreement absent. This pattern is counter to what one

would expect typical ratio data to show, hence ${}_c\alpha_{\text{ratio}} = 0.797 < {}_c\alpha_{\text{interval}}$ while responsive to some ordering, hence ${}_c\alpha_{\text{ratio}} > {}_c\alpha_{\text{nominal}}$.

When the distribution of observed coincidences in the off-diagonal cells exhibit no ordering:

$${}_c\alpha_{\text{nominal}} = {}_c\alpha_{\text{ordinal}} = {}_c\alpha_{\text{interval}} = {}_c\alpha_{\text{ratio}}$$

Any deviation from this equality signals the preponderance or absence of specific metric qualities in the reliability data. This mathematical fact enables ${}_c\alpha$ to indicate the prevalence of a particular metric. However, this possibility depends on the existence of sufficient numbers of off-diagonal mismatches or unreliabilities, that is, ${}_c\alpha_{\text{metric}} < 1$. As agreement approaches perfection, the ability to identify the prevalence of a particular metric disappears. This said, I need to warn against letting the highest ${}_c\alpha_{\text{metric}}$ -coefficient decide on the metric that underlies the data. The choice of difference functions cannot be made dependent on the highest ${}_c\alpha_{\text{metric}}$ -value measured. It is to be determined by two factors. It has to correspond to how given coding instructions define the values of a variable, and the metric of the agreement coefficient must not exceed in strength the metric of the analytical technique to be applied to the data.

The latter point goes back to Stevens's (1946) definitions of levels of measurement, the four most common difference functions ${}_{\text{metric}}\delta_{ck}^2$ defined for ${}_c\alpha$ in this section impose an ordering of the strengths of ${}_c\alpha_{\text{metric}}$, successively responding to qualities of data that a weaker one omits:

FIGURE 12.7 ■ ${}_c\alpha_{\text{metric}}$ Ordered by What Adds to Their Strength



Relative to the weakest ${}_c\alpha_{\text{nominal}}$, the strength of the ordinal metric difference function exceeds that of the nominal one by being responsive to the ordering of values, to which the interval metric difference function adds the strength of their algebraic differences, and to which the ratio metric difference adds its sensitivity to how far values deviate from their absolute zero points (as in ages, lengths, proportions, and incomes). If the reliability of data is measured by an ${}_c\alpha$ -coefficient that exceeds the metric strength required for an intended data analysis, it responds to data qualities that do not play a role in their subsequent analysis. Relying on an agreement measure whose strength exceeds that of how the data are analyzed in fact can be deceptive. If the reliability measure is weaker than that required by the intended analytical method, it would omit data qualities that may be crucial in affecting the outcome of a research project, but omission is at least not deceptive. In sum, try to use reliability measures that respond to all the qualities an analysis attends to but never more.

12.3.4 Some Typical Examples and Important Properties

12.3.4.1 Binary Data, None Missing

Binary data are generated when observers decide whether a quality is present or not, which o alternative applies, say yes or no to a question, here described by recording 0 or 1 to each of a predefined set of N units of analysis. For example, when a political scientist is merely interested in whether any kind of references to the United States are made in an article printed, say, in the Croat press. Unable to read Croatian, the political scientist hires three supposedly native speakers, Jon, Han, and Lee, to identify the presence or absence of such references in 12 issues of a Croat newspaper. They are given the same instructions to mark each of the 12 issues “0” for absent and “1” for present after reading and judging the newspaper issues independently from each other. Their reliability data might look as tabulated in their canonical form of a 3 Observer-by-12 Units matrix, containing the records of their judgments:

Units u :	1	2	3	4	5	6	7	8	9	10	11	12
Jon:	0	1	0	1	1	0	0	0	1	1	1	1
Han:	0	1	0	1	0	0	0	0	1	1	0	1
Lee:	0	1	1	0	0	1	0	0	1	0	0	0

This reliability data matrix tells us that Jon and Han agree on their identification of references 10 out of 12 times, or 83%. Han and Lee agree on 6 out of 12 times, or 50%, and Jon and Lee also agree 50% but on different units. Overall agreement is not so obvious. Moreover, %-agreement does not tell us anything about how reliable these data are and what could be expected when the assigned values are the product of chance. One could expect chance if the hired coders may not be as linguistically competent as they claim to be, do not attend to required details of the phenomena to be identified, or throw dice. Statistically, 0% agreement is about as unlikely to achieve as is 100%, and 50% equals chance only when 0s and 1s are equally likely. In this example, this is not the case. Percent-agreement is often used to talk about the reliability between two observers, but it has no valid reliability interpretations, not even when it measures 100%. Without reference to the variance in the data and to chance, %-agreement is simply uninterpretable as a measure of reliability—regardless of its popularity in the literature.

Whether one follows the direct path to ${}_c\alpha$ or the path via coincidence matrices, both call for constructing a values-by-units matrix:

Units u :	1	2	3	4	5	6	7	8	9	10	11	12	$n_{\cdot c}$
Values: 0	3	0	2	1	2	2	3	3	0	1	2	1	20
1	0	3	1	2	1	1	0	0	3	2	1	2	16
m_u :	3	3	3	3	3	3	3	3	3	3	3	3	$36 = n_{\cdot\cdot} = mN = 3 \cdot 12$

Because no values are missing, the column sums are the same for each unit: $m_u = 3 = m$ observers. There are $n_{\cdot 0} = 20$ units valued 0, $n_{\cdot 1} = 16$ units valued 1, and $n_{\cdot\cdot} = 36$ values in total.

The political scientist may be convinced that the three coders know the many ways the United States can be referred to but has no clue whether they could understand her English coding instructions and apply them to check texts, whether they read the newspapers carefully, or whether they had an unarticulated agenda. However, all of these considerations are secondary to their effects on the agreement achieved among their claims to have identified references to the United States.

Following the direct path to ${}_c\alpha$, suggested in section 12.3.1, and recognizing that for binary data all metrics produce the same ${}_c\alpha$, under these extremely simple conditions, applying (5) to these data yields:

$${}_c\alpha_{\text{binary}} = 1 - (36 - 1) \frac{0 + 0 + \frac{2 \cdot 1}{3 - 1} + \frac{1 \cdot 2}{3 - 1} + \frac{2 \cdot 1}{3 - 1} + \frac{2 \cdot 1}{3 - 1} + 0 + 0 + 0 + \frac{1 \cdot 2}{3 - 1} + \frac{2 \cdot 1}{3 - 1} + \frac{1 \cdot 2}{3 - 1}}{20 \cdot 16} = 0.234$$

Alternatively, after constructing the two coincidence matrices, the observed coincidence o_{ek} from the entries in the preceding 2 values-by-12 units matrix by (6):

$$\begin{aligned} o_{00} &= \frac{3(3-1)}{3-1} + 0 + \frac{2(2-1)}{3-1} + \frac{1(1-1)}{3-1} + \frac{2(2-1)}{3-1} + \frac{1(1-1)}{3-1} = 13 \\ o_{01} = o_{10} &= 0 + 0 + \frac{2 \cdot 1}{3-1} + \frac{1 \cdot 2}{3-1} + \frac{2 \cdot 1}{3-1} + \frac{2 \cdot 1}{3-1} + 0 + 0 + 0 + \frac{1 \cdot 2}{3-1} + \frac{2 \cdot 1}{3-1} + \frac{1 \cdot 2}{3-1} = 7 \\ o_{11} &= 0 + \frac{3(3-1)}{3-1} + \frac{1(1-1)}{3-1} + \frac{2(2-1)}{3-1} + \frac{1(1-1)}{3-1} + \frac{1(1-1)}{3-1} + \dots + \frac{2(2-1)}{3-1} = 9 \end{aligned}$$

and the expected coincidences by (8):

$$e_{00} = \frac{20(20-1)}{36-1} = 10.857 \quad e_{01} = e_{10} = \frac{20 \cdot 16}{36-1} = 9.143 \quad e_{11} = \frac{16(16-1)}{36-1} = 6.857$$

yields the following two coincidence matrices:

Observed coincidences				Expected coincidences			
Values:	0	1		Values:	0	1	
0	13	7	20	0	10.86	9.14	20
1	7	9	16	1	9.14	6.86	16
	20	16	36		20	16	36

The ${}_c\alpha$ -agreement in (10) reduces to its binary form:

$${}_c\alpha_{\text{binary}} = 1 - \frac{{}_cD_o}{{}_cD_e} = 1 - (n_{..} - 1) \frac{o_{01}}{n_{0.}n_{.1}} = 1 - \frac{o_{01}}{e_{01}} \quad (16)$$

and for our example:

$${}_c\alpha_{\text{binary}} = 1 - \frac{{}_cD_o}{{}_cD_e} = 1 - (36 - 1) \frac{7}{20 \cdot 16} = 1 - \frac{7}{9.1429} = 0.234$$

Obviously the two computational paths come to the same conclusions. However, I would contend that when comparing the entries in the two coincidence matrices, one

could already see that the numbers of observed matches are only slightly larger than those expected, suggesting that reliability for this binary coding task is very low, by all standards not sufficient to rely on the data these coders generated. In this case, one can say that for the $(7+7)/36$ or 39% disagreements in the data, the political scientist would not know whether newspapers contained or did not contain references to the United States, and of the $(13+9)/36$ or 61% agreements, $(10.86+6.86)/36$ or 49% are due to chance. Worse, our political scientist would never know which of the matching references had merits and which should be discounted as the product of chance, totally unrelated to whether the United States was referred to. Although from a small sample of data like these, one could not possibly draw any serious conclusions (see section 12.8.3 for required sample sizes). If the reliability data would have been of an adequate sample size, decisions among answers to a research question based on such data would most likely have been close to chance. This is what $\alpha = 0.234$ indicates.

There are two general properties of α that binary data can convincingly demonstrate: the relationship between coincidences and contingencies and how α partitions the observed coincidences.

12.3.4.2 Coincidences and Contingencies

The coincidence matrices defined in section 12.3.2 and shown in Figures 12.5 and 12.6 should not be confused with the better-known contingency matrices, which are at home in the tradition of assessing statistical associations or correlations between two variables:

- Contingency matrices tabulate units of analysis in terms of the values assigned to them. Their totals sum to the number of units observed. By contrast, coincidence matrices tabulate pairs of values, and their sums are the total number of values used by observers to record what they observed, read, valued, or measured on the units they were given.
- The entries in contingency matrices are asymmetrical: A unit valued c by one observer i and k by another observer j is not considered the same when i values that unit k and j takes it to be c . It follows that contingencies are $x_{ck} \neq x_{kc}$. By contrast, the entries of coincidence matrices are symmetrical. Coincidences are $o_{ck} = o_{kc}$, which takes observers to be interchangeable.
- Calculating expected agreements like expected contingencies from the unequal marginal sums of contingency matrices counts observers' unequal proclivities for particular values as agreement! Calculating expected agreements from the common marginal sums of coincidence matrices counts these differences as disagreement.
- Contingency matrices preserve the identity of two observers, coincidence matrices do not. Confusing the purpose of establishing data reliability with how observers correlate (or agree) is perpetuated by the semantic entailments of calling such coefficients "*inter-coder agreement coefficients*." This can and has indeed diverted researchers' attention from the reliability of data to the ease with which observers agree (e.g., Zhao, Liu, & Deng 2012; Feng, 2015).
- The marginal sums of contingency matrices are unrelated to the population of data that reliability needs to assess. By contrast, the marginal sums of

coincidence matrices are the best estimates one can possibly get of the probability distribution of the data whose reliability is in question.

- While it is possible to enumerate observed agreements from contingency matrices for two observers, this possibility is not obviously extendable to many observers, least of all when their contributions to the reliability data are uneven, when data are incomplete, or values are missing.
- The variables between which correlations are calculated are considered conceptually independent. Inasmuch as observers are expected to follow common coding instructions, they conceptually depend on being given identical instructions. Coincidence matrices acknowledge observers to apply common instructions. The requirement that observers work independent of each other is a prerequisite of enumerating matches or disagreements.
- Contingency-like matrices are legitimate representations of reliability data when establishing accuracy or surrogacy, distinguished from replicability in Table 12.1 and developed in section 12.5.

When reliability data are complete, that is all units are coded by all observers, there is a simple relationship between contingency matrices and coincidence matrices. The latter is the sum of all contingencies between all pairs of observers, including their inverses, averaged so that the coincidence matrices sum to the number of pairs in the reliability data. To start with the example of 3 observers, judging the presence or absence of references to the United States in 12 publications, the contingencies between Han and Lee are:

		Contingencies					Coincidences		
Han's values:		0	1		Values:		0	1	
Lee's values	0	5	3	8	0	10	5	15	
	1	2	2	4	1	5	4	9	
		7	5	12			15	9	24

With x_{ck} as the number of units categorized c by one observer and k by another, as tabulated in contingency matrices, and o_{ck} as the number of coincidences of c and k in coincidence matrices, one should be able to verify from Han and Lee's contingencies that their coincidences are the sums of contingencies and their inverses:

$$o_{ck} = o_{kc} = x_{ck} + x_{kc}$$

For the binary reliability data, generated by the three observers Jon, Han, and Lee, there are $3(3 - 1) = 6$ possible contingency matrices between pairs of observers including their inverses. To result in a contingency matrix that enumerates pairs of values, we have to divide the cell sums by $(m - 1) = (3 - 1)$, visualized in the following:

$$\begin{array}{c} \text{Han} \end{array} \begin{array}{|c|c|} \hline \text{Jon} \\ \hline 5 & 2 \\ \hline 0 & 5 \\ \hline \end{array} + \begin{array}{c} \text{Jon} \end{array} \begin{array}{|c|c|} \hline \text{Han} \\ \hline 5 & 0 \\ \hline 2 & 5 \\ \hline \end{array} + \begin{array}{c} \text{Lee} \end{array} \begin{array}{|c|c|} \hline \text{Jon} \\ \hline 3 & 5 \\ \hline 2 & 2 \\ \hline \end{array} + \begin{array}{c} \text{Jon} \end{array} \begin{array}{|c|c|} \hline \text{Lee} \\ \hline 3 & 2 \\ \hline 5 & 2 \\ \hline \end{array} + \begin{array}{c} \text{Lee} \end{array} \begin{array}{|c|c|} \hline \text{Han} \\ \hline 5 & 3 \\ \hline 2 & 2 \\ \hline \end{array} + \begin{array}{c} \text{Han} \end{array} \begin{array}{|c|c|} \hline \text{Lee} \\ \hline 5 & 2 \\ \hline 3 & 2 \\ \hline \end{array} = \frac{\begin{array}{|c|c|} \hline 26 & 14 \\ \hline 14 & 18 \\ \hline \end{array}}{3-1} = \frac{\begin{array}{|c|c|} \hline 26 & 14 \\ \hline 14 & 18 \\ \hline \end{array}}{2} = \begin{array}{|c|c|} \hline 13 & 7 \\ \hline 7 & 9 \\ \hline \end{array}$$

12.3.4.3 $\epsilon\alpha$'s Decomposition of Observed Coincidences

For nominal data, $\epsilon\alpha$ can be conceptualized as a function of the entries of three coincidence matrices: of the observed coincidences, of the expected coincidences, and of the ideal coincidences with all values perfectly matching. Graphically, it appears as follows:

$$\begin{bmatrix} o_{00} & o_{01} \\ o_{10} & o_{11} \end{bmatrix} = \epsilon\alpha \begin{bmatrix} n_0 & \\ & n_1 \end{bmatrix} + (1 - \epsilon\alpha) \begin{bmatrix} e_{00} & e_{01} \\ e_{10} & e_{11} \end{bmatrix} \quad (17)$$

(17) suggest a third interpretation of $\epsilon\alpha$:

$\epsilon\alpha$ is the extent to which the ideal of perfect reliability is present in the observed coincidences.

Its algebraic relationship can be illustrated with our binary reliability data:

$$\begin{bmatrix} 13 & 7 \\ 7 & 9 \end{bmatrix} = 0.234 \begin{bmatrix} 20 & \\ & 16 \end{bmatrix} + (1 - 0.234) \begin{bmatrix} 10.86 & 9.14 \\ 9.14 & 6.86 \end{bmatrix}$$

$$\begin{bmatrix} 13 & 7 \\ 7 & 9 \end{bmatrix} = \begin{bmatrix} 4.68 & \\ & 3.74 \end{bmatrix} + \begin{bmatrix} 8.32 & 7.00 \\ 7.00 & 5.26 \end{bmatrix}$$

With this interpretation of $\epsilon\alpha$, the data suggest that of the 36 possible values that could be agreed upon, only $4.68 + 3.74 = 8.42$ out of a total of 36 pairs of values can be trusted, without knowledge of which they are. The algebraic relationship (17) applies to all nominal data. It should be noted that when more than two values are involved, $(1 - \epsilon\alpha)$ does not predict the individual coincidences in off-diagonal cells, only their sums, and $\epsilon\alpha$ does not predict the distribution of matching pairs in the diagonal cells, only how many will be there.

12.3.4.4 Two Observers, Several Nominal Values, No Missing Data

Let this rather common situation be exemplified by the canonical form of the following reliability data:

Units u =	1	2	3	4	5	6	7	8	9	10	11	12
Mary=	<i>a</i>	<i>a</i>	<i>c</i>	<i>c</i>	<i>c</i>	<i>c</i>	<i>c</i>	<i>b</i>	<i>b</i>	<i>b</i>	<i>b</i>	<i>d</i>
Dave=	<i>a</i>	<i>c</i>	<i>c</i>	<i>c</i>	<i>c</i>	<i>c</i>	<i>b</i>	<i>b</i>	<i>b</i>	<i>b</i>	<i>b</i>	<i>d</i>

By summing over the two observers, we obtain the values-by-units matrix containing the frequencies n_{uc} with which value *c* occurs in unit *u* as in Figure 12.4:

Units u :		1	2	3	4	5	6	7	8	9	10	11	12	$n_{.c}$
Values:	a	2	1	0	0	0	0	0	0	0	0	0	0	3
	b	0	0	0	0	0	0	1	2	2	2	2	0	9
	c	0	1	2	2	2	2	1	0	0	0	0	0	10
	d	0	0	0	0	0	0	0	0	0	0	0	2	2
m_u :		2	2	2	2	2	2	2	2	2	2	2	2	$24 = n_{..}$

Because there are $m=2$ observers, each assigns one value to each of $N=12$ units u , and no data are missing, the total number of pairable values in these matrices is $n_{..}=2N=24$. With (5), ${}_c\alpha_{\text{nominal}}$ can be obtained directly from this values-by-units matrix:

$${}_c\alpha_{\text{nominal}} = 1 - (24 - 1) \frac{0 + \frac{11}{2-1} + 0 + 0 + 0 + 0 + \frac{11}{2-1} + 0 + 0 + 0 + 0 + 0}{3(9+10+2) + 9(10+2) + 10 \times 2} = 0.760$$

That there are only two positive quantities in the numerator and six in the denominator may be seen more clearly in the off-diagonal cells of the following observed coincidence matrix summing the units in the values-by-units matrix used in this example:

		Observed coincidences						Expected coincidences			
Values:		a	b	c	d	Values:		a	b	c	d
	a	2	0	1	0		a	0.26	1.17	1.30	0.26
	b	0	8	1	0		b	1.17	3.13	3.91	0.78
	c	1	1	8	0		c	1.30	3.91	3.91	0.87
	d	0	0	0	2		d	0.26	0.78	0.87	0.09
		3	9	10	2			3	9	10	2
		24						24			

Apparently, the expected coincidences exceed the observed, in fact by a ratio of 4 to 1, which is also the ratio of the observed to the expected matches, providing intuitive evidence of agreement well above chance. Quantitatively and with expression (10):

$${}_c\alpha_{\text{nominal}} = 1 - \frac{1+1}{1.174+1.304+0.261+3.913+0.783+0.870} = 0.760$$

For complete nominal data, and only for these, there exists a well-known expression which is computationally convenient for hand calculations. It results from the algebraic fact that for such data, the total number $n_{..}$ of pairs of values equals the number of agreements A = the sum of the matching pairs in the diagonal of a coincidence matrix—plus the number of disagreements D = the sum of the mismatching

pairs in the off-diagonal triangles of a coincidence matrix. This enables defining ${}_c\alpha_{\text{nominal}}$ in terms of three agreements A :

$$A_{\text{max}} = A + D = n.. = \sum_{c=1}^v \sum_{k=1}^v o_{ck} = \sum_{c=1}^v \sum_{k=1}^v e_{ck}$$

$$A_o = A_{\text{max}} - {}_cD_o = \sum_{c=1}^v o_{cc} = \text{The sum of the diagonal entries in the matrix of observed coincidences}$$

$$A_e = A_{\text{max}} - {}_cD_e = \sum_{c=1}^v e_{cc} = \text{The sum of the diagonal entries in the matrix of expected coincidences}$$

Whereupon:

$${}_c\alpha_{\text{nominal}} = 1 - \frac{{}_cD_o}{{}_cD_e} = \frac{A_o - A_e}{A_{\text{max}} - A_e} \quad (18)$$

For complete nominal data, (18) offers a fourth interpretation of ${}_c\alpha$:

${}_c\alpha$ is the proportion of the above chance observed agreement to the largest possible above chance agreement.

In terms of the 3-by-12 reliability data, which satisfy the condition for (18) to be applicable—nominal and complete data:

$${}_c\alpha_{\text{nominal}} = 1 - \frac{{}_cD_o}{{}_cD_e} = \frac{A_o - A_e}{A_{\text{max}} - A_e} = \frac{\sum_{c=1}^v o_{cc} - \sum_{c=1}^v e_{cc}}{n.. - \sum_{c=1}^v e_{cc}} = \frac{(2+8+8+2) - (0.261+3.130+3.913+0.087)}{24 - (0.261+3.130+3.913+0.087)} = 0.760$$

Owing to the fact that the diagonals in coincidences always contain fewer cells than their off-diagonal triangles, (18) calls for fewer additions, hence its usefulness in hand calculations and checks.

For just two observers, (18) resembles the form of Scott's (1955) $\pi = \frac{P_o - P_e}{1 - P_e}$ (pi)

and Cohen's (1960) popular but misleading κ (kappa)—misleading because its P_e is calculated from contingencies, not coincidences. This is addressed in section 12.3.5. The generalization of the form used by Scott and Cohen to more than two observers is fairly limited which is why ${}_c\alpha$ is pursuing the generalizable form (1).

12.3.4.5 Uninformative Values in Ordered Variables

In efforts to define a variable exhaustively—that is, so that all units of analysis can be coded, and none are left out just because the coding instrument does not have a place for them—it is customary to add a category to code all units with unanticipated, undefined, or uncertain qualities, typically labeled “NA,” “Not applicable,” “No answer,” “Other,” “None of the above,” “Ambiguous,” or “Undecidable.” Such a category does not provide information about what a content analyst had in mind studying, exhibits no metric qualities, cannot be correlated with any aspect of a guiding research question, and worse, the ability to assess the reliability of such data is limited to the nominal metric. For example, when rating objects with semantic differential scales, which have six equal intervals ranging from -3 to $+3$, and deserving to be assessed as such, the addition of “NA” would prevent the calculation of interval agreements, and if ${}_c\alpha$ would apply the nominal difference function, the reliabilities that neighborhood relations of scale values could provide are ignored. To compute statistics on data that contain no-information values, analysts use a variety of ways

to get “NA” at least into a software, for example by replacing it with an odd number such as “999.” Then one indeed could calculate $c\alpha_{\text{interval}}$ but obtain an uninterpretable reliability measure. It would respond to great differences between the legitimate values, for example, between -3 and $+3$, rendering the resulting agreement measure uninterpretable.

When such uninformative values occur in ordered variables, in data of ordinal, interval, or ratio metric qualities, then it is advisable to obtain two separate agreement coefficients, evaluating $c\alpha_{\text{binary}}$ for the reliability of distinguishing between relevant and irrelevant qualities and $c\alpha_{\text{metric}}$ of the needed ordered data, in the expectation that data coded as not applicable will not enter subsequent analyses.

This can be accomplished by breaking the reliability data in Figure 12.4 into two tables, one containing just two values, NA, and all relevant values lumped into a single category, and the other containing only the informative values wanted for subsequent analysis.

In terms of coincidence matrices, this task can be demonstrated by the following numerical example:

Original observed coincidences

	NA	-2	-1	0	1	2
NA	12	4	8	4	10	6
-2	4	18				
-1	8		20	3		
0	4		3	10	1	
1	10			1	43	2
2	6				2	17

$c\alpha_{\text{nominal}} = 0.521$

(a)

NA	I={-2, -1, 0, 1, 2}
12	32
32	120

$c\alpha_{\text{binary}} = 0.067$

(b)

I

	-2	-1	0	1	2
-2	18				
-1		20	3		
0		3	10	1	
1			1	43	2
2				2	17

$c\alpha_{\text{interval}} = 0.972$

Evaluating these reliability data, including NA, limits the computation of agreement to the use of the nominal metric difference function. It yields $c\alpha_{\text{nominal}} = 0.521$. Visual inspection of the distribution of its observed coincidences suggests that the no-information value NA is used rather often and disagrees with all information bearing values, suggesting that observers have great uncertainties with the information bearing values as defined. A distribution of this kind is not atypical. Some observers select NA as soon they are a bit uncertain about which other values to choose. Others struggle harder to avoid that category but increase the disagreement among the information bearing values. My recommendation is to avoid NA-type categories and instead provide for all possibilities of a variable. Nevertheless, to get to the reliability of the values that count, we (a) construct a 2-by-2 coincidence matrix with values NA and I, which lumps all information bearing values into one category, $I = \{-2, -1, 0, 1, 2\}$ and (b) obtain a submatrix from the original matrix, containing only information bearing values. As it turns out, the binary distinction

between NA and all information bearing values I measures $c\alpha_{\text{binary}} = 0.067$, whereas the information bearing interval values yields $c\alpha_{\text{interval}} = 0.972$. The decision to eliminate the NA-value from any further analysis is obvious. However, its omission also means that the distinction among relevant and irrelevant units is not reportable as a finding. It precludes making reliable claims about the proportions of the information bearing values in the data whose reliability is being tested.

The above procedure is not limited to the removal of conceptual default categories. It is possible to disregard values that the analyst defined in order for a variable to be exhaustive of all of its possible occurrences, but answering a research question requires not the whole set. The point is that values that matter need to be demonstrably reliable.

12.3.4.6 Trading Information for Reliability

The conceptual advantage of reliability data in the form of coincidence matrices should by now be demonstrated. It is relatively easy to identify patterns of disagreement in these matrices and decide which values need better definitions, who is confusing which, and whether some values may be dropped for the reliability gains of the remaining set.

I want to comment on a way to trade information that a researcher may have thought to be important but could be sacrificed for higher reliability: Semantic ambiguity or the polysemy of values. Ambiguity leaves observers uncertain as to which of several equally applicable values to use to record a phenomenon. Most words are polysemous in the sense of having several distinct meanings—see the number of meanings a dictionary lists of any one word. Polysemy tends to be resolved by considerations of the context in which words occur. However, in the content analysis of predefined units, the context needed to disambiguate polysemous expressions may be too small, leaving behind typical patterns of confusions. The following coincidence matrix exhibit several kinds:

Values:

	a	b	c	d	e	f	g
a	54	0	0	0	0	0	0
b	0	12	1	16	0	0	1
c	0	1	30	0	0	28	0
d	0	16	0	22	0	0	0
e	0	0	0	0	122	0	0
f	0	0	28	0	0	34	1
g	0	1	0	0	0	1	16

$c\alpha_{\text{nominal}} = 0.699$

Lumped values:

	a	b&d	e	c&f	g
a	54	0	0	0	0
b&d	0	66	0	1	1
e	0	0	122	0	0
c&f	0	1	0	120	1
g	0	1	0	1	16

$c\alpha_{\text{nominal}} = 0.979$

The matrix on the left measures $c\alpha_{\text{nominal}} = 0.699$ —generally considered unacceptable. Visual inspection reveals disagreements in several categories that could be the cause of the low agreement. Five categories could have been responsible: **b**, **c**, **d**, **f**, and **g** but only minimally. Disagreement is absent for **a** and **e**. However, the observed disagreements are far from randomly distributed. Disagreements between categories

b and **d** are localized by occurring mostly among themselves, that is, in the **c-d** and **d-c** cells whose coincidences are highlighted in bold. Lumping truly ambiguous categories tends to improve reliability while reducing the variety of categories available for subsequent analysis—a loss of information. The other ambiguity is evident in categories **c** and **f**, whose coincidences are highlighted in italics. Evidently, the observers involved fail to distinguish between these two pairs of values as well and use them as if they were synonymous. Instead of discarding the whole variable as unreliable, or eliminating its unreliable values, lumping categories, **b&d** and **c&f** in the coincidence matrix on the right, eliminates these two systematic confusions and yields a respectable $\epsilon\alpha_{\text{nominal}} = 0.979$ but also a smaller set of analyzable values.

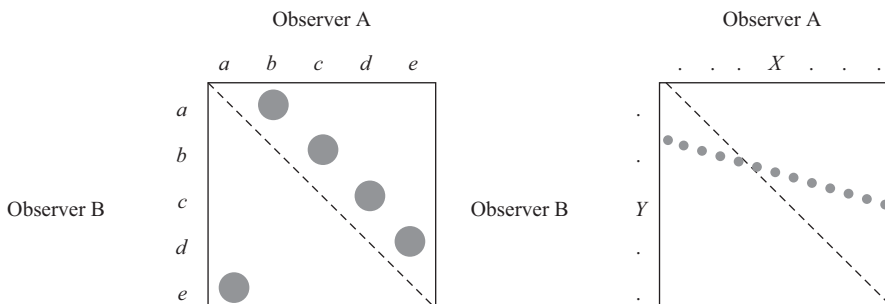
This example concerns nominal data, which allow categories to be freely permutable and lumpable out of any order. When lumping ordered data, one has to take their metric seriously. For one, only neighboring values can be lumped. If this is justified for interval data, lumping neighboring values destroys the interval metric nature of the data and requires their subsequent analysis to consider them ordinal values, which have no problem with merging neighboring ranks. There are other options to salvage data in which disagreements are recognizably systematic (Krippendorff, 1970b, 2008), discussion of which goes beyond this presentation of $\epsilon\alpha$.

12.3.5 Some Contested Coefficients and Correspondences

In a survey of content analyses published in *Journalism & Mass Communication Research* from 1971 through 1995, Riffe and Freitag (1996; cited in Riffe, Lacy, & Fico, 1998) found that only 56% reported assessments of reliability. In their review of the consumer research literature from 1981 through 1990, Kolbe and Burnett (1991) noted similar deficiencies: 31% of the published content analyses showed no concerns for reliabilities; 19% mentioned reliability without revealing any method of calculation; 35% reported %-agreement, including Holsti's; 8% used one of several less well-known measures (each mentioned only once or twice in the sample); and 7% used Krippendorff's α . (The share of α has increased drastically since the above-mentioned software became available; Hayes & Krippendorff, 2007). Lombard et al. (2002) found reliability discussed in 69% of 200 content analyses indexed in *Communication Abstracts* from 1994 through 1998. These are discouraging findings. In this section, I review the most popular agreement indices found in the literature and discuss their shortcomings and relationships to the $\epsilon\alpha$ coefficient.

Some content analysts make the common but serious mistake of considering the performances of individual observers as variables; applying readily available association, correlation, or consistency measures to their contingency matrix representations; and interpreting these as measures of reliability. Correlation coefficients—Pearson's product-moment r_{ij} , for example—measure the extent to which two logically independent interval variables, say X and Y , covary in a linear relationship of the form $Y = a + bX$. They indicate the degree to which the values of one variable predict the values of the other. Agreement coefficients, by contrast, must measure the extent to which $Y = X$. High correlation means that data approximate *any* regression line, whereas high agreement means that they approximate the 45-degree line. If one observer is consistently, say, two points behind the other, or they follow a regression line as suggested by the gray dots in the right of the following two contingency matrices, correlation is perfect, but agreement is not. The same

holds for cross-tabulations of nominal data. In the contingency matrix on the left, ● signifies a nonzero frequency of co-occurring categories. There is a 1-to-1 relationship between the two sets of categories, association is perfect, and the use of categories by one observer is perfectly predictable from that by the other. But as categories do not match, there is no agreement whatsoever.



Although correlation coefficients have been used as reliability measures, even recommended (Potter & Levine-Donnerstein, 1999, p. 277), especially in the literature on psychological testing (see the otherwise informative exposition of reliability and validity issues by Carmines & Zeller, 1979; and a review by Salgado & Moscoso, 1996), the above should make clear that in content analysis and other data generating efforts, their use is seriously misleading.

In section 12.3.1, I already criticized the use of %-agreement—or “crude” agreement, as this measure is sometimes called. Yet %-agreement measures creep into the literature in different guises. Holsti (1969, p. 140), for example, describes Osgood’s (1959, p. 44) reliability index as $2M/(N_1 + N_2)$, wherein M is the number of units on whose categorizations two readers agree, N_1 is the number of units identified by one reader, and N_2 is that number identified by another. Although Holsti presents pertinent criticism of %-like agreement indices, citing Bennett, Alpert, and Goldstein’s (1954) arguments and building up to his recommendation of Scott’s (1955) π (pi), it is amazing how many content analysts still overlook the by now widely published objections to this uninterpretable agreement measure. For example, Neuendorf (2002) and Lombard et al. (2002), instead of discouraging %-agreement and its relatives, discuss it as a (perhaps too) “liberal” alternative.

Bennett et al. (1954) were probably the first to realize that %-agreement A_o was the more difficult to achieve the more categories there were available for coding. They proposed a coefficient $S = (A_o - 1/K)/(1 - 1/K)$ that corrects for the number K of categories available for coding. While coder difficulty should not be a criterion for judging data to be reliable or not, it is remarkable that this coefficient has been reinvented at least five times since it was originally proposed: as Guilford’s G (Holley & Guilford, 1964), as the RE (random error) coefficient (Maxwell, 1970), as Janson and Vegelius’s (1979) C , as k_n (Brennan & Prediger, 1981), and, most recently, within the intercoder reliability coefficient I_r (Perreault & Leigh, 1989). Perreault and Leigh (1989) were at least aware of S . Proponents of this coefficient cite reasons ranging from the mounting difficulties of selecting among increasing number categories, fairness to each category, consistency with the research traditions of their disciplines, to the

absence of hard knowledge about the true distribution of categories in the population from which reliability data were sampled. In treating all categories as equally likely, S inflates agreement when used unevenly, especially when some categories are not used at all. The latter enables researchers to manipulate S in their favor by adding unused or rarely used categories to the set. Perreault and Leigh argue that chance-corrected agreement coefficients, such as Cohen's κ , are too conservative, whereas S (or their I_r), they say, is not. For arguments against this assessment, see Krippendorff (2004b).

In response to S 's shortcomings, Scott (1955, also in Krippendorff & Bock, 2009, pp. 347–349) proposed his reliability index π (pi), which resembles one of the forms of α_{nominal} in (18), limited to two observers and assuming infinite sample sizes:

$$\pi = \frac{P_o - P_e}{1 - P_e}$$

where $P_o = A_o / n..$ is the proportion of units with matching categories (%-agreement), and $P_e = A_e / n.. = \sum_{k=1}^v \bar{p}_k^2$ is the proportion of pairs of values that are expected to match by chance. It is expressed in terms of the average proportions $\bar{p}_k$ of values used by two observers.

In effect, $\bar{p}_k$ estimates the proportion of units valued k in the population of the data whose reliability is under evaluation. P_e treats observers as interchangeable, as required for assessing the reliability of data, and does consider the collective judgment of two observers as the best estimates of the population proportions. Thus, Scott's P_e is the agreement that can be expected to occur in the population when chance prevails.

Subsequent to the publication of Scott's π , Cohen (1960) introduced an unfortunate modification of π into the literature, trying to bring agreement measurement closer to conventional contingency approaches, as he said, and calling it κ (kappa). This coefficient is popular in biomedical and educational research, where most of its proponents' work, but is inappropriate in assessing reliability in content analysis, as we shall see. Cohen merely replaced Scott's expected agreement P_e with a proportion that conforms to the tradition of association statistics, which I call $p_c = \sum_{k=1}^v p_{Ak} p_{Bk}$, A referring to one observer and B to the other. κ is defined by:

$$\kappa = \frac{p_o - p_c}{1 - p_c} = \frac{\sum_{k=1}^v p_{kk} - \sum_{k=1}^v p_{Ak} p_{Bk}}{1 - \sum_{k=1}^v p_{Ak} p_{Bk}}$$

where $P_o = p_o$ is the proportion of units with matching categories, equal to the %-agreement in Scott's π , but p_c is unlike Scott's P_e . Cohen's p_c is the sum of the products of the proportion p_{Ak} of values k used by observer A , and the proportion p_{Bk} of the same value k used by the other observer, B . p_c is the %-agreement that can be expected when the two observers are statistically independent, whereas P_e is the %-agreement that can be expected when the values assigned by two observers are statistically unrelated to the units coded. Statistical independence among observers is only marginally related to the independence of values and the phenomena valued.

Fleiss (1971) sought to generalize Cohen's κ to a fixed number of raters, but, perhaps unknowingly, generalized Scott's π instead, calling it *kappa*, here denoted K , as suggested by Siegel and Castellan (1988, p. 286). For two observers, K equals π , requiring no special attention to K here.

A numerical example may demonstrate how P_o , π , κ , and ${}_c\alpha$ differ in what they respond to. Consider two contingency tables containing the frequencies of units recorded by two observers:

		Observer A						Observer A			
Categories:		a	b	c		Categories:		a	b	c	
Observer B	a	12	9	9	30	Observer B	a	12	18	18	48
	b	9	14	9	32		b	0	14	18	32
	c	9	9	20	38		c	0	0	20	20
		30	32	38	100			12	32	56	100
		$P_o = 0.460$						$P_o = 0.460$			
N assumed ∞		$\kappa = 0.186$						$\kappa = 0.258$			
N assumed ∞		$\pi = 0.186$						$\pi = 0.186$			
$N \geq 600$:		${}_c\alpha = 0.186$						${}_c\alpha = 0.186$			
$N = 100$:		${}_c\alpha = 0.190$						${}_c\alpha = 0.190$			

Both tables show reliability data that have the same %-agreement $P_o = 0.460$ out of $n_{..}=100$ in their diagonal. But they differ in how mismatches are distributed, and in the two observers' marginal sums of frequencies. In the table on the left, observers agree on marginal frequencies and Scott's π and Cohen's κ are the same, as they should be. But when the marginal frequencies of the two coders disagree, as is apparent in the table on the right, κ exceeds π , suggesting that there would be more agreement, which is far from so. There are still only 46 units in the diagonal cells. How can κ be so mistaken? Note that the 54 mismatches in the left table populate both off-diagonal triangles, whereas in the right table they are found in only one. The increase is not in %-agreement but in an increase in the predictability of matching values from unequal observer proclivities. Unlike κ , π and α are not affected by where the mismatching values occur. When generating content analysis data, it should not matter indeed who contributed which disagreements and, when data are nominal, which categories are confused. Moreover, predictability has nothing to do with reliability. Thus, when the distributions in the two marginal sums in a contingency table are unequal, κ adds a measure of this disagreement to the agreement it is supposed to measure! κ overestimates reliability whenever coders have unequal preferences for coding categories.

It should be pointed out that Cohen (1960), in his proposal of κ , falsely criticized Scott for ignoring "one source of disagreement between a pair of judges [due to] their proclivity to distribute their judgments differently over the categories" (p. 41). His proposal to modify π achieved just the opposite. His κ counts disagreements among observer preferences for available categories as agreements, contrary to what Cohen claimed. This is a major conceptual flaw. Brennan and Prediger (1981) describe this property of κ by pointing out that "two judges who independently, and without prior knowledge, produce similar marginal distributions must obtain a much higher agreement rate to obtain a given value of kappa, than two judges who

produce radically different marginals.” The first two judges “are in a sense penalized” for agreeing on marginal frequencies (p. 692). Many proponents of κ reproduce Cohen’s false claim without verification. Zwick (1988), citing others, mentions this flaw as well and suggests testing for marginal homogeneity before computing κ , but this merely patches up κ ’s obvious inadequacies.

The above reliability data also demonstrate the effects of small sample sizes. k and π assume samples to be fixed and of infinite size while α is sensitive to various sample sizes. If n values are used by the coders, Scott’s and Fleiss’s expected %-agreement, $P_e = \sum_k \bar{p}_k^2$ includes the proportion $1/n..$ of values paired with themselves. That proportion is not due to chance but an unwanted artifact of P_e ’s definition. It follows that P_e overestimates the expected agreement and artificially reduces π (and Fleiss’s K as well) below the actual observer accomplishments. By contrast, ${}_c\alpha$ excludes the proportion $1/n..$ of self-paired values in its expected disagreement D_e (see expression (4) in section 12.3.1, (10) in section 12.3.2, and (15) in section 12.3.3) and is not biased by the way expectations are estimated. ${}_c\alpha$ pairs values without replacement in both D_o and D_e . π and K pair values for P_o without replacements as well but pair values for P_e with replacement. As sample sizes increase, in this example with α expressed in three significant digits, already by $N \geq 600$, ${}_c\alpha$ and π become asymptotically indistinguishable.

The structural differences between the most popular agreement coefficients (Kolbe & Burnett, 1991; Lombard et al., 2002; Neuendorf, 2002) plus a recent proposal by Gwet (2002) can be seen most clearly when reduced to their simplest binary or dichotomous forms. I state these in terms of a 2-by-2 contingency matrix (not the coincidence matrices discussed in section 12.3.2) containing proportions a , b , c , and d of N units, coded 0 or 1 by the two observers.

Observer A’s Values:		0	1	
Observer B’s Values:	0	a	b	p_B
	1	c	d	q_B
		p_A	q_A	1

Where: $a + b = p_B = 1 - q_B$ and $a + c = p_A = 1 - q_A$ are proportions of the number N of units coded

$\bar{p} = (p_A + p_B)/2 = (1 - \bar{q})$ are population estimates

$1/2$ is the logical probability of 0 and of 1.

	Agreement = 1 –	Observed / Expected Disagreement
%-agreement	$P_o = 1 -$	$(b + c)$
Bennett et al. (1954)	$S = 1 -$	$(b + c)/2 \cdot 1/2 \cdot 1/2$
Scott (1955)	$\pi = 1 -$	$(b + c)/2 \bar{p} \bar{q}$
Cohen (1960)	$\kappa = 1 -$	$(b + c)/p_A q_B + p_B q_A$
Krippendorff (1970a)	${}_c\alpha = 1 - \frac{n-1}{n}$	$(b + c)/2 \bar{p} \bar{q}$, n = Number of pairable values
Gwet (2002)	$AC_1 = 1 -$	$(b + c)/\bar{p}^2 + \bar{q}^2$

Evidently, all of these measures contain the proportion of mismatches ($b + c$). The measure of %-agreement P_o stops there, making no allowances for expected disagreements and implying nothing about the categories that are available for coding (here only two) and about the distributional characteristics of data being categorized.

S acknowledges expectations but states them in terms of the number of categories available in a coding instrument. In its binary form, with the two categories being equally likely logically, the expected disagreement is $2 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$, or 50%. Otherwise, the expected disagreement of S is sensitive to the number of categories available but not responsive to the distribution of values used by observers, and it makes no reference to the population of data whose reliability is at stake.

In both $\epsilon\alpha$ and π , the expected disagreement in the two cells b and c is $2 \bar{p} \bar{q}$ which is the product of the population estimates $\bar{p}$ for 0s and $\bar{q}$ for 1s. π and $\epsilon\alpha$ are alike except for the factor $(n - 1)/n$ by which $\epsilon\alpha$ acknowledges small sample sizes, whereas π does not. As numerically demonstrated above, when sample sizes increase, $\epsilon\alpha$ and π become asymptotically indistinguishable.

Cohen's κ , by contrast, reveals itself as a hybrid coefficient (Krippendorff, 1978). Its observed disagreement ($b + c$) conforms to all the other agreement coefficients, but its expected disagreement, $p_A q_B + p_B q_A$, resembles that of correlation and association measures. In fact, it calculates expected disagreements just as for the familiar χ^2 statistic. It calculates the statistical independence of the two observers' preferences for these categories— p_A and q_A for one observer and p_B and q_B for the other. However, in agreement coefficients for reliability assessments, observers need to be considered interchangeable, not uncorrelated. Association between and predictability of the individual preferences for categories by one observer from those of the other are not at issue. Evidently, the expected agreement in κ is a function of the relationship between the two observers, not of the population of values that end up being assigned to the sampled units.

In terms of the behavior of coders or the reliability of data, Gwet's AC_1 is even more difficult to interpret. Unlike all other coefficients, it subtracts from $AC_1=1$ the proportion of the observed *disagreement* $1 - P_o = (b+c)/n..$ and the expected *agreement* $P_e = (\bar{p}^2 + \bar{q}^2)$ in cells a and d . For binary data, as in this binary example, $AC_1 = (P_e + P_o - 1)/P_e$. Ostensively developed to give %-agreement P_o more weight when variation is low (see section 12.6.1), the meaning of AC_1 's scale is not at all clear. $AC_1=1$ when disagreements are absent, which is fine. However, $AC_1=0$ can happen only if $a=b=c=d=1/4$, which is a questionable reference point for reliability assessments. I am not the only one for whom the difference between the expected agreement P_e and observed disagreement $1 - P_o$ makes no sense.

In relating the structure of these agreement coefficients to the conception of reliability adopted here, I want to be clear: Reliability concerns arise when the trustworthiness of data is in question and doubts about how well the phenomena of interest have been described needs to be dispelled. It is the coding of units drawn from a population of phenomena that is of interest here, not how difficult the coding task may be or how observer preferences for categories correlate. Without privileged access to the population of phenomena from which reliability data are sampled, researchers must find a way to estimate how well they can trust the data to be surrogated of the phenomena they intend to study. It is a fundamental assumption of reliability concerns that the perceptions of many are trusted more than the perception of any one

of them. Consequently, ${}_c\alpha$ estimates the distribution of categories in the population of phenomena from the judgments of as many observers as possible (at least two), making the common assumption that observer differences wash out in their average. Evidently, π , K , and ${}_c\alpha$ express their expected disagreements by reference to the estimated distribution of values in a population of interest, P_o , S , and κ do not and AC_1 perpetrates a category conflict with that reference.

This brings us to the fifth interpretation of ${}_c\alpha$. According to the above, ${}_c\alpha$'s reliability scale is anchored at two hypothetical points: the condition of all observers applying perfectly shared conceptions to the same set of phenomena, and yielding—sampling considerations aside—a perfectly reliable assignment of categories to otherwise nondescript phenomena; and the condition of individual observers showing no evidence of sharing conceptions of the coding task or not responding to the same phenomena by scrambling all given units and all the categories assigned by all observers to these units. In the absence of knowledge of the population of phenomena, the collectively obtained distribution of categories of the units sampled from that population can be regarded as the best estimate of the categories in the population of phenomena of interest. On this scale,

*${}_c\alpha$ is the degree to which independent and interchangeable observers
respond to unstructured qualities of phenomena of analytical interest
identically rather than randomly.*

Thus, ${}_c\alpha$ honors the collectively established population estimates by expressing the degree to which its categories are reliably assigned to a given set of units or distributed over these units by chance. P_o , S , and κ do not estimate the population characteristics of the data that reliability needs to address. They therefore cannot speak to the reliability of data.

Probably because the aforementioned Cronbach's (1951) alpha is called a measure of reliability as well, it has found its way into the content analysis literature. However, this coefficient was never intended to assess coding efforts, and, in fact, it cannot. In its binary form, it is Kuder and Richardson's (1937, p. 158) KR-20 and measures what in our context could be called the consistency of individual coders' judgments. It takes the variances of individual observers ($\sum_i p_i q_i$) as the variance of the "true" scores and expresses it as a proportion of the total variances ($\sigma^2 T$), which is the sum of the true score and the measurement error (Carmines & Zeller, 1979). In its familiar terms, and in the form of the above coefficients, it is defined as follows:

$$\begin{aligned}\text{Cronbach's alpha} &= \frac{m}{m-1} \left(1 - \frac{\sum_i p_i q_i}{\sigma_T^2} \right) \\ &= 2 \left(1 - \frac{p_A q_A + p_B q_B}{a(p_A - p_B)^2 + (b+c)(p_A - p_B)^2 + d(q_A + q_B)^2} \right)\end{aligned}$$

It belongs to the family of correlation coefficients and must not be construed as an agreement measure. Popularity and use in other empirical domains notwithstanding, %-agreement P_o , Bennett et al.'s S , Cohen's κ , Cronbach's alpha, and all correlation coefficients are simply not appropriate for assessing the reliability of coded data.

With the exception of Krippendorff's α , all of the above agreement coefficients are limited to nominal data generated by just two observers. This is because they are all conceptualized as corrections of %-agreement, a measure of matching or not matching categories. As has been demonstrated in the foregoing, α can handle any number of observers, not just two, is applicable to all common levels of measurement, not just nominal data, copes with missing data, and accommodates small sample sizes. As sample sizes become large, α_{nominal} for two coders converges to Scott's π , and for a fixed number of coders it also converges to Fleiss's K . For two observers, the D_o in α_{ordinal} is identical to D^2/N in Spearman's rank correlation coefficient ρ (rho); and α_{interval} turns out to equal Pearson et al.'s (1901) intraclass correlation coefficient R_p which is the correlation coefficient r_{ij} applied to coincidence rather than contingency matrices (Krippendorff, 1970a). These correspondences establish α 's roots in well-established statistics and attest to α 's generality. In fact, π , κ , and R_p are special cases of α . Its seamless applicability to a wide range of data enables researchers to employ uniform reliability standards to a great variety of data.

P_o , S , κ , AC_1 , and a host of other coefficients, not reviewed here, are limited to two observers, due largely to their various correction of %-agreement, which limits agreement on categories that are either the same or different. Unfortunately, this mere conceptual limitation of %-agreement has given rise to the dominant practice of testing coding reliability with just two coders. This is woefully inadequate. α overcomes this artificial and methodologically hardly justifiable limitation. There have been other proposals to extend agreement coefficients to several observers. I once pursued a blind alley (Krippendorff, 1971). Fleiss's (1971) generalization of π for nominal data to a fixed number of observers has been mentioned, including that he mistakenly called it *Kappa*, which created much confusion in the literature (Siegel & Castellan, 1988, p. 286). π is a special case of K , and K is just another special case of α .

Because the mathematical structures of these coefficients are not easily transparent, most researchers apply different coefficients to different kinds of reliability data. Others, Landis and Koch (1977) for example, have considered κ -type agreements in terms of majority opinions. Hubert (1977) has taken the approach of accepting only perfect consensus. Craig (1981) has proposed a modification of Scott's π to account for majorities among observer judgments. My development of α for many observers was initially guided by Spiegelman, Terwilliger, and Fearing's (1953b) effort to rank different patterns of multi-observer disagreement in nominal data by subjective judgments. The pairwise sum of differences $\sum \sum \delta^2$ in D_o and in D_e of α approximated their subjective ranks nearly perfectly and was moreover consistent with variance notions, which gave me the confidence to apply this function to any number of observers. Indeed, for multiple observers, α_{interval} can be expressed in terms of variance analysis (Krippendorff, 1970b).

There are scholars who rank agreement coefficients according to whether they are more liberal or more conservative (e.g. Lombard et al. 2002), suggesting that the choice among coefficients is attitudinal and has to do with the numbers researchers feel comfortable reporting. This conception fails to recognize the reasons for numerical differences among coefficients. For example, as demonstrated above, κ tends to be larger than π and for large samples, also larger than α . This is not a matter of κ being more liberal than π , rather of omitting unreliabilities that π and α account for. As noted in the literature, κ punishes coders who agree on their marginal distributions, π and α do not. Researchers can improve S by adding unused categories

to a variable. %-agreement has no place for indicating the absence of reliability. In other words, there are good mathematical reasons for some coefficients to be higher than others when applied to the same data. Actually, the absolute magnitudes of coefficients do not matter (see section 12.8.4 on standards), their comparability does, and so does the effect of unreliability on the validity of research results. By applying one general mathematical form to a great variety of kinds of reliability data, α makes it possible to adopt uniform reliability standards for evaluating diverse data and the measuring instruments that generate them.

The multitude of uses of α is still in development. In the first edition of *Content Analysis*, I started to sketch several diagnostic devices—devices for computing the reliability of individual units, for identifying unreliable observers within a group of observers, for determining the metric in use by observers, and for tracing the flow of disagreement through coding decision hierarchies (Krippendorff, 1980b, pp. 148–154). Subsequent advances include separating random and systematic disagreements (Krippendorff, 2008) and the reliability of unitizing textual continuities, discussed in sections 12.4. Most of these approaches concern the single-valued coding of units and segments of text. In 1992, I proposed ways to measure the reliability of multi-valued coding of phenomena (Krippendorff, 1992). It gave rise to reliability measures and software (Krippendorff & Craggs, 2016) that turned out to be too complex to be described here. However, a vastly simplified approach that generalizes α to multi-valued coding is described in section 12.7. It opens the possibility of obtaining reliabilities for the multi-valued coding of freely unitized text, data that computer-aided qualitative data analysis (QDA; see Chapter 11, section 11.5) generate. A presentation of the many analytical capabilities of α would go beyond the scope of this book and must await other publications.

12.4 ACCURACY, SURROGACY, AND THE DECISIVENESS OF MAJORITIES

All but the preceding section 12.3.4 dealt with measuring replicability, the backbone of measuring the reliability of data for subsequent analysis. As discussed in section 12.2.1, the coefficient that evaluates the replicability of data is also applicable to evaluating the stability of an observer or measuring device that which is obtained from repetitions of the data making process by the same individual or mechanism. According to Table 12.1, stability is a reliability measure weaker than replicability as it omits all inter-observer disagreements. It is compromised by the all-too-human abilities of memorizing previous coding decisions, learning shortcuts, and thereby changing coding practices over time, while trying to be consistent. Mechanical instruments are not affected by such human biases. Both have their own biases that measures of stability will not detect: in the case of humans, prejudices and preferences for particular outcomes, and in the case of mechanical devices, wear and tear and systematic distortions. But Table 12.1 also shows a measure stronger than replicability: accuracy or surrogacy. The data leading to this measure require references to a standard against which data making processes are compared.

Accuracy is the degree to which data correspond to an adopted standard. Accuracy is as strong as the chosen standard. When it comes to coding texts, it is customary to let a panel of experts create a standard by which individual observers'

valuation of units can be compared. The use of standards aids in training observers but also in judging their competencies. From my own experiences, judgments by a panel of experts may well include mistakes, discoverable by a majority of observers calling for revisions.

When the chosen standard is taken to be objectively true, accuracy can also serve as a measure of the validity of data. In content analysis objective truth criteria are rarely available, except for standardized measures like measures of times or lengths, the reading of instruments, or entering data into a computer. Some content analysts have compared the data they generated with validated research results obtained by independent methods. Even when used for measuring validity, accuracy measures cannot be trusted beyond the adopted standard.

Surrogacy is the degree to which one observer, device, or algorithm can substitute for a group of observers. Surrogacy is an important complement of accuracy. Both are measured by the same coefficient but aim in opposite directions. Whereas accuracy evaluates the quality of data generating processes against a given standard, surrogacy evaluates a preferred way of generating data against the performance of a group of observers. In content analysis, measuring the accuracy tends to be limited by the usually small number of units for which a standard is established. Obviously, if a standard were available for all units of analysis, there would be no point in asking observers to code them. Surrogacy takes the performance of a set of observers as the standard for answering the question of whether an expert or measuring device could serve as a substitute for the work that human coders could do. Measures of surrogacy are needed when one wants to know whether a physical instrument for identifying wanted features in text is an acceptable substitute for human coding, for example, whether the computational assignment of meanings by the use a dictionary (section 11.4.1) can substitute for how humans read and code the same text or whether an algorithm developed during machine learning has sufficiently improved to let it identify relevant textual features without human guidance. If the question is posed whether a search engine can substitute for readers to go through a large database and categorizes its contents, then the agreement coefficient developed in the previous section is one of surrogacy.

This section develops a family of three accuracy or surrogacy measures that can be considered the stronger relatives of the α -coefficients designed to evaluate the reliability of replicated data. References to crowdcoding suggest research designs that rely on a multitude of observers, coders, or judges. “The wisdom of the crowd” (Surowiecki, 2005) is a metaphor commonly used in justifications of public opinion surveys, voting in democracies, theories in marketing, and what has recently become fashionable as crowdsourcing. These practices tend to be motivated by the belief that aggregates of many are superior to the judgment of a few, experts included. The family of three agreement coefficients tabs into these beliefs.

Common to the definition of crowds is the large number of variously qualified and unequally participating individuals reading, evaluating, and interpreting texts. Crowdcoding in content analysis amounts to disseminating texts and coding instructions to numerous anonymous readers or judges who meet certain testable qualifications, largely for their ability to follow coding instructions, understand the texts they are to code, and return their judgments to the researcher. Crowdcoding omits the kind of biases that coder training can introduce in the form of surreptitious

interpretations of coding instructions. Crowdsourced individuals respond exclusively to the written instruction and the material given to them. However, crowdcoding exhibits other uncertainties that arise from not knowing who the coders, readers, or judges really are and how they work.

The reliability data we are here concerned with are shown in Figure 12.8. It adds a standard to Figure 12.4.

FIGURE 12.8 ■ Values-by-Units Matrix of Reliability Data for Accuracy

Units =	1	2	.	.	<i>u</i>	.	.	.	.	.	<i>N</i>	
A Standard =	<i>k</i> 1	.	.	.	<i>k_u</i>	.	.	.	.	.	<i>k_N</i>	
Values:	1	<i>n</i> 11	.	.	<i>n_{u1}</i>	.	.	.	.	.	<i>n_{N1}</i>	<i>n_{.1}</i>
	:	:	.	.	:	.	.	.	.	.	:	:
	<i>c</i>	<i>n</i> 1 <i>c</i>	.	.	<i>n_{uc}</i>	.	.	.	.	.	<i>n_{Nc}</i>	<i>n_{.c}</i> = $\sum_{u=1}^N n_{uc} \mid k_u \neq \phi$
	:	:	.	.	:	.	.	.	.	.	:	:
	<i>v</i>	<i>n</i> 1 <i>v</i>	.	.	<i>n_{uv}</i>	.	.	.	.	.	<i>n_{Nv}</i>	<i>n_{.v}</i>
Numbers of values pairable with <i>k</i> =	<i>m</i> 1	.	.	.	$m_u = \sum_{c=1}^v n_{uc} \mid k_u \neq \phi$	.	.	.	.	.	<i>m_N</i>	$n_{..} = \sum_{u=1}^N m_u \leq mN$
For replicability crowd members $c\alpha_{\text{metric}}$: $m_u \geq 2$ (See Section 12.3) For accuracy of crowd members $c\alpha_{\text{metric}}^s$: $m_u \geq 1$ and k_u For accuracy of majorities and aggregates $c\alpha_{\text{metric}}^{sm}$: $m_u \geq 3$ and k_u For decisiveness of crowds $c\alpha_{\text{metric}}^m$: $m_u \geq 3$												

12.4.1 Accuracy of or Surrogacy for Crowd Members' Judgments

By analogy to (3), the observed disagreement with the standard *s* (noted in its superscript) is defined rather simply by:

$${}_c D_o^s = \sum_{u=1}^N \sum_{c_u=1}^v n_{uc \text{ metric}} \delta_{ck_u}^2 \quad (19)$$

Note that observed disagreements enumerate the differences of contingencies between m_u values c_u and the standard k_u . They are contingencies, not coincidences as for replicability. The frequency $n_{.c}$ of values *c* is found in the applicable reliability data of Table 12.8—applicable meaning $m_u \geq 1$ and k_u . The frequency with which *k* is applied to values *c* is:

$$n_k = \sum_{u=1}^N m_u \text{ iff } k_u; \quad n_{..} = \sum_{k=1}^v n_k = \sum_{c=1}^v n_{.c}$$

The expected disagreement is defined analogous to (4), but given the non-overlapping sets of values defining the rows and columns, without the need to discount for the pairing of values with themselves:

$${}_c D_e^s = \frac{1}{n_{..}} \sum_{k=1}^v n_k \sum_{c=1}^v n_{.c} \text{metric} \delta_{ck}^2 \quad (20)$$

The accuracy or surrogacy relative to a given standard s becomes:

$${}_c \alpha_{\text{metric}}^s = 1 - \frac{{}_c D_o^s}{{}_c D_e^s} \quad (21)$$

12.4.2 Accuracy of or Surrogacy for Majorities or Aggregates

As already mentioned, “the wisdom of the crowd” consists of the belief that the aggregate judgment of a crowd is superior to that of its individual members and hence more reliable. Trial by jury can be considered a legal operationalization of that “wisdom.” Public voting for representatives in government is justified by the conviction that majorities serve the public better than the dictates of privileged individuals. Numerous psychological experiments in which subjects were asked to estimate the sizes of objects, appraise the behavior of others (the third-person effect), or predict the winner of an election for political office have shown that the aggregate of individual judgments is often closer to independently known facts than that of single experts. In the face of such findings, there may well be a benefit in ignoring the diversity of individual judgments in favor of majorities or aggregates of them. In many research projects, judgments by majorities are adopted as the gold standard.

However, crowdcoding may not be reliable when interpretations of texts require specialized knowledge and literary competencies. In practice, crowdsourcing a large population for their readings of given texts is limited to the kind of questions one can ask most readers to answer. Where the coding task is complex, prejudices are rampant and opinions are charged with divisive emotions, and one may have to find other means of obtaining reliable data. Content analysis is not to be confused with survey research.

To determine the degree to which a majority or aggregate is trustworthy calls for a reliability measure that pairs a given standard, not with all individual members of the crowd as measured by ${}_c \alpha_{\text{metric}}^s$, but with their aggregates. What counts as an aggregate depends on the metric of the crowd generated data. For any one unit u of the reliability data in Figure 12.8, we define the aggregates $\bar{c}_u$ for:

$$\text{Nominal data, } r_u \text{ majorities: } \left\{ \bar{c}_{u1}, \dots, \bar{c}_{ug}, \dots, \bar{c}_{uv} \right\} \mid \left\{ n_{c_{u1}} = \dots = n_{c_{ug}} = \dots = n_{c_{uv}} \right\} = \max(n_{u1}, \dots, n_{uc}, \dots, n_{uv}) \quad (22)$$

$$\text{Ordinal data, the median: } \bar{c}_u = \begin{cases} \text{The } \left(\frac{m_u + 1}{2} \right)^{\text{th}} \text{ rank between the } c_{\text{lowest}} \text{ and } c_{\text{highest}} \text{ rank} & \text{-- iff } m_u \text{ is uneven} \\ \frac{1}{2} \left[\text{The } \left(\frac{m_u}{2} \right)^{\text{th}} + \left(\frac{m_u}{2} + 1 \right)^{\text{th}} \text{ rank on that list} \right] & \text{-- iff } m_u \text{ is even} \end{cases}$$

$$\text{Interval data, the means: } \bar{c}_u = \frac{1}{m_u} \sum_{c=1}^v n_{uc} c$$

$$\text{Ratio data (proportions), the average: } \bar{c}_u = \frac{1}{m_u} \sum_{c=1}^v n_{uc} c \text{ of positive numbers.}$$

Multiple majorities defined in (22) deserve comment. Whereas medians, means, and averages have single values, $\bar{c}$, when data are unordered, nominal, or categorical

values c_u with the largest frequencies n_{uc} could be tied. Statistically, single majorities, indicated by $r_u=1$, are most likely. Ties between $r_u \geq 2$ majorities are less likely, but they do occur, especially when the m_u values in unit u are small. For example, among four observers there could be:

- ○ ○ ○; ○ ○ ○ □; ○ ○ □ ✂ Three undisputable majorities of decreasing decisiveness
- ○ □ □ Two tied “majorities”
- □ ✂ ◆ Four tied “majorities” or unique values

When pairing standards with majorities or an aggregate, the m_u values in units u are replaced with their majorities or aggregates $\bar{c}_u$. This is in keeping with the conception that minorities do not matter, and individual deviations from the aggregated values should not count. When multiple majorities $\bar{c}_{u1}, \dots, \bar{c}_{ug}, \dots, \bar{c}_{ur_u}$ in u are encountered, they are weighted equally, by $1/r_u$ to assure that their contributions to the accuracy of majorities are comparable to those of single $\bar{c}_u$ -values. With these conceptions, the observed disagreements for this accuracy (standard s and majority m indicated in its superscript) are defined by:

$$\begin{aligned} \text{For } r_u \text{ majorities } \bar{c}_{u1}, \dots, \bar{c}_{ug}, \dots, \bar{c}_{ur_u} : \quad {}_c D_o^{sm} &= \sum_{u=1}^N \sum_{g=1}^{r_u} \frac{m_u}{r_u} \text{nominal } \delta_{\bar{c}_{ug} k_u}^2 \\ \text{For one majority or aggregate } \bar{c}_u : \quad {}_c D_o^{sm} &= \sum_{u=1}^N m_u \text{metric } \delta_{\bar{c}_u k_u}^2 \end{aligned} \quad (23)$$

The expected disagreements call for the frequencies with which values k in the standard and the m_u values represented by their majorities $\bar{c}$:

$$\begin{aligned} n_k &= \sum_{u=1}^N m_u \quad \text{iff } k_u \\ \text{For } r_u \text{ majorities } \bar{c}_{u1}, \dots, \bar{c}_{ug}, \dots, \bar{c}_{ur_u} : \quad n_{\bar{c}} &= \sum_{u=1}^N \sum_{g=1}^{r_u} \frac{m_u}{r_u} \quad \text{iff } \bar{c}_{ug} \\ \text{For one majority or aggregate } \bar{c}_u : \quad {}_c D_o^{sm} &= \sum_{u=1}^N m_u \text{metric } \delta_{\bar{c}_u k_u}^2 \\ n_{..} &= \sum_{\bar{c}} n_{\bar{c}} = \sum_k n_k \end{aligned}$$

Following (20), the expected disagreement between standards k and majorities or aggregates $\bar{c}$ is:

$${}_c D_e^{sm} = \frac{1}{N} \sum_{\bar{c}} n_{\bar{c}} \sum_k n_k \text{metric } \delta_{\bar{c} k}^2 \quad (24)$$

And the accuracy of or surrogacy for majorities or aggregates of crowds becomes:

$${}_c \alpha_{\text{metric}}^{sm} = 1 - \frac{{}_c D_o^{sm}}{{}_c D_e^{sm}} \quad (25)$$

As for all α -reliability coefficients, when majorities or aggregates and the standard match in all units u , ${}_c \alpha_{\text{metric}}^{sm} = 1$. When matches are mere chance events, ${}_c \alpha_{\text{metric}}^{sm} = 0$.

Multiple majorities add to ${}_c D_o^m$ and reduce ${}_c \alpha_{\text{metric}}^m$. Just as for ${}_c \alpha_{\text{metric}}$ and ${}_c \alpha_{\text{metric}}^s$, ${}_c \alpha_{\text{metric}}^m$ can be negative when disagreements are overwhelmingly systematic. The property of ${}_c \alpha_{\text{metric}}^m$ to override minorities and ignore deviations from averages or medians is in keeping with the belief that they do not matter, a belief that the measure of decisiveness is to qualify.

12.4.3 The Decisiveness of Majorities or Aggregates

Whenever standards are available, measures of accuracies or surrogacies in (21) and (25) can be computed, but their relationship to reliability is not obvious. While ${}_c \alpha_{\text{nominal}}^s$ pairs all values with their given standards, which counts variance within units as inaccuracies, ${}_c \alpha_{\text{nominal}}^m$ ignores all minorities within units. The common justification for interpreting such a measure as one of reliability stems from the aforementioned belief that majorities deliver more reliable data. This may not be justifiable. For example, when the distribution of values in units is 51% and 49%, there surely is a majority. If that majority matches with the standard this would count as in perfect agreement, and if true for all units, ${}_c \alpha_{\text{nominal}}^m = 1$. However, taking such slim majorities as reliable representation of the phenomena that data are meant to represent obviously needs to be questioned. Any minor perturbation in the process of generating these data could easily flip one majority to another, attesting to the near arbitrariness of such close majorities. The fact that a measure of the accuracy of majorities, ${}_c \alpha_{\text{nominal}}^m$, cannot detect the near arbitrariness of slim majorities is sufficient ground to question the use of majorities as gold standards for reliability. Worse, if a unit is categorized, say, in terms of five categories with frequencies of 22%, 21%, 20%, 19% and 18%, the category with 22% is the unquestionable majority, but claiming this to be a crowd's superior judgment of the phenomenon is most certainly misleading.

The same concern applies to the accuracy of aggregates for ordered values—medians and averages. Empirically, it may well happen that the averages of the judgments by members of large crowds are close to, sometimes even superior to, individual expert judgments. However, such findings say nothing about the variance that all averages ignore by definition. Only in the absence of variance can an accurate aggregate be interpreted as perfectly reliable. As the variances of a distribution increases, accepting averages as reliable data becomes increasingly indefensible.

The reliability measure of the decisiveness of majorities or aggregates presented below qualifies the above mentioned blindness to what matters. It is another relative of the accuracy measure. It does not compare reliability data to a standard, it puts the majorities or aggregates $\bar{c}_u$ within units into the position of a standard k_u as in Figure 12.8, (19), and (20) and it pairs them with the individual values c_u assigned by members of the aggregated crowd members, observers, coders, or judges. Thus, the observed indecisiveness of majorities or aggregates (with m in its superscript) becomes analogous to (23):

$$\begin{aligned} \text{For } r_u \text{ majorities } \bar{c}_{u_1}, \dots, \bar{c}_{u_g}, \dots, \bar{c}_{u_{r_u}} : \quad {}_c D_o^m &= \sum_{u=1}^N \sum_{c=1}^v \frac{n_{uc}}{r_u} \sum_{g=1}^{r_u} \text{nominal} \delta_{c\bar{c}_{u_g}}^2 \\ \text{For one majority or aggregate } \bar{c}_u : \quad {}_c D_o^m &= \sum_{u=1}^N \sum_{c=1}^v n_{uc} \text{metric} \delta_{c\bar{c}_u}^2 \end{aligned} \quad (26)$$

To obtain the corresponding expected indecisiveness measure, the frequency $n_{\cdot c}$ of values used by all observers is taken from the applicable reliability data in Figure 12.8—applicable meaning $m_u \geq 2$ for all units u . The frequencies $n_{\bar{c}}$ of majorities or aggregates $\bar{c}$ paired with values c that define them are as in the above:

$$\text{For } r_u \text{ majorities } \bar{c}_{u1}, \dots, \bar{c}_{ug}, \dots, \bar{c}_{ur_u} : \quad n_{\bar{c}} = \sum_{u=1}^N \sum_{g=1}^{r_u} \frac{m_u}{r_u} \quad \text{iff } \bar{c}_{ug}$$

$$\begin{aligned} \text{For one majority or aggregate } \bar{c}_u : \quad n_{\bar{c}} &= \sum_{u=1}^N m_u \quad \text{iff } \bar{c}_u \\ n_{\cdot\cdot} &= \sum_{\bar{c}} n_{\bar{c}} = \sum_{c=1}^v n_{\cdot c} \end{aligned}$$

And just as in (20), the expected indecisiveness becomes:

$${}_c D_e^m = \frac{1}{n_{\cdot\cdot}} \sum_{\bar{c}} n_{\bar{c}} \sum_{c=1}^v n_{\cdot c} \text{metric} \delta_{c\bar{c}_u}^2 \quad (27)$$

Which leads to the decisiveness of majorities or aggregates:

$${}_c \alpha_{\text{metric}}^m = 1 - \frac{{}_c D_o^m}{{}_c D_e^m} \quad (28)$$

When all observers agree on the values assigned to units, majorities are unanimous, and there is no deviation from the computed aggregates, decisiveness ${}_c \alpha_{\text{metric}}^m = 1$. When values assigned to units are evenly split, including when all m_u values are unique, ${}_c \alpha_{\text{metric}}^m = 0$. These are the extremes of decisiveness.

It follows that decisiveness measures the extent to which majorities or other aggregates can serve as reliable data for subsequent analysis. Only when the decisiveness of reliability data is assured can majorities be relied upon in subsequent analysis. When data are nominal or ordinal, it should be noted that their majorities or medians $\bar{c}$ are among the observed values c . However, when data possess interval or ratio metric properties, their averages $\bar{c}$ may be fractions. Most statistical methods should be able to handle them.

To repeat the motivation of the decisiveness measure, I argued that assessing ${}_c \alpha_{\text{metric}}^m$, the accuracy of majorities or aggregates, can be inconclusive regarding the reliability of data. Whereas the replicability ${}_c \alpha_{\text{metric}}$ counts the variance within units against interpreting data as reliable, the decisiveness of majorities or aggregates ${}_c \alpha_{\text{metric}}^m$ answers the additional question of whether majorities or aggregates could be buried in the data that could justify their use in subsequent analysis. Whenever researches seek to select values among diverse descriptions of the phenomena of analytical interest, measuring a crowd's decisiveness is critical. All reliable data need to be decisive.

12.4.4 Numerical Examples

Instead of exemplifying the three coefficients by whole data sets, I rely on six units with varying patterns of disagreements, chosen to convey how they contribute

variously to the observed disagreements D_o of the three coefficients presented above. The expected disagreements D_e are computed for the six units:

Units:	1	2	3	4	5	6	.	.	u	.
Standard: $k_u =$	2	5	6	3	1	4	.	.	k_u	.
Values: $c = 1$	.	.	8	.	22	.	.	.	.	.
2	40	.	.	49	21	1	.	.	.	.
3	.	.	.	51	20	1	.	.	n_{cu}	.
4	.	.	.	.	19	1	.	.	.	.
5	.	10	.	.	.	1	.	.	.	.
$v = 6$	.	1	80	.	18	1	.	.	.	.
$m_u =$	40	11	88	100	100	5	.	.	m_u	.

When values are unordered:

For comparison, the replicability according to (5) with ${}_c D_e$ obtained as in (3):

$${}_c \alpha_{\text{nominal}} = 0.413.$$

$$\text{Accuracy of values: } {}_c \alpha_{\text{nominal}}^s = 1 - \frac{140}{275.218} = 0.491.$$

Unit 1 exhibits no inaccuracy and contributes nothing to ${}_c D_o^s$ (true for all other coefficients)

Unit 2 contributes $n_{26} = 1$

Unit 3: $n_{31} = 8$

Unit 4: $n_{42} = 49$

Unit 5: $n_{52} + n_{53} + n_{54} + n_{56} = 21 + 20 + 19 + 18 = 78$

Unit 6: $n_{62} + n_{63} + n_{65} + n_{66} = 1 + 1 + 1 + 1 = 4$

Visual inspection reveals unit 5 to be the most inaccurate, followed by unit 4.

$$\text{Accuracy of majorities: } {}_c \alpha_{\text{nominal}}^{sm} = 1 - \frac{4}{275.637} = 0.985.$$

The majorities in units 1 through 5 match the standard and contribute nothing to ${}_c D_o^{sm}$.

In unit 6, of the $r_6 = 5$ majorities only n_{64} matches the standard $k_6 = 4$. The others contribute 4 to ${}_c D_o^{sm}$

$$\text{Decisiveness of majorities: } {}_c \alpha_{\text{nominal}}^m = 1 - \frac{140}{274.334} = 0.490. \text{ It challenges the adequacy of } {}_c \alpha_{\text{nominal}}^{sm} \text{ as majorities turn out not to be sufficiently decisive.}$$

Unit 2: $\bar{c}_2 = 5 \neq c_6$, contributes $n_{26} = 1$ to ${}_c D_o^m$

Unit 3: $\bar{c}_3 = 6 \neq c_1$, $n_{31} = 8$

Unit 4: $\bar{c}_4 = 3 \neq c_2$, $n_{42} = 49$

Unit 5: $\bar{c}_5 = 1 \neq n_{52} + n_{53} + n_{54} + n_{56} = 21 + 20 + 19 + 18 = 78$

Unit 6 matches $r_6 = 5$ majorities with the five values, weighted $1/r_6$ each. Of the 25 pair comparisons $n_{6c} \cdot n_{6k} / r_u$, five match and do not count, hence unit 6 contributes $5(5-1)/5 = 4$

Because majorities in the first five units match with the standard and the 6th unit reduces both coefficients alike, it should not be surprising that ${}_c\alpha_{\text{nominal}}^m$ and ${}_c\alpha_{\text{nominal}}^s$ barely differ.

When values are assumed to have interval characteristics:

For comparison, the replicability: ${}_c\alpha_{\text{interval}} = 0.538 > {}_c\alpha_{\text{nominal}} = 0.413$. It suggests that values are ordered.

$$\text{Accuracy of values: } {}_c\alpha_{\text{interval}}^s = 1 - \frac{982}{2425.297} = 0.595.$$

$$\text{Unit 2 adds } n_{26}(c_2 - k_2)^2 = 1(6-5)^2 = 1 \text{ to } {}_cD_o^s$$

$$\text{Unit 3: } n_{31}(c_3 - k_3)^2 = 8(1-6)^2 = 200$$

$$\text{Unit 4: } n_{42}(c_2 - k_4)^2 = 49(2-3)^2 = 49$$

$$\text{Unit 5, with } k_5=1: n_{52}(c_2 - k_5)^2 + n_{53}(c_3 - k_5)^2 + n_{54}(c_4 - k_5)^2 + n_{56}(c_6 - k_5)^2 = 722$$

$$\text{Unit 6, with } k_6=4: (c_{62} - k_6)^2 + (c_{63} - k_6)^2 + (c_{64} - k_6)^2 + (c_{65} - k_6)^2 + (c_{66} - k_6)^2 = 10$$

$$\text{Accuracy of means: } {}_c\alpha_{\text{interval}}^{sm} = 1 - \frac{442.295}{2425.297} = 0.818.$$

Note that unit 6, with $\bar{c}_6 = 4 = k_6$, contributes nothing to ${}_cD^{sm}$.

Decisiveness of means: ${}_c\alpha_{\text{interval}}^m = 1 - \frac{507.717}{1720.538} = 0.705$. This too reveals ${}_c\alpha_{\text{interval}}^{sm}$, to be insufficiently decisive.

$$\text{Unit 2 adds } n_{25}(c_5 - \bar{c}_2)^2 + n_{26}(c_6 - \bar{c}_2)^2 = 10(5-5.0909)^2 + 1(6-5.0909)^2 = 0.909 \text{ to } {}_cD_o^m$$

$$\text{Unit 3: } n_{31}(c_1 - \bar{c}_3)^2 + n_{36}(c_6 - \bar{c}_3)^2 = 8(1-5.546)^2 + 80(6-5.546)^2 = 181.8182$$

$$\text{Unit 4: } n_{42}(c_2 - \bar{c}_4)^2 + n_{43}(c_3 - \bar{c}_4)^2 = 49(2-2.52)^2 + 51(3-2.52)^2 = 25$$

$$\text{Unit 5: } n_{51}(c_1 - \bar{c}_5)^2 + n_{52}(c_2 - \bar{c}_5)^2 + n_{53}(c_3 - \bar{c}_5)^2 + n_{54}(c_4 - \bar{c}_5)^2 + n_{56}(c_6 - \bar{c}_5)^2 = 290$$

$$\text{Unit 6: } (c_2 - \bar{c}_6)^2 + (c_3 - \bar{c}_6)^2 + (c_4 - \bar{c}_6)^2 + (c_5 - \bar{c}_6)^2 + (c_6 - \bar{c}_6)^2 = 10$$

As already noted, unit 5 accounts for most disagreements in all reliability measures—except in ${}_c\alpha_{\text{nominal}}^m$ where the slim majority of $n_{51}=22$ overwhelms all 78 deviations. Also, unit 6, despite its five unique values adds nothing to ${}_c\alpha_{\text{interval}}^{sm}$; it exemplifies the need to measure the decisiveness of majorities whenever reliance on majorities or averages are considered indicators of reliability.

This example was not chosen to draw conclusions about the reliabilities of data. It merely serves to show what different patterns of disagreements in units and metrics can make in what these coefficients highlight and hide. The finding that the interval reliabilities exceed their nominal versions would suggest that there is an ordering manifest in the reliability data. The finding that decisiveness ${}_c\alpha_{\text{interval}}^m$ exceeds accuracy ${}_c\alpha_{\text{interval}}^s$ would suggest that there are aggregate values $\bar{c}_u$ that are potentially more reliable than the given standard k_u , worthy of further explorations.

12.5 THE RELIABILITY OF TEXT MINING AND INFORMATION RETRIEVAL

The literature on content mining and information retrieval, pattern recognition, and binary classification has pursued two measures to assess the qualities of search

engines: *Precision* and *Recall* (van Rijsbergen, 1979; Manning, Raghavan, & Schütze, 2008). These two measures relate only marginally to reliability concerns but respond to errors in retrieving documents, abstracts, authors, references, and images from large databases that reliability considerations need to embrace as well. In the face of very large collections of texts, there is full consensus that ideal search engines retrieve all and only relevant items and ignore all and only irrelevant ones. A measure of the extent to which this ideal is achieved is akin to a measure of accuracy. However, it is confined to a binary kind.

Measuring deviations from this ideal is marred by one epistemological confusion and three problems. The epistemological confusion is related to the notion of information. One should think that the information retrieval literature distinguishes between information and what the search of a textual database brings to a researcher's attention. However, much of this literature takes the concept of information as obvious and limits its attention to the design of algorithms and their ability to locate items containing character strings that match the chosen search terms. I maintain that information is relative of what researchers want to gain from texts. To determine whether something informs an analysis or not requires human interpretative abilities that search engines can rarely replace. It is simply misleading, amply discussed in Chapter 2, for information retrieval literature to assume that information is contained in texts and to ignore the human interpretive abilities involved. These abilities are involved in two of the three problems that can negatively affect the quality of information retrievals.

The first problem occurs prior to any search, when the research questions that analysts hope to explore need to be translated into the terms that a search engine can work with. A search engine is designed to scan a large textual database and identify and retrieve texts that contain character sequences corresponding to specified search terms, see sections 11.3.2 and 11.3.3. Unless analysts are interested in the mere occurrence of distinct character strings, for example, the names of particular authors, products, or diseases, this is not an easy task. It forces analysts to find literal textual manifestations of what would answer their often abstract research questions, an ability that is the inverse of reading.

The second problem resides in how the design of a search engine matches the organization of the database to be searched. The use of information retrieval algorithms makes sense mainly when textual databases are beyond the scope of human readers' ability to interpret what they contain. As texts can be variously unitized—words, phrases, paragraphs, documents, images, web pages, or journals—and be viewed in terms numerous patterns among them—adjacencies, synonyms, grammatical or typographical relationships—there never is a single number that could quantify the size of a database and what a search engine has to work through. In content analyses, the inability of analyzing a whole population of phenomena of scholarly interest is overcome by applying sampling strategies to research-specific units of phenomena in order to assure their equal representation in the sample actually analyzed. I have not seen a search engine that employs sampling schemes on selected text types. Instead, they tend to be designed to scan everything that a chosen database contains but cannot avoid being at a mercy of the way texts are broken down into units and indexed. Retrieving items is always limited to their most obvious and literal characteristics. Unusual texts, handwritten documents, charts, paintings,

and videos, even texts with unorthodox typographies escape the attention of search engines and may never be found.

The third problem arises after a search engine lists its results. It is rare that search engines retrieve all and only relevant texts. The degree to which this happens is only in part due to the analysts' ability to formulate suitable search terms. Unless the search is limited to particular character sequences, judgments on whether retrieved items could inform a research question, are useful to a scholarly inquiry, or serve as reliable data for further analysis requires considerable understanding of what is needed. Judges need to know the language of the texts and possess specialized competencies to relate a retrieved text to the guiding research questions. This is why judgments on the relevance of received items tend to be made by the very analysts who intend to make use of what is retrieved. Generally, judging the relevance of each received item is certainly easier than formulating search queries, however, it is not a mechanical task.

None of these three problems are correlated with why a search is undertaken. Each accounts for different losses in the qualities of what is mined or retrieved and brought into a research project. Let me briefly discuss the two traditional qualities for evaluation information retrievals, *Precision* and *Recall*, and then propose two proper reliability measures.

For any one search, *Precision* is defined as the following proportion:

$$\text{Precision} = \frac{\text{Number of Relevant Items Retrieved}}{\text{Total Number of (Relevant + Irrelevant) Items Retrieved}}$$

In terms of Table 12.1, *Precision* is an extremely simple accuracy measure. It measures the extent to which a search result approaches the ideal of providing only relevant information. As above mentioned, relevance is always a researcher's judgment relative to his or her analytical objectives. However, inasmuch as the researcher is the author of the search terms that guided the search engine to the retrieved items—often without full knowledge of how the chosen search engine works—and judge of the extent to which a search provided relevant information, *Precision* has to be interpreted as a measure of a researcher's ability to use a given search engine so as to bring only relevant items to his or her attention. Faced with low measures of *Precision*, researchers tend to literalize their initially abstract research questions, for example, searching for an author known to have addressed a certain issue instead of searching for texts that address the issue in question. However, even this strategy is limited by the lack of transparency of how a search algorithm works and databases are organized. Although the literature on information retrieval presents *Precision* as a measure of the quality of a search engine, a significant part of it has to do with the ability of a researcher to handle an usually clumsy search engine with its considerable limitations.

As such, *Precision* responds to only one of three kinds of information retrieval errors, the error of using search terms that retrieve irrelevant items, also called the Type I error. In addition, it objectifies the researcher's ability to identify what is relevant.

Recall is a conceptual companion of *Precision*. It is defined as the following proportion:

$$\text{Recall} = \frac{\text{Number of Relevant Items Retrieved}}{\text{Total Number of Relevant Items in the Database}}$$

Recall measures the extent to which all relevant items in a textual database are actually retrieved. It is conceived to capture information retrieval errors of the second kind: Omissions of relevant items, also called the Type II error. What makes this proportion problematic, however, is the presumption of reliable knowledge of the number of relevant items in the searched database. This number remains unknown until someone actually unitizes and examines the totality of its contents and reliably separates relevant from irrelevant items. *Recall* is an easily conceived but empirically paradoxical measure. The benefit of information retrieval lies in its ability to search databases whose size and composition is far beyond human knowledge. Yet, *Recall* requires the very knowledge the absence of which motivates what it seeks to evaluate. Statisticians have proposed estimates of the number of relevant items in a database from the search results, which are the only data actually available. This calls for accepting a highly questionable assumption, which leads me to conclude that *recall* is not measurable, leaving the measure in an epistemological limbo.

Different search engines rarely perform alike. Some may not retrieve the same set of items when applied repeatedly. This happens most obviously when search engines are designed to learn from their users' search histories, as for Google's search engines, or Amazon's. This well-intended feature makes the results of search engines not replicable and essentially unreliable. Neither *Precision* nor *Recall* are defined to respond to disagreements across different search engines and among researchers assessing the relevance of the retrieved items. There are a few papers nominally concerned with the reliability of information retrievals, but they either rank the efficiency of different search engines or compare the ability of researchers to identify relevant items with that of a search engine—which is part of the measure of *Precision*.

I am aware of several proposals to combine *Precision* and *Recall* into a single measure, for example, by averaging them. However, averaging two rather different measures has little conceptual justifications. To my knowledge, none of these proposals have been able to solve the paradox of *Recall* and handle disagreements among several information retrieval efforts of a common database.

12.5.1 For Replications of Searches

By relying on data from separate searches of the same database for comparison, it is not too difficult to define a measure of the reliability of information retrievals capable of responding to the disagreements on the relevance of retrieved items and staying entirely within available data.

Terminologically, let the reliability data for m searches be indicated: $1, 2, \dots, i, j, \dots, m$:

R_i = The set of relevant items retrieved in search i

I_i = The set of irrelevant items retrieved in search i

$R_i \cup I_i$ = The set of items retrieved in search i

$\bigcup_i (R_i \cup I_i) = T$ = The total set of items retrieved by m searches (without duplications)

$|A|$ = The cardinality of the set A = the number of elements contained in A

$\sum_i \sum_{j \neq i} |R_i \cap R_j|$ = The sum of all pairwise intersections among all relevant items in m research results

The symmetrical matrix of all coincidences of the reliability data from m searches defined above is depicted in Table 12.2.

TABLE 12.2 ■ Matrix of Coincidences in Reliability Data for m Retrievals

$\sum_i \sum_{j \neq i} R_i \cap R_j $	$\sum_i \sum_{j \neq i} R_i \cap I_j $	$\sum_i \sum_{j \neq i} R_i \cap \phi_j $	$(m-1) \sum_i R_i $
$\sum_j \sum_{j \neq i} I_i \cap R_j $	$\sum_i \sum_{j \neq i} I_i \cap I_j $	$\sum_i \sum_{j \neq i} I_i \cap \phi_j $	$(m-1) \sum_i I_i $
$\sum_i \sum_{j \neq i} \phi_i \cap R_j $	$\sum_i \sum_{j \neq i} \phi_i \cap I_j $	$\sum_i \sum_{j \neq i} \phi_i \cap \phi_j $	$(m-1) \sum_i \phi_i $
$(m-1) \sum_i R_j $	$(m-1) \sum_j I_j $	$(m-1) \sum_j \phi_j $	$m(m-1) T $

Information retrievals are perfectly reliable when all and only relevant items are retrieved. Under these conditions, the sum of pairwise agreements on relevant items would equal the total: $\sum_i \sum_{j \neq i} |R_i \cap R_j| = m(m-1) |T|$. In Table 12.2, these two sums are highlighted. Their proportion, P^R , of agreements among relevant items is defined by (29), also expressed as the product of two other proportions:

$$P^{IR} = \frac{\sum_i \sum_{j \neq i} |R_i \cap R_j|}{m(m-1) |T|} = \frac{(m-1) \sum_i |R_i|}{m(m-1) |T|} \cdot \frac{\sum_i \sum_{j \neq i} |R_i \cap R_j|}{(m-1) \sum_i |R_i|} \quad (29)$$

Evidently, if the m searches retrieve only relevant items, $P^{IR} = 1$. Otherwise P^{IR} is reduced by the sum of the remaining eight cells of this three-by-three coincidence matrix in Table 12.2, which contain all possible retrieval errors. Formula (29) differentiates among these eight retrieval errors by equating P^{IR} with the product of two proportions, each is sensitive to different imperfections.

The second of the pair of proportions of P^{IR} is the proportion of agreements among relevant items. Neither *Precision* nor *Recall* is able to address this unreliability. In Table 12.2, the disagreements that negatively affect this proportion are found in the row and column whose sums intersect with relevant matter. This proportion essentially responds to the disagreements among judges about the relevance of items.

The first of the two proportions is the proportion of relevant items among all of them. It responds to the retrieval of irrelevant items and the exclusion items not retrieved by all m searches. This proportion generalized the measure of *Precision* = $R_i / (R_i \cup I_i)$ from any one i to the sum of m searches and includes one part of the error that *Recall* is designed to assess—the failure to retrieve relevant items—for which data are actually available.

Evidently, the cell contents in Table 12.2 that do not enter the second proportion are captured by the first, hence their product, P^{IR} , accounts for all.

However, P^{IR} is not corrected by what can be expected when chance governs the identification of relevant items. Chance would have to correct only the agreement among relevant items in the second proportion, nothing in the first. The following agreement coefficient includes that correction and comes back to (29):

$${}_c\alpha^{IR} = \frac{(m-1)\sum_i |R_i|}{m(m-1)|T|} \left(\frac{\sum_i \sum_{j \neq i} |R_i \cap R_j| - \frac{((m-1)\sum_i |R_i|)^2}{m(m-1)|T|}}{(m-1)\sum_i |R_i| - \frac{((m-1)\sum_i |R_i|)^2}{m(m-1)|T|}} \right); \text{ by definition: } \left(\frac{0}{0} \right) = 1 \quad (30)$$

The parenthesis of (30) formally resembles the chance corrected agreement coefficient (18). However, unlike all other ${}_c\alpha$ -coefficients, it corrects only the observed agreement on the relevance of items, not agreements on irrelevant items and those not retrieved by all, which are retrieval, not coding errors. This correction causes the parenthesis to equal 0 when coincidences are indistinguishable from chance events and 1 when all m relevant items coincide perfectly.

So, when agreements on relevant items are without exceptions, ${}_c\alpha^{IR}$ becomes a measure of the degree to which relevant items are retrieved (= a generalization of *precision*). When the expected agreement is small or insignificant, ${}_c\alpha^{IR} = P^{IR}$. When the agreement on relevant items is due to chance, ${}_c\alpha^{IR} = 0$. And when all retrieved items are relevant without exception ${}_c\alpha^{IR} = 1 \cdot (0/0) = 1$. Just as all other reliability coefficient, ${}_c\alpha^{IR}$ can become negative when agreements among relevant items are systematically below chance.

12.5.2 For Replications of Judgments on the Relevance of Retrieval Results

Note that (30) defines an agreement coefficient from reliability data consisting of m independent retrievals. In compliance with the literature on information retrieval research, it assumes that the reliability of searches depends entirely on their retrieving relevant items, taking everything that deviates from this criterion—retrieving irrelevant items and failing to retrieve items that other searches found—as distracting from their reliability.

If one conceives search engines to retrieve information, not texts that analysts need to evaluate for their relevance, the problem of evaluating whether analysts are able to reliably distinguish relevant from irrelevant items among those retrieved does not appear to exist. It is conveniently projected onto the search engine. I suggest this to be an epistemological mistake that can be rectified by reference to Table 12.2.

To assess the reliability with which analysts distinguish between relevant and irrelevant items, the nonretrieved items labeled ϕ cannot contribute to answering this question. It follows that the third row and column in Table 12.2 has to be ignored or zeroed. One way to zero the row and column is to rely on only one search or ignore which searches the items come from. This has the effect of reducing the 3-by-3 coincidence matrix of Table 12.2 to a 2-by-2 coincidence matrix. Suppose further that this 2-by-2 coincidence matrix is generated by m analysts independently judging any total of $|T|$ retrieved items. Then the binary ${}_c\alpha_{\text{binary}}$ -coefficient (16) becomes applicable without modifications (except for algebraic simplifications of the sums listed in Table 12.2):

$${}_c\alpha_{\text{binary}} = 1 - \frac{m|T| - 1}{m-1} \frac{\sum_i \sum_{j \neq i} |R_i \cap I_j|}{\sum_i |R_i| \cdot \sum_j |I_j|} \quad (31)$$

12.6 AGREEMENT ON UNITIZING AND CODING FINITE CONTINUA

Most content analyses sample large and coherent texts: books, speeches, narratives, TV shows, video recordings, transcripts of interviews, streams of e-mail messages, or websites. These essentially are or can be conceived of as textual sequences, as finite continua. While one tends to have no trouble reading such continua and extracting meaningful information from them, without the ability to establish the reliability of identifying relevant sections in such continua, the reliabilities of subsequent coding, or analyzing such data remain on uncertain ground. This section seeks to extend reliability considerations to the processes of unitizing given continua.

Tabulating, comparing, or counting natural units seems unproblematic. Chapter 8 addresses various easily recognizable units. When such units exist, attention can be limited to the reliability of coding. However, predefined units are not always available. In such situations, some content analysts have standardized their units of analysis to enable statistical accounts of them. Pages of books, 30-second intervals of conversations, scenes in film, or the first 100 words of articles are easily separable and countable. Limiting attention to a physically identifiable section of a continuum may give researchers an objective definition of units but one that is sure to create meaningless data. For example, topics can hardly be distinguished by book pages. Analyzing talk shows on radio in 30-second intervals may cut question-answer sequences into uninterpretable segments, and distinguishing camera shots in film may make sense to film editors but rarely corresponds to how viewers conceive of what they see. Relying on units that do not fit the nature of the phenomena of interest is likely to introduce uncertainties into the data without the analysts' knowing whether, why, or to what extent.

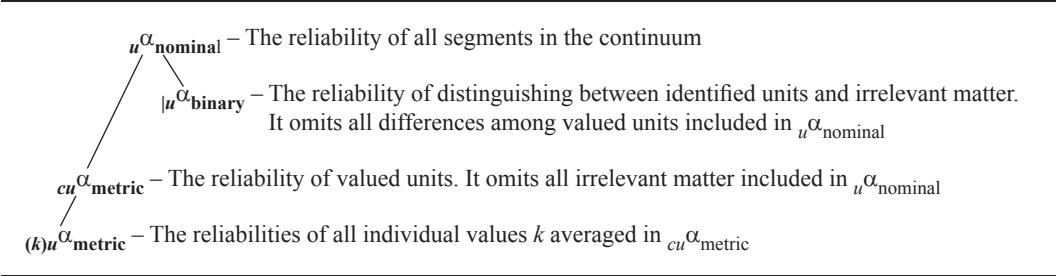
On the other extreme, qualitative scholars have advocated relying on units of texts that emerge in the process of reading. For understanding larger narratives, it is quite natural to select key quotes from speeches, clip informative paragraphs from newspapers, segment conversations by turns taken, or highlight and code analytically relevant passages of text as enabled by qualitative data analysis software. However, without testing the reliability of such emerging unitizations, inferences drawn from them stand on epistemologically shaky ground.

I contend that the lack of concerns for the reliability of unitizing is due largely to the absence of suitable agreement measures. Guetzkow (1950) was the first to address the reliability of unitizing. Unfortunately, his coefficient measures the extent to which two observers disagree on the number of units they identify, not on whether units are of the same kind, leaving open the question of what the coders actually counted. Osgood's (1959, p. 44) and Holsti's (1969, p. 140) %-like index for whether two observers selected the same or different kinds of predefined units cannot recognize intersections of unequal units and, most importantly, fail to consider chance. I proposed a way to overcome these deficiencies (Krippendorff, 1995a, 2004a) with a coefficient for unitizing that was intended to share the basic properties of the α -family of coefficients. Unfortunately, its complexity has discouraged potential users. In the first printing of the 3rd edition of this book, I offered a simplification, but a French research group¹, running numerous tests with artificial reliability data, discovered an undesirable insensitivity that compels me to hereby withdraw the 2004 proposal.

¹Stéphane Bouvry, Yann Mathet, and Antoine Widlöcher of the Université de Caen Basse-Normandie, Caen, France. Personal communication 2012.12.6

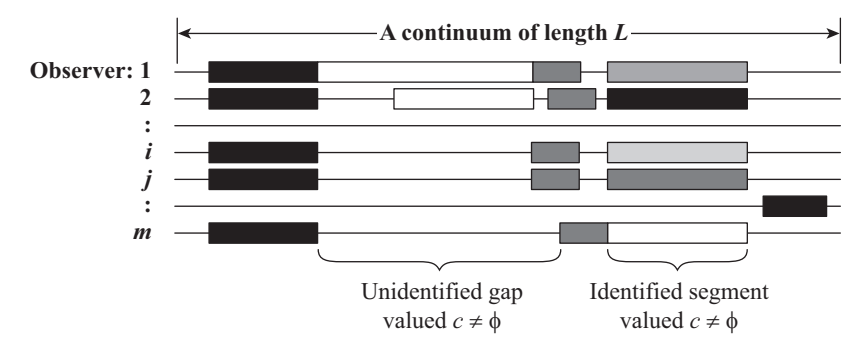
As described in section 12.2.3, unitizing a continuum can be conceptualized in terms of three analytical steps. First, introducing distinctions into a continuum, effectively partitioning it into mutually exclusive segments, second identifying among them the units of analysis that are expected to answer given research questions, and third, categorizing or valuing these units. This section presents a family of four coefficients for unitizing continua that have proven successful and for which software is available (Krippendorff, Mathet, Bouvry, and Widlöcher, 2016). Their relationships of inclusion are best illustrated with Figure 12.9.

FIGURE 12.9 ■ Relationships Within the u^α -family of Unitizing Coefficients



12.6.1 Reliability Data From Unitizing

As required for any reliability test, data for assessing the reliability of unitizing must be generated by two or more independently working observers, judges, annotators, coders, or analysts and have sufficient diversity to represent the phenomena of analytical interest. Unitization may yield relevant and irrelevant segments in a larger continuum. The irrelevant segments are similar to the missing data of the α -coefficients for predefined units. The relevant segments are categorized, valued, or coded by observers. One important consideration is that relevant segments are of unequal lengths. Following is a graphical illustration of the reliability data obtained by unitizing a given continuum by m observers:

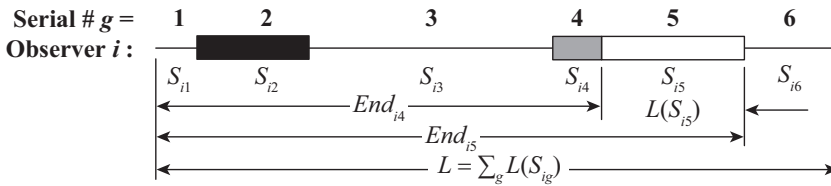


Terms used in this section (see Figure 12.10 for one observer):

- Observers are labeled: 1, 2, ..., i, j , ..., $m \geq 2$.
- Each observer partitions a common continuum into mutually exclusive segments S .

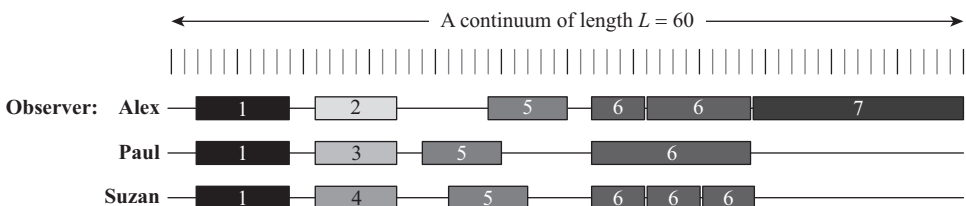
- Segments S_{ig} and S_{jh} are consecutively numbered:
For observer i : 1, 2, 3, ..., g , ...
For observer j : 1, 2, 3, ..., h , ...
- On the continuum, segments S_{ig} are located by where they end: End_{ig} .
Their beginning follows the end of their preceding segment S_{ig-1} : End_{ig-1} .
- Lengths are measured in integers
Segments S_{ig} : $L(S_{ig}) = End_{ig} - End_{ig-1}$
Intersections $S_{ig} \cap S_{jh} \neq \{\}$: $L(S_{ig} \cap S_{jh}) = \min(End_{ig}, End_{jh}) - \max(End_{ig-1}, End_{jh-1})$
Unions $S_{ig} \cup S_{jh}$: $L(S_{ig} \cup S_{jh}) = \max(End_{ig}, End_{jh}) - \min(End_{ig-1}, End_{jh-1})$
The continuum: $L = \sum_g L(S_{ig}) = \sum_h L(S_{jh})$
- Values c (or k) are assigned to each segment S_{ig} valued c :
For identified units (relevant matter), c is assigned by observers.
For gaps between units (irrelevant matter), $c = \phi$ is assigned by definition.

FIGURE 12.10 ■ Locating Segments—Units and Gaps Between Them—On a Continuum



Reliability is perfect when all segments pairwise agree regarding their length, location in the continuum, and kind or assigned value. Disagreements arise either when segments pairwise intersect in parts, when they are unequally valued, or both. Figure 12.11 depicts a simple numerical example that we refer to in the following:

FIGURE 12.11 ■ Numerical Example of a Simple Unitized Continuum



It depicts three observers' unitizations of a common continuum. The units in the first group agree perfectly, in location, length, and kind. The three units in the second group agree in location and length but not in kind. The units in the third group are of the same length and kind but disagree in their location on the continuum, intersecting only partly with one another. In the fourth group, all units are adjacent to each other, of the same kind, collectively occupying the same stretch of the continuum, but observers disagree on the number and lengths of the units they identified. The last group contains just one unit, identified by only one observer, not recognized by any of the other two.

12.6.2 Observed and Expected Coincidences

The following pursues the path to ${}_u\alpha$ via coincidence matrices. Figure 12.11 provides numerical examples for its development. Whereas ${}_c\alpha$ pairs all values that observers have assigned to predefined units, because unitizing allows for the valuation of segments unequal lengths, coincidences among segments will have to reflect the unequal lengths of intersections of segments among pairs of observers. To capture the intersections of segments, we define the sum of coincidences ℓ_{ck} as the sum of all pairs of intersecting segments, valued c and k , adjusted so that the sum of all coincidences equals the sum of all segments that the m participating observers identified:

$$\ell_{ck} = \frac{1}{m-1} \sum_i^m \sum_{j \neq i}^m \sum_{g,h} L(S_{ig \text{ valued } c} \cap S_{jh \text{ valued } k}) \quad (32)$$

The *matrix of the observed coincidences* in Figure 12.12 tabulates the sums of the lengths of these intersections:

FIGURE 12.12 ■ Matrix of Observed Coincidences Among All Intersecting Segments

Values:	ϕ	1	.	k	.	.	v	
ϕ	$\ell_{\phi\phi}$	$\ell_{\phi 1}$	.	$\ell_{\phi k}$	.	.	$\ell_{\phi v}$	$\ell_{\phi \cdot}$
1	$\ell_{1\phi}$	ℓ_{11}	.	ℓ_{1k}	.	.	ℓ_{1v}	$\ell_{1 \cdot}$
.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.
c	$\ell_{c\phi}$	ℓ_{c1}	.	ℓ_{ck}	.	.	ℓ_{cv}	$\ell_{c \cdot} = \sum_{k=\phi}^v \ell_{ck}$
.	.	.	.	.	.	.	.	.
v	$\ell_{v\phi}$	ℓ_{v1}	.	ℓ_{vk}	.	.	ℓ_{vv}	$\ell_{v \cdot}$
	$\ell_{\cdot \phi}$	$\ell_{\cdot 1}$	.	$\ell_{\cdot k}$	.	.	$\ell_{\cdot v}$	$\ell_{\cdot \cdot} = \sum_{c=\phi}^v \sum_{k=\phi}^v \ell_{ck} = mL$

One implication of summing and averaging coinciding segments with (32) is that the contiguities of individual segments are not preserved but acknowledged in the expression of the expected coincidences.

All α -reliability coefficients are defined in terms of the observed disagreements and the expected disagreements. In the case of unitizing, the observed disagreement ${}_u D_o$ is a function of the observed coincidences ℓ_{ck} in Figure 12.12. The expected disagreements ${}_u D_e$ are a function of the marginal sums in Figure 12.12, corrected by the lengths of segments that go into these sums. The expected coincidences ε_{ck} are as follows:

$$\varepsilon_{ck} = \ell_{..} \frac{\ell_c \cdot \ell_{.k} - \left(\sum_i^m \sum_g \begin{cases} L(S_{ig} \text{ valued } c=\phi) \\ \left(L(S_{ig} \text{ valued } c \neq \phi) \right)^2 \text{ iff } c = k \end{cases} \right)}{\ell_{..}^2 - \sum_i^m \sum_g \begin{cases} L(S_{ig} \text{ valued } = \phi) \\ \left(L(S_{ig} \text{ valued } \neq \phi) \right)^2 \end{cases}} \quad (33)$$

When data are small and segments are very uneven in lengths, the matrix of expected coincidences has margins that may deviate slightly from those in the matrix of observed coincidences in Figure 12.12:

$$\varepsilon_{c.} = \sum_{k=\phi}^v \varepsilon_{ck} \approx \ell_c.$$

$$\varepsilon_{.k} = \sum_{c=\phi}^v \varepsilon_{ck} \approx \ell_{.k}$$

$$\text{but } \varepsilon_{..} = \sum_{c=\phi}^v \sum_{k=\phi}^v \varepsilon_{ck} = \ell_{..}$$

For the numerical example in Figure 12.12 the two matrices defined by (32) and (33) are as follows:

Matrix of all observed coincidences										Matrix of all expected coincidences									
	ϕ	1	2	3	4	5	6	7			ϕ	1	2	3	4	5	6	7	
ϕ	45	.	.	.	.	10	.	16	71		28.46	8.54	2.44	2.44	2.44	7.32	14.63	6.50	72.77
1	.	21	.	.	.	.	.	.	21		8.54	1.68	0.72	0.72	0.72	2.16	4.33	1.92	20.80
2	.	.	0	3	3	.	.	.	6		2.44	0.72	0	0.21	0.21	0.62	1.24	0.55	5.98
3	.	.	.	3	0	3	.	.	6		2.44	0.72	0.21	0	0.21	0.62	1.24	0.55	5.98
4	.	.	.	3	3	0	.	.	6		2.44	0.72	0.21	0.21	0	0.62	1.24	0.55	5.98
5	10	.	.	.	.	.	8	.	18		7.32	2.16	0.62	0.62	0.62	1.24	3.71	1.65	17.93
6	.	.	.	.	.	.	.	36	36		14.63	4.33	1.24	1.24	1.24	3.71	5.86	3.30	35.54
7	16	.	.	.	.	.	.	.	16		6.50	1.92	0.55	0.55	0.55	1.65	3.30	0	15.02
	71	21	6	6	6	18	36	16	180		72.77	20.80	5.98	5.98	5.98	17.93	35.54	15.02	180.00

To comment briefly on the construction of the observed coincidences, notice in Figure 12.11 the three segments of length 7 and valued 1. They appear as their sum in the 1-1 cell. The three segments in the second group are of equal length but unequal value. Pair comparisons involve six pairs of mismatching values, which can be found in the matrix of observed coincidences in rows-by-columns numbered 2, 3, and 4. With no other segments valued 2, 3, or 4, their length of 6 is the only length that appears in the row and column sums. The next group of identified segments are equal in length and valuation but not in location on the continuum. The intersection between Alex and Paul is 1 unit in length, between Alex and Suzan it is 3, and between Paul and Suzan it is 4, or 8 in total. This number is found in the 5-5 cell of the matrix of observed coincidences. The remaining intersections of these three segments with irrelevant parts of the continuum is found in the ϕ -5 and 5- ϕ cells. The last segment, identified and valued 7 by Alex but by nobody else is found in the ϕ -7 and 7- ϕ cells.

The group of segments valued 6 is instructive for another reason. Summing and averaging intersections as defined in (32) means that distinctions matter only when they separate unequally valued segments, as in the third group of units with irrelevant segments. The distinctions within this fourth group are not recognizable in the sums of their lengths of agreements, found in the 6-6 cell of the above matrix.

In the matrix of expected coincidences, I want to point to two important properties. First, in general, expectations are a function of the product of two marginal sums. This will be recognized in the numerator of ε_{ck} in (33). However, just as we do not pair observed segments with themselves, (33) has to prevent pairing of segments with themselves as well. In $\varepsilon_{c=k}$, what would amount to self-pairing is subtracted from the product $\ell_c \cdot \ell_k$. The second point has to do with acknowledging the lengths of the segments involved. When segments are relevant units of analysis, $\varepsilon_{c=k}$ preserves their contiguity by squaring their length whereas irrelevant matter between identified units are not considered as having contiguities or holistic qualities, only lengths. The effects of not matching units with themselves may be seen when there is only one of a kind to be paired with all the others, this is the case with the three segments valued 2, 3, and 4. Each is paired with all others but the cells 2-2, 3-3, and 4-4 are appropriately zero.

12.6.3 Agreement on the Partitions of a Continuum Into Unequal Segments

Inasmuch as partitions of a continuum may and typically do include irrelevant segments, left-over matter between relevant segments, without any metric qualities, the only applicable metric for partitions is the nominal metric. It distinguishes only whether intersecting segments are of the same or a different kind. Its limitation to nominal data, that is, treating all values as unordered categories offers algebraic simplifications. Following (2), with reference to the observed and expected coincidences defined by (32) and (33) and without the latter, the *agreement coefficient* ${}_u\alpha$ become:

$$\begin{aligned} {}_u\alpha_{\text{nominal}} &= 1 - \frac{{}_u D_o}{{}_u D_e} = 1 - \frac{\sum_{c=\phi}^v \sum_{k=\phi}^v \ell_{ck} \text{ nominal} \delta_{ck}^2}{\sum_{c=\phi}^v \sum_{k=\phi}^v \varepsilon_{ck} \text{ nominal} \delta_{ck}^2} = 1 - \frac{\ell_{..} - \sum_{c=\phi}^v \ell_{cc}}{\ell_{..} - \sum_{c=\phi}^v \varepsilon_{cc}} = \\ &= 1 - \left(\frac{\ell_{..} - \frac{1}{\ell_{..}} \sum_i^m \sum_g \left\{ \begin{array}{l} L(S_{ig} \text{ valued} = \phi) \\ \left(L(S_{ig} \text{ valued} \neq \phi) \right)^2 \end{array} \right\}}{\ell_{..}^2 - \sum_{c=\phi}^v \ell_c^2} \right) \frac{\ell_{..} - \sum_{c=\phi}^v \ell_{cc}}{\ell_{..}^2 - \sum_{c=\phi}^v \ell_c^2} \end{aligned} \quad (34)$$

The first is the general form of (1). The second enumerates all observed and expected disagreements as in (10). The third takes advantage of nominal data according to which the number of disagreements equals the total minus the number of agreements, and the last form replaces the expected coincidences by their definition.

Because the two matrices of coincidences are known, we can use the simplest form of (34), which is the third, and obtain the ${}_u\alpha$ -reliability in the terms found in these matrices:

$${}_u\alpha_{\text{nominal}} = 1 - \frac{{}_u D_o}{{}_u D_e} = 1 - \frac{\ell_{..} - \sum_{c=\phi}^v \ell_{cc}}{\ell_{..} - \sum_{c=\phi}^v \epsilon_{cc}} = 1 - \frac{180 - (45 + 21 + 8 + 36)}{180 - (28.46 + 1.68 + 1.24 + 5.80)} = 0.510$$

12.6.4 Agreement on the Distinction Between Relevant and Irrelevant Segments

Especially when research needs to report on the volumes of relevant matter found in texts, it is important to know the reliability of this distinction. It turns out to be a very simple modification of (32), (33), and (34). The ${}_u\alpha$ -coefficient is obtained by applying any of the definitions of ${}_u\alpha$ to a coincidence matrix that eliminates the distinctions among individually valued units by summing all of them into a single category. It effectively reduces Figure 12.12 along the dotted lines to the 2-by-2 matrix of observed coincidences in Figure 12.13, merely distinguishing among two kinds of values, $c=\phi$ and $c\neq\phi$, or ϕ and $\neq\phi$ for short:

FIGURE 12.13 ■ Matrix of Observed Coincidences Among Relevant and Irrelevant Segments

Values:		ϕ	$\neq\phi$	
ϕ	$\ell_{\phi\phi}$	$\ell_{\phi\neq\phi}$	$\ell_{\phi.}$	
$\neq\phi$	$\ell_{\neq\phi\phi}$	$\ell_{\neq\phi\neq\phi}$	$\ell_{\neq\phi.} = \sum_{c=1}^v \sum_{k=\phi}^v \ell_{ck}$	
	$\ell_{.\phi}$	$\ell_{. \neq\phi}$	$\ell_{..} = \sum_{c=\phi}^v \sum_{k=\phi}^v \ell_{ck} = mL$	

Applying (34) to these data, simplifying its form by taking advantage of the binary nature of the data, omitting references to the nominal difference function δ_{ck}^2 , and stating ${}_u\alpha_{\text{binary}}$ in terms of disagreements, observed and expected coincidences and without the latter yields:

$${}_u\alpha_{\text{binary}} = 1 - \frac{{}_u D_o}{{}_u D_e} = 1 - \frac{\ell_{\phi\neq\phi}}{\epsilon_{\phi\neq\phi}} = 1 - \left(\ell_{..} - \frac{1}{\ell_{..}} \sum_i^m \sum_g \left\{ L(S_{ig} \text{ valued} = \phi) \right\} \right) \frac{\ell_{\phi\neq\phi}}{\ell_{\phi.} \cdot \ell_{. \neq\phi}} \quad (35)$$

By summing the 8-by-8 matrices of observed and expected coincidences from our example along the dotted lines yields the following two 2-by-2 matrices.

Observed coincidences

Values:

	ϕ	$\neq\phi$	
ϕ	45	26	71
$\neq\phi$	26	83	109
	71	109	180

Expected coincidences

Values:

	ϕ	$\neq\phi$	
ϕ	28.46	44.31	72.77
$\neq\phi$	44.31	72.77	107.28
	72.77	107.23	180.00

And:

$$\begin{aligned}
 {}_{|u}\alpha_{\text{binary}} &= 1 - \frac{{}_{|u}D_o}{{}_{|u}D_e} = 1 - \frac{26}{44.31} = 0.413 \\
 &= 1 - \left(180 - \frac{1}{180} (71 + 3 \cdot 7^2 + 6 \cdot 6^2 + 4 \cdot 4^2 + 8^2 + 12^2 + 16^2) \right) \frac{26}{71 \cdot 109} = 0.413 \\
 &= 1 - (180 - 5.344) \frac{26}{7739} = 0.413
 \end{aligned}$$

12.6.5 Agreement on the Coding of Single-Valued Segments Independent of Unitizing

After establishing the reliability ${}_u\alpha$ of partitioning a continuum into mutually exclusive parts of various kinds, and assuring themselves with the help of ${}_u\alpha$ that relevant segments are reliably distinguished and their volumes are reportable, content analysts face the problem of assessing the reliability of coding relevant segments of a continuum. Unfortunately, most content analysts omit assessing the reliability of unitizing by standardizing units and caring only to test the reliability of categorizing, valuing, or coding them. To assure reliability without disregarding the unequal segmentation of continua, an α -coefficient for coding the identified units, called ${}_{cu}\alpha$, is recommended—the subscript cu is to indicate that the coding is conditional on the prior unitization, whose unreliability is now ignored.

The most important feature of ${}_{cu}\alpha$ is its ability to respond to any metric properties of the reliability data, not just the nominal metric to which ${}_u\alpha$ is limited. This means that it can only be applied to intersections of valued segments, not irrelevant ones, labeled ϕ . All of the observed coincidences needed to obtain the reliability of coding unitized matter are contained in the v -by- v submatrix of values in the lower right quadrant of Figure 12.12. This submatrix has its own row and column sums:

$$\ell_{ck}^* = \sum_{k=1}^v \ell_{ck} = \ell_{c\cdot} - \ell_{c\phi}$$

$$\ell_{\cdot k}^* = \sum_{c=1}^v \ell_{ck} = \ell_{\cdot k} - \ell_{\phi k}$$

$$\text{and } \ell^{*..} = \sum_{c=1}^v \sum_{k=1}^v \ell_{ck} = \ell_{\neq\phi \neq\phi}$$

The *expected coincidences* ε_{ck}^* for this sub-matrix are obtained by analogy to (33) but limited to the intersections of relevant matter:

$$\varepsilon_{ck}^* = \frac{\ell_{c..}^* \ell_{..k}^* - \left(\sum_i^m \sum_g \left(L \left(S_{ig} \text{ valued } c \neq \phi \cap \bigcup_{j \neq i, h} S_{jh} \text{ valued } \neq \phi \right) \right)^2 \text{ iff } c = k \right)}{\ell_{..}^* - \frac{1}{\ell_{..}^*} \sum_i^m \sum_g \left(L \left(S_{ig} \text{ valued } \neq \phi \cap \bigcup_{j \neq i, h} S_{jh} \text{ valued } \neq \phi \right) \right)^2} \quad (36)$$

Applied to the observed and expected coincidences of our example would yield the following v -by- v matrices:

Sub-matrix of observed coincidences

ϕ	1	2	3	4	5	6	7	
ϕ								
1	21	.	.	.	.	.	.	21
2	.	0	3	3	.	.	.	6
3	.	3	0	3	.	.	.	6
4	.	3	3	0	.	.	.	6
5	.	.	.	.	8	.	.	8
6	.	.	.	.	.	36	.	36
7	.	.	.	.	.	.	.	.
	21	6	6	6	8	36	.	83

Sub-matrix of expected coincidences

ϕ	1	2	3	4	5	6	7	
ϕ								
1	3.85	1.65	1.65	1.65	2.20	9.90	.	20.91
2	1.65	0	0.47	0.47	0.63	2.83	.	6.05
3	1.65	0.47	0	0.47	0.63	2.83	.	6.05
4	1.65	0.47	0.47	0	0.63	2.83	.	6.05
5	2.20	0.63	0.63	0.63	0.50	3.77	.	6.05
6	9.90	2.83	2.83	2.83	3.77	13.41	.	35.59
7	.	.	.	.	.	.	.	.
	20.91	6.05	6.05	6.05	6.05	35.59	.	83.00

Note that the observed coincidences among intersecting segments are found in the larger matrix while the expected coincidences are computed anew.

In these terms, the $_{cu}\alpha$ -agreement for coding relevant segments of a continuum becomes just as in the general form of (34), which maintains reference to the difference function:

$$_{cu}\alpha_{metric} = 1 - \frac{_{cu}D_o}{_{cu}D_e} = 1 - \frac{\sum_{c=1}^v \sum_{k=1}^v \ell_{ck}^{metric} \delta_{ck}^2}{\sum_{c=1}^v \sum_{k=1}^v \varepsilon_{ck}^{metric} \delta_{ck}^2} \quad (37)$$

Replacing the expected coincidences by their definition becomes unwieldy and distracts from the simplicity of the general form. Applied to the two v -by- v coincidences of our example, treating the values as categorical yields:

$$_{cu}\alpha_{nominal} = 1 - \frac{\sum_{c=1}^v \sum_{k=1}^v \ell_{ck}^{nominal} \delta_{ck}^2}{\sum_{c=1}^v \sum_{k=1}^v \varepsilon_{ck}^{nominal} \delta_{ck}^2} = 1 - \frac{\ell_{..}^* - \sum_{c=1}^v \ell_{cc}}{\ell_{..}^* - \sum_{c=1}^v \varepsilon_{cc}^*} = 1 - \frac{83 - (21 + 8 + 36)}{83 - (3.85 + 0.50 + 13.41)} = 1 - \frac{18}{65.24} = 0.724$$

$$_{cu}\alpha_{interval} = 1 - \frac{\sum_{c=1}^v \sum_{k=1}^v \ell_{ck}^{interval} \delta_{ck}^2}{\sum_{c=1}^v \sum_{k=1}^v \varepsilon_{ck}^{interval} \delta_{ck}^2} = 1 - \frac{2[(3+3)1^2 + 3 \cdot 2^2]}{2[(1.65 + 0.47 + 0.47 + 0.63 + 3.77)1^2 + \dots + 9.90 \cdot 5^2]} = 1 - \frac{36}{806.56} = 0.955$$

Evidently, $_{cu}\alpha_{interval} > _{cu}\alpha_{nominal}$. The reason for this inequality is due to the distribution of observed coincidences. The algebraic differences among units 2, 3,

and 4 are relatively small to what could be and weigh proportionally less than if they were further apart. The nominal difference function, by contrast, merely distinguishes whether the valuation of coincidences agree or not. It does not respond to the distribution of algebraic differences.

There is another quality of ${}_{cu}\alpha$ worth noting. For metric differences to be applicable, intersections with irrelevant segments, valued ϕ are omitted from ${}_{cu}\alpha$. In our example, this pertains to the unit valued 7 and the intersections of the three segments in the third group with irrelevant matter, valued ϕ . We can report the

$$\% \text{ of } {}_{cu}\alpha\text{'s coverage of relevant segments} = 100 \frac{\ell^*}{\ell_{..} - \ell_{\phi}}, \text{ here } 100 \frac{83}{180 - 71} = 76.15\%$$

12.6.6 Agreement on Individual Values Among Valued Intersections

It is often important to evaluate the agreement of individual values and compare their reliability to determine which of them is too ambiguous, may require more work, or cannot be relied upon in reporting research findings. For any value k , we obtain the agreement coefficient ${}_{(k)u}\alpha_{\text{metric}}$ from the observed and expected disagreements in the k^{th} row or column:

$${}_{(k)u}\alpha_{\text{metric}} = 1 - \frac{\varepsilon^* \cdot k \sum_{c=1}^v \ell_{ck} \text{ metric } \delta_{ck}^2}{\ell^* \cdot k \sum_{c=1}^v \varepsilon_{ck}^* \text{ metric } \delta_{ck}^2} \quad (38)$$

And express the percent covered among all segments valued k by:

$$\% \text{ of } {}_{(k)u}\alpha\text{'s coverage of all segments valued } k = 100 \frac{\ell^* \cdot k}{\ell_{\cdot k}}$$

Applying (38) to the individual values in the two v -by- v coincidence matrices of our example, yields these reliabilities:

$${}_{(1)u}\alpha_{\text{nominal}} = 1 - \frac{20.91}{21} \frac{21 - 21}{20.91 - 3.85} = 1.000 \quad \text{-- covering } 100 \frac{21}{21} = 100\% \text{ of all units valued } 1$$

$${}_{(2)u}\alpha_{\text{nominal}} = 1 - \frac{6.05}{6} \frac{6 - 0}{6.05 - 0} = 0.000 \quad \text{-- covering } 100 \frac{6}{6} = 100\% \text{ of all units valued } 2$$

$${}_{(3)u}\alpha_{\text{nominal}} = {}_{(4)u}\alpha_{\text{nominal}} = {}_{(2)u}\alpha_{\text{nominal}} = 0.000$$

$${}_{(5)u}\alpha_{\text{nominal}} = 1 - \frac{8.36}{8} \frac{8 - 8}{8.36 - 0.50} = 1.000 \quad \text{-- covering } 100 \frac{8}{18} = 44\% \text{ of all units valued } 5$$

$${}_{(6)u}\alpha_{\text{interval}} = 1 - \frac{35.58}{36} \frac{36 - 36}{35.58 - 13.41} = 1.000 \quad \text{-- covering } 100 \frac{36}{36} = 100\% \text{ of all units valued } 6$$

These results are quite obvious from the matrix of observed coincidences. The only values that deviate from perfect reliability are values 2, 3, and 4, exhibiting no agreement and measuring zero each. The remaining values are perfectly reliable, of which ${}_{(5)u}\alpha_{\text{nominal}}$ covering only 44% of the segments valued 5. The latter intersect with irrelevant segments ϕ which ${}_{cu}\alpha$ and ${}_{(k)u}\alpha$ ignore but the percent of covering assesses.

12.7 AGREEMENT ON MULTI-VALUED CODING

While the practice of assigning single values to given units is common, and reliability tests are worked out as in the previous sections, there are many situations in which phenomena are more naturally described in terms of multiple values. Literary theory has argued for some time that all texts have multiple interpretations, not one. Word dictionaries list semantically similar meanings among which readers may choose depending on their syntactical function, context of use, or the discourse being pursued in a text. A challenge of computational linguistics is to disambiguate sentence structures and identify all alternative readings. In open-ended interviews, it is quite common that simple questions yield multiple answers. In psychology, Rorschach tests expect subjects to give several accounts of their perceptions to deliberately ambiguous images, yielding psychological insights about the respondent. All human beings play diverse roles in their lives, not one. Fictional characters tend to possess complex personalities that make for novels to be fascinating reading. Even very result oriented marketing research interested in the hobbies of customers needs to recognize that people have more than one. In everyday use of language, multi-valuedness is probably more prevalent than single-valuedness.

One reason for the preference of representing units of analysis in terms of single categories is the preponderance of available statistical techniques that require quantifying the frequencies of single-valued accounts. Statistical techniques for analyzing multi-valued data are more complex, yet quite common in contemporary information technology where reliability issues seem less of a problem, however. Qualitative scholars have no problem with generating multiple interpretations of texts or observations but shy away from evaluating the reliability of their choices. To my knowledge, the developers of computer-aided qualitative text analysis software have not addressed issues of the reliability of the multi-valued coding their software supports.

Social research would be severely limited if multi-valued phenomena would be reduced to single-valued ones—just because statistical techniques for analysis are simpler or more readily available. This applies to assessing the reliability of multi-valued phenomena as well. Some researchers who have faced multi-valued data have computed them, including their reliability, by regarding the presence or absence of individual categories as binary variables. Treating multi-valued data as multi-dimensional data assumes an enormous space of possibilities, ($2^{\text{Number of all values used in size}}$), mostly empty, not only making the estimation of expected disagreements nearly meaningless but also blatantly ignoring the relationships between values that observers may have had in mind when describing a phenomenon in terms of more than one category. This is most obvious when values are ordered. The four readings {20, 24, 21, 23} deviate from the average of 22. By contrast, if these four readings would be regarded as nominal categories, which are either unlike or the same, in this case have nothing in common, hence they would omit references to their proximities.

Following an earlier proposal (Krippendorff, 1992), Krippendorff and Craggs (2016) have developed multi-valued conceptions of reliability and provided software to compute it for categorizing predefined units. Unfortunately, when the number of combinatorially available values exceeds ten, this software runs into computational limitations.

The following outlines a computationally efficient approach to obtain the reliability of multi-valued coding by sacrificing some of the alternatives provided by but not implemented in the above-mentioned work.

12.7.1 Two Reasons for Multi-Valued Coding

One reason for the need of multi-valued coding resides in the experience of concurrently valid multitudes—not to be confused with the difficulty of deciding among several alternatives—but actually facing phenomena for which a single description would be obviously wrong. People assume several roles in their lives, have several hobbies, several children, and a number of bank accounts. Students take more than one class per semester, and people employ multiple qualitative terms to describe each other’s abilities and personal qualities. Forcing observers to pick one category to describe such multi-valued phenomena renders the resulting data not only arbitrary but typically unreliable as well. Observers should be able to use sets values on their accounts of observations, readings, or interpretations:

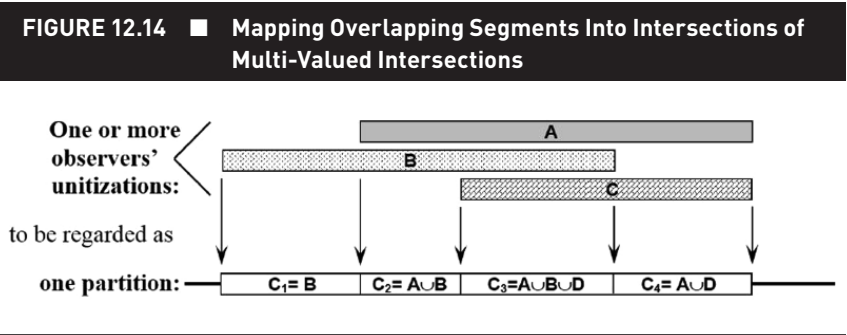
$\{c_1, c_2, \dots\} = C$ and $\{k_1, k_2, \dots\} = K$ – excluding empty sets $\{\}$

to describe these phenomena. Single-valued coding is of course always a special case of multi-valued coding.

A second reason for multi-valued data arises in situations governed by multiple and equally valid perspectives, for example, any merchandise always requires at least two perspectives, that of a seller and of a buyer, and perhaps of witnesses including scientific observers trying to describe a transaction. A court may have to consider the view of victims, perpetrators, and witnesses. When multiple perspectives are considered equally valid constituents of social phenomena, their union needs to describe them.

This is also the path to understand judging overlapping segments of a continuum. Suppose one or several observers unitize a text in terms of overlapping and separately valued segments, section 12.5 would not be able to handle their reliability data. First, because the $_{u}\alpha$ -family of agreement coefficient is defined for partitions of a continuum into mutually exclusive segments, and second, it is defined only for single-valued coding of these segments. However, just as the $_{cu}\alpha$ -agreement measure handles intersections of pairs of observers, one can extend the manner in which it handles such situations to overlapping and separately valued segments by one or more observers in terms of their intersections and valuing them by the unions of their segments’ values. This creates a way to compare groups of observers, each taken as one.

Figure 12.14 illustrates a situation in which an observer or group of three identifies three overlapping segments separately valued A, B, and D, which may designate sets of values, including containing only one.



12.7.2 Reliability Data for Multi-Valued Coding

For simplicity, we represent multi-valued reliability data in Figure 12.15 in terms of the matrices of observed coincidences. For coding predefined units, coincidences are obtained from reliability data analogue to Figure 12.4 and (6), however not as o_{ck} between single coincidences between values c and k but as coincidences o_{CK} between sets C and K of values, with m_u as the number of multi-valued sets describing unit u :

$$o_{CK} = \sum_{u=1}^N \frac{\begin{cases} n_{uC}(n_{uK} - 1) & \text{iff } C = K \\ n_{uC}n_{uK} & \text{iff } C \neq K \end{cases}}{m_u - 1} \quad (39)$$

Coincidences o_{CK} are tabulated analogue to Figure 12.5 and shown on the left of Figure 12.15

For the multi-valued coding of unitized segments, coincidences ℓ_{CK} among sets C and K are defined analogue to (32) but exclusive of intersections with irrelevant matter or blanks, labeled ϕ :

$$\ell_{CK} = \frac{1}{m-1} \sum_i^m \sum_{j \neq i}^m \sum_{g,h} L(S_{ig} \text{ valued } C \neq \phi \cap S_{jh} \text{ valued } K \neq \phi) \quad (40)$$

and tabulated in the coincidence matrix to the right of Figure 12.15. It corresponds to the coincidence matrix for computing α_{cu} , except for referring to multi-valued sets C and K instead of single values c and k :

FIGURE 12.15 ■ Matrices of Observed Coincidences Among Multi-Valued Sets of Values

	1	2	...	K	.	
1	o_{11}	o_{12}	...	o_{1K}	.	$n_{1\cdot}$
C	o_{C1}	o_{C2}	...	o_{CK}	.	$n_{C\cdot}$
.	.	.	...	.	.	.
.	.	.	...	.	.	.
.	.	.	...	.	.	.
	$n_{\cdot 1}$	$n_{\cdot 2}$	...	$n_{\cdot K}$	.	$n_{\cdot\cdot} = \sum_C \sum_K o_{CK}$

	1	2	...	K	.	
1	ℓ_{11}	ℓ_{12}	...	ℓ_{1K}	.	$\ell_{1\cdot}^*$
C	ℓ_{C1}	ℓ_{C2}	...	ℓ_{CK}	.	$\ell_{C\cdot}^* = \sum_K \ell_{CK}$
.	.	.	...	.	.	.
.	.	.	...	.	.	.
.	.	.	...	.	.	.
	$\ell_{\cdot 1}^*$	$\ell_{\cdot 2}^*$	...	$\ell_{\cdot K}^*$	.	$\ell_{\cdot\cdot}^* = \sum_C \sum_K \ell_{CK}$

In the matrix on the left, o_{CK} is the number of C - K pairs found within units, in the matrix on the right, ℓ_{CK} is the sum of the lengths of intersections valued C and K .

12.7.3 Multi-Valued Difference Functions

For nominal data, the multi-valued difference function enumerates the number of pairs of values across C and in K on which there is no agreement:

$$\text{nominal } \Delta_{CK} = |C \parallel K| - |C \cap K|^2 \quad (41)$$

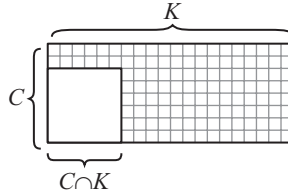
Where:

$|C|$ = C 's cardinality = the number of elements in the set C

$|C||K|$ = The number of possible C - K pairs between values in C and values in K

$|C \cap K|^2$ = The number of C - K pairs whose differences are zero.

Graphically, ${}_{\text{nominal}}\Delta_{CK}$ regards any two sets as follows:



Generalizing (41) for multi-valued nominal data to a multi-valued nonnominal data, calls for applying the metric difference functions ${}_{\text{metric}}\delta_{ck}^2$ for single values (12) through (15) to nonmatching values and yields:

$${}_{\text{metric}}\Delta_{CK} = \sum_{c \in C} \sum_{k \in K} {}_{\text{metric}}\delta_{ck}^2 - \sum_{c \in C \cap K} \sum_{k \in K \cap C} {}_{\text{metric}}\delta_{ck}^2 \quad (42)$$

When the metric difference functions ${}_{\text{metric}}\delta_{ck}^2$, in (42) are replaced by ${}_{\text{nominal}}\delta_{ck}^2$, (42) becomes (41).

12.7.4 Disagreements for Multi-Valued Data

The observed disagreements follow (10) but are defined in the above terms:

$${}_CD_o = \frac{\sum_C \sum_K o_{CK} {}_{\text{metric}}\Delta_{CK}}{\sum_C \sum_K o_{CK} |C||K|} \quad (43)$$

$${}_{Cu}D_o = \frac{\sum_C \sum_K \ell_{CK} {}_{\text{metric}}\Delta_{CK}}{\sum_C \sum_K \ell_{CK} |C||K|} \quad (44)$$

The expected disagreements, analogue to (8) and (9) are:

$${}_CD_e = \frac{\sum_C n_C \cdot \sum_K n_K {}_{\text{metric}}\Delta_{CK}}{(\sum_C n_C \cdot |C|)^2 - \sum_C n_C \cdot |C|} \quad (45)$$

Where:

$\sum_C n_C \cdot |C|$ = The number of values used, regardless of the sets in which they occur. It is also the number of values not to be paired with themselves.

$(\sum_C n_C \cdot |C|)^2$ = The number of possible pairs of values

For multi-valued intersection of variously unitized segments of a continuum, the numbers of values, of pairs of values in (45) are replaced by the sum of the lengths of intersecting segments, yielding the expected disagreement:

$$c_u D_e = \frac{\sum_{C \neq \phi} \ell_C^* \cdot \sum_{K \neq \phi} \ell_K^* \cdot K \text{ nominal } \Delta_{CK}}{(\sum_{C \neq \phi} \ell_C^* \cdot |C|)^2 - \sum_i^m \sum_g \left(L(S_{ig} \text{ valued } C \neq \phi \cap \bigcup_{j \neq i, h} S_{jh} \text{ valued } \neq \phi) \right)^2 |C|} \quad (46)$$

Where

$L(S_{ig} \text{ valued } C \neq \phi \cap \bigcup_{j \neq i, h} S_{jh} \text{ valued } \neq \phi)$ = The pairable length of segment S_{ig}

$(L(S_{ig} \text{ valued } C \neq \phi \cap \bigcup_{j \neq i, h} S_{jh} \text{ valued } \neq \phi))^2 |C|$ = The result of pairing the above with itself
and for all of the $|C|$ values assigned to it.

Subtracting the sum of all wrongful inclusions of pairing segments with themselves from $(\sum_{C \neq \phi} \ell_C^* \cdot |C|)^2$ yields the largest expected disagreement that correctly exclude segments paired with themselves.

12.7.5 $c\alpha$ for Multi-Valued Coding

$$c \alpha_{\text{metric}} = 1 - \frac{c D_o}{c D_e} \quad (47)$$

$$c_u \alpha_{\text{metric}} = 1 - \frac{c_u D_o}{c_u D_e} \quad (48)$$

12.8 STATISTICAL PROPERTIES OF α

All agreement coefficients α are statistical measures. In the following, I briefly discuss four conditions and precautions for accepting or rejecting them as indications of reliability: insufficient variation, sampling considerations, statistical significance, and standards for the reliability of data.

12.8.1 Insufficient Variation

A frequently puzzling condition arises when reliability data show insufficient variation. Consider this almost extreme example:

Reliability Data									Observed coincidences			Expected coincidences				
Units:	1	2	3	4	5	6	7	8	0	1		0	1			
Observer A:	0	0	0	0	0	0	0	0	0	14	1	15	0	14	1	15
Observer B:	0	0	0	0	0	0	0	1	1	1	0	1	1	1	0	1
									15	1	16		15	1	16	

Here, the resulting matrices of observed and expected coincidences are identical, $D_o = D_e$ and ${}_c\alpha = 0$. Time and time again, statistical novices find this condition difficult to accept. They have argued that there evidently is considerable agreement on the value “0,” in fact in 7 out of 8 units, or 88%. How could ${}_c\alpha$ suggest reliability to be absent? This argument not only confuses percent agreement with agreement above what is expected by chance but also overlooks the necessity for all data to exhibit variation in the form of information about the phenomena to be attended to, not only to answer a research question but also to shed light on the reliability of data or of the instrument used in coding. In the above case, perhaps the material coded was mostly of the same kind, perhaps the observers found their task boring and settled on scoring habitually, perhaps the measuring instrument they were using was broken—we do not know. However, in the 8th unit, in which one observer noticed something out of line, in the only unit that seems to be at variance with the others, the two observers fail to agree, and ${}_c\alpha = 0$ as it should. When analyzing rare phenomena, sample sizes may have to be larger than for phenomena that are widely available. What applies for data in general applies to reliability data in particular. When the reliability data are to inform the analysts of the trustworthiness of the data, rare incidences are particularly important.

Consider the reliability of a medical diagnosis. Most people are healthy. The symptoms for any one known disease surely is a statistically rare occurrence. If a test for any one disease would yield the above kind of reliability data, that test cannot possibly be trusted, and ${}_c\alpha$ near zero is perfectly justified.

Regarding the numerical example, suppose observer B had assigned a “0” to unit 8 as well. Then we would have 100% agreement but even less evidence that the observers did their job. They could have been inattentive or stuck in preconceptions—assurances to the contrary aside—or, to avoid being judged unreliable, coders may have agreed in advance simply to label everything “0.” In effect, they functioned not unlike a broken measuring instrument, such as a broken thermometer showing the same temperature all the time. Any working measuring instrument needs to demonstrate its responsiveness to external variations. Facing the total absence of variation, $D_o = D_e = 0$, and ${}_c\alpha = 1 - 0/0$ is mathematically indeterminate, as it could be either 0 or $-\infty$. I define $0/0 = 1$ as it is the limiting case to which all agreement coefficients converge as the variance in the reliability vanishes.

Suppose further that the two observers agree on a “1” for unit 8, then observer A’s change of heart concerning this value would cause ${}_c\alpha$ to jump from 0 to 1. This might seem surprising. Although the two 1s are still rare occurrences, by chance alone, they could show up either in one unit, as observed, and yield ${}_c\alpha = 1.00$, or in two different units and yield ${}_c\alpha = -0.071$. There is at least some variation to which ${}_c\alpha$ does respond.

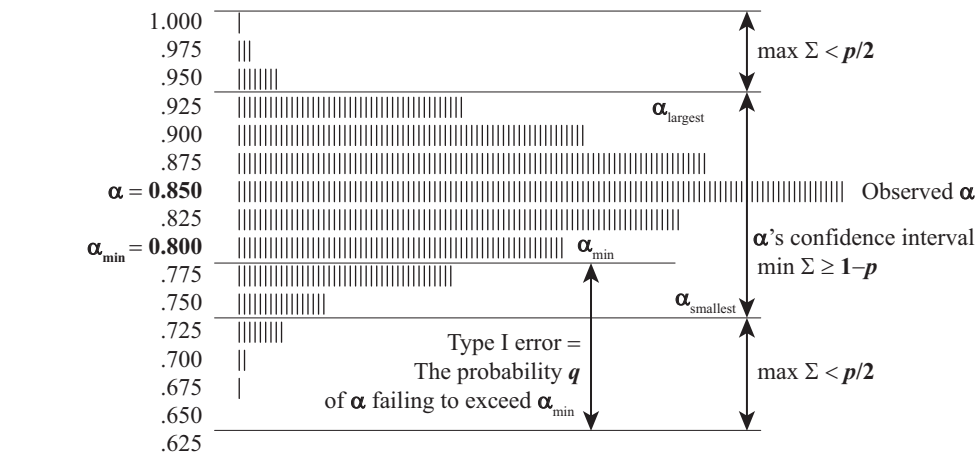
Data without variation not only fail to pass reliability tests, they cannot be correlated with anything either, their analytical meanings are largely void, and they cannot convey sufficient information about the analyzed phenomena to answer research questions. Where variation is low or absent, content analysts who want to be assured that their coding or unitizing instructions yield reliable data may have to undertake special efforts to sample alternatives to the usual or create alternatives to test whether they are recognizable as such. In a recent paper I discuss these borderline examples of reliability data and argued that the trustworthiness of data depends not only on high levels of observed agreement among observers but also on adequate information about that agreement (Krippendorff, 2011a). Feinstein and Cicchetti (1990) and Cicchetti and Feinstein (1990) have discussed the problem of low variation

in Cohen's (1960) κ (kappa) as paradoxical and proposed a modification of k , much like Gwet (2008) did of Scott's (1955) π (pi). However, I prefer to assess the variation or information in the data separately from how well data represent the phenomena that observers agreed upon above what can be expected by chance.

12.8.2 Statistical Significance

A common mistake is to accept or reject data as reliable when the null hypothesis that agreement occurs by chance can be rejected with statistical confidence. However, the whole reason for measuring the reliability of data is to ensure that they do not deviate too much from perfect agreement, not that they deviate from chance. In the definition of α , chance agreement merely serves as the point on the reliability scale at which there is no statistical connection between the data and the phenomena of interest, the other point being where there is no ambiguity that the data represent real phenomena, indicated by agreement is perfect, without exception and reliability is without doubt. While the distribution of α resembles a χ^2 distribution, it is basically unknown, especially because the phenomena to be coded may have odd characteristics. Because mathematical estimations make many assumptions and are mere approximations, we have resorted to generating distributions by bootstrapping. Bootstrapping involves drawing several thousand subsamples from the available reliability data, computing α for each, and thereby generating a probability distribution of hypothetical α values that could occur within the constraints of the observed coincidences (see Krippendorff, 2011b; and Hayes & Krippendorff, 2007, for its implementation in an SPSS and SAS macro). Figure 12.16 depicts a typical distribution of bootstrapped α values.

FIGURE 12.16 ■ Bootstrapped Distribution of α and Areas of Interest



In determining the reliability of data, two statistical parameters of α 's distribution are important:

- α 's *confidence interval* for the chosen level of significance p (two-tailed):

$$\alpha_{\text{largest}} \geq \alpha \geq \alpha_{\text{smallest}}$$

- The probability q of failing to reach the smallest acceptable reliability $\alpha_{\min}$:

$$q \mid \alpha < \alpha_{\min}$$

The confidence interval spells out the range within which the computed ${}_c\alpha$ is likely to vary with $(1-p)$ % certainty. The probability q of failing to reach the minimum required reliability $\alpha_{\min}$ is the probability of accepting data as reliable when they are not, also called the Type I error. The latter is not related to the null hypothesis but expresses the probability that the data are unreliable.

For the numerical example in section 12.3.1, we calculated ${}_c\alpha_{\text{nominal}}=0.743$. An algorithm (Krippendorff, 2016) which is a part of software for the basic ${}_c\alpha$ (Hayes & Krippendorff, 2007) yields the 99% confidence interval between $\alpha_{\text{smallest}}=0.615$ and $\alpha_{\text{largest}}=0.860$ and the probability $q=0.198$ of the failure to exceed $\alpha_{\min}=0.667$.

12.8.3 Sampling Considerations

A frequent question concerns how large the sample of units and of coders should be for α to be statistically meaningful. This question has no simple answer. I consider two distinct issues of sampling reliability data below and then make a practical recommendation.

To establish the *reliability of particular data*, reliability data need to be representative of the population of the data whose reliability is in question. The size of adequate samples is inversely related to the proportion of units in different categories. Rare but important categories must occur in sufficient numbers so as to make the same difference in the reliability calculations as in subsequent analyses (see the example discussed in section 12.3.1). As a rule of thumb, each category of units should occur often enough to yield at least five agreements by chance (I will be more specific below, however).

Where the research results are particularly important or data are not very numerous, content analysts have resorted to coding all data. This assures that the measured ${}_c\alpha$ is the α of the population of data, but this additional effort can be time-consuming and costly (see section 12.8.4 for other benefits). Coding all data solves only one of two sampling problems, however, leaving the selection of an adequate number of generally available coders unattended.

To obtain *required sample sizes* for either situation, we rely on Bloch and Kraemer's (1989, p. 276) Formula 3.7, which estimates the variance for binary data in 2-by-2 tables. For two observers, the minimum *number N of units* required for distinguishing value c from values *not- c* ,

$$N_{c \mid \text{not-}c} = z_p^2 \left(\frac{(1 + \alpha_{\min})(3 - \alpha_{\min})}{4(1 - \alpha_{\min})P_c(1 - P_c)} - \alpha_{\min} \right)$$

turns out to be a function of the probability P_c of value c (not to be confused with the expected %-agreement P_e in Scott's π or p_c in Cohen's κ), the lowest acceptable $\alpha_{\min}$ below which data would have to be rejected as unreliable, and the desired level of statistical significance p seen in the z_p -value for one-tailed tests.

Because reliability concerns the categories or values used in coding, Table 12.3 lists the required *number $2N_{c \mid \text{not-}c}$ of values*—Observers $\times$ Units—for three minimally acceptable reliabilities $\alpha_{\min}$, four levels of statistical significance p , and several probabilities P_c , also expressed in terms of the numbers v of equally likely values.

When assessing the *reliability of data*, $P_c = P_{\min}$ is the probability of the least frequent value in the population of values contributed by all observers. For example,

TABLE 12.3 ■ Required Number $n_{..}$ of Values Used in Coding														
Smallest acceptable $\alpha_{\min}$:			.667				.800				.900			
Significance level p :			.100	.050	.010	.005	.100	.050	.010	.005	.100	.050	.010	.005
$P_c = .500$;	$v = 2$	values	36	60	119	146	62	103	206	252	128	211	422	518
= .333;	= 3		41	67	135	165	71	116	233	285	144	238	477	584
= .250;	= 4		49	81	161	198	84	139	277	340	172	283	566	694
= .200;	= 5		58	95	190	233	99	163	326	400	202	332	667	815
= .167;	= 6		67	110	220	270	114	189	377	462	233	384	768	941
= .143;	= 7		76	125	251	307	130	214	429	526	265	436	872	1069
= .125;	= 8		85	141	281	345	146	241	481	590	297	489	980	1198
= .111;	= 9		95	156	312	383	162	267	534	654	329	542	1083	1328
= .100;	= 10		104	172	344	421	178	293	587	719	361	595	1190	1459
= .050;	= 20		200	329	657	806	340	560	1119	1372	657	1131	2263	2775
= .020;	= 50		487	802	1604	1966	825	1360	2719	3334	1675	2759	5534	6765
= .010;	= 100		966	1591	3182	3901	1640	2701	5403	6624	3307	5447	10896	13359

suppose this probability is found to be $P_{min} = 0.125$. If reliability is to exceed the smallest acceptable $\alpha_{min} = 0.800$ and 95% of all alphas (at the 0.05 level of significance) should satisfy this condition, the table suggests a minimum number of 241 values. Two observers would have to code at least 121 units, three at least 81, and so on.

By contrast, when assessing the *reliability of coding instructions*, one aims for a uniform distribution over all values of a variable and $P_c = 1/v$. Suppose $v = 5$ values are available for coding that variable, then $P_c = 1/5 = 0.2$, and the required number of values would be only 163, or about 33 per value, regardless of how many observers are involved.

A priori determination of the least likely value presents an epistemological dilemma: How could one know this frequency before coding the data? One practical way out of this dilemma is to (a) start by assuming that all available values to occur with equal frequency, (b) let the chosen observers code the then-required number of units, (c) find the actual probability of the least frequent value, have the missing number of units coded, and (d), with that result, repeat (c) until the sample is adequate.

The required number of values may be generated by any combination of the numbers of observers $\times$ units. However, in balancing these two numbers, it is important to keep the purpose of reliability tests in mind. Not only must the sample of coded units be representative of the data or the coding instructions whose reliability is in question, but also the qualifications of the observers employed in the process need to be sufficiently common to be selected by other researchers in order to replicate the reliability tests or build on the reported findings.

12.8.4 Standards for the Reliability of Data

The ultimate aim of testing reliability is to ensure that unreliabilities are negligible so as to justify continuing the coding of data, or when data are generated, to decide whether they are worth analyzing in the hope to be able to answer the guiding research questions. In the following, I answer four commonly asked questions regarding reasonable standards.

What is an acceptable level of reliability? Facing the real difficulties of obtaining perfect agreement, can one require that α be at least 0.95, 0.90, or 0.80? Unfortunately, although every content analyst faces this question, there is no set answer. To shed light on how different levels of reliability can be interpreted, Marten Brouwer, a colleague of mine from The Netherlands, designed an experiment. He gave coders who spoke only English a set of complicated Dutch words and asked them to describe U.S. television characters using those words. The Dutch words had no resemblance to any words in English, and the English speakers could hardly pronounce them, but the words must have invoked some consistent associations with perceived personality characteristics because the agreement was $\alpha = 0.44$. Knowing the observers' unfamiliarity with the meanings of these words, nobody in their right mind could claim the ability to infer from the data they generated the qualities of the TV characters they recorded. Surprisingly, the measured agreement was above chance, but on account of entirely unknown associations in the observers' minds, associations that the researcher and the users of findings based on them can hardly imagine. This finding gives us another reference point on the scale of α 's values, one that one should not approach. After further explorations of the relationship between achieved agreement and understanding of the categories involved, we adopted the following policies:

- Aim at $\alpha = 1.000$ —obviously and ideally.
- Rely only on variables with reliabilities $\alpha \geq 0.800$.
- Consider variables with reliabilities between $\alpha = 0.667$ and $\alpha = 0.800$ only for drawing tentative conclusions.

These standards have been adopted in numerous content analyses in the social sciences, and they might continue to serve as guidelines. Similar guidelines have been proposed for other coefficients. But because measured agreements are affected by their sample sizes, the distribution of values in the data whose reliability is to be assessed, and by the variance they exhibit, it is important to judge the reliability of data by the probability distribution of the measured agreement. Relying on α 's distribution offers criteria that are more justifiable than those based on parametric assumptions that do not reflect observers' responses to odd occurrences of values and unequal contributions sample sizes. In these terms, the recommendations could be rephrased:

- Do not accept data with reliabilities whose confidence interval reaches below the smallest acceptable reliability α_{min} , for example, of 0.800, but no less than 0.667.
- Ensure that the probability q of the failure to exceed the smallest acceptable reliability α_{min} is reasonably small, for example, 0.050 = the tolerable risk of drawing wrong conclusions.

I recommend such levels with considerable hesitation. The choice of reliability standards should always be related to the validity requirements imposed on the research results, specifically to the costs of drawing wrong conclusions. If the outcome of a content analysis will affect someone's life—such as in court proceedings or going to war over it—the analyst should not rely on data whose probability of leading to a wrong decision is less than what is commonly accepted (for example, the probability of being killed in a car accident). The results of most content analyses do not have drastic consequences, however, and researchers tend to adopt far lower standards. Even a cutoff point of $\alpha = 0.800$ —meaning only 80% of the data are coded or transcribed to a degree better than chance—is a pretty low standard by comparison to standards used in engineering, architecture, and medical research.

Whether a content analysis is exploratory or intended to be decisive, no researcher should ignore reliability, set reliability standards so low that findings cannot be taken seriously, use deceitful ways of generating reliability data, or apply deceptive agreement measures to prop up the appearance of reliability. In content analysis, the famous phrase “beyond reasonable doubt” has an operational meaning.

I wish to comment briefly on the psychology of agreements measuring less than perfect. Throughout this chapter I insisted on taking the consequences of unreliable data as the criteria for interpretations of α 's computed values. There are other approaches. Landis and Koch (1977) proposed to put words to their numerical values:

Kappa Statistic	Strength of Agreements
< 0.00	Poor
0.00 to 0.20	Slight
0.21 to 0.40	Fair
0.41 to 0.60	Moderate
0.61 to 0.80	Substantial
0.81 to 1.00	Almost perfect

I am suggesting that benchmarks like these are largely misleading. Kappa values less than 0.00 are not “Poor,” they are below chance. And to characterize the agreement of 0.44 we measure for data that were totally uninterpretable, “Moderate” would be a deceptive attribution. To be fair, other researchers have proposed slightly more realistic verbal

substitute for measured agreements. Fleiss (1981) called reliability “excellent” when his K exceeded 0.75; “intermediate to good” when it measured between 0.40 and 0.75; and “poor” when below 0.40. It is worth noting that none of the attempts to replace quantities of observed agreements by words address what really matters: the consequences of imperfect data.

Lombard et al. (2002) are not alone in distinguishing between conservative reliability coefficients (κ and α) and liberal ones (% agreement). The criteria for these categorizations are not hard to guess. They have to do with how difficult it is to achieve respectable values. Conservative agreement coefficients are supposedly too tough, and liberal ones are too easy to satisfy—as if the ease of getting high agreement values mattered. That the difficulty or ease of reaching acceptable reliability levels has much to do with what agreement measures respond to or omit is not understood, and labels like these prevent researchers from exploring the source of their difficulties.

The search for coefficients that produce higher reportable agreements comes in various guises. Feinstein and Cicchetti (1990) give the higher percent agreement measure a weight it does not deserve. Their paradox, “high (percent) agreement but low kappa,” describes an extreme condition under which agreement coefficients that assess the amount of noise in the data do not correlate with percent agreement—as if percent agreement provided a reasonable reliability standard. Gwet (2008) sought to avoid this paradox by publishing a coefficient AC_1 that produces higher values than any of the chance corrected agreement coefficients. To be able to report higher numbers, some statistically naïve researchers have been attracted to AC_1 . However, its values provide no clue about the extent to which imperfect data might still be worth analyzing. Another proposal that would prop up reportable agreement is Perreault and Leigh’s (1989) proposal to square root Bennett et al.’s (1954) S . To be clear, the square roots of any valid agreement coefficient (between zero and one) yields larger values. Square roots affect the linearity of a coefficient but not what it measures. Any applicable standard would have to be transformed as well: $\sqrt[3]{0.667} = 0.816$, $\sqrt[3]{0.800} = 0.894$, and $\sqrt[3]{0.900} = 0.949$. The square root of chance-corrected agreement coefficients may well be psychologically appealing but has absolutely no methodological benefits.

Given the α -values of separate variables, how reliable are the data as a whole? Consider that α is defined for separate variables, while most content analyses rely on many. Ideally, the variables of a data language are logically independent, free of conceptual redundancies, and disagreements in the data affect the research results equally. If every variable counts equally, every variable must also be reliable.

It is a serious mistake to average the reliabilities of the variables of a complex coding instrument and take this average as a measure of overall data reliability. Computing averages assumes that higher values compensate for lower ones. Typical content analyses include clerical variables, publication dates, lengths, and mechanically obtained measures that tend to be perfectly reliable, whereas the variables that typically matter most tend to be more difficult to code and end up being less reliable. Researchers who average such reliabilities may deceive themselves believing that they have reasons to trust their data and may unwittingly lead astray.

A condition in which averaging may make sense arises when the values of several variables are subsequently summed or averaged to form a composite index. Averaging their reliabilities is justifiable only when this index is such that scoring on one account is as good as scoring on another, and omissions in one variable compensate for commissions in another, so that agreements in one variable balance disagreements in another. These rather stringent conditions are rarely ever met.

Generally, when variables are equally important to the research effort, any unreliable variable can become the bottleneck for confidence in the data as a whole and the results. Thus:

For data with many variables, the lowest α among them is the reliability of all.

This standard might appear a stringent criterion, but it is entirely consistent with the practice of dropping unreliable variables from further analysis. Compromising by abandoning the information that one had hoped to analyze in variables that turned out to be unreliable is the only way to assure adequate reliability of the data as a whole.

Is there an ultimate criterion for judging the reliability of data? There are good reasons to measure reliability at the front end of any analysis. Creating analyzable data by reading, unitizing, coding, or transcribing of texts typically is the most uncertain part of a content analysis, in fact of any empirical study. The analytical transformations of data into conclusions are expected to be transparent and explicitly stated, so that the analytical process employed remains examinable and replicable by others. In section 12.1, I stated that unreliability limits the chance of valid results. If the reliability of all variables is perfect, the burden of proving conclusions to be valid is entirely due to the appropriate choice and flawless execution of the analytical methods employed in answering the research questions by reference to the given data. If the data are somewhat unreliable, it would help to be able to trace the consequences of unreliabilities from the data through the analytical process to the computed answers to a research question. This is made difficult, however, for at least two reasons. First, agreement coefficients, such as κ , assess the extent to which coders agree above chance. These measures alone cannot distinguish which part of agreed upon units of analysis is due to chance and which is not. Once data are generated, it is difficult to separate information from extraneous variations or noise and to trace the latter through an analytical process. Second, analytical procedures differ in how variations in data—information and noise—affect them. Some are robust in the sense that unreliabilities do not matter much, while others may be sensitive to noise in the data and can flip one conclusion to its alternative.

For example, suppose that in the course of an analysis several categories of a variable are lumped into one; then, any confusion between these categories may no longer affect the results. In fact, lumping unreliable categories into one is a recommended way to improve the reliability of data, as long as one can afford to lose the information their distinctions were expected to provide—discussed in section 12.3.4.6. Suppose the analyst decides between competing explanations based on computed correlations or a composite index obtained by averaging several variables. This may well have the effect of absorbing some unreliability in the data. Shannon's information theory (Shannon & Weaver, 1949) provides for the possibility of eliminating certain amounts of noise that surreptitiously resides in the original data provided the data possess enough redundancy. Many analytical methods take advantage of this possibility, for example, when an analysis reduces the considerable diversity in the data to its far simpler but more abstract answers to the guiding research question. The only way to find out whether this is so or whether the analyst unwittingly allows noise to determine the research results is to carry reliability considerations from the beginning to the end of the analysis.

Hence the ultimate test of the reliability of data and its analysis is to take the following steps:

1. Replicate the data-making process for all or for a large enough proportion of data being analyzed.

2. From these somewhat unreliable data, generate several possible versions of data by permuting the values on which observers disagreed, and apply the intended analysis to each version.
3. Obtain the distribution of the likely conclusions (answers to research questions) obtained from each.

Concurrent conclusions manifest adequate reliability, notwithstanding unreliabilities that may have been evident during coding and may have been eliminated in processing the data. Lacking concurrence, a distribution of results demonstrates the uncertainty that unreliable data carry into the conclusions. Usually, the ultimate effect of unreliable data is a loss of statistical significance.

Duplicating the data-making effort may already be costly, but if reliability really matters at the end, this is what ultimately is required. Generating possible versions of the data from unreliable coding is a computationally intensive task, for which software is required. Of course, the analyst may invent shortcuts, replicating only a subsample of the data and/or applying the analyses on permutations of each observers' data, not of all their disagreeing values, or applying the procedure only to an intermediate part of a larger analysis.

However, analyzing many versions of data is not only a computationally extensive task, it can be done only in situations where data analysis is formalized, algorithmic, and repeatedly executable without variation. It results in a probability distribution, similar to Figure 12.16 in which the confidence intervals of the various answers to the research questions become determinable.

Again: How does the reliability of data relate to their analytical results? To be clear, replicability tests merely assure that the data generated by a procedure—enacted by observers or embodied in measuring instruments—responds to intersubjectively real phenomena. Agreement measures say nothing about the phenomena of analytical interest, only about the extent to which the data-making process is free of extraneous variations or noise. It does not address the fundamental relationship between data and their sources, between texts and their interpretations or contexts of use. It is the coding instructions that observers are asked to follow or the known design of the measuring device employed in that process that allows analysts to infer what observers saw or read or what a measuring instrument responded to. The degree to which observed agreement among replications lacks perfection is the extent to which researchers are uncertain of what the data they intend to analyze are about. Obviously, this uncertainty translates into the uncertainty of what the analytical conclusions derived from them represent.

While unreliable data can only reduce the probability that their analyses lead to conclusions about phenomena a researcher is concerned with, reliable data cannot guarantee that the information about the phenomena of interest is carried into the research results. All data analytical methods selectively preserve, eliminate, or unintentionally distort information about the phenomena in the generated data. The ability to trace that information through an analytical method requires competencies beyond measuring the agreements among replications. Computing a probability distribution of possible results from the versions implicit in unreliable data is a way to bypass examinations of the analytical process but cannot reveal the inadequacies of a chosen analytical method. In other words, there are severe practical and computational limits of relating the reliability of data to what the results of their analysis actually represent.

Validity is a related issue that is addressed in Chapter 13.

13

VALIDITY

Validation provides compelling reasons for taking the results of scientific research seriously. It can serve as the ground for developing theories and the basis of successful interventions. This chapter develops a typology of validation efforts that content analysts may use in justifying their research. It also shows ways in which analysts can quantitatively assess at least some of these efforts.

13.1 VALIDITY DEFINED

Validity is that quality of research results that leads us to accept them as true, as speaking about the real world of objects, people, phenomena, events, processes, and actions. A measuring instrument is considered valid if it measures what its user claims it measures. A content analysis is valid if the inferences drawn from the available texts withstand the test of independently available evidence, of separate observations, of competing theories or interpretations, or of being able to inform successful actions.

Riffe, Lacy, and Fico (1998) suggest, “The essence of the validity problem in content analysis as well as in other research . . . is that research should speak as truthfully as possible to as many as possible” (p. 150). The meaning of *truthful* is the central focus of this chapter. The idea that research should speak “to as many people as possible” leads to useful distinctions among face validity, social validity, and empirical validity.

Face validity is “obvious” or “common truth.” We appeal to face validity when we accept research findings because they “make sense”—that is, they are plausible and believable “on their face”—usually without the expectation of having to give detailed justifications. It makes sense, indeed, to measure public attention to an issue by the relative frequency with which the issue is mentioned in mass media. It makes sense to measure the quality of political deliberations by the number of alternative reasons brought into a discussion. After subsequent empirical scrutiny, face validity may well prove untenable, but at the time the research is published it appears just right. Face validity does not equal expectations. For example, it did not occur to anyone in the 1970s that members of minority groups were targets of jokes in U.S. television fiction more often than were members of the majority until content analysts thought to pursue this topic and found that correlation. Findings like these make sense in retrospect. Although face validity has its roots in common sense, in widely shared consensus, it is fundamentally an individual’s judgment with the assumption that everyone else would agree with it.

The reason content analysts rely on face validity perhaps more than do researchers who use other methods of inquiry is that content analysis is fundamentally concerned with readings of texts, with what symbols mean, and with how images are interpreted, all of which are largely rooted in common sense, in the shared culture in which such interpretations are made, which is difficult to measure but often highly reliable at a particular time. This is not to say that the reliance on face validity is absent in other research endeavors. In fact, even the most rigorous researchers would not use methods or publish findings that violate their common sense. Face validity is the gatekeeper for all other kinds of validity. It is difficult to explain how face validity works, yet it is omnipresent.

Social validity is that quality of research findings that leads us to accept them on account of their contribution to public discussion of important social concerns, such as violence on television, antisocial messages in rap music, racism in sermons, hate speech on radio talk shows, lack of civility in political campaign advertisements, and the creation of disinformation. Research examining such public issues is socially validated by proponents and antagonists who worry about these issues and are eager to translate research findings into actions. In Riffe et al.'s (1998) terms, social validity is "the degree to which the content analysis categories created by the researchers have relevance and meaning beyond an academic audience" (p. 137). When applying for funding, the ability to claim that a proposed research has social validity is attractive to most funding agencies. Unlike face validity, the social validity of content analysis studies is often debated, negotiated, and of public concern by definition. A content analysis that is socially valid can attract public attention, propose practical solutions, and generate funding. Publicly acknowledged authorities on the subject of research are key to the social validity of the findings. Arguing from the privileged position of scientists, content analysts may well inadvertently become such authorities, especially when they are able to explain their findings in seemingly irrefutable quantitative terms, for example at congressional hearings or to advocacy groups with particular public agendas. The line between accepting research because of the reputation of the researcher and accepting it because of the evidence it brought forth is often blurred. Even empirically oriented psychologists have started to take social issues increasingly seriously. A significant part of the latest edition of the *Standards for Educational and Psychological Testing* established by the American Educational Research Association, the American Psychological Association, and the National Council on Measurement in Education (2014) is concerned with fairness, with the responsibilities of the researcher for the test taker, and with testing and public policy. In these publications, concerns for the social validity testing appear in their possible social or psychological consequences. Although most researchers accept this kind of validity, it is distinct from empirical validity, which is the focus of the remainder of this chapter.

Empirical validity is the degree to which available evidence and established theory support various stages of a research process, the degree to which specific inferences withstand the challenges of additional data, of the findings of other research efforts, of evidence encountered in the domain of the researcher's research question, or of criticism based on observations, experiments, or measurements as opposed to logic or process. Campbell (1957) calls the latter "internal validity." Empirical validity cannot deny intuition (face validity), nor can it divorce itself entirely from social, political, and cultural factors (social validity)—after all, scientific research is reviewed by the researchers' peers, who may have their own theoretical agendas and are hardly

immune to social and political concerns. However, in the following, I separate empirical validity from the face and social validities and consider it to be established largely within the scientific community and to be based on rational arguments that bring empirical evidence to bear on the research results, the research process, and the conditions under which data were acquired.

In discussing empirical validity, several content analysis textbooks follow the American Psychological Association's *Technical Recommendations for Psychological Tests and Diagnostic Techniques* (1954), a landmark publication that defined the kinds of validity concerns that psychologists face when they are developing tests of individual characteristics or abilities. In addition to face validity, the chief types of validities distinguished in the 1954 *Recommendations* were content validity, construct validity, criterion-related validity, and predictive validity. These *Recommendations* focused narrowly on evidence and so did not mention social validity, which concerns questions regarding the larger context of psychological testing.

Content validity is the extent to which a psychological test captures all the features that define the concept that the test claims to measure. For example, measurement of an applicant's aptitude for a job would require a complete inventory of job requirements, not just IQ tests or motivational measures.

Construct validity acknowledges that many concepts in the social sciences—such as self-esteem, alienation, and ethnic prejudice—are abstract and cannot be observed directly. To validate a measure of self-esteem, for example, one would first have to spell out the observable behaviors and verbal responses that the self-esteem concept entails, then measure and correlate these with the proposed measure, and finally examine whether or not each correlation supports what a theory of self-esteem predicts.

Criterion-related validity, sometimes called *instrumental validity*, is the degree to which a measure correlates with or estimates something external to it; for example, IQ may be shown to correlate with grade point average. Customarily, criterion-related validity is divided into two kinds, concurrent and predictive. *Concurrent validity* is demonstrated by correlations that concur with the test in question, and *predictive validity* concerns variables that estimate features that may become available sometime in the future.

These classical distinctions have undergone various transformations. The above-mentioned *Recommendations* gave way to the first version of the *Standards for Educational and Psychological Testing* in 1985, subsequently followed by the 1999 and 2014 *Standards*, which no longer support distinctions between validity types, and refer instead to types of “validity evidence” (see American Educational Research Association et al., 1999, p. 11). They also recognize that theoretical constructs underlie all measurements, and this recognition led to a classification of validating evidence based on *test content*, *response process*, *internal structure*, *relations to other variables*, and, as mentioned above, the *consequences of testing* (pp. 11–17). The debate about these conceptions of validity is ongoing, but it is important to recognize that the above focuses narrowly on only one theory of scientific inquiry, measurement theory, and is informed mainly by one disciplinary orientation, the psychological testing of individuals. Measurements are only part of what content analyses can provide, and inferences about psychological characteristics of individuals or populations of individuals are rare in content analysis, although not excluded.

To understand the conceptions of validity in conducting content analyses, one must keep in mind that all empirical validation efforts enlist evidence and established

theories to ensure that research results are taken seriously. When the goal is merely the construction of theory, a research project may matter only to a small scientific community. But when research is intended to have policy implications—when findings are to aid business decisions, provide evidence in court, critically evaluate media outlets, or affect the lives of individual human beings in other ways—wrong conclusions may have costly consequences. Validation reduces the risk of making decisions based on misleading research findings. Content analysts and psychologists concerned with testing have to cope with different risks.

Content analysts face at least three kinds of obstacles when they try to apply traditional methods of validation: substantive, conceptual, and methodological. We return to our conceptual framework of content analysis (as depicted in Figure 2.2) to understand all three.

Substantively, probably most important to content analysts is the acknowledgment that all texts exist because they mean something to people other than the analysts, starting with producing, publishing, reading, and interpreting them and using them in processes of constructing, maintaining, or undoing social realities. The object of content analysis is far more complicated than analyzing how individuals respond to test questions with preformulated answers. Highlighting some of these differences, Potter and Levine-Donnerstein (1999) have introduced useful distinctions among three kinds of content analyses, liberally rephrased as content analyses that aim to describe manifest content, content analyses that provide inferences about latent patterns, and content analyses that provide interpretations (or “make projections,” as they say; p. 261). The first kind conforms to a measurement conception of content analysis and is unproblematic as far as the application of the above kinds of validity is concerned, although it conflicts with conclusions from Chapter 2. The second refers to a context—of experts, as Potter and Levine-Donnerstein suggest, and established theory in relation to which validity needs to be established. Figure 2.2 depicts just this, with the scientific community or any chosen stakeholder group providing this context. The third kind of content analysis these scholars describe allows more freedom of imagination on the part of content analysts but is constrained by cognitive schemata and inferential rules that are embodied in a designated population of text users whose conceptions are both the focus of analysis and the source of validity. This ethnographic and interpretivist-sounding conception does not differ from that outlined in Chapter 2 as a definition of content analysis of the second kind—except for the authors’ allowing research results to be more freewheeling. In this kind of content analysis, Potter and Levine-Donnerstein affirm, validity standards cannot be divorced from chosen contexts, but they limit it to the conceptions of individuals for whom the analyzed texts have the meanings they have. Sensitivity to a context distinguishes content analysis from other methods of inquiry and provides the criteria for acceptance of its results.

A *conceptual obstacle* to the validation of content analyses derives from the already criticized definition of *content as inherent in text* (see Chapter 2) and the attendant commitment merely to describe *it*. To be sure, all descriptions are abstract and, as such, arbitrary, and conceiving content as inherent to texts has enabled content analysts to apply any category schemes they please, provided the schemes are reliable. Content analysts with such a conception of content in mind confuse their descriptions of content with how others may read or use the same texts, seemingly needing no further validation. For example, suppose a content analysis of mass-media entertainment concludes that there has been a shift in the United States from material to spiritual

values. There is no way to validate this finding unless the analysts are willing to take responsibility for their definitions and make clear where this claimed shift should be observable independent from the analyzed media, in whose lives this shift is expected to make a difference, or which variables are expected to correlate with this abstraction in order for it to be considered as describing something real. As mentioned in Chapter 2, one reason for the popularity of the conceptions of content as inherent to text and of content analysis as merely descriptive of this content is the virtual impossibility of empirically (in)validating such findings. In the absence of specificity about what could validate or invalidate the findings of a content analysis, appeals to face and social validity, which researchers can more easily control through their own rhetoric, public testimony, and publication, seem to be the only recourse.

The *methodological obstacle* to validation is more difficult to overcome. Consider the following trilemma: (1) If content analysts happen to have no independent evidence about what they are inferring, then validity or invalidity cannot be established, at least not until pertinent data show up. Content analysis shares this epistemological constraint with all efforts to predict future events. (2) If these analysts had evidence about the context of their texts—that is, about phenomena related to the analyzed texts—but used it in the development of their analytical constructs, then this evidence would no longer be independent of the research results and hence cannot be used to validate the findings. And finally, (3) if these analysts had concurrent evidence that could validate their inferences but kept it away from their analysis, there would be no point in their conducting the content analysis. It would at best add one incident to vindicate an analysis. Content analysts can resolve this trilemma, at least in part, by relying on various forms of imperfect and indirect validating evidence about the phenomena of interest. I address how this could happen in the following section.

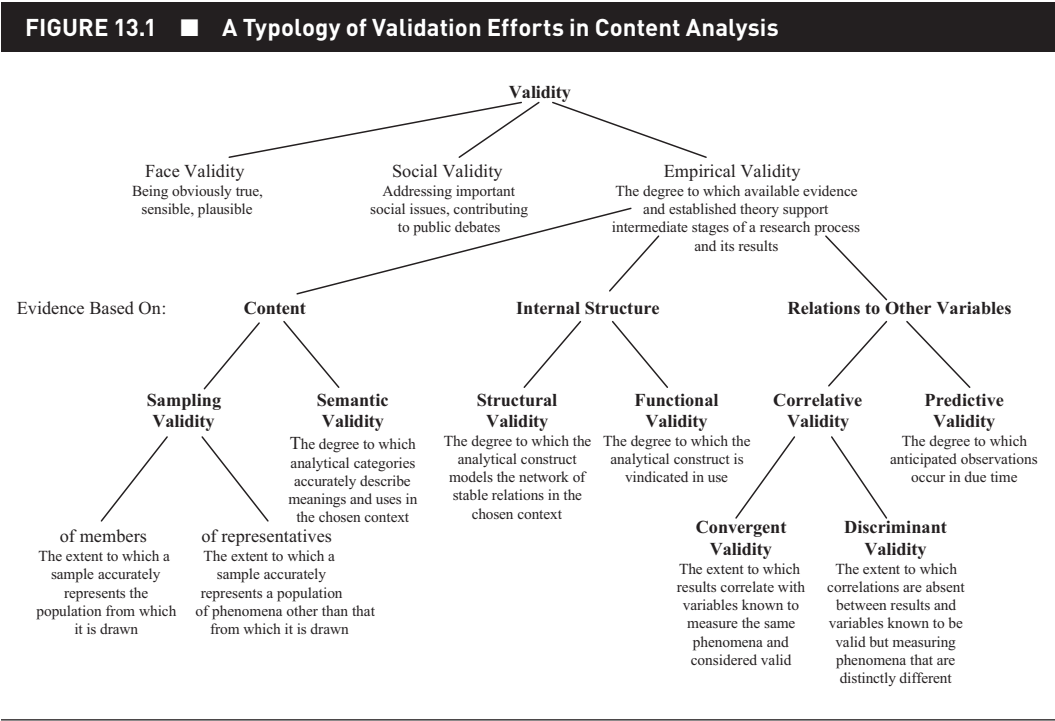
13.2 A TYPOLOGY FOR VALIDATING EVIDENCE

A fundamental difference between psychological testing and content analysis is that the latter is concerned with bodies of text that are meaningful in relation to a chosen context (see Chapter 2), whereas the former does not acknowledge that relationship and the inferential step it entails. It follows that content analysts must empirically demonstrate the context sensitivity of their research. In addition, it is important to bear in mind that content analysis data, texts, images, and findings, although unquestionably mediated by human individuals as language-using beings, readers/writers, conceptualizers, and actors, are not necessarily about individuals. Everything said is said to someone else. Although the first reader of any text is its author, if nobody else read, it we wouldn't know of it. Texts reside in relationships. This means that psychological measurement theoretical conceptions of validity have to be expanded by model theoretical conceptions, as graphically outlined in Figure 9.1. This epistemological shift calls for validating evidence that differs, albeit in only some categories, from the evidence defined by the above-mentioned *Standards*, whose conceptions content analysis can adopt only in parts. To start, three kinds of validating evidence may enter a content analysis:

- Evidence that justifies the treatment of text, what it is, what it means, and what it represents (This is loosely related to what the *Standards* refer to as “evidence based on test content”; American Educational Research Association et al., 1999, p. 11.)

- Evidence that justifies the abductive inferences that a content analysis is making (Here, analysts are concerned with the validity of the analytical constructs on which they rely. This is loosely related to what the *Standards* call “evidence based on [the] internal structure” of a test; p. 13.)
- Evidence that justifies the results, whether a content analysis contributes answers to the research questions of other researchers or is borne out in fact (This is loosely related to the older “criterion-related validity,” which the *Standards* discuss in terms of “evidence based on relations to other variables.”)

These and the following distinctions are depicted in Figure 13.1.



Evidence that justifies the treatment of texts concerns largely the sampling and recording phases of a content analysis. Such evidence may be divided into two kinds:

- Evidence on *sampling validity* concerns the degree to which a sample of texts accurately represents the population of phenomena in whose place it is analyzed. Ideally, content analysts actively sample a population, using sampling plans that ensure representativeness. But in many practical situations, texts become available by their sources’ choice and contain intentional biases in representing the phenomena of their interest. Content analysts may not be able to control such biases but will want to know whether such samples can be trusted and how they could mislead them.
- Evidence on *semantic validity* ascertains the extent to which the categories of an analysis of texts corresponds to the meanings these texts have within the chosen context. The anthropological preference for emic or indigenous rather than etic or imposed categories of analysis and ethnographers’ efforts

to verify their interpretations with their informers demonstrate concern for semantic validity, here with reference to the lives of populations of interviewees. In content analysis, other contexts are considered as well, but contexts they must be. Other kinds of research efforts (psychological testing for one, but also survey research) tend to avoid semantic validity by controlling the range of permissible answers to questions and not really exploring what these questions could mean to their subjects. Semantic validity is allied largely with content analysis.

Evidence that justifies the abductive inferences of a content analysis sheds light on how well the analytical construct in use actually does model what it claims to model. Again, two kinds of such evidence may be distinguished:

- Evidence on *structural validity* demonstrates the structural correspondence between available data or established theory and the modeled relationships or the rules of inference that a content analysis is using.
- Evidence on *functional validity* demonstrates a functional correspondence between what a content analysis does and what successful analyses have done, including how the chosen context is known to behave. If these behaviors covary repeatedly and over a variety of situations, one can suspect that they share an underlying construct.

This distinction between structural validity and functional validity is motivated by Feigl's (1952) distinction between

two types of justification . . . validation and vindication. In this context, *validation* is a mode of justification according to which the acceptability of a particular analytical procedure is established by showing it to be derivable from general principles, . . . theories (or data) that are accepted quite independently of the procedure to be justified. On the other hand, *vindication* may render an analytical method acceptable on the grounds that it leads to accurate predictions (to a degree better than chance) regardless of the details of that method. The rules of induction and deduction are essential to (construct) validation while the relation between means and particular ends provides the basis for (construct) vindication. (Krippendorff, 1969b, p. 12)

In the 1980 edition of *Content Analysis*, I called these two kinds of validity *construct validation* and *construct vindication* (Krippendorff, 1980b). However, because construct validity has a slightly different definition in the psychological test literature, my use of this label caused some confusion, hence the current change in terms. Incidentally, Selltitz, Jahoda, Deutsch, and Cook (1964) call vindication *pragmatic validity*, as "the researcher then does not need to know *why* the test performance is an efficient indication of the characteristics in which he is interested" (p. 157).

Finally, one could be concerned with the validity of the results of a content analysis, criterion-based validity or instrumental validity, and consider two ways of supporting the results:

- Evidence on *correlative validity* ascertains the degree to which the findings obtained by one method correlate with findings obtained by other

variables that are considered more valid than the method in question. To be correlated, all variables must be presently and simultaneously available. The result, therefore, is also called *concurrent validity*. Correlative validity requires a demonstration of both *convergent validity*, or high correlation with measures of the contextual characteristics it claims to indicate, and *discriminant validity*, or low correlation with measures of contextual characteristics it intends to exclude.

- Evidence for *predictive validity* establishes the degree to which the answers of a content analysis accurately anticipate events, identify properties, or describe states of affairs, knowledge of which is absent or did not enter that analysis. Analogous to selecting among the possible answers to a research question, predictions spell out what can be anticipated and what is ruled out. Predictions may concern phenomena that precede, are concurrent to, or follow the texts used in making them.

13.2.1 Sampling Validity

Sampling validity becomes an issue whenever a sample of texts differs from the population of phenomena of interest, not just in size, which often is convenient, but also in composition, which can bias the content analysis of the sampled texts. As already stated, sampling validity is the degree to which a population is accurately represented in the sample. To begin with, two situations need to be distinguished:

- (a) The sample consists of a *subset* of members of the population of interest.
- (b) The sample consists of *representations* of phenomena that lie outside the sample and the population from which the sample is drawn.

Evidence in the first situation—drawing a sample from the very population of interest—is well understood by means of *statistical sampling theory*, as discussed in Chapter 6. Whether one is interested in a sample's medial, proportional, variational, or distributional accuracy, statistical theory provides measures of the sampling error. This error is a measure of a sample's *invalidity*. For purposes of this discussion:

$$\text{Sampling Validity (a)} = 1 - \text{sampling error} = 1 - \frac{\sigma}{\sqrt{N}} \sqrt{\frac{n-N}{n-1}},$$

where σ is the standard deviation of the population, which is a measure of its diversity of categories; n is the size of the population; and N is the size of the sample.

In traditional sampling theory, sampling errors are a function of three factors. First and most important is the sample size, N . The larger the sample, the smaller the sampling error and the larger the sampling validity (a). Second is the diversity of categories within the population, represented by the standard deviation σ . Given two samples of equal size, the sample drawn from a more diverse population has larger sampling errors and is less likely to be valid than a sample drawn from a less diverse population. Third is the proportion of the population sampled. As samples become more inclusive, $(n - N) \rightarrow 0$, the sampling error shrinks and sampling validity (a) grows.

Evidence in the second situation—drawing samples of representations in view of what they represent—is not easy to obtain, yet it is a frequent concern for content analysts. To be clear, as I noted in Chapter 2, content analysts do not study texts, images,

or distributional characteristics for their own sake or to generalize to other texts, images, or distributional characteristics; rather, they use texts as a means to get to what the texts' users have in mind, what the texts are about, what they mean or do and to whom. There rarely exists a one-to-one correspondence between meanings, references, uses, or contents and units of texts. Sampling theory offers no simple test to establish whether a sample of textual units fairly represents the phenomena that a content analyst intends to access through these texts. Additionally, the texts that reach the content analyst usually are *presampled* by others—by organizations following institutionalized rules; by individuals with particular intentions, who highlight some kinds of information and downplay others; or by media of communication that have their own built-in technological or economical filters. Communication researchers have long studied how reality is constructed, represented, and misrepresented in the mass media, but they have rarely used these findings in validation efforts. Concepts such as gatekeeping in news flows, ideological/racial/gender biases in writing, the positive spin that affected parties put on politically embarrassing stories, and the attention paid by institutions of journalism to particular stories (i.e., to whom journalistic institutions grant a voice and, by implication, what or whom they ignore) are well established and often quantified.

If the phenomena of interest need to be accurately represented in the texts that researchers are analyzing, then sampling must undo the biases that result from the selective ways texts are made available. Validating evidence for sampling validity (b) can be of two kinds:

- Knowledge of the population of phenomena with which one of its samples is to be compared
- Knowledge of the self-sampling practices of the source of the available texts

To measure the degree to which a population of phenomena is fairly represented in a sample of textual units, a simple percentage measure that derives from the well-known coefficient of contingency C is useful. In its incarnation as a validity measure, $1 - C^2$, it concerns proportions only and has two versions that correspond to the above two kinds of validating evidence:

$$\text{Sampling Validity (b)} = 1 - C^2 = \frac{1}{1 + \sum_i \frac{(P_i - p_i)^2}{p_i}} = \frac{1}{1 + \sum_i P_i \frac{b_i^2}{1 - b_i}},$$

where P_i is the proportion that is observed to represent phenomena of category i in the sample and p_i is the proportion representing phenomena of category i in the population from which the sample is drawn. When known or ascertainable, the proportions p_i serve as validating evidence in the first version of the formula for sampling validity (b). When the biases b_i of representing categories i are known, algebraically equivalent to $(1 - p_i/P_i)$, b_i serves as validating evidence in the second version of the formula for sampling validity (b). Studies of biases, assessing or estimating b_i , are more common, easier to conduct, and designed differently from those assessing the proportions p_i , hence the two versions.

The first version of the above-stated sampling validity (b) is a function of the observed proportion P_i in the sample and the proportion p_i in the population, which is the validating evidence and must be obtained independently of the sample. When $P_i = p_i$ for all categories i , sampling validity is unity. It declines with increasing differences between the two proportions.

The second version of sampling validity (b) is a function of the observed proportion P_i in the sample and the self-sampling bias $b_i (= 1 - p_i/P_i)$, which is the extent to which the source of the sampled text over- or underrepresents categories i . Here, the bias b_i serves as validating evidence, which must be obtained independently of the sample as well. If this bias $b_i = 0$ for all categories i , then sampling can proceed as usual. If this bias deviates from zero in either direction, then sampling validity (b) is reduced. If the source biases b_i are known, one can approximate a valid sample either by using the technique of varying probability sampling (Chapter 6, section 6.2.4), which compensates for the known biases, or by transforming the proportions P_i in a biased sample by $P'_i = (1 - b_i)P_i$, which corrects for the biases in representing the phenomena in question.

13.2.2 Semantic Validity

Semantic validity is the degree to which the analytical categories of texts correspond to the meanings these texts have for particular readers or the roles they play within a chosen context. Virtually all content analyses respond to texts according to their meanings: denotations, connotations, insinuations, implications, associations, metaphors, frames, uses, symbolic qualities, and so on. Readers or users able to interpret given texts could be employed as sources of validating evidence for the categories that a content analysis employs. In older definitions of content analysis, “accurate descriptions of meanings” were the only aim mentioned, whether they referred to classifications of sign-vehicles (Janis, 1943/1965), descriptions of the “manifest content of communication” (Berelson, 1952, p. 16; Berelson & Lazarsfeld, 1948, p. 6), coding (Cartwright, 1953, p. 424), or “putting a variety of word patterns into [the categories] of a classification scheme” (Miller, 1951, p. 96). Although it is widely recognized that accurate descriptions of these meanings are the key to the success of content analyses, despite their ultimately inferential aims, what counts as accurate and particularly whose meanings are taken to be valid depend on the chosen context of an analysis.

In Chapter 2, I noted that even content analysts involved in purely descriptive efforts ought to acknowledge a context that they or others could consult to validate those efforts. If content analysts claimed to describe meanings without reference to any context of specific uses and users—authors, readers, newspaper editors, professional experts, professionals with specialized perspectives, social institutions, standard thesauri or dictionaries, even analysts’ own discourse communities—there would be no way to know what could validate or invalidate these descriptions, and analysts would be left to appeal to face validity or to play on their scientific (social or political) authority. Although semantic validity is an issue that most content analysts take seriously, it is rarely formally tested.

It is easy for researchers to take an objectivist stance and consider meanings as universal and as defined in general dictionaries or to take an ethnographic stance and delegate decisions on meanings to the authors of given texts. However, both of these extremes deny the fact that virtually all analytical descriptions simplify or abstract how ordinary readers may interpret texts, usually in the interest of the analyst’s research questions. Categories are always more general than the objects they categorize. In fact, content analysts rarely take the literal meanings of analyzed texts as the basis for their inferences; instead, they operate on a level of abstraction above that of ordinary talk. Concepts such as speech acts, monologue, self-esteem, ethnic prejudices, sexual harassment, and libel, as well as such distinctions as between

pro- and antisocial behavior, are all fairly abstract and not necessarily understood by the authors or readers of the analyzed texts. Distinctions among the functions of political campaign discourse in terms of acclaiming, attacking, and defending (Benoit, Blaney, & Pier, 1998) are analytically useful but may not help political candidates to organize their campaigns. It is the use of abstract categories that makes semantic validation of content analyses analogue to content validation of psychological tests. For example, the content validity of a test designed to determine the aptitude of a job candidate for a particular kind of employment is the extent to which the test includes all demands of that job, not just a few outstanding qualifications. Analogously, the semantic validity of the categories “acclaiming,” “attacking,” and “defending” should be the extent to which these categories embrace all functions of political campaign discourse and clearly distinguish among the three categories.

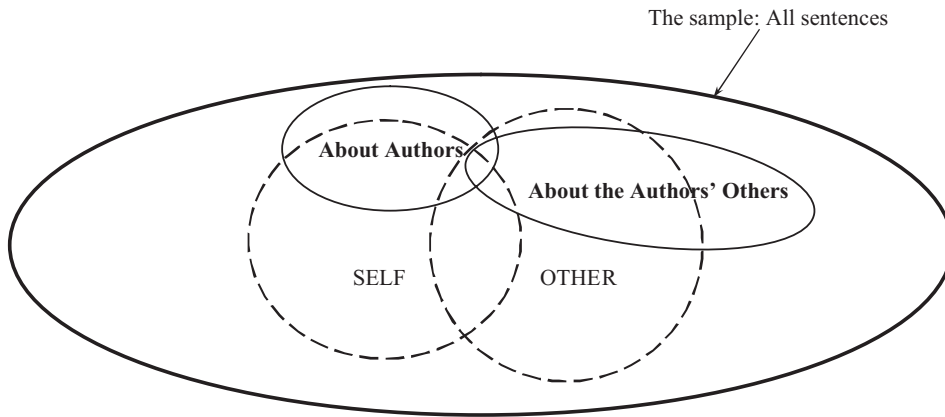
The preparations for an analysis of values in political documents may serve as an example of an iterative use of semantic validity criteria. In this study, we started with a collection of what a panel of political scientists could easily identify as “value-laden statements” of the kinds we would find in the documents to be examined for the values their authors expressed. To reproduce these experts’ distinctions, we formulated explicit recording instructions. The coders varied greatly in their ability to make the same distinctions, and a computer program we had hoped to employ turned out to be virtually useless. The whole history of this effort is too long to relate, but we began by developing a list of political values the documents contained—democracy, freedom, progress, and the like—and allowed others to be added. This turned out to be far from satisfactory. A large number of value-laden statements contained none of the values on our list. We then looked into various modes of reasoning that led us to implicit goals, preferences for processes, criteria for decision making, and so on, added them to our emerging instructions, and reapplied them to our collection. Slowly, we narrowed the gap between the distinctions that our instructions suggested between value-laden and value-neutral statements and those that our panel of experts made (Krippendorff, 1970c). One might question our use of experts as providing the evidence for our semantic validation efforts. We could have used another population for reference, but because we were working under typical time constraints and our analysis was to make a contribution to the political science literature, the theoretical motivations seemed to justify the choice of this context, and we were satisfied that the instrument reasonably approximated the consensus reached by those we trusted to know what they were talking about.

Semantic validity arguments come in numerous guises. To validate the dictionary entries of the General Inquirer, a computer program for tagging texts and then analyzing the tags, Dunphy used a KWIC (keyword in context) list to explore the various meanings of tagged words. For example, the keyword *play* (see Figure 11.1) showed numerous senses that were not initially anticipated and that would have been confused by any computer program that tagged only single words (Dunphy, 1966, p. 159). To address such semantic problems, Stone, Dunphy, Smith, and Ogilvie (1966) developed so-called disambiguation procedures that looked into the linguistic environments of homonyms for clues to how their meanings could be identified more correctly. These rules improved the General Inquirer’s ability to distinguish between textual units according to how they would be read by ordinary English readers. In effect, these researchers’ efforts were aimed at improving the semantic validity of the computer tagging of text.

Consider the distinction between *self* and *other* references made by a dictionary approach to computer text analysis (see Chapter 11, section 11.4.1). Suppose the words

self, *I*, *my*, *myself*, *me*, *mine*, *we*, *our*, *ourselves*, and *us* are tagged *self*, and the words *other*, *you*, *your*, *yourself*, *he*, *his*, *him*, *himself*, *she*, *hers*, *her*, *herself*, *them*, and *themselves* are tagged *other*. Suppose sentences are the units of analysis. These two tags would identify two subsets of sentences within the set of all sentences in the sample. One could obtain evidence for the semantic validity of these dictionary entries by asking competent readers, working independently, to classify the sampled sentences into those that speak about the “self” of the authors and those that speak about what these authors consider “others.” Figure 13.2 depicts two sets of tagged sentences surrounded by dashed lines and the sets that would serve as validating evidence within solid lines. The degree to which these two kinds of sets overlap is a qualitative indication of the semantic validity of the two tags that this dictionary is using. In computer-generated categorizations of text, the semantic validity is rarely as good as the figure suggests, but we are interested here only in what is meant by semantic validity: the complete overlap between a classification of uncertain validity with one we have reasons to trust.

FIGURE 13.2 ■ A Qualitative Illustration of the Semantic Validity of Two Categories



A more traditional way for scholars to challenge the semantic validity of a content analysis is by creating counterexamples. This strategy is well established in linguistics, where as soon as one linguist claims that a proposed grammar accounts for all grammatically correct sentences, another comes up with examples of sentences that this grammar will misidentify. Closer to content analysis, in the critique of his own contingency analysis, Osgood (1959, pp. 73–77) employs this strategy to discount exaggerated claims of what his analysis does. According to Osgood’s contingency analysis, the following statement would be counted as an incident of association between *love* and *mother*:

1. I love my mother.

But, so would these statements:

2. I loved my mother more than anyone else.
3. Mother loved me.
4. I don’t love my mother.
5. Have I always loved my mother?—Hell no!
6. My beloved father hated mother.

Because *love* and *mother* co-occur in all six statements, contingency analysis would group them into the same equivalence class of *love-mother* co-occurrences. However, relative to statement 1, statement 2 shows contingency analysis to be insensitive to verbal qualifications of an expressed association—“more than anyone else” is not a frequency. Statement 3 shows this analysis to be unable to distinguish between active and passive constructions—that is, the target of love. Statement 4 shows it to be insensitive to negation, statement 5 shows it to be insensitive to irony, and statement 6 shows it to be insensitive to grammatical constructions. Osgood did not use these observations to argue against his method, of course; rather, he used them to clarify what it registers: statistical, not logical, associations among pairs of concepts, presumably in someone’s mind. Here, even denying a relationship between two concepts would be considered evidence that they have something to do with each other.

A content analysis in a legal context may serve as a final example. In a libel case recorded at a Texas court as *Wood et al. v. Andrews et al.* (Dallas County, Cause No. 94–8888-C, 1997), the plaintiffs, 20 psychiatrists who worked for a mental health institution, hired a content analyst to establish objectively the libelous and defamatory nature of publicity they had received in the press. That publicity was attributed largely to one of the defendants, a lawyer who represented a number of former patients of the hospital where the plaintiffs worked (one of whom was a codefendant) who were suing that institution for patient abuse, malpractice, illegal profiteering, and insurance fraud. This content analyst retrieved all 52 newspaper articles published between 1993 and 1996 that contained references to that mental institution; 36 of these articles mentioned the lawyer defendant, and 16 did not. She examined the 36 articles that mentioned the defendant and used for comparison the articles that did not. She then unitized all of the articles into 970 assertions, assigning each to one of 16 categories that emerged from her reading of the articles, focusing on kinds of bad publicity. She then drew her conclusions from a statistic of these categorizations.

In response, the defendant hired an expert to examine the content analyst’s research and findings. The expert raised the issue of the semantic validity of the categories the analyst had used to make her point. Although she had conducted her analysis carefully, with categories traceable to the original assertions, her conclusions were irrelevant because her categories ignored the context that mattered in the case: the terms of the law, the legal definition of libel. Texas law defines libel in fairly specific terms. To be considered libelous, an assertion has to meet the following criteria:

- i. Made with the intent to harm a person publicly or financially or in disregard of injurious consequences to another person
- ii. Knowingly untrue
- iii. Read and understood as stating facts
- iv. Causing its readers to alter their speaking in ways that blacken a person’s public image, impeaches that person’s honesty, integrity, virtue, or reputation, and
- v. Actually incurring financial injury to that person or expose that person to debilitating hatred, contempt, and ridicule

In other words, assertions in the category of libel have to be (i) made with the intent to harm, (ii) known to be untrue, and (iii) read as stating facts. Evidence on criteria iv and v would presumably require observations or testimony. The content analyst's categories traced bad publicity about the plaintiffs to the defendant but failed to provide answers in the legally required categories. For example, accusations of insurance fraud, if true, are not libelous, regardless of how often they are mentioned. And assertions critical of the plaintiffs may not have been made with the intent to harm. In this context, a semantically valid content analysis would have to let the articles answer the questions in categories to which applicable law could apply—not how an average reader might interpret the newspaper articles. One could conclude that this analyst's categories had no semantic validity in the prescribed context.

One way of testing the semantic validity of content analyses consists of matching their operationalized categories, to the extent they determine the published findings, with how competent readers or users of these findings understand them. Faced with difficulties of achieving adequate levels of reliabilities for coding complex concepts, content analysts are encouraged to become increasingly specific in defining their coding instructions. However, in the reality of introducing such improvements, as the definition of categories shift to what is easier to code, their names tend to stay the same. Analysts, being heavily involved in operationalizing their categories, often do not notice these shifts and report their findings in terms they started out with, not how they ended up defining them. For research to stay relevant, to have actionable implications, its stakeholders need to be able to understand the categories in terms of which data were generated, analyzed, and published. The original meanings of the analyzed texts ought to relate to published findings. This is what semantic validity is to assure.

One case in point was a class action suit against an automobile manufacturer whose cars had a serious defect that caused numerous accidents. The suit claimed that instead of acknowledging and correcting this defect, the manufacturer denied it and misinformed future customers by continuing to advertise the safety of its automobiles. In response, that manufacturer commissioned a hugely expensive content analysis of its TV ads and publications to counter the claim of misleading advertisements. It “proved” that safety was rarely mentioned. However, because safety claims can be made in various implicit ways, which are difficult to code, the coding instructions were designed to start by identifying mentions of various, easy-to-code, beneficial features of its automobiles, making the category safety available mainly after all other categories were ruled as absent. The content analysis was commissioned to disprove deceptive claims of car safety, and, naturally, safety showed up rarely. The content analysis employed an army of coders and reported high reliabilities for their coding, however, it was measured after violating protocols to assure the independence of their judgments.

The content analysis also failed semantic validity tests. Evidence of it emerged when the principal investigator, who had not been involved in the training of coders, was given several TV ads that had been included in the study and was asked to identify their categories, with safety being the focal point. His judgments were accurate not much better than chance. It demonstrated that he barely knew what he was reporting to the courts, and he got lost in pie charts and statistical comparisons that were semantically questionable.

In this case, doubts of the semantic validity of his content analysis were supported by comparing only a few coded units of analysis and were limited to the principal investigator as a proxy for other stakeholders. Merely questioning the semantic validity of research may suffice in a court of law, in dissertation defenses, and in public hearings that appeal to a shared understanding. However, the logic of such comparisons—establishing whether interpretations are correct to a degree better than chance—is prototypical to measuring reliability in terms of accuracy and quantifiable by the ${}_c\alpha^s$ -agreement coefficient presented in Section 12.4.1.

Obtaining ${}_c\alpha^s$ calls for several $1, 2, \dots, u, \dots, N$ units of textual data, for example, TV ads, each coded k_u , which serve as the standard against which the interpretations cu of these data by various stakeholders—ordinary readers, viewers, critics, judges, and juries—are matched. The nature of these reliability data are reproduced here for convenience:

Data=	1	2	.	.	u	.	.	.	.	N	
Semantic criteria=	k_1	.	.	.	k_u	.	.	.	.	k_N	
1	n_{11}	.	.	.	n_{u1}	.	.	.	.	n_{N1}	$n_{.1}$
Interpretations of data by	:	:	.	.	:	.	.	.	.	:	:
their stakeholders:	n_{1c}	.	.	.	n_{uc}	.	.	.	.	n_{Nc}	$n_{.c} = \sum_{u=1}^N n_{uc} \mid k_u \neq \phi$
	:	:	.	.	:	.	.	.	.	:	:
Sum of semantically	n_{1v}	.	.	.	n_{uv}	.	.	.	.	n_{Nv}	$n_{.v}$
interpretable data=	m_1	.	.	.	$m_u = \sum_{c=1}^v n_{uc} \mid k_u \neq \phi$	.	.	.	.	m_N	$n_{..} = \sum_{u=1}^N m_u \leq mN$

In which terms semantic validity can be measured as in (19), (20), and (21):

$$\text{Semantic validity: } {}_c\alpha_{\text{metric}}^s = 1 - \frac{{}_cD_o^s}{{}_cD_e^s} = 1 - n_{..} \frac{\sum_{u=1}^N \sum_{c=1}^v n_{uc} \text{ metric } \delta_{ck_u}^2}{\sum_{k=1}^v n_k \sum_{c=1}^v n_{.c} \text{ metric } \delta_{ck}^2}$$

$$\text{where } n_k = \sum_{u=1}^N m_u \text{ iff } k_u$$

Note that semantic validity, when actually measured as the accuracy of stakeholders' interpretations of textual data or findings of content analyses, cannot be expected to approximate the standards of data reliability. Not only are stakeholders' interpretations of what content analysis results claim to accomplish more loose than we expect for content analysts to proceed, those who have reasons to care about the semantic validity of published findings may well prefer plausible demonstrations to quantitative accounts. The main purpose of establishing the semantic validity of content analyses, qualitatively or quantitatively where possible, is to assure that analytical uses of meaningful texts remain arguably consistent with the chosen context from which these texts are sampled, and stakeholders of reported results are not misled by the terms in which they are reported and understood. As pointed out in Chapter 2, this context is a content analyst's choice, but choice it has to be. Spelling out where evidence for semantic validity could be obtained encourages content analysts to articulate the ground on which they chose to stand, not to get lost in their own abstractions.

13.2.3 Structural Validity

Structural validity is at issue when content analysts argue over whether the analytical constructs they have adopted accurately represent the known uses of available texts, the stable meanings, language habits, signifying practices, and behaviors in the chosen context. Thus, structural validity assesses the backing (see Chapter 2, section 2.4) of an analyst's abductive inferences primarily from categorized text and secondarily in processes of categorizations, provided the latter involves coders that serve as a backing for inferences or interpretations implicit in the coding/recording process. This evidence may consist of unquestionable incidences of the stable relationships between potentially available texts and the targets of content analysis and valid theories about them. When a content analysis is designed *de novo*, and thus has no history of successes or failures, structural validation is the only way to lend credibility to its inferences.

The work of historians is most clearly of this kind. Although it is said that history never repeats itself, it may well repeat certain patterns that can be accounted for through generalizations, especially about human/social nature. For historians to rely on such patterns, they must be conceived as relatively permanent within a particular historical context. Dibble (1963), who analyzed arguments by historians in support of and against inferences about the factual nature of events drawn from historical documents, distinguished four kinds of evidence or generalizations for the structural validity of historical accounts. One kind of evidence concerns the roles and practices of the social institutions that create the records to be validated, using their own codes of conduct and preserving certain documents and not others. These are sociological generalizations about what Dibble calls "social bookkeeping" practices. A second kind concerns the characteristics of witnesses who describe what they experienced or report on what they heard. These are psychological generalizations about the working of memory, the influence of interests, and the influence of emotional or ideological involvement with the events in question. A third kind of evidence concerns how the structure of narratives relates to the narrated events. These are linguistic or literary generalizations about how texts are organized and what they mean to readers at that time. Finally, there are physical generalizations of how documents tend to travel, who accesses or reproduces them, how they reach their destinations, how they are filtered or filed, and how they fade or drop out of circulation. Dibble suggests that historians use such generalizations to validate or invalidate their inferences. They exemplify what content analysts do as well, perhaps a bit more systematically.

The inferences that Leites, Bernaut, and Garthoff (1951) made from speeches delivered on the occasion of Stalin's birthday, discussed in Chapter 9, serve as another particularly transparent example of structural validation in an essentially unique situation. Once the researchers' analytical construct was in place, the inferences from available speeches followed. The validity of their construct was established by experienced Sovietologists who argued by references to generalizations about how political discourse functioned in the Soviet Union, especially how politburo members close to Stalin would have to avoid showing interpersonal closeness. With the structural validity of their construct demonstrated, the results of its application were accepted on this ground—and later proved to be correct.

Osgood, Saporta, and Nunnally (1956) fashioned their evaluative assertion analysis according to then-prevailing theories of affective cognition, cognitive dissonance theory in particular, which had been substantiated in numerous controlled experiments with subjects. This was the only validating evidence used in this case, and through it the researchers sought to establish that the computations built into their analytical procedure structurally corresponded to what was known about individual cognition. Evaluative assertion analysis has been extended—for example, by Cuilenburg, Kleinnijenhuis, and De Ridder (1986) and Kleinnijenhuis, De Ridder, and Rietberg (1997)—based on the structural validity provided earlier and only occasionally reexamined.

13.2.4 Functional Validity

Functional validity is the degree to which analytical constructs are vindicated in use rather than in structure. A content analysis is vindicated by reference to its history of use, particularly by its absence of significant failures. *Usefulness* and *success* may mean many things, of course, and these concepts make sense only in the presence of alternative methods competing with each other in the same empirical contexts. To vindicate a content analysis, one must demonstrate that its analytical constructs, which account for the analyst's proceeding from available texts to the answers to given research questions, are useful over time and in many empirical situations. Whereas evidence for structural validity is based on a correspondence between what one knows about a context and how that knowledge is built into the analytical procedure, functional validity is grounded in whether or not or how well it works.

Functional validity has long been recognized, although it has been known by different names. Janis (1943/1965) suggested that because meanings unobservably mediate between texts (or “signs,” as he preferred to call them) and observable behaviors, one can establish the validity of semantically oriented content analyses only indirectly, by “inferring validity from [their] productivity” (p. 65). He noted, “A content analysis procedure is productive insofar as the results it yields are found to correlate with other variables” (p. 70). In effect, Janis argued that because there is no validating evidence for how audience members understand given messages, accounting for references, attributions, assertions, and speculating about probable effects are justifiable only when the “categories . . . occur as variables in many true empirical propositions” (p. 65). He essentially bypassed answering the question of how or why an analysis produces the results it does as long as they connect with other phenomena that are considered interesting.

An example might be the use of neuronal network theory in the computation of word co-occurrences by the software system CatPac (Woelfel, 1993, 1997). The designers of this system incorporated several ideas into its operation: that concepts are represented by single words, that the textual proximity of pairs of words and the frequency of their co-occurrences affect the way they are stored/recalled in an author's or reader's brain, that the strength of their pairwise relationships is dynamically adjusted with use, that recent co-occurrences overshadow earlier ones, and so on. These propositions, individually convincing, could be regarded as validating the procedure structurally, and its proponents claim as much when they call CatPac a

system that performs a “semantic network” analysis. However, the computations that yield results are so complex, and so little is known about how the human brain develops concepts, that the connection between how people conceptualize and how the computational procedure gets to its results remains obscure.

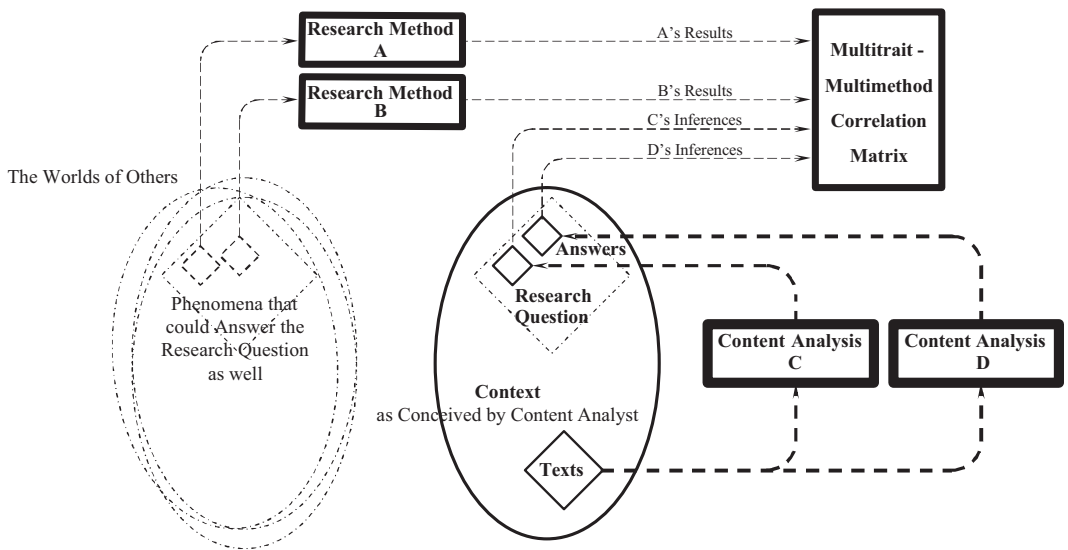
However, CatPac has been used extensively and applied to a variety of data by researchers interested in mass communication, marketing, politics, bibliographic citations, and many more areas. Improvements have been introduced over time. Occasional lack of face validity caused the developers to make a variety of adjustments, such as excluding function words and stemming, lemmatizing the vocabulary to reduce grammatical variations considered meaningless in the context of the system’s use. CatPac applications naturally migrated into areas that seemed most promising, particularly where the results correlated with other phenomena of interest or aided practical decisions. It found niches in which it proved itself of practical value, useful, and successful. Does CatPac really compute semantic networks? Not the way linguists and researchers in the artificial intelligence community conceptualize them. Does it replicate what neurons do in the brain? Surely not structurally. However, the very fact that it finds users and uses in competition with other computational techniques can be regarded as vindicating evidence, demonstrating its functional validity.

13.2.5 Correlative Validity

It is an epistemological fact (but a fact not always recognized in the literature of psychological testing) that correlations between test results and other variables relate measures of phenomena to each other, not phenomena. Correlations cannot bridge the epistemological gap between measures and the phenomena they claim to measure. Correlations do not predict, either—as I shall demonstrate below. They merely weigh the extent to which one measure can substitute for another. If the measures in a battery of measures are correlated with each other, those that correlate perfectly are substitutable without question, and those that correlate less than perfectly are substitutable to the degree of their correlation. The basic idea of correlational validity is that validity travels along high correlations. Validity always comes from somewhere—one or more trusted variables whose validity is established prior to or outside of efforts to establish correlational validity. If the results of a content analysis of crime reports in newspapers and the results of public opinion polls about perceived crime correlate higher with each other than either of the two variables with official crime statistics, as found by Zucker (1978), then content analysis and public opinion polling might well be substitutable for each other, but neither can replace crime statistics. If none of these variables can be trusted to begin with, validity cannot be an issue. Figure 13.3 depicts the general idea of correlational validity schematically.

Campbell and Fiske (1959) developed the idea of validation by correlation statistics into a full-fledged methodology. Taking Popper’s idea of falsification to heart, they argued that correlative validity of a new method requires not only high correlation with established measures of the trait it intends to measure but also low or zero correlation with established measures of traits it intends to distinguish. They call the former *convergent validity* and the latter *discriminant validity*. It follows that a research result can fail to be correlative valid in two ways: by low correlations with measures that are known to measure the phenomena of interest and by high correlations with measures that are known to measure distinctly different phenomena or phenomena independent of the one intended to be measured.

FIGURE 13.3 ■ Correlative Validation



To show that a measure possesses both convergent and discriminant validity, one must compute a battery of correlation coefficients between measures of a number of traits, each obtained by several independent methods. These are tabulated in what Campbell and Fiske (1959) call a multitrait-multimethod matrix (see also Alwin, 1974). A detailed discussion of this method is beyond the scope of this chapter, but Table 13.1 provides an example of such a matrix (for more on this method, see

TABLE 13.1 ■ Multitrait-Multimethod Matrix for Three Content Analyses

Method	Item	HE	HP	HA	OE	OP	OA	SE	SP	SA
Method 1 (Holsti)	HE	1.00								
	HP	.04	1.00							
	HA	-.08	.53*	1.00						
Method 2 (Osgood)	OE				1.00					
	OP				.39*	1.00				
	OA				.32*	.37*	1.00			
Method 3 (Saris)	SE							1.00		
	SP							-.01	1.00	
	SA							.01	.03	1.00

Krippendorff, 1980b). It compares three computer implementations of Osgood's main semantic differential scales of affective meaning, evaluative (E), potency (P), and activity (A) (see Chapter 7, section 7.4.4), by three researchers, Holsti (H), Osgood (O), and Saris (S) (see Saris-Gallhofer & Morton, 1978).

In this table, all correlations with themselves are listed on the main diagonal. These are unities, of course, and uninformative as such. Convergent validity would be demonstrated by high correlations, ideally unities, in the diagonals of the three heteromethod blocks. In these submatrices, they are all significant, as indicated by asterisks, but they differ regarding the affects being compared. The measures of the evaluative dimension correlate highest with each other, followed by the measures of potency; the measures of the activity dimension are lowest in all three instances. This is consistent with the findings of other research, suggesting that judgments of good and bad—heroes and villains, successes and failures, beauty and ugliness—yield better scales, turn out to account for more variance in semantic differential studies than the other two, and are also more reliable in content analyses generally. Discriminant validity is indicated when the off-diagonal correlations in the heteromethod blocks are lower than the correlations in their diagonals, ideally zero. Here they are lower but still far from the ideal. In fact, the surprisingly significant correlations between OA and HP, OA and SP, and OP and SE suggest that the three dimensions of affective meanings are not clearly differentiated across these methods. However, the culprit in this lack of discrimination is found in the three monomethod triangles. Within Osgood's method, all three dimensions correlate significantly—that is, it does not discriminate too well among the three kinds of meanings. In Holsti's method, it is the correlation between the activity and potency dimensions that signals a lack of discrimination, whereas in Saris's method, off-diagonal correlations are near the ideal of zero, independent of each other, and show high discriminant validity.

This discussion is not intended to generalize about the three methods. They may differ for a variety of reasons (see Saris-Gallhofer & Morton, 1978). My only aim here is to show how convergent and discriminant validity can play out in correlative validations.

Another example is the validation of an index of argument quality by Cappella, Price, and Nir (2002). These researchers developed their index in the course of a study of online deliberations during the U.S. presidential elections in the year 2000. It counts the number of arguments that participants in this study could give in support of their own positions and, what is perhaps more interesting, the number of arguments that these participants could imagine others would have against their positions. Cappella et al. wisely refrain from claiming to measure opinion quality, as a single dimension would probably fail a semantic validity test. The name they give their measure reflects more closely what it actually measures, "argument repertoire." To test its convergent validity, they show that it correlates highly with numerous variables that could be construed as assessing aspects of a common construct, including political knowledge, political interest, flexibility (subjects' willingness to participate in various discussion groups), and mass-media exposure. They note that

those with the capacity to write out reasons for their opinions and to identify relevant reasons for opposed opinions also express interest in politics, are more accurate in their factual political knowledge, and use the print and broadcast media as sources of their political news. Even their personal communication is more political and diverse. Coupled

with . . . data . . . indicating higher argument repertoire for those with more education and more commitment to their ideology and party, [we] have good evidence of convergent validity. (pp. 83–84)

Cappella et al. (2002) do not show discriminant validity, however, and it is therefore not so clear what their measure distinguishes. It may well embrace general communication and social skills as well as intelligence, which would be far beyond the intention to define a measure that adds to the vocabulary of public opinion researchers.

13.2.6 Predictive Validity

Predictions extend available knowledge to as yet unobserved domains. The predicted phenomena may have existed somewhere in the past (e.g., historical events, characteristics of late authors, antecedent conditions of received communications), may be concurrent with the texts being analyzed (e.g., attitudes, psychopathologies, individual aptitudes, the extent to which someone is plagued by problems, the makeup of cultural climates) or may occur in near or distant futures (e.g., the consequences of persuasive messages, the success of future employees, the continuations of trends). I am suggesting two defining criteria for predictive validity. The first emphasizes the nature of evidence as in the *Standards* (American Educational Research Association et al., 1999): For correlational validity, validating evidence must be concurrent, whereas for predictive validity it need not be and in fact typically is not. The second defining criterion requires predictions to be specific, to select a set of observations that is smaller than the set of all conceivable ones—just as any answer to a research question must exclude some of the logically possible answers. Eventually, predictions are validated when the validating evidence stays within the set of predicted observations.

To draw a clear line between correlational and predictive validity, I return to Cappella et al.'s (2002) argument repertoire measure. These researchers found high correlations of argument repertoire not only with variables that belong to the same construct (as noted above) but also with variables that they conceptualized as caused by what argument repertoire measures: participation. They observed two kinds of participation: willingness to attend group deliberations about political topics and willingness to get involved in substantive exchanges while attending (p. 89). Both correlated highly with argument repertoire. Because all data of this study were concurrent and these variables correlated, the validity thereby established is correlational, even if one could argue, and hence conceptualize, that participation is an effect and not a cause.

However, conceptions of causality aside, once researchers state findings so as to be selective among conceivable alternatives and open to the consideration of evidence concerning these alternatives, predictive validation can take place. In fact, when Cappella et al. (2002) report that those with greater argument repertoires are more willing to participate in political deliberations, they make rather specific predictions that could be checked against future data. Establishing the validity of their predictions would require agreement with subsequent observations—not correlations, but observations concerning whether and how often people with high argument repertoires do indeed participate in one or both ways.

A classic example of predictive validation is George's (1959a) attempt to evaluate the U.S. Federal Communications Commission's (FCC) predictions made from German domestic propaganda during World War II. All of the FCC analysts'

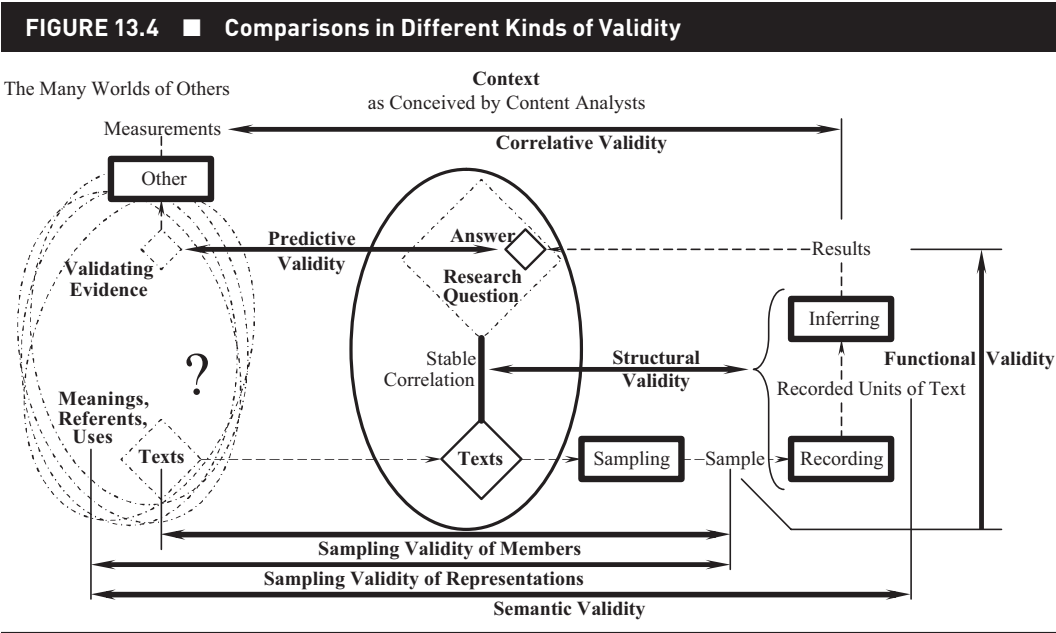
inferences were available in the form of reports they had written. After the war, George was able to match the inferences, one by one, with documents that had then become available. He judged each of the inferences for which validating evidence was available as correct, nearly so, or wrong. He demonstrated that the FCC analysts' predictions were accurate to a degree better than chance. George's research (which was not as simple as this brief description makes it seem) suggests how analysts can bring subsequent evidence to bear on predictions.

To recap, predictions cannot be validated by correlation. A watch that runs slow correlates highly with standard time but is incorrect nevertheless. Unless one knows the bias of the watch, one cannot tell the correct time with it. The infamous body counts disseminated by the U.S. government through the mass media during the Vietnam War may have correlated highly with military activity, but after a while, nobody could trust the exaggerated numbers. To predict the author of an unsigned document, it is not enough to demonstrate that signed documents show a correlation between word choices and the identities of authors; the unsigned document must be traced to one author, ideally excluding all others.

Predictions of past, present, or future happenings from texts must also avow exclusions, happenings that are not expected. If a content analysis said yes to all possible answers to a research question, it would be as worthless as if it said no to all of them. The more selective a content analysis is, the more information it provides. Subsequent observations validate predictions when they occur within the set of observations that had been predicted, not outside that set (always or at least to a degree better than chance).

To quantify predictive validity, the measures that are appropriate are the same as those used to assess semantic validity. Both concern valid representations—in the case of semantic validity, of the meanings, referents, or uses of texts; and in the case of predictive validity, of whether the answers to research questions are borne out in fact. The appropriate measure of predictive validity is not correlation but agreement.

Figure 13.4, which is an overlay of Figure 2.2, locates the validation efforts discussed in this chapter within the components of content analysis (as discussed in Chapter 4).



14

A PRACTICAL GUIDE

This final chapter discusses three starting points for content analyses. For each, it recommends procedural steps, raises issues that might come up during a research project, notes the junctures at which content analysts need to make decisions, and suggests what they need to take into consideration in making those decisions. It gives an overview of the entire content analysis process, from conceptualizing the research questions that content analysts are called on to answer to reporting their results, providing ample references to related material in the foregoing chapters.

In the preceding chapters, I have introduced the concepts involved in content analysis one by one and have suggested solutions to conceptual and methodological problems that content analysts need to address. In this chapter, I rearticulate these concepts with practice in mind, so that readers who have particular research problems that content analysis might solve will get an idea of what they can or need to do and why.

Like most social research, content analysis involves four kinds of activities:

- Designing an analysis (section 14.1)
- Writing a research proposal (section 14.2)
- Applying the research design (section 14.3)
- Narrating the results (section 14.4)

These activities are neither mutually exclusive nor entirely sequential. Surely, all inquiries start with some conceptualization of the research process, but as content analysis requires a great deal of preparatory effort, the design phase may well become part of a research proposal. Researchers who have their own resources, including students working on their theses, may not need to write formal research proposals, but they are nevertheless well advised to have such proposals in mind. Often, the relationship between the design of a content analysis and the application of that design is circular. A seemingly perfect design may reveal flaws in its application that bring the researcher back to the drawing board to sample more or different kinds of data, reconceptualize the data language, improve the coding instructions, and even radically change the approach taken initially. This is the hermeneutic circle in scientific inquiry.

14.1 DESIGNING AN ANALYSIS

Any design proposes something that would not come into being without guided efforts—here, a procedure for moving from potentially available observations, texts, sounds, and images to the narrated answers to a research question (see Figure 4.2). A research design consists of the detailed specifications that guide the handling of data and make the research reproducible and critically examinable at a later point in time. Aside from getting involved in available text, the development of a research design is the most intellectually challenging part of a content analysis. In the course of designing a study, analysts clarify their research interests, learn to respect their own readings as different from those of others, explore the questions they would like to see answered, delve into available literature for insights about the contexts of their analysis, and play with analytical possibilities—until step-by-step specifications emerge that promise to bring the analysis to a worthwhile conclusion (see Chapter 4).

Researchers may enter content analysis from different starting points. The possible starting points may not be equally desirable, but circumstances for research rarely are. Below, I discuss content analyses in terms of three points of entry:

- *Text-driven* content analyses (14.1.1) are motivated by the availability of texts rich enough to stimulate the analysts' interests in them. As the research questions emerge, as analysts are becoming involved with such texts, text-driven analyses are also called "fishing expeditions."
- *Problem-driven* content analyses (14.1.2) are motivated by epistemic questions about currently inaccessible phenomena, events, or processes that the analysts believe texts are able to answer. Analysts start from research questions and proceed to find analytical paths from the choice of suitable texts to their answers.
- *Method-driven* content analyses (14.1.3) are motivated by the analysts' desire to apply known analytical procedures to areas previously explored by other means.

14.1.1 Text-Driven Analyses

A text-driven analysis starts with a body of text, as noted above: an interesting set of personal letters, a collection of taped interviews, the diary of a famous person, a compilation of comic books, transcripts of naturally occurring conversations, conference proceedings (a collection of presented papers), a significant period of publications (newspapers, professional journals, movies), election campaign speeches, news accounts of a particular crime, notes made during anthropological fieldwork, a collection of family photographs, advertisements in magazines published in different countries, reports to the shareholders of a corporation, library listings of a particular institution, articles mentioning a particular drug, telephone books, and so on.

Without an explicit research question in mind, researchers typically start by familiarizing themselves with the chosen body of texts. They may begin with "housekeeping" chores—cataloging the texts, unitizing the body of text into "packages" that can be handled more or less independent of each other. When the researchers have a sense of how many texts there are, they may want to explore apparent intertextualities: quotes, references, overlaps, rearticulations, elaborations,

and sequential orderings. They may also note how the texts reproduce, respond to, or elaborate on each other. Intertextualities form dependency networks among texts, which in turn suggest possible paths for reading the given body of texts—for example, along lines of narrative coherences, genres, or historical sequences, or according to how various sources respond to each other. Constructions of intertextualities amount to ways of reading, but—to be clear—these always are an analyst's readings.

Next comes a careful reading of the texts for the purpose of summarizing what they collectively mean to the analyst, what they denote, connote, or suggest, or how they are or could be used as a whole. If the body of texts is so large that it becomes difficult for a single individual to keep track of all the relevant details while reading, content analysts may take advantage of a variety of computer aids, including qualitative data analysis (QDA) software such as QSR's NVivo, R's RQDA, or word frequencies and KWIC lists with software such as Concordance, VBPro, WordStat, or TextQuest. Even the search functions of ordinary word processing programs, such as Microsoft Word, may be used for simple content analyses. ATLAS.ti is especially well suited to the creation of networks of texts. The advantage of such analytical aids lies in the assurances they can provide that text explorations are systematic, effectively countering the natural tendency of humans to read and recall selectively. Text-driven analyses are also called *interpretive* or *qualitative*, as opposed to *quantitative* (see Chapter 4, section 4.1.2), discussed as QDA in Chapter 11 with reference to computer aids (see section 11.5); such analyses are indebted to the tradition of grounded theory beginning with Glaser and Strauss (1967).

Even when aided by QDA software, it is important to note, such text explorations are essentially limited to a single analyst's conceptions and ability to read. Such software is convenient, but it does not assure the replicability of the process.

When the volume of texts exceeds individual abilities as well, when teamwork becomes essential, the problems of coordinating the work of several analysts begin to dominate the research effort. It then becomes important that the analysts working on a project agree on the terms of the analysis; in addition, as differences in their individual readings will inevitably surface, the analysts need to render these readings in comparable categories. This will bring the analysis in line with the procedures described in Chapter 4—whether the analysis is called qualitative or quantitative. I continue this thread below, in section 14.1.2.

Problems of coordination among content analysts on a large project also challenge the analysts to overcome the temptation to take their own readings as the only ones that count, which often coincides with the belief that “content” is contained in text, an inherent quality of text, and that everyone ought to know what *it* is (see Chapter 2). Obviously, texts can be read variously and mean different things to different people. Recognizing this leads analysts to listen, while reading texts, to the voices of other readers—the texts' writers, audiences, and users—to understand what the texts mean to them, how language is implicated in their interpretation, the roles the texts do or could play in the lives of their users, and whose voice is represented, who listens, who responds, and who is silenced in these texts. Unless they consider alternative readings, content analysts are limited to analyzing their own understanding and nothing outside it. By reading with alternative voices in mind (see Krippendorff, 2006), analysts begin to expand their horizons, to get a feeling for the context they need to construct to make room for diverse people associated with the texts and for the institutions that govern the texts' antecedent conditions, concurrent interpretations, and practical consequences.

As this context becomes conceptually clear, so do the questions that analysts feel they could answer. Figure 4.3, which depicts this text-driven entry to content analysis, suggests that such questions arise not only from the virtual voices of other readers or users but also from what the authors of pertinent literature on the context say.

Qualitative content analysts typically stop at their own interpretations of texts, however, and the recommendation I have just made is intended to undermine the very notion that texts can drive an analysis. Contrary to what lawyers are fond of saying, documents never speak for themselves—interpretations are always made by intelligent readers. And texts inevitably have multiple meanings. Although the seeming objectivity of computer searches tends to hide this fact, the queries that inform such searches are always formulated by the very analysts who also interpret the search results. When analysts acknowledge their own conceptual contributions and, by implication, the possibility of diverse readings, especially when they have to train coders to comply with written recording/coding instructions, their analyses can no longer be conceived as text driven. This brings us to the next, perhaps more preferable, way of entering a content analysis.

14.1.2 Problem-Driven Analyses

A problem-driven analysis derives from epistemic questions, from a desire to know something currently inaccessible and the belief that a systematic reading of potentially available texts and other data could provide answers. Content analysts who start at this point tend to be involved in real-world problems: psychoanalysts seeking to diagnose the pathology of a patient; lawyers trying to generate evidence that will support or refute an accusation of plagiarism; historians hoping to clarify how a historical event unfolded; literary or forensic scholars aspiring to know who wrote an unsigned document; educators attempting to predict the readability of textbooks; mass-media researchers intending to substantiate claims of undesirable effects on civic society of TV election coverage; propaganda analysts looking for military intelligence hidden in enemy domestic broadcasts; rhetoricians desiring to measure how much civic knowledge candidates for political office bring to a debate. All of these are epistemic problems—that is, they are problems of not knowing something deemed significant. Content analysts must convert such problems into research questions, which they then attempt to answer through a purposive examination of texts. In Chapter 3, I discuss the different kinds of inferences that analysts may draw from texts; here, the focus is on the steps that analysts may want to take to get to those inferences:

- Formulating research questions
- Ascertaining stable correlations
- Locating relevant texts
- Defining and identifying sampling units among relevant texts
- Sampling a sufficiently large number these units
- Developing coding categories and recording instructions
- Selecting appropriate analytical procedures

- Adopting standards for the reliability of generated data and statistical significance levels for the results
- Allocating resources for each step of the proposed analysis

14.1.2.1 Formulating Research Questions

Formulating research questions is by far the most important conceptual task that analysts face, for it is the key to a successful research design. As noted in Chapter 2, the research questions posed in content analysis have the following characteristics:

- They concern currently unobserved phenomena in the problematized context of available texts.
- They entail several possible answers.
- They provide for at least two ways of selecting from among these answers—if not in practice, then at least in principle.

The first of these defining features rearticulates the aim of content analysis: to make abductive inferences from texts to phenomena outside those texts. In Chapter 2 (section 2.4), I discuss how abduction is distinguished from induction (moving from particulars to generalizations) and deduction (moving from generalizations to particulars) as a form of reasoning that moves from particular texts, through context-sensitive explanations of these texts, to particular answers to research questions (i.e., from particulars to particulars). Such answers need to solve a problem in the widest possible sense, which may range from solving an important mystery to informing a decision to act.

The second characteristic above demands of a research question that it entail several conceivable answers—neither an open field in which anything can happen nor a single answer that the analyst intends to prove, or worse, what is already known.

The third feature suggests that it would not be sufficient simply to answer a research question by means of a content analysis. The researcher should think of at least one other way to answer that question, one that could validate the answer that the content analysis suggests, at least in principle: additional observations, correlations with measures already known to be valid, even observations of improved success in acting on the information the content analysis provided (see Chapter 13). This feature insists that it is methodologically unsound to obtain results that have nothing to do with the context of the analyzed texts, an abstract representation without empirical basis. The answers that a content analysis provides should be validatable in principle, and this entails an alternative method or independently available evidence.

Novices in content analysis need to understand that not all questions qualify as research questions. For example, a question concerning whether a computer program is suitable for analyzing a body of text is a question about the properties of the software relative to available texts—it does not address anything in the context of the analyzed texts. The question of whether one can measure a desired quality of texts does not qualify either, because it concerns an analyst's ability and is answerable by that analyst's doing it. The latter has nothing to do with the chosen context of analysis. Nor are questions concerning how often an author uses a certain expression appropriate research questions. Counting is an operation performed on a body of text. Its result says nothing other than that someone has counted something. This argument applies to all computational

abstractions: type/token ratios, vocabulary sizes, and various correlations within a text. Unfortunately, the history of content analysis is full of examples of researchers who have merely declared counts to be indices of phenomena, usually of social or political significance, without spelling out how their claims could be validated. For example, a count of the number of violent acts on television means nothing unless one has at least a hunch that high numbers of such acts are correlated with violence in the streets or induce fears in viewers. More pernicious are pseudoquestions that exclude alternative answers. In the 1930s, many journalists quantified categories of content as a way of “objectifying” preconceived public concerns. Content analysts must not confuse proving one’s point in a public debate with pursuing a research question that has several possible answers.

To be validatable in principle (see Chapters 2 and 13) has to be distinguished from being validated in practice. For example, history always happened in the past. It no longer is observable, although it arguably was present some time ago. The traces of historical events that do survive are nothing more than indirect indicators of the events themselves. Texts are no exception. Inferences about historical events are best correlated with other traces left behind. To find relevant traces, assess their trustworthiness, and relate them to appropriate dimensions of a content analysis, analysts require considerable insight into the context of their analysis. Psychotherapeutic diagnosis provides a different example. No therapist can enter a patient’s mind, so the first definitional feature of research questions is satisfied. The American Psychiatric Association’s *Diagnostic and Statistical Manual of Mental Disorders* (2013) lists all legitimate mental disorders for therapists to choose from. One may disagree with this list, but it does satisfy the second feature. The diagnosis, an informed selection among known mental disorders, might be confirmed by other therapists, validated by independent tests, or vindicated by a successful treatment. Similarly, the answer to the question of whether an author committed plagiarism may not be known for sure without witnesses or admission by the accused. The latter could validate or invalidate the conclusion derived from a systematic comparison of two texts. Some of the more traditional questions that content analysts have answered concerned what different audiences could learn from exposure to particular mass-media messages and the political leanings of newspaper editors. To validate inferences from these kinds of media messages, content analysts have compared their results with survey results, with information from interviews with experts, and with focus group data—all sources of validation on which social scientists rely heavily. Content analyses may also be validated by informing successful actions. Analyses of motivations has informed campaigns to reduce dependencies on tobacco. The point to be kept in mind is that content analysts should be specific about what could validate or invalidate their results before collecting data to answer their research questions.

Well-formulated research questions not only guide a research design, they also constitute one-third of a content analyst’s world construction, the framework within which the analysis is to be undertaken. Research questions concern the uncertain or variable part of that world (see Figure 2.2). The analysts’ next two steps involve the stable parts of their world.

14.1.2.2 Ascertaining Stable Correlations (With the Research Questions)

Content analysts answer research questions by analyzing texts, which should be understood quite generally to include images, sound, websites, symbolic events, even

numerical data, provided they mean something in the chosen context. The abductive inferences that content analysis relies on presupposes some knowledge on the analysts' part of how the research questions relate to available texts. This knowledge need not be, and typically is not, exclusively linguistic or semantic. Content analysts are rarely interested in what is said literally, in dictionary definitions or accounts by a standard reader, if such a person exists. Content analysts are interested in what is said as much as what is not said—that is, they are interested in what texts reveal about phenomena of interest. Ethnic prejudices and unquestionable ideological commitments are rarely made explicit but may become evident in their consequences. Insiders may not recognize them but statistical comparisons that extend the boundaries of particular communities may well reveal what is stable in one community and unlike another. In Chapter 2, I describe these connections as stable correlations, stable or enduring because only if they can be assumed to remain invariant, at least during the conduct of an analysis, can they justify the inferences that content analysts are making. Questions whose answers are not correlated with anything observable, readable, or accomplishable with texts cannot be answered.

Traditionally, content analysts have focused on *linguistic references*, *expressions* of attitudes, and *evaluations*. These assume a one-to-one correlation between textual units and the phenomena referred to, expressed, articulated, or, in a naive sense, “contained” in them. More recently, content analysts have relied on texts as *statistical correlates* of the phenomena of interest, using the wear and tear shown by library books to assess the popularity of certain topics, for example, or relying on the correlation between expressed public concerns and voting. Typically, such correlates stem from other methods of inquiry—such as public opinion research, media effects studies, perception experiments, and theories of cultural cognition—that often are concerned with frequencies, contingencies, and variances. Texts may be seen also as *by-products* of the phenomena of interest, such as when researchers use mass-media coverage to infer how the mass-media industry is organized; as *causes*, such as when researchers attempt to infer audience perceptions or media-induced anxieties; as *consequences*, such as when researchers analyze medical records to determine the population characteristics of patients; or as *instrumental*, such as when researchers take texts as evidence of manipulation efforts by their authors, as in political or public health campaigns. Webb, Campbell, Schwartz, and Sechrest (1966) add to this list of possible connections *physical traces* that the phenomena of interest leave behind and *actuarial records* that institutions maintain for reasons other than analysts' interest in them. Dibble (1963) recognized the latter as well, and George (1959a) has reported the use of complex sociopolitical networks of stable correlations to answer research questions (see Chapter 9 and Figure 9.2). Artificial intelligence models account for still other stabilities in the world of content analysts, enabling them to obtain answers from sparse textual evidence.

The analysts' task at this step of a research project is to ascertain a reliable network of these correlations, correlations that researchers can rely on to be stable and general (invariant over time and in various situations), certain (able to determine or be determined), and selective (able to narrow the set of possible answers to a research question). Content analysts may use all kinds of sources for this knowledge. Past empirical research about text-context relationships is one such source. In Chapter 4 (section 4.2), I suggest several designs for the generation of empirical knowledge in preparation for a content analysis. Available theories and models of how the

phenomena of interest are communicated within a social system can also serve as sources of needed evidence. Chapter 9 gives examples of analytical constructs that are as simple as the aforementioned one-to-one correlations and as complex as a model of how a government controls public opinion through its publications, such as in preparatory propaganda efforts. Causality is not the only path, as suggested by the above list of correlations, nor do correlations need to be direct. Content analysts often lament the absence of general theories and the naive simplicity of theories that claim universality. Indeed, literature on the contexts of particular texts, another important source of knowledge of prevailing stabilities, often includes situation-specific descriptions, temporally limited “mini-theories,” and highly qualified propositions, all of which could well support a content analyst’s efforts. These if-then propositions may be derived from ordinary readings, known stereotypical reactions, widely used folk sayings, metaphors, colloquialisms, and so on. The hope is that such a network of propositions contains a path that connects available texts to the needed answers. A well-substantiated path is all that content analysts need to create.

In assuming such a stable network of correlations, analysts may need to keep track of the conditions under which these stabilities are warranted and when they become unreliable. For example, correlations that have been found to hold for undergraduate subjects may not be generalizable to other populations, at least not without qualifications. In crisis situations, organizational rules may break down or be replaced by others. The meanings of verbal expressions may change over time and/or become variable from one social situation to another, or from one culture to another. Some content analysts are tempted to assume linguistic universality or to assume that correlations once found do not change under conditions other than those studied (see Figure 2.2), but assuming such correlations to be stable when they are not can seriously mislead content analysts.

Obviously, knowledge of the needed correlations is informed by how a context of the analyzed texts is defined. Political scientists look for correlations that differ from those that psychiatrists are able to consider. Sociologists and communication researchers approach content analyses with different constraints in mind, and the worlds that they respectively construct for given texts may well be incommensurate with one another.

A network of stable correlations constitutes the second third of the content analyst’s world construction (see Figure 4.2). Its purpose is to channel, almost in an information theoretical sense, the diversity encountered in texts to the possible answers to a research question. The analysts’ next step is one that is often mentioned as the starting point for content analysis: locating relevant texts.

14.1.2.3 Locating Relevant Texts

In content analysis, texts inform analysts’ questions and so must be sampled from populations of texts that can be informative in this sense. A text is relevant if there is evidence for or an assumption of stable correlations between that text and answers to the research question. By backtracking along the path of the intended inferences, moving from the phenomena of interest along the stable correlations to potentially available texts, content analysts can justify the relevance of a population of texts to given research questions.

As noted above, traditionally content analysts have made their analytical efforts easy by assuming one-to-one relationships between textual units and the phenomena of

interest, what they are assumed to refer to, express, or “contain.” With this assumption, selecting (including reading and counting) texts virtually substitutes for selecting (including observing and enumerating) the phenomena addressed by the research questions. This hides the usually complex roles that texts play in social situations. For example, to infer variations in achievement motives over various periods of a culture, McClelland (1958) searched for messages in which achievements were *created*, *negotiated*, or *celebrated*. This led him not only to popular literature, biographies, and expressions in art but also to images on Greek vases and postage stamps. Content analysts have a long history of analyzing the newspapers read by political elites in various countries to infer the politics of the citizens of those countries. This choice is grounded in the assumption that political agendas are set and public debates are spearheaded by certain leading newspapers, so-called prestige papers, rather than by local ones, which are more likely to reproduce what the prestige papers print and are, hence, less informative. To reveal an author’s suspected racial or ethnic prejudices, content analysts may have to sample from the writings of that author not intended for publication, such as personal diaries, letters to close acquaintances, or texts written specifically for the author’s ethnic in-group. When a source of texts has a stake in the outcome of the analysis, the content analysts need to consider what that source knows about how it might be analyzed and focus instead on textual characteristics that the source is not aware of or cannot control. This rules out instrumental aspects of communication.

Although the logic of such choices is pretty clear, it is not easy to be more specific about how content analysts go about deciding on the informativeness of the texts they propose to analyze. Reading a small sample is a good start. Examining headlines or abstracts for clues to the relevance of texts is a common practice. Pursuing citation networks to the key publications is a strategy familiar as snowball sampling (see Chapter 6, section 6.2.6). Alleged expertise could also lead analysts to suitable populations, provided such attributions can be trusted. The reputation of a publication is another criterion for locating relevant texts. Often, however, content analysts have to be satisfied with what is made available to them. Propaganda analysts during wartime and analysts working for institutes that monitor international agreements have to start with what they can intercept. Scholarship on literary figures is limited to what is written by and about them and their times. Conversation analysts can record only with permission, which excludes many privileged conversations (and can introduce unwanted biases).

The Internet as well as large, full-text electronic databases and digital libraries have vastly expanded the availability of content-analyzable texts. Browsers, text retrieval engines, and computer-aided text analysis tools (see Chapter 11, section 11.3) can locate relevant texts in stages. Starting perhaps with vague hunches about what is relevant, text mining may begin a search with queries that cast a deliberately wide net over all conceivable texts just to get a sense of how much is there. Making good use of abduction, analysts typically develop increasingly detailed explanations of the body of texts they have so far scanned, become clearer about how available text may be correlated with the research question, and iteratively narrow the search. During such explorations, content analysts develop conceptions of the available texts and how to analyze them while simultaneously reducing the sample of texts to a manageable size. Search engines typically are severely limited in what they can identify (see Chapter 11, section 11.3). There almost always remains a considerable gap between what they retrieve and what is relevant (see the discussion of semantic validity in

Chapter 13, section 13.2.2). Nevertheless, techniques for sampling from electronic databases, Web pages, and online exchanges are improving, promising content analysts an increasingly rich source of textual data.

Increasingly, content analysis is applied to texts obtained from Internet discussion groups, e-mail exchanges, websites, blogs, Facebook, and Twitter. This can pose ethical issues. True, most content analyses rely on public documents, newspapers, books, radio broadcasts, corporate accounts to their shareholders, or historical documents in the public domain, many of which have been generated by institutions or authors long deceased. In content analysis, ethical issues have been largely delegated to those generating data from textual matter. Therapeutic interviews, for example, typically are recorded by therapists, whose professional code of conduct prevents them from revealing the identities of their clients when publishing their findings. Although the distinction between public and private spheres is not always clear, I distinguish two kinds of Internet data of interest to content analysts. The analysis of websites, blogs, court records, and documents that are meant to be public is ethically neutral. However, content analysts can easily and unintentionally violate the privacy of individuals whose writing is not intended to be public, are unaware that their texts are being analyzed for unintended purposes, and would take offense at such a violation of their privacy. On the Internet, even assigning pseudonyms rather than real names to individual voices may no longer provide sufficient protection of the identities of writers or speakers, as powerful search engines may be able to find any given quote's source. Eysenbach and Till (2001) have distinguished three methods of gathering Internet data from discussion groups, which may be described briefly as unobtrusive, with consent, and through explicit participation in deliberations; they discuss the methodological pitfalls of all three. Sixsmith and Murray (2001) go further in discussing the ethical issues associated with analyzing Internet posts and archives in terms of consent, privacy, anonymity, and the issue of interpreting the meanings of others' voices.

Despite these constraints, with well-formulated research questions, a good sense of the network of correlations operating in the chosen context, and respect for the ethical issues of using the voices of others in ways they did not intend, content analysts can collect a wide range of data to explore. The analysts' decision on the population of texts to be considered completes the construction of the world in which the content analysis can proceed (see Chapter 2). We now turn to the components of content analysis (see Figure 4.2).

14.1.2.4 Defining and Identifying Sampling Units Among Relevant Texts

One way to make a content analysis of a large volume of text manageable is to break it into smaller units and deal with each separately. In Chapter 5, three kinds of units are distinguished according to the functions they serve within the content analytical process: *sampling units* are mutually exclusive units of text that are selectively included in an analysis (see Chapter 6); *recording units* are also mutually exclusive, either equal to or contained in the sampling units but separately described, coded, or recorded in the terms of a data language; and *context units* set limits on the amount of text to be consulted in determining what a recording unit means (see Chapter 7). Chapter 5 also mentions *units of enumeration*, which usually coincide with recording units, sometimes in the form of numerical measurements: column inches, type sizes, picture sizes, ratios of different kinds of words, and other numerical scales.

In text mining, units can be defined by proximity operators (see Chapter 11, section 11.3.2): a delineated stretch of text, a document, an article, or a paragraph that contains a match with the query. Search results may serve as sampling units or as recording units. It is conceivable that text searches provide answers to analysts' research questions directly, but this is rare.

The definitions of units of analysis have important implications. When the units are mutually exclusive, counting them leads to comparable frequencies; when they overlap, it does not. Separating units also severs all relations among them, omitting information that resides between neighboring words or phrases. For example, taking single words as units disregards their roles in sentences, so that their syntactical meanings are lost; units in the form of sentences omit the roles that sentences play in paragraphs, thought sequences, the points made in longer arguments, and so on. Thus, *unitizing a text is justifiable only if the relationships between units do not inform the research question*. The above-mentioned context units are intended to preserve at least some of the information that surrounds the recording units. Generally, if units are too small (such as words or short expressions), semantic validity suffers, and a content analysis tends to become shallow. If units are too large (e.g., whole documents, Web pages, books, TV shows), the content analysis tends to become unreliable (see Chapter 12, section 12.1).

14.1.2.5 Sampling a Sufficiently Large Number of These Units

If the population of relevant texts is too large, content analysts may select representative samples of these texts. A heuristic approach to sampling is to start with any arguably unbiased sample of text, analyze it for how well it answers the research questions, and, if it fails to meet acceptable standards, either continue to sample until the questions are answered with reasonable certainty or declare the process hopeless. The latter may signal the need for a redesign of the content analysis.

Chapter 6 suggests suitable sampling strategies. However, because texts as well as the targets of content analyses are about phenomena outside or surrounding the texts, sampling in content analysis differs from sampling in other research techniques—for example, sampling of individuals for public opinion surveys. In content analysis, researchers need to sample texts with two populations in mind: the “population” phenomena that correlate with and hence lead to answers to a research question and the population of texts that represents these phenomena. In sampling the texts for a content analysis, researchers must give both phenomena a fair chance of contributing to the answer to the research question.

Chapter 6 also addresses the problem of texts that come into analysts' hands for reasons unrelated to the research questions (see also the discussion of text-driven analyses in section 14.1.1). If texts are made available rather than purposefully sampled, their representativeness cannot be assured. They may be biased on account of their sources' selectivity. For example, historical documents survive for a variety of reasons that are usually unrelated to why they are of interest. Politicians have good reasons to hide embarrassing information from public view, and television news is not about what happens in the world but rather represents what the mass-media institutions deem newsworthy and what they can fit into available programming space. When the population of available texts is small, content analysts may not have the luxury of sampling and so face the problem of rectifying the sampling biases inherent in these texts; see sampling validity (b) in Chapter 13, section 13.2.1.

14.1.2.6 Developing Appropriate Coding Categories and Recording Instructions

As I have noted above, when the volume of text exceeds a single researcher's analytical capabilities and analysts must therefore work in teams, or, even more important, when their results are to satisfy scientific standards and need to be replicable elsewhere, the analysts involved need to work not only together but also alike or else their results will not be comparable. The coordination this requires is accomplished through the formulation of clear instructions for coders (see Chapter 7) to describe the same textual units in the same analytical terms, a data language (see Chapter 8). To ensure replicability, such instructions may include the following:

- A list of the qualifications that coders (observers, interpreters, judges) need for the task
- Descriptions of training procedures and instructional materials used to calibrate coders' conceptions
- Operational definitions of the recording and context units and rules on how to distinguish them
- Operational definitions of the syntax (form) and semantics (meanings) of the data language (the categories or analytical terms) that coders are to apply in describing, translating, or categorizing each textual unit (Ideally, these definitions inform the cognitive operations that coders are asked to employ in reading and recording the texts. The definitions may be supplemented by examples of what should not be coded and why, including examples of what should not be included. See MacQueen, McLellan, Kay, & Milstein, 1998; see also Krippendorff & Bock, 2009, Chapter 4.1.)
- Copies of the form(s) or electronic records to be used in creating records and entering data for processing: spreadsheets, examples of completed questionnaires, and initial tabulations

Typically, before these instructions are applied by several coders and to a large body of text, the analysts need to pretest them on a small sample of texts and then modify and retest them until they satisfy reasonable reliability standards (see Chapter 12, section 12.6.4).

There are several well-known strategies for developing suitable coding categories and recording instructions. Unfortunately, many content analysts use categories that are uniquely tailored to available texts, in effect starting each content analysis from scratch, almost in the spirit of text-driven approaches. Although this strategy eases the coding task and increases reliability, it creates content analyses whose results are not comparable with each other and therefore rarely advance theory. Although ingenuity is always welcome, content analysts who rely on conceptualizations that have proven successful elsewhere have a better chance of contributing to existing knowledge.

A second strategy that many analysts use is to rely on the recording instructions of published content analyses with similar aims. I have mentioned various systems of categories in Chapters 7 and 8, and readers can find other examples in the works of authors such as Berelson (1952), Holsti (1969), Weber (1990), Gottschalk (1995), Roberts (1997), Riffe, Lacy, and Fico (1998), and Neuendorf (2002) and in *The Content*

Analysis Reader (Krippendorff & Bock, 2009). In addition, there are the categories built into computer programs, especially dictionary approaches (see Chapter 12, section 12.5.1). Some content analyses rely on only a few variables, whereas others define very many. Some require only a page of instructions; the instructions for others fill whole books (e.g., Dollard & Auld, 1959; Gottschalk & Gleser, 1969; Smith, 1992b). If existing coding categories have proven reliable and if they lead to answers to the research question under consideration, there is no need to invent a new scheme.

A third strategy is to draw from available literature on or theories of the context of the analysis. If the descriptive accounts or theories about this context can be operationalized into categories for coding texts, then analysts can gain immediate access to what the literature suggests the stable correlations are. This is the path that Osgood, Saporta, and Nunnally (1956) took repeatedly, for example, in developing evaluative assertion analysis. This analysis operationalized theories of cognitive balance, which led to such theoretical concepts as “attitude objects,” “connectors,” and “common meaning terms.” Where theories are unavailable, content analysis categories may be found in official classifications or taxonomies. If analysts need to describe occupational categories, it would make sense for them to consult official Federal Trade Commission listings or surveys of sociological studies of occupational status and prestige. For psychoanalytic research, the American Psychiatric Association’s *Diagnostic and Statistical Manual of Mental Disorders* (2000) is indispensable. As these categories are widely used, content analyses that use them can tap into empirical findings that are cast in these terms. Along the same lines, if a content analysis is to provide variables for testing particular hypotheses—for example, about participation in political deliberation—then the analysts might take their categories from previous research in this area, as Cappella, Price, and Nir (2002) did, relying for their argument repertoire index on Kuhn’s (1991) work on practical reasoning in everyday life. By deriving categories from established theories of the contexts of their analyses, researchers can avoid simplistic formulations and tap into a wealth of available conceptualizations.

In addition to having to be reliable, the categories of a data language should be tested, where possible, for their semantic validity (see Chapter 13, section 13.2.2). This is especially important for computer dictionaries, which, while perfectly reliable, may tag or transform text in incomprehensible ways. In Chapter 11 (section 11.4.1), I discuss several computer dictionaries and give one account of the development of categories suitable for computer processing (see also Krippendorff & Bock, 2009, Chapter 5.2). It may serve as a template for programming computer coding dictionaries.

14.1.2.7 Selecting Appropriate Analytical Procedures

The best analytical procedures parallel what is going on in the context of the available texts. Figure 9.1 depicts the inferential component of content analysis as a procedural model of the presumed stable text–context relationships. Evaluative assertion analysis (see Chapter 9, section 9.2.3), for example, models the transfer of attitudes from common meaning terms to objects and from one attitude object to another. It operationalizes a set of psychological propositions, amounting to a rather particular analytical construct. This analysis is appropriate only where attitudes are the target of research questions and the context conforms to how the process is theorized. Semantic network analysis (see Chapter 11, section 11.4.3) has a very

different structure. It is an outgrowth of computational theories of cognition and is appropriate where these theories are proven valid.

The catalog of well-formulated analytical procedures from which analysts can choose is not very large. In making an informed choice from among several canned computer programs, assembling an analytical procedure from available components, or constructing one from scratch, content analysts are advised to consider three things: (a) be clear about the network of stable correlations in the analytical context chosen for the texts being analyzed, and (b) find out exactly how texts are regarded, processed, or transformed in the analytical procedures available, in order to (c) select the procedure whose computations or analytical steps provide the best model of the network of stable correlations and are therefore most likely to yield valid answers to the research questions.

Selecting among analytical procedures is not easy. Analysts should be aware that promoters of text analysis software tend to overstate their claims about what their software does—for instance, promising that it can extract concepts from text when all it does is calculate statistically outstanding word co-occurrences. For example, on close examination, claims that a software package mines content, models text, or develops and tests theories automatically often boil down to disappointingly simple procedural tricks that are far removed from what these claims suggest. Similarly, theorists often generalize the analytical powers of their own projects beyond available evidence. And, what is even more disheartening, most canned computer programs seal the assumptions built into their algorithms against outside inspection. These are some of the difficulties that analysts face in trying to make informed choices.

The LIWC (Linguistic Inquiry and Word Count) software (Pennebaker, Francis, & Booth, 2001; Pennebaker & Stone, 2001) is a notable exception. It is a dictionary approach (see Chapter 11, section 11.4.1). It clearly documents its rich, multidimensional dictionary, offers users numerous choices, and allows them to apply their own analytical constructs on top of the frequencies of dictionary entries it provides (for an application, see Krippendorff & Bock, 2009, Chapter 7.7).

14.1.2.8 Adopting Standards for the Reliability of Generated Data and Statistical Significance Levels for the Results

Given that the answers to content analysis research questions are inferences from texts about not-yet-observed phenomena, these answers are always of hypothetical validity. Standards serve to limit the uncertainty associated with such answers. This uncertainty is a function of three elements:

- The nature of the context of the texts being analyzed
- The extent of the analysts' knowledge of the text-context correlations
- The care with which the analysis is conducted

The true nature of the context from which texts are taken is not really under the analysts' control. Some contexts are highly institutionalized, whereas others are open and situationally determined, if not deliberately ambiguous. In some, the connections between texts and the answers to research questions are linear and direct; in others, these connections are chaotic or probabilistic. This limits the certainty of

the inferences that analysts can make. There is a danger that content analysts may oversimplify the construction of the context they end up using in their analysis.

Knowledge of these correlations is another matter. No content analysis can be justified without some knowledge of this kind. But complete ignorance rarely exists. Content analysts are competent readers, at least in their own language, and do not let pass what seems incomprehensible to them. Beyond the ever-present face and social validity, content analysts may test for sampling, semantic, structural, and functional validity (see Chapter 13) and argue for the validity of their findings from the strengths of these tests, weaving information from appropriate literature and their own practical experiences into their rhetoric. In arguing for the validity of content analysis results, both proponents and critics rely on scientific standards of plausible reasoning. Such standards, although permeating discussions of scientific accomplishments, may not be quantifiable, attesting to the rhetorical nature of scientific research.

The third source of uncertainty, carelessness in conducting an analysis, shows up in at least two ways: as unreliability at the front end of an analysis, where it is measured with the help of suitable reliability coefficients (see Chapter 12), and in the way an analysis is designed to proceed to its result, also called internal validity (see Chapter 13). The recording/coding phase of content analysis is especially vulnerable to disagreements among coders, which show up in reliability tests.

But how high should standards be set? Obviously, when the results of a content analysis affect the life or death of a defendant in a court of law, when major business decisions with large price tags are based on them, for example, or when whole populations, say during wartime, are affected, standards need to be significantly higher than for scholarly work where the most that is at stake is the content analyst's reputation (see Chapter 12). The attainment of higher standards, although always desirable, tends to be more costly. It may require more careful preparatory investigations (see Chapter 4, section 4.2), a larger body of data (see Chapter 6), more sophisticated techniques of analysis, and so on. Content analysts may not wish to undertake projects whose standards are beyond their reach, both in terms of the needed resources and in terms of their responsibilities for disastrous consequences of mistaken inferences from texts.

Standards for sampling, semantic, structural, and functional validity should be related to the level of validity demanded of the results. To decide on such standards, researchers may want to work backward from how certain, general, or selective the results need to be to how often-unavoidable imperfections can affect them.

The relationship between reliability—a function of the agreement between two or more analytical processes, coders, or devices (see Chapter 12)—and validity (see Chapter 13) is quite transparent. High reliability is a prerequisite of high validity but does not guarantee it. Computer content analyses, while perfectly replicable, can be completely invalid. In Chapter 12 (see section 12.6.4), I discuss reliability standards and a way to test the consequences of unreliable data. The evaluation of the design of a content analysis, internal validity, is more qualitative in nature.

14.1.2.9 Allocating Resources

Content analysts have much to organize: analytical procedures, personnel, and scarce resources. Some activities may be reserved for the principal investigator, whereas others may be delegated to research assistants, requiring training and instructions, or

to professional research companies. Some must be executed sequentially—for example, the sampling of texts will have to take place before their coding, and coding must be done before analysis—and others may be done in parallel. Some take up short moments of time (e.g., running a computer program); others may be tedious (e.g., the reading and manual coding of text, most preparatory work, and the cleaning of dirty data). Limited resources—whether in qualified personnel, analytical devices, or funds—can impose organizational constraints on a project as well. Unless a content analysis is small and exploratory, analysts have to develop ways to organize their work efficiently.

There are numerous tools available to help analysts organize the processes of research. Most of these tools analyze the interconnected activities in a research project as a network. In such a network, arrows represent activities that one person or group can perform. By associating times and costs with each arrow, researchers can calculate needed resources, see potential bottlenecks, assign people to parallel or sequential activities, and estimate minimum and maximum amounts of time to completion.

Among the planning tools that content analysts may find useful are flowcharts such as those used in computer programming, the Program Evaluation and Review Technique (PERT), the Critical Path Method (CPM), and Gantt charts (interested readers can find information on all of these on the Internet). These methods enable researchers to find the least expensive or the fastest paths to achieving research results and match available skills and resources with possible ways of organizing the analytical work.

14.1.3 Method-Driven Analyses

Method-driven analyses are suspect when they are motivated by what Abraham Kaplan (1964, p. 28) calls the “Law of the Instrument”: When a child discovers how to use a hammer, everything seems to be in need of hammering. Analogously, when researchers get hooked on one analytical technique, when they become experts in its use, they may well end up applying that technique to everything in sight—and not without pleasure. Technologies have this attraction, and content analysts, especially those employing computer-aided text analysis software, are not immune to this lure. Typically, mastering any reasonably complex analytical technique requires analysts to invest so much of their time that they find it increasingly difficult to shift gears, even to see alternatives outside their expertise. Instead of starting from real-life problems, content analysts can be tempted by this technological expertise to look for areas of research where their preferred methods are arguably applicable. This raises the possibility that the insights gained from method-driven analyses are more reflective of what particular methods can produce than of how the objects of inquiry operate.

On the positive side, when researchers conduct method-driven content analysis, especially with validity concerns in mind, they simultaneously expand their method’s areas of application while encountering its limitations. For example, the use of CatPac, a software package that fuses two theories—the self-organization of neuronal networks and the movement of diverse social objects within abstract spaces (Woelfel & Fink, 1980)—migrated from tracking advertising and public relations campaigns to optimizing mass-media messages to analyzing communication, bibliographic, and vocabulary networks in social settings (Barnett & Doerfel, 1997). CatPac is now known largely as a clustering program for qualitative data, for texts in particular. In the path it took to arrive at this point, the software encountered failures and critics but also demonstrated successes and gained proponents. It found its niche.

Method-driven content analyses face fewer design issues than do problem-driven analyses, largely because once a method is chosen, analytical options are limited. For example, in CatPac, recording units are unalterably defined as character strings, usually single words. Instead of ascertaining their meanings, CatPac applies an algorithm directly to these words. It clusters words using information about their co-occurrences within specified stretches of text. The clusters resulting from such an analysis are interpreted as representing conceptions in the minds of speakers, in the culture of an organization, or in the public at large. Proponents of CatPac consider this automatic mapping to be the software's most important virtue, whereas critics miss the use of human intelligence. Once a method is chosen, the research questions that can be answered are usually fixed. In CatPac, they concern the forming of clusters of textual units, interpreted as processes of conceptualization on several levels of abstraction, and this process is thought to be manifest in the sources of the analyzed texts.

In the design of method-driven content analyses, usually only five preparatory steps remain for analysts to accomplish:

- Locating and sampling units of relevant texts
- Ascertaining stable correlations in the contexts of these texts
- Preparing the texts in method-specific and context-sensitive ways
- Adopting standards or criteria
- Allocating resources

Locating and sampling relevant texts in method-driven analyses is less an issue of finding texts that correlate with the answers to a research question than one of locating texts that are easily amenable to processing by the chosen method.

Regarding the second step, method-driven content analysts are less inclined to explore correlations in a context for the directions an analysis could be taking than they are to ascertain whether the correlations surrounding the texts are compatible with the assumptions built into the chosen method. To continue the example above, CatPac assumes one-to-one relationships between single words or phrases and concepts in the minds of speakers. CatPac users are advised to examine literature or other sources of knowledge to determine whether its assumptions are warranted, whether the social phenomena of interest warrant this one-to-one relationship, and, more important, whether co-occurrences in texts are valid determinants of their meanings.

The preparation of texts in method-driven analyses resembles the previously described development of recording instruments but not necessarily for use by coders. In CatPac, for example, analysts prepare a text by removing words deemed irrelevant, largely function words, and eliminating mere grammatical variations through stemming or lemmatization. Other text analysis software packages distinguish between go-words and stop-words, apply dictionaries that assign tags to words or phrases by which they are subsequently recognized, or parse sentences into components (see Chapter 11). Less automated analytical techniques, such as contingency analysis, require extensive manual editing of text. Justifications for these text transformations rely heavily on the analysts' judgments of what is relevant and what is not. Semantic validity is one applicable standard, reliability for manual editing is another, and computability by the method is a final and, in this approach, often primary criterion.

The adoption of standards in method-driven analyses essentially follows the arguments presented above, although some may not apply. For example, in computer analyses, reliability is not an issue. However, because computers are unable to comprehend texts the way humans do, semantic, structural, and functional validities are all the more important. Many computer aids fail semantic validity criteria but make up for this defect by processing huge volumes of text.

Finally, researchers must allocate their resources whether they are conducting problem- or method-driven analyses. The use of computational methods, once mastered, tends to be much cheaper than the use of methods that are chosen for their ability to answer method-independent research questions.

Method-driven content analysts tend to justify their use of methods by vindication (see Chapter 13). A method of analysis is vindicated when it consistently produces interpretable results. When a method crosses the limits of its usefulness, it produces obscure and uninterpretable results. The users of CatPac have had such experiences. Unfortunately, many researchers are hesitant to report their failures, even though content analysts can learn more about the limits of particular analytical methods from their failures than they can from their successes.

14.2 WRITING A RESEARCH PROPOSAL

A research proposal puts forth the plan of a content analysis for consideration by a sponsor, dissertation committee, or teacher—someone who is able to grant permission, provide resources, or command time for the content analyst to engage in the proposed inquiry. As such, a proposal has both a rhetorical function and a contractual function.

14.2.1 Rhetorical Function

The rhetorical function of the research proposal is to convince the sponsor(s) of two things:

- That the proposed research is worthwhile or beneficial to the sponsor
- That proponents of the research project are capable of delivering what they propose

In academic research, scholars tend to accomplish the first of these by citing relevant literature to demonstrate a gap in knowledge or in method that the proposed research can be expected to narrow. Ideally, this gap is of social significance, widespread, and instrumental to other advances, not just of personal interest to the researchers. However, all funding agencies have their own missions, which are manifest in their histories of funding certain research projects and not others, just as scholars in positions to approve research proposals have their theoretical concerns and epistemological commitments. For a proposal to succeed, it needs to address these. In applied research, clients tend to seek information with practical implications. To be successful, proposals should demonstrate that the benefits of the research outweigh its costs.

Often, researchers face competing expectations when they set out to write a research proposal. Such expectations may take the form of conflicting criteria from different departments of a funding agency, differing perspectives on the part

of the members of a dissertation committee, or hidden agendas being pursued by decision makers who cannot talk about them publicly. In such a case, the researchers' best strategy is to write a proposal that enrolls all decision makers into the project, giving each a reason to support it. In dissertation research, this sometimes means that a student needs to write one chapter for each committee member. A commercial research proposal may want to show how each stakeholder in the proposed research could benefit from its results, or at least not be disadvantaged by them, and perhaps even how all stakeholders could be brought together on account of the proposed research project.

Past accomplishments are clearly the best recommendations of researchers' abilities. Sponsors examine proposals for researchers' academic degrees and lists of their publications as well as reviews of the analysts' previous research, especially those written by reputable critics, and letters of support from respected authorities. Researchers without relevant previous research records may compensate for this deficiency by providing compelling literature reviews in which they demonstrate familiarity with both the issues involved and how other researchers have solved or failed to solve similar research problems.

A research proposal does not merely discuss issues, however. It also needs to spell out the steps that the researchers intend to take and explain why. Indeed, probably the most convincing demonstration of the researchers' ability is a detailed research plan that evaluators can follow and critically examine for its likely success. The proposal should also report on any preparatory work the researchers have completed, spelling out or suggesting how the challenging problems of the proposed research are being solved. In content analysis, this often means that researchers should present evidence of the reliability of the proposed recording instructions and the capability of the chosen analytical or computational techniques to advance the aim of the research, as well as accounts of the relevance to the research question of the texts to be sampled.

14.2.2 Contractual Function

A research proposal, once approved, entails the expectation that the sponsor or funding agency will provide what it has agreed to make available—financial resources, organizational assistance, or legal support—and that the researchers will deliver what they have proposed. The approval of a proposal creates contractual obligations whether the proposed research is intended to qualify an individual for an academic degree, to contribute theoretical insights, or to provide business or political intelligence.

One of the characteristics of scientific research is that its results cannot be guaranteed before the research is completed. Just as researchers who propose to test a hypothesis must consider evidence in favor of that hypothesis as well as against it, so must content analysts who propose to answer a certain research question keep all of its possible answers open for the analysis to decide among them. Nature does what it does, and texts do not always yield what sponsors like to see and analysts hope to show.

In fact, lack of "cooperation" of texts often stimulates new insights and opens unanticipated turns, which brings us to the second peculiarity of scientific research: serendipity. A research proposal must outline at least one path to answering the research questions but at the same time preserve the analysts' ability to deviate from that path when unanticipated shortcuts become apparent, when new methods turn up, or when unforeseen findings shift the focus of attention to something even more

exciting—provided the research objective stays within expectations and scientific standards are not compromised.

The inability to guarantee particular results and serendipity in the conduct of research can be anathema to funding agencies that have vested interests in preferred outcomes. Proposal writers may need to address this peculiarity of research and convince the sponsors that all legitimate research questions have alternative answers or formulate their research questions so that the sponsors see virtue in the possibility of less preferred answers.

14.2.3 Outline for a Research Proposal

A typical proposal for a content analysis includes all of the following parts:

- *A statement of the general epistemic or methodological issue* that the proposed analysis will address: what that issue is, why and to whom it is important, its social validity (Chapter 13)
- *A review of available literature on the context* in which this issue resides, showing the kinds of questions that have been asked and answered, the kinds of research methods previously applied, and what has worked and what has not, including the analysts' own research or experiences, if relevant
- *A formulation of the specific research questions* to be answered by the proposed research. This should be embedded in an account of the analytical framework adopted, the *context chosen by the analysts* to make sense of these questions, and the *body of text* by means of which the analysts expect to answer these questions (see Chapter 2)
- *A step-by-step description of the procedure to be followed*, including accounts of any preparatory research already undertaken or to be carried out, the hypotheses to be tested (see Chapter 4, section 4.2) as well as how and why they are to be tested, the proposed analytical steps (Figure 4.2), and the standards adopted for each. This description should cover the following:
 - The *units of analysis* (see Chapter 5) proposed, defined, and distinguished, and what they respectively contain and omit from the body of text
 - The *sampling strategies* (see Chapter 6) to be used, where the population of relevant texts is located, how easily available it is, criteria for adequate sample sizes, methods for correcting self-sampling biases, and the sampling validity to be achieved (see Chapter 13, section 13.2.1). This includes the computational tools employed to search and retrieve relevant texts from promising databases.
 - The *recording/coding categories* and *data language* (see Chapters 7 and 8) to be used, whether in the form of instructions to coders, available computer dictionaries, or queries for text mining (see Chapter 11) or to be derived from theories, literature (see Chapter 9, section 9.3), or the texts themselves (see Chapter 1, section 1.7; see also section 14.1.1); what these categories preserve or omit; semantic validity to be achieved (see Chapter 13, section 13.2.2); the reliability to be guaranteed (see Chapter 12); and the results of any pretests of the recording instructions

- The computational (statistical or algebraic) *techniques for reducing* or summarizing the body of recorded text and the justifications for these techniques relative to what is known about the context of the texts
- The *inferential procedures* that the analysts will ultimately use to answer research questions from texts (see Chapter 9): the analytical constructs that underlie these and any evidence for their structural and functional validity (see Chapter 13, sections 13.2.3 and 13.2.4), available computer programs to be used, and evidence of their previously established correlative or predictive validities, if any (sections 13.2.5 and 13.2.6)
- How and to whom the research results are to be made available: the *narrative forms* in which the answers to the research questions will be presented, numerical arrays, visualizations, or computed indices, for example, and planned publications, presentations to conferences, or reports to sponsors; the *kinds of conclusions* drawn from the results, whether they are expected to advance theoretical knowledge, make recommendations for actions, or settle a controversy; and (often overlooked) a *self-critical assessment* of the uncertainties that are likely to remain associated with the larger issues that the results are to address
- *An account of the specific time periods and resources needed* to complete the proposed analysis (the costs of personnel, equipment, and outside services) presented in the form of a timeline of the phases of research showing the milestones to be achieved and the resources needed at each phase
- *References to style manuals* that the sponsor of the proposal or targeted publications of its findings accepts
- *Appendices* containing material pertinent to the proposal but not central to the potential sponsor's understanding of what is being proposed, such as examples of the kinds of texts to be analyzed, lists of texts to be sampled, the proposed categories of the analysis, its data language and/or recording instructions if already available, preliminary reliabilities achieved so far, specifications of the software to be used, preliminary analytical results, and testimony by experts supporting the proposed research

14.3 APPLYING THE RESEARCH DESIGN

Ideally, the work involved in carrying out a well-designed content analysis becomes routine—even when a content analysis is qualitative, text-driven, or hermeneutic and exploratory to begin with. With all intellectual and methodological problems solved during the design phase, the analysis could be turned over to a research organization. In practice, however, problems are bound to emerge: Needed texts may turn out to be unavailable, scarcely relevant, or biased by incorrigible self-sampling practices of their sources, and software may turn out not to work as expected. However, the most frequent disruptions stem from the inability to meet accepted standards of reliability (see Chapter 12). As solutions to such emerging problems cannot be specified in advance, short of discontinuing a content analysis altogether, researchers may have to be prepared to go back and modify the defective parts of a research design (see Figure 4.2), keeping the overall research objective in mind. It is not unusual for a

content analysis to need several iterations of locating unreliabilities, correcting their causes, and repeating these steps until applicable standards are satisfied. Often researchers get stuck with coding instructions that turn out to be unreliable, and they introduce small modifications instead of radically reconceptualizing their approach. In addition to unreliabilities, new empirical findings and literature about the context of the analyzed texts can prompt reconstructions of the context an analysis was presupposing. Content analysts should not be afraid to go back to solid ground, as the issue always is to select potentially valid answers to the research question.

14.4 NARRATING THE RESULTS

Research proposals are written to convince sponsors, but research reports typically are addressed to other readers. Also, research reports usually go far beyond the mere statement of findings of fact. Such reports account for how analysts have accomplished what they set out to do; describe to which literature, judgments, or decisions the research results contribute; and raise questions for further exploration. In highly institutionalized settings—such as laboratories or public opinion polling organizations—where research questions are codified, researchers are well-known, and analytical procedures are well established, research reports may be limited to information about where the analyses deviated from the typical. Generally, a research report should offer details sufficient to convince at least three kinds of addressees of the importance of the results:

- The sponsor, agency, or client who approved and/or supported the research
- The content analysts' peers in the scientific community
- The public at large

These addressees may have conflicting agendas, which may have to be attended to separately.

Sponsors are interested, first, in whether the researchers fulfilled their contractual obligations. A research report needs to demonstrate that they did, and, where applicable, should also justify where and why they deviated from the original proposal. Second, funding agencies are keenly aware of the publicity they gain from having supported worthwhile research and often look for the social validity or political significance of the results. Political or commercial clients might be more interested in the benefits they can reap from the research, whereas academics may look for the knowledge advanced by the results. A research report may need to address these concerns. Third is the issue of empirical validity: Can the analysts' claimed results be trusted? Nonscientific users may be inclined to accept content analysis results on account of the analysts' scientific credentials or the reputations of the institutions with which they are associated. Users in the legal and scientific communities, by contrast, may approach a research report more critically, looking for and needing to find relevant details and evidence in support of findings.

For members of the scientific community, the ability to reproduce research results elsewhere is the most widely used standard. As the burden of proof is laid on the researchers, the research report must provide convincing evidence that the results can

be reproduced. Measures of the reliability of the potentially unreliable components of an analysis—recording/coding, for example—may provide the needed assurances. Many scholarly journals require evidence for the reliability and statistical adequacy of research findings. But reliabilities merely put an upper limit on the potential validity of research results, as discussed above.

Because the answers that content analyses yield are obtained through abductive inferences, analysts need to establish the validity of their inferences. They may accomplish this by retracing the analytical steps taken and justifying each step in terms of whether it models or represents what is known about the context of the texts. Analysts can use sampling, semantic, structural, and functional validities (see Chapter 13) to support such arguments, especially when their results seem counterintuitive. However, content analysts should expect that even plausible research results may be received with a healthy dose of suspicion. The face validity they perceive may not be clear to all those who are touched by the research findings, forcing the analysts to go the extra mile to provide arguments that are compelling, even for their critics, even if the validity of their findings may seem obvious within their community of peers.

When research results are published, they enter the conversations of diverse readers, if not public debate. To succeed in the view of the public, analysts may want to provide compelling narratives that emphasize the impacts their findings will have on the readers' lives, using concepts, comparisons, or metaphors that are meaningful in the readers' own worlds and not misleading.

14.4.1 Outline for a Research Report

Generally, a research report should contain the following parts (many of which are also found in the research proposal; see section 14.2.3):

- *A summary or abstract* of the research for decision makers who have little time to concern themselves with details (This part of the report often decides whether readers will continue with the remainder.)
- A table of contents
- A statement of the epistemic or methodological issues that informed the analysis
- *A review of the literature* on the context in which these issues reside
- *An account of the framework adopted* for the content analysis, including the research questions addressed (with an explanation of the sponsor's or analysts' interest in these questions), the texts analyzed to answer the questions, and the context chosen to justify the analysis (see Chapter 2)
- *A description of the research design* actually followed, including the preparatory research undertaken (see Chapter 4), any complications encountered in the process, and how emerging problems were solved, with specific information in the following areas to enable critical evaluation of the process:
 - The *body of texts* sampled: what it consists of, what motivated the analysts to choose it, by which strategy it was selected (see Chapter 6), and how the analysts dealt with biases (see Chapter 13, section 13.2.1)

- The *data language* used (see Chapter 8): the system of descriptive categories and measurements the analysts employed to bridge the gap between raw texts and the computational techniques applied
- The *units of analysis* (see Chapter 5): their operational definitions, how they were used in the process, and what they preserved and ignored from the texts
- The *recording/coding process*: whether built into computer dictionaries or search queries (see Chapter 11) or enacted by human coders (see Chapter 7), the reliability (see Chapter 12) and, where evaluated, the semantic validity (see Chapter 13, section 13.2.2) of each variable
- The *computational* (statistical or algebraic) *techniques employed to summarize, simplify, or reduce* the volume of records obtained from the body of texts
- The *inferential techniques* (computer programs [see Chapter 11] or other analytical procedures [see Chapter 9]) used to answer the research questions and, where available, evidence of structural validity (see Chapter 13, section 13.2.3) or functional validity (section 13.2.4)
- The *research results (the answers to the research questions)* in the form of data files, summaries, statistical accounts, propositions of a factual nature, recommendations for actions, or judgments (all in view of their potential validity), crafted in a compelling narrative that the anticipated readers of the report can easily understand
- A *self-critical appraisal* of the analysis (Did it really yield something new?), of the time and resources spent (Was it worth the effort?), of the methods used in relation to other methods (Could there have been more appropriate techniques?), of the computational procedures used (Did they accomplish what was expected of them?), and of the meta-accomplishments of the analysis (Did it contribute to the content analyst's field? For example, did it raise new questions for further explorations?)
- Additional matter
 - The complete *list of references* to cited literature
 - *Appendices* containing materials that enable interested readers to read beyond the report, such as the recording instructions, computer dictionaries used, the list of reliabilities obtained, and tables of numerical findings, even data that could be used by other researchers (Scientific research is open, and data may need to be made available for reanalysis elsewhere.)
 - *Acknowledgments* of contributors to the research effort (All research projects proceed in networks of interpersonal relations. Because assistants, coders, consultants, advisers, librarians, and teachers do not expect to be acknowledged in official research reports, it is a gesture of generosity when researchers name those who have been involved.)

GLOSSARY

Abductive inference: The process of proceeding from true propositions in one logical domain to propositions in another logical domain, believed to be true on account of presumed empirical relationships between them—that is, from particulars to particulars without generalizations governing both. (2.4.5)

Accuracy: One of three kinds of *Reliability*. Accuracy is a measure of the extent to which a data-making instrument (e.g., a *Coding* instruction, measuring device, observational accounting practice) produces data that are accurate relative to a trusted standard. *Standards* may be established by a panel of experts, a proven method, or a norm. If the standard is considered true, accuracy measures the extent to which the generated data are valid. (12.2.1)

Alpha (α): A generalized coefficient of agreement used to assess the *Reliability* of *Unitizing*, *Coding*, and *Data*. Alpha measures the agreement among independent *Observers*, coders, or analysts relative to what can be expected by chance, that is, the agreement that would be observed when they randomly assign the values from a population estimate of values to given *Units* or randomly segment a given continuum into units of different lengths and kinds. It is applicable to any number of observers, all standard *Metrics*, missing data, and small *Sampling* sizes. Alpha generalizes several special-purpose coefficients but goes far beyond either of them. (12.2.3)

Analytical construct: An operationalization (formalization) of the content analyst's knowledge of how *Text* is used in a chosen *Context*. It is the ground on which *Abductive inferences* from given text to unobserved features of content analytical interests are justifiable. An analytical construct is akin to a computable model of the network of stable correlations between available texts and the analyst's *Research question*. (2.4.4; 4.2.1; 9)

Association: At least two kinds are distinguished: (a) co-occurrences of words, concepts, or symbols from which cognitive proximities can be inferred (10.6; 11.4.3), and (b) linguistically expressed connections between attitude objects considered either associative (e.g., is, has, likes, belongs) or dissociative (e.g., is not, dislikes, opposes) in degrees. (9.2.3; 11.4.3)

Association structures: Sets of objects, concepts, or symbols plus their *Associations* of various strengths between all pairs (11.4.5); not to be confused with

Semantic networks, whose relations have diverse meanings. (10.5; 11.4.4)

Attention: A common *Inference* from the amount or relative frequency of a source's coverage of an issue.

Attribution: The use of language, including adjectives and stories, in declaring objects, events, or people to be of a certain kind or to have certain qualities. (10.5)

CATA: See Computer-aided text analysis.

Categorizing: The process of grouping ideas, objects, and data into mutually exclusive sets. In *Content analysis*, categorizing reduces a diversity of *Recording/coding units* into convenient kinds—the values of a nominal *Variable*.

Cluster sample: A sample that follows a hierarchical conception of the population, proceeding from general to subordinate *Units*; for example, *Sampling* newspapers from a list of all major newspapers of a country, then sampling particular issues from the newspapers sampled, then sampling particular articles from the issues sampled, and so on. At each step, all units available on that level should have the same chance of being included. (6.2.5)

Coder: See *Observer*.

Coding: The process of *Categorizing*, describing, evaluating, judging, or measuring descriptively undifferentiated *Units* of analysis, thereby rendering them analyzable in well-defined terms. (12.2.3)

Coding instructions: Instructions given to *Observers* to standardize processes of *Recording*, categorizing, or valuing phenomena of analytical interest. They enable several independent observers to generate reliable (*Reliability*) data. And they are instrumental for data analysts and the stakeholders in research results to infer from the data what observers saw, read, and recorded.

Coefficient of imbalance: A measure of the degree to which favorable attributes of an issue outweigh unfavorable ones. (3.3.2)

Coincidence matrix: A tabulation of the pairwise coincidences of categories or numerical values assigned to each of a given set of units by any number of *Observers*. Rows and columns of this matrix bear the names of the categories or values used. The matrix is square and symmetrical. Its diagonal cells

Note: Numbers in parentheses within and following glossary entries refer to sections or whole chapters of this volume in which the concepts are discussed in depth.

contain perfect matches and its off-diagonal cells the disagreements. Its entries sum to the number of values paired. It visualizes the *Reliability* of the *Data* it tabulates. (12.3.2; 12.4.4) Not to be confused with a *Contingency matrix*.

Computer-aided text analysis (CATA): *Text analysis* in which the analytical task is divided between humans doing what they do best and computers employed for what they do better: searching through huge bodies of literature and processing character strings at high speeds but with limited ability to identify meanings. (11.4)

Concordance: A literary account of words, phrases, and their contexts and locations in the work of an author. Software designed for creating concordances has been used in *Content analysis* as well, especially in surveying the vocabulary of a body of *Texts* in view of their contextual meanings.

Content: Something that, according to a popular metaphor, is contained in messages or *Texts* regardless of who reads them. Content analysts avoid the use of this content-objectifying metaphor by acknowledging both the analysts' informed attention to available texts as well as the many worlds of readers, writers, and users for whom the analyzed texts could have all kinds of meanings. (2.2)

Content analysis: A *Research* technique for making replicable and valid *Inferences* from *Texts* (or other meaningful matter) to the *Contexts* of their use. (for contested definitions, 2.1)

Context: Generally, the world in which a body of *Text* plays a role or has meanings to those attending to it. In *Content analysis*, one distinguishes the many worlds of others in which texts play various roles, from which texts are sampled (decontextualized) for analysis, and the world of the analyst, in which particular *Research questions* are posed and answered with the help of available texts. The context of the analyst need not coincide with the worlds of others; however, its construction must be justifiable to the content analyst's peers and users of the content analysis results. (2.4.3) Most specifically, the linguistic context of a word is its textual environment, which enables readers to make context-specific sense of that word.

Context units: Textual matter that is defined to limit what *Observers* or readers are to consider when *Categorizing*, or describing *Recording/coding units*. The size of context units is important. If allowed to be too large, coding tends to become unreliable. If defined too small or not at all, recording units may lose the information that stems from the textual environment of these units. For example, personal pronouns are meaningless without reference to the contexts in which they occur. (5.2.3)

Contingency analysis: Analysis of the co-occurrences of words, concepts, or symbols, especially relative to their co-occurring by chance, in a body of *Text* to infer the *Association structures* of authors or readers of that text. (10.6)

Contingency matrix: A cross-tabulation of observed *Units* of analysis created in order to analyze statistical dependencies between two categorical or nominal *Variables* describing these units. Contingencies tend to be analyzed by association statistics, such as χ^2 or product-moment correlations. When *Reliability* data are recorded in contingency matrices, these matrices are square, rows and columns refer to different *Observers*, perfect matches of categories occur in their diagonal cells, and systematic mismatches occur in their off-diagonal cells. However, analyzing reliability with association statistics would be a mistake, as reliability requires observers to be interchangeable, and statistical dependencies among the proclivities of individual observers are misleading indicators of reliability. The entries of contingency matrices sum to the number of units described. (12.3.4.2; 12.3.5) A contingency matrix must not be confused with a *Coincidence matrix*, whose entries sum to the values assigned to units and is used in evaluating reliability. (12.3.2; 12.4.4)

Conversation: Fundamentally open and self-governed human interactions that rely, but not exclusively, on linguistic utterances or *Texts*, each responding to preceding utterances or texts. The prototypical conversation is interpersonal. However, content analysts have extended the notion of conversation when making *Inferences* from diplomatic exchanges, deliberations in small groups, and therapeutic sessions regarding how relationships develop interactively and interactive processes unfold. (3.6)

Crowdcoding: A version of crowdsourcing that invites anonymous Internet users to code or evaluate texts for subsequent analysis. Besides savings in cost and time to train observers to generate analyzable data, crowdcoding is often justified by appealing to the wisdom of crowds, the belief that collective judgments are better than individual ones (7.4)—tested by the *Decisiveness* of majorities (12.4.3)

Data (plural of datum): Semidurable records taken as the unquestioned basis for reasoning, discussion, or calculation. As such, data are not found but generated at least by declaration more typically by experiments, surveys, or measurements. Data must show some diversity (convey information—hence the plural of *datum*), must be comparable with each other, and, in the case of scientific data, must be obtained by replicable methods or be validatable. The data of *Content analyses* are *Texts*. (4.1; 8)

Data language: The formal organization of *Data*—for example, a system of categories and measurements.

A data language has a syntax and a semantics. Its syntax must render the data amenable to a proposed analysis (e.g., be computable). Its semantics links the data to the phenomena they are thought to represent, often formalized in *Coding* instructions that specify the relationship between each datum and what *Observers* saw or read, or embodied in measuring instruments. [8]

Data reduction: Amounts to simplifying typically large volumes of *Data*—for example, by counting, abstracting, computing indices, offering statistical accounts of analytical results, or describing a statistical distribution by its parameters. In more qualitative approaches, data reduction means selecting representative quotes or prototypical examples from the analyzed *Text*, summarizing or abstracting an analyst's reading. All content analyses start with volumes of texts and reduce them to the answers to *Research questions*. [4.1.1; 10]

Decisiveness of majorities or aggregates: One of four *Reliability* measures that assesses the degree to which individual observer's judgments, evaluations, or descriptive accounts of phenomena deviate from their majority or average. It is based on data obtained by replication of data generating processes, but unlike *Replicability*, it answers the question of whether a majority can be trusted more so than individual observers. [12.4.4]

Deductive inference: The process of proceeding from a general proposition, considered true, to particular propositions that are logically implied and are therefore considered true as well. [2.4.5]

Dictionary (CATA): A sometimes user-specifiable computational tagging of single or compound words in a body of *Text*. Tags have meanings that are simpler and more general than those occurring in the text, ideally preserving the meanings that are relevant for answering a given *Research question*. CATA dictionaries make use of thesauri, *Stemming*, *Lemmatization*, and disambiguation rules that rely on the linguistic environment of tagged *Units* of text. They aim at *Data reduction* by providing frequencies of tags in text. The relationship of tags to what they tagged demonstrates their *Semantic validity*. [11.4.1]

Difference functions: In the calculation of the agreement coefficient *Alpha*, difference functions weigh the frequencies of coinciding values by the *Metric* differences between these values. A difference function can be stated mathematically or tabularly. In matrix form, a difference function is defined by a list of all differences between two sets of categories, ranks, values, [12.3.4] or sets of values. [12.7.2]

Direct/indirect indicators: *Analytical constructs* in the form of correlations between textual attributes and the phenomena of interest are called direct indicators

(see *Index*), and those that take the form of a network of correlates, including intervening *Variables*, are called indirect indicators. The history of *Content analysis* shows a move from direct to indirect indicators of the answers to *Research questions*. [9.3.3]

Evaluative assertion analysis: The use of two-valued predicates to infer the valuation of an object from how it is linguistically associated (see also *Association*). [9.2.3]

Extrapolations: Inferences about the unobserved gaps between observations, such as between data points (interpolations), or beyond a recorded history of observations (predictions). Extrapolations assume the continuity of observed phenomena, trends, patterns, or differences. [3.2; 9.3.1]

General Inquirer: The historically first operational dictionary approach to computational *Content analysis*; accepts several *CATA Dictionaries*. [11.4.1]

Go-words: A list of words to be included in an analysis. [11.3.1]

Index: A *Variable* that is computed on textual attributes for its correlation with extratextual phenomena or variables of interest. An index empirically relates logically distinct kinds and may therefore be used to infer particular phenomena or compare several *Texts* in their own terms. [3.4; 9.3.3]

Inductive inference: The process of proceeding from particular propositions, such as a sample of observations, to general propositions, such as to statistical generalizations of that sample, accounting for these observations in most if not all respects. [2.4.5]

Inference: The process of passing from accepted propositions, statements, or data to other propositions or statements whose validity is believed to be preserved in that process. One can distinguish *Deductive*, *Inductive*, and *Abductive inferences*. In *Content analysis* inferences are typically abductive, proceeding from available *Text* to the analyst's *Research question*, which pertains to unavailable features of the analyst's chosen *Context*. [2.1; 2.4.5]

Institutions: Habitual social practices that are enacted within a community and serve normative functions for how members organize themselves and construct the realities they live by. Content analysts may infer how the exchange of textual matter encourages, constructs, or undermines particular institutional realities. [3.7]

KWIC (keyword in context) lists: The tabulation of character strings of a specified length, within which selected keywords or phrases occur in a *Text*. KWIC lists consist of the contexts in which chosen words or phrases occur in analyzed texts suggesting their meanings or roles they play within these texts. [11.3]

Latent content: See its opposite, *Manifest content*. [2.1]

Lemmaization: The process of *Stemming* a word to its stem or root form after considering the grammatical function of the word in a sentence. [11.3; 11.4.1]

Levels of measurement: See *Metric*.

LexisNexis: A company that makes probably the largest English-language electronic text archive available to subscribers. Lexis provides a searchable database of U.S. laws and legal decisions. Nexis provides access to published data from more than 20,000 multilingual news sources, journals, biographical and reference materials, business reports, and regulatory findings.

LIWC (Linguistic Inquiry and Word Count): A *CATA Dictionary* approach. [11.4.1]

Manifest content: *Texts* that are easy to read, generally understood, unambiguously interpretable, and therefore yield high agreement, even among untrained coders. Early definitions of *Content analysis* sought to exclude *Latent content* for its methodological difficulties, requiring experts to identify and code. [2.1]

Matrix: A rectangular array of numbers or symbols, arranged in rows and columns, often summed to its marginal totals. (See *Coincidence matrix*, *Contingency matrix*, and *Difference functions*.)

Meme: A unitary idea, message, behavior, or style that spreads throughout a community of minds. Analogous to genes, which transmit biological information, memes transmit cultural information and are thought to affect the content and organization of human minds. [11.4.5]

Metric: An explicit ordering of the values of a *Variable*, distinguished by the kinds of operations on these values that preserve their ordering [8.7]. Most common are binary, nominal, ordinal, interval, and ratio metrics, also called *levels of measurement* [8.6]. Analytical methods tend to make specific demands on the metric of the data they can process. Conversely, the metric of given data limits the choices among analytic/computational techniques available. In the analysis of reliability, metrics are operationalized in the form of *Difference functions*.

Networked texts: Whereas writing is continuous, and many social phenomena, ranging from speech to historical developments, proceed linearly, hypertext can be read in sequences at their readers' digressions, and users of social media generate large volumes of messages and dynamically link them across many. All posts by Internet users respond to previously received messages and acquire their meanings by what happens after they are posted, commented on, reproduced, and acted upon. Individual posts are short and quite uninformative of what

happens. Social media texts are networked, and their content analyses cannot ignore the contexts of their connections. [2.3; 11.3.4]

Observer: A generic term for a coder, unitizer, transcriber, judge, rater—anyone who has access to undifferentiated phenomena and transforms them into analyzable *Data*. (qualifications, 7.2; training, 7.3; reliability, 12.2)

Plagiarism: An established resemblance of a work by one author and a previously published work by another, controlled for common cultural backgrounds or joint exposure to events. [10.5]

Precision: A measure of the extent to which the items retrieved by a search algorithm are relevant to a research question. This proportion is extended to a reliability coefficient for text retrieval. [12.5]

Propaganda analysis: In *Content analysis*, the term applies to three different conceptions: (a) Using printed matter and speeches to identify propagandists in pursuit of doctrinaire intents or in the unacknowledged service of a foreign government. Keys to this identification are the use of devious persuasion techniques. (b) Making *Inferences* from an enemy country's domestic broadcasts about planned military actions, political support of the governing elite by its population, prediction of leadership succession, and opposition to the government. (c) Assessing the effectiveness of one's own propaganda efforts. [1.4]

QDA software: See *Qualitative data analysis software*.

Qualitative content analysis: Conventionally contrasted with *Quantitative content analysis*—involves a close reading of textual matter, reorganizing relevant parts of it into analytical categories, and creating interpretations, narratives of scholarly interest relating to the meanings and uses of the analyzed *Text*. Any interpretation of a text is an *Inference* that should have *Semantic validity*—that is, it must be plausibly related to the original text. To demonstrate that connection, qualitative content analysts prefer to support their inferences by quoting from the analyzed text. Qualitative content analysts are not opposed to counting, but their inferences rarely rely on frequencies. [1.7]

Qualitative data analysis (QDA) software: Software that provides a means to efficiently manage bodies of *Text* as well as graphical, audio, and video data in the pursuit of text-driven *Qualitative content analysis*. It accounts for analyst-introduced unitizing, coding, annotating, retrieving, sorting, counting, and graphing of various relationships among textual elements. [11.5]

Quantitative content analysis: Conventionally contrasted with *Qualitative content analysis*—relies

heavily on enumerating coded textual matter. However, all *Coding of Text* involves qualitative judgments or identifications, so that the distinction between qualitative and quantitative approaches is largely one of emphasis, often falsely identified with being interpretive versus scientific. (2.1)

Quantitative newspaper analysis: A precursor to *Content analysis* developed around the end of the 19th century. It measured the amount of space in newspapers devoted to various subject matters, largely to objectify and criticize developments in newspaper publishing at that time. (1.2)

Random sample: A sample of *Units* of analysis selected from a known population where each unit has the same probability of being included. (6.2.1)

Recall: A measure of the extent to which a search algorithm retrieves all relevant items contained in a database. This proportion entails an epistemological paradox as the sole motivation for using search algorithms, which is the human inability to know what a data base contains. (12.5)

Recording: Creating somewhat durable records from transient or otherwise unanalyzable phenomena. Recording ranges from the use of mechanical devices (voice recorders) and transcriptions to *Unitizing* and *Coding of Text*. Durability is a prerequisite of all re-search efforts and *Data* in particular, which moreover require analyzability by available techniques. (7)

Recording/coding units: Identified within unanalyzed *Text*, recording units are independently describable, transcribable, categorizable, or codable units. Each recording or coding unit becomes represented by an enumerable record, code, or datum (see *Data*). To include information from the textual surroundings of such units without abandoning their analytically required independence, content analysts often ask observers to describe them as a function of *Context units* within which they occur. (5.2.2)

Relational content analysis: A form of content analysis that seeks to set itself apart from categorical accounts of textual elements, focusing on relations, contingencies, or semantic connections between them. It underlies *Contingency analysis*, *Evaluative assertion analysis*, *Semantic networks*, and *Webgraphs*.

Reliability: The attribute of *Data* on which researchers can rely in answering their *Research questions*. Unreliability results from two factors: disagreements among independent ways of generating data about the phenomena in question and lack of information (variation) in the resulting data to answer given research questions about these phenomena. In *Content analysis*, the first is assessed by agreement coefficients, such as *Alpha*, applied to how several *Observers* unitize the same continuum

and/or record, describe, or code the same set of *Units*. The second becomes evident by tracing the information in data throughout an analytical process to its results. (When variation is absent, data cannot be correlated with anything; and when undetected noise in the data is not eliminated during an analysis, the answers to *Research questions* may well be unrelated to the phenomena of interest.) Four kinds of reliability are distinguished: *Stability*, *Replicability*, *Accuracy* or *Surrogacy*, (12.1) and *Decisiveness of majorities*. (12.4) High reliability is a prerequisite of *Validity* but does not guarantee it. (12.1)

Replicability: One of four kinds of *Reliability*. Replicability measures the extent to which data-making instruments (*Coding* instructions, measuring devices, or observational methods) can be relied upon to generate the same *Data* from the same set of phenomena in diverse circumstances, employing different and independently working *Observers*, coders, judges, or measuring instruments. It is a function of the degree to which independently obtained data of the same phenomena agree. (12.2.1)

Re-presentation: A text-invoked recall of something previously experienced, making that experience present again, as distinct from *representation*, which suggests some objective relationship between *Text* and what it is about. (3.5)

Research or re-search: A repeated and replicable search within *Data* for patterns that explain them. Content analysts move outside such searches by relying on identifiable properties in *Texts* to answer *Research questions* pertaining to the *Contexts* of these texts.

Research design: The network of procedural steps a researcher plans to take to proceed from generating *Data* to research results. Research designs should be explicit so that they can be replicated or critically evaluated for the conclusiveness of the findings. (4; 14.1)

Research questions: In *Content analysis*, research questions must satisfy three requirements: (a) They must pertain to currently unknown phenomena in the *Context* of analyzed *Texts*, (b) they must allow several possible answers among which the analysis selects, and (c) they must provide the possibility—even if only in principle—of alternative ways to answer the chosen question. The requirement (a) is implicit in the definition of content analysis, (b) prevents researchers from merely generalizing from or confirming what they already know, and (c) assures that the answers of research questions are validatable in principle, by means other than a content analysis, subsequent observations, correlations with related variables, or successful actions based on the results. (2.4.2; 14.1.2.1)

Sampling: The process of selecting a representative sample from a larger population of *Units* so that

an analysis of the sample enables the researcher to draw conclusions about that population. The *Inference* involved is inductive—not deductive or abductive. {6}

Sampling units: Units distinguished for selective inclusion in an analysis. {5.2.1}

Sampling validity: The assurance that a sample is large enough to represent a population of phenomena in its composition so that the sample can be studied in place of that population. Because *Content analysis* makes *Inferences* from *Texts* to nontextual phenomena, two populations are important: (a) the population of texts actually sampled and analyzed, and (b) the population phenomena to be inferred from the sampled texts. Sampling validity is assured when texts are sampled so that the phenomena of interest are fairly represented in the sample. {13.2.1}

Semantic differential scales: Seven-point scales between polar-opposite attributes (good-bad, strong-weak, active-passive), usually labeled from -3 to +3, and widely used in assessments of sources' evaluations of word meanings or attitude objects referred to. {7.4.4}

Semantic networks: Sets of objects, concepts, or symbols connected by linguistically meaningful binary relations, such as two-place predicates {11.4.4}; not to be confused with *Association structures*, whose binary relations are merely of varying strengths.

Semantic validity: One *Validity* criterion unique to content analysis, measuring the degree to which relevant readings of a *Text* are preserved in processes of *Unitizing*, *Coding*, and analyzing the text. {13.2.2}

Sentiment analysis: A computational *Content analysis* using a dictionary of pleasant-unpleasant, active-passive, and easy to imagine-hard to imagine attributes of selected attitude objects or issues. It is intended to infer their emotional value. {11.4.1}

Snowball sample: A multistage sample obtained by examining an initial sample of *Units* for leads to other *Sampling units* and so on, until some practical closure is reached. {6.2.6}

Stability: One of four kinds of *Reliability*. Stability measures the extent to which a researcher or data-making instrument repeatedly generates the same *Data* from the same set of phenomena. Stability is not sufficient to establish the reliability of content analysis data, as consistent biases, prejudices, or misunderstandings of *Coding* instructions would not become manifest in repetitions. *Replicability*, which can detect individual idiosyncrasies, provides a stronger measure. {12.2.1}

Standards: Established rules, principles, or measures against which deviations can be measured. *Reliability* and *Validity* are two standards against

which the quality of a *Content analysis* is judged {12.1}. Reliability determines the degree to which data represent the phenomena to be analyzed and validity the extent to which its inferences are borne out by independent evidence. Content analysis may also use standards in its *Research questions*, for example, in establishing *Plagiarism*, to evaluate biases (*Coefficient of imbalance*), or to judge an account as true or false. {3.3; 9.3.2}

Stemming: The process of reducing grammatical variations of words to their common stems, bases, or root forms, used to simplify *CATA Dictionaries*. {11.3}

Stop-words: A list of words to be excluded from an analysis, largely because they are judged irrelevant to the *Research question*. {11.3.1}

Stratified sample: A sample whose *Units* are selected (randomly or systematically) from several known subpopulations (strata) in a population so that each subpopulation is represented in the sample either equally or in proportion to its size. {6.2.3}

Surrogacy: The degree to which a measure or model can serve as a surrogate of other phenomena. It is computationally equivalent to *Accuracy*, one of four forms of reliability. {12.2.1} {12.4}

Systematic sample: A sample that consists of every *kth Unit* from a serial population or list of textual units. {6.2.2}

TAT: See Thematic Apperception Test.

Text analysis: Provides accounts of the character strings, compositions, coincidences, syntactical structures, and layouts of given *Text*. Word counts and *KWIC* lists are basic. *Text mining* and *Webgraphs* make use of search engines and more complicated algorithms for analyzing texts. Computer text analysis has little place for meanings and leaves the burden of relating its findings to the given *Research question* to the content analyst. {11.3}

Text mining: The largely computerized search and identification of usually rare textual elements in a large bodies of *Texts*. The textual elements searched and found should not be confused with information that could answer the *Research questions* content analysts ask. Although there are occasions where names of individuals, places, or concepts lead to such answers, the latter always requires human judgments. {11.3.2}

Texts: Anything variable and textured that has meaning to somebody, including to the analyst, and can be examined or read, interpreted, and acted upon repeatedly: letters, e-mails, blogs, literature, images, video recordings, transcripts of conversations, political speeches, historical records, police reports, accounts of legal proceedings, corporate statements, advertisements, medical documents,

photographic images, posters, cultural artifacts, and so on. [2.4.1]

Thematic Apperception Test (TAT): A psychological test involving the content analysis of interpretations of ambiguous images by subjects in order to reveal hidden attitudes, prejudices, motives, and personality variables.

Trends: See *Extrapolations*.

Unitizing: Identifying contiguous and nonoverlapping sections within a descriptively undifferentiated continuum of *Text* or other spatially organized matter, typically generating *Recording* or *coding units*—words, sentences, utterances, rhetorical moves, metaphors, themes, images, anything—that may thereafter be coded whole and subsequently analyzed. [5.3; 12.2.3]

Units: Decontextualized but information-bearing textual wholes that are distinguished within an otherwise undifferentiated continuum and thereafter considered separate from their context and independent of each other [5.1]. Content analysts distinguish three kinds of units by the functions they serve within an analysis: *Sampling*, *Recording* or *Coding*, and *Context units* [5.2]. Units may be distinguished physically [5.3.1], syntactically [5.3.2], categorially [5.3.3], propositionally [5.3.4], and thematically [5.3.5].

Units of enumeration: The units that are actually counted or measured. Although *Recording/coding units* may be described numerically, for example, in terms of their size, their location, or their statistical properties, enumeration typically follows *Coding*, and units of enumeration often coincide with *Recording/coding units*. [5.2.3]

Unobtrusiveness: The attribute of a *Research* method that uses *Data* about which their sources or object of study have no knowledge, and the collection of which does not affect these sources or objects. *Content analysis* is mostly unobtrusive, as it relies on *Text* not generated for the purpose of analysis. By contrast, survey research and psychological experiments are obtrusive, as asking questions or instructing subjects affects those studied. [2.5]

Unstructuredness: The attribute of phenomena that appear in a form that available analytical techniques cannot (yet) handle. Unstructured phenomena may

be transient, as is human speech, or possess properties that are variable and complex, as in most human communications. Much of content analysis involves transforming unstructured matter into analyzable *Text*. [2.5]

Validating evidence: *Data* that substantiate analysts' results by means other than those used in the analysis. In *Content analysis*, validating evidence is largely ex post facto, as its *Research questions* are posed in the very absence of direct evidence for what it seeks to infer from available *Text*. [2.4.6]

Validity: The quality of a claim to be as stated, true, or correct. A *Content analysis* is valid to the extent the *Inferences* about the *Context* of the analyzed *Texts* withstand the test of independently obtained *Validating evidence*. [13]

Variable: A conceptual unity that embraces alternatives, something that can vary. Variables are essential to any *Data language*. Open variables are placeholders for *Units* as observed or found in *Text*. Closed variables have a predefined range of or consist of a list of conceptually possible values [8.3]. The values of a variable may be unordered [8.4], or ordered variously [8.5], and possess diverse *Metrics* [8.6].

Varying probability sample: A sample whose *Units* are selected according to known probabilities of their relevance to a *Research question*. It compensates for known inequalities in the units' informativeness, importance, or self-sampling biases. [6.2.4]

Webgraphs: Graphical depictions of the links between Web pages available on the Internet. Inasmuch as links are established by website designers for directing readers' attention to other relevant matter, they represent potential traffic and networks of meanings within communities of users. [11.3.4]

WordNet: A lexical database for the English language. It groups English words into sets of synonyms, similar to a thesaurus, provides short definitions for these sets, and orders them according to the semantic relations between them. Its organization is based on common uses of language. It supports *Text mining* and computational linguistics approaches to information processing and has been used in the development of content analysis instruments. [11.3.2]

REFERENCES

- Abell, Peter. (1987). *The syntax of social life: The theory and method of comparative narratives*. New York, NY: Oxford University Press.
- Abell, Peter. (1993). Some aspects of narrative method. *Journal of Mathematical Sociology*, 18, 93–134.
- Abelson, Robert P. (1963). Computer simulation of hot cognition. In Silvan S. Tomkins & Samuel Messick (Eds.), *Computer simulation of personality* (pp. 277–298). New York, NY: Wiley.
- Abelson, Robert P. (1968). Simulation of social behavior. In Gardner Lindzey & Elliot Aronson (Eds.), *The handbook of social psychology* (pp. 274–356). Reading, MA: Addison-Wesley.
- Abelson, Robert P., & Rosenberg, Milton J. (1958). Symbolic psychologic: A model of attitude cognition. *Behavioral Science*, 3, 1–13.
- Adorno, Theodor W. (1960). Television and the patterns of mass culture. In Bernard Rosenberg & David M. White (Eds.), *Mass culture* (pp. 474–488). New York, NY: Free Press.
- Ahn, Luis von; & Dabbish, Laura. (2004). Labeling images with a computer game. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 319–326.
- Albig, William. (1938). The content of radio programs 1925–1935. *Social Forces*, 16, 338–349.
- Albrecht, Milton C. (1956). Does literature reflect common values? *American Sociological Review*, 21, 722–729.
- Alexa, Melina. (1997). *Computer-assisted text analysis methodology in the social sciences* (Arbeitsbericht 7). Mannheim, Germany: Zentrum für Umfragen, Methoden und Analysen.
- Alexa, Melina, & Züll, Cornelia. (1999). *A review of software for text analysis* (Nachrichten 5). Mannheim, Germany: Zentrum für Umfragen, Methoden und Analysen.
- Allen, Liska E. (1963). Automation: Substitute and supplement in legal practice. *American Behavioral Scientist*, 7, 39–44.
- Allport, Gordon W. (1942). *The use of personal documents in psychological science*. New York, NY: Social Science Research Council.
- Allport, Gordon W. (Ed.). (1965). *Letters from Jenny*. New York, NY: Harcourt Brace Jovanovich.
- Allport, Gordon W., & Faden, Janet M. (1940). The psychology of newspapers: Five tentative laws. *Public Opinion Quarterly*, 4, 687–703.
- Alpaydin, Ethem (2004). *Introduction to machine learning*. Cambridge, MA: MIT Press.
- Altheide, David L. (1987). Ethnographic content analysis. *Qualitative Sociology*, 10, 65–77.
- Alwin, Duane F. (1974). Approaches to the interpretation of relationships in the multitrait-multimethod matrix. In Herbert L. Costner (Ed.), *Sociological methodology 1973–1974* (pp. 79–105). San Francisco, CA: Jossey-Bass.
- American Educational Research Association, American Psychological Association, and National Council on Measurement in Education. (1985). *Standards for educational and psychological testing*. Washington, DC: American Psychological Association.
- American Educational Research Association, American Psychological Association, and National Council on Measurement in Education. (1999). *Standards for educational and psychological testing*. Washington, DC: American Psychological Association.
- American Educational Research Association, American Psychological Association, and National Council on Measurement in Education. (2014). *Standards for educational and psychological testing*. Washington, DC: American Psychological Association.
- American Psychiatric Association. (2000). *Diagnostic and statistical manual of mental disorders* (4th ed., rev.). Washington, DC: Author.
- American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders* (5th ed., rev.). Washington, DC: Author.
- American Psychological Association. (1954). Technical recommendations for psychological tests and diagnostic techniques. *Psychological Bulletin*, 51(Suppl. 2), 200–254.
- Andsager, Julie L., & Powers, Angela. (1999). Social or economic concerns: How news and women's magazines framed breast cancer in the 1990s. *Journalism & Mass Communication Quarterly*, 76, 531–550.
- Armstrong, Robert P. (1959). Content analysis in folkloristics. In Ithiel de Sola Pool (Ed.), *Trends in content analysis* (pp. 151–170). Urbana: University of Illinois Press.

- Arnheim, Rudolf, & Bayne, Martha C. (1941). Foreign language broadcasts over local American stations. In Paul F. Lazarsfeld & Frank N. Stanton (Eds.), *Radio research 1941* (pp. 3–64). New York, NY: Duell, Sloan & Pearce.
- Aron, Betty. (1950). The Thematic Apperception Test in the study of prejudiced and unprejudiced individuals. In Theodor W. Adorno; Else Frenkel-Brunswick; Daniel J. Levinson; & R. Nevitt Sanford, *The authoritarian personality*. New York, NY: Harper.
- Ash, Philip. (1948). The periodical press and the Taft-Hartley Act. *Public Opinion Quarterly*, 12, 266–271.
- Asheim, Lester. (1950). From book to film. In Bernard Berelson and Morris Janowitz (Eds.), *Reader in public opinion and communication* (pp. 299–306). New York, NY: Free Press.
- Atkinson, J. Maxwell, & Heritage, John. (Eds.). (1984). *Structures of social action: Studies in conversation analysis*. Cambridge, MA: Cambridge University Press.
- Averill, James R. (1985). The social construction of emotions with special reference to love. In Kenneth J. Gergen & Keith E. Davis (Eds.), *The social construction of the person* (pp. 89–107). New York, NY: Springer-Verlag.
- Baldwin, Alfred L. (1942). Personal structure analysis: A statistical method for investigating the single personality. *Journal of Abnormal and Social Psychology*, 37, 163–183.
- Bales, Robert F. (1950). *Interaction process analysis*. Reading, MA: Addison-Wesley.
- Barcus, Francis E. (1959). Communications content: Analysis of the research 1900–1958: A content analysis of content analysis. Unpublished doctoral dissertation, University of Illinois.
- Barnett, George A., & Doerfel, Marya L. (1997, May). *A semantic network analysis of the International Communication Association*. Paper presented at the 47th Annual Meeting of the International Communication Association, Montreal, Canada.
- Barton, Allen H. (1968). Bringing society back in: Survey research and macro-methodology. *American Behavioral Scientist*, 12(2), 1–9.
- Bateson, Gregory. (2000). *Steps to an ecology of mind*. New York, NY: Basic Books.
- Bauer, Christian, & Scharl, Arno. (2000). Quantitative evaluation of Web site content and structure. *Internet Research*, 10, 31–41.
- Baxter, Leslie A., & Montgomery, Barbara M. (1996). *Relating: Dialogues and dialectics*. New York, NY: Guilford Press.
- Becker, Howard P. (1930). Distribution of space in the *American Journal of Sociology*, 1895–1927. *American Journal of Sociology*, 36, 461–466.
- Becker, Howard P. (1932). Space apportioned forty-eight topics in the *American Journal of Sociology*, 1895–1930. *American Journal of Sociology*, 38, 71–78.
- Bengston, David N., & Xu, Zhi. (1995). *Changing national forest values: A content analysis* [Research Paper NC-323]. St. Paul, MN: U.S. Department of Agriculture, Forest Service, North Carolina Forest Experimentation Station.
- Bennett, Edward M.; Alpert, R.; & Goldstein, A. C. (1954). Communications through limited response questioning. *Public Opinion Quarterly*, 18, 303–308.
- Benoit, William L.; Blaney, Joseph R.; & Pier, P. M. (1998). *Campaign '96: A functional analysis of acclaiming, attacking, and defending*. Westport, CT: Praeger.
- Berelson, Bernard. (1949). What “missing the newspaper” means. In Paul F. Lazarsfeld & Frank N. Stanton (Eds.), *Communications research 1948–1949* (pp. 111–129). New York, NY: Harper Brothers.
- Berelson, Bernard. (1952). *Content analysis in communications research*. New York, NY: Free Press.
- Berelson, Bernard, & Lazarsfeld, Paul F. (1948). *The analysis of communication content*. Chicago, IL: University of Chicago Press.
- Berelson, Bernard, & Salter, Peter J. (1946). Majority and minority Americans: An analysis of magazine fiction. *Public Opinion Quarterly*, 10, 168–190.
- Berelson, Bernard, & Steiner, George A. (1964). *Human behavior: An inventory of scientific findings*. New York, NY: Harcourt Brace Jovanovich.
- Berger, Peter L., & Luckmann, Thomas. (1966). *The social construction of reality: A treatise in the sociology of knowledge*. Harmondsworth, England: Penguin.
- Berkman, Dave. (1963). Advertising in *Ebony* and *Life*: Negro aspirations vs. reality. *Journalism Quarterly*, 40, 53–64.
- Bermejo, Fernando. (1997). *Press self-presentation: Self-reference and reflexivity in newspapers*. Unpublished master's thesis, University of Pennsylvania, Annenberg School for Communication.
- Bernard, H. Russell, & Ryan, Gery W. (1998). Text analysis, qualitative and quantitative methods. In H. Russell Bernard (Ed.), *Handbook of methods in cultural anthropology* (pp. 595–646). Walnut Creek, CA: AltaMira.
- Best, Michael L. (1997). Models for interacting populations of memes: Competition and niche behavior. *Journal of Memetics—Evolutionary Models of Information Transmission*, 1, 80–96.
- Best, Michael L. (1998). Corporal ecologies and population fitness on the Net. *Journal of Artificial Life*, 3, 261–287.
- Bishop, Stephanie. (1998). *Driving dangerously in the Prozac nation—Road rage: The making of a modern*

- malady. Unpublished manuscript, University of Pennsylvania, Annenberg School for Communication.
- Blalock, Herbert M. (1964). *Causal inferences in non-experimental research*. Chapel Hill: University of North Carolina Press.
- Bloch, Daniel A., & Kraemer, Helena Chmura. (1989). 2×2 kappa coefficients: Measures of agreement or association. *Biometrics*, 45, 269–287.
- Bonfantini, Massimo, & Proni, Giampaolo. (1988). To guess or not to guess? In Umberto Eco & Thomas A. Sebeok (Eds.), *The sign of three: Dupin, Holmes, Peirce* (pp. 119–134). Bloomington: Indiana University Press.
- Boot, N. (1980). Homography and lemmatization in Dutch texts. *ALLC Bulletin*, 8, 175–189.
- Boulding, Kenneth E. (1978). *Ecodynamics*. Beverly Hills, CA: Sage.
- Brennan, Robert L., & Prediger, Dale J. (1981). Coefficient kappa: Some uses, misuses, and alternatives. *Educational and Psychological Measurement*, 41, 687–699.
- Brin, Sergey, & Page, Lawrence. (1998). The anatomy of a large-scale hypertextual Web search engine. *Computer Networks and ISDN Systems*, 30, 107–117. Accessed at <http://infolab.stanford.edu/pub/papers/google.pdf>
- Broder, David P. (1940). The adjective-verb quotient: A contribution to the psychology of language. *Psychological Record*, 3, 310–343.
- Broom, Leonard, & Reece, Shirley. (1955). Political and racial interest: A study in content analysis. *Public Opinion Quarterly*, 19, 5–19.
- Brouwer, Marten; Clark, Cedric C.; Gerbner, George; & Krippendorff, Klaus. (1969). The television world of violence. In Robert K. Baker & Sandra J. Ball (Eds.), *Mass media and violence* (Vol. 9, pp. 311–339, 519–591). Washington, DC: Government Printing Office.
- Bruner, Jerome S., & Allport, Gordon W. (1940). Fifty years of change in American psychology. *Psychological Bulletin*, 37, 757–776.
- Bruns, Axel. (2007). Methodologies for mapping the political blogosphere: Explorations using the IssueCrawler research tool. *First Monday*, 12, 5. <http://firstmonday.org/ojs/index.php/fm/article/view/1834/1718>
- Budd, Richard W. (1964). Attention score: A device for measuring news “play.” *Journalism Quarterly*, 41, 259–262.
- Budge, Ian; Robertson, David; & Hearl, Derek. (1987). Ideology, strategy and party change: Spatial analyses of post-war election programmes in 19 democracies. Cambridge, MA: Cambridge University Press.
- Cahnman, Werner J. (1948). A note on marriage announcements in the *New York Times*. *American Sociological Review*, 13, 96–97.
- Campbell, Donald T. (1957). Factors relevant to the validity of experiments in social settings. *Psychological Bulletin*, 54, 297–311.
- Campbell, Donald T., & Fiske, Donald W. (1959). Convergent and discriminant validation by the multitrait-multimethod matrix. *Psychological Bulletin*, 56, 81–105.
- Cappella, Joseph N.; Price, Vincent; & Nir, Lilach. (2002). Argument repertoire as a reliable and valid measure of opinion quality: Electronic dialogue during campaign 2000. *Political Communication*, 19, 73–93.
- Cappella, Joseph N.; Turow, Joseph; & Jamieson, Kathleen Hall. (1996). *Call-in political talk radio: Background, content, audiences, portrayal in mainstream media* (Report series no. 5). Philadelphia: University of Pennsylvania, Annenberg Public Policy Center.
- Carletta, Jean; Isard, Amy; Isard, Stephen; Kowtko, Jacqueline C.; Doherty-Sneddon, Gwyneth; & Anderson, Anne H. (1997). The reliability of a dialogue structure coding scheme. *Computational Linguistics*, 23, 13–31.
- Carley, Kathleen M. (1997). Network text analysis: The network positions of concepts. In Carl W. Roberts (Ed.), *Text analysis for the social sciences: Methods for drawing statistical inferences from texts and transcripts* (pp. 79–100). Mahwah, NJ: Erlbaum.
- Carmines, Edward G., & Zeller, Richard A. (1979). *Reliability and validity assessment*. Beverly Hills, CA: Sage.
- Cartwright, Dorwin P. (1953). Analysis of qualitative material. In Leon Festinger & Daniel Katz (Eds.), *Research methods in the behavioral sciences* (pp. 421–470). New York, NY: Holt, Rinehart & Winston.
- Chung, Cindy K., & Pennebaker, James W. (2007). The psychological function of function words. In K. Fiedler (Ed.), *Social communication: Frontiers of social psychology* (pp. 343–359). New York: Psychology Press.
- Cheng, Na; Chandramouli, R.; & Subbalakshmi, K. P. (2011). *Author gender identification from text*. Accessed at <http://www.ece.stevens-tech.edu/~mouli/gender-id.pdf>
- Chomsky, Noam. (1959). Review of B. F. Skinner *Verbal Behavior*. *Language*, 35, 26–58.
- Cicchetti, Domenic V., & Feinstein, Alvan R. (1990). High agreement but low kappa: II. Resolving the paradoxes. *Journal of Clinical Epidemiology*, 43(6), 551–559.
- Clark, Cedric, C. (1966). *Cultural differences in reaction to discrepant communication: An exploratory study*. Dissertation. East Lansing: Michigan State University.
- Cohen, Bernard C. (1957). The political process and foreign policy: The making of the Japanese peace settlement. Princeton, NJ: Princeton University Press.

- Cohen, Jacob. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20, 37–46.
- Cohen, Jacob; Cohen, Patricia; West, Stephen G.; & Aiken, Leona S. (2003). *Applied multiple regression/correlation analysis for the behavioral sciences*, 3rd ed. Mahwah, NJ: Erlbaum.
- Content analysis: A new evidentiary technique. (1948). *University of Chicago Law Review*, 15, 910–925.
- Coombs, Clyde H. (1964). *A theory of data*. New York, NY: Wiley.
- Cooren, François. (2010). *Action and agency in dialogue: Passion, incarnation and ventriloquism*. Amsterdam, The Netherlands: John Benjamins.
- Corsaro, William, & Heise, David R. (1990). Event structure models from ethnographic data. In Clifford Clogg (Ed.), *Sociological methodology, 1990* (pp. 1–57). Oxford, England: Blackwell.
- Council on Interracial Books for Children. (1977). *Stereotypes, distortions and omissions in U.S. history textbooks*. New York, NY: Racism and Sexism Resource Center for Educators.
- Craig, Robert T. (1981). Generalization of Scott's index of intercoder agreement. *Public Opinion Quarterly*, 45, 260–264.
- Cronbach, Lee J. (1951). Coefficient alpha and the internal structure of tests. *Psychometrika*, 16, 297–334.
- Cuilenburg, Jan J. van. (1991). Inhoudsanalyse en computer. In Roel Popping & Jules L. Peschcar (Eds.), *Goed Geïnformeerd* (pp. 71–82). Houten, The Netherlands: Bohn Stafleu Van Loghum.
- Cuilenburg, Jan J. van; Kleinnijenhuis, Jan; & De Ridder, Jan A. (1986). A theory of evaluative discourse: Toward a graph theory of journalistic texts. *European Journal of Communication*, 1, 65–96.
- Cuilenburg, Jan J. van; Kleinnijenhuis, Jan; & De Ridder, Jan A. (1988). Artificial intelligence and content analysis: Problems of and strategies for computer text analysis. *Quality and Quantity*, 22, 65–97.
- Dale, Edgar. (1937). The need for the study of newsreels. *Public Opinion Quarterly*, 1, 122–125.
- D'Andrade, Roy. (1991). The identification of schemas in naturalistic data. In Mardi J. Horowitz (Ed.), *Person schemas and maladaptive interpersonal patterns* (pp. 279–301). Chicago, IL: University of Chicago Press.
- D'Andrade, Roy. (1995). *The development of cognitive anthropology*. Cambridge, MA: Cambridge University Press.
- Danielson, Wayne A.; Lasorsa, Dominic L.; & Im, Dal S. (1992). Journalists and novelists: A study of diverging styles. *Journalism Quarterly*, 69, 436–446.
- Danowski, James A. (1982). A network-based content analysis methodology for computer-mediated communication: An illustration with a computer bulletin board. In Michael Burgoon (Ed.), *Communication yearbook 6* (pp. 904–925). Beverly Hills, CA: Sage.
- Danowski, James A. (1993). Network analysis of message content. In William D. Richards, Jr., & George A. Barnett (Eds.), *Progress in communication sciences* (Vol. 4, pp. 197–221). Norwood, NJ: Ablex.
- Darnton, Robert. (1999, June 12). No computer can hold the past. *New York Times*, p. A15.
- Dawkins, Richard. (1976). *The selfish gene*. New York, NY: Oxford University Press.
- Dawkins, Richard. (1982). *The extended phenotype: The long reach of the gene*. San Francisco, CA: W. H. Freeman.
- Deerwester, Scott C.; Dumais, Susan T.; Furnas, George W.; Landauer, Thomas K.; & Harshman, Richard A. (1990). Indexing by latent semantic analysis. *Journal of the American Society for Information Science*, 41, 391–407.
- Denzin, Norman K., & Lincoln, Yvonna S. (2000). Introduction: The discipline and practice of qualitative research. In Norman K. Denzin & Yvonna S. Lincoln (Eds.), *Handbook of qualitative research* (2nd ed., pp. 1–28). Thousand Oaks, CA: Sage.
- DeWeese, L. Carroll, III. (1977). Computer content analysis of "day-old" newspapers: A feasibility study. *Public Opinion Quarterly*, 41, 91–94.
- Dibble, Vernon K. (1963). Four types of inferences from documents to events. *History and Theory*, 3, 203–221.
- Diefenbach, Donald L. (2001). Historical foundation of computer-assisted content analysis. In Mark D. West (Ed.), *Theory, method, and practice in computer content analysis* (pp. 13–41). Westport, CT: Ablex.
- Dodds, Peter S.; Harris, Kameron D.; Kloumann, Isabel M.; Bliss, Catherine A.; & Danford, C. M. (2011). Temporal patterns of happiness and information in a global social network: Hedonometrics and Twitter. *PLoS ONE*, 6, 12, e26752. <http://arxiv.org/abs/1101.5120v3>, accessed 2017.9.15.
- Dollard, John, & Auld, Frank, Jr. (1959). *Scoring human motives: A manual*. New Haven, CT: Yale University Press.
- Dollard, John, & Mowrer, O. Hobart. (1947). A method of measuring tension in written documents. *Journal of Abnormal and Social Psychology*, 42, 3–32.
- Doreian, Patrick. (Ed.). (1993). Narrative methods [Special issue]. *Journal of Mathematical Sociology*, 18(1–2).
- Dovring, Karin. (1954–1955). Quantitative semantics in 18th century Sweden. *Public Opinion Quarterly*, 18, 389–394.

- Dumais, Susan T. (1992). LSI meets TREC: A status report. In Donna K. Harman (Ed.), *The First Text REtrieval Conference (TREC-1)* (NIST Special Publication 500-207). Washington, DC: U.S. Department of Commerce, National Institute of Standards and Technology.
- Dumais, Susan T. (1993). Latent semantic indexing (LSI) and TREC-2. In Donna K. Harman (Ed.), *The Second Text REtrieval Conference (TREC-2)* (NIST Special Publication 500-215). Washington, DC: U.S. Department of Commerce, National Institute of Standards and Technology.
- Dunphy, Dexter C. (1966). The construction of categories for content analysis dictionaries. In Philip J. Stone; Dexter C. Dunphy; Marshall S. Smith; & Daniel M. Ogilvie, *The General Inquirer: A computer approach to content analysis* (pp. 134-168). Cambridge, MA: MIT Press.
- Durig, Alex. (1995). The event frame. *Research Studies in Symbolic Interactionism*, 7, 243-266.
- Dziurzynski, Patricia S. (1977). Development of a content analytic instrument for advertising appeals used in prime time television commercials. Unpublished master's thesis, University of Pennsylvania.
- Eco, Umberto. (1994). *The limits of interpretation*. Bloomington: Indiana University Press.
- Eigen, Manfred J.; McCaskill, John; & Schuster, Peter. (1988). Molecular quasi-species. *Journal of Physical Chemistry*, 92, 6881-6891.
- Ekman, Paul, & Friesen, Wallace V. (1968). Non-verbal behavior in psychotherapy research. In John Shlien (Ed.), *Research in psychotherapy* (Vol. 3, pp. 179-216). Washington, DC: American Psychological Association.
- Ekman, Paul; Friesen, Wallace V.; & Taussig, Thomas G. (1969). Vid-R and SCAN: Tools and methods for the automated analysis of visual records. In George Gerbner; Ole R. Holsti; Klaus Krippendorff; William J. Paisley; & Philip J. Stone (Eds.), *The analysis of communication content: Developments in scientific theories and computer techniques* (pp. 297-312). New York, NY: Wiley.
- Ellison, John W. (1965). Computers and the testaments. In *Proceedings, Conference on Computers for the Humanities* (pp. 64-74). New Haven, CT: Yale University Press.
- Esbensen, Kim H. (2010). *Multivariate data analysis in practice* (5th ed.). Esbjerg, Denmark: Aalborg University.
- Eshbaugh-Soha, Matthew (2010). The tone of local presidential news coverage. *Political Communication*, 27, 2: 121-140.
- Evans, William. (2002). Computer environments for content analysis: Reconceptualizing the roles of humans and computers. In Orville Vernon Burton (Ed.), *Computing in the social sciences and humanities* (pp. 67-83). Urbana: University of Illinois Press.
- Eysenbach, Gunther, & Till, James E. (2001). Ethical issues in qualitative research in Internet communities. *British Medical Journal*, 323(7321), 1103.
- Fan, David P. (1988). *Predictions of public opinion from the mass media: Computer content analysis and mathematical modeling*. New York, NY: Greenwood.
- Fan, David P., & Cook, R. Dennis. (1997). *Predictions of the Michigan Index of Consumer Sentiment from the press*. Unpublished manuscript, University of Minnesota, St. Paul.
- Farrall, Kenneth, & Delli Carpini, Michael X. (2005, October). *Issue politics, social networks, and the Web graph*. Paper presented at the annual meeting of the Association of Internet Researchers, Chicago, IL.
- Feigl, Herbert. (1952). Validation and vindication: An analysis of the nature and the limits of ethical arguments. In Wilfried Sellars & John Hospers (Eds.), *Readings in ethical theory* (pp. 667-680). New York, NY: Appleton-Century-Crofts.
- Feinstein, Alvan R., & Cicchetti, Domenic V. (1990). High agreement but low kappa: I. The problem of two paradoxes. *Journal of Clinical Epidemiology*, 43(6), 543-549.
- Fellbaum, Christiane. (Ed.). (1998). *WordNet: An electronic lexical database*. Cambridge, MA: MIT Press.
- Feng, Guangchao C. (2015). Mistakes and how to avoid mistakes in using intercoder reliability indices. *Methodology*, 11, 1:13-22.
- Fenton, Frances. (1910). The influence of newspaper presentations on the growth of crime and other anti-social activity. *American Journal of Sociology*, 16, 342-371, 538-564.
- Fleck, Ludwik. (1979). *Genesis and development of a scientific fact*. Chicago: University of Illinois Press. (Original work published 1935)
- Fleiss, Joseph L. (1971). Measuring nominal scale agreement among many raters. *Psychological Bulletin*, 76, 378-382.
- Fleiss, Joseph L. (1981). *Statistical methods for rates and proportions*. New York, NY: Wiley.
- Flesch, Rudolph. (1948). A new readability yardstick. *Journal of Applied Psychology*, 32, 221-233.
- Flesch, Rudolph. (1951). *How to test readability*. New York, NY: Harper & Row.
- Flesch, Rudolph. (1974). *The art of readable writing* (Rev. ed.). New York, NY: Harper & Row.
- Foltz, Peter W. (1990). Using latent semantic indexing for information filtering. In *Proceedings of the*

- 5th Conference on Office Information Systems [Special issue]. *ACM SIGOIS Bulletin*, 11(2-3).
- Ford, Sarah; Fallowfield, Leslie; & Lewis, Shon. (1996). Doctor-patient interactions in oncology. *Social Science & Medicine*, 42, 1511-1519.
- Foster, Charles R. (1938). *Editorial treatment of education in the American press*. Cambridge, MA: Harvard University Press.
- Freeman, Steven F. (2001). *Patterns of executive attention in U.S. auto industry letters to shareholders 1963-1987*. Unpublished manuscript, University of Pennsylvania, Wharton School.
- Furnas, George W.; Deerwester, Scott C.; Dumais, Susan T.; Landauer, Thomas K.; Harshman, Richard A.; Streeter, Lynn A.; & Lochbaum, Karen E. (1988). Information retrieval using a singular value decomposition model of latent semantic structure. In Yves Chiaramella (Ed.), *Proceedings of the 11th International ACM SIGIR Conference on Research and Development in Information Retrieval* (pp. 465-480). New York, NY: Association for Computing Machinery.
- Garfield, Eugene. (1955, July 15). Citation indexes for science: A new dimension in documentation through association of ideas. *Science* (New Series), 122(3159), 108-111. Accessed at <http://scimaps.org/static/docs/Garfield1955cit.pdf>
- Garfield, Eugene. (1979). Citation indexing: Its theory and application to science, technology and humanities. New York, NY: Wiley.
- Geller, A.; Kaplan, D.; & Lasswell, Harold D. (1942). An experimental comparison of four ways of coding editorial content. *Journalism Quarterly*, 19, 362-370.
- Gentzkow, Matthew, & Shapiro, Jesse M. (2010). What Drives Media Slant? Evidence from U.S. Daily Newspapers. *Econometrica*, 78, 1, 35-71.
- George, Alexander L. (1959a). Propaganda analysis: A study of inferences made from Nazi propaganda in World War II. Evanston, IL: Row, Peterson.
- George, Alexander L. (1959b). Quantitative and qualitative approaches to content analysis. In Ithiel de Sola Pool (Ed.), *Trends in content analysis* (pp. 7-32). Urbana: University of Illinois Press.
- Gerbner, George. (1958). The social role of the confession magazine. *Social Problems*, 6, 29-40.
- Gerbner, George. (1964). Ideological perspectives and political tendencies in news reporting. *Journalism Quarterly*, 41, 495-508.
- Gerbner, George. (1969). Toward "cultural indicators": The analysis of mass mediated public message systems. In George Gerbner; Ole R. Holsti; Klaus Krippendorff; William J. Paisley; & Philip J. Stone (Eds.), *The analysis of communication content: Developments in scientific theories and computer techniques* (pp. 123-132). New York, NY: Wiley.
- Gerbner, George. (1985). Mass media discourse: Message system analysis as a component of cultural indicators. In Teun A. Van Dijk (Ed.), *Discourse and communications: New approaches to the analyses of mass media discourse and communication* (pp. 13-25). Berlin, Germany: Walter de Gruyter.
- Gerbner, George; Gross, Larry; Morgan, Michael; & Signorielli, Nancy. (1994). *Television violence profile no. 16: The turning point: From research to action*. Philadelphia: University of Pennsylvania, Annenberg School for Communication.
- Gerbner, George; Gross, Larry; Morgan, Michael; & Signorielli, Nancy. (1995). Growing up with television: The cultivation perspective. In Jennings Bryant & Dolf Zillmann (Eds.), *Media effects: Advances in theory and research* (pp. 17-41). Mahwah, NJ: Erlbaum.
- Gerbner, George; Gross, Larry; Signorielli, Nancy; Morgan, Michael; & Jackson-Beeck, Marilyn. (1979). *Violence profile no. 10: Trends in network television drama and viewer conceptions of social reality, 1967-1978*. Philadelphia: University of Pennsylvania, Annenberg School of Communications.
- Gerbner, George; Holsti, Ole R.; Krippendorff, Klaus; Paisley, William J.; & Stone, Philip J. (Eds.). (1969). *The analysis of communication content: Developments in scientific theories and computer techniques*. New York, NY: Wiley.
- Gerbner, George, & Marvanyi, George. (1977). The many worlds of the world's press. *Journal of Communication*, 27(1), 52-75.
- Gergen, Kenneth J. (1985). Social constructionist inquiry: Context and implications. In Kenneth J. Gergen & Keith E. Davis (Eds.), *The social construction of the person* (pp. 3-18). New York, NY: Springer-Verlag.
- Gergen, Kenneth J. (1991). The saturated self: Dilemmas of identity in contemporary life. New York, NY: Basic Books.
- Gieber, Walter. (1964). News is what newspapermen make it. In Lewis A. Dexter & David M. White (Eds.), *People, society, and mass communication* (pp. 173-182). New York, NY: Free Press.
- Glaser, Barney G., & Strauss, Anselm L. (1967). *The discovery of grounded theory: Strategies for qualitative research*. Chicago, IL: Aldine.
- Goodenough, Ward H. (1972). Componential analysis. In James P. Spradley (Ed.), *Culture and cognition* (pp. 327-343). San Francisco, CA: Chandler.
- Goodwin, Charles. (1977). *Some aspects of the interaction of speaker and hearer in the construction of the turn at talk in natural conversation*. Unpublished doctoral dissertation, University of Pennsylvania, Annenberg School of Communications.

- Goodwin, Charles. (1981). *Conversational organization: Interaction between speakers and hearers*. New York, NY: Academic Press.
- Gottschalk, Louis A. (1995). *Content analysis of verbal behavior: New findings and clinical applications*. Hillsdale, NJ: Erlbaum.
- Gottschalk, Louis A., & Bechtel, Robert J. (1982). The measurement of anxiety through the computer analysis of verbal samples. *Comprehensive Psychiatry*, 23, 364–369.
- Gottschalk, Louis A., & Gleser, Goldine C. (1969). *The measurement of psychological states through the content analysis of verbal behavior*. Berkeley: University of California Press.
- Graham, Todd, & Witschge, Tamara. (2003). In search of online deliberations: Towards a new method for examining the quality of online discussions. *Communications*, 28, 173–204.
- Groth, Otto. (1948). *Die Geschichte der deutschen Zeitungswissenschaft, Probleme und Methoden*. Munich, Germany: Konrad Weinmayer.
- Guetzkow, Harold. (1950). Unitizing and categorizing problems in coding qualitative data. *Journal of Clinical Psychology*, 6, 47–58.
- Guo, Lei; Mays, Kate; Sameki, Mehrnoosh; & Betke, Margit. (2017, May). *From crowdsourcing to crowd-coding: An alternative approach to annotate big data in communication research*. Paper presented at the 67th Annual Conference of the International Communication Association, San Diego, CA.
- Gwet, Kilem (2002). Kappa statistic is not satisfactory for assessing the extent of agreement between raters. *Statistical Methods for Inter-Rater Reliability Assessment*, 1. Accessed at http://www.agreestat.com/research_papers/kappa_statistic_is_not_satisfactory.pdf
- Gwet, Kilem Li. (2008). Computing inter-rater reliability and its variance in the presence of high agreement. *British Journal of Mathematical and Statistical Psychology*, 61, 29–48.
- Hackett, Robert, & Zhao, Yaezhi. (1994). Challenging the master narratives: Peace protest and opinion/editorial discourse in the U.S. during the Gulf War. *Discourse & Society*, 5, 509–511.
- Harary, Frank; Norman, Robert Z.; & Cartwright, Dorwin P. (1965). *Structural models: An introduction to the theory of directed graphs*. New York, NY: Wiley.
- Harris, Karen L. (1996). Content analysis in negotiation research: A review and guide. *Behavioral Research Methods, Instruments, & Computers*, 28, 458–467.
- Hart, Roderick P. (1985). Systematic analysis of political discourse: The development of DICTION. In Keith R. Sanders; Lynda Lee Kaid; & Dan Nimmo (Eds.), *Political communication yearbook 1984* (pp. 97–134). Carbondale: Southern Illinois University Press.
- Harwood, Jake, & Giles, Howard. (1992). Don't make me laugh: Age representations in a humorous context. *Discourse & Society*, 3, 403–436.
- Hatch, David L., & Hatch, Mary. (1947). Criteria of social status as derived from marriage announcements in the *New York Times*. *American Sociological Review*, 12, 396–403.
- Hawk, Lee. (1997). "Listenability" of network newscasts: The change over sixteen years. *Feedback: Broadcast Education Association*, 38(2), 18–21.
- Hayes, Andrew F. (n.d.). My macros and code for SPSS and SAS. Accessed at <http://www.afhayes.com/spss-sas-and-mplus-macros-and-code.html>
- Hayes, Andrew F., & Krippendorff, Klaus. (2007). Answering the call for a standard reliability measure for coding data. *Communication Methods and Measures*, 1, 77–89. Accessed at <http://www.afhayes.com/public/cmm2007.pdf>
- Hays, David C. (1960). *Automatic content analysis*. Santa Monica, CA: Rand Corporation.
- Hays, David C. (1969). Linguistic foundations for a theory of content analysis. In George Gerbner; Ole R. Holsti; Klaus Krippendorff; William J. Paisley; & Philip J. Stone (Eds.), *The analysis of communication content: Developments in scientific theories and computer techniques* (pp. 57–67). New York, NY: Wiley.
- Heclo, Hugh. (1978). Issue networks and the executive establishment. In Anthony King (Ed.), *The New American Political System*. Washington, DC: American Enterprise Institute for Public Policy Research.
- Heise, David R. (1965). Semantic differential profiles for 1,000 most frequent English words. *Psychological Monographs*, 79(8, Whole No. 601).
- Heise, David R. (1995). *Specifying event content in narratives*. Unpublished manuscript, Indiana University. Accessed at <http://www.indiana.edu/~socpsy/papers/EventContent.html>
- Hellsten, Iina, & Leydesdorff, Loet. (2017). *Automated analysis of topic-actor networks on Twitter: New approach to the analysis of socio semantic networks*. Unpublished manuscript, University of Amsterdam, Amsterdam School of Communication Research (ASCoR). Preprint available at <https://arxiv.org/abs/1711.08387>
- Herdan, Gustav. (1960). *Type-token mathematics: A textbook of mathematical linguistics*. The Hague, The Netherlands: Mouton.

- Herma, Hans; Kriss, Ernst; & Shor, Joseph. (1943). Freud's theory of the dream in American textbooks. *Journal of Abnormal and Social Psychology*, 38, 319–334.
- Herrnstein, Richard J., & Murray, Charles. (1994). *The bell curve; Intelligence and class structure in American life*. New York, NY: Simon and Schuster.
- Hillman, James. (1995). *Kinds of power: A guide to its intelligent uses*. New York, NY: Doubleday.
- Holley, W., & Guilford, J. P. (1964). A note on the G-index of agreement. *Educational and Psychological Measurement*, 24, 749–754.
- Holsti, Ole R. (1962). *The belief system and national images: John Foster Dulles and the Soviet Union*. Unpublished doctoral dissertation, Stanford University.
- Holsti, Ole R. (1969). *Content analysis for the social sciences and humanities*. Reading, MA: Addison-Wesley.
- Holsti, Ole R.; Brody, Richard A.; & North, Robert C. (1965). Measuring affect and action in international reaction models: Empirical materials from the 1962 Cuban crisis. *Peace Research Society Papers*, 2, 170–190.
- Hopper, Robert; Koch, Susan; & Mandelbaum, Jennifer. (1986). Conversation analysis methods. In Donald G. Ellis & William A. Donahue (Eds.), *Contemporary issues in language and discourse processes* (pp. 169–186). Hillsdale, NJ: Erlbaum.
- Houle, Paul. (2002). [Contribution to a discussion of neural network content analysis programs on CON-TENT@sphinx.gsu.edu], March 30.
- Howe, Jeff. (2006, June 1). The rise of crowdsourcing. *Wired*.
- Howe, Jeff. (2008). *Crowdsourcing: How the power of the crowd is driving the future of business*. New York, NY: Random House.
- Hsueh, Pei-Yun; Melville, Prem; & Sindhwani, Vikas. (2009). Data quality from crowdsourcing: A study of annotation selection criteria. *Proceedings of the NAACL HLT Workshop on Active Learning for Natural Language Processing*. Boulder, CO. Pages 27–35.
- Hubert, Lawrence. (1977). Kappa revisited. *Psychological Bulletin*, 84, 289–297.
- Iker, Howard P. (1974). Select: A computer program to identify associationally rich words for content analysis: I. Statistical results. *Computers and the Humanities*, 8, 313–319.
- Iker, Howard P. (1975). *Words system manual*. Rochester, NY: Computer Printout.
- Iker, Howard P., & Harway, Norman I. (1969). A computer system approach toward the recognition and analysis of content. In George Gerbner; Ole R. Holsti; Klaus Krippendorff; William J. Paisley; & Philip J. Stone (Eds.), *The analysis of communication content: Developments in scientific theories and computer techniques* (pp. 381–405). New York, NY: Wiley.
- Innis, Harold A. (1951). *The bias of communication*. Toronto, Canada: University of Toronto Press.
- Institute for Propaganda Analysis. (1937). How to detect propaganda. *Propaganda Analysis*, 1, 5–8.
- Jamieson, Kathleen Hall. (1984). *Packaging the presidency: A history and criticism of presidential campaign advertising*. New York, NY: Oxford University Press.
- Jamieson, Kathleen Hall. (1998). *A preliminary report on so-called "negative" ads and campaigning, October 7, 1998*. Philadelphia: University of Pennsylvania, Annenberg School for Communication.
- Janda, Kenneth. (1969). A microfilm and computer system for analyzing comparative politics literature. In George Gerbner; Ole R. Holsti; Klaus Krippendorff; William J. Paisley; & Philip J. Stone (Eds.), *The analysis of communication content: Developments in scientific theories and computer techniques* (pp. 407–435). New York, NY: Wiley.
- Janis, Irving L. (1965). The problem of validating content analysis. In Harold D. Lasswell; Nathan Leites; & Associates (Eds.), *Language of politics: Studies in quantitative semantics* (pp. 55–82). Cambridge, MA: MIT Press. [Original work published 1943]
- Janis, Irving L., & Fadner, Raymond H. (1965). The coefficient of imbalance. In Harold D. Lasswell; Nathan Leites; & Associates (Eds.), *Language of politics: Studies in quantitative semantics* (pp. 153–169). Cambridge, MA: MIT Press. [Original work published 1943]
- Janson, Svante, & Vegelius, Jan. (1979). On generalizations of the G index and the phi coefficient to nominal scales. *Multivariate Behavioral Research*, 14, 255–269.
- Jefferson, Gail. (1978). Sequential aspects of storytelling in conversation. In Jim Schenkein (Ed.), *Studies in the organization of conversational interaction* (pp. 219–248). New York, NY: Free Press.
- Jensen, Klaus Bruhn. (2006). News as ideology: Economic statistics and political ritual in television network news. *Journal of Communication*, 37(1), 8–27.
- Johnson-Laird, Philip N.; Herrmann, Douglas J.; & Chaffin, Roger. (1984). Only connections: A critique of semantic networks. *Psychological Bulletin*, 96, 292–315.
- Josephson, John R., & Josephson, Susan G. (1994). Introduction. In John R. Josephson & Susan G. Josephson (Eds.), *Abductive inference: Computation, philosophy, technology*. New York, NY: Cambridge University Press.

- Kamada, Tomihisa, & Kawai, Satoru. (1989). An algorithm for drawing general undirected graphs. *Information Processing Letters*, 31(1), 7-15.
- Kaplan, Abraham. (1964). The conduct of inquiry: Methodology for behavioral science. San Francisco, CA: Chandler.
- Kaplan, Abraham, & Goldsen, Joseph M. (1965). The reliability of content analysis categories. In Harold D. Lasswell; Nathan Leites; & Associates (Eds.), *Language of politics: Studies in quantitative semantics* (pp. 83-112). Cambridge, MA: MIT Press.
- Katz, Elihu; Gurevitch, Michael; Danet, Brenda; & Peled, Tsiyona. (1969). Petitions and prayers: A content analysis of persuasive appeals. *Social Forces*, 47, 447-463.
- Katz, Elihu; Gurevitch, Michael; Peled, Tsiyona; & Danet, Brenda. (1969). Exchanges with clients: A diagnostic approach to organizations and professions. *Human Relations*, 22, 309-324.
- Katz, Jonathan N. (1995). *The invention of heterosexuality*. New York, NY: Dutton.
- Kelly, Edward F., & Stone, Philip J. (1975). *Computer recognition of English word senses*. Amsterdam, The Netherlands: North-Holland.
- Kenski, Kate (with Jamieson, Kathleen Hall, & Romer, Dan). (1999). *Public smarter than press, pundits, or academics about so-called "negative" ads and campaigning* [Survey, October 6]. Philadelphia: University of Pennsylvania, Annenberg Public Policy Center.
- Kim, Joohan, & Gamson, William A. (1999, May). *Computer assisted frame analysis (CAFA) of abortion issues: A pilot study*. Paper presented at the 49th Annual Meeting of the International Communication Association, San Francisco, CA.
- Klausner, Samuel Z. (1968). *Two centuries of child-rearing manuals* [Technical report to the Joint Commission on Mental Health of Children]. Philadelphia: University of Pennsylvania, Department of Sociology.
- Klein, Hans K. (1999). Tocqueville in cyberspace: Using the Internet for citizen associations. *The Information Society*, 15(4), 213-220. Accessed at <http://www.indiana.edu/~tisj/readers/full-text/15-4%20klein.pdf>
- Klein, Malcolm W., & Maccoby, Nathan. (1954). Newspaper objectivity in the 1952 campaign. *Journalism Quarterly*, 31, 285-296.
- Kleinnijenhuis, Jan. (1990, June). *Applications of graph theory to cognitive communication research*. Paper presented at the 40th Annual Meeting of the International Communication Association, Dublin, Ireland.
- Kleinnijenhuis, Jan; De Ridder, Jan A.; & Rietberg, Edwald M. (1997). Reasoning in economic discourse: An application of the network approach to the Dutch press. In Carl W. Roberts (Ed.), *Text analysis for the social sciences: Methods for drawing statistical inferences from texts and transcripts* (pp. 191-207). Mahwah, NJ: Erlbaum.
- Klir, Jiri, & Valach, Miroslav. (1965). Language as a means of communication between man and machine. In Jiri Klir & Miroslav Valach, *Cybernetic modeling* (pp. 315-373). Princeton, NJ: D. Van Nostrand.
- Kolbe, Richard H., & Burnett, Melissa S. (1991). Content-analysis research: An examination of applications with directives for improving research reliability and objectivity. *Journal of Consumer Research*, 18, 243-250.
- Koppel, Moshe; Argamon, Shlomo; & Shimoni, Anat Rachel. (2002). Automatically categorizing written texts by author gender. *Literary and Linguistic Computing*, 17(4), 401-412.
- Kracauer, Siegfried. (1947). Propaganda and the Nazi war film. In *From Caligari to Hitler: A psychological history of the German film* (pp. 308-331). Princeton, NJ: Princeton University Press.
- Kracauer, Siegfried. (1952-1953). The challenge of quantitative content analysis. *Public Opinion Quarterly*, 16, 631-642.
- Krendel, Ezra S. (1970). A case study of citizen complaints as social indicators. *IEEE Transactions on Systems Science and Cybernetics*, 6, 267-272.
- Krippendorff, Klaus. (1967). An examination of content analysis: A proposal for a general framework and an information calculus for message analytic situations. Unpublished doctoral dissertation, University of Illinois, Urbana.
- Krippendorff, Klaus. (1969a). Models of messages: Three prototypes. In George Gerbner; Ole R. Holsti; Klaus Krippendorff; William J. Paisley; & Philip J. Stone (Eds.), *The analysis of communication content: Developments in scientific theories and computer techniques* (pp. 69-106). New York, NY: Wiley.
- Krippendorff, Klaus. (1969b). Theories and analytical constructs: Introduction. In George Gerbner; Ole R. Holsti; Klaus Krippendorff; William J. Paisley; & Philip J. Stone (Eds.), *The analysis of communication content: Developments in scientific theories and computer techniques* (pp. 3-16). New York, NY: Wiley.
- Krippendorff, Klaus. (1970a). Bivariate agreement coefficients for reliability data. In Edgar F. Borgatta & George W. Bohrnstedt (Eds.), *Sociological methodology 1970* (pp. 139-150). San Francisco, CA: Jossey-Bass.
- Krippendorff, Klaus. (1970b). Estimating the reliability, systematic error and random error of interval data. *Educational and Psychological Measurement*, 30, 61-70.

- Krippendorff, Klaus. (1970c). The expression of value in political documents. *Journalism Quarterly*, 47, 510–518.
- Krippendorff, Klaus. (1970d). On generating data in communication research. *Journal of Communication*, 20(3), 241–269. Accessed at http://repository.upenn.edu/asc_papers/273
- Krippendorff, Klaus. (1971). Reliability of recording instructions: Multivariate agreement for nominal data. *Behavioral Science*, 16, 222–235.
- Krippendorff, Klaus. (1978). Reliability of binary attribute data. *Biometrics*, 34, 142–144. Accessed at <http://www.jstor.org/pss/2529602>
- Krippendorff, Klaus. (1980a). Clustering. In Peter R. Monge & Joseph N. Cappella (Eds.), *Multivariate techniques in communication research* (pp. 259–308). New York, NY: Academic Press.
- Krippendorff, Klaus. (1980b). *Content analysis: An introduction to its methodology*. Beverly Hills, CA: Sage.
- Krippendorff, Klaus. (1986). *Information theory: Structural models for qualitative data*. Beverly Hills, CA: Sage.
- Krippendorff, Klaus. (1987). Association, agreement and equity. *Quality and Quantity*, 21, 109–123.
- Krippendorff, Klaus. (1991). Reconstructing (some) communication research methods. In Frederick Steier (Ed.), *Research and reflexivity* (pp. 115–142). Newbury Park, CA: Sage.
- Krippendorff, Klaus. (1992, May). *Recent developments in reliability analysis*. Paper presented at the 42nd Annual Meeting of the International Communication Association, Miami, FL. Accessed at http://repository.upenn.edu/asc_papers/44
- Krippendorff, Klaus. (1993). The past of communication's hoped-for future. *Journal of Communication*, 43(3), 34–44.
- Krippendorff, Klaus. (1995a). On the reliability of unitizing continuous data. In Peter V. Marsden (Ed.), *Sociological methodology, 1995* (Vol. 25, pp. 47–76). Cambridge, MA: Blackwell.
- Krippendorff, Klaus. (1995b). Undoing power. *Critical Studies in Mass Communication*, 12, 101–132.
- Krippendorff, Klaus. (2004a). *Content analysis* (2nd ed.). Thousand Oaks, CA: Sage.
- Krippendorff, Klaus. (2004b). Measuring the reliability of qualitative text analysis data. *Quality and Quantity*, 38, 787–800. Accessed at http://repository.upenn.edu/asc_papers/42
- Krippendorff, Klaus. (2005). Writing: Monologue, dialogue, and ecological narrative. In Michael B. Hinner (Ed.), *Introduction to business communication* (pp. 119–159). Frankfurt, Germany: Lang. Accessed at http://repository.upenn.edu/asc_papers/93
- Krippendorff, Klaus. (2006). The dialogical reality of meaning. *American Journal of SEMIOTICS*, 19(1–4), 17–34. Accessed at http://repository.upenn.edu/asc_papers/51
- Krippendorff, Klaus. (2008). Systematic and random disagreement and the reliability of nominal data. *Communication Methods and Measures*, 2(4), 323–338. Accessed at http://repository.upenn.edu/asc_papers/205
- Krippendorff, Klaus. (2011a). Agreement and information in the reliability of coding. *Communication Measures and Methods*, 5(2), 93–112. Accessed at http://repository.upenn.edu/asc_papers/278
- Krippendorff, Klaus. (2011b). A dissenting view on so-called paradoxes of reliability coefficients. *Communication Yearbook 36*: Chapter 20, pp. 481–499 in C. T. Salmond, (Ed.). New York, NY: Routledge.
- Krippendorff, Klaus. (2016). *Algorithm for bootstrapping a distribution of α* . Retrieved from <http://web.asc.upenn.edu/usr/krippendorff/boot.c-Alpha.pdf>
- Krippendorff, Klaus. (2017). Three concepts to retire. *Annals of the International Communication Association* 41(1), 92–99.
- Krippendorff, Klaus, & Bock, Mary Angela. (Eds.). (2009). *The content analysis reader*. Thousand Oaks, CA: Sage.
- Krippendorff, Klaus, & Craggs, Richard. (2016). The reliability of multi-valued coding of data. *Communication Methods and Measures* 10(4), 181–198.
- Krippendorff, Klaus, & Eleey, Michael. (1986). Monitoring the symbolic environment of organizations. *Public Relations Review*, 12(1), 13–36.
- Krippendorff, Klaus; Mathet, Yann; Bouvry, Stéphane & Widlöcher, Antoine (2016). On the reliability of unitizing textual continua: Further developments. *Quality & Quantity* 50, 2347–2364. Retrieved from <https://link.springer.com/article/10.1007/s11135-015-0266-1>
- Kuder, G. F., & Richardson, M. W. (1937). The theory of the estimation of test reliability. *Psychometrika* 2(3), 151–160.
- Kuhn, Deanna. (1991). *The skills of argument*. New York, NY: Cambridge University Press.
- Labov, William. (1972). *Sociolinguistic patterns*. Philadelphia: University of Pennsylvania Press.
- Laffal, Julius. (1993). *A concept dictionary of English*. Essex, CT: Galley.
- Lakoff, George. (1987). *Women, fire, and dangerous things: What categories reveal about the mind*. Chicago, IL: University of Chicago Press.

- Landis, H. H., & Burt, Harold E. (1924). A study of conversations. *Journal of Comparative Psychology*, 4, 81–89.
- Landis, J. Richard, & Koch, Gary H. (1977). An application of hierarchical kappa-type statistics in the assessment of majority agreement among multiple observers. *Biometrics*, 33, 363–374.
- Lashner, Marilyn A. (1990). Content analysis: A method for analyzing communications. *The Expert and the Law*, 10(1), 2–5.
- Lasswell, Harold D. (1927). *Propaganda technique in the world war*. New York, NY: Knopf.
- Lasswell, Harold D. (1938). A provisional classification of symbol data. *Psychiatry*, 1, 197–204.
- Lasswell, Harold D. (1941). The World Attention Survey: An exploration of the possibilities of studying attention being given to the United States by newspapers abroad. *Public Opinion Quarterly*, 5, 456–462.
- Lasswell, Harold D. (1960). The structure and function of communication in society. In Wilbur Schramm (Ed.), *Mass communications* (pp. 117–130). Urbana: University of Illinois Press.
- Lasswell, Harold D. (1963). *Politics: Who gets what, when, how*. New York, NY: Meridian.
- Lasswell, Harold D. (1965a). Detection: Propaganda detection and the courts. In Harold D. Lasswell; Nathan Leites; & Associates (Eds.), *Language of politics: Studies in quantitative semantics* (pp. 173–232). Cambridge, MA: MIT Press.
- Lasswell, Harold D. (1965b). Why be quantitative? In Harold D. Lasswell; Nathan Leites; & Associates (Eds.), *Language of politics: Studies in quantitative semantics* (pp. 40–52). Cambridge, MA: MIT Press. [Original work published 1949]
- Lasswell, Harold D., & Kaplan, Abraham. (1950). *Power and society: A framework for political inquiry*. New Haven, CT: Yale University Press.
- Lasswell, Harold D.; Leites, Nathan; & Associates. (Eds.). (1965). *Language of politics: Studies in quantitative semantics*. Cambridge, MA: MIT Press.
- Lasswell, Harold D.; Lerner, Daniel; & Pool, Ithiel de Sola. (1952). *The comparative study of symbols*. Stanford, CA: Stanford University Press.
- Latour, Bruno, & Woolgar, Steve. (1986). *Laboratory life: The construction of scientific facts*. Princeton, NJ: Princeton University Press.
- Lazarsfeld, Paul F.; Berelson, Bernard; & Gaudet, Hazel. (1948). *The people's choice: How the voter makes up his mind in a presidential campaign*. New York, NY: Columbia University Press.
- Lee, Chin-Chuan; Chan, Joseph Man; Pan, Zhong-dang; & So, Clement Y. K. (2002). *Global media spectacle: News war over Hong Kong*. Albany: State University of New York Press.
- Leetaru, Kalev H. (2011). *Data mining methods for the content analyst: An introduction to the computational analysis of content*. New York, NY: Routledge.
- Lehmann, Pola; & malisazobel (2017). Chancen und Grenzen der Schwarmintelligenz. Werkstattbericht aus dem Crowd-Coding-Projekt. Wissenschaftszentrum Berlin für Sozialforschung. March 9, 2017.
- Leites, Nathan; Bernaut, Elsa; & Garthoff, Raymond L. (1951). Politburo images of Stalin. *World Politics*, 3, 317–339.
- Leites, Nathan, & Pool, Ithiel de Sola. (1942). *Communist propaganda in reaction to frustration* (Document No. 27). Washington DC: Library of Congress, Experimental Division for Study of Wartime Communications.
- Leskovec, Jure; Backstrom, Lars; & Kleinberg, Jon. (2017). *Meme-tracking and the dynamics of the news cycle*. Paper presented at the 67th Annual Conference of the International Communication Association, May 25–29, San Diego, CA.
- Lindsay, Robert K. (1963). Inferential memory as the basis of machines which understand natural language. In Edward A. Feigenbaum & Julian Feldman (Eds.), *Computers and thought* (pp. 217–233). New York, NY: McGraw-Hill.
- Lippmann, Walter. (1922). *Public opinion*. New York, NY: Macmillan.
- Löbl, Eugen. (1903). *Kultur und Presse*. Leipzig, Germany: Duncker & Humblot.
- Loeventhal, Leo. (1944). Biographies in popular magazines. In Paul F. Lazarsfeld & Frank N. Stanton (Eds.), *Radio research 1942–1943* (pp. 507–548). New York, NY: Duell, Sloan & Pearce.
- Lombard, Matthew; Snyder-Duch, Jennifer; & Bracken, Cheryl Campanella. (2002). Content analysis in mass communication research: An assessment and reporting of intercoder reliability. *Human Communication Research*, 28, 587–604.
- Lorr, Maurice, & McNair, Douglas M. (1966). Methods relating to evaluation of therapeutic outcome. In Louis A. Gottschalk & Arthur H. Auerbach (Eds.), *Methods of research in psychotherapy* (pp. 573–594). Englewood Cliffs, NJ: Prentice Hall.
- Lowe, Will. (2003). *Software for content analysis: A review*. Cambridge, MA: Harvard University. Accessed at http://kb.ucla.edu/system/datas/5/original/content_analysis.pdf
- Lynch, Kevin. (1965). *The image of the city*. Cambridge, MA: MIT Press.
- Maccoby, Nathan; Sabghir, F. O.; & Cushing, B. (1950). A method for the analysis of news coverage of industry. *Public Opinion Quarterly*, 14, 753–758.

- MacQueen, Kathleen M.; McLellan, Eleanor; Kay, Kelly; & Milstein, Bobby. [1998]. Codebook development for team-based qualitative analysis. *Cultural Anthropology Methods*, 10(2), 31–36.
- Mahl, George F. (1959). Exploring emotional states by content analysis. In Ithiel de Sola Pool (Ed.), *Trends in content analysis* (pp. 89–130). Urbana: University of Illinois Press.
- Mann, Mary B. (1944). Studies in language behavior: III. The quantitative differentiation of samples of written language. *Psychological Monographs*, 56(2), 41–47.
- Manning, Christopher D.; Raghavan, Prabhakar; & Schütze, Hinrich. [2008]. *Introduction to information retrieval*. Cambridge, MA: Cambridge University Press.
- Markov, Andrei A. (1913). Essai d'une recherche statistique sur le texte du roman *Eugene Onegin* illustrant la liaison des épreuves en chaîne (Russian). *Bulletin de L'Académie Impériale des Sciences de St. Pétersbourg* 6(7), 153–162.
- Martin, Helen. (1936). Nationalism and children's literature. *Library Quarterly*, 6, 405–418.
- Martindale, Colin. [1990]. *The clockwork muse: The predictability of artistic change*. New York, NY: Basic Books.
- Maruyama, Margoroh. (1963). The second cybernetics: Deviation-amplifying mutual causal processes. *American Scientist*, 51, 164–179.
- Mathews, Byron C. [1910]. A study of a New York daily. *Independent*, 68, 82–86.
- Maxwell, A. E. (1970). Comparing the classification of subjects by two independent judges. *British Journal of Psychiatry*, 116, 651–655.
- McClelland, David C. (1958). The use of measures of human motivation in the study of society. In John W. Atkinson (Ed.), *Motives in fantasy, action and society* (pp. 518–552). Princeton, NJ: D. Van Nostrand.
- McClelland, David C.; Atkinson, John W.; Clark, Russell A.; & Lowell, Edgar L. (1992). A scoring manual for the achievement motive. In Charles P. Smith (Ed.), *Motivation and personality: Handbook of thematic content analysis* (pp. 153–178). Cambridge, MA: Cambridge University Press.
- McCombs, Maxwell, & Shaw, Donald L. (1972). The agenda-setting function of mass media. *Public Opinion Quarterly*, 36, 176–187.
- McCombs, Maxwell; Shaw, Donald L.; & Weaver, David. (1997). *Communication and democracy: Exploring the intellectual frontiers in agenda-setting theory*. Mahwah, NJ: Erlbaum.
- McDiarmid, John. (1937). Presidential inaugural addresses: A study in verbal symbols. *Public Opinion Quarterly*, 1, 79–82.
- McKnight, Katherine S., & Walberg, Herbert J. (1994). Neural network analysis of student essays. *Journal of Research and Development in Education*, 32, 26–31.
- McMillan, Sally J. (2000). The microscope and the moving target: The challenge of applying content analysis to the World Wide Web. *Journalism and Mass Communication Quarterly*, 77, 80–98.
- McTavish, Donald D.; Litkowski, Kenneth C.; & Schrader, Susan. (1997). A computer content analysis approach to measuring social distance in residential organizations for older people. *Social Science Computer Review*, 15, 170–180.
- McTavish, Donald D., & Pirro, Ellen B. (1990). Contextual content analysis. *Quality and Quantity*, 24, 245–265.
- Merrill, John C. (1962). The image of the United States in ten Mexican dailies. *Journalism Quarterly*, 39, 203–209.
- Merritt, Richard L. (1966). *Symbols of American community, 1735–1775*. New Haven, CT: Yale University Press.
- Merten, Klaus. (1991). *Inhaltsanalyse: Eine Einführung in Theorie, Methode und Praxis*. Opladen, Germany: Westdeutscher Verlag.
- Miles, Josephine. (1951). The continuity of English poetic language. In *University of California publications in English* (pp. 517–535). Berkeley: University of California Press.
- Miller, George A. (1951). *Language and communication*. New York, NY: McGraw-Hill.
- Miller, George A.; Beckwith, Richard; Fellbaum, Christiane; Gross, Derek; Miller, Katherine J.; & Tengi, Randee I. (1993). *Five papers on WordNet* [CSL Report 43]. Princeton, NJ: Princeton University, Cognitive Science Laboratory.
- Miller, Kevin J.; Fullmer, Steven L.; & Walls, Richard T. (1996). A dozen years of mainstreaming literature: A content analysis. *Exceptionality*, 6(2), 99–109.
- Mokotoff, Gary. (1997). Soundexing and genealogy. *Avotaynu*. Accessed at <http://www.avotaynu.com/soundex.htm>
- Monge, Peter R., & Contractor, Noshir S. (2003). *Theories of communication networks*. New York, NY: Oxford University Press.
- Monge, Peter R., & Eisenberg, Erik M. (1987). Emergent communication networks. In Fredric M. Jablin; Linda L. Putnam; Karlene H. Roberts; & Lyman W. Porter (Eds.), *Handbook of organizational and management communication* (pp. 204–242). Newbury Park, CA: Sage.
- Montgomery, Michael. (1989). Protocol for using LAGS CodeMap and LagsMap microcomputer programs: CodeMap for a lexical item. Accessed at <http://hyde.park.uga.edu/lags/protocol.txt>

- Montgomery, Michael. (1993). Review: The linguistic atlas of the Gulf states. *American Speech*, 68(3), 263–318.
- Morton, Andrew Q. (1963, November 3). A computer challenges the church. *Observer*.
- Morton, Andrew Q., & Levinson, Michael. (1966). Some indications of authorship in green prose. In Jacob Leed (Ed.), *The computer and literary style* (pp. 141–179). Kent, OH: Kent State University Press.
- Mosteller, Frederick, & Wallace, David L. (1963). Inference in an authorship problem. *Journal of the American Statistical Association*, 58, 275–309.
- Mosteller, Frederick, & Wallace, David L. (1964). *Inference and disputed authorship: The Federalist*. Reading, MA: Addison-Wesley.
- Murray, Edward J.; Auld, Frank, Jr.; & White, Alice M. (1954). A psychotherapy case showing progress but no decrease in the discomfort-relief quotient. *Journal of Consulting Psychology*, 18, 349–353.
- Murray, Henry A. (1943). *Thematic Apperception Test manual*. Cambridge, MA: Harvard University Press.
- Nacos, Brigitte L.; Shapiro, Robert Y.; Young, John T.; Fan, David P.; Kjellstrand, Torsten; & McCaa, Craig. (1991). Content analysis of news reports: Comparing human coding and a computer-assisted method. *Communication*, 12, 111–128.
- Namenwirth, J. Zvi. (1973). Wheels of time and the interdependence of value change in America. *Journal of Interdisciplinary History*, 3, 649–683.
- Namenwirth, J. Zvi, & Weber, Robert P. (1987). *Dynamics of culture*. Boston, MA: Allen & Unwin.
- Naureckas, Jim (1995, January 1). Racism resurgent: How media let the *Bell Curve*'s pseudo-science define the agenda on race. *Fair*.
- Neuendorf, Kimberly A. (2002). *The content analysis guidebook*. Thousand Oaks, CA: Sage.
- Newell, Allen, & Simon, Herbert A. (1956). The logic theory machine. *IRE-Transactions on Information Theory* 2(3), 61–79.
- Newell, Allen, & Simon, Herbert A. (1963). General Problem Solver: A program that simulates human thought. In Edward A. Feigenbaum & Julian Feldman (Eds.), *Computers and thought* (pp. 279–293). New York, NY: McGraw-Hill.
- Niederer, Sabine, M. C. (2013). Global warming is not a crisis! Studying climate change skepticism on the web. *Necsus*, 3 (Spring).
- Niederer, Sabine, M. C. (2016). *Networked content analysis*. Amsterdam, The Netherlands: Dissertation defended at the Amsterdam University of Applied Sciences.
- Niederer, Sabine, M. C. (2018, forthcoming). The study of networked content: Five considerations for digital research in the humanities. In G. Schiuma & D. Carlucci (Eds.), *Big data in the arts and humanities: Theory and practice* (pp. 89–100). Boca Raton, FL: CRC Press-Taylor & Francis Group.
- Nixon, Raymond B., & Jones, Robert L. (1956). The content of non-competitive newspapers. *Journalism Quarterly*, 33, 299–314.
- North, Robert C.; Holsti, Ole R.; Zaninovich, M. George; & Zinnes, Dina A. (1963). *Content analysis: A handbook with applications for the study of international crisis*. Evanston, IL: Northwestern University Press.
- Ogden, Charles Kay. (1937). *Basic English and grammatical reform*. Cambridge, MA: Orthological Institute.
- Osgood, Charles E. (1959). The representation model and relevant research methods. In Ithiel de Sola Pool (Ed.), *Trends in content analysis* (pp. 33–88). Urbana: University of Illinois Press.
- Osgood, Charles E. (1974a). Probing subjective culture: Part 1. Cross-linguistic tool-making. *Journal of Communication*, 24(1), 21–35.
- Osgood, Charles E. (1974b). Probing subjective culture: Part 2. Cross-cultural tool using. *Journal of Communication*, 24(2), 82–100.
- Osgood, Charles E.; Saporta, Sol; & Nunnally, Jum C. (1956). Evaluative assertion analysis. *Litera*, 3, 47–102.
- Osgood, Charles E.; Suci, George J.; & Tannenbaum, Percy H. (1957). *The measurement of meaning*. Urbana: University of Illinois Press.
- O'Sullivan, Thomas C., Jr. (1961). Factor analysis concepts identified in theoretical writings: An experiment design. Lexington, MA: Itek Laboratories.
- Paisley, William J. (1964). Identifying the unknown communicator in painting, literature and music: The significance of minor encoding habits. *Journal of Communication*, 14(4), 219–237.
- Palmquist, Michael E.; Carley, Kathleen M.; & Dale, Thomas A. (1997). Applications of computer-aided text analysis: Analyzing literary and nonliterary texts. In Carl W. Roberts (Ed.), *Text analysis for the social sciences: Methods for drawing statistical inferences from texts and transcripts* (pp. 171–189). Mahwah, NJ: Erlbaum.
- Parsons, Talcott. (1951). *The social system*. New York, NY: Free Press.
- Parsons, Talcott, & Bales, Robert F. (1953). The dimensions of action-space. In Talcott Parsons; Robert F. Bales; & Edward A. Shills (Eds.), *Working papers in the theory of action* (pp. 63–109). New York, NY: Free Press.
- Patterson, Brian R.; Neupauer, Nichola C.; Burant, Patricia A.; Koehn, Steven C.; & Reed, April T. (1996). A preliminary examination of conversation analytic

- techniques: Rates of inter-transcriber reliability. *Western Journal of Communication*, 60, 76–91.
- Pearson, Karl et al. (1901). Mathematical contributions to the theory of evolution: IX. On the principle of homotyposis and its relation to heredity, to variability of the individual, and to that of race. Part I: Homotyposis in the vegetable kingdom. *Philosophical Transactions of the Royal Society*, 197(Series A), 285–379.
- Pederson, Lee; McDaniel, Susan Leas; Adams, Carol; & Liao, Caisheng. (Eds.). (1989). *Linguistic atlas of the Gulf states: Vol. 3. Technical index*. Athens: University of Georgia Press.
- Péladeau, Normand. (1996). *SimStat for Windows: User's guide*. Montreal, Canada: Provalis Research.
- Pennebaker, James W. (1997). Writing about emotional experiences as a therapeutic process. *Psychological Science*, 8, 162–166.
- Pennebaker, James W.; Francis, Martha E.; & Booth, Roger J. (2001). *Linguistic inquiry and word count (LIWC)* (2nd ed., PDF manual). Mahwah, NJ: Erlbaum.
- Pennebaker, James W., & Stone, Lori D. (2001, May). *LIWC 2001*. Contribution to a CATA workshop presented at the 51st Annual Meeting of the International Communication Association, Washington, DC.
- Pennings, Paul, & Keman, Hans. (2002). Towards a new methodology of estimating party policy positions. *Quality and Quantity*, 36, 55–79.
- Perreault, William D., & Leigh, Lawrence E. (1989). Reliability of nominal data based on qualitative judgments. *Journal of Marketing Research*, 26, 135–148.
- Pescosolido, Bernice A.; Grauerholz, Elisabeth; & Milkie, Melissa A. (1996). Culture and conflict: The portrayal of blacks in U.S. children's picture books through the mid- and late twentieth century. *American Sociological Review*, 62, 443–464.
- Peter, Jochen, & Lauf, Edmund. (2002). Reliability in cross-national content analysis. *Journalism & Mass Communication Quarterly*, 79, 815–832.
- Phillips, David P. (1978). Airplane accident fatalities increase just after newspaper stories about murder and suicide. *Science*, 201, 748–749.
- Piault, Collette. (1965). A methodological investigation of content analysis using electronic computers for data processing. In Dell Hymes (Ed.), *The use of computers in anthropology* (pp. 273–293). The Hague, The Netherlands: Mouton.
- Pierce, Bessie L. (1930). *Civic attitudes in American school textbooks*. Chicago, IL: University of Chicago Press.
- Ploughman, Penelope. (1995). The American print news media "construction" of five natural disasters. *Disasters*, 19, 308–326.
- Pocklington, Richard, & Best, Michael L. (1997). Cultural evolution and units of selection in replicating text. *Journal of Theoretical Biology*, 188, 79–87.
- Pool, Ithiel de Sola. (1951). *Symbols of internationalism*. Stanford, CA: Stanford University Press.
- Pool, Ithiel de Sola. (1952a). *The prestige papers: A survey of their editorials*. Stanford, CA: Stanford University Press.
- Pool, Ithiel de Sola. (1952b). *Symbols of democracy*. Stanford, CA: Stanford University Press.
- Pool, Ithiel de Sola. (Ed.). (1959a). *Trends in content analysis*. Urbana: University of Illinois Press.
- Pool, Ithiel de Sola. (1959b). Trends in content analysis today: A summary. In Ithiel de Sola Pool (Ed.), *Trends in content analysis* (pp. 189–233). Urbana: University of Illinois Press.
- Pool, Ithiel de Sola; Abelson, Robert P.; & Popkin, Samuel L. (1964). *Candidates, issues and strategies: A computer simulation of the 1960 presidential election*. Cambridge, MA: MIT Press.
- Popping, Roel. (1988). On agreement indices for nominal data. In William E. Saris & Irmtraut N. Gallhofer (Eds.), *Sociometric research: Data collection and scaling* (Vol. 1, pp. 90–105). New York, NY: St. Martin's.
- Popping, Roel. (1997). Computer programs for the analysis of texts and transcripts. In Carl W. Roberts (Ed.), *Text analysis for the social sciences: Methods for drawing statistical inferences from texts and transcripts* (pp. 209–221). Mahwah, NJ: Erlbaum.
- Popping, Roel. (2000). *Computer-assisted text analysis*. London, England: Sage.
- Posen, Solomon. (1997). The portrayal of the doctor in non-medical literature: The impaired doctor. *Medical Journal of Australia*, 166, 48–51.
- Potter, W. James, & Levine-Donnerstein, Deborah. (1999). Rethinking reliability and reliability in content analysis. *Journal of Applied Communication Research*, 27, 258–284.
- Potter, W. James, & Vaughan, Misha. (1997). Antisocial behaviors in television entertainment: Trends and profiles. *Communication Research Reports*, 14(1), 116–124.
- Quillian, M. Ross. (1968). Semantic memory. In Marvin L. Minsky (Ed.), *Semantic information processing* (pp. 216–270). Cambridge, MA: MIT Press.
- Rainoff, T. J. (1929). Wave-like fluctuations of creative productivity in the development of West-European physics in the eighteenth and nineteenth centuries. *Isis*, 12, 287–307.
- Rapoport, Anatol. (1969). A system-theoretic view of content analysis. In George Gerbner; Ole R. Holsti; Klaus Krippendorff; William J. Paisley; & Philip J. Stone (Eds.), *The analysis of communication content*:

- Developments in scientific theories and computer techniques* (pp. 17–38). New York, NY: Wiley.
- Reynolds, Henry T. (1977). *Analysis of nominal data*. Beverly Hills, CA: Sage.
- Rice, Ronald E. (1990). Hierarchies and clusters in communication and library and information science journals, 1978–1987. In Christine Borgman (Ed.), *Scholarly communication and bibliometrics* (pp. 138–153). Newbury Park, CA: Sage.
- Rice, Ronald E.; Borman, Christine L.; & Reeves, Bryan. (1988). Citation networks of communication journals, 1977–1985. *Human Communication Research*, 15, 256–283.
- Rice, Ronald E., & Richards, William D., Jr. (1985). An overview of network analysis methods and programs. In Brenda Dervin & Melvin J. Voigt (Eds.), *Progress in communication sciences* (Vol. 6, pp. 105–165). Norwood, NJ: Ablex.
- Richards, William D., Jr., & Rice, Ronald E. (1981). NEGOPY network analysis program. *Social Networks*, 3, 215–223.
- Riffe, Daniel, & Freitag, Alan. (1996). *Twenty-five years of content analyses in Journalism & Mass Communication Quarterly*. Paper presented at the annual meeting of the Association for Education in Journalism and Mass Communication, Anaheim, CA.
- Riffe, Daniel; Lacy, Stephen; & Fico, Frederick G. (1998). *Analyzing media messages: Using quantitative content analysis in research*. Mahwah, NJ: Erlbaum.
- Rijsbergen, Cornelis J. (1979). *Information retrieval*. London, England: Butterworth.
- Roberts, Carl W. (1989). Other than counting words: A linguistic approach to content analysis. *Social Forces*, 68, 147–177.
- Roberts, Carl W. (Ed.). (1997). *Text analysis for the social sciences: Methods for drawing statistical inferences from texts and transcripts*. Mahwah, NJ: Erlbaum.
- Rogers, Everett M., & Kincaid, D. Lawrence. (1981). *Communication networks: Toward a new paradigm for research*. New York, NY: Free Press.
- Rogers, Everett M., & Shoemaker, Floyd F. (1971). *Communication of innovations*. New York, NY: Free Press.
- Rogers, Richard. (2002). The Issue Crawler: The makings of live social science on the Web. *EASST Review*, 21, 3–4, 8–11. Retrieved from <http://www.easst.net/review/sept2002/issuecrawler.shtml>. Accessed 2017.9.9.
- Rogers, Richard. (2004). *Information politics on the Web*. Cambridge, MA: MIT Press.
- Rogers, Richard. (2010). Mapping public Web space with the IssueCrawler. Pp. 89–99 in B. Reber & C. Brossaud (Eds.). *Digital cognitive technologies: Epistemology and the knowledge economy*. London, England: Wiley.
- Rogers, Richard. (2017). Foundations of digital methods: Query design. Pp. 75–95 in M. T. Schäfer & K. van Es (Eds.). *The datafied society: Studying culture through data*. Amsterdam, The Netherlands: Amsterdam University Press.
- Rogers, Richard; Jansen, Fieke; Stevenson, Michael; & Weltevrede, Esther. (2010). *Mapping democracy: 2009—Access to online information and knowledge*. Amsterdam, The Netherlands: Digital Methods Initiative. Retrieved from <http://www.giswatch.org/fr/node/158>
- Romme, A. Georges L. (1995). Boolean comparative analysis of qualitative data. *Quality and Quantity*, 29, 317–329.
- Rosch, Eleanor. (1983). Prototype classification and logical classification: The two systems. In Ellin Kofsky Scholnick (Ed.), *New trends in conceptual representation: Challenges to Piaget's theory?* (pp. 73–86). Hillsdale, NJ: Erlbaum.
- Rosenberg, Stanley D.; Schnurr, Paula P.; & Oxman, Thomas E. (1990). Content analysis: A comparison of manual and computerized systems. *Journal of Personality Assessment*, 54, 298–310.
- Ruesch, Jürgen, & Bateson, Gregory. (1951). *Communication: The social matrix of psychiatry*. New York, NY: W. W. Norton.
- Sacks, Harvey. (1974). An analysis of the course of a joke's telling in conversation. In Joel Sherzer & Richard Bauman (Eds.), *Explorations in the ethnography of speaking* (pp. 337–353). London, England: Cambridge University Press.
- Salgado, Jesus F., & Moscoso, Silvia. (1996). Meta-analysis of interrater reliability of job performance ratings in validity studies of personnel selection. *Perceptual and Motor Skills*, 83, 1195–1201.
- Salmond, Ann. (1982). Theoretical landscapes. In David Parkin (Ed.), *Semantic anthropology* (pp. 65–87). New York, NY: Academic Press.
- Samarel, Nelda; Fawcett, Jacqueline; Krippendorff, Klaus; Piacentino, Jayne C.; Eliasof, Barbara; Hughes, Phyllis; . . . Ziegler, Elaine. (1998). Women's perceptions of group support and adaptation to breast cancer. *Journal of Advanced Nursing*, 28, 1259–1268.
- Saris-Gallhofer, Irmtraut N., & Morton, E. L. (1978). A validity study of Holsti's content analysis procedure. *Quality and Quantity*, 12, 131–145.
- Schank, Roger C., & Abelson, Robert P. (1977). *Scripts, plans, goals and understanding: An inquiry*

- into human knowledge structures. Hillsdale, NJ: Erlbaum.
- Schnurr, Paula P.; Rosenberg, Stanley D.; & Oxman, Thomas E. (1992). Comparison of TAT and free speech techniques for eliciting source materials in computerized content analysis. *Journal of Personality Assessment*, 58, 311–325.
- Schnurr, Paula P.; Rosenberg, Stanley D.; & Oxman, Thomas E. (1993). Issues in the comparison of techniques for eliciting source material in computerized content analysis. *Journal of Personality Assessment*, 61, 337–342.
- Schrodtt, Philip A.; Davis, Shannon G.; & Weddle, Judith L. (1994). KEDS: A program for the machine coding of event data. *Social Science Computer Review*, 12, 561–588.
- Schutz, William C. (1958). On categorizing qualitative data in content analysis. *Public Opinion Quarterly*, 22, 503–515.
- Scott, William A. (1955). Reliability of content analysis: The case of nominal scale coding. *Public Opinion Quarterly*, 19, 321–325.
- Searle, John. (1969). *Speech acts: An essay in the philosophy of language*. Cambridge, MA: Cambridge University Press.
- Sebeok, Thomas A., & Orzack, Louis H. (1953). The structure and content of Cheremis charms. *Anthropos*, 48, 369–388.
- Sebeok, Thomas A., & Zeps, Valdis J. (1958). An analysis of structured content with application of electronic computer research in psycholinguistics. *Language and Speech*, 1, 181–193.
- Sedelow, Sally Y. (1967). *Stylistic analysis*. Santa Monica, CA: SDC.
- Sedelow, Sally Y. (1989). The interlingual thesaurus model for global technical communication: Research results. (ERIC Document Reproduction Service No. ED324936).
- Sedelow, Sally Y., & Sedelow, Walter A., Jr. (1966). A preface to computational stylistics. In Jacob Leed (Ed.), *The computer and literary style* (pp. 1–13). Kent, OH: Kent State University Press.
- Sedelow, Sally Y., & Sedelow, Walter A., Jr. (1986). Thesaural knowledge representation. In *Advances in lexicology: Proceedings of the Second Annual Conference of the UW Centre for the New Oxford English Dictionary* (pp. 29–43). Waterloo, ON, Canada: UW Centre for the New Oxford English Dictionary.
- Selltiz, Claire; Jahoda, Marie; Deutsch, Morton; & Cook, Stuart W. (1964). *Research methods in social relations*. New York, NY: Holt, Rinehart & Winston.
- Shanas, Ethel. (1945). The American Journal of Sociology through fifty years. *American Journal of Sociology*, 50, 522–533.
- Shannon, Claude E., & Weaver, Warren. (1949). *The mathematical theory of communication*. Urbana: University of Illinois Press.
- Shapiro, Gilbert. (1997). The future of coders: Human judgments in a world of sophisticated software. In Carl W. Roberts (Ed.), *Text analysis for the social sciences: Methods for drawing statistical inferences from texts and transcripts* (pp. 225–238). Mahwah, NJ: Erlbaum.
- Shapiro, Gilbert, & Markoff, John. (1997). A matter of definition. In Carl W. Roberts (Ed.), *Text analysis for the social sciences: Methods for drawing statistical inferences from texts and transcripts* (pp. 9–31). Mahwah, NJ: Erlbaum.
- Shneidman, Edwin S. (1963). The logic of politics. In Leon Arons & Mark A. May (Eds.), *Television and human behavior* (pp. 178–199). Englewood Cliffs, NJ: Prentice Hall.
- Shneidman, Edwin S. (1966). *The logics of communication: A manual for analysis*. China Lake, CA: U.S. Naval Ordnance Test Station.
- Shneidman, Edwin S. (1969). Logical content analysis: An explication of styles of concluding. In George Gerbner; Ole R. Holsti; Klaus Krippendorff; William J. Paisley; & Philip J. Stone (Eds.), *The analysis of communication content: Developments in scientific theories and computer techniques* (pp. 261–279). New York, NY: Wiley.
- Shneidman, Edwin S., & Farberow Norman L. (1957). Some comparisons between genuine and simulated suicide notes. *Journal of General Psychology*, 56, 251–256.
- Shuman, Ronald B. (1937). Identification elements of advertising slogans. *Southwestern Social Science Quarterly*, 17, 342–352.
- Siegel, Sydney, & Castellan, N. John. (1988). *Non-parametric statistics for the behavioral sciences* (2nd ed.). Boston, MA: McGraw-Hill.
- Simon, Herbert A. (1957). *Models of man*. New York, NY: Wiley.
- Simonton, Dean K. (1994). Computer content analysis of melodic structure: Classical composers and their compositions. *Psychology of Music*, 22, 31–43.
- Simpson, George E. (1934). *The Negro in the Philadelphia press*. Unpublished doctoral dissertation, University of Pennsylvania.
- Singer, J. David. (1964). Soviet and American foreign policy attitudes: A content analysis of elite articulations. *Journal of Conflict Resolution*, 8, 424–485.
- Sixsmith, Judith, & Murray, Craig D. (2001). Ethical issues in the documentary data analysis of Internet posts and archives. *Qualitative Health Research*, 11(3), 423–432.

- Skalski, Paul D. (2002). Message archives. In Kimberly A. Neuendorf, *The content analysis guidebook* (pp. 215–218). Thousand Oaks, CA: Sage.
- Sloman, Steven, & Fernbach, Philip. (2017). *The knowledge illusion; Why we never think alone*. New York, NY: Riverhead Books.
- Sluga, Hans D., and David G. Stern (Eds.). (1996). *The Cambridge companion to Wittgenstein*. Cambridge, MA: Cambridge University Press.
- Smith, Charles P. (1992a). Introduction: Inferences from verbal material. In Charles P. Smith (Ed.), *Motivation and personality: Handbook of thematic content analysis*. (pp. 1–17). Cambridge, MA: Cambridge University Press.
- Smith, Charles P. (Ed.). (1992b). *Motivation and personality: Handbook of thematic content analysis*. Cambridge, MA: Cambridge University Press.
- Smythe, Dallas W. (1954). Some observations on communications theory. *Audio-Visual Communication Review*, 2, 248–260.
- So, Clement Y. K. (1988). Citation patterns of core communication journals. *Human Communication Research*, 15, 236–255.
- So, Clement Y. K. (1995). *Mapping the intellectual landscape of communication studies: An evaluation of its disciplinary status*. Unpublished doctoral dissertation, University of Pennsylvania, Annenberg School for Communication.
- Spearman, Charles (1904). The proof and measurement of association between two things. *American Journal of Psychology*, 15, 72–101.
- Speed, Gilmer J. (1893). Do newspapers now give the news? *Forum*, 15, 705–711.
- Spiegelman, Marvin; Terwilliger, Charles; & Fearing, Franklin. (1953a). The content of comics: Goals and means to goals of comic strip characters. *Journal of Social Psychology*, 37, 189–203.
- Spiegelman, Marvin; Terwilliger, Charles; & Fearing, Franklin. (1953b). The reliability of agreement in content analysis. *Journal of Social Psychology*, 37, 175–187.
- Stempel, Guido H., III. (1952). Sample size for classifying subject matter in dailies: Research in brief. *Journalism Quarterly*, 29, 333–334.
- Stempel, Guido H., III. (1961). The prestige press covers the 1960 presidential campaign. *Journalism Quarterly*, 38, 157–170.
- Stevens, Stanley S. (1946). On the theory of scales of measurement. *Science*, 103, 677–680.
- Stone, Philip J. (1975). Report on the Workshop on Content Analysis in the Social Sciences, Pisa, 1974. *Social Science Information*, 14, 107–111.
- Stone, Philip J.; Dunphy, Dexter C.; Smith, Marshall S.; & Ogilvie, Daniel M. (1966). *The General Inquirer: A computer approach to content analysis*. Cambridge, MA: MIT Press.
- Stone, Philip J., & Hunt, Earl B. (1963). A computer approach to content analysis using the General Inquirer system. In E. C. Johnson (Ed.), *American Federation of Information Processing Societies, conference proceedings* (pp. 241–256). Baltimore, MD: American Federation of Information Processing Societies.
- Street, Arthur T. (1909, July 25). The truth about newspapers. *Chicago Tribune*.
- Strodtzoff, Glenn G.; Hawkins, Robert P.; & Schoenfeld, A. Clay. (1985). Media roles in a social movement: A model of ideology diffusion. *Journal of Communication*, 35(2), 134–153.
- Sullivan, Dan. (2001). *Document warehousing and text mining*. New York, NY: Wiley.
- Surowieki, James. (2005). *The wisdom of crowds*. New York, NY: Doubleday.
- Szykiersky, Dorit, & Raviv, Amiram. (1995). The image of the psychotherapist in literature. *American Journal of Psychotherapy*, 49, 405–415.
- Tannenbaum, Percy H., & Greenberg, Bradley S. (1961). *J. Q. references: A study of professional change*. *Journalism Quarterly*, 38, 203–207.
- Taylor, Wilson L. (1953). “Cloze procedure”: A new tool for measuring readability. *Journalism Quarterly*, 30, 415–433.
- ten Have, Paul. (1999). *Doing conversation analysis: A practical guide*. Thousand Oaks, CA: Sage.
- Tenney, Alvan A. (1912). The scientific analysis of the press. *Independent*, 73, 895–898.
- Tesch, Renata. (1990). *Qualitative research: Analysis types and software tools*. Bristol, PA: Falmer.
- Thome, Helmut, & Rahlf, Thomas. (1996). Dubious cycles: A mathematical critique of the Namenwirth/Weber thesis on cultural change with an introduction into filter design methods. *Quality and Quantity*, 30, 427–448.
- Thompson, Stith. (1932). Motif-index of folk literature: A classification of narrative elements in folktales, ballads, myths, fables, mediaeval romances, exempla, fabliaux, jest-books, and local legends. Bloomington: Indiana University Studies.
- Tijssen, Robert J. W., & Van Raan, Anthony F. J. (1994). Mapping changes in science and technology: Bibliometric co-occurrences analysis of the R&D literature. *Evaluation Review*, 18, 98–115.

- Todorov, Tzvetan. [1988]. *Mikhail Bakhtin: The dialogical principle*. Wlad Godzich (Trans.). Minneapolis: University of Minneapolis Press.
- Toulmin, Stephen E. (1958). *The uses of argument*. Cambridge, MA: Cambridge University Press.
- Treichler, Paula A. (1988). AIDS, homophobia, and biomedical discourse: An epidemic of signification. In Douglas Crimp (Ed.), *AIDS, cultural analysis, cultural activism* (pp. 31–70). Cambridge, MA: MIT Press.
- Truzzi, Marcello. (1988). Sherlock Holmes, applied social psychologist. In Umberto Eco & Thomas A. Sebeok (Eds.), *The sign of three: Dupin, Holmes, Peirce* (pp. 55–80). Bloomington: Indiana University Press.
- Tuggle, C. A. (1997). Differences in television sports reporting of men's and women's athletics: ESPN SportsCenter and CNN Sports Tonight. *Journal of Broadcasting & Electronic Media*, 41, 14–24.
- Tukey, John W. (1980). Methodological comments focused on opportunities. In Peter R. Monge & Joseph N. Cappella (Eds.), *Multivariate techniques in human communication research* (pp. 489–528). New York, NY: Academic Press.
- Turow, Joseph. (1989). *Playing doctor: Television, story-telling, and medical power*. New York, NY: Oxford University Press.
- Valente, Thames W. (1993). Diffusion of innovations and policy decision-making. *Journal of Communication*, 43[1], 30–45.
- Van Dijk, Teun A. (1977). Text and context: Explorations in the semantics and pragmatics of discourse. New York, NY: Longman.
- Van Dijk, Teun A. (1991). *Racism and the press*. New York, NY: Routledge.
- Van Dijk, Teun A. (1993). Principles of critical discourse analysis. *Discourse & Society*, 4, 249–283.
- Van Rijsbergen, Cornelius J. (1979). *Information Retrieval*. London, England; Boston, MA: Butterworths.
- Vernadsky, Vladimir I. (1945). The biosphere and the noosphere. *Scientific American*, 33[1], 1–12.
- Wallace, Anthony F. C. (1961). *Culture and personality*. New York, NY: Random House.
- Walworth, Arthur. (1938). School histories at war: A study of the treatment of our wars in the secondary school history books of the United States and in those of its former enemies. Cambridge, MA: Harvard University Press.
- Wang, Rong. (2017). Marginality or strategic selection? Collaborative crowdsourcing through team building. Paper presented at the 67th Annual Conference of the International Communication Association, May 25–29, San Diego, CA.
- Waples, Douglas, & Berelson, Bernard. (1941). *What the voters were told: An essay in content analysis*. Unpublished manuscript, University of Chicago, Graduate Library School.
- Watzlawick, Paul; Beavin, Janet H.; & Jackson, Don D. (1967). *Pragmatics of human communication: A study of interaction patterns, pathologies, and paradoxes*. New York, NY: W. W. Norton.
- Weaver, Donald H.; Buddenbaum, Judith M.; & Fair, Jo Ellen. (1985). Press freedom, media, and development, 1950–1979: A study of 134 nations. *Journal of Communication*, 35[2], 104–117.
- Webb, Eugene J.; Campbell, Donald T.; Schwartz, Richard D.; & Sechrest, Lee. (1966). *Unobtrusive measures: Nonreactive research in the social sciences*. Chicago, IL: Rand McNally.
- Weber, Max. (1911). "Geschäftsbericht" in Verhandlungen des Ersten Deutschen Soziologietages vom 19.–22. Oktober 1910 in Frankfurt A. M. In *Schrift der Deutschen Gesellschaft für Soziologie* (pp. 39–62).
- Weber, René; Mangus, J. Michael; Huskey, Richard; Amir, Ori; Swanson, Reid; Gordon, Andrew; Khoosha-beh, Peter; Hahn, Lindsay; & Tamborini, Ron. (2017). *Extracting moral foundations from text narratives: Relevance, challenges, and solutions*. Paper presented at the 67th Annual Conference of the International Communication Association, May 25–29, San Diego, CA.
- Weber, Robert P. (1984). Computer-generated content analysis: A short primer. *Qualitative Sociology*, 7, 126–174.
- Weber, Robert P. (1990). *Basic content analysis* (2nd ed.). Newbury Park, CA: Sage.
- Weick, Karl E. (1968). Systematic observational methods. In Gardner Lindzey & Elliot Aronson (Eds.), *The handbook of social psychology* (pp. 357–451). Reading, MA: Addison-Wesley.
- Weitzman, Eben A., & Miles, Matthew B. (1995). *Computer programs for qualitative data analysis: A software sourcebook*. Thousand Oaks, CA: Sage.
- Wellman, Barry, & Berkowitz, Stephen D. (Eds.). (1988). *Social structures: A network approach*. Cambridge, MA: Cambridge University Press.
- Wells, Robert A., & King, Erika G. (1994). Prestige newspaper coverage of foreign affairs in the 1990 congressional campaign. *Journalism Quarterly*, 71, 652–664.
- Whissell, Cynthia M. (1980). The Dictionary of Affect in Language. In Robert Plutchik & Henry Kellerman (Eds.), *Emotion: Theory, research, and experience* (pp. 113–131). New York, NY: Academic Press.

- Whissell, Cynthia M. (n.d.). *Whissell's Dictionary of Affect in Language technical manual and user's guide*. Accessed at <http://www.hdcus.com/manuals/wdalman.pdf>
- Whissell, Cynthia M.; Fournier, M.; Pelland, R.; Weir, D.; & Makarek, K. A. (1986). Dictionary of Affect in Language: IV. Reliability, validity, and applications. *Perceptual and Motor Skills*, 62, 875–888.
- White, David M. (1964). The “gatekeeper”: A case study in selection of news. In Lewis A. Dexter & David M. White (Eds.), *People, society and mass communication* (pp. 160–172). New York, NY: Free Press.
- White, Paul W. (1924, May 31). Quarter century survey of press content shows demand for facts. *Editor and Publisher*, 57.
- White, Ralph K. (1947). Black Boy: A value analysis. *Journal of Abnormal and Social Psychology*, 42, 440–461.
- Wilcox, Dennis F. (1900). The American newspaper: A study in social psychology. *Annals of the American Academy of Political and Social Science*, 16, 56–92.
- Wiley, Malcolm M. (1926). *The country newspaper: A study of socialization and newspaper content*. Chapel Hill: University of North Carolina Press.
- Winograd, Terry, & Flores, Fernando. (1986). *Understanding computers and cognition: A new foundation for design*. Norwood, NJ: Ablex.
- Wittgenstein, Ludwig. (1922). *Tractatus Logico-Philosophicus*. New York, NY: Harcourt, Brace & Co.
- Wittgenstein, Ludwig. (1958). *Philosophical investigations* (English text of the 3rd edition). G. E. M. Anscombe (Trans.). New York, NY: Macmillan.
- Woelfel, Joseph. (1993). Artificial neural networks for policy research. *Journal of Communication*, 43(1), 63–80.
- Woelfel, Joseph. (1997). Attitudes as nonhierarchical clusters in neural networks. In George A. Barnett & Franklin J. Boster (Eds.), *Progress in communication sciences* (Vol. 13, pp. 213–227). Norwood, NJ: Ablex.
- Woelfel, Joseph, & Fink, Edward L. (1980). *The measurement of communication processes: Galileo theory and method*. New York, NY: Academic Press.
- Wolfers, Justin. (2017, August 20). Exposing a toxic milieu for women in economics. *New York Times*, p. 3.
- Wonsek, Pamela. (1992). College basketball on television: A study of racism in the media. *Media, Culture, and Society*, 14, 449–461.
- Woodward, Julian L. (1934). Quantitative newspaper analysis as a technique of opinion research. *Social Forces*, 12, 526–537.
- Wright, Charles E. (1964). Functional analysis and mass communication. In Lewis A. Dexter & David M. White (Eds.), *People, society, and mass communication* (pp. 91–109). New York, NY: Free Press.
- Wu, Alice H. (2017). *Gender stereotyping in academia: Evidence from economics job market rumors forum*. Retrieved from https://www.dropbox.com/s/v6q7g-fcbv9feef5/Wu_EJMR_paper.pdf?dl=0
- Yule, George U. (1944). *The statistical study of literary vocabulary*. London, England: Cambridge University Press.
- Zeldow, Peter B., & McAdams, Dan P. (1993). On the comparison of TAT and free speech techniques in personality assessment. *Journal of Personality Assessment*, 60, 181–185.
- Zhao, Xinshu; Liu, Jun S.; & Deng, Ke. (2012). Assumptions behind inter-coder reliability indices. In C. T. Salmon (Ed.), *Communication Yearbook 36*. New York, NY: Routledge.
- Zillmann, Dolf. (1964). *Konzept der Semantischen Aspektanalyse*. Unpublished manuscript, Institut für Kommunikationsforschung, Zurich, Switzerland. (Mimeo)
- Zipf, George Kingsley. (1935). *The psycho-biology of language: An introduction to dynamic philology*. Boston, MA: Houghton Mifflin.
- Zucker, Harold G. (1978). The variable nature of news media influence. In Brent D. Ruben (Ed.), *Communication yearbook 2* (pp. 225–240). New Brunswick, NJ: Transaction.
- Züll, Cornelia; Weber, Robert P.; & Mohler, Peter. (1989). *Computer-assisted text analysis for the social sciences: The General Inquirer III*. Mannheim, Germany: Zentrum für Umfragen, Methoden und Analysen.
- Zwick, Rebecca. (1988). Another look at interrater agreement. *Psychological Bulletin*, 103, 347–387.

INDEX

- Abductive inference, 43–45
 - alternative criteria and, 93
 - analytical constructs and, 179
 - correlations, based on
 - research question formulation and, 387, 389
 - structural validity and, 376
 - validating evidence for, 366–367
 - warrants and, 89, 405
- Abelson, R. P., 259
- Above-chance co-occurrences of nouns, 65
- Abstracting, statistical, 255
- Academia software, 34
- Academic Search Complete, 50
 - Accuracy, surrogacy, and decisiveness
 - overview, 323–325:
 - crowd members' judgments, 325–326
 - majorities or aggregates, 326–329
 - numerical examples, 329–331
 - standards, 199
- Actionable knowledge structures, 259
- Activity, semantic differential scales of, 380
- Actor-action-target-effect statements, 156
- Actor-action-target units, 112, 259
- Actor Networks Analysis, 35
- Adjective-verb ratio, 65
- Administrative information for recording/coding, 151
- Adorno, T. W., 78
- Advertising:
 - Appeals, classification of, 213–214
 - differentiation in
 - content analysis and, 49
 - mass media programming and, 77
 - method-driven analyses and, 398
 - one-way communication, 79
 - sentiment analysis and, 254
 - statistical association approaches to, 258
 - subliminal messages in, 15
 - thematic trends in, 56
 - word association databases and, 34
- Affective cognition theories, 377
- Affective meaning, semantic differential
 - scales of, 380
- Agenda setting, 81
- Agreement coefficient, 277. *See also* Reliability
- Ahn, L. von, 134
- AI (artificial intelligence), 255, 259
- Albig, W., 108
- Alexa, M., 219
- Alife software, 266
- Allen, L. E., 71, 259
- Allport, G. W., 13, 17, 32, 58, 70, 120
- Alpert, R., 316
- Alpha coefficients, 285
 - agreement for coding predefined units, 287–309
 - agreement for unitizing, 337–346
 - bootstrapped alpha values and, 353
 - Cronbach's *alpha*, 291, 321
 - Krippendorff's *alpha*, 277, 315, 319, 322
 - reliability data
 - reliability of data, 356–360
 - sample size, required, 354–356
 - statistical properties of, 351–360
 - statistical significance and, 353–354
 - variation required, 351–353*See also* Reliability
- Al-Qaeda terrorists, 253
- Alternative voices, 385
- Altheide, D. L., 22
- Amazon.com, 197
- Amazon's Mechanical Turk, 135–137
- American Educational Research Association, 362
- American Journal of Sociology*, 56
- American Psychiatric Association, 83, 388, 395
- American Psychological Association, 362–363
- American Reference Library, 230
- Analysis of Communication Content, The* (Berelson and Lazarsfeld), 14
- Analytical constructs, 178–193
 - certainty, sources of, 180–186
 - content analysis framework and, 42–43
 - confidence in
 - discriminant function
 - embodied practices
 - established theories
 - exper knowledge/experiences and
 - extrapolations
 - hypothesis testing of, 95
 - indices/symptoms
 - networked text, 240
 - previous successes/failures
 - recursive/autocorrelative functions, 268
 - role of, 178–179
 - trial and error development of, 96–98
 - types of, 186–191
 - uncertainty sources, 191–193
- Analytical/representational techniques.
 - See* Representations
- Anderson, A. H., 144
- Andsager, J. L., 206
- Annenberg School for Communication, 19
- Annenberg Television Script Archive, 230

- Antecedents of communication, 53
- Anxiety index, 65, 67
- AQUAD software, 269, 270
- Arafat, Y., 261
- Argamon, S., 223
- Argumentation theory, 1, 45
- Argument quality index, 380
- Aristotelian logic, 1
- Aristotle, 171
- Armstrong, R. P., 112
- Arnheim, R., 49
- Artificial intelligence (AI), 4, 19, 71, 255–259, 389
- Artificial life. *See* Alife software
- ASCII/ANSI characters, 247, 269,
- Ash, P., 62
- Asheim, L., 58
- Assertions content analysis, 52
- Associated Press, 255
- Associations, correlations, and cross-tabulations, 200–203
- Association structures:
 - collective associative memory, 20
 - contingency analysis and, 210–212
 - networks of, 264
 - spatial representation of, 212
 - word co-occurrences and, 211
 - See also* Statistical association approaches
- Atkinson, J. M., 111
- ATLAS.ti data analysis software, 20, 153, 385
- Attention measure (Budd), 164
- Attitude, 13–14, 16, 26, 32–34, 51–52, 66. *See also* Sentiment analysis
- Attitude object, 110–111
- Attribution analysis, 52, 207
- Attributions:
 - content analysis designs for, 99
 - content analysis of, 52, 207
 - facts in content analyses as, 81
- Auld, F., Jr., 67
- Authors, absence of, 4
- Authorship, 4, 10, 68, 187, 223
- Autocorrelative functions, 186
- Automatic Content Analysis* (Rand Corporation), 19
- Autonomous systems, dynamics of, 56

- Backstrom, L., 50
- Bakhtin, M., 72, 275
- Baldwin, A. L., 17, 188, 256
- Bales, R. F., 18, 19, 57, 110, 205, 251
- Barnett, G. A., 264
- Bateson, G., 58, 73
- Bell Curve* (Herrenstein and Murray), 60
- Bengston, D., 251–252
- Bennett et al.'s S, 358
- Berelson, B., 5, 12, 14–15, 17, 25–26, 33, 49, 52–54, 58–59, 63, 68, 76, 91, 98, 107, 137, 183, 188, 197, 199, 394
- Berger, P. L., 75–76
- Bermejo, F., 233
- Bernaut, E., 181, 376
- Best, M. L., 266–267
- Bias:
 - class, U.S. Constitution and, 12
 - content analysis and, 47, 104
 - context units and, 106
 - editorial, 44
 - English-language bias, 19
 - Inferences
 - Institutionalized messages
 - journalism and media, 33, 62, 63
 - politics and, 33, 62
 - sampling, 115–117
 - systematic sampling and, 119
 - varying probability sampling and, 120
 - See also* Interpretive standards
- Big data, 50. *See also* Text mining
- Bishop, S., 233
- Bliss, C. A., 50
- Blogosphere, 50
- Bock, M. A., 7, 10, 11, 12, 14, 15, 56, 68, 95, 100, 109, 119, 129, 155, 164, 169, 181, 183, 187, 190, 197, 221, 223, 224, 226, 227, 251, 253, 255, 258, 261, 317, 394, 395, 396
- Boole, G., 1
- Boolean accounts
 - analytical steps towards
 - cross-tabulations of
 - logic of operators, 227, 236–239
 - proximity operators
 - search engines, use in
 - See also* Text mining/ Information retrieval
- Bootstrapping probability distributions, 353
- Branching values, in tree diagrams, 171
- British cultural studies, 22
- Broadcasts:
 - analytical constructs about, 190–191
 - audience interpretation of, 15–16
 - preparatory propaganda and, 15
 - propaganda analysis
 - publicity efforts, inferences about, 34
 - See also* Mass media; Television; Television programming
- Broder, D. P., 65
- Brody, R. A., 57, 189
- Brouwer, M., 356
- Browsers, intelligent, 272
- Bruner, J. S., 120
- Budd, R. W., 164
- Buddenbaum, J. M., 204
- Bush, G. W., 226, 235, 242

- Cameron, W. J., 211
- Campbell, D. T., 47, 362–363, 378–379, 389
- Cappella, J. N., 108, 380–381, 395
- Carletta, J., 143
- Carley, K. M., 208, 261–264

- CATA (computer-aided text analysis). *See* Computer-aided text analysis (CATA)
- Categorical distinctions in unitizing, 109–110, 113
- Categorization:
- auto-categorization, 219
 - automatic coding and, 270
 - computer-aided content analysis and, 42
 - data language
 - hierarchical/nonhierarchical categorization, 175, 257, 270
 - measurement theory and, 129
 - nominal differences
 - reciprocal categorizations, 76, 78
 - recording units and, 104
 - semantic validity and, 366–367, 370–375
 - structural validity and, 367
- CatPac software, 256–257, 377–378, 398–400
- Causal inference modeling, 259
- Census data, 123
- CETA (Computer-aided Evaluative Text Analysis) software, 247–248, 260
- Chains of symbols theory, 11
- Chains of values, 168–169
- Chance
- estimates:
 - statistical independence of coders/observers, 317, 320
 - statistical independence of data and coded phenomena
 - population of pairs of values, 317
 - population estimates
 - standards, as 200–203
- Chandramouli, R., 223
- Channels, constraints of, 2
- Character string analysis, 220–227
- Cheng, Na, 223
- Cheremis charms, 56–57
- Cheremis folktales, 227
- Chicago Tribune*, 123
- Children's literature analysis, 13
- Chomsky, Noam, 66
- Chung, Cindy 253
- Cicchetti, D. V., 352, 358
- Circular causal feedback loops, 169
- Citation index, 57, 121, 274
- Citation network analysis, 57, 275
- Citizen dissatisfaction index, 65
- Clark, C. C., 59, 111
- Clark, R. A., 111
- Clinton, H., 242
- Closed variables, 162–163, 197
- Closure, constructs for, 148–149
- Cloze procedure, 11
- Clustering, 212–214
- Cluster sampling, 118, 120–122, 151–152
- Code-A-Text software package, 274
- CodeMap CATA software, 224
- Coders, 16, 128
- availability, frequency of
 - background knowledge and, 129
 - calibration of coders and, 134
 - cognitive abilities of, 130
 - independence of, 133
 - inter-coder agreement
 - literary competencies
 - qualifications for, 130–131
 - recording instructions
 - training for, 131–134
- Coding:
- computational content analysis by, 248–254
 - crowd coding
 - discriminate function
 - double coding
 - encoding/decoding paradigm
 - errors, sources of
 - latent meaning/silences, inferences from
 - multi-valued, 347–351
 - outsourcing, 4
 - problem-driven content analysis, 394–395
 - replicability of
 - single-valued
 - stability/consistence of
 - See also* Recording/coding
- Coding finite continua: overview, 337–338
- data from, 338–340
 - intersections of valued segments, 346
 - observed and expected coincidences, 340–342
 - partitions of, 342–343
 - relevant and irrelevant segments, distinction between, 343–344
 - single-valued segments independent of unitizing, 344–346
- Coding predefined units: overview, 289–293
- contested coefficients and correspondence, 315–323
 - difference functions for data with various matrices, 299–305
 - direct path to α -agreement, 293–296
 - examples of, 306–315
 - indirect path to α via coincidence matrices, 296–299
- Coefficient of imbalance, 13
- Coefficients. *See* Alpha coefficients; Coefficient of imbalance; Correlation coefficients; Reliability
- Cognitive processes:
- simulations of, 71
 - algorithmic nature of, 3
 - personal structure analysis and, 17
- Cognitive Science Laboratory, Princeton University, 235
- Cohen, J., 255, 312
- Cohen's κ statistic, 353
- Coincidence matrices:
- intersecting segments of, 340–341
 - retrievals reliability data in, 335
 - multi-valued sets of values in, 349

- pairs of values
 - path to α via, 296–299
 - single-valued
- Coincidences,
 - expected, 340–342, 344–345
 - observed
 - perfect
- Common platforms for computer aids, 272–273
- Communication:
 - antecedents of, 53
 - audits of, 64
 - channels/media of
 - digital
 - history of
 - institutional processes for inferences reinforced by, 79–80
 - mathematical theory of
 - meaning, container metaphor of
 - message content in context of, 53
 - networks
 - one-way
 - relations established by, 2–3
 - theories of networks of, 259
 - See also* Mass media
- Community-specific vocabularies, 35
- Community television projects, 63
- Components of content analysis
 - See* answering research questions about the contexts of texts
 - See* designing a
 - See* developing analytical constructs
 - See* narrating answers to research questions
 - See* recording phenomena in data
 - See* reducing data statistically or conceptually
 - See* sampling texts
 - See* unitizing texts
- Computational content analyses, 220, 246–268
 - additional information, provision of, 247
 - clerical text cleaning and, 246–247
 - CATA/dictionary approaches and, 248–254
 - computer-aided text entry and, 247
 - DICTION software, rhetorical qualities
 - General Inquirer
 - lemmatization
 - machine learning, 254–255
 - memetic approaches and, 264–268
 - networked text analysis, 239–246, 274–275
 - packaging textual material and, 247
 - preediting of raw text and, 247–248
 - semantic network approaches and, 258–264
 - sentiment analysis
 - statistical association approaches and, 256–258
 - See also* Computer-aided text analysis (CATA)
- Computational theory of a context, 40, 90
- Computer-aided Evaluative Text Analysis (CETA)
 - software, 247–248, 260
- Computer-aided text analysis (CATA):
 - dictionaries in, 249
 - image and portrayal analysis by, 208
 - justifying use of, 219
 - Keyword in Context (KWIC) analysis
 - readability measure in, 224
 - software packages for, 219–220
 - text cleaning for, 247
 - word accounts in, 221–222
 - word co-occurrences, 227
- Qualitative Data Analysis (QDA)
 - software, 268–272
- Computer aids for content analysis, 215–276;
 - overview, 215–220, 246–248
- character string accounts, 220–227
- coding approaches for, 248–254
- common platforms, 272–273
- intelligent browsers, 272
- intertextualities, 274
- machine learning, 254–255
- meaning, computational theories of, 273–274
- memetic approaches for, 264–268
- natural interfaces, 275–276
- semantic network approaches for, 258–264
- statistical association approaches for, 256–258
- text mining. *See* Text mining
- Conceivable worlds, 70
- Concept dictionaries, 235
- Concepts, co-occurrences of, 65, 99
- Concordance, 19, 106, 216, 222, 224–225, 270
- Software for, 222, 224, 385
- Confidence levels, 192
- Congressional Record*, 232
- Consequences of communication, 53
- Constants, definition of, 159
- Constructs for closure, 148–149
- Construct validity, 363, 367
- Construct vindication, 367
- Container metaphor, for messages, 4
- Contemporary content analysis, 4–6
- Content:
 - authorship and, 3, 4
 - communication history and, 2–3
 - electronic media and, 5
 - metaphors of
 - mining, 227–239, 287, 288, 331, 391, 393
 - picture theory of, 33
 - popular/simplistic notion of, 3
 - See also* Text
- Content Analysis* (journal), 49
- Content Analysis* (Krippendorff), 367
- Content analysis, conceptual foundations of, 24–50
 - analysts' context
 - analytical constructs for, 42–43
 - context of worlds of others, 40–42
 - contrasts and comparisons in, 46–50
 - definition, 24–27
 - epistemological foundations of, 27–31
 - examples of, 31–37
 - inferences for, 43–45
 - research questions for, 39–40
 - texts in, 38–39

- Unobtrusive/obtrusive
 - validating evidence for, 45–46
- Content analysis, history of, 10–23
 - computer text, 18–21
 - early, 12–14
 - generalized, 17–18
 - precursors to, 10–11
 - propaganda analysis, 14–17
 - qualitative approaches to, 21–23
 - quantitative newspaper, 11–12
- Content analysis, introduction to, 1–8
 - book grouping, 6–8
 - contemporary, 4–6
 - definition of, 1
 - empirically grounded method of, 1–2
 - symbols, contents, and intents in, 2–4
- Content analysis, practical guide to, 383–406
 - method-driven, 398–400
 - narration of results, 404–406
 - overview, 383–384
 - problem-driven. *See* Problem-driven content analysis
 - research design, application of, 403–404
 - research proposals, 400–403
 - text-driven, 384–386
- Content analysis design, 86–101
 - beyond content analysis, 98–101
 - components of, 87–91
 - preparatory, 94–98
 - quantitative and qualitative, 91–94
- Content Analysis in Communications Research* (Berelson), 14
- Content Analysis Reader, The* (Krippendorff and Bock), 7, 155, 225–226, 394–395
- Content validity, 363
- Context, 1, 4
 - analytical constructs for model of, 179
 - content analysis framework and, 40–42
 - operationalizing available knowledge of, 94–95
 - units in, 105–107, 113
- Context-insensitivity, 48
- Contingencies, 100, 210–212
 - correlations
- Contingency analysis, 11, 210–212, 248, 372
- Continuum (textual):
 - coding relevant segments of, 344–345
 - partition of, 338, 342–343, 348
 - unitizing, 286, 338, 339, 351
- Contractual function of research
 - proposals, 401–402
- Contributing conditions, correlations affected by, 41
- Convergent validity, 366, 368, 378, 380, 381
- Conversation:
 - analysis of, 22, 41
 - analytical constructs as, 189
 - coding categories for, 144
 - content analysis of, 100–101
 - inferences from, 73–74
 - texts as by-products of, 80–81
- Conversation analysis, 22, 28–29, 338
 - accuracy checks, 281
 - decision scheme for, 143
 - interaction triplets and, 111
 - recorded interactions, 189
 - speech, recording/transcription of, 73–74
 - syntactical distinctions and, 108
 - transcription conventions and, 73, 281
- Conversationalist* content analysis, 72
- Co-occurrences of concepts, 65, 99
- Cook, R., 367
- Coombs, C. H., 157
- Coordinator software package, 274
- Cooren, F., 275
- Cornebise, J., 137
- Corpus Linguistics, 230
- Correlation coefficients:
 - association and, 202
 - covariance measure, 315
 - extensional lists for, 141
 - information loss with, 89
 - linearity standard and, 201, 205
 - as misleading, 316
 - Spearman's rho, 301
 - "strength of relation" as, 140
- Correlations, 41–42, 200–203
- Correlative validity, 367–368, 378–381
- Council on Interracial Books for Children, 64, 112
- Country Newspaper, The* (Willey), 12
- Counts, 197–200
- Criterion-related validity, 363
- Critical discourse analysis, 71
- Critical Path Method [CPM], 398
- Cronbach, L. J., 321
- Cronbach's *alpha*, 291, 321
- Cross-tabulations, associations, and
 - correlations, 200–203
- Crowdcoding, 134–138, 326
- Crowd members' judgments, 325–326
- Crowdsourcing, 135
- Cryptography, 1
- Cuban missile crisis, 57, 189
- Cubes, to depict multidimensional
 - variations, 169–170
- Cuilenburg, J. J. van, 377
- Cultivation hypothesis, 100
- Culture:
 - commonalities/differences, comparisons of, 18
 - content, significance of, 2
 - cultural artifacts, analysis of, 28
- Curve fitting, 205
- Cushing, B., 120
- Cycles. *See* Recursions of values
- Dabbish, L., 134
- DAL (Dictionary of Affect in Language), 253
- Dale, E., 108
- Dale, T. A., 208
- Danet, B., 111

- Danford, C. M., 50
- Danielson, W. A., 65
- Danowski, J. A., 71, 258
- Darnton, R., 233
- Data:
- big data
 - categories
 - defined, 86
 - interval data/scales, 175, 177, 300, 302, 304, 315, 326
 - large volumes of, 5, 48–49, 72, 89
 - nominal data, 292, 304
 - ordinal data, 301, 326, 340
 - qualitative data, 17, 220, 268–272
 - quantitative data
 - ratio data, 299, 304–305, 326
 - reliability of, 332–336
 - type/token statistic, 89
 - unstructured data, 51
- See also Content analysis design; Data languages; Recording/coding; Sampling
- Data consortia, 21
- Databases, for text mining, 229–232
- Data languages, 157–177
- analytical efforts and, 157–159
 - constants
 - definitions of, 159–161
 - grammar of
 - levels of, 129
 - logic of
 - mathematical operations for, 176–177
 - metrics for, 172–176
 - values
 - variables in, 161–167
 - nominal, 167
 - ordered, 167–172
- See also Coders; Recording/coding; Semantics of data
- Data making, 88, 360
- Data points, 48, 54, 138, 158, 206
- Data reduction, 88, 90
- Davis, K. E., 248
- Davis, S. G., 248
- Dawkins, R., 264–265
- Decision schemes, 142–143
- Decision trees, 143, 171, 174–175
- Decisiveness. See Accuracy, surrogacy, and decisiveness
- Declarative definitions, 68
- Deductive inference, 43
- Delli Carpini, M. X., 243
- Denzin, N. K., 93
- De Ridder, J. A., 377
- Derivations of theorems, 54
- Design. See Content analysis designs; Research design
- Designations content analysis, 51–52
- Deutsch, M., 367
- DeWeese, L. C., III, 21, 216
- Diagnostic and Statistical Manual of Mental Disorders* (American Psychiatric Association), 82–83, 109, 388, 395
- Dibble, V. K., 61, 376, 389
- Dictionaries:
- CATA dictionaries, 248–249
 - computational content analyses and, 248–249
 - construction of, 249
 - customizable dictionaries, 250
 - Dictionary of Affect in Language (DAL), 253
 - DICTON content analysis software, 250
 - discourse-specific dictionary, 188
 - functional validity of, 253
 - lemmatization and, 248, 276, 399
 - Linguistic Inquiry and Word Count (LIWC)
 - Regressive Imagery Dictionary (RID), 250–251
 - rhetorical quality categories, 250
 - semantic validity of, 252
- See also Computational content analyses
- Diefenbach, D. L., 20
- Differences analysis, 58–61
- Digital communications, 3. See also Computer-aided text analysis (CATA); Electronic media; Internet
- Discomfort-relief quotient, 67
- Discourse analysis, 21–22
- Discriminant functions, developing, 96–98
- Discriminant validity, 378
- Discursively co-constructed reality, 3
- Dissonance theory, 183
- Distributional indices, 223
- Dodds, P. S., 50
- Doerfel, M. L., 264
- Dollard, J., 67
- Dovring, K., 10
- Dow Jones, Inc., 49
- Doyle, A. C., 44
- dtSearch CATA software, 222
- Dulles, J. F., 58
- Dunphy, D. C., 19, 30, 53, 149, 222, 225, 371
- Dynamic interdependency model, 57
- Dziurzynski, P. S., 213
- Eco, U., 44
- Ecological validity
- Economic explanations for inferences, 77
- Economics Job Market Rumors (EJMR) forum, 60
- Educational Resources Information Center (ERIC), 49, 230
- Efficiency:
- research questions for, 39
 - unitizing and, 112–114
- Eisenberg, E. M., 264
- EJMR (Economics Job Market Rumors) forum, 60
- Ekman, P., 108, 227
- Electronic media:
- early content analysis of, 13
 - complex content analysis and, 5
- See also Computer-aided text analysis (CATA); Internet; Search engines

- Eleey, M., 34
 Ellison, J. W., 68
 Emic analysis, 27
 Empirical grounding:
 Analytical constructs
 content analysis method of, 1–2
 inferred answers to research questions, 39–40
 Empirical validity, 362
 Enumeration, units of, 106
 Epi-search.com service, 236
 ERIC (Educational Information Resources Center), 49, 230
 ESA (Event Structure Analysis), 248, 260
 Ethnographic content analysis, 22, 27, 39
 Ethnomethodology, 22
 Etic analysis, 27
Eugene Onegin, 11
 European Consortium for Political Research, 56, 57
 Evaluations, as inference, 62–64
 Evaluative assertion analysis, 110, 185, 259, 260, 377, 395
 Evaluative semantic differential scales, 380
 Event Structure Analysis (ESA), 248, 260
 Evidentiary technique, 62
 Exhaustive data categories, in coding, 138–139
 Exhaustive values of variables, 162
 Experiments in sample size, 125–127
 Expert systems, 72
 Explicit variables, 163
 Ex post facto validation, 46
 Extensional lists, 140–141
 Extrapolations: overview, 54–55
 analytical constructs as, 186
 differences found in, 58–61
 patterns in, 56–57
 trends in, 55–56
 Extratextual phenomena, 65
 Eyewitness accounts, 278
 Eysenbach, G., 392

 Facebook.com, 34
 Face recognition software, 227
 Face validity, 361–362
 Factiva.com, 49
 Factor analysis, 158, 205–207
 Faden, J. M., 13, 58
 Fadner, R. H., 13, 62–63
 Fair, J. E., 204
 Fan, D. P., 20, 267–268
 FAO (United Nations' Food and Agriculture Organization), 35–36
 Farrall, K., 242–243
 Federal Communication Commission (FCC), 15, 46, 62, 64, 96, 148, 190, 381–382
 Federal Trade Commission (FTC), 395
 Feigl, H., 367
 Feinstein, A. R., 352, 358
 Feminist theory, 22
 Fico, F. G., 26, 361, 394

 "Filtering" methodology, 56
 "Fishing expeditions," 39
 Fiske, D. W., 378–379
 Flesch, R., 65, 68, 164, 224
 Focus group data, content analysis of, 32–33
 Folklore, 56, 112
 Foreign Broadcast Intelligence Service, of Federal Communications Commission (FCC), 15
 Formal data languages, 158
 Fourier analysis, 205
 Frequency indices, 66–68
 Friesen, W. V., 108, 227
 FTC (Federal Trade Commission), 395
 Functional validity, 253, 367, 377–378

 Gamson, W. A., 248–249
 Gantt charts, 398
 Garfield, E., 57
 Garthoff, R. L., 181, 376
 Geller, A., 106
 General Inquirer computer content analysis software, 19, 222, 249, 251, 253, 371
 George, A. L., 15, 17, 46, 148, 179, 187–188, 191, 381–382, 389
 Gerbner, G., 17, 19, 26, 28, 46, 58–59, 65, 70–71, 100, 119, 164, 193, 198, 251, 265–266
 German Sociological Society, 11
 Glaser, B., 385
 Gleser, G. C., 253
 Goebbels, J., 16, 47, 108, 188, 256, 261
 Goldstein, A. C., 316
 Goodwin, C., 22
 Google Books, 50
 Google.com, 230, 241
 Gottschalk, L. A., 253, 394
 Govcom.org Foundation, 241
 Go-words, 222, 224, 247
 GPoliTxts digital archive, 230z
 Graham, T., 174
 Grammar, in data languages, 159–160
 Graph theory, 259
 Gross, D., 164, 198
 Gross, L., 17, 46, 59, 100, 119, 164, 193, 198, 265
 Groth, O., 121
 Grounded theory, 1–2, 385
 Grouping values, 173–174
 Guetzkow, H., 337
 Gurevitch, M., 111
 Gwet, K., 353, 358
 Gwet's AC, 320

 Habitualization, 75
 Hacker concept, 208, 263
 HAL (hyperdimensional analogue to language), 257
 Harris, K. L., 50
 Hart, R. P., 250
 Harvard Psychosociological Dictionary, 251
 Harway, N. I., 256
 Hatch, D. L., 119

- Hawk, L., 65, 164
- Hawkins, R. P., 198
- Hayes, A. F., 293, 315, 353, 354
- Hays, D. C., 19, 69, 72, 258
- Heclo, H., 240
- Heise, D. R., 248, 253, 260
- Heisenberg's uncertainty principle, 47
- Hellsten, I., 35, 275
- Herma, H., 164–165
- Hermeneutical approaches, 39
- Herrnstein, R. J., 60
- Holsti, O. R., 19, 26, 53, 57, 104, 112, 189, 251, 337, 380, 394
- Homer (ancient Greek poet), 4
- Hoover Institution, 58
- Houle, P., 255
- Howe, J., 134
- Hsueh, P.-Y., 137
- Hubert, L., 322
- Hunt, E. B., 96–97, 180–181, 255
- Hyperdimensional analogue to language (HAL), 257
- Hypertextuality, 72, 240–241
- Hypotheses:
 - content analysis and, 52, 67, 99
 - inferences vs., 43
 - scientific, 39, 87
- Hypothesis testing:
 - analytical constructs, 95
 - content analysis results compared to other variables, 100
 - simulation, 145–146
- ICA (International Communication Association), 70, 264
- Identi.ca, 34
- Identification, as inference, 61–62
- "If-then" statements, 42
- Iker, H. P., 256
- Iliad* (Homer), 4
- Im, D. S., 65
- Image recognition software, 227
- Images, 207–210
- Imbalance, coefficient of, 63
- Implicit variables, 163
- Impressionistic content analysis, 197, 221
- Indices:
 - analytical constructs as, 186–188
 - argument quality, 380
 - latent semantic indexing (LSI), 257
 - readability
 - symptoms for inferences and, 65–69
 - variables mapped to, 161
- Indirect method of validation, 46
- Inferences, 51–83
 - abductive, 88–89, 365
 - comparing similar phenomena for, 99
 - content analysis framework of, 43–45
 - content analysis success in, 80–83
 - conversations for, 73–74
 - deductive
 - data languages and, 159
 - extrapolations for, 54–61
 - indices and symptoms for, 65–69
 - inductive, 43
 - institutional processes for, 74–80
 - linguistic re-presentations of, 69–72
 - psychodiagnostic, 108
 - rules of, 42, 261
 - standards for measurement of, 61–64
 - testing relationships for, 99–101
 - traditional overviews for, 51–54
- Inferential statistics, 103
- Information theory
 - Information about phenomena of
 - analytical interest
 - Mathematical Theory of Communication*
 - Noise/pollution of data
 - Redundancy, 75
- Information retrieval. *See* Text mining
- Innis, H. A., 78
- Institute for Propaganda Analysis, 15
- Institute for the Study of Coherence and Emergence (ISCE), 236
 - Institutional processes for inferences:
 - overview, 74–77
 - analytical constructs as, 189–191
 - communications reinforcement of, 79–80
 - explanations as, 77–78
 - generalizations for, 78–79
- Institutional realities, as facts in content analyses, 82
- Instrumental validity, 363
- Intelligent browsers, 272
- Interaction process analysis, 18, 57, 251
- Interactions, 74, 189
- Interactive-hermeneutic approaches, 20, 23
- Interfaces, natural, 275–276
- International Communication Association (ICA), 70, 264
- International Standard Book Numbers (ISBNs), 118
- Internet, 117–118, 121. *See also* Search engines
- INTERNET ARCHIVE, 50
- Interpretation, logic of, 44
- Interpretive approaches, 22, 39
- Interpretive standards, 198
- Intertextuality, 70, 116, 121–122, 274, 384–385
- Interval metrics, 175–177, 302–303, 305, 315
- Interviewing, 32–33, 146–148
- Isard, A., 144
- Isard, S., 144
- ISBNs (International Standard Book Numbers), 118
- ISCE (Institute for the Study of Coherence and Emergence), 236
- IssueCrawler software suite, 239, 241–243
- Jackson-Beeck, M., 164, 198
- Jahoda, M., 367

- Jamieson, K. H., 22, 65, 108
 Janda, K., 227, 230
 Janis, I. L., 13, 30, 46, 53, 61–63, 128–129, 377
 Jansen, F., 50
 Janson, S., 316
 Janson and Vegelius's *C*, 316
 Joke telling, 22
 Josephson, J. R., 44
 Josephson, S. G., 44
 Journalism:
 Accuracy in reporting, evaluation of
 bias and, 161
 evaluations of, 62–64
 fairness/balance standard, 13
 misrepresentations and,
 newspapers, early 20th century, 11–12
 objectivity and, 97
 privacy rights and,
 public opinion and, 82
 sampling validity and, 369
 surveillance function of, 79
 truthful reporting standard and, 64
 See also Mass media; Quantitative
 newspaper analysis
Journalism & Mass Communication Research, 315
 Judgments:
 algorithm vs. human, 282
 coders and, 129, 164, 165, 253, 321, 324, 374
 consensus vs. individual, 283
 content analysis definition, 128–129
 crowd, 138, 325–326
 decision schemes and, 142–143
 expert panels, 324
 legitimized standards, 64
 majority, 326, 328
 quantitative vs. qualitative, 107
 replications of, on relevance of retrieval
 results, 336
 research results, 406
 semantic validity and, 399
 Justification clause, 110

 Kansas Events Data System (KEDS), 248
 Kaplan, A., 169–170, 398
 Kaplan, D., 106
 Kappa statistic, 357
 Katz, E., 111
 Kay, K., 155
 KEDS (Kansas Events Data System), 248
 Kenski, K., 232
 Keywords in context (KWIC) analyses, 20, 99, 208,
 224–225, 276, 371
 Kim, J., 248–249
 King, E. G., 123
 Kinship terminology, 54
 Klausner, S. Z., 147
 Kleinberg, J., 50
 Kleinnijenhuis, J., 260–261, 263, 377

 Kloumann, I., 50
 Koch, G. H., 357
 Koppel, M., 223
 Kowtko, J. C., 144
 Kracauer, S., 17, 109
 Kraemer, H. C., 354
 Krendal, E. S., 65
 Krippendorff, K., 3, 7, 10, 11, 12, 14, 15, 19, 26, 31, 34,
 37, 56, 61, 68, 77, 82, 95, 100, 109, 119, 129, 155,
 159, 164, 169, 181, 182, 183, 187, 188, 190, 197,
 200, 213, 219, 221, 223, 224, 226, 227, 251, 253,
 255, 258, 261, 271, 274, 291, 293, 315, 317, 319,
 320, 322, 323, 337, 338, 347, 352, 353, 354, 367,
 371, 380, 385, 394, 395, 396
 Krippendorff's *alpha*, 277, 291, 315, 322.
 See also Reliability
 Kriss, E., 164
 Kuder, G. F., 321
 Kuder and Richardson's *KR-20*, 321
 KWIC (keywords in context) analyses, 20, 99, 208,
 224–225, 276, 371

 Labov, W., 57
 Lacy, S., 26, 122, 315, 361, 394
 Laffal, J., 235, 252
 Lakoff, G., 141
 Language
 Artificial intelligence, research on
 Data
 Conscious use of
 Grammatical patterns in
 Greek interest in
 Linguistic variables
 Naming and
 Speech and
 System of
 Verbal corpora, systems theory of
 Landis, H. H., 357
 Lasorsa, D. L., 65
 Lasswell, H. D., 13–15, 17, 56, 67, 79–80, 91, 106–108,
 123, 146, 159, 169–170, 197, 251, 259, 265
 Latent semantic indexing (LSI) algorithm, 257
 Lazarsfeld, P. F., 5, 12, 14, 33, 49, 91, 370
 Learning theories, 67, 69
 Leetaru, K. H., 235–236
 Legal explanations for inferences, 77
 Lehmann, P., 137
 Leigh, L. E., 358
 Leites, N., 52, 181–182, 376
 Lemmatization process, 233, 248, 276, 399
 Leskovec, J., 50
Letters From Jenny (Allport), 70
 Levine-Donnerstein, D., 364
 Levinson, M., 68
 Lexical database, 235
 LexisNexis, Inc., 49, 230–233, 239
 Leydesdorff, L., 35, 275
 LibraryThing software, 34

- Lincoln, Y. S., 93
- Lindsay, R. K., 259
- Linguistic Inquiry and Word Count (LIWC) dictionary, 250–251, 253, 396
- Linguistic re-presentations, 69–72
- Lippmann, W., 243
- “Lippmannian device” to map controversies, 243–246
- Listenability scores for TV programming (Hawk), 164
- Literary theory, 22, 171
- LIWC (Linguistic Inquiry and Word Count) software, 250–251, 253, 396
- Löbl, E., 11
- Loeventhal, L., 56
- Logic, in data languages, 159–160
- Logical fallacies, 149
- Logic of content analysis designs. *See* Content analysis designs
- Logic theory machine (Newell and Simon), 71
- Lombard, M., 358
- Lorr, M., 133
- Los Angeles Times*, 123
- Lowe, W., 220
- Lowell, E. L., 111
- LSI (latent semantic indexing) algorithm, 257
- Luckmann, T., 75–76
- Lynch, K., 70
- Maccoby, N., 120
- Machine learning, 61, 96, 254–255
- MacQueen, K. M., 155
- Magnitudes and scales for data, 143–145
- Mahl, G. F., 73, 139, 143
- Malisazobel, 137
- Manifest characteristics of communication, 53
- Map development, 70
- Markoff, J., 26
- Markov, A., 11
- Martindale, C., 250
- Marvanyi, G., 70–71
- Marxist approaches, 22
- Massachusetts Institute of Technology (MIT), 263
- Mass communication:
- content analysis of, 33–34
 - indices used in, 65–66
 - message system analysis of, 58
 - one-way theory of communication conformity with, 79
 - public opinion poll correlation with, 81
- Mass media, 1, 3
- accurate representation standard and, 199
 - bias, evaluation of, 62–63
 - cultivation hypothesis and, 100
 - democracy, challenges to, 13
 - ethical standards and, 77
 - ideological/class orientations, reproduction of, 58
 - industrial producers, imprint of, 26
 - institutionalization of behavior and, 75
 - interactive media and, 79
 - mass communication research, indices in, 65–66
 - message systems analysis and, 59
 - new mass media, 15
 - one-way theory of communication and, 79
 - ownership, effects of, 77
 - political election campaigns and, 62
 - public agendas and, 42
 - public opinion, 15
 - representativeness, operationalization of, 63
 - social norms and, 33
 - technological perspective on, 77–78
 - See also* Journalism; Quantitative newspaper analysis
- Mathematical operations, data languages and, 176–177
- Mathews, B. C., 11–12
- Matrices:
- canonical form of reliability data, 289–290
 - coincidence matrix, 293, 311–315, 335–336, 343
 - contingency matrix, 258, 309, 315, 316, 319
 - expected coincidences, matrix of, 297–299, 304, 307, 310–312, 340–345, 349, 351–352
 - multitrait-multimethod matrix, 379–380
 - observed coincidences, matrix of, 296–299, 303–305, 307, 310–313, 340–341
 - reliability data matrix, 294, 325
 - values-by-units matrix, 294
 - See also* Reliability
- MAXQDA qualitative data analysis software, 153
- McAdams, D. P., 20
- MCCA (Minnesota Contextual Content Analysis), 250
- McClelland, D., 111, 114, 391
- McDiarmid, J., 13
- McLellan, E., 155
- McMillan, S. J., 110, 117
- McNair, D. M., 133
- Meaning:
- computational theories of, 273–274
 - container metaphor for, 4, 26, 28, 70, 80
 - contextual factors in, 35
 - extratextual meaning, 39
 - human interactions, circularity of, 169
 - inherent meaning, 16, 28
 - repetition, semantic satiation and, 69
 - representational theory of, 272
 - See also* Content analysis entries
- Measurement theory
- MECA software, 262
- Media
- Digital
 - electronic
 - interactive
 - Mass

- One-way/two-ways Technologies
- Melville, P., 137
- Memes
- Memetic approaches, 264–268
- Merriam-Webster's Collegiate Dictionary*, 161
- Merrill, J. C., 63
- Merritt, R. L., 280
- Merten, K., 31
- Messages, 2, 4
- Message system analysis, 58
- Method-driven content analysis, 384, 398–400
- Metrics for data languages, 172–176
- Metrics/scales of measurement, 168
 - chains, 168
 - circular, 169
 - difference functions and,
 - grouping unordered sets of values and, 173–174
 - interval metric, 173, 175–176, 302–303
 - mathematical operations and, 176–177
 - nominal metric,
 - ordinal metric, 173–175, 301–302
 - polar attributes, 145, 163
 - ratio metric, 176–177, 299–300, 303–304
 - strengths, 305
 - verbally anchoring, 163
 - See also* Data languages; Ordering of values; Variables
- Microsoft Word, 249, 385
- Miles, J., 219
- Miller, G. A., 20, 129
- Military intelligence
 - FCC predictions of Nazi Germany's, 15–16
- Milstein, B., 155
- Minnesota Contextual Content Analysis (MCCA), 250
- MIT (Massachusetts Institute of Technology), 263
- Mohler, P., 251
- Monge, P. R., 264
- Montgomery, M., 225, 226
- Moravian Brethren, 11
- Morgan, M., 17, 46, 59, 100, 119, 164, 193, 198, 265
- Morton, A. Q., 68
- Morton, E. L., 380
- Moscoso, S., 316
- Mosteller, F., 68, 187, 223
- Mowrer, O. H., 67
- Multidimensional scaling, factor analysis and, 205–207
- Multiple regression analysis, 203
- Multiple-unit content analyses, 152
- Multitrait-multimethod matrix, 379–380
- Multiuser discussions (MUDs), 21
- Multi-valued coding, 347–351
- Multivariate analysis techniques, 203–205
- Murray, C. D., 364, 392
- Murray, Charles, 60, 67
- Murray, E. J., 64
- Murray, H. A., 20, 111
- Mutually exclusivity of values in variables, 138–139, 162
- Nacos, B. L., 20
- Namenwirth, J. Z., 56, 169, 204–205
- Narratives:
 - content analysis results as, 404–406
 - recording units for, 104
 - research question answers as, 88–90
- National Council on Measurement in Education, 362
- Natural interfaces, 275–276
- Natural language, 19
- NEGOPY software program, 258
- Negotiation studies, 22, 74, 133, 189
 - Networked text analysis: overview, 239–240
 - analytical constructs for, 240
 - hypertextuality in, 240–241
 - IssueCrawler software suite for, 241–243
 - "Lippmannian device" for, 243–246
 - website linkages in, 274–275
- Neuendorf, K. A., 394
- Newell, A., 19, 71
- New School for Social Research, 15
- Newsgroups, 21
- Newspapers:
 - quantitative content analysis of, 11–14, 21, 62, 120
 - See also* Journalism, Mass media
- New York Times*, 33, 119, 123, 145, 232–233
- Niederer, S. M. C., 189, 243, 246, 275
- Nir, L., 380, 395
- Nominal variables, 167
- "Noosphere," 55
- North, R. C., 57, 189
- Nunnally, J. C., 110, 183, 377, 395
- NVivo data analysis software, 20, 153, 270, 385
- Obama, B., 242–243
- Observations:
 - contaminated observations, 47
 - direct, inability to use, 31
 - research questions and, 39–40
 - inter-observer disagreement
 - intra-observer disagreement, 278
- Ogilvie, D. M., 19, 30, 53, 149, 222, 371
- One-way theory of communication, 79
- Open variables, 162–163, 197
- Open-ended interview questions
- Operationalizing available knowledge of context, 94–95
- Ordered variables, 167–172
- Ordering of values, 167–168, 304–305. *See also* Data languages; Metrics/scales of measurement; Variables
- Ordinal metrics, 173–175, 177
- Organizational information for recording/coding, 151–153

- Organizations
 - Organizational communication audits
 - Structures of
 - See also Institutions
- Orzack, L. H., 56
- Osgood, C. E., 11, 26, 18, 57, 65, 95–97, 100, 108, 110, 143, 144, 183–185, 188, 205, 210, 211, 212, 251, 253, 256, 259, 260, 264, 337, 372, 373, 377, 380, 395
- O'Sullivan, T. C., Jr., 139
- Outsourcing coding, 4
- Oxford Text Archive, 230
- Oxman, T. E., 20
- Packaging the Presidency* (Jamieson), 22
- Page, L., 241
- Paisley, W. J., 19, 187, 251
- Paley, W. B., 224
- Palmquist, M. E., 208
- Parsons, T., 79, 205
- Path analysis, 204
- Patterns analysis, 56–57, 75–76
- Patterns, extrapolations to infer, 56–57
- PBS (Public Broadcasting Service), 34
- Pearson's product moment correlation coefficient
- Pearson's intraclass correlation coefficient (coefficient of racial likeness)
- Peirce, C. S., 65
- Peircean logic of abduction, 44
- Péladeau, N., 249
- Peled, T., 111
- Pennebaker, J. W., 223, 253
- Perception clause, 110
- Perreault, W. D., 358
- Personal structure analysis, 17
- PERT (Program Evaluation and Review Technique)
 - flowcharts, 398
- Phenomena
 - analysed and inferred from data
 - of content analytical interest
 - reliably encoded in data
 - unstructured, inferences from, 99–100
- Physical distinctions in unitizing, 107–108, 113
- Piault, C., 160–161
- Pierce, B. L., 49
- Plagiarism case analysis, 209–210, 265
- Political campaigns:
 - conversation analysis of, 189
 - lack of civility in ads, 362
 - negative ads in, 65, 232
 - semantic validity and, 371
 - simulations applied to, 71
 - speeches delivered during, 103
- Political explanations for inferences, 77
- Political speech analysis, 58, 102–103
 - Fourier analysis and, 204–205
 - implications of political speech and, 137–138
 - KWIC lists and, 208
 - logical fallacies and, 149
 - sentiment analysis and, 254
 - Soviet leadership succession study and, 182
 - structural validity and, 376
- PoliTxs digital archive, 230
- Pool, I. de S., 18, 52, 58–59, 68, 188
- Popping, R., 219
- Portrayals, 207–210
- Potency, semantic differential scales of, 380
- Potter, W. J., 364
- Powers, A., 206
- Practical guide to content analysis. *See* Content analysis, practical guide to
- Pragmatical content analysis, 51
- Pragmatic validity, 367
- Precision, measure of, 331–333, 335
- Predictive validity, 368, 381–382
- Prediger, D. J., 316, 318
- Preparatory content analysis designs, 94–98
- Price, V., 380, 395
- Princeton University, 235
- Probability sampling, 118–120
 - Problem-driven content analysis: overview, 384, 386–387
 - analytical procedures for, 395–396
 - coding categories and recording instructions for, 394–395
 - relevant texts for, 390–392
 - reliability standards and statistical significance levels for results of, 396–397
 - research questions for, 387–388
 - resource allocation for, 397–398
 - sampling units in relevant texts for, 392–393
 - stable correlations for, 388–390
- Productivity, unitizing and, 112–114
- Profiles, 207–210
- Program Evaluation and Review Technique (PERT)
 - flowcharts, 398
- Project Gutenberg, 230
- Propaganda,
 - content analysis of, 14–17, 31–32, 46
 - intelligence
 - preparatory
- Propaganda Analysis* (Berelson), 15
- Propositional distinctions in unitizing, 110–111, 113–114
- Proquest NewspaperARCHIVE, 50
- Psychodiagnostic inferences, 108
- Psychology:
 - association psychology, 256
 - attitude, concept of, 13
 - content analysis, utilization of, 18, 32, 53, 144, 169
 - data sources for research, 32–33
 - psychological tests, 46, 95, 175, 316, 362, 363, 365, 367, 371
 - Rorschach tests, 347
 - rumor transmission studies and, 13
 - Thematic Apperception Test (TAT), 17, 20, 32, 111

- Psychology of Newspapers: Five Tentative Laws, The* (Allport and Faden), 13
- Public behaviors, 82
- Public Broadcasting Service (PBS), 34
- Public opinion:
- mass communications correlation with, 81
 - media crime reports correlation to, 100
 - memetic approach to predict, 267–268
 - political reality dependency on, 82
 - social stereotypes and, 13
- Publizistik* (newspaper science), 11
- Purposive sampling, 122–123
- Pushkin, A., 11
- Qualitative content analysis:
- designs for, 91–94
 - discourse analysis
 - ethnographic content analysis
 - history of, 21–23
 - interpretation
 - nominal variables in, 167
 - reflexivity of the analyst
 - rhetorical analysis
 - software for, 153
 - texts as connected discourse in, 70–71
- Qualitative data analysis (QDA) software, 268–272, 385–386
- Quantitative content analysis:
- counts
 - designs for, 91–94
 - statistical indices
 - newspaper, 11–12, 16–17
- Quantitative newspaper analysis, 11–14, 21, 62, 120
- Quantitative Newspaper Analysis as a Technique of Opinion Research* (Woodward), 13
- Quasi-species, 266
- Queries, 228
- RADIR (Revolution and the Development of International Relations, Hoover Institution), 58–59
- Rahlf, T., 56
- Rainoff, T. J., 14
- Rand Corporation, 19
- Random sampling, 118, 122
- Ranke, L. von, 4–5
- Rapoport, A., 55, 65–66, 265–266
- Ratio metrics, 176–177
- Re-presentations
- Alternative readings of texts
 - Analytical constructs
 - Cognitive processes, simulation of
 - Conceivable worlds, provision of
 - Conversationalist, text implications
 - Recalling from memory
 - Social cognition/public mind, exploration
- Reactive situations, 47
- Readability yardstick (Flesch), 65, 68, 108, 164, 224
- “Reality” television shows, 64
- Recall, measure of, 331–335
- Reciprocal categorizations, 76
- Recording/coding, 128–156 overview, 149–151
- administrative information in, 151
 - crowdcoding, 134–138
 - definition of, 88–89
 - function of, 128–130
 - organizational information in, 151–153
 - qualifications for, 130–131
 - relevant information in, 153–156
 - representing phenomena of analytical interests
 - semantics of data. *See* Semantics of data
 - semantic validity
 - training for, 131–134
 - units for, 104–105
- Records, 149–153. *See also* Recording/coding
- Recursions of values, 169
- Recursive analytical constructs, 268
- Reducing data, 88–89
- Redundancy, information theoretical notions of (Shannon), 75
- Regression analyses, 42, 137, 187, 201–203, 238, 255, 315. *See also* Multiple regression analysis
- Regressive Imagery Dictionary (RID), 250–251
- Relevance sampling, 122–123
- Reliability, 24, 73, 277–360
- accuracy, surrogacy, and decisiveness, 323–331
 - of coding predefined units, 289–315
 - of coder/observer performance, 137
 - conditions for generating reliability data, 283–289
 - context units and, 106, 113
 - crowdcoding and, 136–137
 - decision schemes and, 143
 - decisiveness, 287, 296, 325, 327–329
 - distinctions, of
 - majorities, 296, 327–329
 - measured by agreement among observers
 - of multi-valued coding 347–351
 - paradox, alleged, 358
 - predefined units, 285–286, 337, 338, 340, 347, 349
 - productivity, efficiency and, 112–113
 - purpose of, 277–280
 - relationship to validity
 - reliable data, defined, 277
 - replicability
 - reproducibility, 384
 - scholarly journal requirements for, 405
 - statistical properties of *alpha*, 351–360
 - surrogacy and, 320, 323–327
 - tests of, uses of, 136–137
 - text mining and information retrieval, 331–336
 - types of, 280–283
 - ultimate test of, 359
 - of unitizing (partitioning) finite continua, 112–114
 - of unitizing and coding finite continua 337–346

Reliability coefficients:

- Bennett et al.'s *S*, 316, 319, 321, 358
- Cohen's *kappa*, 312, 317, 318, 319, 320, 353, 354
- correlation and,
- Cronbach's *alpha*, 291
- expected percent agreement,
- Fleiss' *K*
- Gwet's *AC*, 320
- Krippendorff's *alpha*, 277, 315, 319, 322
- observed/expected agreement,
- Pearson's product-moment coefficient, 315, 322
- Pearson's intraclass correlation coefficient (coefficient of racial likeness)
- percent agreement,
- Perreault and Leigh's *Ir*
- Scott's *pi*, 312, 316, 317, 318, 319, 322, 353, 354
- Spearman's rank correlation coefficient *rho*, 301, 322
- See also Alpha agreement; Disagreement; Reliability

Reliability data:

- test-retest, (stability/consistency) 280, 281
- test-test, (replicability) 280, 281, 283
- test-standard, (accuracy or surrogacy) 280, 281

Replicability:

- accuracy vs.
- alpha coefficients and, 287, 329–331
- coding, of, 134, 136
- defined, 281
- ensuring, 394
- judgments on relevance of retrievals, 336
- memes and, 265
- reliability and, 280, 359
- research, 5, 24–25
- research design, 86
- search, 334–336
- stability/consistency vs., 279, 323
- surrogacy vs., 309, 325
- tests of, 360
- unitizing, of

Representations, 196–214

- analytical constructs as, 188–189
- clustering, 212–214
- contingencies and contingency analysis, 210–212
- counts, 197–200
- cross-tabulations, associations, and correlations, 200–203
- factor analysis and multidimensional scaling, 205–207
- images, portrayals, semantic nodes, and profiles, 207–210
- multivariate techniques of, 203–205

Research design:

- application of, 403–404
- collaborative teamwork and, 5
- design logic and, 86
- methodology, purpose of, 4–5
- See also Content analysis designs

Research questions:

- content analysis framework and, 39–40
- problem-driven content analysis, 387–388
- variability in targets of, 191–192

Research reports:

- outline for, 405–406
- results, narration of, 404–406
- purpose of, 404
- searching for, 228

Resource allocation for problem-driven content analysis, 397–398

Results narration, in content analysis, 404–406

Reuters news wire service, 49, 231

Revolution and the Development of International Relations (RADIR, Hoover Institution), 58–59

Rhetorical analyses, 22

Rhetorical function of research proposals, 400–401

Rhetorical theory, 1

RID (Regressive Imagery Dictionary), 250–251

Richardson, M. W., 321

Rietberg, E. M., 377

Riffe, D., 26, 125–126, 361–362, 394

Rijsbergen, C. J., 222, 332

Roberts, C.W., 110, 394

Rogers, E. M., 50, 117

Rogers, R., 241, 243

Roget's *Thesaurus*, 141

Romme, A., 239

Rosch, E., 142

Rosenberg, M. J., 259

Rosenberg, S. D., 20, 251

RQDA content analysis software, 385

Ruesch, J., 73, 82

Rules of inference, 42

Rumor transmission, 13

Sabghir, F. O., 120

Sacks, H., 22

Salgado, J. F., 316

Salter, P. J., 58–59, 63, 199

Sampling, 115–127

- bias
- census, 123
- cluster, 120–121
- convenience, 124
- definition of, 88–89
- probability, 119–120
- random, 118
- relevance, 122–123
- relevant textual units for, 392–393
- representative sample
- sample size, 124–127
- sampling units
- snowball, 121–122
- statistical properties of *alpha* and, 354–356
- stratified, 119
- split-half
- systematic, 118–119

- theory in, 115–117
- units of, 103–104
- Sampling validity, 366, 368–370
- San Francisco Suicide Prevention Center, 96
- Saporta, S., 110, 183, 377, 395
- Saris, W. E., 380
- Saris-Gallhofer, I. N., 380
- Saussure, F. de, 54
- Scales and magnitudes for data, 143–145
- Schank, R. C., 259
- Schizophrenia, index of, 65
- Schnurr, P. P., 20
- Schoenfeld, A. C., 198
- Schrodt, P. A., 248
- Schutz, W. C., 143
- Schwartz, R. D., 47, 389
- Science Citation Index*, 121
- Scientific hypotheses, 39
- Scott, W. A., 312, 316, 317, 318, 319, 322, 353, 354
- Scott's *pi* statistic, 312, 316–319, 322, 353, 354
- Search engines, 228–229
 - Boolean specifications
 - fuzzy search options and, 235
 - intelligent browsers and, 272
 - Internet searches and, 23, 110
 - phonetic similarities and, 235
 - precision/quality measure for, 333–336
 - recall/quality measure for, 333–336
 - reliability of, 336
 - search language, query formulation and, 233–236
 - stemming similarities and, 236
 - text mining and, 228–229, 236–239
 - thesaurus similarities and, 236
 - typographical similarities and, 235
 - See also Internet; Text mining
- Searches, replication of, 334–336
- Searle, J., 163
- Sebeok, T. A., 19, 56, 227
- Sechrest, L., 47, 389
- Sedelow, S. Y., 20, 225, 248
- Sedelow, W. A., Jr., 20
- Selltiz, C., 367
- Semantical content analysis, 51
- Semantically valid queries for text
 - mining, 232–235
- Semantic aspect scales, 145, 168
- Semantic differential dimensions, 57
- Semantic differential scales, 18, 143, 163, 251, 380
- Semantic network analysis, 35, 71, 258–264, 395–396
- Semantic nodes, 207–210
- Semantics of data:
 - constructs for closure, 148–149
 - decision schemes, 142–143
 - examples of, 141–142
 - exhaustive and mutually exclusive, 138–139
 - extensional lists, 140–141
 - hypothesis testing simulation, 145–146
 - interviewing simulation, 146–148
 - magnitudes and scales, 143–145
 - verbal designations, 139–140
 - See also Data languages
- Semantic network analysis, 35, 71, 258–264, 377.
 - See also Computational content analyses
- Semantic validity:
 - definition of, 218
 - dictionary, 253
 - evidence on, 366–367
 - explanation of, 370–375
 - method-driven analyses and, 399
 - measurement of
- Semio software package, 257
- Semiotics, theory of signs, 188
- Sentiment analysis, 253–254
- Shamir, Y., 261
- Shanas, E., 14, 56
- Shannon, C. E., 11, 75
- Shapiro, G., 26, 129
- Shared meaning, 28, 249
- Sharing of attributions standard, 207
- Shimoni, A., 223
- Shneidman, E. S., 96, 149
- Shor, J., 164
- Shuman, R. B., 49
- Siegel, S., 317, 322
- Signorielli, N., 17, 46, 59, 100, 119, 164, 193, 198, 265
- Sign-vehicle content analysis, 52, 61, 128
- Simon, H. A., 19, 71
- Simonton, D. K., 58
- SimStat content analysis software, 249
- Simulations:
 - cognitive processes, 71
 - hypothesis testing, 145–146
 - interviewing, 146–148
- Sindhvani, V., 137
- Singer, J. D., 132–133
- Single-valued coding, 344–346
- Sixsmith, Judith, 392
- Skalski, Paul D., 230
- Skinner, B. F., 66
- Smith, C. P., 19, 30, 53, 110
- Smith, M. S., 371
- Smythe, D. W., 17
- Snowball sampling, 121–122
- So, C. Y. K., 223
- Social constructivist analyses, 22
- Social deviance, 64
- Social media, 34–37, 50
- Social networks, 50, 117, 122, 240, 246, 259, 265
- Social processes, 4
 - algorithmic nature of, 3
 - collective associative memory, 20
 - computer systems and, 4
 - social grammars, social relationships and, 82
 - social realities, rearticulations of texts and, 264
 - society, functional requirements for, 205

- Social relationships, 81–82
- Social Science Research Council's Committee on Linguistics and Psychology, 18
- Social stereotypes, 13
- Social structures
- Social Systems
- Social validity, 362
- Socio-conceptual networks, 261–262
- Software. *See*
 - Computational content analysis
 - Computer-aided text analysis
 - Qualitative data analysis
 - Statistical association approaches
- Songs of Zion*, 10–11
- Soviet leadership succession study, 181
- Speech-disturbance ratio, 65
- Speed, G. J., 55
- Speirer, H., 15
- Spiegelman, M., 322
- Split-half technique, for sampling, 127–126
- SPSS software, 271, 273, 293, 353
- Stability of analytical constructs model, 192–193
- Stable correlations for problem-driven content analysis, 388–390
- Stable temporal pattern, as interpretive standard, 198
- Standards, 61–64
 - accurate representation standard, 199
 - applications of, 186
 - adoption of, in method-driven analysis, 400
 - analytical constructs and, 186
 - chance standard, 200–204
 - ethical standards, 11, 77
 - evaluations and, 62–64
 - fairness/balance standard, 13, 71
 - identifications and, 61–62
 - journalistic presentations and, 62–64
 - judgments and, 64
 - linearity standard, 201
 - reliability of data, standards for, 356–360
 - sharing of attributions standard, 207
 - stable temporal pattern standard, 198
 - uniform distribution standard, 198
- Standards for Educational and Psychological Testing* (American Educational Research Association, American Psychological Association, and National Council on Measurement in Education), 362, 381
- Stability
- Statistical abstracting, 255
- Statistical association approaches to computational content analysis, 256–258
 - Abstracts, automatic generation of
 - CatPac software
 - Concept associations, co-occurring words
 - Foundational theories of meaning
 - Hyperdimensional analogo to language
 - Latent semantic indexing
 - Linguistic environment of words
 - NEGOPY software, communication net analysis
 - Neuronal network, model of
 - Non-hierarchical word clustering
 - Support vector machines
 - TextAnalyst software
 - WORD software
- Statistical properties of *alpha*:
 - Information/total variance ratio
 - variability requirement 351–353
 - noise
 - sampling considerations, 354–356
 - standards for data reliability, 356–360
 - statistical significance, 353–354
- Statistical sampling theory, 115–116, 121, 124–125
- Statistical significance, 353–354, 396–397
- Steiner, G. A., 183
- Stemming process, 223
- Stempel, G. H., III, 120, 125–126
- Stevenson, M., 50
- Stone, L. D., 396
- Stone, P. J., 19, 30, 53, 96–97, 149, 180–181, 222, 251, 254, 371
- Stop-words, 223, 247, 399
- Stratified sampling, 119
- Strauss, A. L., 385
- Strength of associations, 66
- Strodthoff, G. G., 198
- Structural validity, 376–377
- Stuart, P., 142
- Study of Wartime Communications, 15
- Subbalakshmi, K. P., 223
- Subliminal messages, 15
- Substantive information for
 - recording/coding, 153–156
- Suci, G. J., 18, 143, 251
- Suicide note analysis, 96–97, 149, 180–181, 254–255
- Sullivan, D., 257
- Support vector machines (SVMs), 255
- Surrogacy. *See* Accuracy, surrogacy, and decisiveness
- Survey research, 5, 12–14, 23, 103, 115, 116
- Symbolic interactionism, 22
- Symptoms and indices:
 - analytical constructs as, 186–188
 - inferences from, 65–69
- Syntactical distinctions in unitizing, 108–109, 113
- Syntax of data languages, 158. *See also* Data languages
- Systematic sampling, 118–119
- Systems, 3, 54–55
- Table of co-occurrences, 226–227
- Taft-Hartley Labor Act, 62
- Tannenbaum, P. H., 18, 143, 251
- TAT (Thematic Apperception Tests), 17, 20, 111

- Taussig, T. G., 227
- Technical Recommendations for Psychological Tests and Diagnostic Techniques* (American Psychological Association), 363
- Technological explanations for inferences, 77–78
- Television:
- advertising appeals on, 213
 - Annenberg Television Script Archive, 230
 - channels, 2
 - cluster sampling and, 121
 - commentators, on political speeches, 145
 - face recognition software and, 227
 - FCC regulation of, 64
 - fictional characters on, 175, 199–201
 - introduction of, 13
 - discourse analysis and, 22
 - listenability index and, 65, 164
 - political debates on, 230
 - sports reporting, cultural values and, 58
 - stratified sampling and, 119
 - systematic sampling and, 119
 - UCLA Film and Television Archive, 230
 - Vanderbilt News Archive and, 49
- Television programming:
- categorizing, 128
 - coding for studies involving, 134, 153
 - content analysis and, 15, 17, 49
 - fiction, hedonism and, 46
 - institutional processes and, 75, 77–78
 - news programs, 22, 49, 50, 65, 393
 - minorities as represented in, 28, 33, 59, 361
 - ratings and, 78, 106
 - “reality” TV, 64
 - statistical “representativeness” in, 63
 - syntactical distinctions and, 108
 - violence in, 46, 65, 100, 134, 193, 198, 286, 362, 388
 - violence index, 65
 - “vocal publics” and, 77
- Tenney, A. A., 12, 55, 265
- Terminal values, in tree diagrams, 171
- Terwilliger, C., 322
- Tesch, R., 219
- Text:
- accounts of character strings, 220–227
 - computer analysis of, 18–21, 220–239
 - content analysis driven by, 384–386
 - content analysis framework for, 38–39
 - defined, 38
 - problem-driven content analysis, 390–392
 - networked, 239–246, 274
 - sampling techniques, 117–124
- TextAnalyst software package, 257
- TextArc website, 224
- Text-driven analyses, 384–386
- Text mining: overview, 227–228, 331–334
 - databases for, 229–232
 - evidence for theories in, 236–239
 - precision, measure of
 - problems in, 235–236
 - reliability of, 331–336
 - recall, measure of
 - replication of judgments on relevance of retrievals, 336
 - replication of searches, 334–336
 - sampling units in, 393
 - search engines for, 228–229
 - semantically valid queries for, 232–235
- TextPack content analysis software, 19, 250
- TextQuest content analysis software, 250–251, 385
- TextSmart software package, 257, 274
- Thematic Apperception Tests (TAT), 17, 20, 111
- Thematic distinctions in unitizing, 111–112, 114
- Theory of argumentation, 45
- Theory of Data, A* (Coombs), 157
- Thome, H., 56
- Thompson, S., 112
- Tijssen, R. J. W., 257
- Till, J. E., 392
- Time-series analysis approach, 56
- Toulmin, S. E., 45
- Transformational grammars, 160
- Tree diagrams of values, 171, 174–175
- Trends, extrapolations to infer, 55–56
- Trends in Content Analysis* (Pool), 18
- Tukey, J. W., 158
- Turow, J., 108
- Twitter.com, 34–36
- UCLA Film and Television Archive, 230
- Uncertainty, sources of, 191–193
- Uniform distribution standard, 198, 356
- Uniform Resource Locators (URLs), 118, 121
- United Nations Conference on Sustainable Development Rio+20, 35
- United Nations’ Food and Agriculture Organization (FAO), 35–36
- Units
- Aggregation of
 - Context, 105–107
 - Distinguished
 - Predefined
 - Recording/coding
 - Sampling
 - Units of enumeration
- Unitizing, 102–114
- categorical distinctions in, 109–110
 - definition of, 88
 - physical distinctions in, 107–108
 - productivity, efficiency, and reliability in, 112–114
 - propositional distinctions in, 110–111
 - syntactical distinctions in, 108–109
 - thematic distinctions in, 111–112

Unitizing finite continua overview, 337–338:
 agreement among intersections of segments in
 unitized continua
 coding of segments independent of unitizing,
 344–346
 data from, 338–340
 observed and expected coincidences of
 intersecting segments, 340–342
 partitions of a continuum into unequal
 segments, 342–343
 relevant and irrelevant segments, distinction
 between, 343–344
 Unstructured phenomena,
 data of, 47–48
 URLs (Uniform Resource Locators) of
 Web pages, 118, 121
 U.S. Department of Education, 230
 Usenet discussion group, 266
 Uses of content analysis. *See* Inferences
 U.S. Library of Congress, 15
 U.S. National Archives and Records
 Administration, 230
 U.S. Supreme Court, 141–142

Validating evidence, 37
 absence of, 96
 classification scheme for, 363
 content analysis framework for, 38,
 45–46, 73–74
 functional validity and, 377
 predictive validity and, 381, 382
 reliance on imperfect and indirect, 36
 sampling validity, 369–370
 semantic validity and, 370, 372
 typology for, 365–368

Validity, 361–382
 agreement measure of
 correlative, 378–381
 definition of, 361–365
 ecological
 functional, 253, 377–378
 predictive, 381–382
 research, 24, 40
 sampling, 368–370
 semantic, 218, 253, 370–375, 399
 structural, 367, 376–377

Values:
 categories
 codes for predefined units
 codes for segments in unitized continua, 346
 definition of, 159
 difference functions for pairs of
 members of multi-valued sets
 metrics of
 unordered/ordered, 330–331

Value studies, 17
 Vanderbilt News Archive, 49
 Van Dijk, T. A., 21–22
 Van Raan, A. F. J., 257

Variability in research question targets,
 191–192

Variables:
 characteristics of, 162–163
 coder interpretations of, 165–166
 definition of, 159, 161
 distinctions within
 implicit and explicit, 163
 nominal, 167
 open/closed
 ordered/unordered, 166–172

Variance analysis, 158
 Variation, insufficient, 351–353
 Varying probability sampling, 119–120
 VBPro content analysis software, 221–222,
 250, 385
 Vegelius, J., 316
 Venn diagrams, 237–238
 Verbal corpuses theory, 55
 Verbal designations for data, 139–140
 Verbastat software package, 274
 Vernadsky, V. I., 55
 Vindication, 367
 Violence index (Gerbner et al.), 164
 Vocabularies, community-specific, 35
 Voice recognition software, 227

Wallace, D. L., 68, 187, 223
 Wang, R., 134
Washington Post, 123
 Weaver, David, 81
 Weaver, D. H., 203, 204, 206
 Weaver, W., 11, 75, 359
 Webb, E. J., 47, 389
 Weber, M., 11
 Weber, R., 136
 Weber, R. P., 56, 169, 208, 251, 394
 Webgraphs, 117, 241–242
Webster's Dictionary of the English Language, 1
 Weddle, J. L., 248
 Weitzman, E. A., 219
 Wells, R. A., 123
 Weltevrede, E., 50
 Western Standard Publishing Company, 230
 Whissell, C. M., 253
 White, A. M., 67
 White, D. M., 77
 White, P. W., 12
 White, R. K., 17
 Willey, M. M., 12
Why Be Quantitative? (Lasswell), 17, 67
 Wikipedia.com, 240, 246
 Willey, M. M., 12
 “Wisdom of crowds,” 135, 137–138
 Witschge, T., 174
 Wittgenstein, L., 3, 189, 240
 Woelfel, J., 211, 257
Wood et al. v. Andrews et al. [1997], 373

- Woodward, J. L., 13
- Word association databases, 34
- Word clouds, 197–198
- Word co-occurrences, 57
- Wordlink program, 258
- WordNet software, 20, 235, 252
- Word processing software, 20
- WORDS program, 256
- WordStat content analysis software, 222, 224, 249–251, 253, 385
- Workshop on Content Analysis in the Social Sciences (1974), 19
- World Attention Survey, 108
- World Wide Web, unitizing, 110
- World Wildlife Foundation (WWF), 35–36
- Wright, C. E., 79
- Wu, A. H., 60–61
- Xu, Z., 251–252
- Yellow journalism, 12, 13
- YouTube.com, 49
- Yule, G. U., 68, 187, 223
- Zaninovich, M. G., 112
- Zeldow, P. B., 20
- Zeps, V. J., 19, 227
- Zillmann, D., 145, 168
- Zinnes, D. A., 112
- Zipf, G. K.,
 - Zipf's law, 222
- Zucker, H. G., 100, 378
- Züll, C., 220, 251
- ZyINDEX CATA software, 222
- ZyLAB CATA software, 222

ABOUT THE AUTHOR

Klaus Krippendorff (Grad. Ulm School of Design 1962; PhD, communication, University of Illinois, Urbana, 1967, and PhDc Linnaeus University, Kalmar, Sweden, 2012) is Professor of Communication and Gregory Bateson Professor Emeritus for Cybernetics, Language, and Culture at the University of Pennsylvania's Annenberg School for Communication. In addition to publishing numerous articles in journals of communication, sociological methodology, cybernetics, system theory, and design, he authored *Information Theory: Structural Models for Qualitative Data*; *A Dictionary of Cybernetics*; *On Communicating: Otherness, Meaning, and Information*; and *The Semantic Turn: A New Foundation for Design*. He edited *Communication and Control in Society* and *Design in the Age of Information: A Report to the National Science Foundation (NSF)* and coedited *The Analysis of Communication Content: Developments in Scientific Theories and Computer Techniques*.

The first edition of *Content Analysis: An Introduction to Its Methodology* was published in 1980, the second in 2004, and the third in 2013. The first edition was translated into Italian, Persian, Japanese, Spanish, Indonesian, and Hungarian, and the third into Chinese. In 2004, it received the International Communication Association (ICA) Fellows Book Award for its lasting contribution to the advancement of communication studies. In 2017, Klaus Krippendorff received the first methodological innovation award from the Mass Communication Division of ICA for his contributions to content analysis and reliability statistics. To broaden the range of examples and solutions to practical problems in the book, in 2009, he and Mary Angela Bock coedited *The Content Analysis Reader*.

Klaus Krippendorff currently pursues five exciting projects: With epistemology in mind, he explores how realities are socially constructed in public, political, professional, and scientific discourses. As a critical scholar, he seeks to understand conditions of social entrapment and possibilities of liberation from various forms of oppression, including by conceptions of power, artificial intelligence as superior human intelligence, and attributing agency to physical phenomena and abstractions used in language. His cybernetic background leads him to recursive conceptions of conversations—interpersonal, public, and organizational—and their erosion into diverse forms communication, discourses, and algorithms. Part of this project is to come to grips with the ongoing algorithmization of social institutions. As a former designer, now design scholar, he made it his mission to move human communication conceptions into the center of design considerations: conceptualizing technology through human interfaces; stakeholder networks within which technologies are realized; how media, texts, talk, and interactions constitute the social nature of artifacts; and the ecological interactions of technological artifacts that their human use sets in motion. In addition to these conceptual contributions, he continues to generalize his reliability statistics, Krippendorff's alpha, to a variety of data-making efforts.